PRACTICAL STRUCTURAL ANALYSIS AND DESIGN OF RESIDENTIAL BUILDINGS USING STAAD.*Pro* V8*i*, CSC ORION™ AND MANUAL CALCULATIONS

(According to Eurocode 2)

A Simplified Comparative Design Book

Volume 1, 1st Edition

- General Arrangement
- Modelling
- Materials
- Floor slabs
- Staircase
- Floor beams
- Columns
- Foundations
- Detailing
- Bending Schedule
- Quantification & Costing
- MATLAB® Programming

UBANI OBINNA U.

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(According to Eurocode 2)

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Structville Engineering

Ranks Michael Group

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This book is dedicated to Engr. Prof. C.H. Aginam, who taught me structural analysis and design in the classroom, and Engr. Adefarasin Adewunmi (MNSE), who taught me structural analysis and design in the office using computer aided means. I sincerely appreciate what you have done for me.

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About the author

Ubani Obinna Uzodimma graduated from the Department of Civil Engineering, Nnamdi Azikiwe University, Awka, with 1st class honours. He has been involved in the analysis and design of many reinforced concrete structures. He has some publications in international journals to his name.

Preface

This book is primarily intended for civil engineers and civil engineering students in universities and polytechnics. The main aim of this publication is for it to serve as a simplified practical guide when carrying out design using *Staad.Pro* or *Orion* software, and also as a first hand exposure to the manual design of reinforced concrete residential buildings. On the other hand, for engineers who are already familiar with BS 8110-1:1997, but intend to switch or have a good idea of the Eurocodes, this notebook can serve as an initial exposure, before considering more technical textbooks or reference manuals. I strongly believe that with this simple publication by your side, you can carry out full analysis and design of reinforced concrete structures manually, or by using computer programs like Orion and Staad.

Chapter 1 of this publication talks about the problems and challenges of housing in Africa, the influx of many design software and programs in the market, the position of engineers, and the implications of such programs. Furthermore, the basis of structural design based on Eurocode 0 (EN 1990) was introduced; including the concept of limit state design. Finally, the building case study we intend to analyse and design was briefly introduced.

Chapter 2 talks about the materials that are employed in reinforced concrete design – steel and concrete. The major ingredients of concrete and their properties were briefly discussed for the purpose of adequately specifying concrete. The behaviour of fresh concrete and the mechanical properties of hardened concrete were also reviewed. For reinforcement, the mechanical properties, types, and available sizes were reviewed. Loading of buildings to Eurocode 1 (EN 1991) was reviewed, and the self weights of very common construction materials were also presented.

Chapter 3 talks about how to prepare the general arrangement (GA) of buildings from architectural drawings. Here also, our case study building was fully presented, including the floor plans, elevations, and GA. Furthermore, a step by step guide to modelling, analysis, and design on Orion and Staad.Pro were also presented in a well detailed manner. The results of the analyses and design from both programs were also briefly presented.

Chapter 4 presented the reinforced concrete section analysis and design formulae in Eurocode 2 for flexural design, shear design, check of deflection and crack control.

In Chapter 5, the full manual analysis of the floor slabs were presented, including design of the staircase. Some references to the results from the computer programs were also highlighted.

The Chapter 6 of this book extensively handled the loading, analyses, and design of the floor beams. As a matter of fact, all the floor beams in the building plan were manually analysed and designed. Also, the internal stresses obtained from computer aided design were compared with the results from manual analysis for the entire thirteen beams.

In Chapter 7, the full loading, and analyses of the columns were carried out manually starting from the roof to the ground floor. Also, the processes of obtaining column design moments at ultimate limit state using sub-frames were presented. The column axial loads obtained from manual analysis were subsequently compared with the results from Staad.Pro and Orion. At the end, the designs of the columns based on the recommendations of Eurocode 2 were carried out.

Chapter 8 talked about the design of foundations. The design processes recommended by Eurocode 7 (EN 1997) were presented, including all the checks necessary for adequate structural performance of the foundation. The result obtained from the manual analysis was compared with result from Orion Software.

In Chapter 9, all the detailing guides according to Eurocode 2 were presented. Subsequently, the practical detailing of the floor slabs, beams, columns, and foundations of the case study building were presented in full.

In Chapter 10, the bending schedule and quantification of all the detailed reinforcements were presented.

Chapter 11 talks about how to quantify materials and cost a building, using our case study building as an example. This detailed chapter talked about how to estimate the quantity of concrete needed to complete a certain phase of construction like foundations, slabs etc. Also, presented is how to obtain the quantity of cement required (in 50 kg bags), the quantity of sand and granite required (in metre cube, tonnes, and number of tipper trips). The chapter also talks about how to estimate the quantity of excavation and fill, and how to estimate excavation and filling cost. Furthermore, the process calculating the number of blocks required for completing a project was presented. Also included in this section is how to estimate the quantity of cement and sand needed to mould blocks (based on the number of blocks to be moulded from one bag of cement). Also, guidance is given on how to estimate the quantity of formwork needed.

In Chapter 12, I discussed the discrepancies I discovered in manual analysis and computer aided analysis. This was followed by conclusion based on the things that I have seen from the work.

As you can see, this is a handbook that you can keep by your side that will enable you to carry out full design and presentation using any method of your choice. I called it a notebook because there are some details that were omitted that can only be found in more standard and specific textbooks. You can see that this book is not specifically devoted to any special topic, but to almost everything about a residential building. In the appendices, I presented how we can write MATLAB programs for flexural design, check of deflection, shear design, calculation of section properties, and analysis of sub-frames for analysis of column moments. Thank you for purchasing this book!

Ubani Obinna U.
March, 2017

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CHAPTER 1

INTRODUCTION

1.1 The Challenges of Housing in Africa

The challenges and problems of adequate housing and accommodation in Africa cannot be overemphasised. In Nigeria, Africa's most populous country, it estimated that 100,000 new houses are being built annually, compared to an annual demand of 700,000 houses, and an accumulated deficit of some 17 million houses. The housing crisis stems mainly from the ever increasing population and high migration to the urban areas. A CNN opinion report has it that Nigeria is expected to have 174 million people between the ages of 15-64 by the year 2035. In addition to that, the housing deficit is estimated at 17 million units annually.

This problem of inadequate housing has led to many incessant erections of buildings especially in the urban areas of the country. In order to achieve economy, substandard materials are employed for construction, with extremely poor construction practice. This has led to the collapse of many structures, thereby costing lives, properties, and endangering the environment. The Nigeria Building Code was published in the year 2006 to set building standards but the effect of this is yet to be fully manifested. Also, a lot of quacks have come into the construction industry, having little or no experience about construction or design of buildings. These quacks sometimes parade themselves as engineers or builders, and they are handling projects here and there. Furthermore, professional boundaries are often breached in the construction industry, and this is a major problem for the built environment of Nigeria. However, with adequate regulation, most of these problems can be efficiently handled, and adequate reformations should be made about the usage of land as we look forward to an interesting future.

1.2 On Design Codes

The most popular structural design code in Nigeria is the BS 8110-1:1997, and this is taught in most classrooms all over Nigeria. While there are no major differences in the flexural design of reinforced concrete elements between BS 8110 and EC2, there are major differences in shear design based on the strut angle inclination which varies between 21.8° to 45° in EC2, but is fixed at 45° in BS 8110. The column reinforcement design results in the two codes are also not so different, despite the more rigorous approach in EC2. I am presenting this design in EC2 because of its more universality especially in the European Union and not just in Britain alone. Through this way, the Nigerian engineering community can interact freely, be able to understand easily, and also be able to present their own designs with very quick understanding irrespective of borders. This is more possible with the EC2 than with BS 8110. The BS 8110 was outdated in Britain in the year 2008, and they have completely switched to the Eurocodes.

1.3 On the Use of computer Software

I feel grateful over the influx of many structural analysis and design software in the international market. Nigerian engineers are not left out in enjoying the benefits from these programs since they have made the job faster and easier. However, the problems associated with these software are mainly not from the software themselves but from the users. On no grounds should the speed, efficiency, and 'reliability' of these computer programs give room to mental laziness. As far as what I know is concerned, there is no substitute for full understanding of theory of structures, and structural design from the first principles. I feel blessed to have passed through an institution where computer programs for analysis and design are assumed to be non-existent. All our analysis and design were carried out manually, and that is why I am confident whenever I use computer programs, because I will be properly guided in interpreting the results, irrespective of the producer of the software.

It is a blessing for an engineer to be able to question and scrutinize results produced from computer software and programs. Sometimes, these programs may have bugs, or we may input wrong values, or make mistakes in our modelling. These programs may not be able to judge our inputs accurately, even though some programs are sophisticated enough to warn us when our inputs are way beyond the standard deviation. For instance, if the applied load is say 300KN, and by error we input 30KN, the program will not detect any error, and will process the results based on the inputs. However, if we input the compressive strength of 250 N/mm^2 for concrete, when we intend to put 25 N/mm^2, a good program should warn us about the value, and it is now left for you to make the judgement when you see the results. In the same vein, a software user should know the range of the magnitude of internal stresses expected from any applied load and so on. For instance, if a span is relatively short but heavily loaded, I expect to combat with heavy shear forces in my design. If I see an unnecessarily heavy bending moment for a short span, I will definitely go back to my inputs, and cross check my values (perhaps I might have input 15m for the span length instead of 1.5m). This same thing goes for the shape and nature of internal stresses diagram produced. Normally, you should know the areas that should be hogging and sagging, and also areas where you expect very little or no bending moment (e.g. central columns in a regular and uniform building, or members of trusses). When you see unlikely results, you can check the used stiffness, modelling errors, input signs (sign conventions are sensitive in most programs) etc.

It is a disaster for someone who does not under the meaning of flexural rigidity (EI), and the possible bending moment diagram of continuous beam of equal span subjected to uniformly distributed load, to be dabbling with structural analysis and design software. The idea of '*garbage in, and garbage out*' should be strongly discouraged. Engineers should be very sound in theory first, before considering computer aided analysis and design.

This idea forms the basis of this design book, whereby manual analysis for structural analysis and design (presented fully) has been carried out, and compared with two other popular analysis and design programs. This will be an eye opener and study guide for beginners, and an additional comparative source of knowledge for already experienced engineers.

1.4 Basic Requirements in Structural Design

There are basic requirements which a structure that has been designed must meet. These are well outlined in Eurocode 0 (EN 1990:2002). EN 1990 can be described as the head of Eurocode. The code establishes principles and requirements for the safety, serviceability and durability of structures, and describes the basis for their design and verification, including guidelines for related aspects of structural reliability.

EN 1990 is normally used in conjunction with other Eurocodes in structural design of buildings and civil engineering works, including geotechnical aspects, structural fire design etc. In a nutshell, EN 1990 demands that a structure should be designed and executed in such a way that it will:

1. During its intended life, with appropriate degrees of reliability and in an economical way, sustain all actions and influences that are likely to occur during execution and use, and also remain fit for the intended use.
2. Have adequate mechanical resistance, serviceability, and durability
3. In the event of fire, have adequate resistance for the required period of fire exposure
4. Not be damaged by accidents (e.g. explosion, impact and consequences of human error) to an extent disproportionate to the original cause.

These requirements are met by employing the popular 'Limit State' design method.

1.5 Limit States

Limit states are defined as states beyond which the structure no longer satisfies an agreed relevant design criteria. Two basic groups of limit states considered are;

(1) Ultimate limit states and
(2) Serviceability limit states.

Ultimate limit states (ULS) are those associated with collapse or failure, and generally govern the strength of the structure or structural members because they correspond to the maximum load carrying resistance of such member. ULS also include loss of equilibrium or stability of the structure as a whole.

Serviceability limit states are states which correspond to conditions beyond which specified service conditions of a structure or structural member are no longer met. Serviceability limit states may be reversible or irreversible. They generally include issues like deformation, cracking and vibration which:

(1) Damage the structure or non-structural elements (finishes, partitions, etc.) or the contents of buildings (such as machinery or sensitive equipment)
(2) Cause discomfort to the occupants of buildings
(3) Affect adversely appearance, durability or water tightness.

Serviceability limit state governs the stiffness of the structure and the detailing of reinforcement within it.

1.6 Actions

Actions in Eurocodes are simply the loadings applied to a structure. Actions from externally applied load (including self weight) are called *direct actions*, while imposed deformations such as temperature differences, settlement of supports, moisture variation etc are called *indirect actions.* The effects of actions are the internal forces (shear, moments, axial, stresses, strains) on the members, or the overall rotation and translation of the structure.

Dead loads are called permanent actions (*G*) in Eurocode, and they are the loads that are likely to act on a building throughout a given period of time. Live loads are called variable actions (*Q*) which are actions for which the variation of magnitude with time is very significant. There are also accidental actions (*A*), which are actions that are of usually short duration but of significant magnitude. Accidental actions are deemed unlikely to occur on a given structure during the design working life. Characteristic valued for loads are well specified in Eurocode 1 (EN 1991), and they are briefly highlighted in Chapter 2 of this book.

The basic classification of actions is given in Table 1.1 below;

Classification of actions

Table 1.1: Classification of actions

Permanent Action	Variable Action	Accidental Action
Self-weight of structures, fittings and fixed equipment	Imposed floor loads	Explosions
Prestressing force	Snow loads	Fire
Water and earth loads	Wind loads	Impact from vehicles
Indirect action, e.g. settlement of supports	Indirect action, e.g. temperature effects	

In limit state design, loads are usually magnified by γ factors called the partial safety factors. The γ factors account for the following;

(1) The possibility of unfavourable deviation of the loads from the characteristic values
(2) Inaccuracies in the analyses
(3) Unforeseen redistribution of stress
(4) Variations in the geometry of the structure and its elements, since this affects the determination of the action effects.

If two variable loads are applied to a structure simultaneously, and the loads are independent of each other, it will be unrealistic to use full partial safety factor as it is unlikely that the two loads will be at their maximum at the same time. Therefore joint probabilities are considered to ensure that the probability of occurrence of the two loads is the same as that of a single load. It will be more reasonable to consider one load at its maximum in conjunction with a reduced value for the other load (*Beeby and Narayanan*, 2009). This leads to multiplication of the load case with a 'combination value' ψ_0.

When a design does not involve geotechnical actions, the strength of elements should be verified using load combination Set B. Two options are given. Either combination (6.10) from EN1990 or the less favourable of equations (6.10a) and (6.10b) may be used:

$$\gamma_{G,j,sup}G_{k,j,sup} + \gamma_{G,j,inf}G_{k,j,inf} + \gamma_{Q,1}Q_{k,1} + \gamma_{Q,i}\psi_{0,i}Q_{k,i} \quad (6.10)$$

$\gamma_{G,j,sup}G_{k,j,sup}$ is used when the permanent loads are unfavourable, and $\gamma_{G,\,j,\,inf}G_{k,j,inf}$ is used when the permanent actions are favourable. Numerically, $\gamma_{G,j,sup}$ = 1.35, $\gamma_{G,j,inf}$ = 1.0, and γ_Q = 1.5 when unfavourable and 0 when favourable (EN 1990,UKNational Annex)

.

$$\gamma_{G,\,j,\,sup}G_{k,\,j,\,sup} + \gamma_{G,\,j,\,inf}G_{k,\,j,\,inf} + \gamma_{Q,\,i}\psi_{0,\,i}Q_{k,\,i} \quad (6.10a)$$

$$\xi\gamma_{G,\,j,\,sup}G_{k,\,j,\,sup} + \gamma_{G,\,j,\,inf}G_{k,\,j,\,inf} + \gamma_{Q,1}\,Q_{k,1} + \gamma_{Q,\,i}\psi_{0,\,i}Q_{k,\,i} \quad (6.10b)$$

Numerically, ξ = 0.925, $\gamma_{G,j,sup}$ = 1.35, $\gamma_{G,j,inf}$ = 1.0, and γ_Q = 1.5 when unfavourable and 0 when favourable (EN 1990, UK National Annex).

The above combinations assume that a number of variable actions are present at the same time. $Q_{k,1}$ is the leading variable action when it is obvious that it is so, otherwise each load is in turn treated as a dominant load and the others as secondary. The dominant load is then combined with the combination value of the secondary loads while both are multiplied by their respective γ values (see Table 1.2).

Table 1.2: Partial safety factors for load combinations in EN 1990 - ultimate limit state (Beeby and Narayanan, 2009)

Load Combination	Permanent Load		Variable Load			
	Adverse	Beneficial	Adverse	Beneficial	Wind	Prestress
Combination 6.10						
Permanent + Imposed	1.35	1.0	1.5	0	-	1.0
Permanent + Wind	1.35	1.0	-	-	1.5	1.0
Permanent + Imposed + Wind	1.35	1.0	1.5	0	1.5 x 0.5 = 0.75	1.0
Combination 6.10(a)						
Permanent + Imposed	1.35	1.0	ψ_0^*1.5	-	-	1.0
Permanent + Wind	1.35	1.0	-	-	$\psi_0 1.5 = 0.9$	1.0
Permanent + Imposed + Wind	1.35	1.0	ψ_0^*1.5	-	$\psi_0 1.5 = 0.9$	1.0
Combination 6.10(b)						
Permanent + Imposed	ξ1.35 = 1.25	ξ1.0 = 0.925	1.5	-	-	1.0
Permanent + Wind	ξ1.35 = 1.25	ξ1.0 = 0.925	-	-	1.5	1.0
Permanent + Imposed + Wind	ξ1.35 = 1.25	ξ1.0 = 0.925	1.5	-	$\psi_0 1.5 = 0.9$	1.0

ψ_0^* varies for the building type.

However, for residential buildings in areas where environmental loads are not adverse, the combination;

$E_d = 1.35G_k + 1.5Q_k$ is usually very adequate for design. However, when it is envisaged that other actions such as temperature difference, settlement of supports, etc may be influential, adequate considerations will have to be made in the combination of actions.

1.7 Design Working Life

The design working life of a building should be specified.

The indicative working life of a building is given in Table 2.1 of EN 1990:2002+A1:2005, and given in Table 1.3 below;

Table 1.3: Working life of a building (Table 2.1, EN 1990:2002+A1:2005)

Design Working Life Category	Indicative Working Life (years)	Example
1	10	Temporary structures
2	10 - 25	Replaceable structural parts e.g. bearings
3	15 – 30	Agricultural and similar structures
4	50	Building structures and other common structures
5	100	Monumental buildings, bridges, and other civil engineering structures

1.8 Durability requirements

According to clause 2.5 of EN 1990:2002+A1, a structure shall be designed such that deterioration over its design working life does not impair the performance of the structure below that intended, having due regard to its environment and the anticipated level of maintenance. To achieve a durable structure, factors like the intended use of the structure, environmental conditions, materials, choice of structural system, structural detailing, quality of control and workmanship, protective measures and intended maintenance during the design life must be taken into account.

In view of environmental and usage conditions, all forms of aggressive or indirect action should be considered including:

chemical attack, arising from e.g.

- the use of the building or the structure (storage of liquids, etc)
- solutions of acids or sulphate salts
- chlorides contained in the concrete
- alkali-aggregate reactions

physical attack, arising from e.g.

- temperature change
- abrasion
- water penetration

1.9 Methods of durability Verification: Concrete Cover

The concrete cover is the distance between the surface of the reinforcement closest to the nearest concrete surface (including links and stirrups and surface reinforcement where relevant) and the nearest concrete surface. The nominal cover shall be specified on the drawings. It is defined as a minimum cover, c_{min}, plus an allowance in design for deviation, Δc_{dev} .

$$C_{nom} = c_{min} + \Delta c_{dev} \text{ ---------- (1.1)}$$

According to clause 4.4.1.2, minimum concrete cover, c_{min}, shall be provided in order to ensure the safe transmission of bond forces, protection of the steel against corrosion (durability), and an adequate fire resistance. C_{min} is given by expression (1.2) below;

$$c_{min} = \max \{c_{min,b}; c_{min,dur} + \Delta c_{dur,\gamma} - \Delta c_{dur,st} - \Delta c_{dur,add}; 10\ mm\} \text{ ------- (1.2)}$$

where:
$c_{min,b}$ minimum cover due to bond requirement
$c_{min,dur}$ minimum cover due to environmental conditions
$\Delta c_{dur,\gamma}$ additive safety element
$\Delta c_{dur,st}$ reduction of minimum cover for use of stainless steel
$\Delta c_{dur,add}$ reduction of minimum cover for use of additional protection

The minimum cover for bond requirement is the diameter of the bar, but if the maximum size of the aggregate is greater than 32mm, the value should be increased by 5mm (see clause 4.4.1.2 (3) of EN 1992). The minimum cover due to environmental conditions can be gotten from the Table 1.4 below;

Table 1.4: Values of minimum cover, $c_{min,dur}$, requirements with regard to durability for reinforcement steel in accordance with EN 10080. (Table 4.4N, EN 1992-1-1:2004)

Environmental Requirement for $c_{min,dur}$ (mm)							
Structural Class	Exposure Class according to Table 4.1						
	X0	XC1	XC2 / XC3	XC4	XD1 / XS1	XD2 / XS2	XD3 / XS3
S1	10	10	10	15	20	25	30
S2	10	10	15	20	25	30	35
S3	10	10	20	25	30	35	40
S4	10	15	25	30	35	40	45
S5	15	20	30	35	40	45	50
S6	20	25	35	40	45	50	55

The exposure classes related to environmental conditions in accordance with EN 206-1 can be found in Table 4.1 of Eurocode 2. The ones most likely to apply to residential buildings are shown in Table 1.5 below.

Table 1.5: Exposure classes related to environmental conditions in accordance with EN 206-1 (Table 4.1, EN 1992-1-1:2004)

Class designation	Description of the environment	Informative examples where exposure classes may occur
1 No risk of corrosion or attack		
X0	For concrete without reinforcement or embedded metal: all exposures except where there is freeze/thaw, abrasion or chemical attack For concrete with reinforcement or embedded metal: very dry	Concrete inside buildings with very low air humidity
2 Corrosion induced by carbonation		
XC1	Dry or permanently wet	Concrete inside buildings with low air humidity Concrete permanently submerged in water
XC2	Wet, rarely dry	Concrete surfaces subject to long-term water contact Many foundations
XC3	Moderate humidity	Concrete inside buildings with moderate or high air humidity External concrete sheltered from rain
XC4	Cyclic wet and dry	Concrete surfaces subject to water contact, not within exposure class XC2

The value of Δc_{dev} for use in a Country may be found in its National Annex. The recommended value is 10 mm.

1.10 Fire Resistance Requirement for Slabs

The fire resistance requirements for solids slabs with beams are given in Table 1.6 below.

Table 1.6: Solid slab fire resistance requirement (Bond et al, 2006)

Standard Fire Resistance	Minimum Dimensions (mm)		
	One-way spanning slab	Two-way spanning slab	
		Ly/Lx ≤ 1.5	1.5 < Ly/Lx ≤ 2.0
1 hour	h = 80, a = 20	h = 80, a = 10	h = 80, a = 15
1 hour 30 minutes	h = 100, a = 30	h = 100, a = 15	h = 80, a = 20
2 hours	h = 120, a = 40	h = 80, a = 20	h = 80, a = 25
4 hours	h = 175, a = 65	h = 175, a = 40	h = 175, a = 50

h = minimum thickness of slab; a = minimum distance from the centre of the main bar to the surface of the member

1.11 The building we are analysing

In order to fully present the simplified processes involved in the design of simple reinforced concrete residential building, the case study of the building with the approach elevation shown below has been considered for full structural analysis and design. The building is a relatively small one required to fit into a 10m x 15m plot of land.

Figure 1.1: Front view of residential building under consideration

From chapter 3, the floor plans, the elevations, the sections, and the general arrangement of the building have been shown completely. The building was analysed using two of the most popular design software in Nigeria, *Staad.Pro*, and *Orion*. Also, full manual calculations were also carried out to compare the answers gotten from the analyses.

CHAPTER 2

LOADING AND MATERIALS SPECIFICATIONS

2.1 Introduction

The primary materials for reinforced concrete construction are concrete and steel. It is generally known in civil engineering that concrete performs poorly in tension, but very good in compression, while steel is very good in compression and tension. Therefore, reinforced concrete is a composite material wherein the steel reinforcement takes up the tensile stresses and concrete takes up the compressive stresses. Sometimes, steel is also employed to assist concrete in handling compressive stresses (for example in doubly reinforced beams).

2.2 Concrete

While steel is manufactured in the industry under controlled conditions, the production of concrete can encounter many variations on site. Such variations can arise from a lot of situations like mixing, transportation, placement, consolidation etc. Concrete is a material that is made from cementitious material (cementing medium) (*Neville and Brooks*, 2008). Normally, such a cementing medium is the product of hydration reaction, but I wish to keep it very simple at this point. *In a popular sense, concrete is a composite mixture of water, cement, fine aggregates, and coarse aggregates, held together by hydration reaction between the cement and water*. Another ingredient of cement that is normally specified to achieve a desired special effect is generally called an *admixture*.

Figure 2.1: All the primary ingredients for the production of concrete

2.2.1 Portland cement

According to *Neville and Brooks* (2008), Portland cement is the name given to a cement obtained by intimately mixing together calcareous and argillaceous, or silica-, alumina-, and iron oxide-bearing materials, burning them at a clinkering temperature, and grinding the resulting clinker. By implication, Portland cement consists mainly of lime, silica, alumina, and iron oxide. The main compounds in Portland cement are given in the Table 2.0 below;

Table 2.0: Main compounds of Portland Cement (Adapted from Table 2.1, Neville and Brooks, 2008)

Name of compound	Oxide composition	Abbreviation
Tricalcium silicate	$3CaO.SiO_2$	C_3S
Dicalcium silicate	$2CaO.SiO_2$-	C_2S
Tricalcium aluminate	$3CaO.Al_2O_3$	C_3A
Tetracalcium aluminoferrite	$4CaO.Al_2O_3.Fe_2O_3$	C_4AF

C_3S and C_2S are the most important compounds since they are responsible for the strength of the hydrated cement paste. C_3A is beneficial in that it facilitates the combination of lime and silica. C_4AF reacts with gypsum to form calcium sulfoferrite and its presence may accelerate the hydration of silicates.

2.2.2 Aggregates

It is always necessary to use the right type and quality of aggregates in the production of concrete. The fine and coarse aggregates generally occupy 60% to 75% of the concrete volume (70% to 85% by mass) and strongly influence the concrete's freshly mixed and hardened properties, mixture proportions, and economy(*Kosmatka, Kerkhoff and Panarese, PCA,* 2003). Aggregates are normally sourced from rivers, natural deposits, or from rock quarrying.

Fine aggregates generally consist of natural sand or crushed stone with most particles smaller than 5 mm (0.2 inches). Coarse aggregates consist of one or a combination of gravels or crushed stone with particles predominantly larger than 5 mm (0.2 in.) and generally between 9.5 mm and 37.5 mm (3/8 in. and 11/2 in.). Coarse aggregates are normally described in shape as irregular, rounded, flaky, angular, elongated, or flaky and elongated. The physical properties of aggregates that are normally specified for the purpose of concrete mix design are specific gravity, bulk density, porosity and absorption, and the moisture content. For fine aggregates, the fineness modulus is normally specified also from sieve analysis which is defined as the sum of the cumulative percentages retained on the sieves of the standard series, divided by 100 (see section 11.2.2.2 of Chapter 11 for mix design).

2.2.3 Mixing water

The recommended mixing water for concrete should be water that is suitable for drinking. The limits of impurities in mixing water according to different codes are given in Table 2.1 below;

Table 2.1: Limit of impurities in mixing water (mg per litre, ppm) (Table 4.1, *Neville and Brooks*, 2008)

Impurity	BS 3148:1980	BS EN 1008:2002	ASTM C 1602/C 1602M-06
Chloride ion:			
Prestressed concrete	500	500	500
Reinforced concrete		1000	1000
Plain concrete		4500	
Sulphate	1000 (SO_3)	2000 (SO_3)	3000(SO_4)
Alkali	1000	1500	600

Sodium carbonate (alkali) can cause very rapid setting, while bicarbonates can either accelerate or retard the setting time. In large concentrations, these salts can reduce the concrete strength. Concern over high chloride content in mixing water is mainly due to the possible corrosion of reinforcing steel

by the chloride ions. Chloride ions attack the protective oxide film formed on the steel by the highly alkaline (pH greater than 12.5) chemical environment present in concrete. Concern over high sulphate content in mix water is due to possible expansive reactions and deterioration by sulphate attack, especially in areas where the concrete will be exposed to high sulphate soils or water (*Kosmatka, Kerkhoff and Panarese, PCA,* 2003).

2.2.4 Admixtures

Admixtures are those ingredients in concrete other than Portland cement, water, and aggregates that are added to the mixture immediately before or during mixing. Admixtures can be classified by function as follows (*Kosmatka, Kerkhoff and Panarese, PCA,* 2003):

1. Air-entraining admixtures
2. Water-reducing admixtures
3. Plasticizers
4. Accelerating admixtures
5. Retarding admixtures
6. Hydration-control admixtures
7. Corrosion inhibitors
8. Shrinkage reducers
9. Alkali-silica reactivity inhibitors
10. Colouring admixtures
11. Miscellaneous admixtures such as workability, bonding, damp-proofing, permeability reducing, grouting, gas-forming, anti-washout, foaming, and pumping admixtures

2.2.5 Fresh Concrete Workability

The hardened properties of concrete largely depend on the degree of compaction achieved while placing the concrete when it is still fresh. *Workability is the amount of useful internal work necessary to produce full compaction (Neville and Brooks,* 2008). Consistence is the ease with which concrete will flow. Workability of concrete depends on a number of interacting factors such as water content, aggregate type and grading, aggregate/cement ratio, presence of admixtures, and fineness of cement. Also, time and temperature affects workability. Any freshly mixed concrete will stiffen with time, but this should not be in any way confused with setting. Workability is popularly measured on site using the slump test. The degree of compaction achieved in concrete is called the *compacting factor*, which is measured by the density ratio (i.e. the ratio of the density actually achieved in test to the density of the same concrete fully compacted).

Also, the water to cement ratio is another very important parameter that influences the workability, and final strength of the hardened concrete. Adequate quantity of water is required for full hydration, but the excess of it evaporates thereby leaving pores in the concrete that reduces the strength. The relationship between water/cement ratio and the compressive strength of concrete is given in Figure 2.2 below;

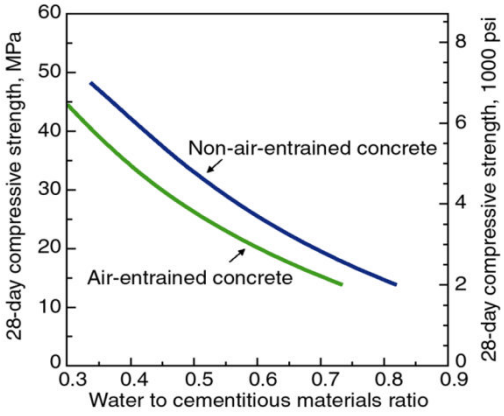


Figure 2.2: Relationship water/cement ratio and 28-day compressive strength (PCA, 2003)

Table 2.2 below gives the relationship between slump, degree of workability, and compacting factor, while Table 2.3 gives the mixing water and target air content requirement for different slumps and nominal size of aggregates.

Table 2.2: Relationship between slump, degree of workability, and compacting factor

Degree of workability	Slump (mm)	Compacting factor
Very low	0 – 25	0.78
Low	25 – 50	0.85
Medium	50 -100	0.92
High	100 - 75	0.95

Table 2.3: Mixing water and target air content requirement for different slumps and nominal size of aggregates (PCA 2010)

Slump (mm)	Water kg/m^3 of concrete for indicated sizes of aggregate (non – air entrained)					
	9.5mm	12.5mm	19mm	25mm	37.5mm	50mm
25 - 50	*207*	*199*	*190*	*179*	*166*	*154*
75 - 100	*228*	*216*	*205*	*193*	*181*	*169*
150 - 175	*243*	*228*	*216*	*202*	*190*	*178*
Approximate amount of entrapped air	*3%*	*2.5%*	*2%*	*1.5%*	*1%*	*0.5%*

In considering the workability, the concrete should not segregate (lack of cohesiveness and homogeneity). This is very necessary if full compaction is to be achieved. *Bleeding* is a form of segregation in which some of the water in the mix tends to rise to the surface of freshly placed concrete. Bleeding is caused by the inability of the solid constituents of the mix to hold all of the mixing water when they settle downwards.

2.2.6 Properties of concrete for use in Eurocode 2

In the design of concrete structures, engineers have the flexibility to specify particular concrete type(s) aimed at meeting the specific performance requirements for their project. For instance where calculated deflections exceed serviceability limits, the designer can achieve lower deflections by increasing the class of concrete and the associated modulus of elasticity, rather than by resizing members (*Bamforth et al*, 2008).

Some mechanical properties of concrete specified by designers are;

- Compressive strength
- tensile and flexural strength
- bond strength
- modulus of elasticity
- tensile strain capacity
- creep

2.2.6.1 Compressive strength

In BS EN 206-1: *Concrete – Specification, performance, production and conformity*, compressive strength is expressed as a strength class. BS EN 1992-1-1 uses the characteristic compressive cylinder strength f_{ck} (based on 2:1 cylinders) as the basis of design calculations. Normally, 28 days compressive strength is specified, but in some cases, higher days strength can be specified. In the UK, and as well as in Nigeria, the compressive strength is usually tested using cubes (usually 150mm x 150mm) rather than cylinders. A higher strength is obtained for cubes because the test machine platens offer greater lateral restraint due to the lower aspect ratio (*Bamforth et al*, 2008). In BS EN 206-1, the 2:1 cylinder strength is taken to be about 20% less than the cube strength for normal structural concrete but with higher strength classes, the cylinder strength achieves a higher proportion of the cube strength. To accommodate these differences, the strength class is defined by both the cylinder and the cube strength (for example, C30/37 **C**cyl/cube).

The design compressive strength (f_{cd}) for concrete according to BS EN 1992-1-1 is given by;

$$f_{cd} = \frac{\alpha_{cc} f_{ck}}{\gamma_c} \text{ ---------- (2.1)}$$

where;
f_{ck} = characteristic cylinder compressive strength of concrete at 28 days
γ_c = partial (safety) factor for concrete
α_{cc} = a coefficient taking account of long-term effects on the compressive strength (which is reduced under sustained load) and unfavourable effects resulting from the way the load is applied.

BS EN 1992-1-1 recommends that α_{cc} = 1. However, α_{cc} is an NDP and the UK National Annex to BS EN 1992-1-1 recommends that α_{cc} should be 0.85 for compression in flexure and axial loading and 1.0 for other phenomena (for example, shear, torsion and web compression – see PD 6687 Clause 2.3). It may also be taken conservatively as 0.85 for all phenomena.

2.2.6.2 Development of strength with time

While design is usually based on the 28-day strength, BS EN 1992-1-1, sub-clause 3.1.2(6) gives an expression for the development of the mean compressive strength of concrete with time at 20°C as follows:

$$f_{cm}(t) = [\beta_{cc}(t)]\, f_{cm} \text{ ---------- (2.2)}$$

where;
$f_{cm}(t)$ is the mean compressive strength at age t days.

f_{cm} = target mean strength

$$f_{ck} = f_{cm} - 1.64SD \text{ ---------- (2.3)}$$

where 1.64SD = 8
Therefore;
SD (Standard deviation) = 8/1.64 ≈ 5MPa

$$\beta_{cc}(t) = \exp\left\{s\left[1 - \left(\frac{28}{t}\right)^{0.5}\right]\right\} \text{ ---------- (2.4)}$$

where;
s is a coefficient which depends on cement type
= 0.20 for cement of strength classes CEM 42.5R, CEM 52.5N and CEM 52.5R (Class R)
= 0.25 for cement of strength classes CEM 32.5R, CEM 42.5N (Class N)
= 0.38 for cement of strength classes CEM 32.5N (Class S)
(where Class R = high early strength; Class N = normal early strength; Class S = slow early strength).

In Nigeria, 42.5R, 42.5N, 32.5R and 32.5N are readily available from major producers like Dangote Cement plc.

2.2.6.3 Tensile strength

In design, tensile strength is used in both serviceability and ultimate limit state calculations. For example in considerations of cracking, deflection, shear, punching shear, bond and anchorage, in evaluation of the cracking moment for prestressed elements, etc

The design tensile strength of concrete, f_{ctd}, according to BS EN 1992-1-1 is taken as:

$$f_{ctd} = \frac{\alpha_{ct} f_{ctk\ 0.05}}{\gamma_c} \text{ ---------- (2.5)}$$

where;
$f_{ctk\ 0.05}$ = characteristic tensile strength of concrete at 28 days
γ_c = partial (safety) factor for concrete = 1.5
α_{ct} = coefficient taking account of long-term effects on the tensile strength, this is an NDP with a recommended value of 1.

For normal structural uses, the mean tensile strength, f_{ctm}, is related to the cylinder strength by the expressions:

Strength classes ≤ C50/60 $f_{ctm} = 0.30\ f_{ck}^{(2/3)}$ ---------- (2.6)

Strength classes > C50/60 $f_{ctm} = 2.12 \log_e [1 + \frac{f_{cm}}{10}]$ MPa ---------- (2.7)

Note that for strength classes ≤ C50/60 f_{ctm} is derived from f_{ck} while for the higher strength classes > C50/60 the tensile strength is derived from f_{cm}.

2.2.6.4 Flexural Tensile strength

The flexural strength is the higher of:

$$f_{ctm,fl} = \left(1.6 - \frac{h}{1000}\right) f_{ctm} \text{ ---------- (2.8)}$$

where;
h is the total member depth in mm

or

$$f_{ctm,fl} = f_{ctm} \text{ ---------- (2.9)}$$

Rearranging Expression (2.8), the f_{ctm} may be estimated from the flexural strength measured on a 100 × 100mm prism in accordance with BS EN 12390-5 and $f_{ctm} = f_{ctm,fl}/1.5$

2.2.6.5 Effect of Age on Tensile strength

BS EN 1992-1-1 provides expressions for calculating tensile strength at different maturities:

$$f_{ctm}(t) = [\beta cc(t)]^{\alpha} f_{ctm} \text{ ---------- (2.10)}$$

where;

$\beta_{cc}(t)$ is defined in Expression (2.4)
α= 1 for t < 28 days
α = 2/3 for t ≥ 28 days.

2.2.6.6 Bond Strength

BS EN 1992-1-1 provides information on bond in relation to anchor lengths. The ultimate bond stress is given by the expression:

$$f_{bd} = 2.25\ \eta_1\ \eta_2\ f_{ctd} \text{ ---------- (2.11)}$$

where;
f_{bd} is the ultimate (design) bond stress
η_1 is a coefficient related to the quality of the bond condition and the position of the bar during concreting
= 1.0 for condition of good bond
= 0.7 for all other cases and for bars in structural elements built with slip forms
η_2 is related to bar diameter
= 1.0 for ϕ ≤ 40mm (NDP)
= (140 – ϕ)/100 for ϕ > 40mm

f_{ctd} is the design tensile strength defined as:

$$f_{ctd} = \frac{\alpha_{ct} f_{ctk\ 0.05}}{\gamma_c}$$

where;
γ_c is the partial safety factor for concrete = 1.5
α_{ct} is a coefficient taking account of long-term effects on the tensile strength and unfavourable effects resulting from the way the load is applied = 1 (NDP).

2.2.6.7 Modulus of elasticity

There are two types of elastic modulus - the **static** modulus, and the **dynamic** modulus. The static modulus is measured by plotting the deformation of a cylinder under an applied load, usually 30–40% of the ultimate load.

In design, the secant modulus, E_{cm} (in GPa), is derived from the mean compressive strength, f_{cm} (in MPa), from the expression:

$$E_{cm} = 22\,[f_{cm}/10]^{0.3}\ \text{GPa} \quad \text{---------- (2.12)}$$

The variation of modulus of elasticity with time is estimated using the expression:

$$E_{cm}(t) = [f_{cm}(t)/f_{cm}]^{0.3}\,E_{cm} \quad \text{---------- (2.13)}$$

2.2.6.8 Creep

Creep is time-dependent deformation (strain) under sustained loading, excluding non load - induced deformations such as shrinkage, swelling, thermal strain. Creep strain is typically two to four times the elastic strain and knowledge of creep is needed for several reasons:

- To estimate long-term deflections in beams and long-term shortening in columns and walls. This may be important, for example, in establishing tolerances for movement when fixing rigid, brittle partitions to a concrete frame.
- To estimate prestress losses.
- To estimate stress relaxation and redistribution over time. This may be beneficial in reducing the risk and/or extent of cracking. Creep in tension may also partly relieve the stresses induced by other restrained movements, for example, drying shrinkage, thermal contraction; or by loading.

The ultimate creep strain is calculated by multiplying the elastic strain by a creep coefficient using the expression:

$$\varepsilon_{cc}(\infty, t_0) = \varphi(\infty, t_0)\,(\sigma_c/E_c) \quad \text{---------- (2.14)}$$

where;
$\varepsilon_{cc}(\infty, t_0)$ = creep deformation at time $t = \infty$
$\varphi(\infty, t_0)$ = creep coefficient at time $t = \infty$
σ_c = constant compressive stress applied at time $t = t_0$
E_c = tangent modulus = 1.05 Ec_m

2.3 Steel and Reinforcement

Reinforcement for concrete generally consists of deformed steel bars, or welded steel mesh fabric. Normal reinforcement relies entirely upon the alkaline environment provided by a durable concrete cover for its protection against corrosion. Under special circumstances, galvanised, epoxy-coated or stainless steel can be used (*Reynolds, Steedman, and Threfall*, 2008).

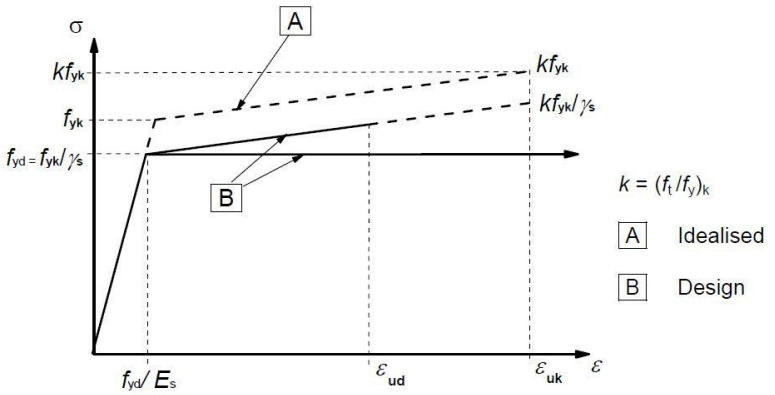


Figure 2.3: Idealised and design stress-strain diagrams for reinforcing steel (Figure 3.8 EC2)

The grade of reinforcement steel denotes the specified characteristic yield stress (f_{yk}). This is obtained by dividing the characteristic yield load by the nominal cross-section area of the bar. For products without a pronounced yield stress, the 0.2% proof stress $f_{0.2k}$ is substituted as the yield stress. The idealised and design stress-strain diagrams for reinforcing steel (for tension and compression) according to EC2 are shown in Figure 2.3.

In general, ductility is inversely related to yield stress. Therefore, in applications where ductility is critical, it is important to ensure that the actual yield strength does not exceed the specified value by a large margin. Ductility is defined using the strain at maximum load (ε_{uk}) and the ratio between the maximum and the yield strengths (f_t/f_{yk}). Ductility is an essential property if advantage is to be taken of the plastic behaviour of structures (*Beeby and Narayanan,* 2009). The greater the ductility, the greater the elongation in axially loaded members, and the greater the rotation capacity in members subjected to flexure. In members where the ultimate strength is governed by the yielding of reinforcement (shallow members with a low percentage of steel), the higher the value of strain at ultimate load, the greater the ductility.

According to Annex C of EN 1992-1-1:2004 (E), the design rules in EC2 are applicable to steel reinforcement with characteristic yield strength in the range 400 – 600 N/mm^2. Details of the actual yield strength of steel available in the UK for the reinforcement of concrete can be found in BS 4449: 2005. This document indicates that steel reinforcement will now be manufactured in three grades, all of 500 N/mm^2 characteristic yield strength, but with differing ductility. Plain round bars of characteristic yield strength 250 N/mm^2 are not covered in EC2. The mechanical properties of reinforcement for use in Eurocode is shown in Table 2.4

Table 2.4: Mechanical properties of reinforcements

Steel reinforcement to BS 4449 (in general conformity with BS EN 10080)			
Description	**Ribbed weldable reinforcing steel in the form of bars, coils or decoiled products.**		
Characteristic yield strength (MPa)	500		
Ductility class	A	B	C
Grade designation	B500A	B500B	B500C
Characteristic tensile/yield strength ratio	1.05	1.08	1.15
Characteristic total elongation at maximum force (%)	2.5	5.0	7.5
Preferred sizes (mm)	8,10,12,16,20,25,32,40		

2.4 Partial Factor for materials

In order to account for the differences between the strength of test specimens of the structural materials and their strength *in situ*, the strength properties will need to be reduced. This is achieved by dividing the characteristic values by partial safety factors for materials (γ_M). Therefore, the design value of materials is given by; $X_d = X_k / \gamma_M$. Uncertainties in the resistance models are also covered by γ_M. Although not stated in the code, γ_M also accounts for local weaknesses and inaccuracies in the assessment of resistance of the section (*Beeby and Narayanan*, 2009).
The partial factor for materials is shown in Table 2.5 below;

Table 2.5: Partial Factor for materials

Combination	**Partial factor for Concrete γ_C**	**Partial Factor for Steel γ_S**
Fundamental	1.5	1.15 (500 Mpa grade)
Accidental (excluding fire situation)	1.2	1.0
Fire Situation	1.0	1.0

For serviceability limit state, γ_M = 1.0

2.5 Selection of area of steel

After carrying out a reinforced concrete design, one of the steps is to select an appropriate area of reinforcement. Table 2.6 below gives the cross sectional areas of reinforcements for selection in columns, beams, piles etc.

When such tables are not available, you can easily calculate it by yourself. Let us consider ϕ16mm.

Area = $\frac{\pi d^2}{4} = \frac{\pi(16)^2}{4}$ = 201.06 mm^2

Therefore, 4 numbers of ϕ16mm will give us 4 x 201.06 mm^2 = 804 mm^2

Table 2.6: Cross-sectional area of number of bars (mm^2) for bar sizes (mm)

Number of bars	Cross-sectional area of number of bars (mm^2) for bar sizes (mm)							
	8	10	12	16	20	25	32	40
1	50	78	113	201	314	491	804	1257
2	101	157	226	402	628	982	1608	2513
3	151	236	339	603	942	1473	2413	3770
4	201	314	452	804	1257	1963	3217	5027
5	251	393	565	1005	1571	2454	4021	6283
6	302	471	679	1206	1885	2945	4825	7540
7	352	550	792	1407	2199	3436	5630	8796
8	402	628	905	1608	2513	3927	6434	10050
9	452	707	1018	1810	2827	4418	7238	11310
10	503	785	1131	2011	3142	4909	8042	12570
11	553	864	1244	2212	3456	5400	8847	13820
12	603	942	1357	2413	3770	5890	9651	15080

The selection of the area and spacing of reinforcements for walls, slabs, footings etc can be picked from Table 2.7 below. We normally present such reinforcements per metre strip. When such Table is not available, you can calculate it following this approach;

Considering ϕ12mm, Area = $\frac{\pi d^2}{4} = \frac{\pi(12)^2}{4}$ = 113.097 mm^2

If we decide to space it at 150mm per metre (1000mm), the equivalent area is given by;
(1000/150) x 113.097 mm^2 = 754 mm^2/m

Table 2.7: Cross-sectional area of bars per unit width (mm^2/m) for bar sizes (mm)

Spacing of bars	Cross-sectional area of bars per unit width (mm^2/m) for bar sizes (mm)							
	8	10	12	16	20	25	32	40
75	670	1047	1508	2681	4189	6545	10720	-
100	503	785	1131	2011	3124	4909	8042	12570
125	402	628	905	1608	2513	3927	6434	10053
150	335	524	754	1340	2094	3272	5362	8378
175	287	449	646	1149	1795	2805	4596	7181
200	251	393	565	1005	1571	2454	4021	6283
225	223	349	503	894	1396	2182	3574	5585
250	201	314	452	804	1257	1963	3217	5027
275	182	284	411	731	1142	1785	2923	4570
300	168	262	377	670	1047	1636	2681	4189

2.6 Loading of residential buildings to EN 1991-1: 2001

Residential buildings are usually subjected to imposed floor loads, and other dead load from self weight of materials, and permanent fittings. Depending on environmental conditions and height of the building, wind load may or may not be considered. The conventional load path is as follows: Slab → Beams → Columns → Foundations.

In this book, the UK National Annex to EN 1991-1:2002 has been adopted for the purpose of specifying the imposed floor load. Residential buildings are under category A, and their sub-categories with appropriate description examples are given in Table 2.8. The characteristic imposed load is given in Table 2.9.

Table 2.8: Sub-categories of residential occupancy classification (Table NA.2 NA to BS EN 1991-1:2002)

Category of loaded area	Specific use	Sub-category	Example
A	Areas for domestic and residential activities	A1	All usages within self-contained dwelling units (a unit occupied by a single family or a modular domestic and student accommodation unit with a secure door and comprising not more than six single bedrooms and an internal corridor Communal areas (including kitchens) in blocks of flats with limited use Note For communal areas in other blocks of flats, see A5, A6 and C3
		A2	Bedrooms and dormitories except those in self-contained single family dwelling units and in hotels and motels
		A3	Bedrooms in hotels and motels; hospital wards; toilet areas
		A4	Billiard/snooker rooms
		A5	Balconies in single family dwelling units and communal areas in blocks of flats with limited use (see Note 1)
		A6	Balconies in hostels, guest houses, residential clubs and communal areas in blocks of flats except those covered by Note 1
		A7	Balconies in hotels and motels

Table 2.9: Imposed load on floors, balconies, and stairs of category A buildings (Table NA.3 NA to BS EN 1991-1:2002)

Category of loaded area	Sub-category	q_k kN/m^2	Qk (kN)
A	A1	1.5	2.0
	A2	1.5	2.0
	A3	2.0	2.0
	A4	2.0	2.7
	A5	2.5	2.0
	A6	Same as the rooms to which they give access but with a minimum of 3.0	2.0 (concentrated at the outer edge)
	A7	Same as the rooms to which they give access but with a minimum of 4.0	2.0 (concentrated at the outer edge)

2.6.1 Self weight of common construction materials

The self weight of some commonly encountered construction materials are given in Table 2.10 below.

Table 2.10: Self weight of common construction materials

Material	Unit Weight
Concrete Plain or lightly reinforced 2% reinforcement 4% reinforcement	 24 KN/m^3 25 KN/m^3 26 KN/m^3
Granite (stone)	27 – 30 KN/m^3
Basalt (stone)	27 – 31 KN/m^3
Hot rolled Asphalt	23 KN/m^3
Block work (200mm) (per unit area) Solid Hollow	 3.8 KN/m^2 2.6 KN/m^2
Glass	26.7 KN/m^2
Plaster	215 N/m^2 per mm thick
Corrugated steel or asbestos-cement sheeting (including bolts, sheeting rails etc)	430 N/m^2 per mm thick
Rendering, screed, terrazo	18.9 – 23.6 N/m^2 per mm thick
Clay floor tiles	575 N/m^2 per mm thick

CHAPTER 3

GENERAL ARRANGEMENT AND MODELLING

3.1 Introduction

In order to carry out a design properly, the engineer should be able to adequately idealise the structure so as to try and obtain the closest theoretical behaviour of the structure. This interpretation is usually made from standard and well prepared architectural drawing of the proposed building. The architectural drawing enables the engineer to prepare what is normally referred to as the 'general arrangement' of the building, popularly called the 'G.A.'. The G.A. clearly specifies the disposition of the structural elements such as the columns, beams, and the panelling of the floor slabs. The G.A. also contains the labelling of the axes and the members, based on the grid lines. After completing the GA, the engineer makes preliminary sizing of the structural elements which may be governed by past experience or by deflection requirements based on the code of practice. After the sizing, the engineer is faced with the challenge of loading the structure. But let us briefly review how we go about the G.A.

3.2 General Arrangement (GA)

There are no spelt out rules about how to select the appropriate general arrangement of a structure. To me, adequate presentation of the general arrangement has more to with years of design experience.

However, I am going to highlight some important guidelines which are very necessary.

1. **Respect the architect's original disposition**: For instance, when arranging your columns, do not place columns where the architect has meant to be a free space, nor should the structural elements interrupt the interaction of spaces. Also your columns and beams should not project out where the architect has intended plane walls or sections etc. Just make your arrangement consistent with the original form of the structure.
2. **Select a stable model**: The model or general arrangement you are presenting should be statically stable, and fully representative of the behaviour of the structure.
3. **Buildability and construction consequences**: The model you adopt should be buildable. Consider the technical capacities of the local contractors that will execute the design from your model. For instance, in a region where reinforcement bending machines and cranes are not available, why should you provide models that will require the provision of Y40mm bars, or recommend the use of precast elements?
4. **Know the economic and structural consequences:** Between more steel and more concrete, which one is more economical? For instance, let us consider an external beam that is 8m long. If there are no openings at the wall panels under the mid span of the beam, you can comfortably hide an intermediate column there, thereby having two spans 4m each. The original 8m beam will require more steel reinforcements, or deeper sections, or both to satisfy ultimate and serviceability limit state requirements. However, if you introduce a column at the

mid-span, there will be redistribution of stresses, with a hogging moment at the propped mid-span, and hence, generally lesser reinforcements and concrete sections. However, now note that you are going to construct a new column and new isolated base (requiring concrete, reinforcement, and additional excavation cost). Between the two options, which one gave you the most economical solution? That is another influencing factor.

In this book, a small residential building on a 10m x 15m plot of land has been presented for the purpose of analysis and design. The ground floor plan of the building is shown in Figure 3.0 below. From the architectural disposition, the building is a two family occupancy arrangement, with each family occupying a floor level. To minimise interruption, the staircase has been placed at the extreme of the left hand side of the building. The 1st floor of the building is shown in Figure 3.1. It has a balcony at the kitchen, a cantilever sit-out/verandah, and a little balcony by the staircase area. Apart from that, the general arrangement is fairly the same.

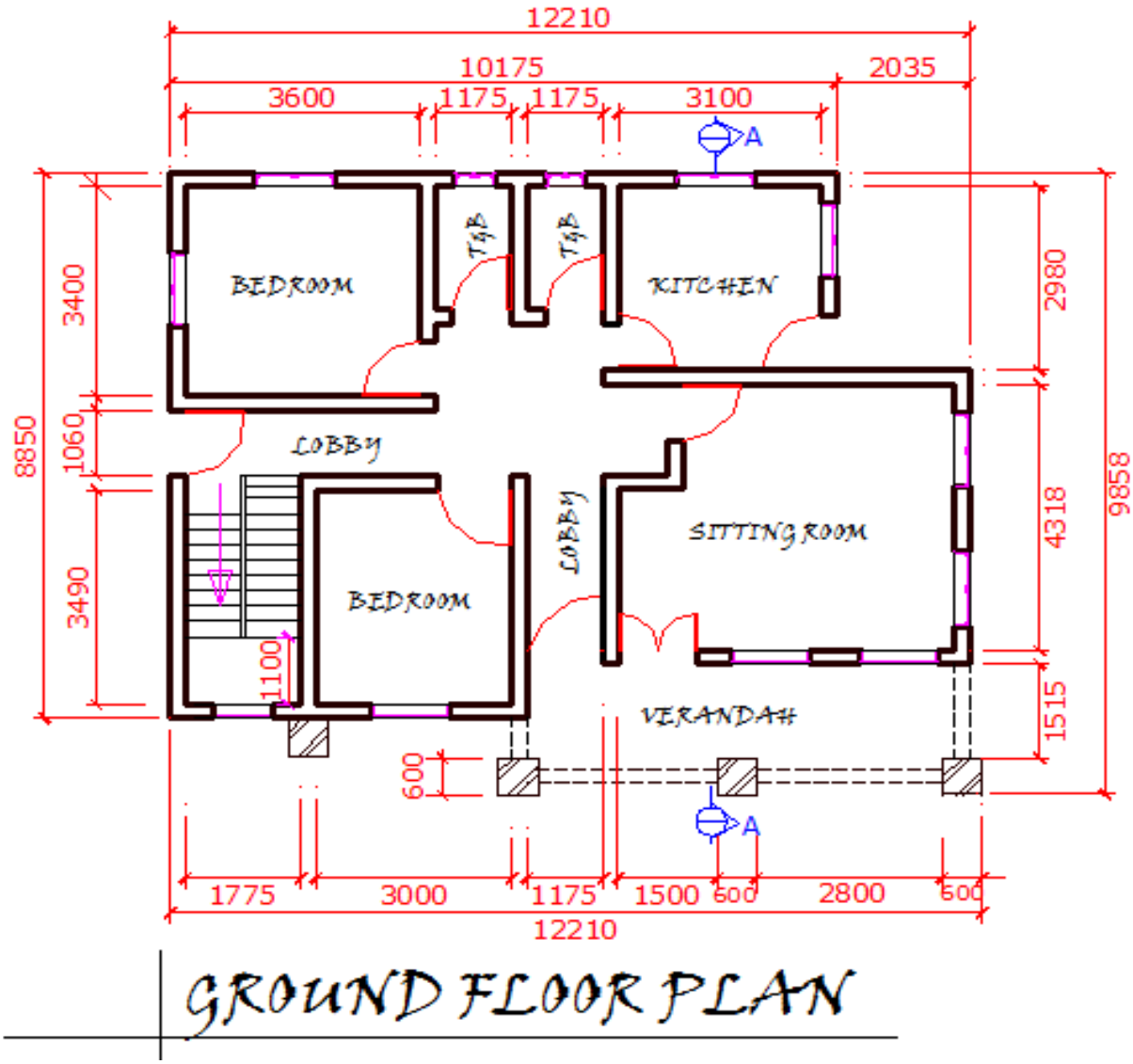


Figure 3.0: Architectural ground floor plan

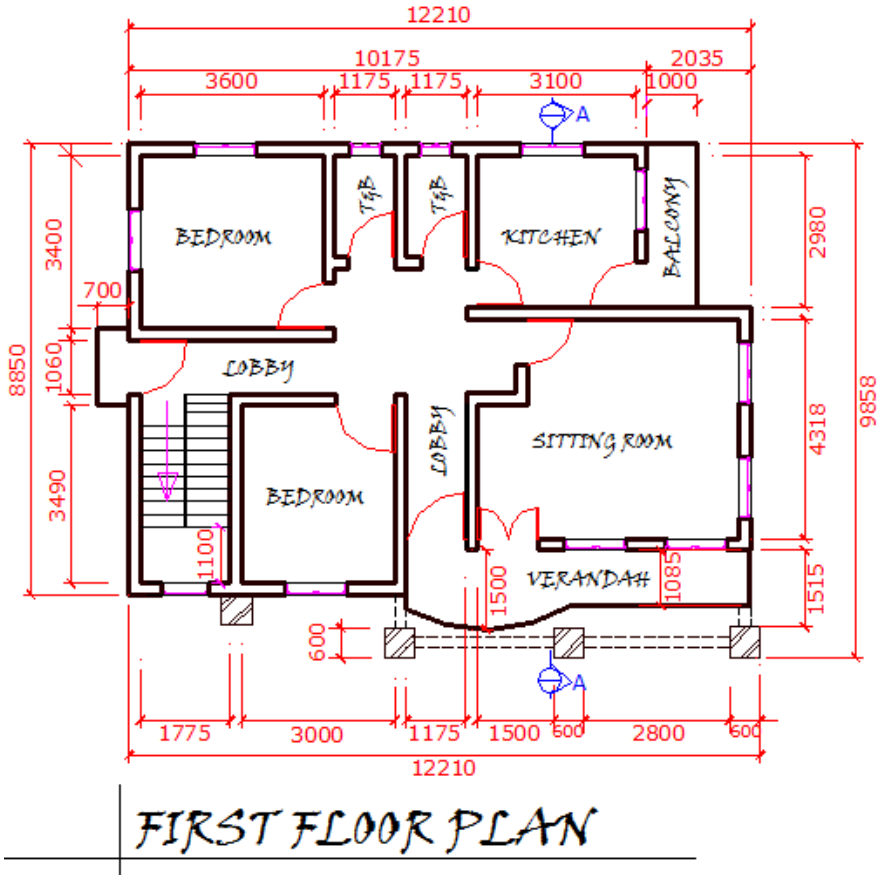


Figure 3.1: Architectural first floor plan

3.2.1 My style of General Arrangement (GA)

The architectural drawing of a building can come in many forms. If it comes as a soft copy of a CAD drawing, then the work is made much easier. It is advisable to place the ground floor plan side by side on the graphical user interface window of your CAD program (e.g. AUTOCAD) with the floor plan of the subsequent stories (see Figure 3.2).

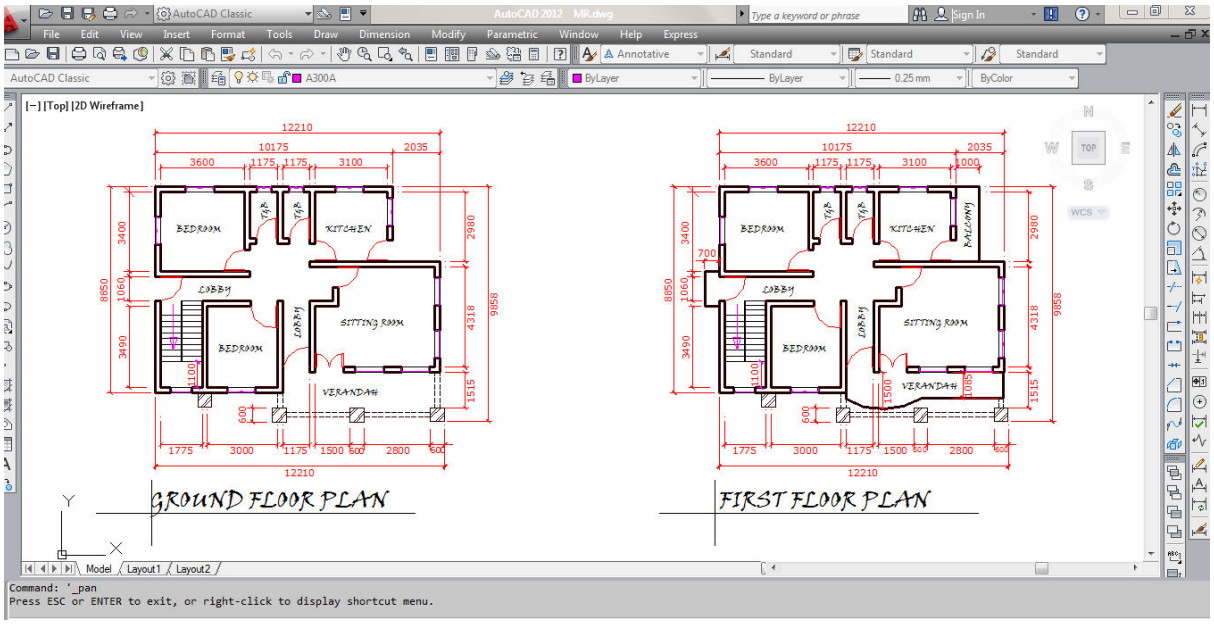


Figure 3.2: Ground floor and first floor plan placed side by side on the same window

All the elevation views of our case study building are as shown below;

Figure 3.3: Approach View

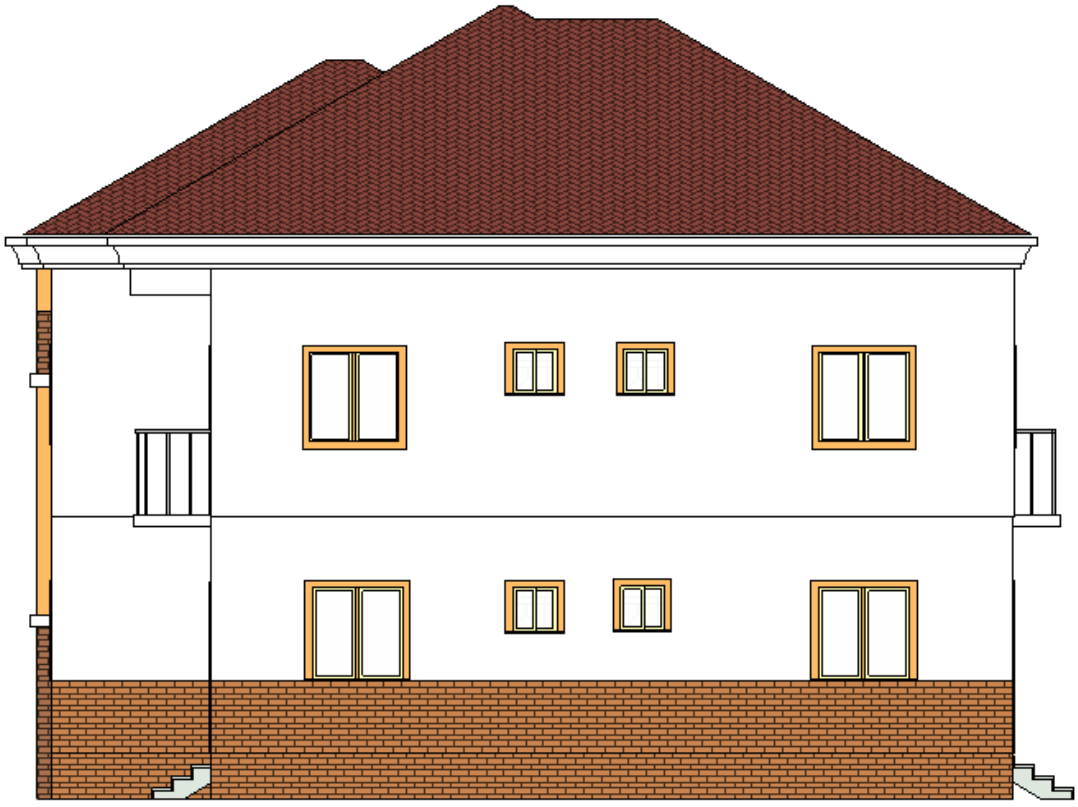


Figure 3.4: Rear view

Figure 3.5: Right Side View

Figure 3.6: Left Side view

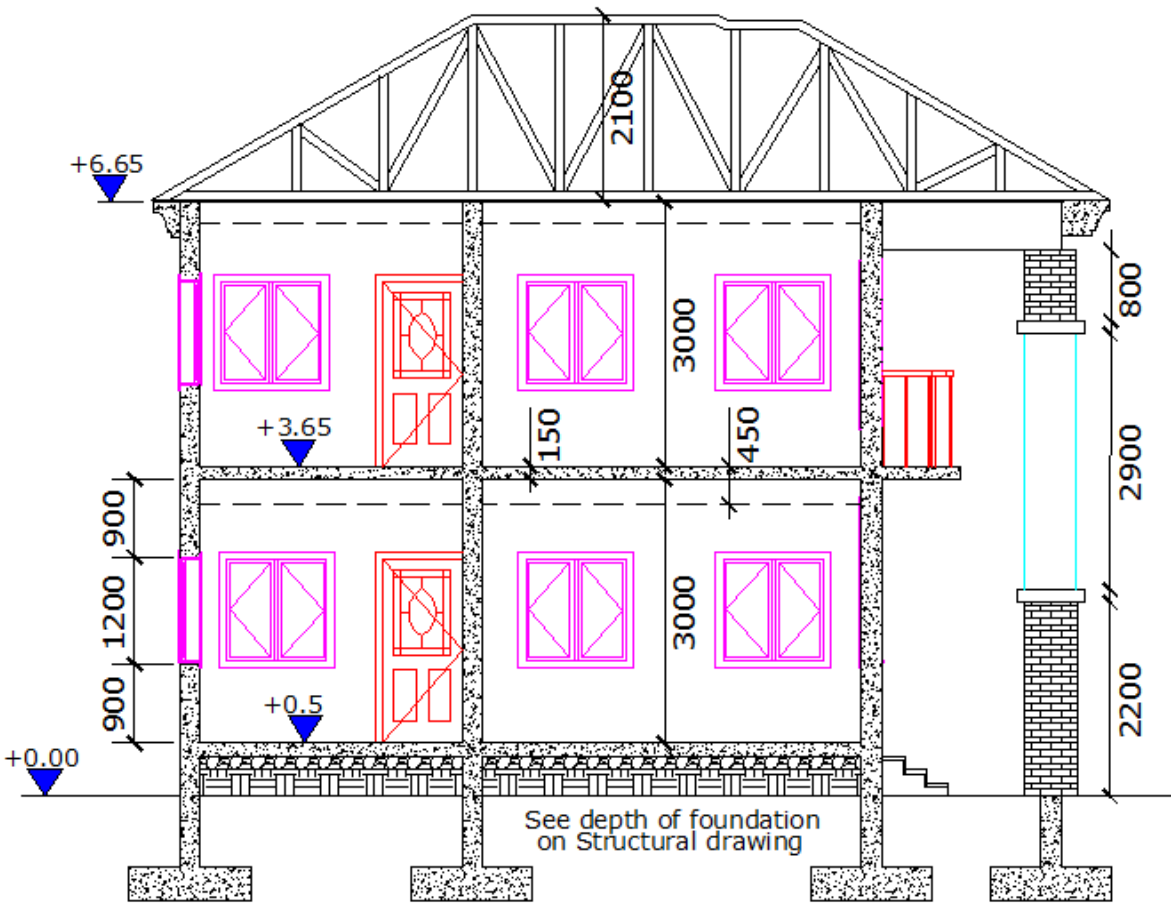


Figure 3.7: Section through the building (section A – A)

Note that architectural drawings come with their own grid lines for different axes. In the current building we are trying to design here, some details (like gridlines) have been removed from the architectural drawing (Figure 3.0 and 3.1) for the purpose of clarity of very necessary details like dimension lines. After placing the floor plans side by side on the window, you can start noticing a few different things about the floor plans immediately. For more ideas on how to proceed, copy the floor plans to another location on the window (still leaving the ones you placed side by side). Now, copy the plan of the first floor, and paste it on the ground floor plan, so that the dimensions and axes are matching perfectly.

HINT: You can choose a prominent corner of the building as a pick up point for your copy and paste operation. For more clarity, you can change the colour or thickness (or both) of the ground floor elements and first floor elements and gridlines, so that you can rightly distinguish between the two.

After you have pasted the first floor on the ground floor, you can now see the interaction of the two floors. All axes that are coincidental will be visible, and all axes that are in first floor and are not in ground floor and vice versa will also become visible. At this point, you can also see the outline of block work on the first floor, and this is more critical for the general arrangement of the floor slab. It will properly guide you on the selection of floor beam axis. The block work axis of the ground floor will aid you on the design of the foundation plan. Do not work on the ground floor alone without looking at the first floor – the last thing you will want to happen is to place a column somewhere on the ground floor, and see it popping out through the lobby of the first floor. So carefully make your selections based on matching axes, and fair uniformity. And as I hinted earlier, your arrangement must be consistent with what the architect has in mind. So this is much like an art, and you have to use your

ingenuity here. (My supervisor during my industrial training once told me that it is the one of the hardest thing to do in structural design). I also believe that years of design experience counts on this also.

As a hint, the next thing to do is to create a rectangular box (say 230 x 230mm) on AUTOCAD, and hatch it with any pattern appealing to you (I normally use SOLID). This represents your columns on the floor plan. Now carefully copy this element and start pasting it at the locations where you have decided to place your columns (this usually occurs at intersections between axes). Also, I normally start at the corners of the building; more often than not, columns must be there irrespective of the arrangement. After that, move to the interiors and place your columns as desired. After you are satisfied with what you have done, carefully check the interaction of the arrangement, and make sure that they are reasonable. Areas where primary and secondary beams will interact will also become very visible.

To make some points clearer, let us look at a portion of the plan we are considering in this text (Figure 3.3)

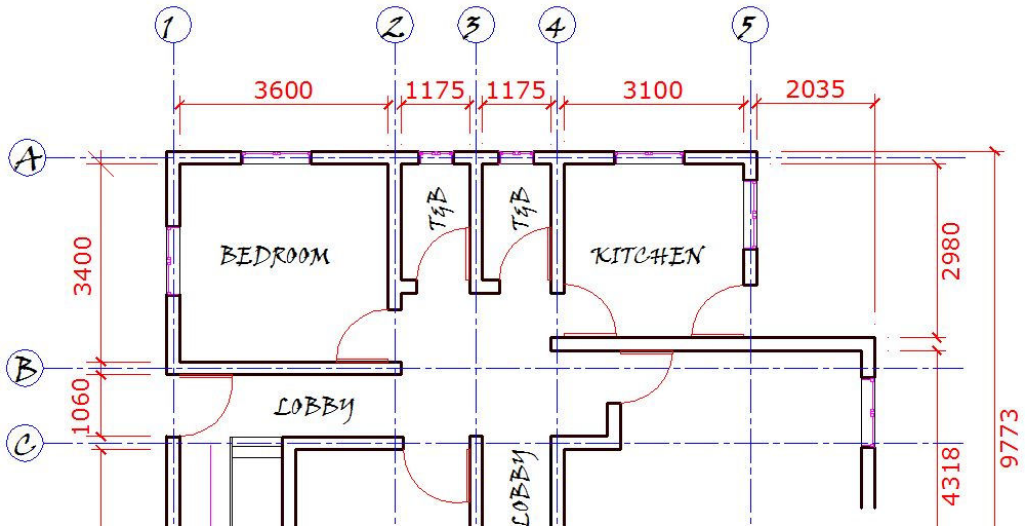


Figure 3.3: A portion of the floor plan

We wish to place columns along axis A. A little consideration will show that we can place columns at points A1, A3, and A5. Also, we can alternatively place columns at points A1, A2, A4, and A5. Without reading further, ponder on that arrangement and see the alternative that you will prefer.

Obviously, A1 and A5 are certain (corner columns). If I should choose to place a column at point A3 (neglecting A2 and A4), these are the implications

1. I will have a larger span for A:1-3 and A:3-5
2. I will have a floor beam running along axis 3 (the beam will probably have to run down to axis C before encountering another support, unless I extend any other beam to act as primary beam to it). That's a complexity on its own.
3. The floor slabs on the bedroom and kitchen will be subjected to block work load from the walls on axis 2 and 4, which some part of it will be subsequently transferred to the beams on axis 3. This is based on the assumption that the ground floor block walls do not carry any load (so I have neglected the effect of the ground floor axis 2 and 4 walls on this assumption)

If I choose the second alternative;

1. I will have shorter spans, and three spans instead of 2. The bending moment on Span A:2 – 4 will probably be hogging.

2. I will have a wall along axis 3, but by proximity and considering load sharing implication, it will not be critical, and will not affect my designs like the pervious alternative.
3. My floor beam at axis 2 will stop at axis B, and my floor beam at axis 4 will stop at the wall close to axis B. So I will not have a complex arrangement to deal with.

Considering all these consequences, I preferred the second alternative. However, if the building is being located in area where the soil is so bad that constructing a single footing will be very expensive, we can settle for alternative 1. The final GA I adopted for the whole model is shown below in Figure 3.4;

However, if the drawing comes in form of a paper work (this is the most typical), then nothing changes in the approach. Place the floor plans side by side, and manually make your decisions, before you go into drafting.

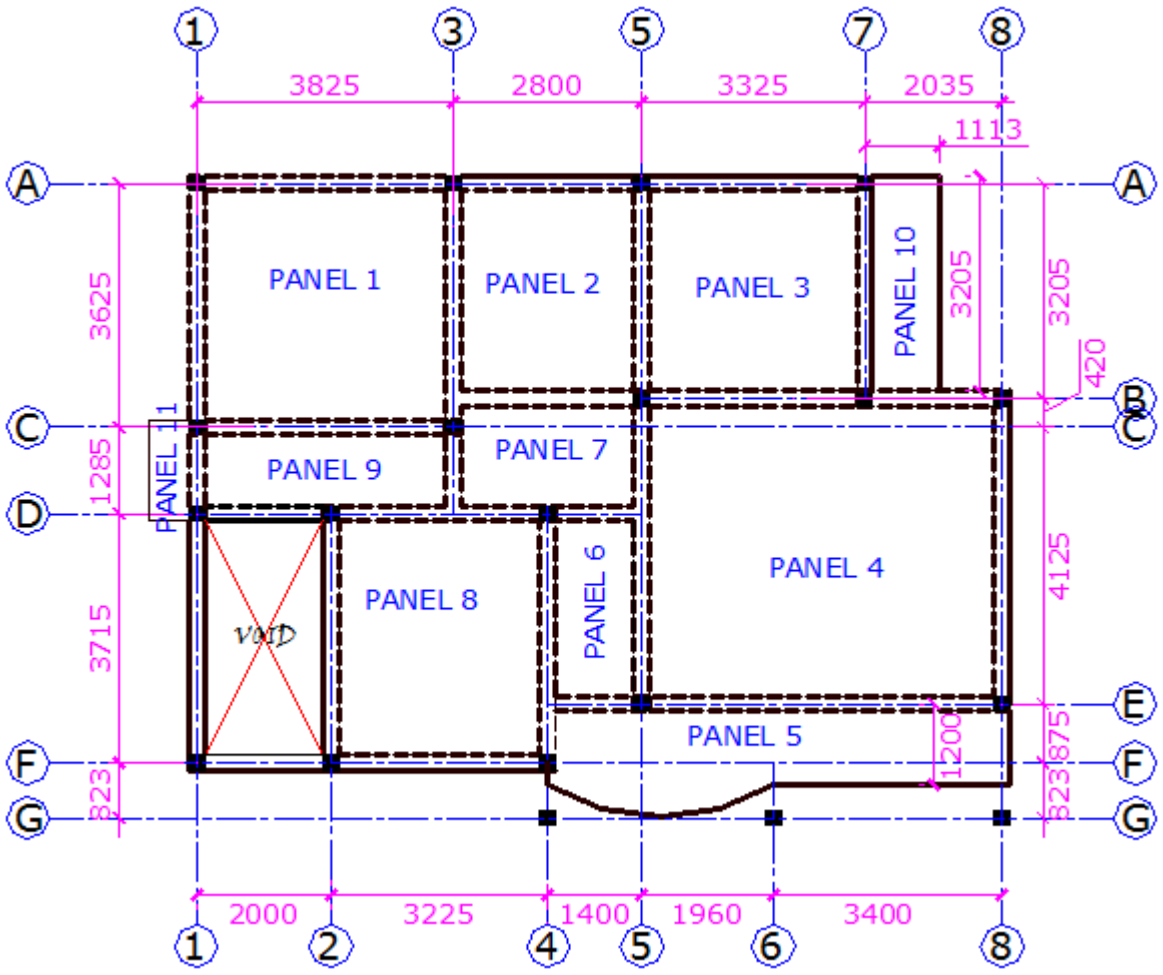


Figure 3.4: Final adopted GA of the building

Member Data

Thickness of slab = 150mm

Dimensions of floor beams = 450mm x 230mm

Dimension of roof beams = 250mm x 230mm

Dimensions of columns = 230 x 230mm

3.3 MODELLING ON CSC ORION

Orion is a structural analysis, design, and drafting program developed for the design of concrete building systems (*CSC Orion Reference Manual*). The program consists of several modules for performing the following tasks;

- 3-D Analysis of the structural model of the building
- Column, Shear wall, Slab, and Beam reinforcement design
- Column, Shear wall, and Beam detailing
- Foundation design and detailing
- Analysis and Design of Stairs
- Concrete and Steel Quantity extractions

Unlike general purpose structural analysis program, Orion is concentrated on accurate analysis, fast data preparation, automated reinforced concrete design, and automated preparation of engineering drawings and details.

In summary, Orion model allows you to;

- Create GA drawing
- Design the floor slabs, and decompose floor loads onto the beams
- Analyse the building frame
- Design continuous beams, columns, walls, and foundation
- Automatically generate RC detail drawings

Having completed your general arrangement, you have to go into modelling of the structure. GA is the first step, irrespective of whether you are undertaking manual or computer aided analysis. It is assumed that you have already purchased and installed Orion software package on your computer. There are many approaches of modelling in Orion like importing from another software etc. But in this text, I am concentrating on modelling from the scratch.

The structure below is our end target.

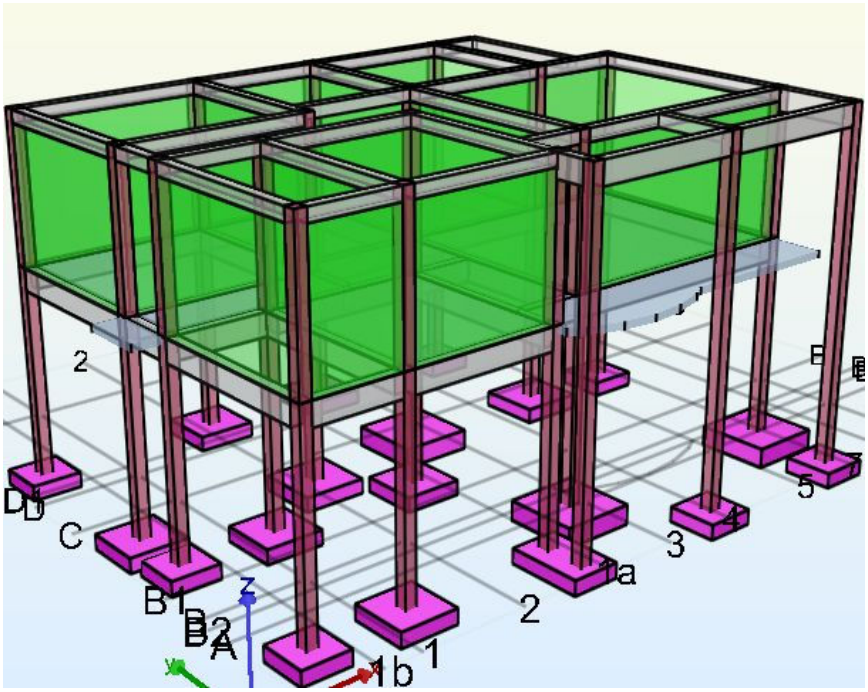
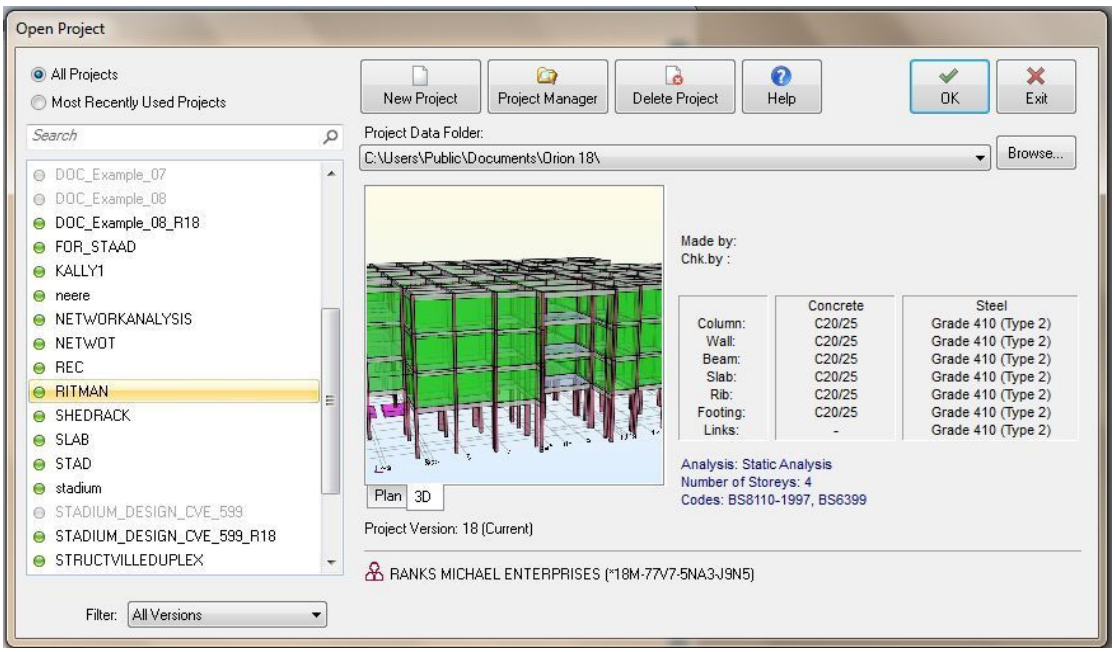


Figure 3.5: Final Target Model on CSC Orion

3.3.1 Getting Started

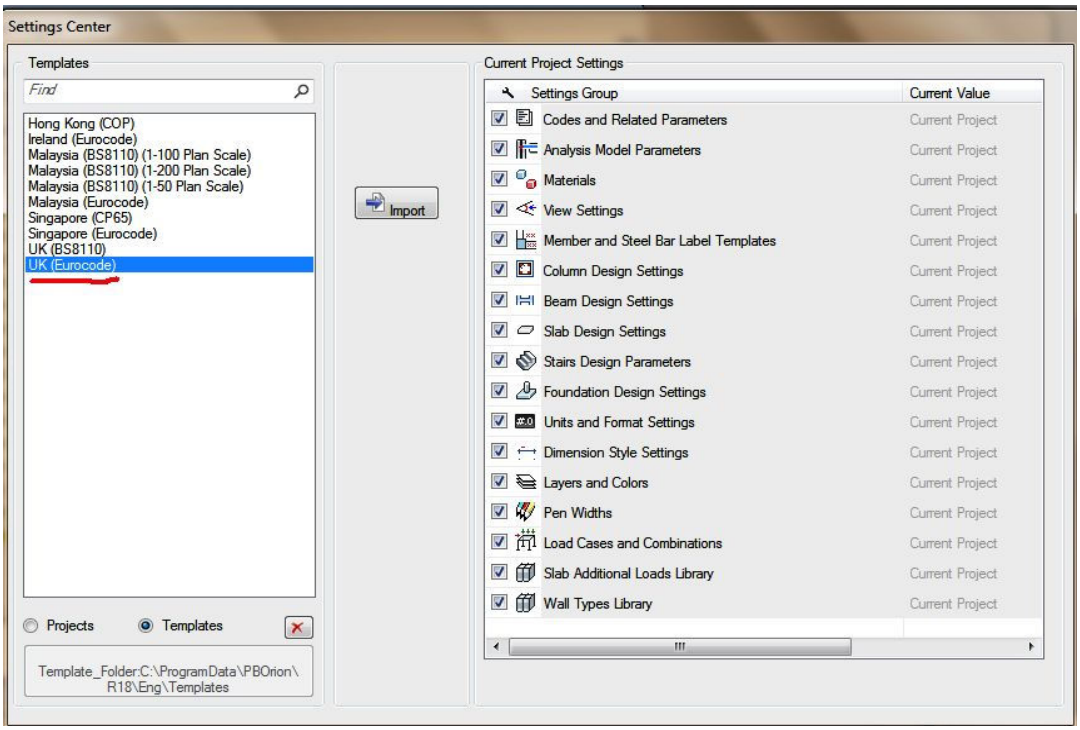
Launch the Orion Software. On launching, this dialog box below comes up;



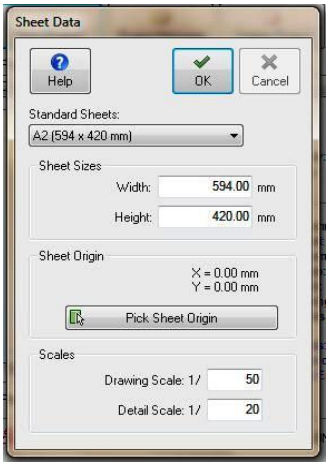
Click on **New Project.**

Enter the **Project Code** of your choice. In this case, note that no spacing is allowed on the project code. You can use underscore to separate words. Hit the **ENTER** button

On hitting the **ENTER** button, the **Settings Centre** Dialog box comes up as shown below. Select **UK(Eurocode),** and ensure that that all the parameters on '**Current Project Settings**' are checked. Then click on **IMPORT**.



Select your 'sheet data as you desire, and click **OK**. Note that various sheet sizes like A2, A3, A4, A5 etc are available from the draw down menu. Specify as desired and click **ok.**



At this point, the graphical user interface (GUI) comes up (Figure 3.6 below). The green rectangle on the screen represents the sheet size you have selected in the previous dialog box. Now, we wish to create the GA of the building. By this time, you should have manually prepared your GA (could be a rough sketch), and have accurately placed your beams and column where you desire them to be. In this case, note that the GA we want to prepare is the one shown in Figure 3.4. Also by this time, you must have carefully specified the preliminary structural dimensions of your elements, and the concrete cover viz;

Dimensions of columns (In this case 230 x 230 mm)
Dimensions of all floor beams (In this case 450 x 230 mm)
Dimensions of roof beams (In this case 250 x 230 mm)
Thickness of slab (h) = 150mm

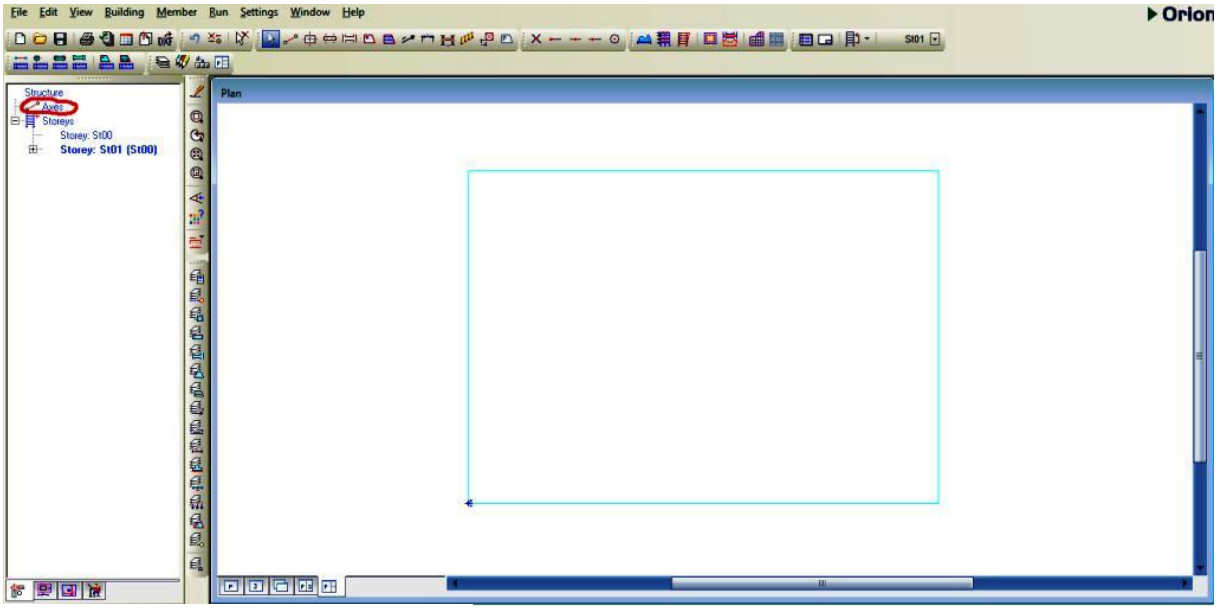


Figure 3.6: CSC Orion main GUI page

3.3.2 Creating axes

Having decided on all these things, go to **Axes (circled in red in Figure 3.6 above)** by the left hand side of the GUI, and right click on it. At this point, we wish to create all our specified axes. So by right clicking on it, select the **Orthogonal Axis Generator**, and click on it. Our cursor at this point changes

to a hair pin cross with thin lines. Now move your mouse to the specified sheet on the GUI (the green rectangle box) and click at a good location on it. The following dialog box comes up;

Dir-1 Axes represent your y-axis (vertical axes on your plan), while **Dir-2 Axes** represent your x-axis (horizontal axes on plan). You can interchange your **Axis Label** as you wish. So to be consistent with our original plan, let us leave it the way it is.

Axis spacing(s) refers to the spacing of the various axes (or our gridlines). Even though it is not specified in Figure 3.4, the edges of our cantilever slab must have its own axis. This is to facilitate the modelling of our Cantilever slabs.

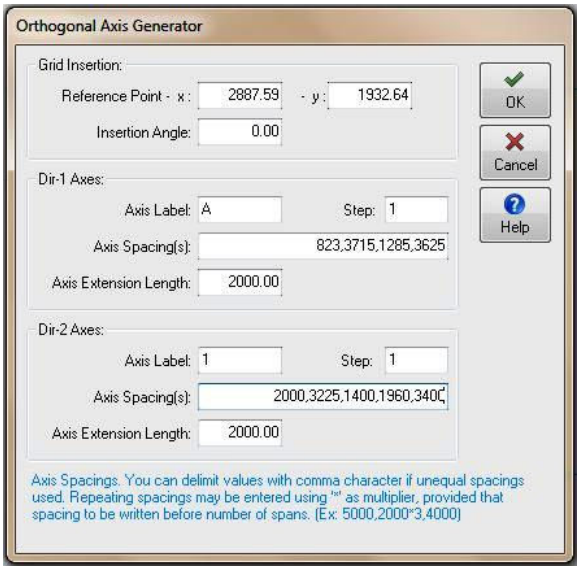
So from our GA, we can specify the axis spacings (each spacing is separated by a comma, all dimensions in mm)

Dir-1 Axes

Axis Spacing(s): 823,3715,1285,3625 (entering values from bottom to the top)

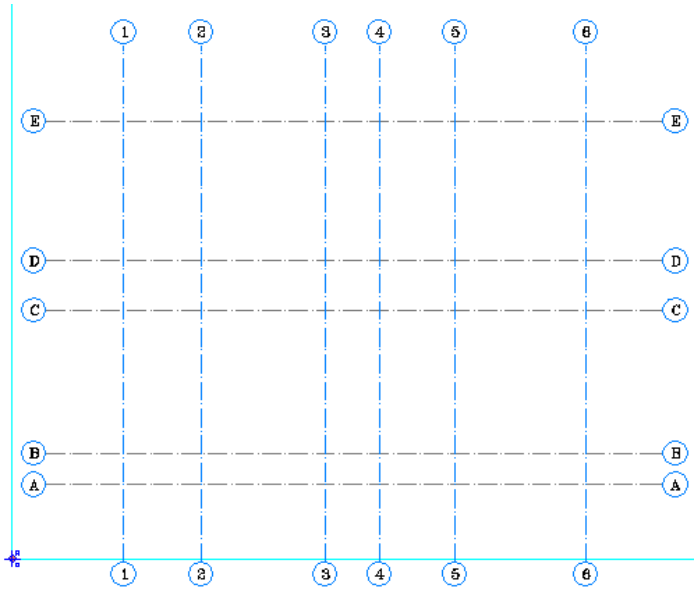
Dir-2 Axes

Axis Spacing(s): 2000, 3225, 1400, 1960, 3400 (entering values from left to the right)

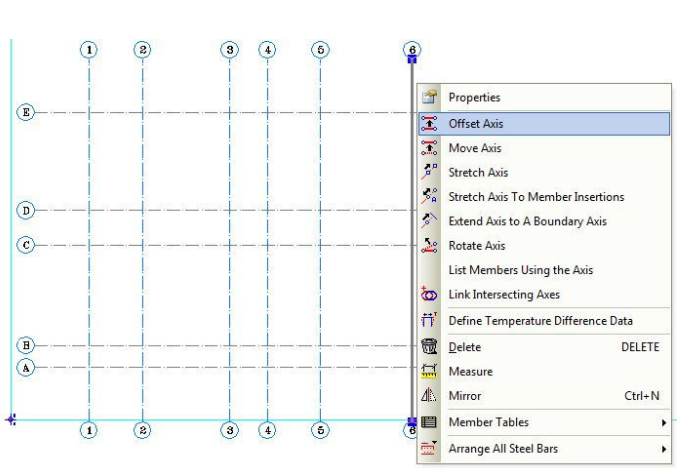


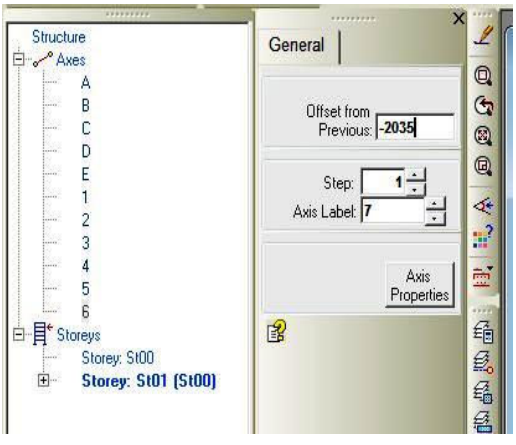
Now you will discover that there are some gridlines that we have not captured. Do not be bothered. If you wish, you can take your time and specify as accurately as possible, the spacing of the gridlines. This is probably the best way. Otherwise, there is a faster way of doing it, which is to offset already existing axis by the distance you want. The only difference is that your labels will become alphanumeric. For instance if you offset axis A, you will probably get axis label A1. I guess this should not be a major concern, since it is subsequently editable. Leave the **axis extension** line the way it is (at 2000mm) and click **OK**.

On clicking **OK**, the figure below is obtained on the user interface, showing the gridlines.



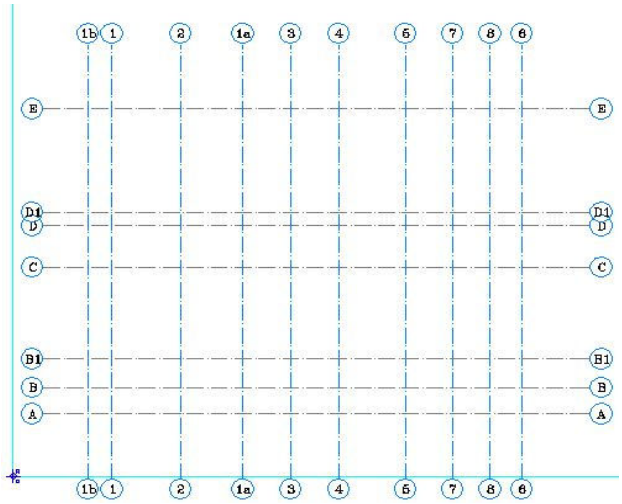
On looking at the figure above, you can verify that Axis E on the interface stands for the Axis A on our GA, and Axis 6 on the GUI stands for the axis 8 on our GA. Now, if you look at our GA, we have axis 7, which we have not yet captured. To create it, we know the distance between axis 7 and 8 to be 2035mm on our manual GA. We wish to create axis 7 on Orion by offsetting axis 8. Through this way also, we create all other axes, including the free edge of our cantilever slabs. So right click on Axis 6 on the Orion GUI, and click on **Offset Axis.**





Now, the dialog box by the right above appears, and you have to specify the offset distance by inputting 2035mm on '**Offset from Previous'**. After inputting the value, click on the direction you wish to make the offset. In this case, you have to click towards the left hand side of axis 6 for the axis to be created. Through this way, we create all other axis remaining on our GA.

The final axes/gridlines that captures the entire model is shown below;

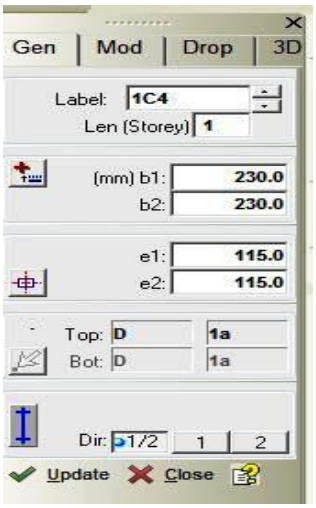


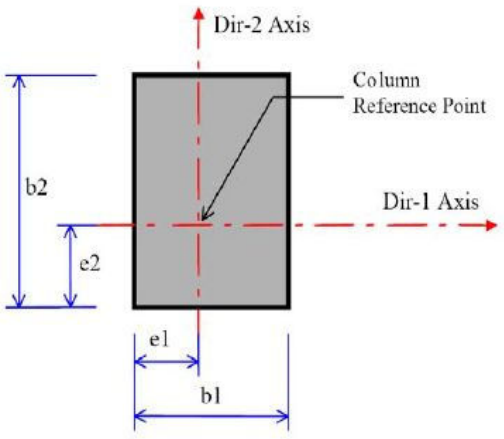
3.3.3 Creation of columns

After creating our axes, the next step is to place the columns at their exact locations. We have to be careful to ensure that the columns are where they are supposed to be.

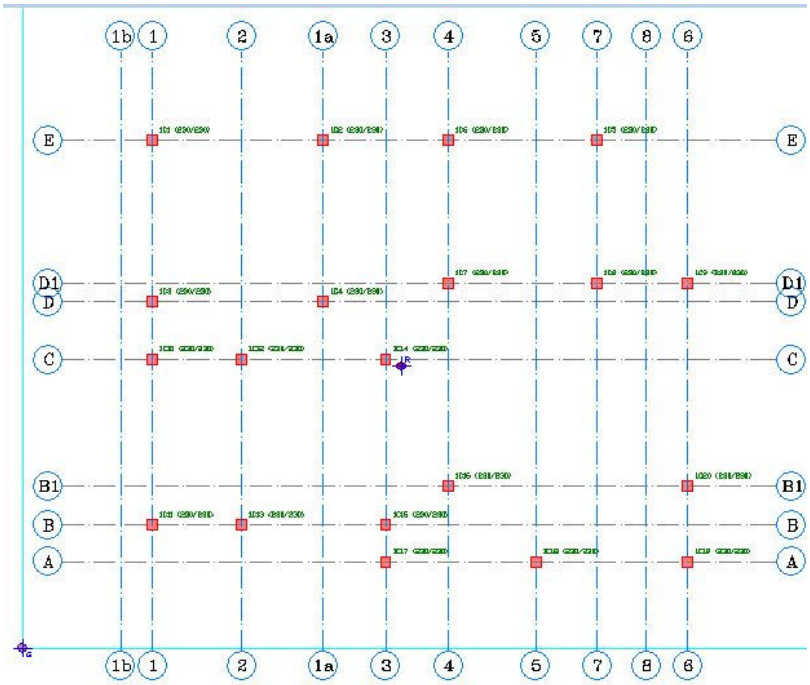
Click on **Column icon** at the menu bar

Input the cross-section of the column (in mm) at the spaces provided at b1 and b2. b1 and b2 are the depth and width of the column as defined below. e1 and e2 defines the eccentricity of the axes with the column. So in the cases where you are modelling a column that will project out of a building, you should have a good idea of how to manipulate your eccentricities. In this case, I want it to pass through the centroid of the column, so I will have to divide the cross-section by two; hence e1 = e2 = 230/2 = 115mm



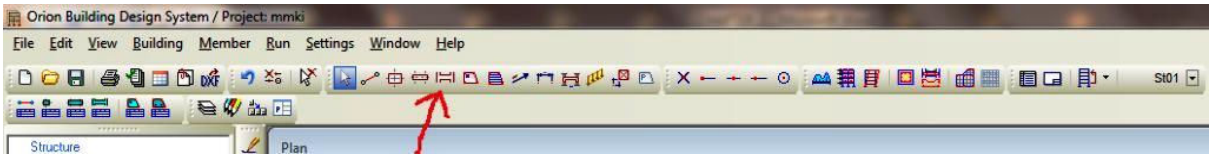


Now, carefully move your columns to the GUI, and place the columns at the locations they belong by clicking your mouse at the intersections where we have our columns. Your final output should be as shown below- carefully verify its consistency with our GA.

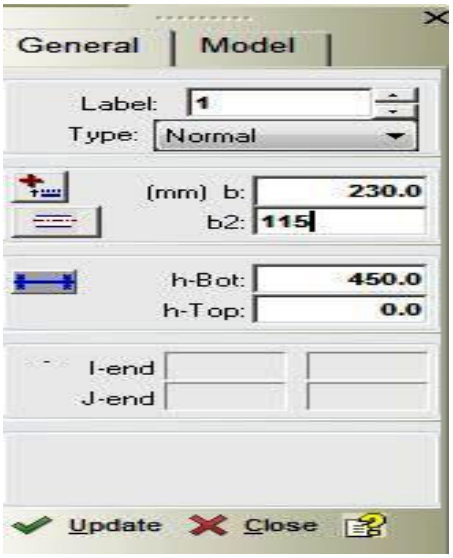


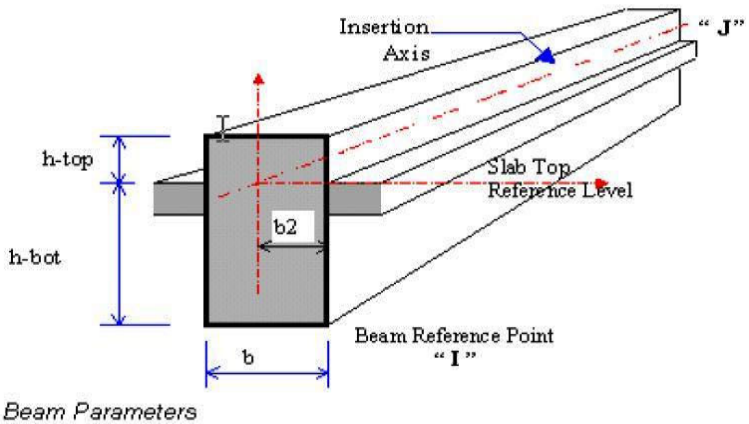
3.3.4 Creation of floor beams

After creating our columns, the next step is to place the floor beams at the axes where they belong.



Click on beam icon, and the dialog box below comes up

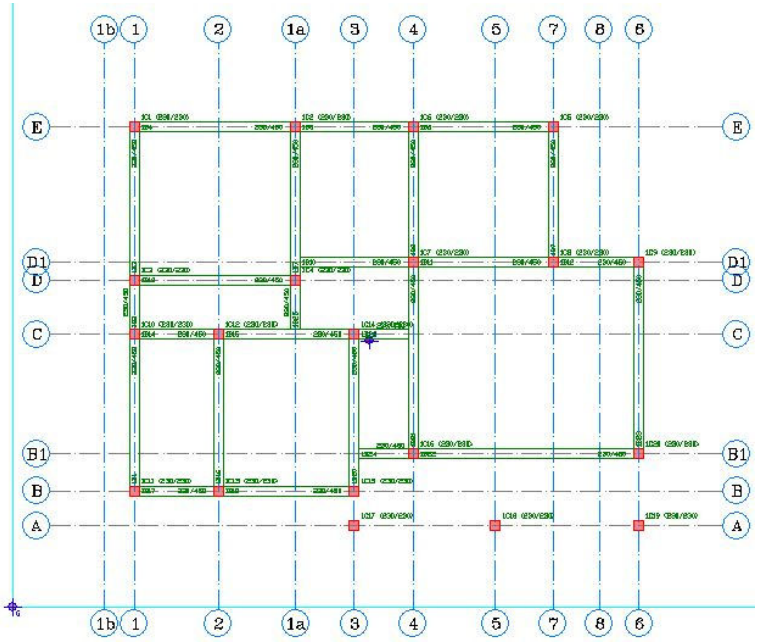




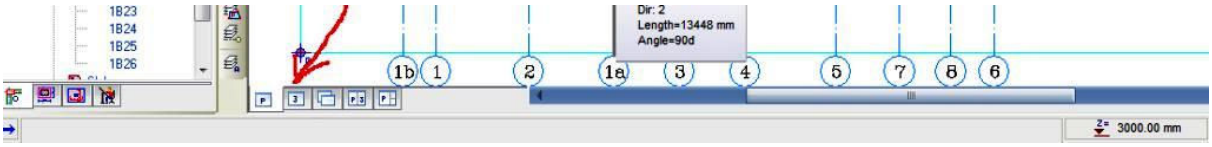
Beam Parameters

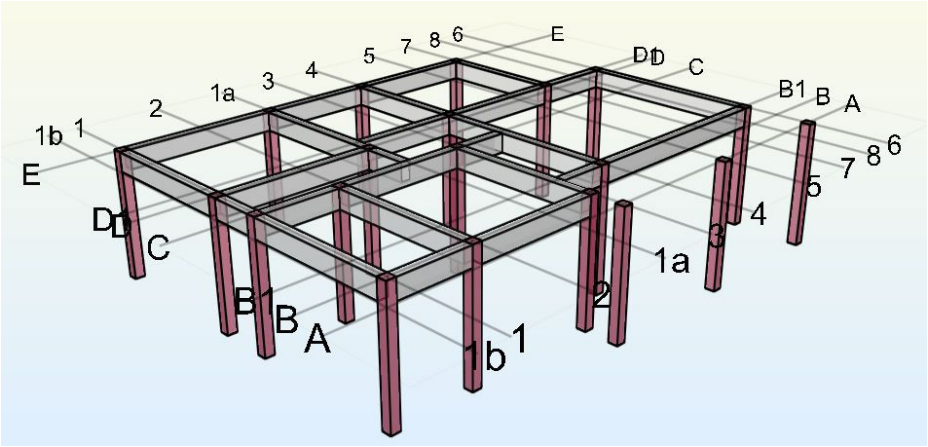
Input **b**, which is the width of the beam. In the case we are treating, b = 230mm. **b2** is the eccentricity of the axis with the beam, which we desire to pass through the centroidal axis at b/2 = 230/2 = 115mm. **hbot** is the total depth of the beam, and the value is 450mm for our present case study. Disregard **htop** by leaving the value as zero.

To add the beams, click at the intersection point of any column you choose as your starting point, and drag it to another column location. Keep clicking and dragging it until you create all the floor beams. Note that a beam that is being dragged can only terminate at a point where two orthogonal axes intersect. Your final arrangement should be as shown below;



To ensure that we are in order, click on 3D icon at the bottom of the GUI to see the 3D view of our creation so far



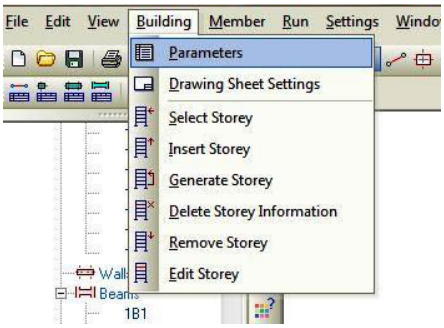


If your arrangement is as shown above, let us proceed. Click on the plan icon and return to the plan view.

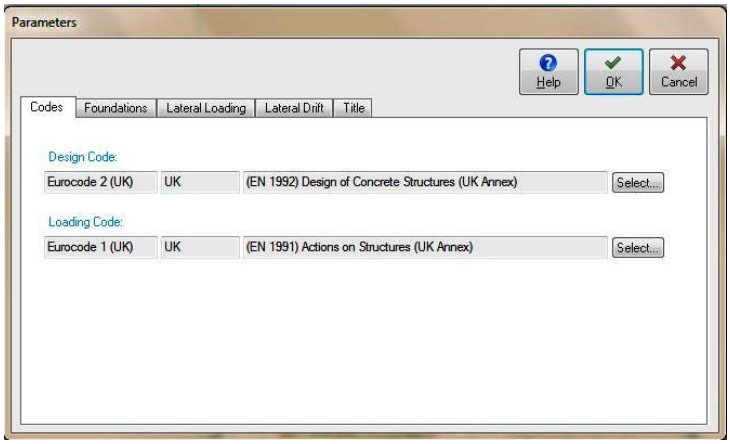
3.3.5 Design Parameters Settings

Ideally, we should just go ahead and create our floor slab, but there are some settings that I will like us to do. It will subsequently affect our results.

Take your mouse and click on **Building** on the menu bar. Select **Parameters**

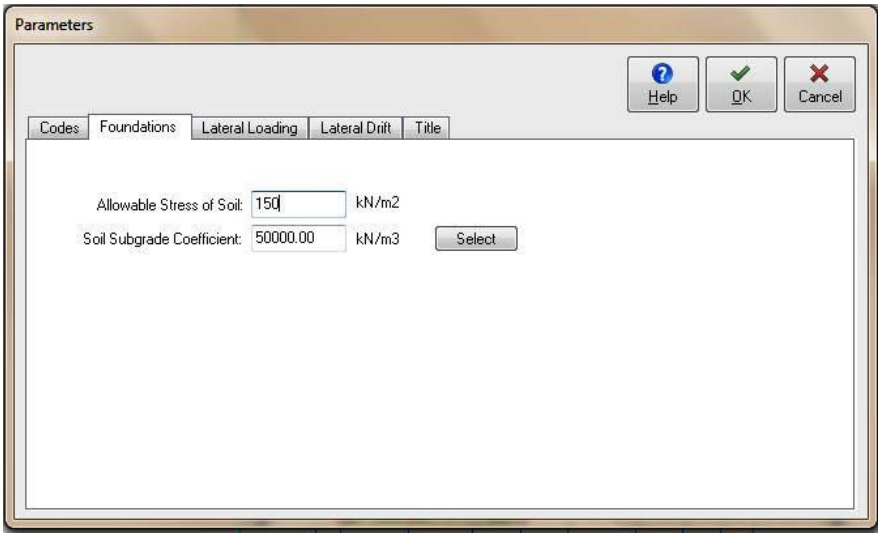


On clicking **Parameters**, the dialog box below comes up;



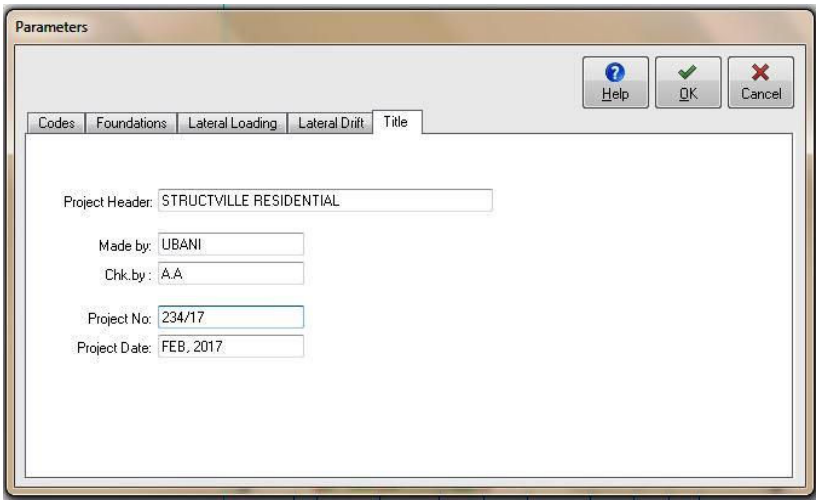
Ensure once again that the **Code** for design is at **Eurocode 2(UK),** and that the Loading Code is **Eurocode 1(UK).**

Click on **Foundations**, and set the allowable stress of soil to 150 KN/m^2 (allowable bearing capacity).

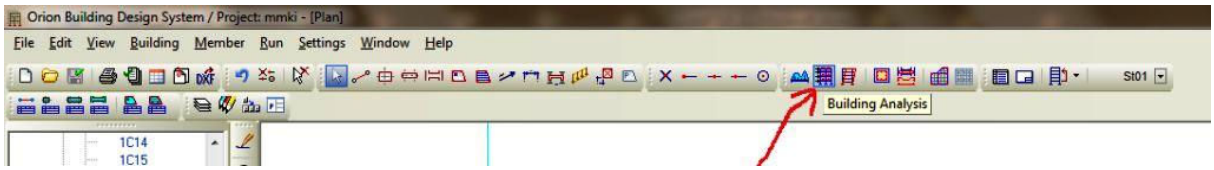


Since we are not bothered about the effect of wind, you can neglect **Lateral Loading** and **Lateral Drift.**

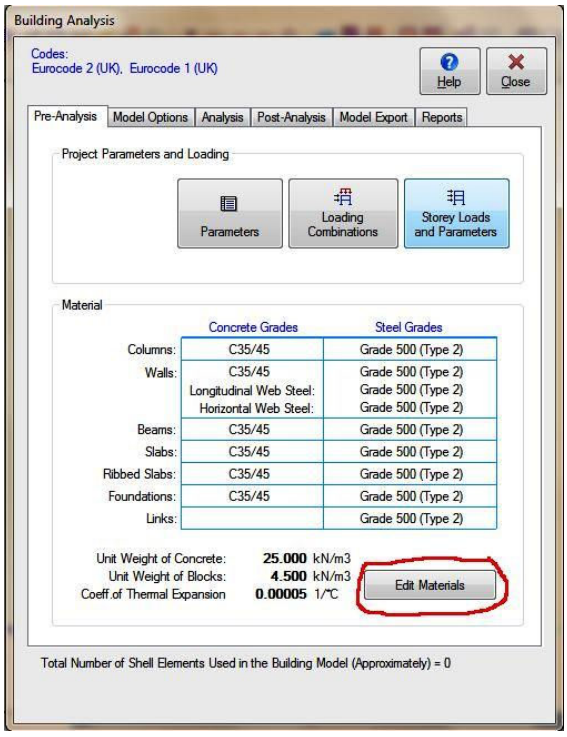
You can go to **Title**, and set project title and other information as you desire.

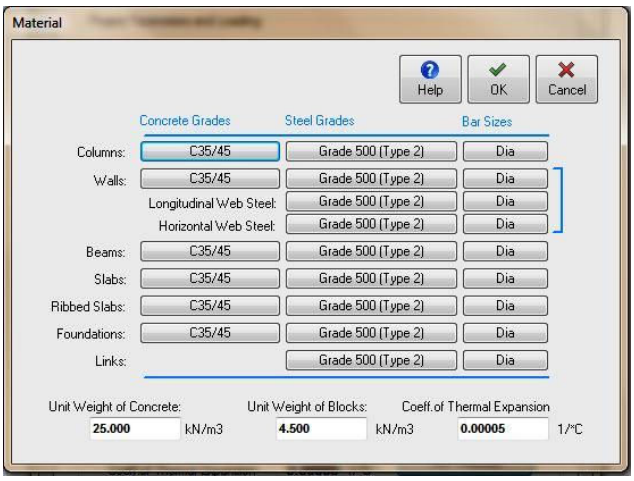


Close the dialog box, and go to **Building Analysis**. We have some important settings to take care of there.

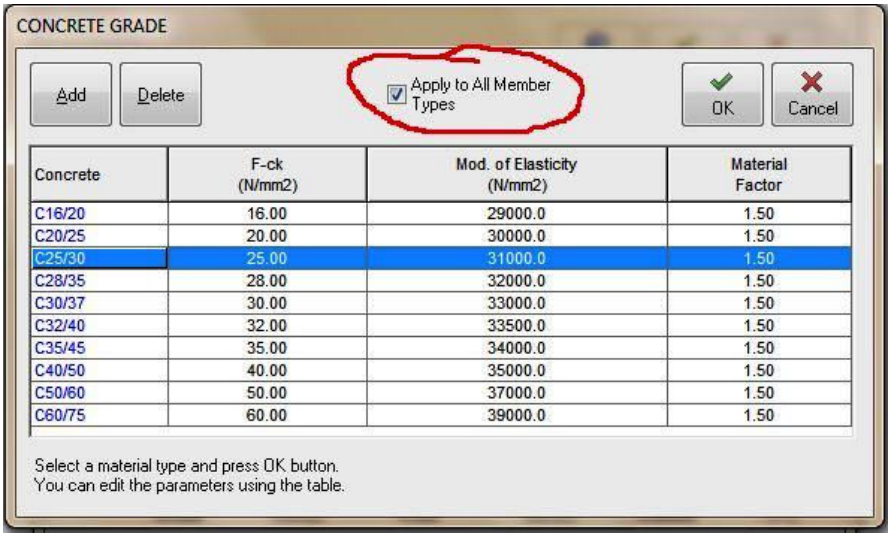


Then click on **Edit Materials**, and another dialog box comes up

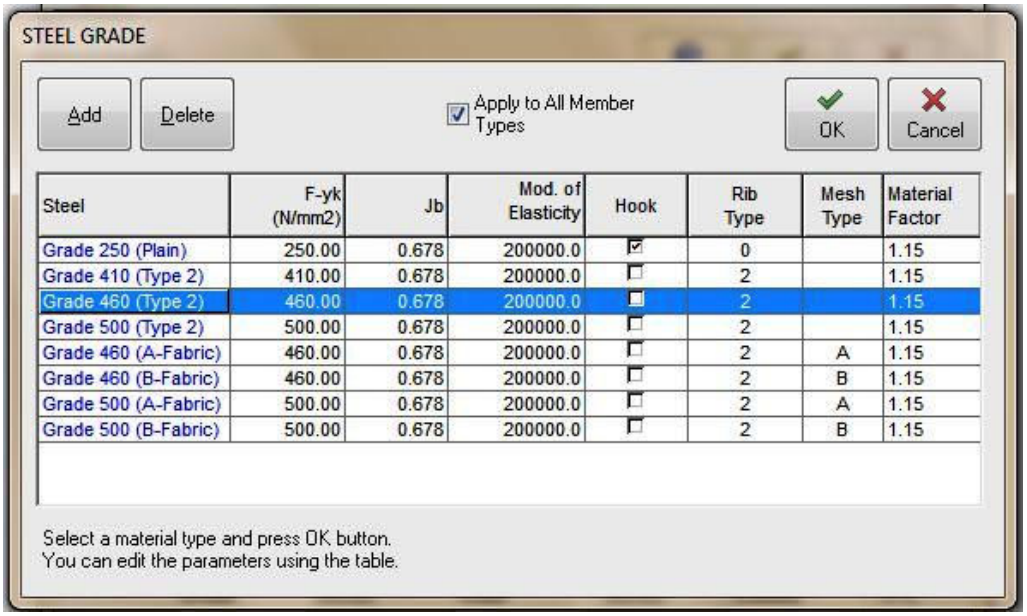




We can select the **concrete grade** we are employing in our design by clicking on the **'concrete grade'** tab of any of the elements. In this case, f_{ck} = 25 N/mm^2 (cylinder strength) for all elements. The default as you can see above is C35/45. So select C25/30 of which the compressive strength is 25 N/mm^2. Now check the box for '**Apply to All Member Types'** to apply it to all the structural Elements.

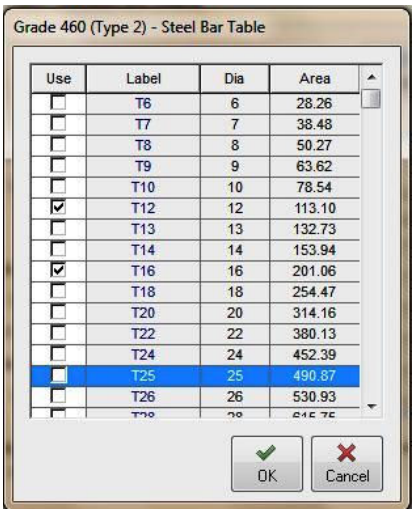


In the same vein, we can select the **steel grade** by clicking on the **'steel grade'** tab of any of the elements. In this case, we are adopting f_{yk} = 460 N/mm^2 for all elements. The default steel grade as you can see above is Grade 500 (Type 2). So select Grade 460(Type 2) of which the yield strength is 460 N/mm^2. Now check the box for '**Apply to All Member Types'** to apply it to all the structural Elements. Click **OK**

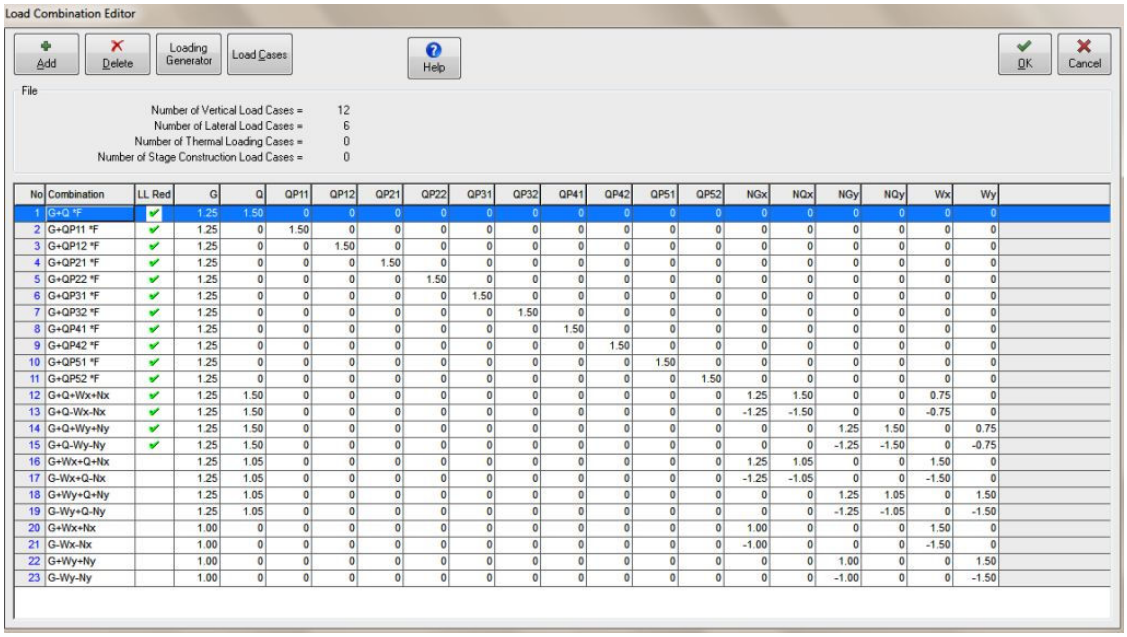


You can verify that the density of concrete is 25 KN/m^3 by default. You can leave the weight of block and coefficient of expansion the same way it is.

You can also select your desired bar sizes for various elements by clicking on the **'Dia'** tab. For instance, if you desire the minimum bar sizes for your columns to be 16mm, you can specify it by checking and un-checking the bar sizes as you desire. Click **OK** after all your selections.

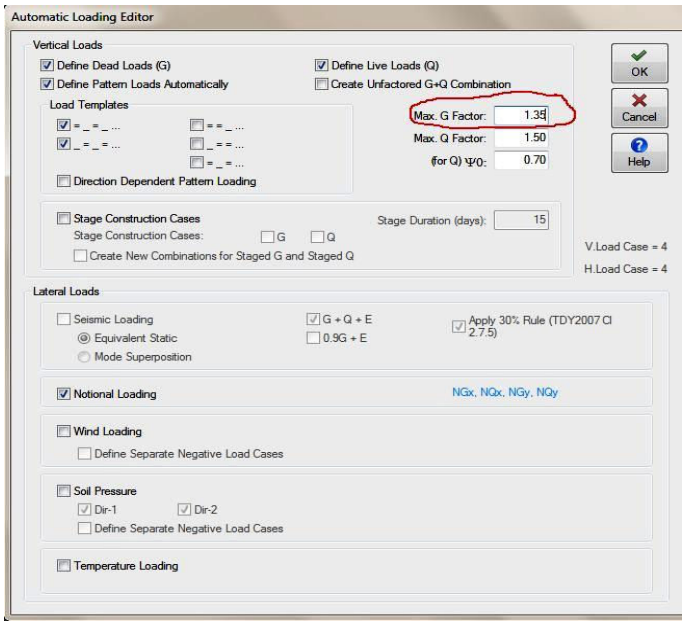


After editing the materials, you can go the **Parameters Tab**. A little consideration will show that we have already taken care of that. Now go to the load combinations Tab. On clicking on it, we the following dialogue box below pops up. We can see that the partial factor for dead loads is 1.25, instead of the 1.35 that we know in Eurocode 2. We have to change that. You can start by editing them one after another, or you click on **'Loading Generator'** Tab.

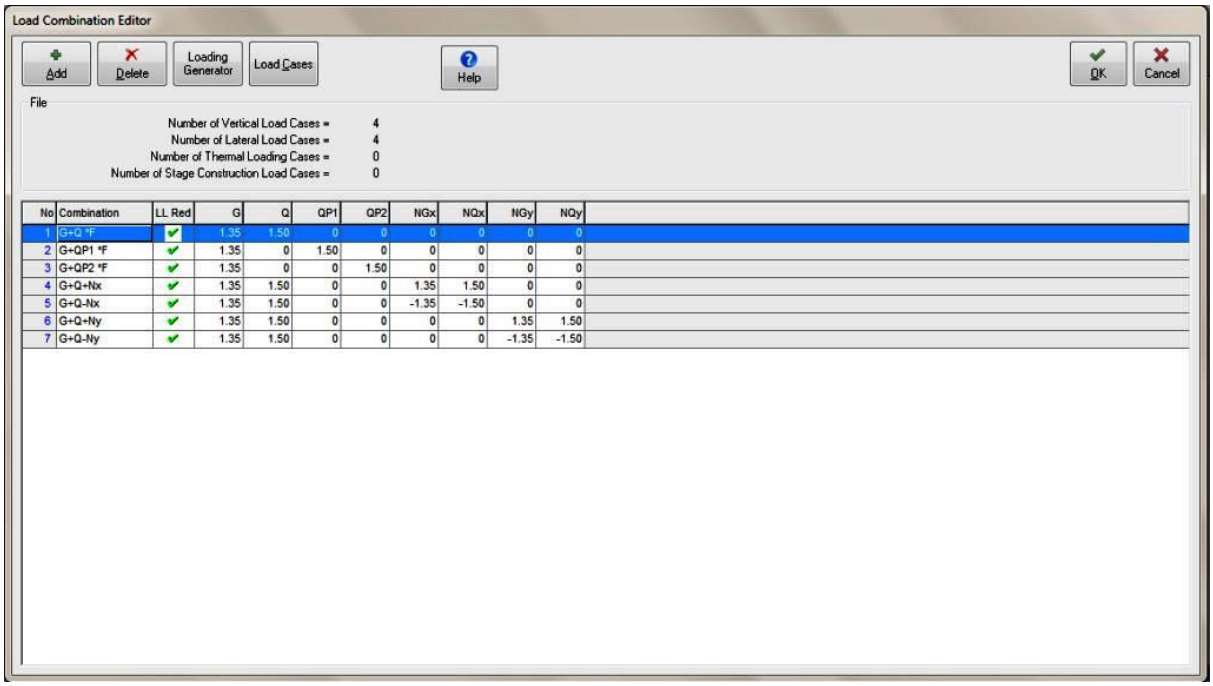


No	Combination	LL Red	G	Q	QP11	QP12	QP21	QP22	QP31	QP32	QP41	QP42	QP51	QP52	NGx	NQx	NGy	NQy	Wx	Wy
1	G+Q *F	✔	1.25	1.50	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2	G+QP11 *F	✔	1.25	0	1.50	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
3	G+QP12 *F	✔	1.25	0	0	1.50	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4	G+QP21 *F	✔	1.25	0	0	0	1.50	0	0	0	0	0	0	0	0	0	0	0	0	0
5	G+QP22 *F	✔	1.25	0	0	0	0	1.50	0	0	0	0	0	0	0	0	0	0	0	0
6	G+QP31 *F	✔	1.25	0	0	0	0	0	1.50	0	0	0	0	0	0	0	0	0	0	0
7	G+QP32 *F	✔	1.25	0	0	0	0	0	0	1.50	0	0	0	0	0	0	0	0	0	0
8	G+QP41 *F	✔	1.25	0	0	0	0	0	0	0	1.50	0	0	0	0	0	0	0	0	0
9	G+QP42 *F	✔	1.25	0	0	0	0	0	0	0	0	1.50	0	0	0	0	0	0	0	0
10	G+QP51 *F	✔	1.25	0	0	0	0	0	0	0	0	0	1.50	0	0	0	0	0	0	0
11	G+QP52 *F	✔	1.25	0	0	0	0	0	0	0	0	0	0	1.50	0	0	0	0	0	0
12	G+Q+Wx+Nx	✔	1.25	1.50	0	0	0	0	0	0	0	0	0	0	1.25	1.50	0	0	0.75	0
13	G+Q-Wx-Nx	✔	1.25	1.50	0	0	0	0	0	0	0	0	0	0	-1.25	-1.50	0	0	-0.75	0
14	G+Q+Wy+Ny	✔	1.25	1.50	0	0	0	0	0	0	0	0	0	0	0	0	1.25	1.50	0	0.75
15	G+Q-Wy-Ny	✔	1.25	1.50	0	0	0	0	0	0	0	0	0	0	0	0	-1.25	-1.50	0	-0.75
16	G+Wx+Q+Nx		1.25	1.05	0	0	0	0	0	0	0	0	0	0	1.25	1.05	0	0	1.50	0
17	G-Wx+Q-Nx		1.25	1.05	0	0	0	0	0	0	0	0	0	0	-1.25	-1.05	0	0	-1.50	0
18	G+Wy+Q+Ny		1.25	1.05	0	0	0	0	0	0	0	0	0	0	0	0	1.25	1.05	0	1.50
19	G-Wy+Q-Ny		1.25	1.05	0	0	0	0	0	0	0	0	0	0	0	0	-1.25	-1.05	0	-1.50
20	G+Wx+Nx		1.00	0	0	0	0	0	0	0	0	0	0	0	1.00	0	0	0	1.50	0
21	G-Wx-Nx		1.00	0	0	0	0	0	0	0	0	0	0	0	-1.00	0	0	0	-1.50	0
22	G+Wy+Ny		1.00	0	0	0	0	0	0	0	0	0	0	0	0	0	1.00	0	0	1.50
23	G-Wy-Ny		1.00	0	0	0	0	0	0	0	0	0	0	0	0	0	-1.00	0	0	-1.50

On clicking the **Load Generator Tab**, the dialog box below comes up. Now change the **Maximum G Factor** from 1.25 to 1.35. You can see that the partial factor for variable loads is 1.5. We will leave that the way it is. Then click **ok**



On doing that, the dialog box below comes up;



No	Combination	LL Red	G	Q	QP1	QP2	NGx	NQx	NGy	NQy
1	G+Q *F	✔	1.35	1.50	0	0	0	0	0	0
2	G+QP1 *F	✔	1.35	0	1.50	0	0	0	0	0
3	G+QP2 *F	✔	1.35	0	0	1.50	0	0	0	0
4	G+Q+Nx	✔	1.35	1.50	0	0	1.35	1.50	0	0
5	G+Q-Nx	✔	1.35	1.50	0	0	-1.35	-1.50	0	0
6	G+Q+Ny	✔	1.35	1.50	0	0	0	0	1.35	1.50
7	G+Q-Ny	✔	1.35	1.50	0	0	0	0	-1.35	-1.50

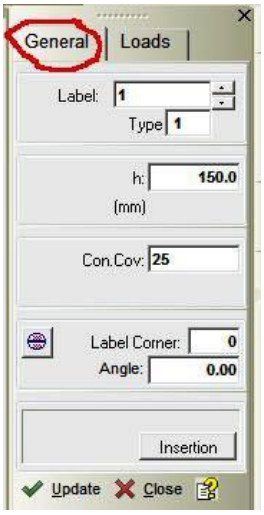
As you can see, the number of load cases has been reduced to just 8. These 8 load cases are sufficient for our analysis. On going to the next tab, you can neglect, '**Storey loads and Parameters'** tab.

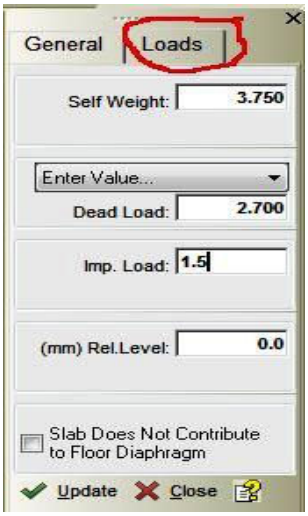
After we are done with this, we can then start modelling of our floor slabs.

3.3.6 Modelling of floor slabs

Click on '**Slab**' icon

On clicking the slab icon, the dialog box below comes up.





On the **General** Tab, Change the thickness of the floor slab (**h**) to 150mm

Specify the concrete cover (**Con. Cov)** = 25mm

On the **Loads Tab**, the self weight of the slab is automatically calculated based on the density of the concrete, and the thickness of the slab. As you can see, it is equal to 3.75 KN/m^2

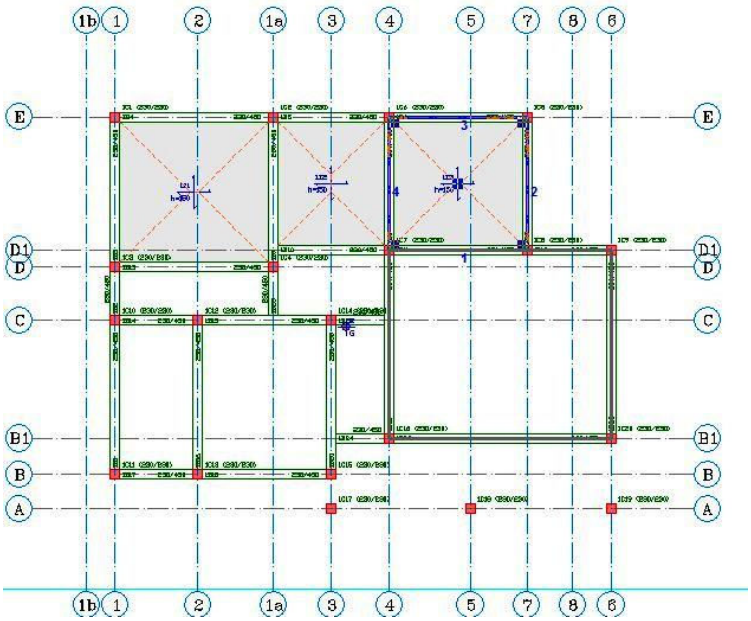
There are additional dead loads to be specified;

Weight of finishes = 1.2 KN/m^2

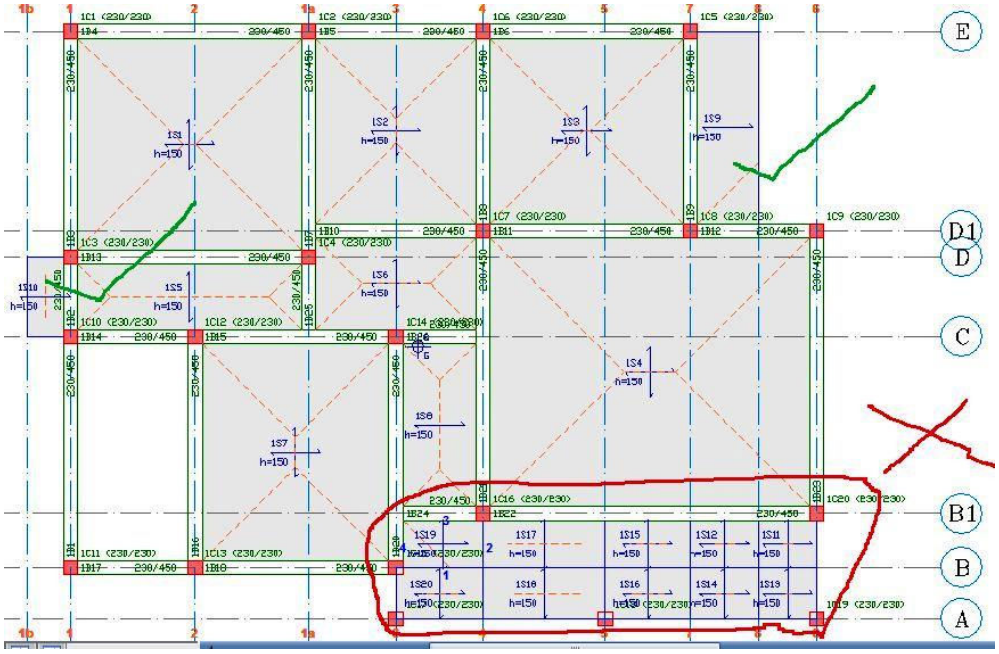
Partition allowance = 1.5 KN/m^2

The total dead load is 2.7 KN/m^2
The specified imposed load (see chapter 2) is 1.5 KN/m^2

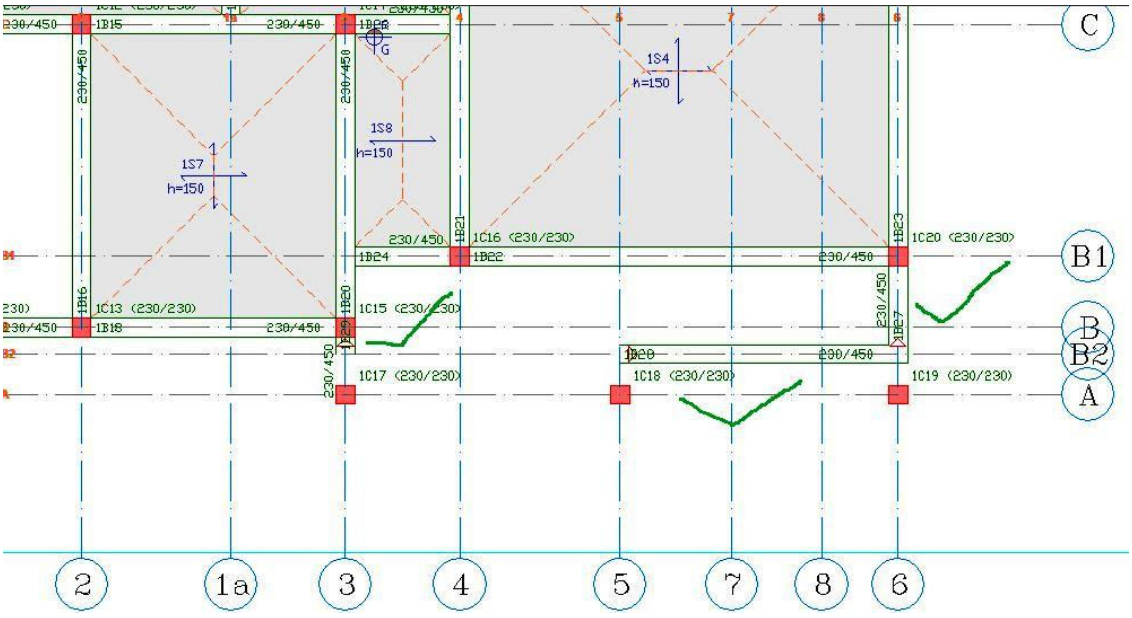
After specifying all the data above, now move the cursor to the GA and start clicking on each panel.



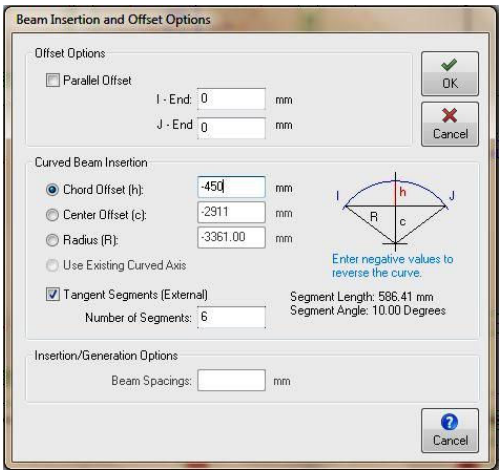
Due to the presence of gridlines at the edge of the cantilever panels (panels 10 and 11), the panels are automatically generated. But a little consideration will show that PANEL 5 is very problematic due to the presence of multiple gridlines, and the fact that we have some curved sections. See the sample difficulty below.



You can verify that the arrangement we have at PANEL 5 is very far from the arrangement that we have in mind. Multiple panels have been created based on the areas created from the gridlines. So how are we going to solve this problem? We are going to apply a little trick. Also note that there is need for a new gridline to adequately capture the edges of the slab of PANEL 5 (see the MANUAL GA). We will offset axis B1 by 1200mm below to create the edge (this creates Axis B2). Now, we have to delete the multiple slabs created in the region of interest (simply click on each of them, and hit the delete button). We will now create beams along the exterior edges of PANEL 5, including a curved beam where it is required (follow the procedure described for beam creation). See what was done below.



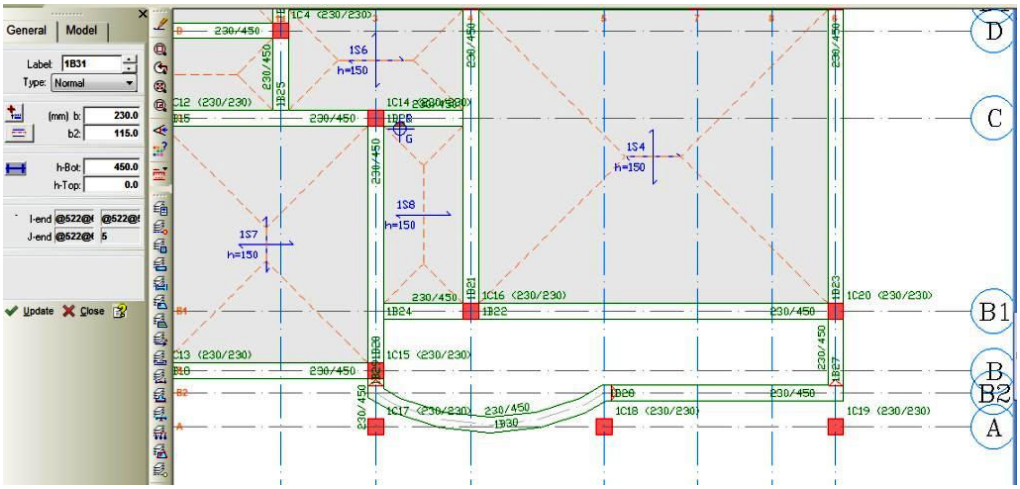
We have created beams from axis **6:B1- B2**, Axis **B2: 5 - 6**, and Axis 3: **B – B2**. Now we will create a curved beam along **Axis A: 3 – 5**. To do this click the beam creation cursor at **3B2** axis intersection, hold the '**shift'** button, and click at the **5B2** intersection. The following dialog box comes up;



Place the **Chord Offset (h**) as - 450mm

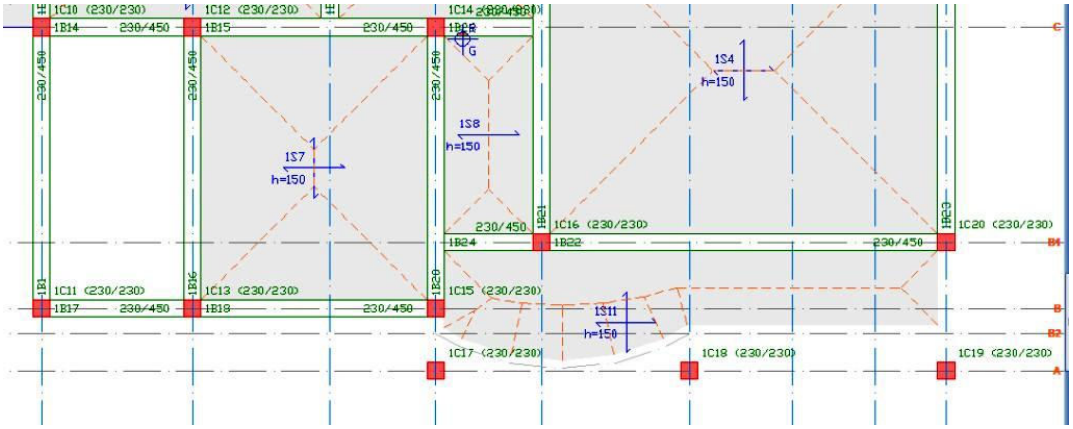
Making it negative implies that you want the chord to curve in the reverse direction. The **centre offset (c)** and **radius (R)** are automatically generated based on the value input.

The result of this activity is shown below;



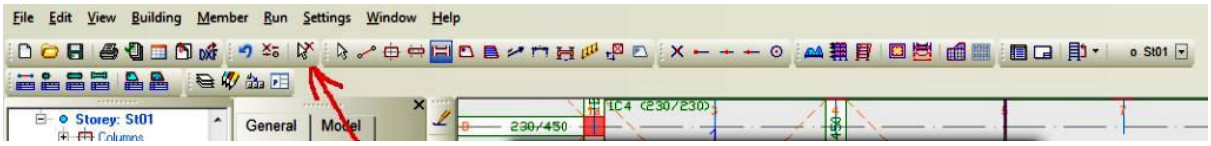
The beam outlining the exterior of the slab has been created as shown above.

Now, create a slab on the area by clicking on the slab icon as described above, and clicking on the panel area. Now **DELETE** the exterior periphery beams that we created for it. By implication, we have created the slab.

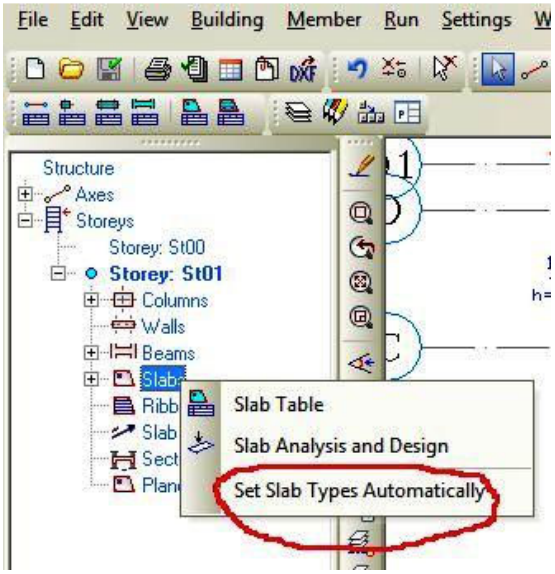


3.3.6.1Setting Slab Types Automatically

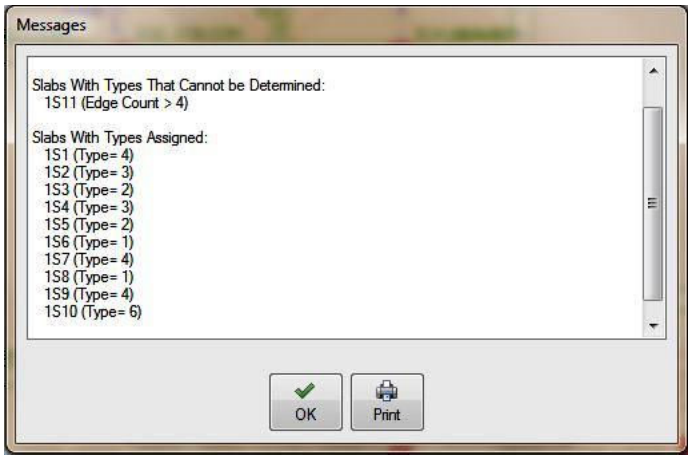
Now clear all selection by clicking on '**clear selection'** icon



Now, go to the slab icon by the left hand side of the screen, and right click on it. On the options, click on '**Set Slab Types Automatically'**. When the '**Slab Type Determination'** dialog box comes up, click **OK**



Now a message pop up comes up (shown below) to show that the slab type has been set (based on edge conditions, and direction of spanning).

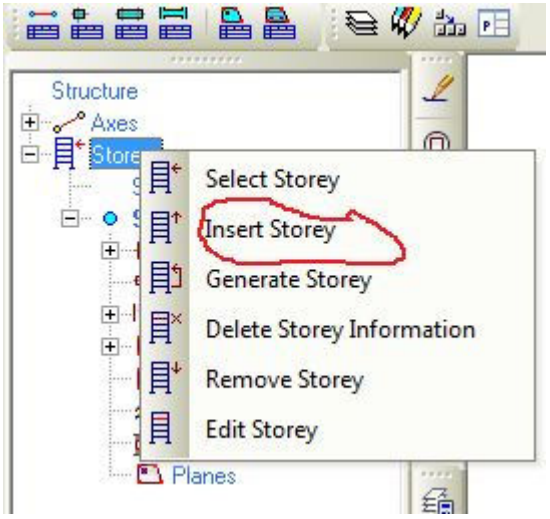


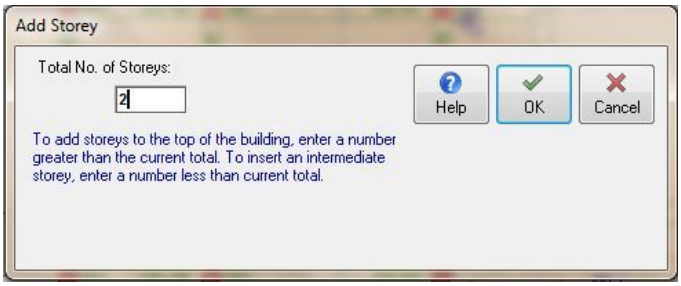
As you can see, 1S11 type (PANEL 9 of our GA) cannot be automatically determined (the curved slab we created). The implication of this is that we will use **Finite Element Analysis** for our floor slab load transfer, instead of **Yield Line method**. Click **OK.**

Kindly refer to Orion Engineer's handbook for more information on Finite Element Analysis and Yield Line Methods of slab load transfer.

3.3.7 Completing the model

We wish to specify the number of storeys on our structure. Go to **Storey** by the left hand side of the GUI, right click on it, and click on **Insert Storey.** On the **Add Storeys** dialog box that pops up, enter the number of storeys as 2. Click ok.

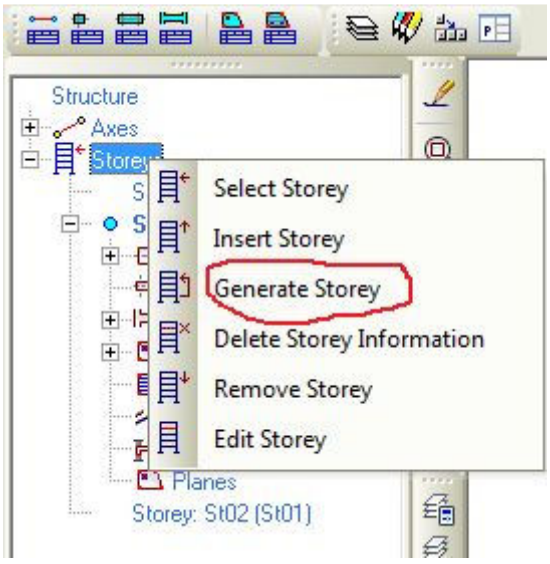


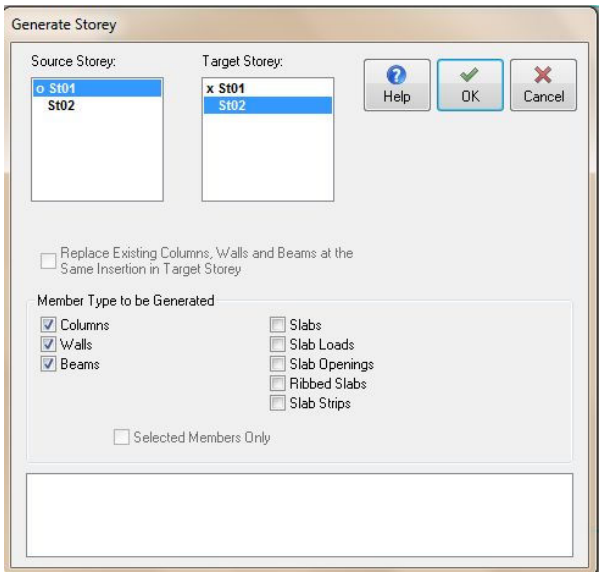


3.3.7.1 Generating Storey

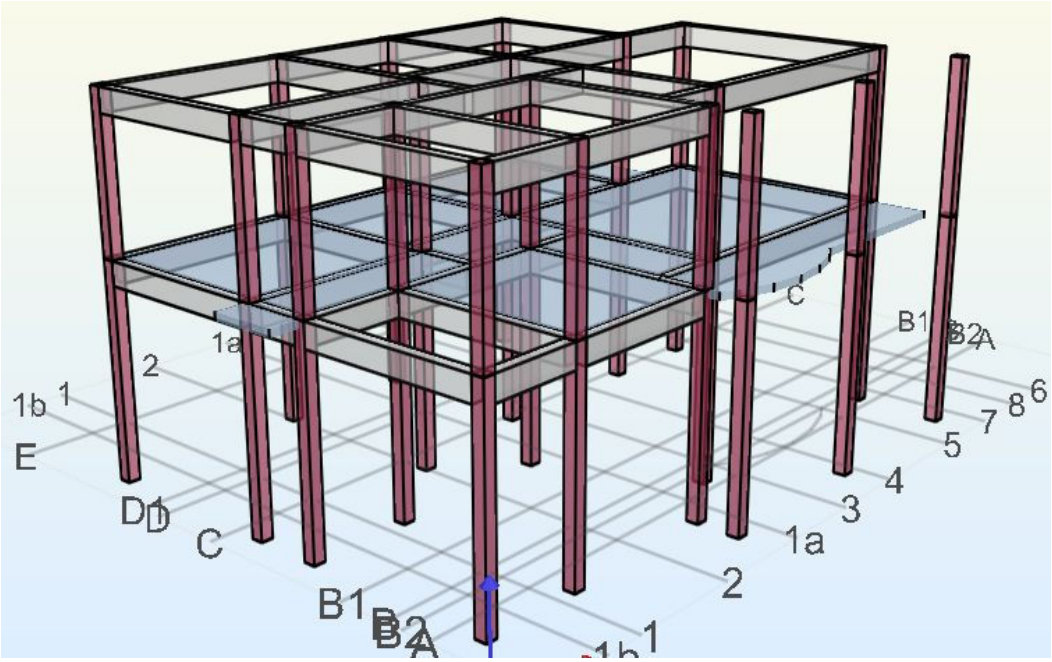
We wish to generate some of the elements on the ground floor to the 1st floor. We will generate only the columns and the floor beams. However, we know that roof beams are of different cross-section with the floor beams. So all we have to do is to change their properties.

Go to **Storey** again, and right click on it. Select **generate storey**. When the '**Generate Storey'** dialog box pops up, click on **St01** (Storey 1) as the **Source Storey**, and **St02** (Storey 2) as the **Target Storey** (see the plates below). That is to say we are moving some elements from the ground floor to the 1st floor. As I hinted earlier, we are generating only **columns** and **beams**. So you will have to uncheck Slabs and its associated elements. Click **OK**



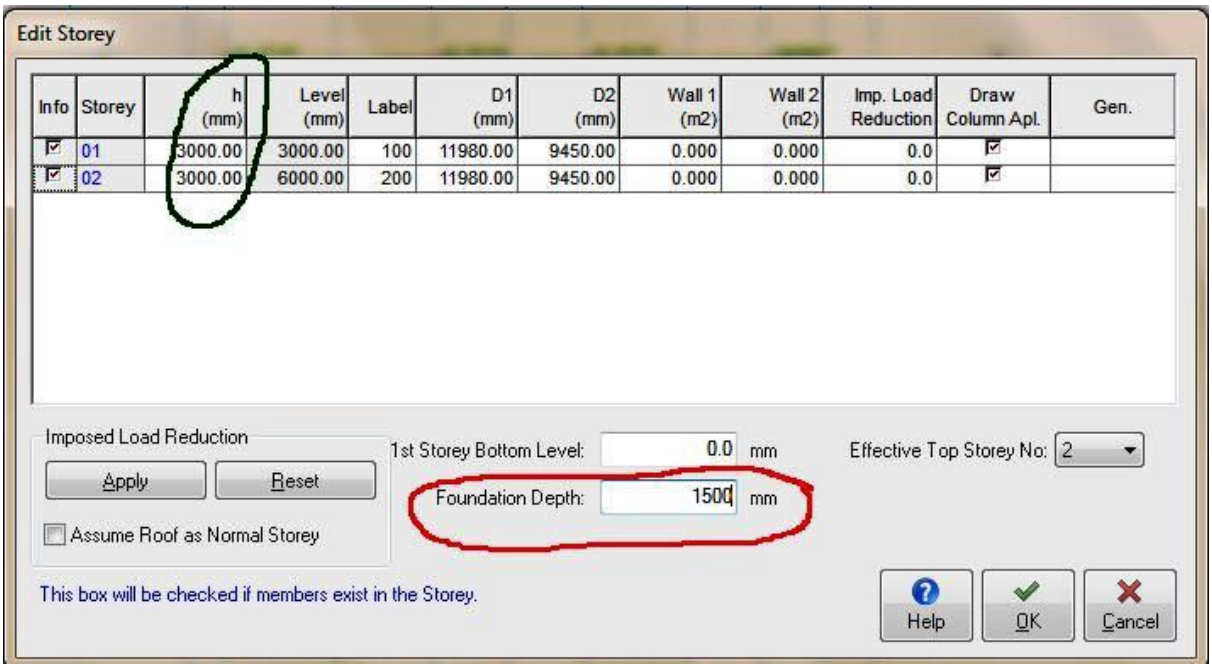


Click on 3D view icon, our output should be like this;



3.3.7.2 Specifying Storey Height

Got to **Storey** again and right click on it. Select '**Edit Storey'**, and the following dialog box comes up.

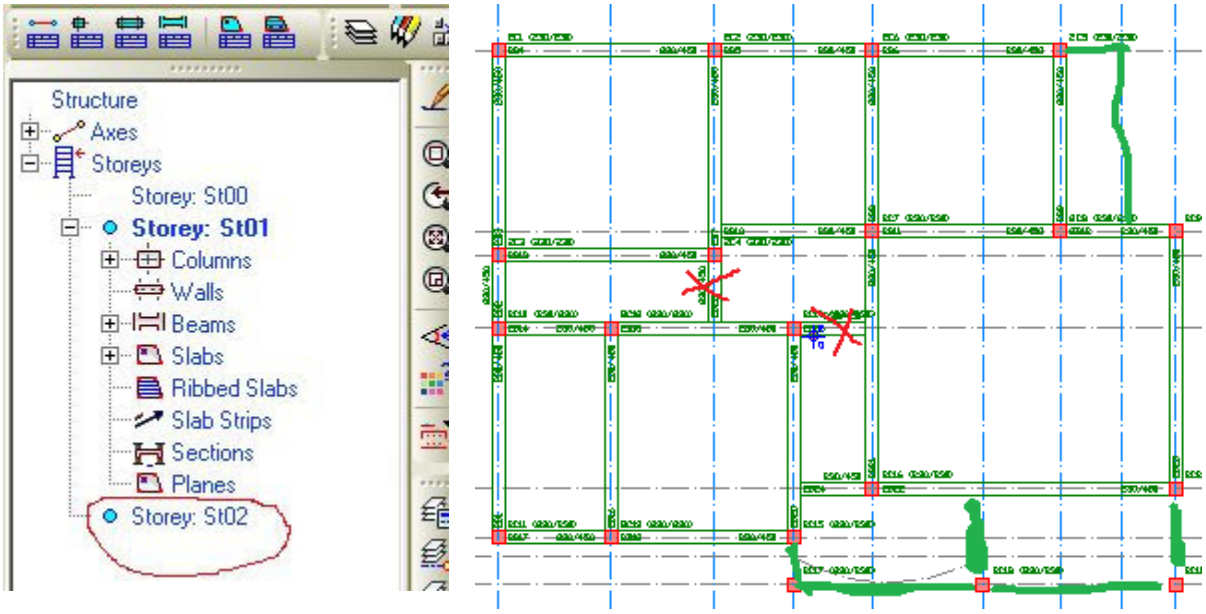


The height of each storey is 3000mm which is desirable, and we are extending the foundation depth to 1500mm. 1000mm is actually the real depth of the foundation, while 500mm is the height of the Damp Proof Course (DPC) above the ground level. We should account for this for the purpose of ground floor column design. Click **OK**. You can view the effect of this on 3D view if you wish.

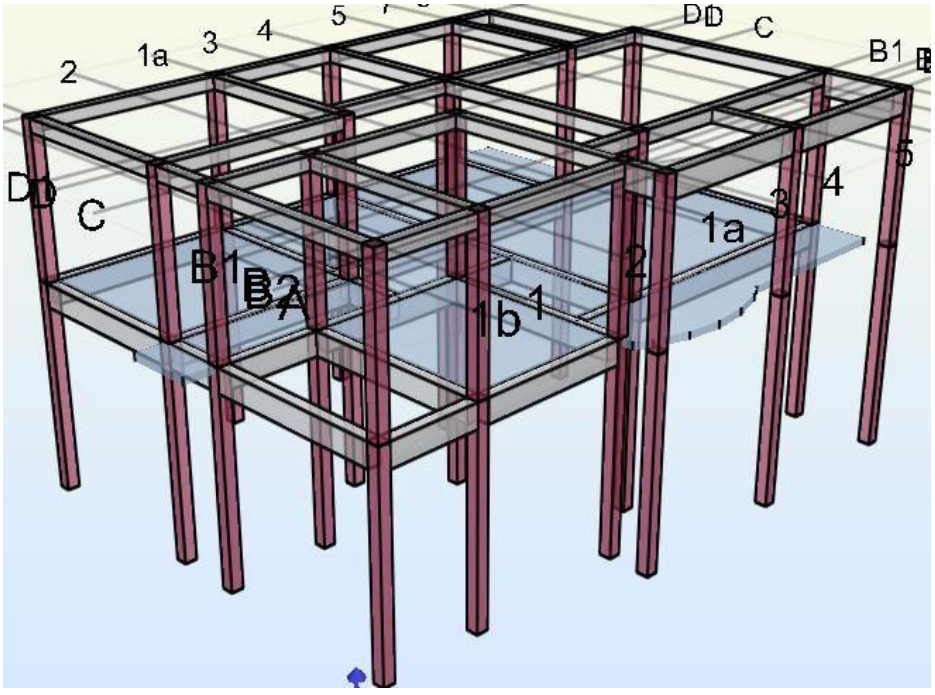
3.3.7.3 Arranging the roof beam

Go back to the plan view;

Click on **St02** twice, to switch your plan to the roof beam level. Here, we have to edit some things, like removing some unnecessary beams, adding some important beams, and editing the depth from 450mm to 250mm. I am removing beams I marked **X**, and drawing some **green lines** in the areas that I am adding new beams.



We need to add the roof beams to the front area to adequately brace the front columns. The roof beams will aid in resisting possible wind loads, even though we are not explicitly accounting for it in this design. Also edit the depth of the beam to 250mm, by right clicking on it, and changing **hbot** to 250mm. For aesthetics and to represent the architectural disposition, leave the beams at the front column as 450mm. The final 3D disposition should be as shown below.



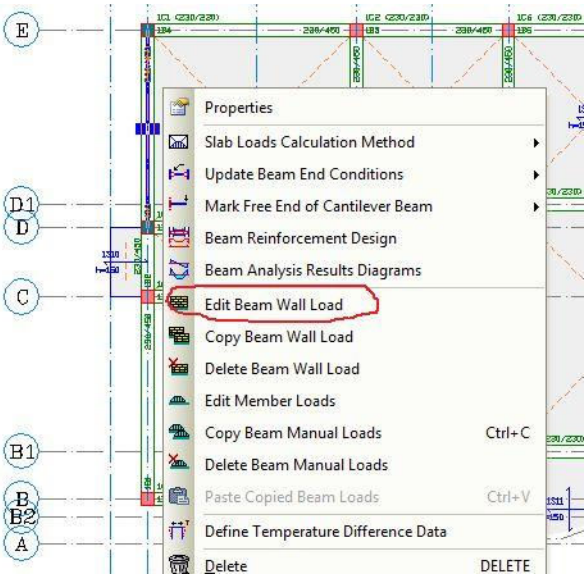
3.3.8 Loading on CSC Orion

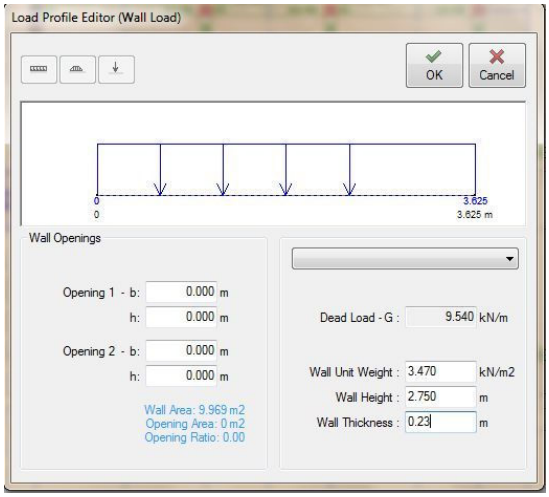
This where we specify the loads acting on the floor beams and roof beams.

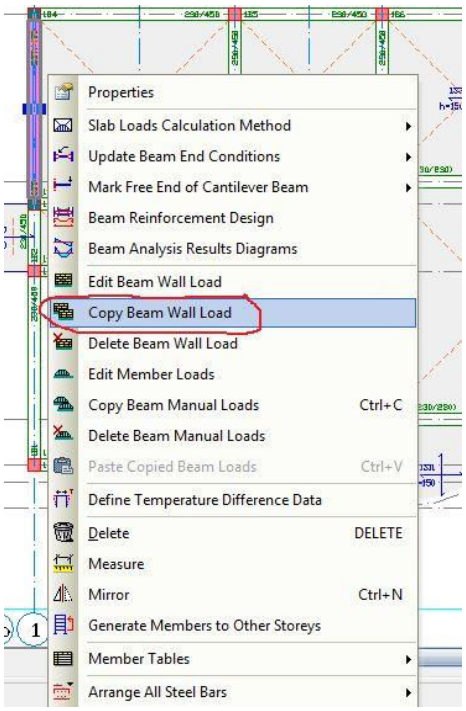
3.3.8.1 Loading the floor beams with wall load

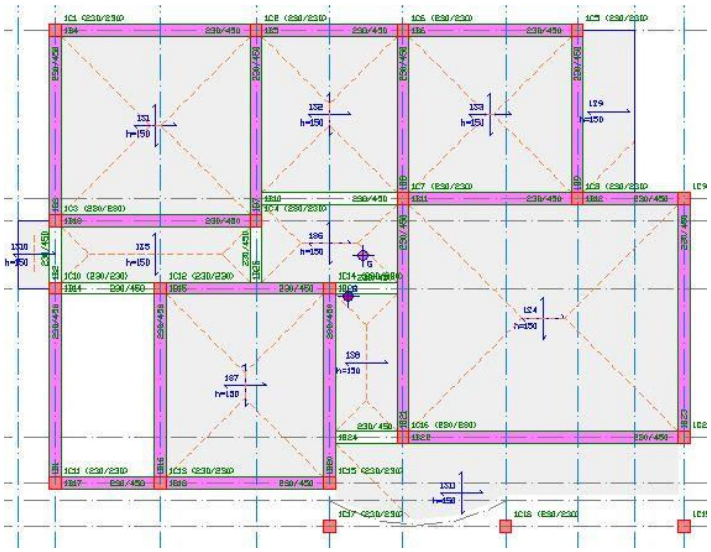
Click on **St01** twice, to return to the first floor plan. Right click on any beam that will carry any block work load. Click on '**Edit Beam Wall Load'**. On Clicking on it, the **Load Profile Editor (Wall Load)** dialogue box comes up. The unit weight of block work and finishes in Nigeria averages 3.47 KN/m^2 (*Oyenuga, 2009*), and the height of the wall is 2.75m. As you know, the height of the storey is 3.0m and the roof beam is dropping by 250mm (0.25m), so the height of the wall is 3.0m – 0.25m = 2.75m. The thickness of our wall is 230mm. ***Note that we are neglecting the effects of opening on the wall load, so as to easily relate with our manual analysis results***. On the other hand, it is more conservative. With the data generated, the dead load is automatically generated. Click OK

Note: You can specify the wall opening dimensions (doors and windows) at the locations specified.









The colour of the beam carrying the wall load now changes to purple (see the pictures above). We can copy the beam wall load to the other beams where it applies. Right click on the beam and click on '**Copy Beam Wall Load'.** Hold the control button and click on all the beams where you want the load to be applied. Then right click your mouse, and click on '**Paste Copied Beam Loads'.** To verify that you are successful, all the beams affected will change to purple. You can verify that I did not load the beams crossing the lobby area with wall load.

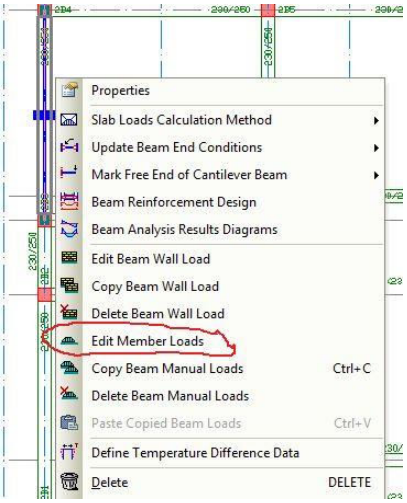
3.3.8.2 Loading the staircase floor beam

The staircase loading being transferred to the floor beam has been manually defined. A uniformly distributed load dead load of 10.837 KN/m, and a live load of 4.9595 KN/m has been manually applied to **Beam D:1-2.** To see how the loading was generated see section 5.4.1 of Chapter 5. To see the steps on how manual loads are applied, see the next section (3.3.8.3).

3.3.8.3 Loading the roof beam

Knowing full well that the beam self weight are calculated automatically. Let us specify the roof dead load as 2 KN/m, and the weight of parapet (concrete fascia) as 2.0 KN/m for the external beams.

Imposed load on roof, let us take 0.85 KN/m.
Therefore total dead load for external roof beams (Gk) = 4 KN/m
Therefore total dead load for internal roof beams (Gk) = 2 KN/m

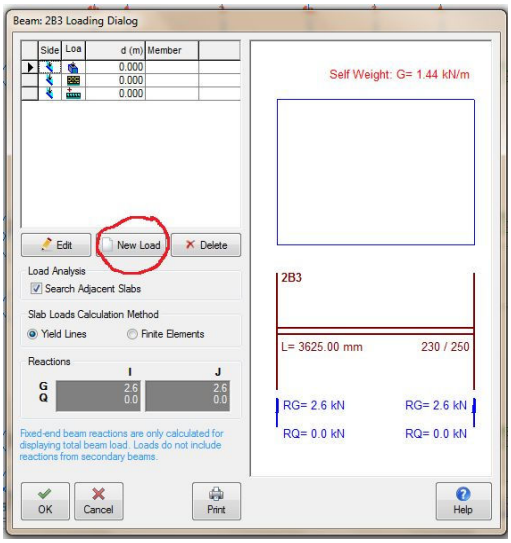


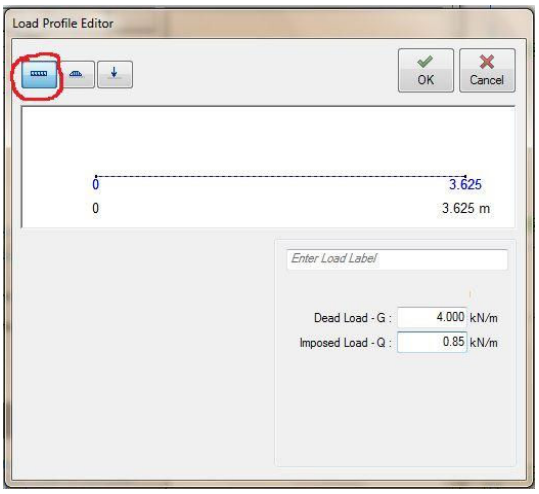
Switch to the **St02** (roof beam) and right click on any external beam of your choice. From the draw down list, select **Edit Member loads**, and the dialogue box below comes up.

Click on '**New Load'** Tab

When the '**Load Profile Editor'** Dialog box comes up, change the load to uniformly distributed load. Enter the Dead Load and Imposed Load as specified. Click **OK**

Now, right click on the beam and click on **Copy Beam Manual Load**. Paste on all external roof beams. Repeat the process for internal beams (remember the dead load is reduced to 2 KN/m).

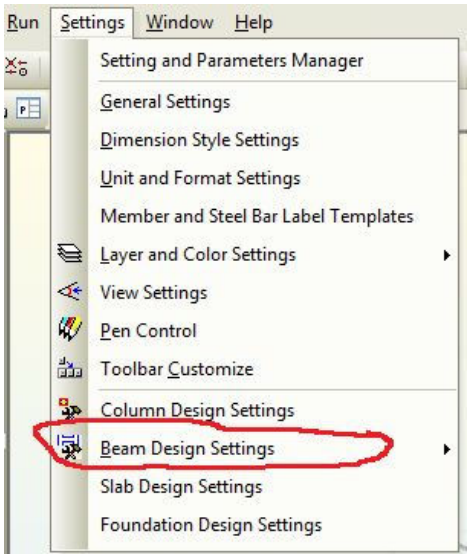


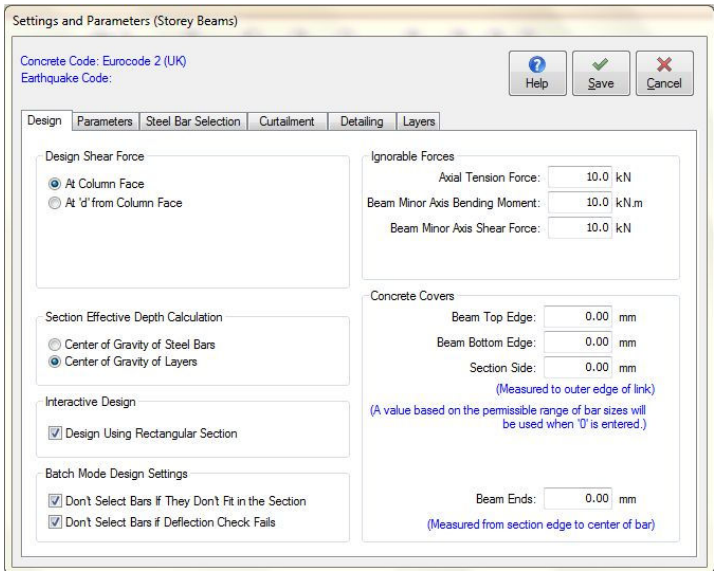


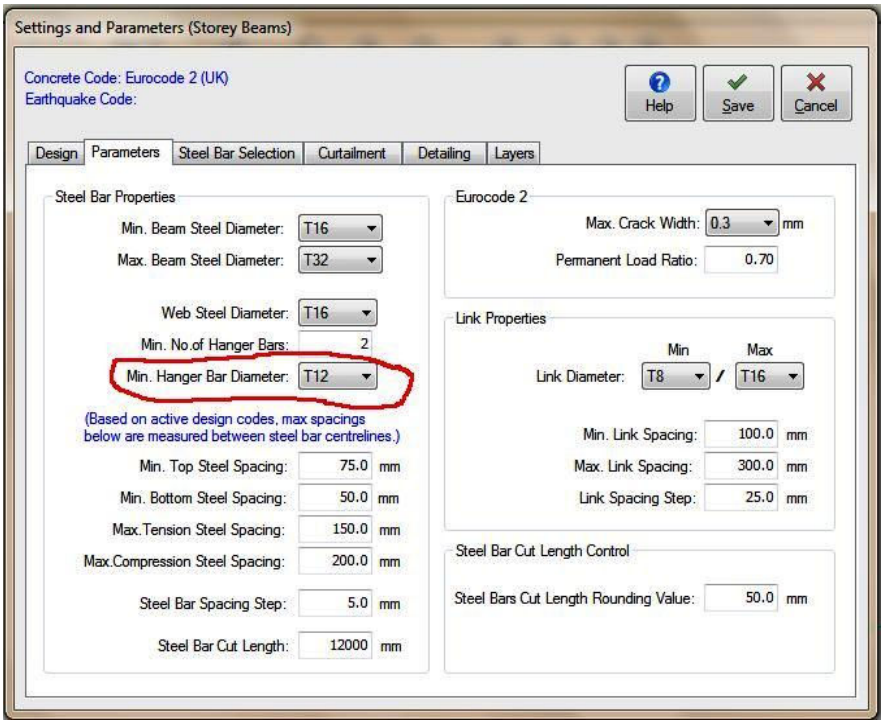
3.3.9 Specific member design Settings

Before we go into analysis, there are some further specific settings that I will like us to look at.

Go to **Settings**, and select **Beam Design Settings**, and Select **Storey beam Settings**







When the '**Settings and Parameters (Storey Beams)**' dialogue box comes up, you can make the adjustments as you wish;

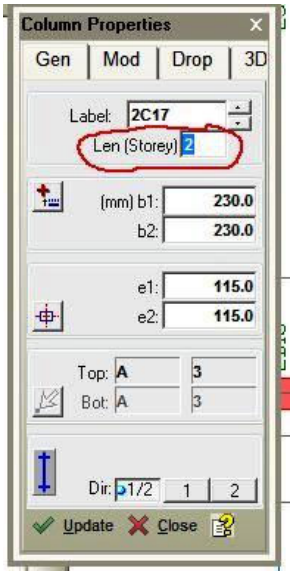
For instance, the option of checking shear at the column face, or at 'd' from the column face is available. It is better to leave other options the way they are. For more information on this, you can read the **Orion Engineer's Handbook**. Also, all other setting regarding detailing, bar curtailment, etc can be adjusted here. Note that leaving some values as zero means that code requirements should be used. Otherwise you can specify yours. The only adjustment I did was to specify my minimum steel for beams as Y16mm and Hanger bars as Y12mm.

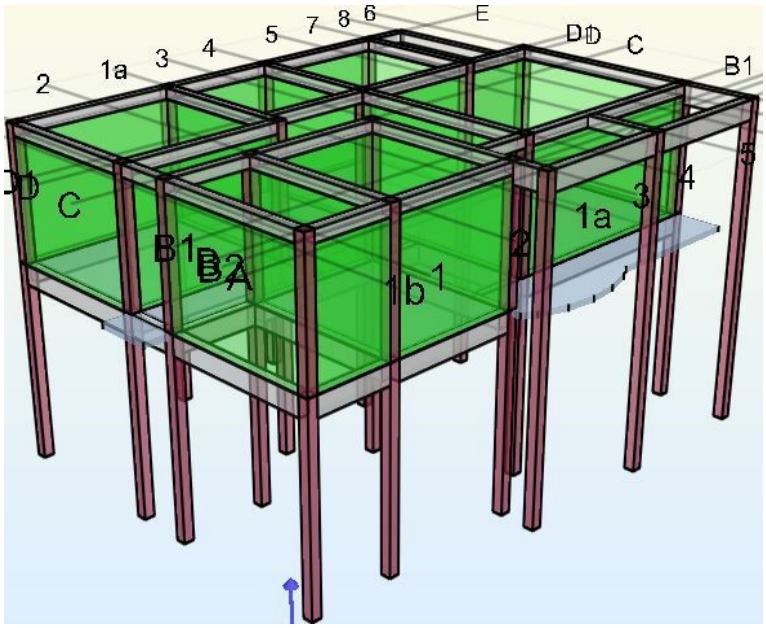
Also, go to **slab design settings** and set the minimum concrete cover to 25mm.

I recommend that you leave column design settings the way it is. You can make adjustments on the steel bar sizes if you wish though. Note that the values we input while editing materials influences the number of options we have for steel bar sizes in specific member design settings.

3.3.9.1 Column spanning more than one storey (Axis G columns)

If you look at the previous 3D views, you will verify that the front columns at axis G of our GA should be single without being broken at the 1st floor level. We can take care of this and change it to a single column length. This is necessary so that in the design of the columns, the appropriate length of the column will be used for the calculation of slenderness effects.





To do this, return to **St02 (Storey 2)** plan view

Right click the column, and go to **Properties**

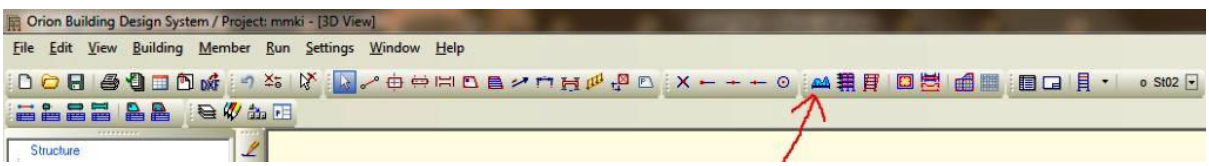
Change the **Len(Storey)** to **2**, so that the column will span for 2 storeys. Update this for the three columns concerned.

Now, go back to **St01**, and delete the columns at that level. The 3D view should be as shown above.

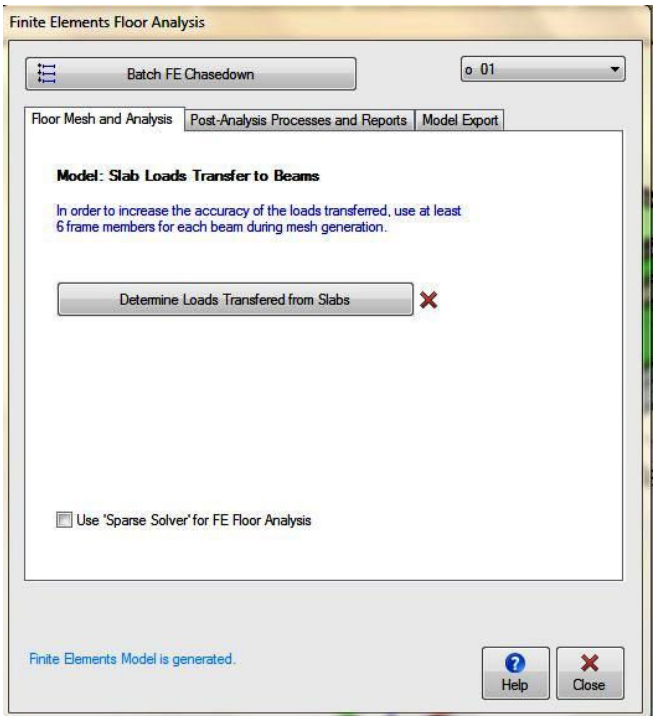
3.3.10 Slab Load Decomposition

Now that the model has been satisfactorily completed, it is time to start the analysis.

Somehow, I normally like to start with the decomposition of the slab load using Finite Element Analysis.

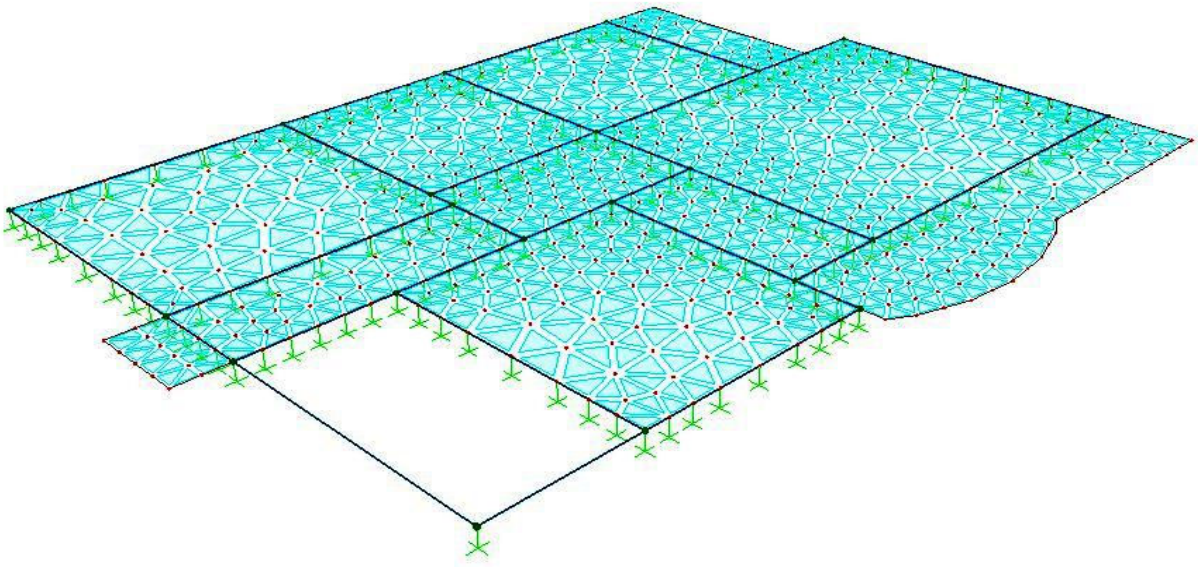


Click on **Load Decomposition by FE** (pointed with red arrow above).



The dialog box by the left hand comes up. Check that you are currently on Storey 1. Click on '**Determine Loads Transferred from Slab'**. Immediately, you are taken to the finite element analysis page (see the picture below). On the FE analysis page, click on '**Generate Model'** to mesh the floor plate. Once the meshing is complete, **close** the page to run the analysis. Click **Close**, and '**Apply to all beams in the current storey**'.

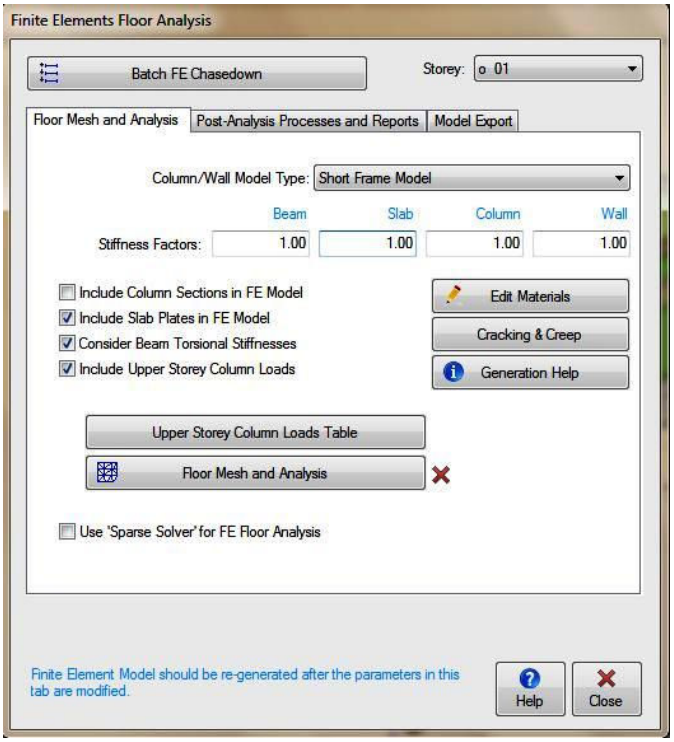
Note that you can adjust the mesh sizes to suit the accuracy level that you are looking for at this page.



Now, all the loads from the floor slab have been transferred to the floor beams.

3.3.11 Finite Element Floor Analysis

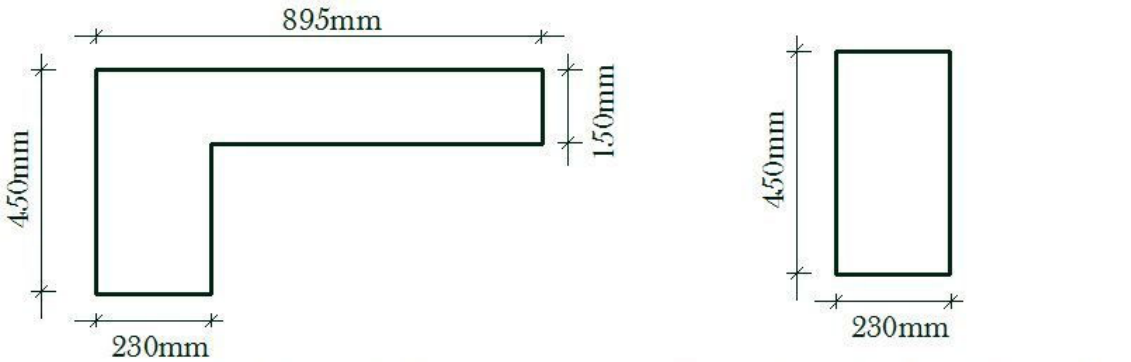
Click on '**Finite Element Floor Analysis'**. The dialog box below comes up.



On the Column/Wall Model Type, select **Short Frame Model**, where the structure is represented as a sub frame with the upper and lower columns fully taken into consideration. For other model types, refer to Orion technical manuals.

Another point of interest is the stiffness factors. Orion carries out Finite element analysis based on the input rectangular sections. Therefore, when you are analysing your floor beams, you may want to increase the stiffness to account for the flanges, which are typical for reinforced concrete buildings.

For instance, consider beam A: 1 – 3 on our GA. The effective flange width according to Eurocode 2 is 895mm. We will do a little comparison to show the modification of stiffness that we are talking about.



(a) Real Section Behaviour (b) Section Used By Orion FE Analysis

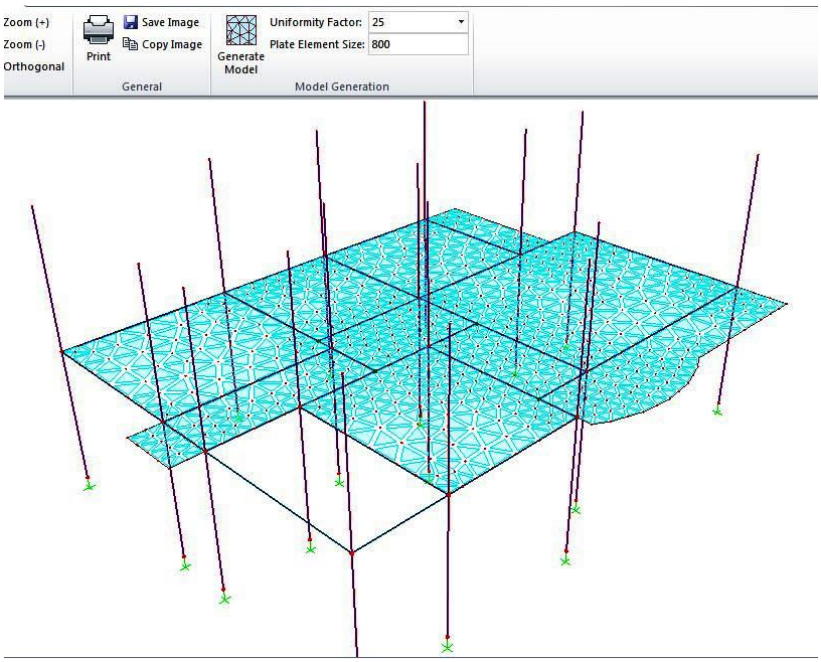
The moment of Inertia (Ixx) of Figure (a) according to the theorem of parallel axis is 0.003 m^4

The moment of Inertia (Ixx) of Figure (b) is 0.001746 m^4

The ratio of the stiffnesses is about 1.76.Therefore, I normally take an approximate beam stiffness factor of 1.5 as adequate. However, this actually calls for more review.

For slabs, Orion advises that a reduction be made in the value of E to account for creep and cracking on the long term, since the gross section is used for the calculation of Ixx. For flat slabs, a range of 0.16 to 0.25 has been recommended. 0.16 is ideal for offices and heavily loaded buildings, while 0.25 is ok for residential buildings. For more information on stiffness factors, kindly read Orion Engineers Handbook (Page 132). So for this particular work, we are increasing beam stiffness by 1.5, and reducing slab stiffness by about 75%.

Now, for this example, check all options except for including 'Column Section in FE Model'. It is more ideal for flat slabs. Now click on **Floor Mesh and Analysis. It is however advisable to do for the 2nd storey first (a kind of chase down)**



Generate the model, and **close** to run the analysis.

After the analysis, check the **AXIAL LOAD COMPARISON REPORT** by clicking on it.

AXIAL LOAD COMPARISON REPORT

SUM OF APPLIED LOADS (Using Un-Decomposed Slab Loads):

G - Dead Loads:

Storey	Column (kN)	Wall (kN)	Beam (kN)	Slab (kN)	Rib (kN)	Total (kN)
St02	91.3	0.0	391.1	0.0	0.0	482.3
St01	67.4	0.0	779.2	626.2	0.0	1472.9
Total						1955.2

Q - Live Loads:

Storey	Column (kN)	Wall (kN)	Beam (kN)	Slab (kN)	Rib (kN)	Total (kN)
St02	0.0	0.0	72.3	0.0	0.0	72.3
St01	0.0	0.0	-6.5	145.6	0.0	139.1
Total						211.5

SUM OF APPLIED LOADS (After Decomposing Slab Loads):

G - Dead Loads:

Storey	Column (kN)	Wall (kN)	Beam (kN)	Slab (kN)	Rib (kN)	Total (kN)
St02	91.3	0.0	391.1	0.0	0.0	482.3
St01	67.4	0.0	1405.4	0.0	0.0	1472.9
Total						1955.2

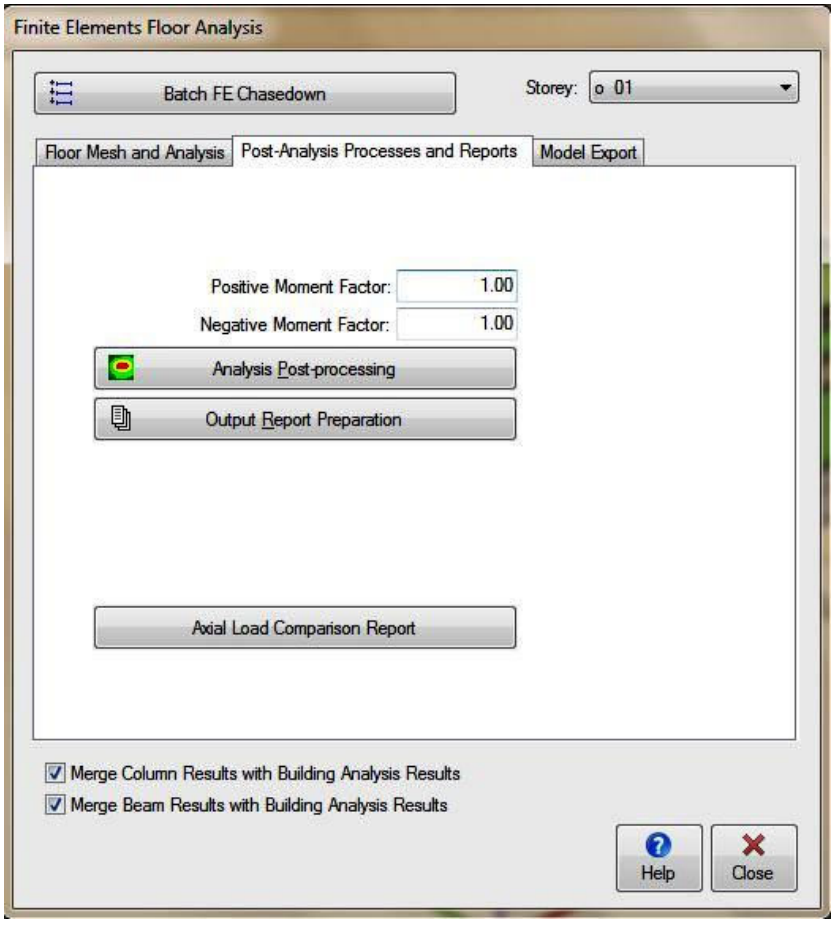
Q - Live Loads:

Storey	Column (kN)	Wall (kN)	Beam (kN)	Slab (kN)	Rib (kN)	Total (kN)
St02	0.0	0.0	72.3	0.0	0.0	72.3
St01	0.0	0.0	139.1	0.0	0.0	139.1
Total						211.5

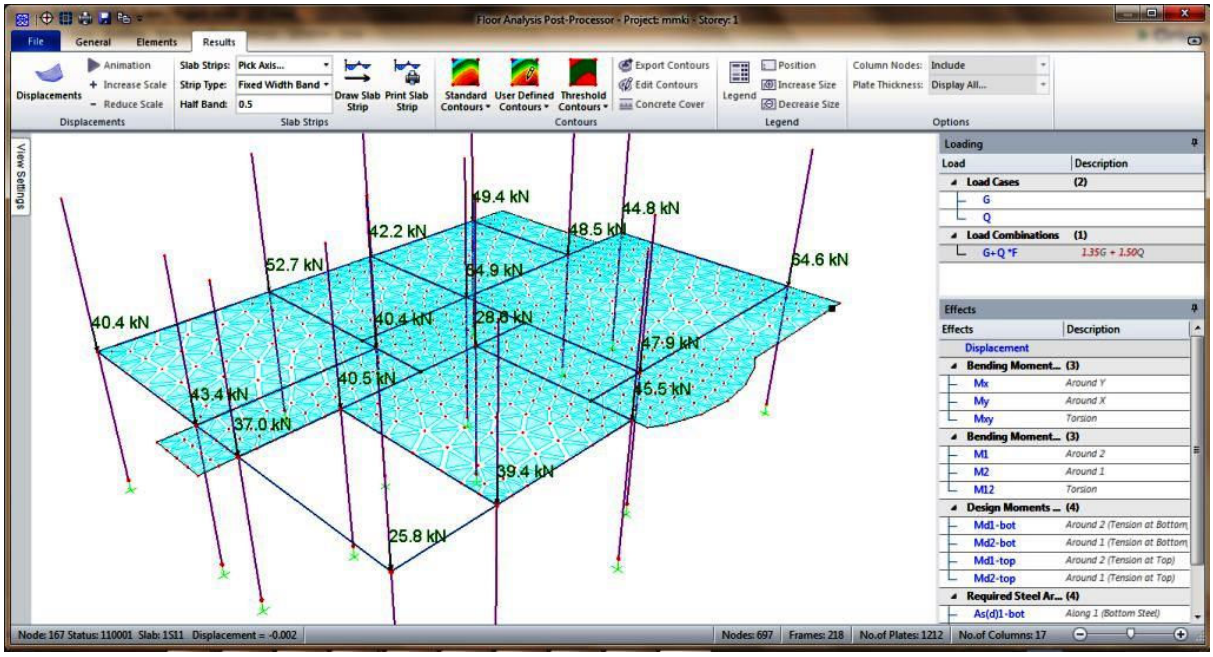
The axial load comparison report is satisfactory.

3.3.12 Floor Slab Post Processing

Click on '**Analysis Post-processing**'



The post processing page comes up.



Change it to plan view at the second icon at the top left corner of the screen. Click on '**standard Contours'**. Now, you can view the stress contour for different planes, such as Mx, My, Mxy, deflection (displacement) etc by clicking the options by the right hand side of the screen etc. I will give a brief description of the results to enhance your understanding.

Mx plot – This contour is displayed relative to a single global coordinate system in the horizontal direction

My plot - This contour is displayed relative to a single global coordinate system in the vertical direction

Mxy plot – This contour displays the torsion (twisting) of the slab panels

M1 plot – This is an Mx plot where the contours are displayed relative to a local coordinate system for each slab panel

M2 plot – This is an My plot where the contours are displayed relative to a local coordinate system for each slab panel

Md1-bot plot – This is the M1 sagging moments adjusted to account for plate torsion using the Wood and Armer adjustments.

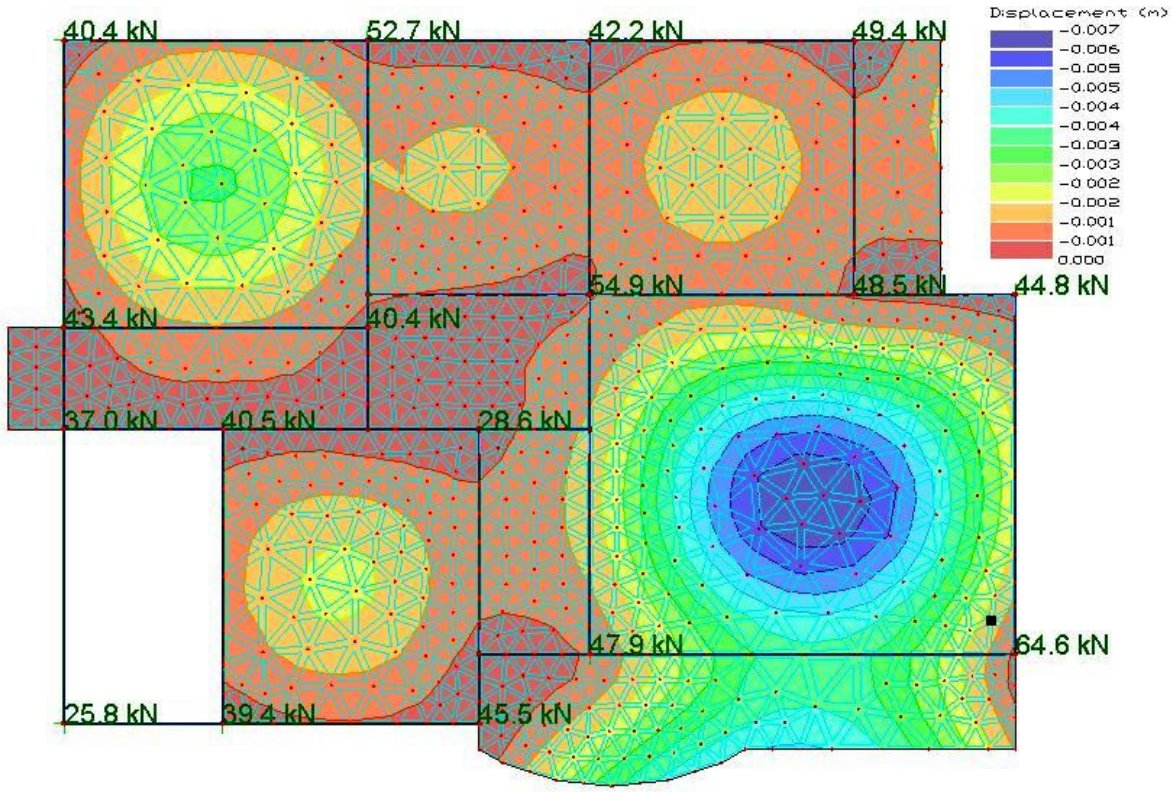
Md1-top plot – This is the M1 hogging moments adjusted to account for plate torsion using the Wood and Armer adjustments.

As1-bot, As2-bot, As1-top, As2-top are the steel areas that are based on the M1 and M2 moments (not including Wood and Armer Adjustments)

As(d)1-bot, As(d)2-bot, As(d)1-top, As(d)2-top are the steel areas that are based on the Md1 and Md2 moments (moments including Wood and Armer Adjustments)

Orion recommends that you work with As(d) contour results

The displacement contour is shown below. Feel free to check all the other contours.

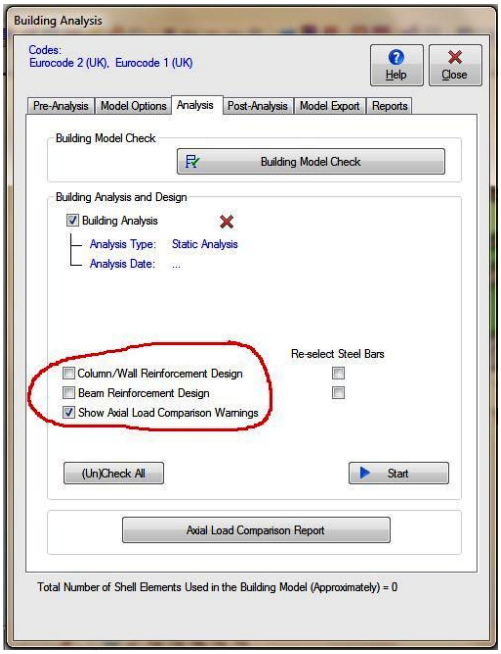


Now close the Post Processing Page.

Now, if you click the **Batch FE Chasedown** Tab, you will see it has been carried out for all the storeys. This is indicated by the green tick that shows that it has been successfully carried out for each storey. Just in case anything suggests that it has not been carried out. Click ok to run the analysis.

3.3.13 Full Building Analysis

Click on **Building Analysis**, and then, go to **Analysis** Tab.



Make sure that you check by clicking on the tiny boxes provided;

Column/Wall Reinforcement Design

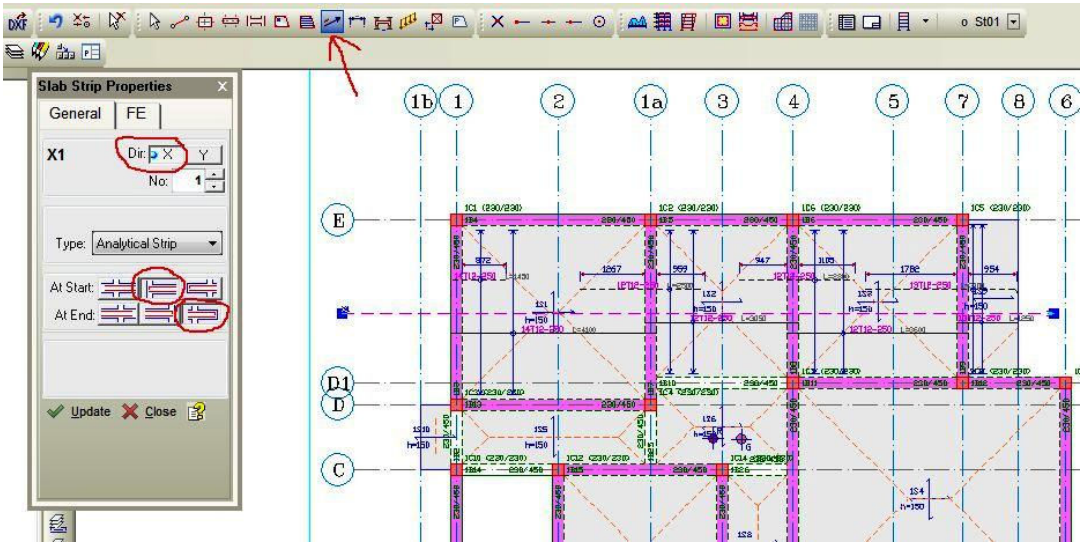
Beam Reinforcement Design

Then click on **Start** to run the analysis.

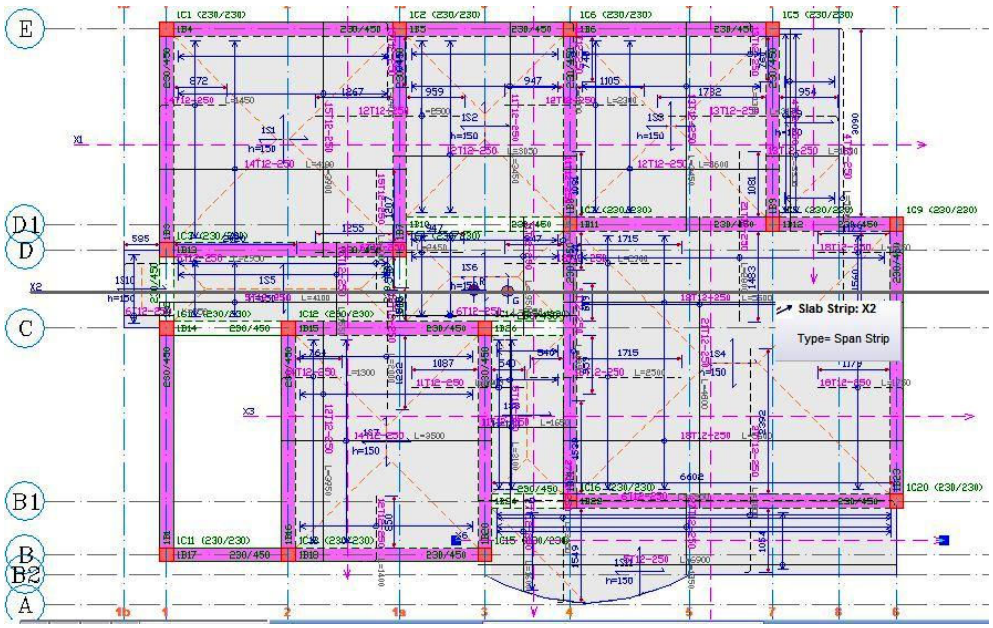
When the Building Analysis is completed successfully, click **OK**.

3.3.14 Slab Design

Click on **Slab strips**. By default, just leave it at **General Strip**. The first direction is the X direction. Now, I wish to draw slab strip across Panels 1, 2, and 3. To start with, I have to select the start and end conditions of the slab. Now, at the start, the slab is simply supported or discontinuous, so I have to select the symbol (bob) at the middle. The end of the slab is a cantilever panel, so I have to select that accordingly by checking the option at the right (just follow the images). To make your strip line straight, hold the **Ctrl** Key. Click somewhere outside Panel 1 by the left, drag the strip line till somewhere outside the cantilever area at Panel 10 and click there. Automatically, the reinforcements and their spacing are displayed. Repeat the process for all the other panels, bearing in mind to respect all the end conditions. Remember that for vertical directions, you have to use the Y strip direction. Remember that you can also use the Finite Element Strip, but you will have to go to FE Post processing for your design results. To do that, you have to change from Analytical Strip to FE Strip. However, the differences in the results for our case will not be pronounced. (See Orion Engineer's Handbook).

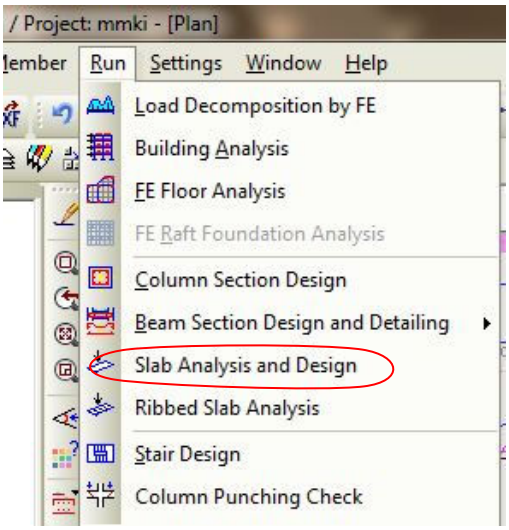


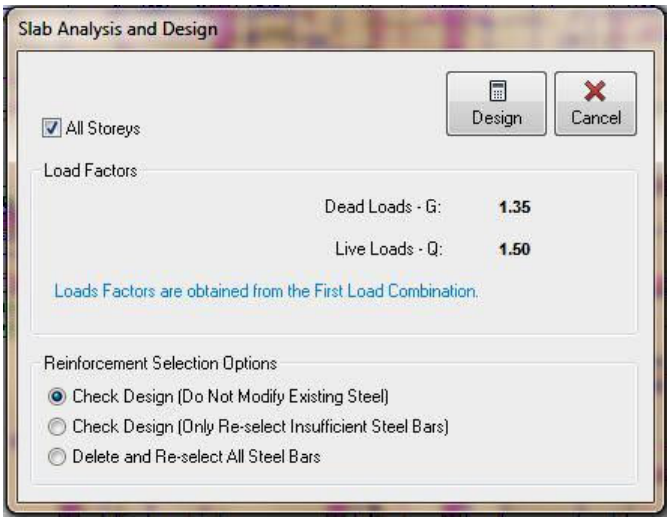
The expected final output is shown below.



Go to **Run**, and Select **Slab Analysis and Design**

Verify the Load Factors and click **Design**



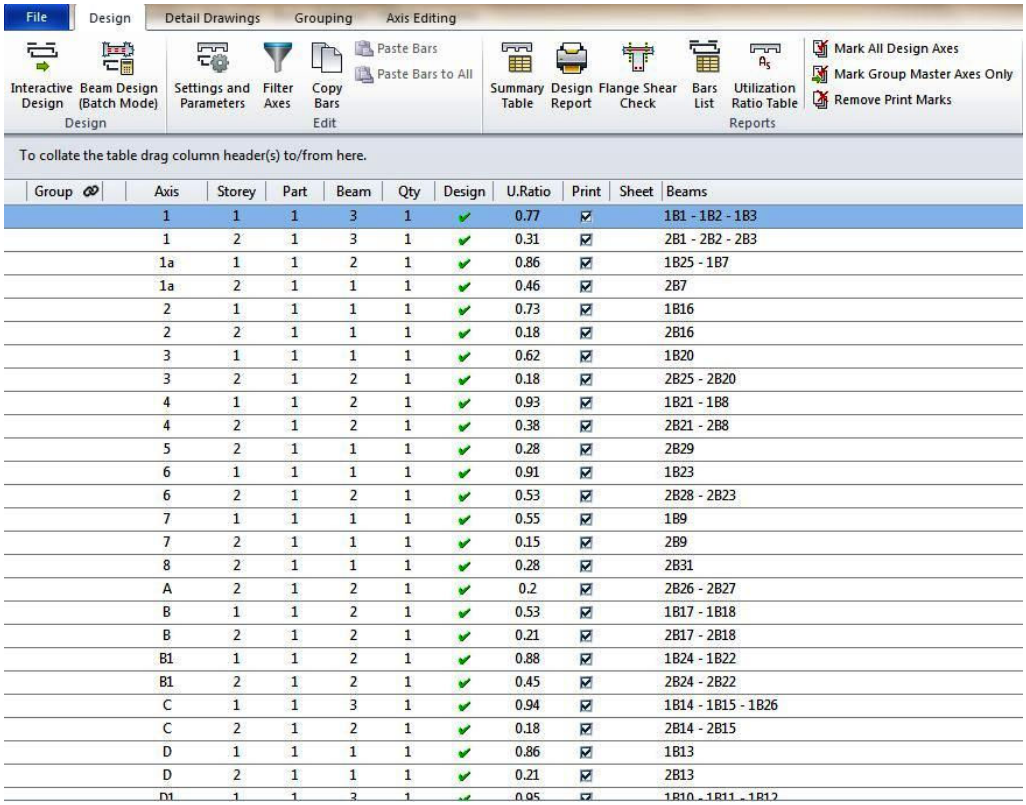


The results of the floor slab are displayed immediately. To export the floor detailing sheet, click on '**Export DXF'** on Orion's main GUI page

3.3.15 Beam Design

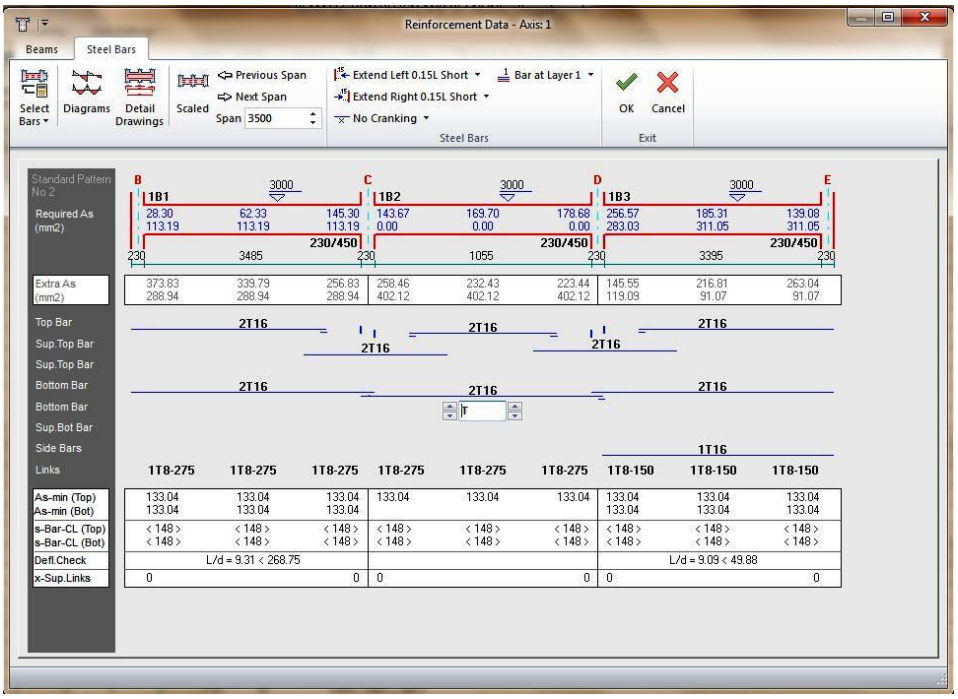
Go back to **Run**, and Select **Beam Section Design and Detailing**, and select **Storey Beams**

The beam design page comes up. Green ticks shows beams where all major requirements are satisfied.

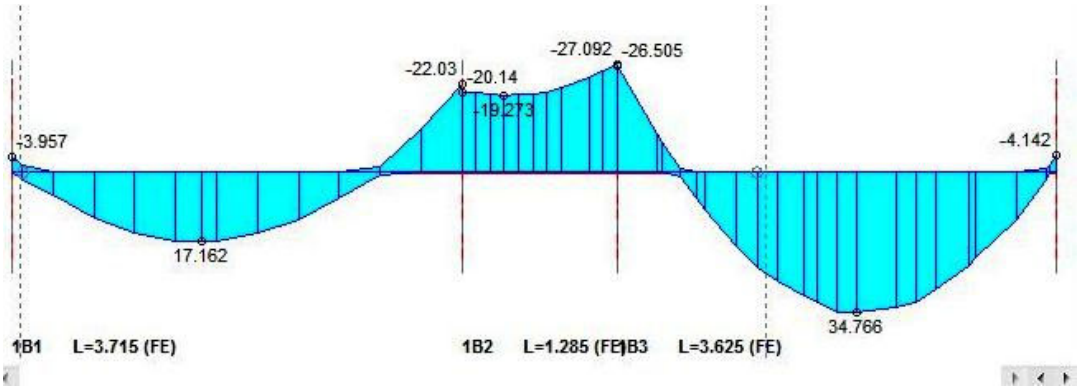


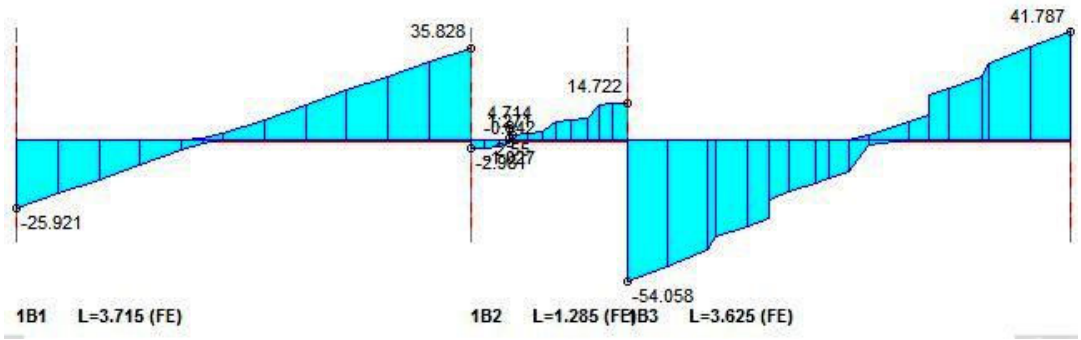
Group	Axis	Storey	Part	Beam	Qty	Design	U.Ratio	Print	Sheet	Beams
	1	1	1	3	1	✔	0.77	☑		1B1 - 1B2 - 1B3
	1	2	1	3	1	✔	0.31	☑		2B1 - 2B2 - 2B3
	1a	1	1	2	1	✔	0.86	☑		1B25 - 1B7
	1a	2	1	1	1	✔	0.46	☑		2B7
	2	1	1	1	1	✔	0.73	☑		1B16
	2	2	1	1	1	✔	0.18	☑		2B16
	3	1	1	1	1	✔	0.62	☑		1B20
	3	2	1	2	1	✔	0.18	☑		2B25 - 2B20
	4	1	1	2	1	✔	0.93	☑		1B21 - 1B8
	4	2	1	2	1	✔	0.38	☑		2B21 - 2B8
	5	2	1	1	1	✔	0.28	☑		2B29
	6	1	1	1	1	✔	0.91	☑		1B23
	6	2	1	2	1	✔	0.53	☑		2B28 - 2B23
	7	1	1	1	1	✔	0.55	☑		1B9
	7	2	1	1	1	✔	0.15	☑		2B9
	8	2	1	1	1	✔	0.28	☑		2B31
	A	2	1	2	1	✔	0.2	☑		2B26 - 2B27
	B	1	1	2	1	✔	0.53	☑		1B17 - 1B18
	B	2	1	2	1	✔	0.21	☑		2B17 - 2B18
	B1	1	1	2	1	✔	0.88	☑		1B24 - 1B22
	B1	2	1	2	1	✔	0.45	☑		2B24 - 2B22
	C	1	1	3	1	✔	0.94	☑		1B14 - 1B15 - 1B26
	C	2	1	2	1	✔	0.18	☑		2B14 - 2B15
	D	1	1	1	1	✔	0.86	☑		1B13
	D	2	1	1	1	✔	0.21	☑		2B13
	D1	1	1	3	1	✔	0.95	☑		1B10 - 1B11 - 1B12

To view more details, let's say for Axis 1 beams (Orion's gridlines), click on it twice, and the dialog box below comes up.

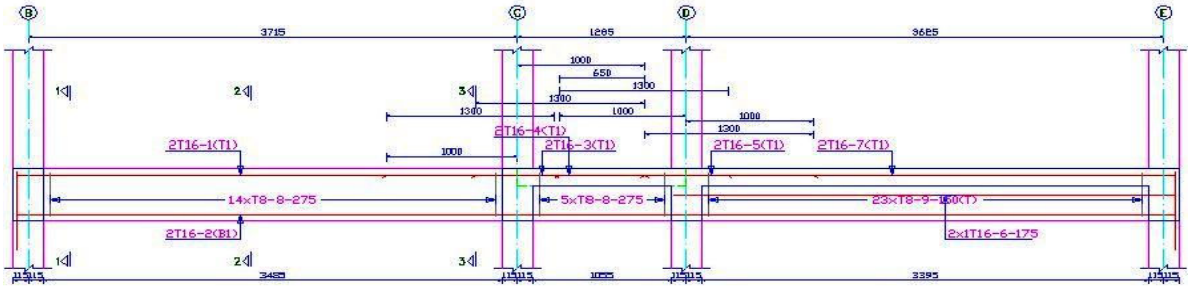


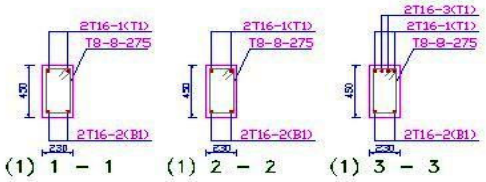
You can view the bending moment diagram, and the detailing of the member at the top left corner by clicking on '**diagrams**'. The typical bending moment and shear force diagram (design envelope) for Axis A beams on our GA is shown below;





If the member is not failing, and you are satisfied with the reinforcements provided, Click on **Detail Drawing** to see the reinforcement sketches.



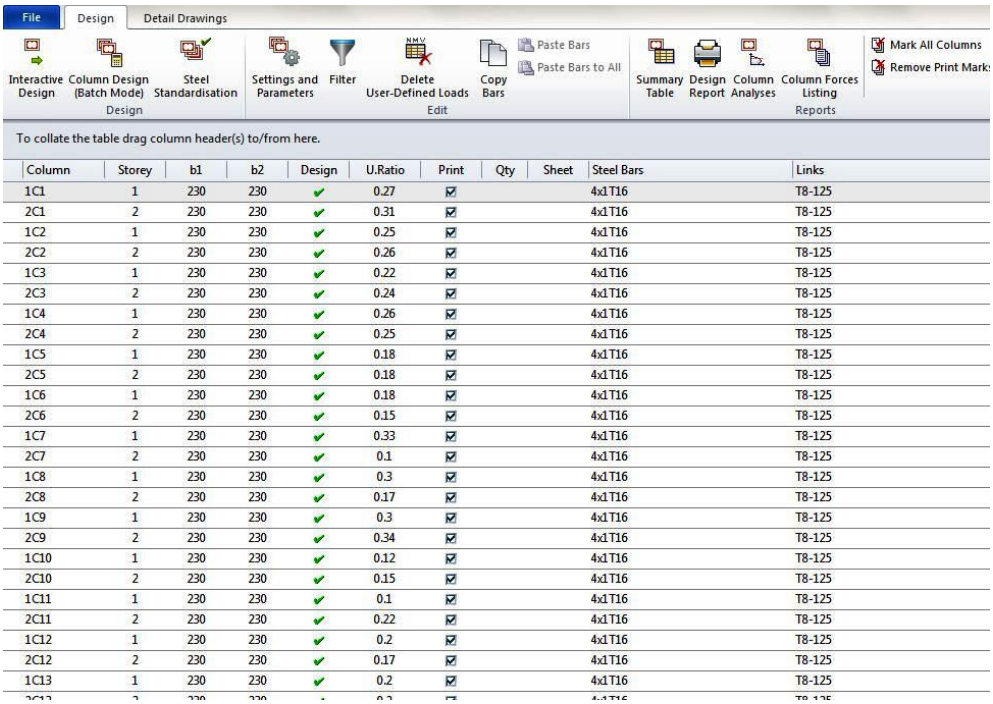


For the entire floor beam or for the entire structure, go back to the main project design and detailing sketches, by closing the Axis 1 Reinforcement Dialog box. Now go to **Detail Drawings**, and Select **Automatic Detail Drawing All Axis**. Select the storey beams you want to be detailed. The reinforcement details are automatically detailed. For a very large building, and when a large number of beams are having problem, you may go to the Batch Design Mode, to automatically re-select the bars provided to take care of the requirements. To view the calculation sheet, click on **Design Report.**

3.3.16 Column Design

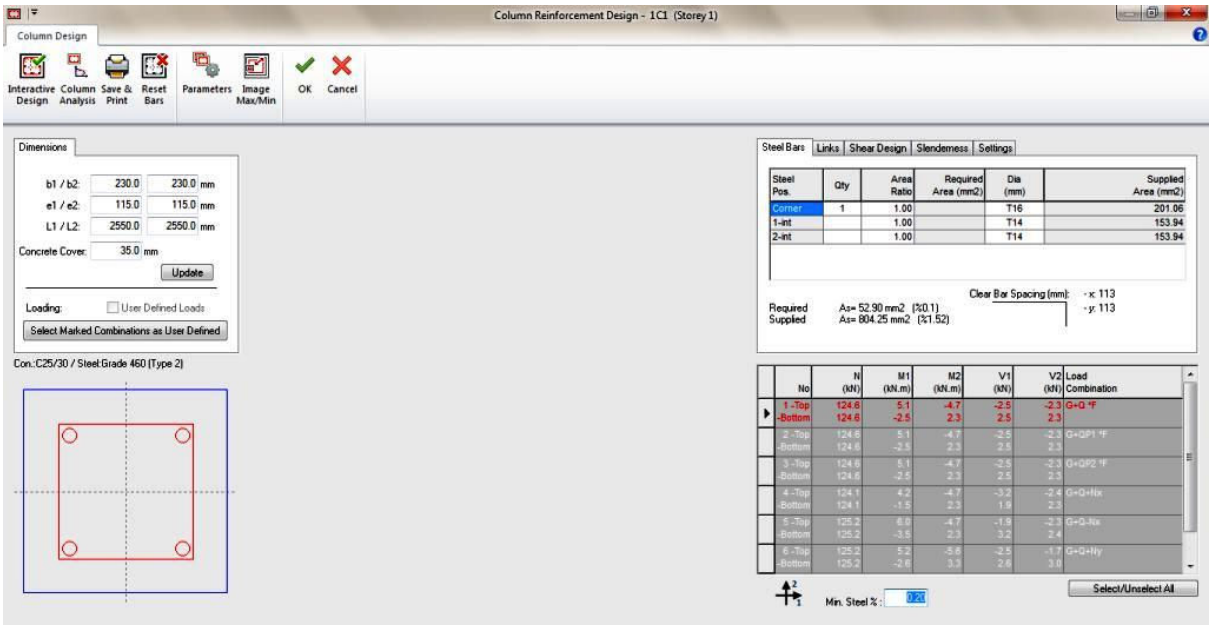
Go back to **Run**, and Select **Beam Section Design and Detailing**, and select **Storey Beams**

The **Column Reinforcement Design** page comes up. Green ticks shows beams where all major requirements are satisfied.



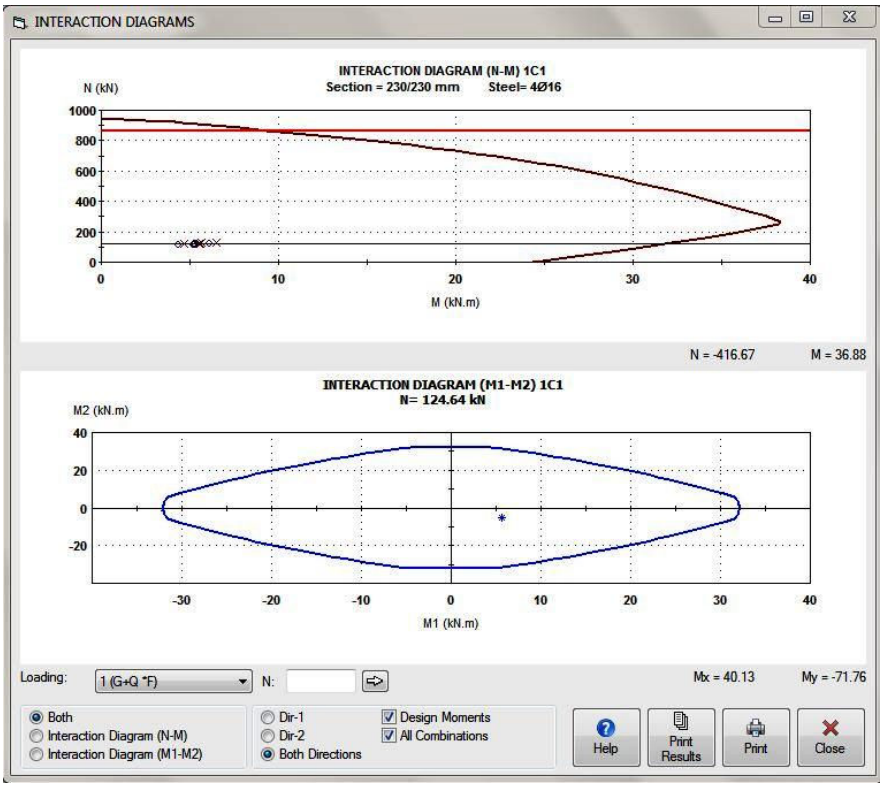
Column	Storey	b1	b2	Design	U.Ratio	Print	Qty	Sheet	Steel Bars	Links
1C1	1	230	230	✔	0.27	☑			4x1T16	T8-125
2C1	2	230	230	✔	0.31	☑			4x1T16	T8-125
1C2	1	230	230	✔	0.25	☑			4x1T16	T8-125
2C2	2	230	230	✔	0.26	☑			4x1T16	T8-125
1C3	1	230	230	✔	0.22	☑			4x1T16	T8-125
2C3	2	230	230	✔	0.24	☑			4x1T16	T8-125
1C4	1	230	230	✔	0.26	☑			4x1T16	T8-125
2C4	2	230	230	✔	0.25	☑			4x1T16	T8-125
1C5	1	230	230	✔	0.18	☑			4x1T16	T8-125
2C5	2	230	230	✔	0.18	☑			4x1T16	T8-125
1C6	1	230	230	✔	0.18	☑			4x1T16	T8-125
2C6	2	230	230	✔	0.15	☑			4x1T16	T8-125
1C7	1	230	230	✔	0.33	☑			4x1T16	T8-125
2C7	2	230	230	✔	0.1	☑			4x1T16	T8-125
1C8	1	230	230	✔	0.3	☑			4x1T16	T8-125
2C8	2	230	230	✔	0.17	☑			4x1T16	T8-125
1C9	1	230	230	✔	0.3	☑			4x1T16	T8-125
2C9	2	230	230	✔	0.34	☑			4x1T16	T8-125
1C10	1	230	230	✔	0.12	☑			4x1T16	T8-125
2C10	2	230	230	✔	0.15	☑			4x1T16	T8-125
1C11	1	230	230	✔	0.1	☑			4x1T16	T8-125
2C11	2	230	230	✔	0.22	☑			4x1T16	T8-125
1C12	1	230	230	✔	0.2	☑			4x1T16	T8-125
2C12	2	230	230	✔	0.17	☑			4x1T16	T8-125
1C13	1	230	230	✔	0.2	☑			4x1T16	T8-125

To view a specific member design detail, double click on the member, Say 1C1, and the dialog box below comes up.



When the section size is not adequate, you can increase it and so on by the bottom left hand side. By the bottom right corner, the Load Combination highlighted in red is the critical one, while at the top right corner, you can reselect the reinforcement and so on. If a column is failing, by clicking the **Interactive Design Tab**, Orion Automatically selects the appropriate reinforcement. In a case where you will need to increase the section size, you will be warned also.

The **Column Analysis** Tab enables you to see the Moment and Axial Force Interaction Diagram. You can also print these details if you so desire.



When you click on **Save and Print**, the following Calculation Sheet is generated. To print the full Analysis Report page for all the columns, return to the main Column Section Reinforcement Page.

1C1 Storey: 1 (Con.:C25/30 / Steel:Grade 460 (Type 2))

Critical Combination:1 - (G+Q *F)

		Min	Design
N (kN)	124.6	-	124.6
M1 (kN.m)	5.1	2.5	5.6
M2 (kN.m)	-4.7	-2.5	-5.2
N-max (kN)	0.0		

Concrete Cover = 35.0 mm

Neutral Axis:
Y = 111.3 mm
Alpha = 20.83 °

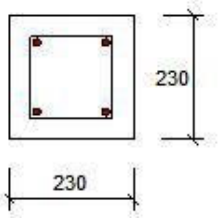
Short Column...
Lambda1/Lim1 = 22.7 < 82.6
Lambda2/Lim2 = 22.7 < 82.7
M-add(1/2)= 0.0 / 0.0 kN.m

V-Ed (1/2)= 3.2 / 3.0 kN
v-Ed = 0.08 / 0.07 N/mm2
v-Rdc = 1.05 / 1.05 N/mm2
v-RdMax = 4.62 / 4.62 N/mm2

As (Required): %0.20 (min) 105.80 mm2
As (Provided): %1.52 804.25 mm2

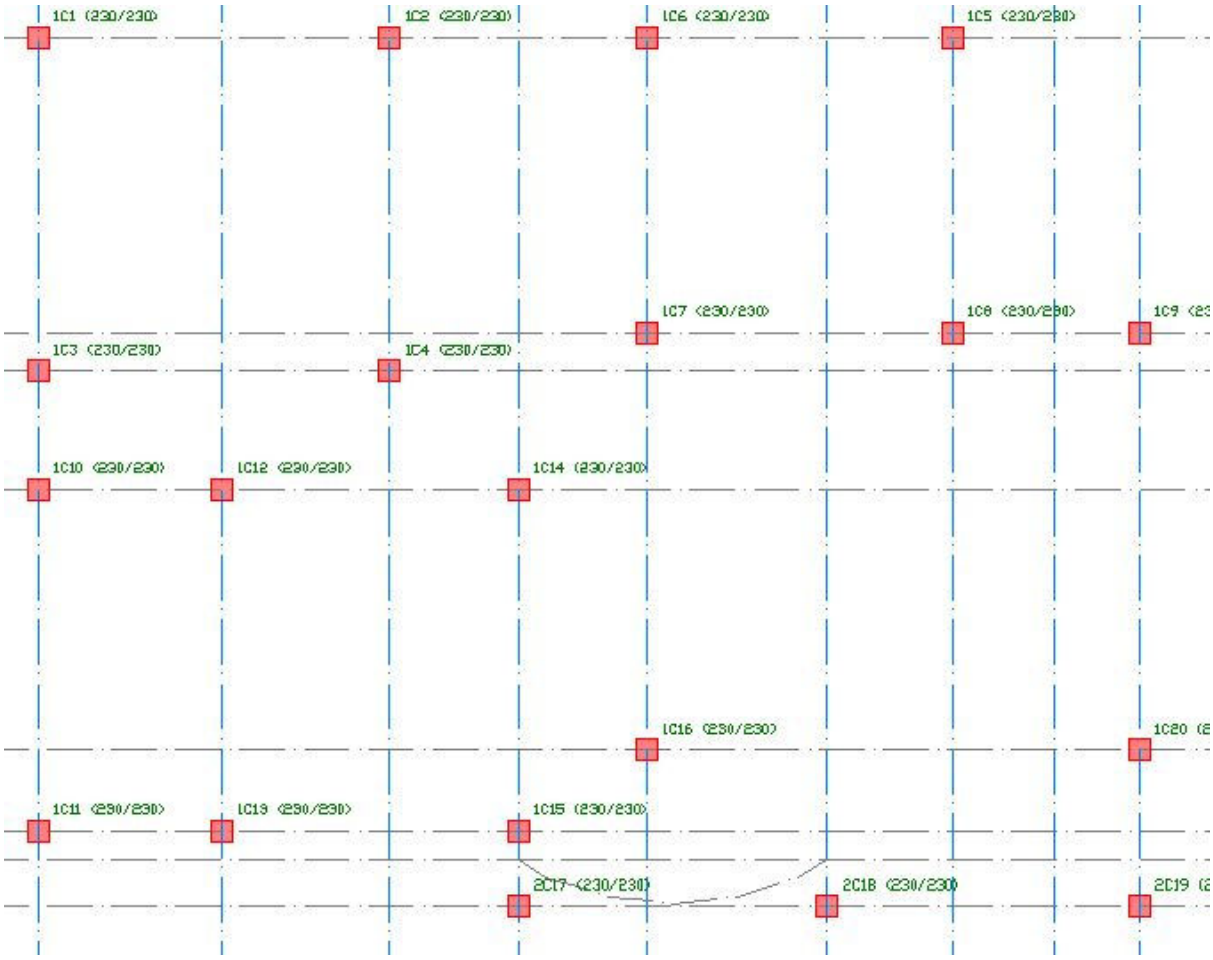
4T16

Links = T8-125



3.3.17 Foundation Design

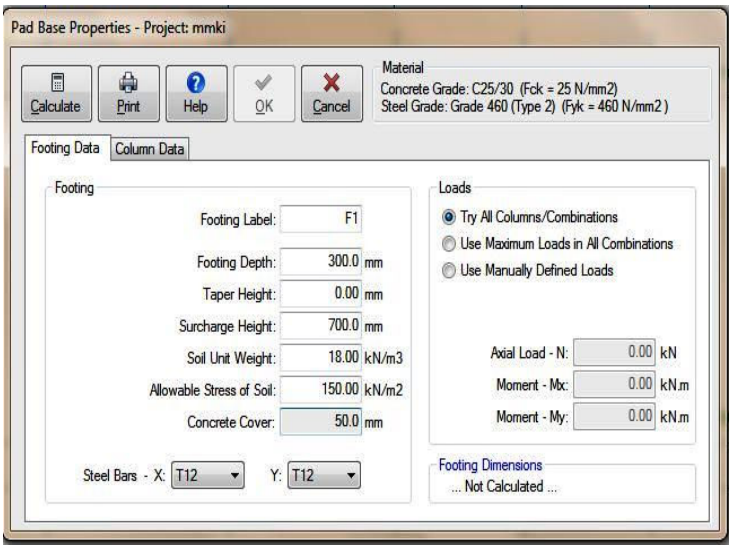
To design the foundations, go to **St00** on the plan view



To design the footing for Column 1C1, click on the column, and right click on it.

Select **Insert Pad Base**

The dialog box below comes up;

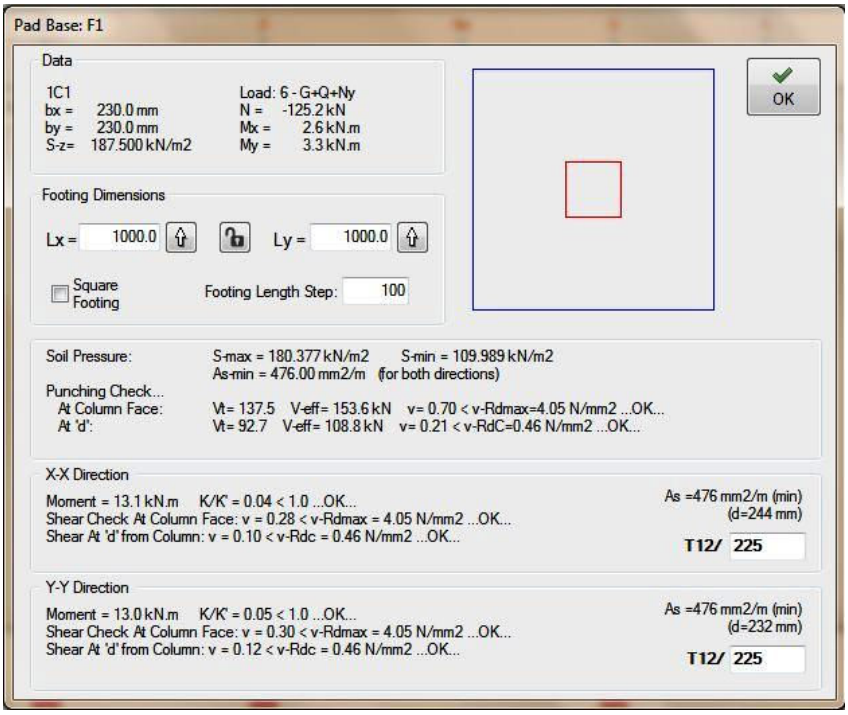


Set the footing depth to a trial depth of 300mm

Select the surcharge height as suitable. In this case, we are taking 700mm

Now, let us try and use Y12mm bars

Click on **Calculate** after adjusting the settings to see the design calculation.



With a bearing capacity of 150 KN/m^2, a square footing of 1000 x 1000mm has been provided.

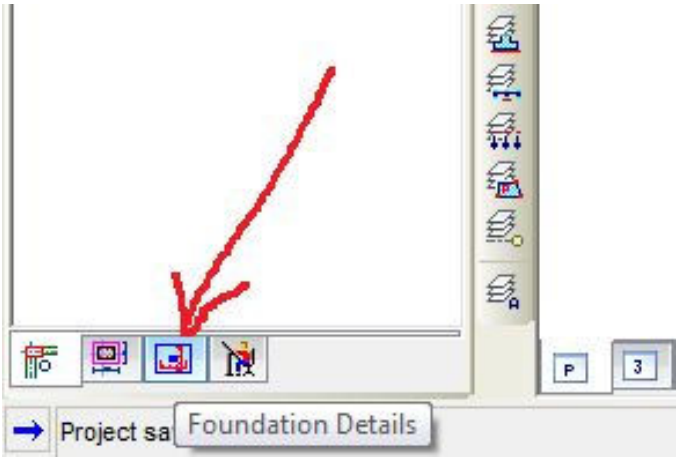
The resultant soil pressure has also been shown with a minimum of 109.989 KN/m^2 and a maximum of 180.377 KN/m^2. With a footing depth of 300mm, all shear requirements at the column face and at d from the column face were all satisfactory. The provided reinforcement was Y12@225mm c/c in both directions. Since this is satisfactory, you click **OK**. Note that you can adjust footing dimensions as you wish, but you cannot go below the minimum required.

Using this procedure you can design all the footings one after another.

3.3.17.1 Combined Footings

Where two columns overlap each other, a combined footing may be the best solution. To do this, click on the two columns involved, and group their foundations together. Then insert Pad Bas as usual. This has been done on columns 1C15 and 2C17. Where the columns are spaced too wide from each other, you can consider designing the base as a strip footing.

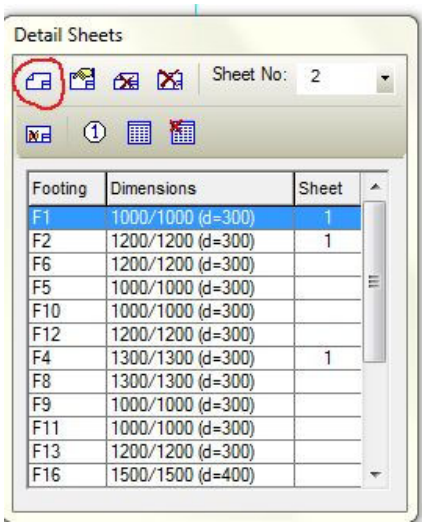
3.3.17.2 Foundation Detailing

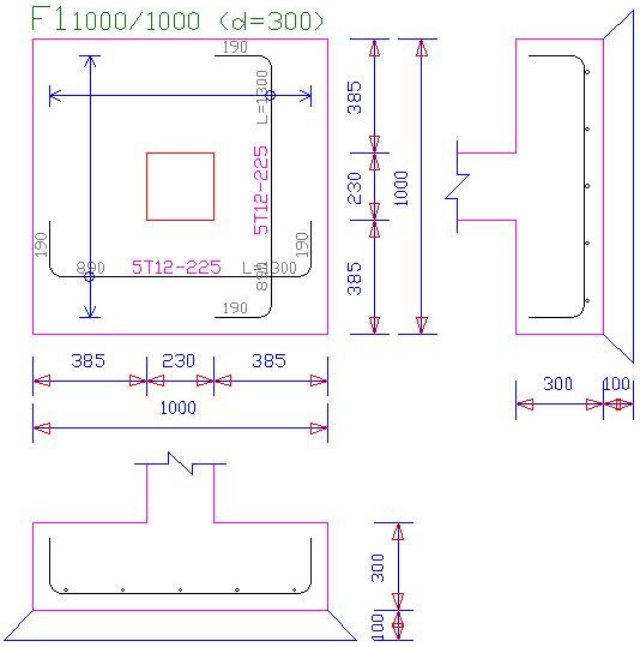


For detailing purposes, it is good adopt a group design. For instance, instead of having footings of variable dimensions all through, you can group the designs for ease in construction, and possible reuse of materials where applicable. For example, the calculation gave the dimension of column base **1C2** as 1200mm x 1200mm, and base **1C6** gave dimensions 1100mm x 1100mm. Instead of having footings of different variable dimensions, I will simply take footing **1C6** to 1200mm x 1200mm. So each group is now assigned a type – Say footing Type 1, Footing Type 2, etc.

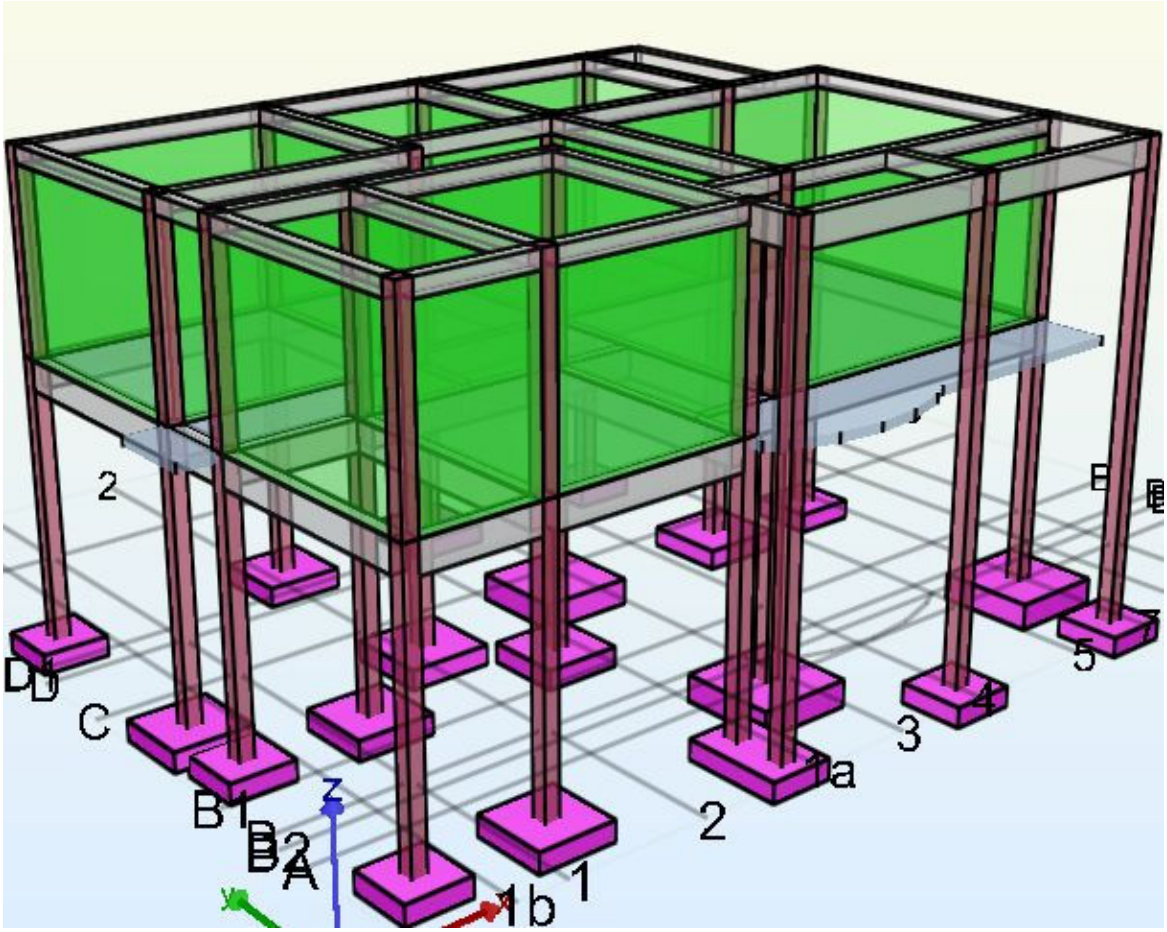
To view the detailing sheet, you can export the plan view just as you export slab details (DXF Export). To view the reinforcement details, click on Foundation details at the bottom left corner of the screen GUI. The Detail Sheets Dialog box comes up. Click on New Detail Sheet, specify the paper size, and click on the GUI for the sheet size to be displayed. Now, if you want the reinforcement details of F1, click on F1, and click on the GUI for the details to be displayed.

The reinforcement detail of F1 is shown below;





The final MODEL of the building in 3D is shown below.



This is the end of our modelling on CSC Orion. I strongly believe that you can now play with it and manipulate it appropriately from now on. Individual results of the structural members will now be displayed as we progress. I strongly recommend that you go through the necessary technical manuals to be more acquainted with Orion modelling and design guides.

3.4 Modelling on Staad.Pro V8*i*

STAAD.Pro is a comprehensive and integrated finite element analysis and design solution, including a state-of-the-art user interface, visualization tools, and international design codes. It is capable of analyzing any structure exposed to static loading, a dynamic response, soil-structure interaction, wind, earthquake, and moving loads. STAAD.Pro is the premier FEM analysis and design tool for any type of project including towers, culverts, plants, bridges, stadiums, and marine structures.

The basic three activities carried out to achieve that goal in Staad are;

a. model generation
b. calculations to obtain the analytical results
c. result verification

All these processes are all facilitated by tools contained in the program's graphical environment.

It is assumed that by now, you must have purchased and successfully installed Staad Pro on your computer.

3.4.1 Creating a Project

Launch the StaadPro software for the start up page to come up.

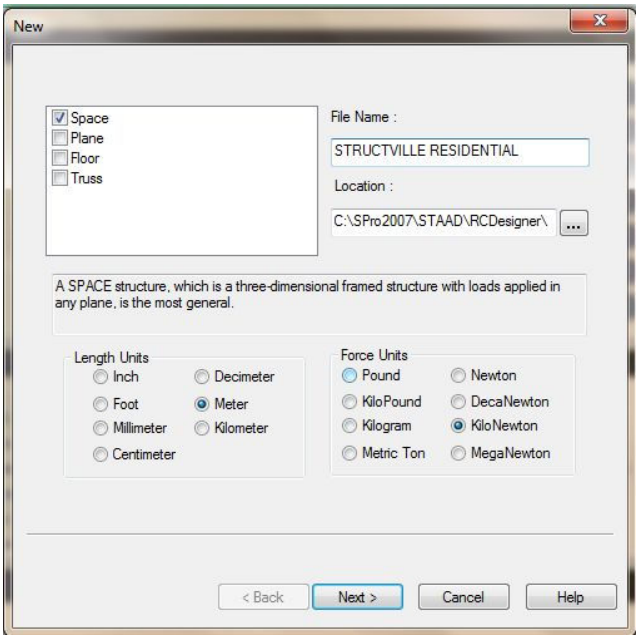
Click on **New Project** to start up a new project

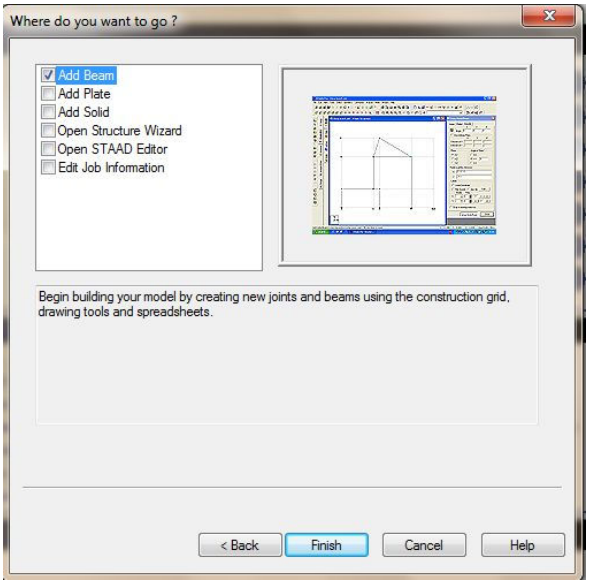
The new project information Dialog Box comes up

Since we are analysing the full building, select the structure as **SPACE** Structure, which has the three dimensions fully specified. Enter the file name as you wish (I entered Structville Residential). Make sure that the Length and Force Units are in **Meter** and **KiloNewton** respectively (see the pictures below).

Click **Next**

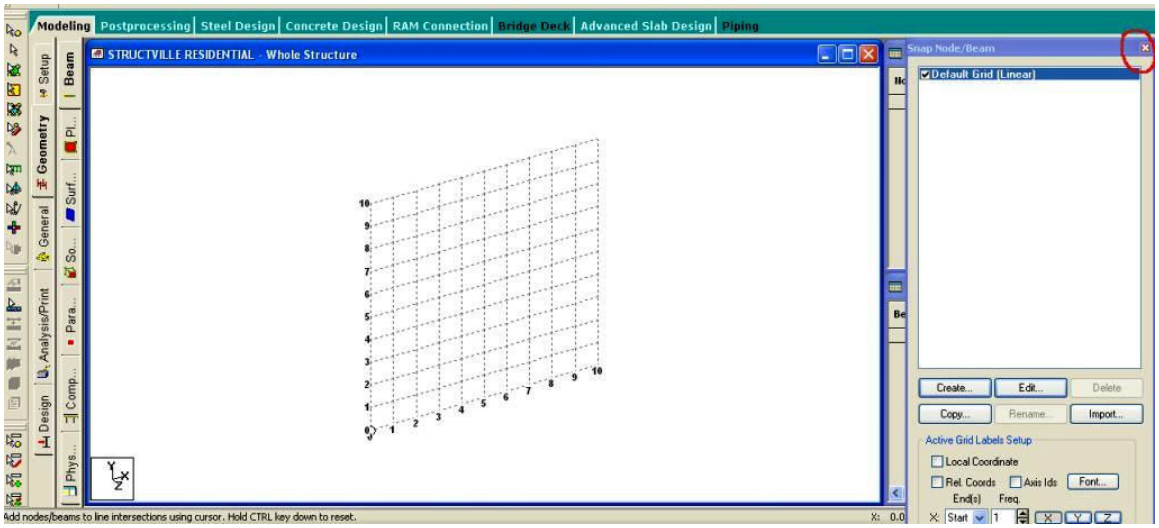
The next dialog box asks you **'where you want to go?'**. Tick '**Add Beam'** and click '**Finish**'



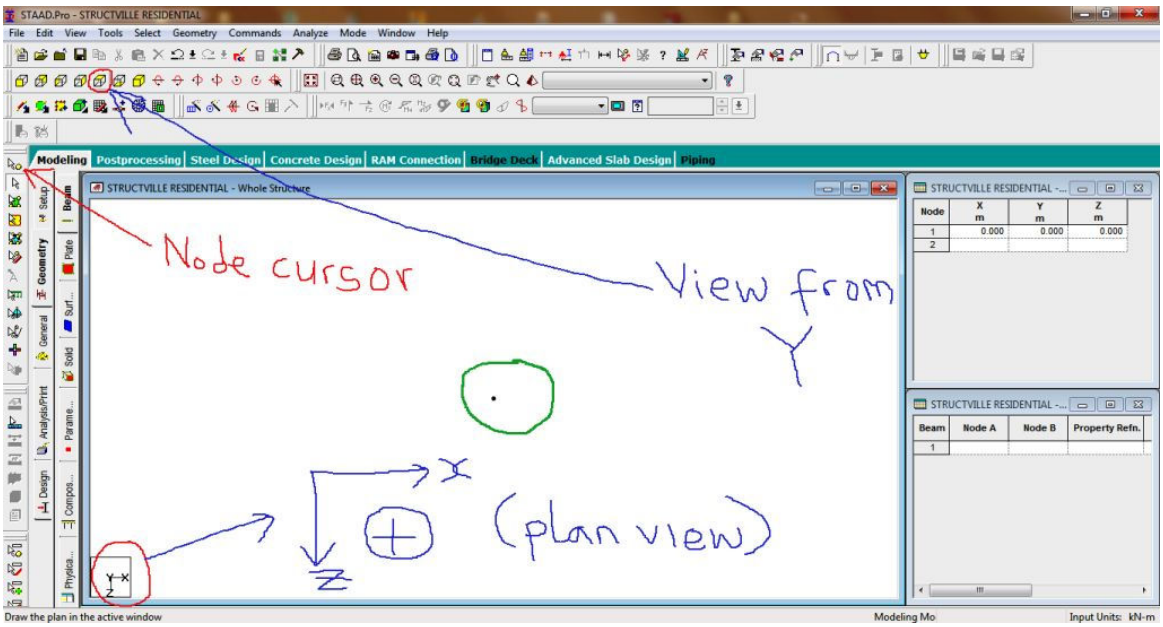


3.4.2 Creating Nodes

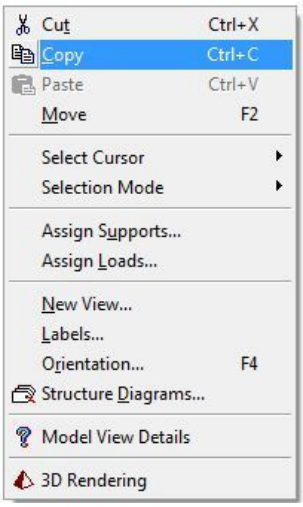
You are now ushered into the graphical user interface (GUI). On the main window, you will see some default gridlines which you can work with. But we are not going to make use of that. So, at the data window of the screen (Right Hand Side of the screen), close the **Default Grid (Linear)** option. This leaves the main window blank, and creates a space for you to input coordinates in the x, y, and z axes.

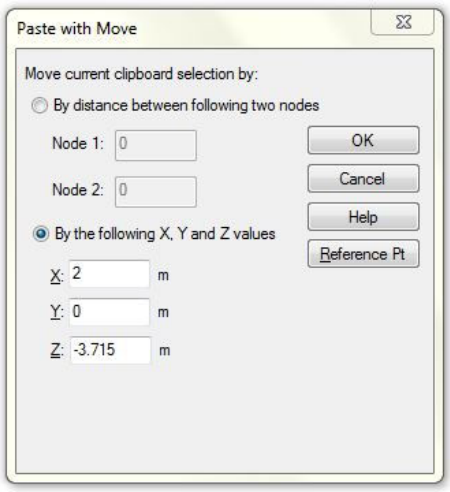


At the data window, enter '**0**' on the *x* coordinate, and press the **ENTER** button. This practically creates the first node (which I can call the benchmark node) with coordinates (0,0,0). This node is circled in green in the picture below. Now, change your view to '**VIEW FROM Y'** (i.e. we are viewing from the top) so that we can get the plan view. This plan view has the *x* direction horizontally, and *z* direction vertically. The sign convention is as shown in the picture below **(z is positive downwards, while x is positive towards the right).** It is very convenient to create our floor plan from this view. After you have done this, now change your cursor to **Node Cursor**. In StaadPro, each element has its own cursor in this order on the screen: **Node Cursor**, **Beam Cursor**, **Plates cursor**, **Solids Cursor**, and **Geometry Cursor** (for selecting all the elements).

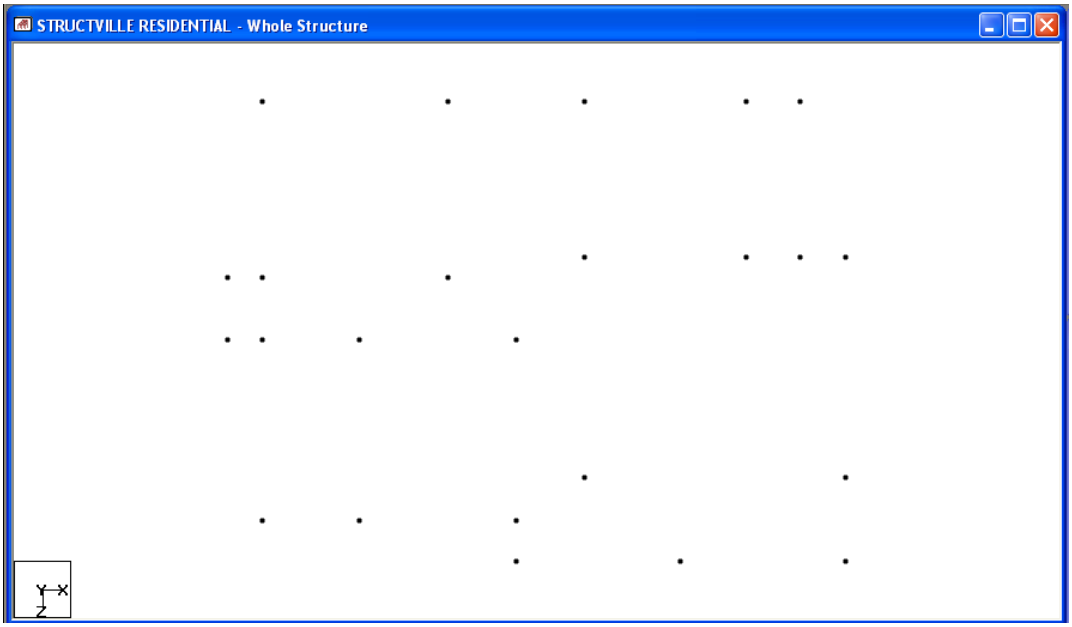


At this point, the manual GA that you have created should be by your side. Now with the **Nodes Cursor**, highlight the benchmark node we have created. You can do this faster by just 'rubber banding' it, than trying to click on it. Now having highlighted the node, right click on it, click on copy, right click on it again, select '**Paste Nodes'**. The dialog box by the right hand side below comes up. For instance, let us assume that the benchmark node represents point F1 on our GA. If we want to create node D2, we have to enter 2 (positive because we are going towards the right) on the x-direction, and -3.715 (negative because we are going upwards) on the z-direction. After which you click **OK**. Also, for us to create Node F2, we simply input 2.0 in the x-direction and click ok. Also, let us assume that we have created Nodes C1 and A1, we can create Nodes A3 and C3 by copying the nodes and pasting them by 3.825m by the right on the x-direction. Note that you only need the 'benchmark node' to take off. After creating additional nodes, you can copy from anywhere and paste to anywhere that is most convenient for you.



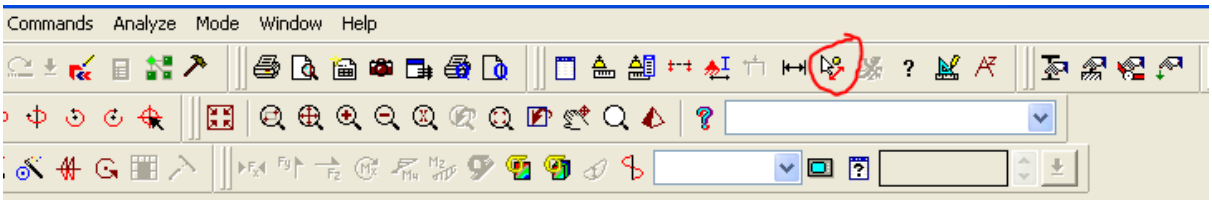


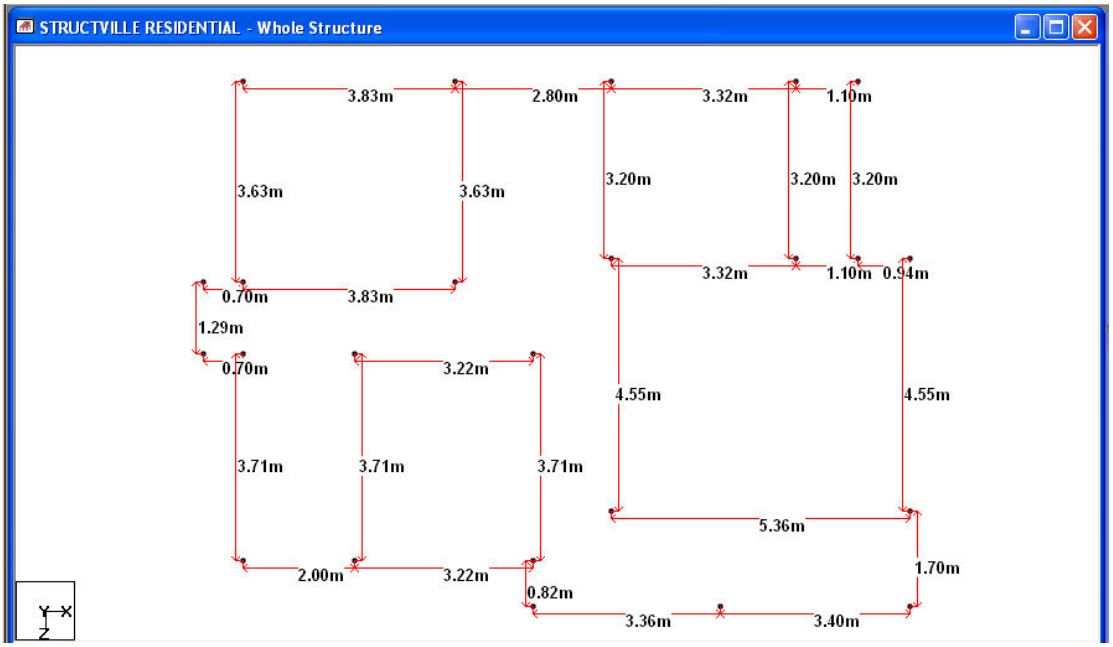
The final result is something like this below for the building we intend to analyse. Make sure that your distances are accurately specified. This will represent our **FIRST FLOOR ARRANGEMENT.**



You can verify the accuracy of what you have done by checking the nodal distances. This can be achieved by activating the '**Display Node to Node Distance Command**' (circled below), and then clicking your cursor at a node, and dragging it to another node and clicking on it to see the nodal distance. Note that Staad displays dimensions to two decimal places. But the real input values are used for analysis.

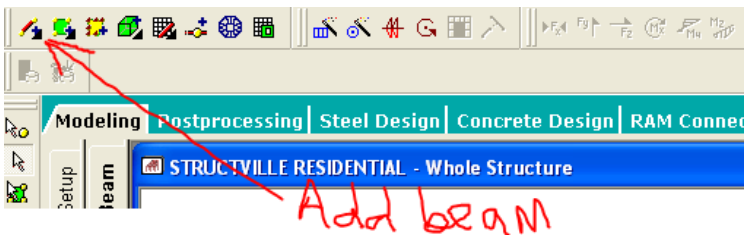
You can later remove the distance display by clicking on the icon by the '**Remove Node to Node distance display'** by the right hand side of the icon circled below.



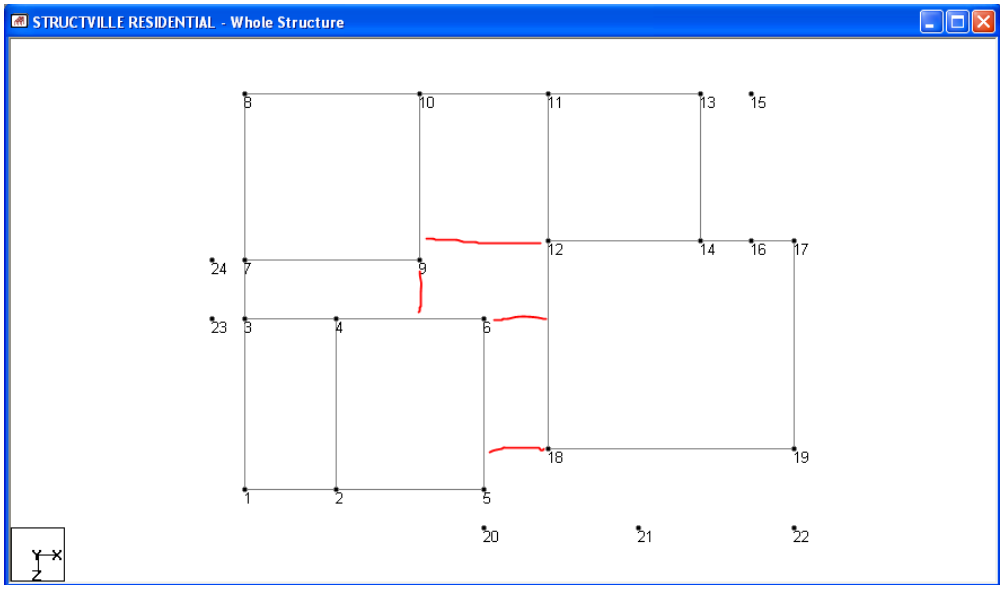


3.4.3 Adding Beams

There are two ways of doing this. Firstly, we can activate the '**Add Beams'** command (see picture below), and connect the beams to the nodes one after another. This is advantageous for small models, and where we have inclined members. Alternatively, we can **highlight all the nodes**, Click on **Geometry**, and then select '**Connect Beams Along'**. We will then specify x and z axis in each case. Staad will tell us the number of beams created in each direction for each case. This is very beneficial for large and fairly regular models. Now, note that Staad will automatically add beams as requested, but for this our case, we will need to delete some beams where they will not be needed, especially around the cantilever areas. I will suggest let us draw one by one. We will definitely use the other alternative later.



So activate the '**Add Beam'** command, click on one node, and drag it to another node as indicated on our GA. Do not skip any node please. In *Orion*, you can drag a beam along axis A from point 1 to 8, and Orion will independently break off the beams at the column points. But on Staad, once you drag the beam from point 1 to 8 on axis A, it will be seen as one member. So drag and click from node to node. See what we came up with below.



Now note that I have I have input **'Node Labels'** in the picture above. To do that, right click on the main window, select '**Labels**', and check the Node Labels option. You can turn it off later.

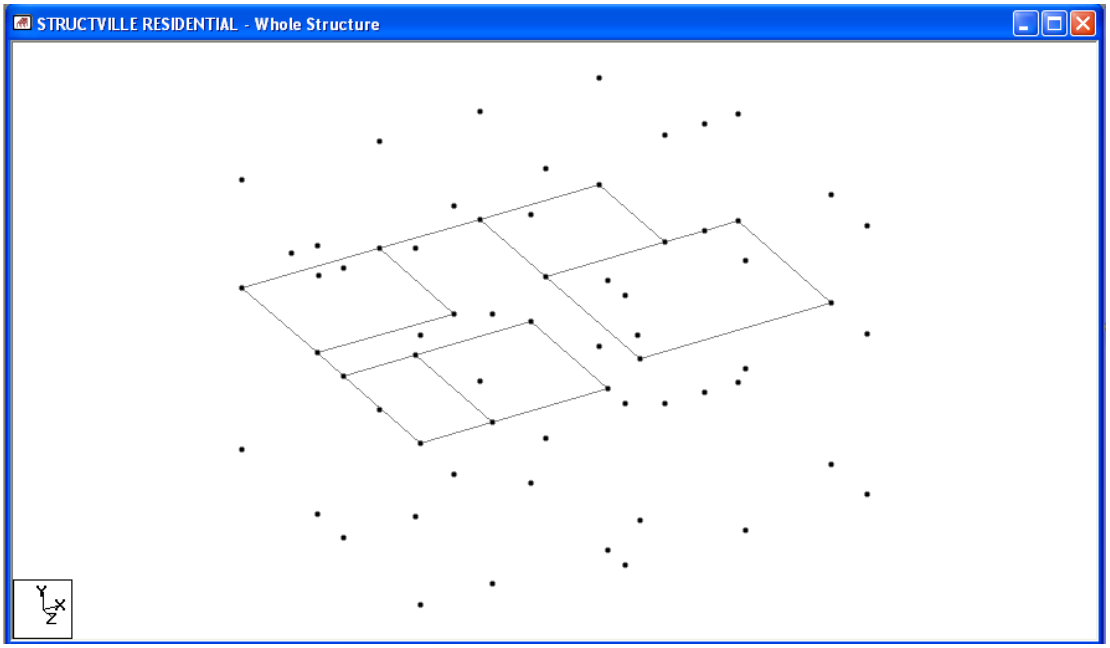
Did you also realise that there some floor beams that I did not create? I highlighted them in red in the picture above. That was simply becase we did not create the nodes intially. Also, realise that nodes for the curved cantilever at the front (PANEL 5) have not been created too. You may ask, 'Why dont we create all these nodes at once?' My reason is simple, if I create so many nodes, I may get confused while connecting my beams, also if I want to create the nodes for my column (given that I have also have primary and secindary beams), it may be a disaster because I will confused with many nodes up and down staring at me. But if I keep it simple and go step by step, I will see everything very clear. Also, you can verify that the beam nodes that we have not created yet are where we have the interaction of primary and secondary beams, and there are no columns at those points. We can take advantage of this now, and create the nodes for our columns. So that we wont have to go back and start deleting later.

3.4.4 Creating the columns

As we previously modelled, our first floor columns are 3m long, and our groundfloor columns 4.5m long. We will also stick to that in this model. I wish to delete the cantilever slab nodes at this point. They are useless for now. We will also follow the 'copy and paste' approach in the creation of the nodes for the column. In this case, we copy and paste to the Y-direction (global vertical axis). To achieve this, hange your view to +Z direction (see the picture below). Note that the copy and paste command will be carried out irrespective of your view. But I just want you to have a view where you will be seeing everything that you ae doing.

Change to Nodes Cursor and rubber band (highlight) all the nodes that we have created at that level. Now right click and copy, then paste 3.0m in the Y-direction (this creates the nodes at the upper level) Copy and paste the same nodes -4.5m in the Y-direction (this creates the nodes at the lower level)

Click on 3D view to see something like this below;

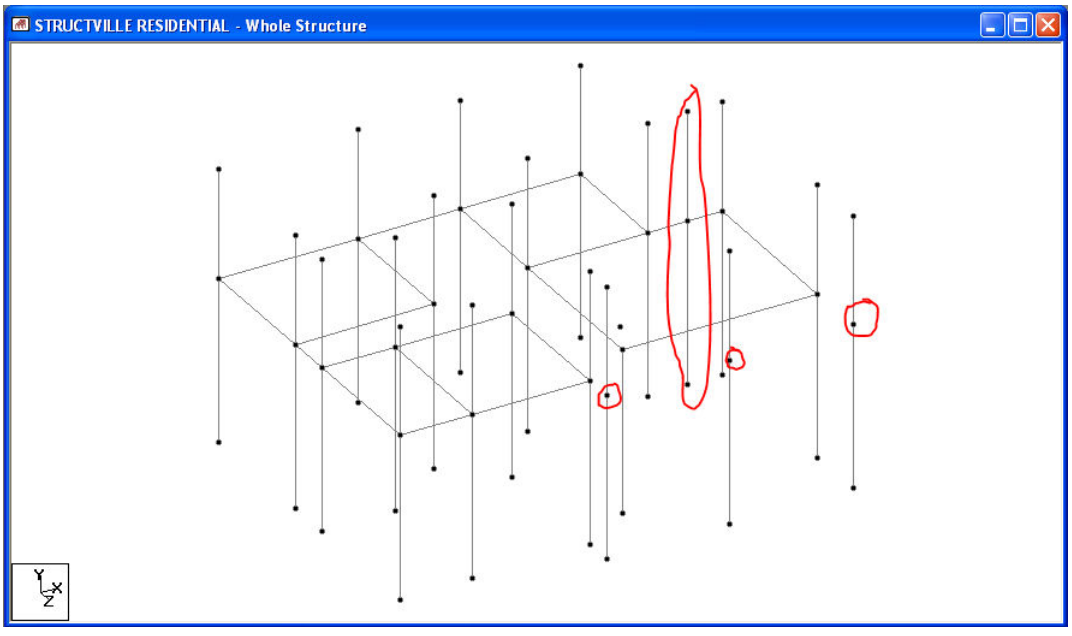


This may appear confusing and that is why you should always keep it simple. But do not worry about connecting the columns by yourself. The columns will be automatically created using the second alternative.

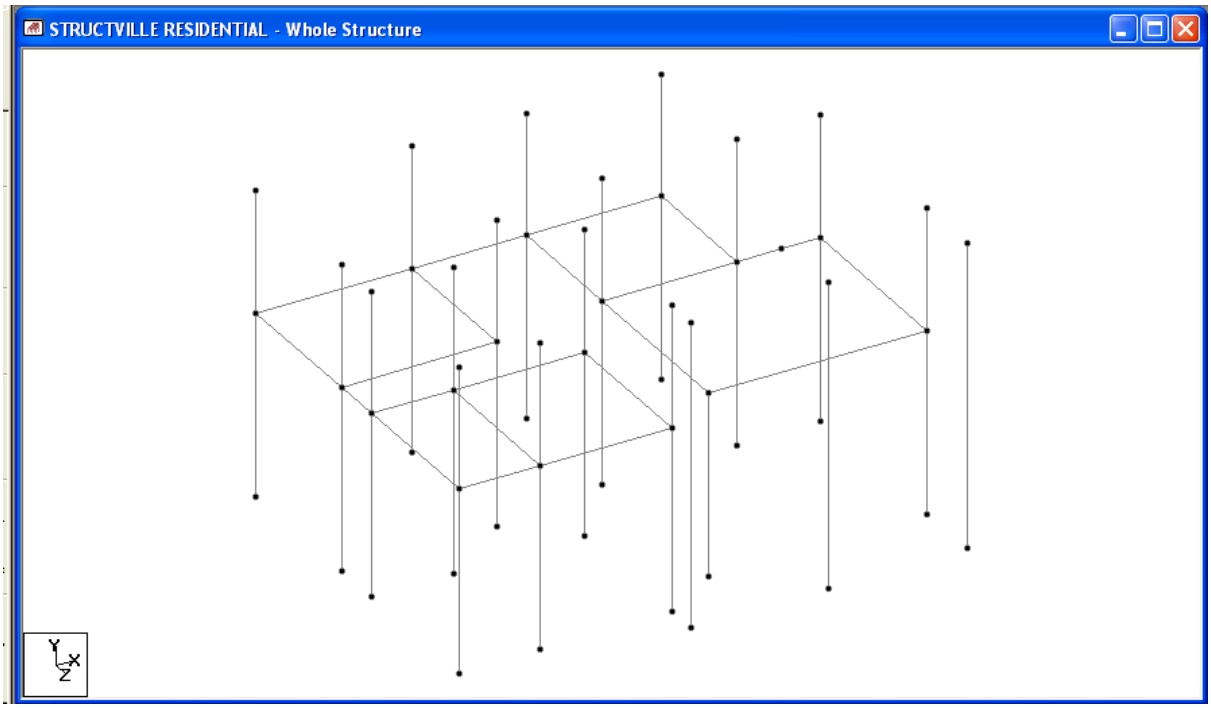
Now, rubber band all the nodes in the model. (Note that we have deleted the cantilever nodes we created).

Now go to **Geometry**, and **'connect all the beams along the Y-direction'**.

The result will be something like this below.

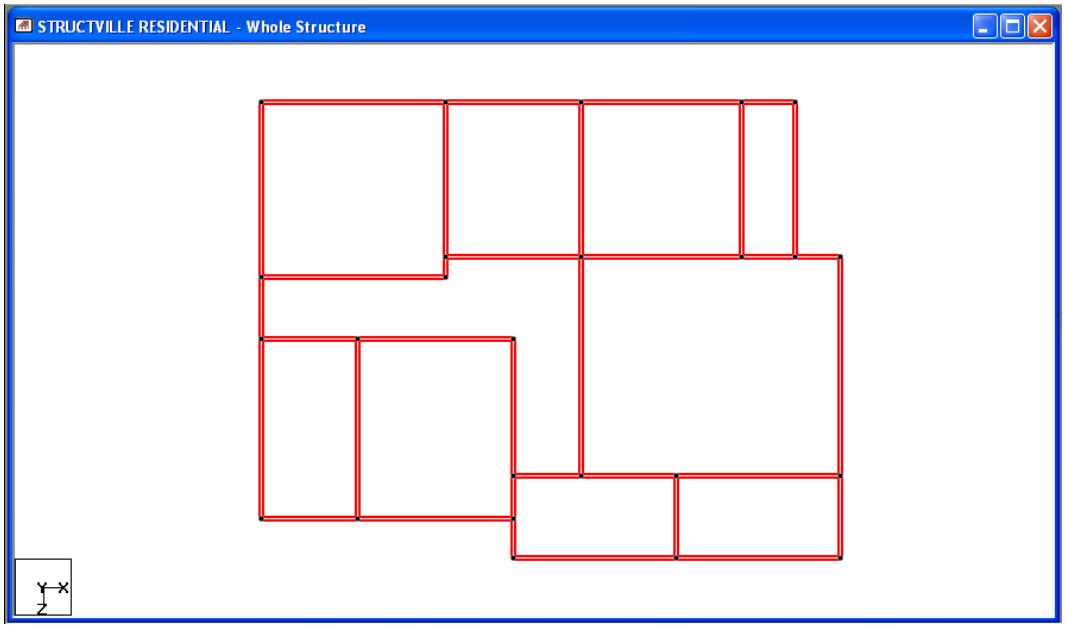


As you can see, there are still some unwanted columns (circled in red above) that are not in the building that we are modelling. Therefore, change your cursor to beam cursor and delete it. Also we want the front columns to be singular, just like we did at section 3.3.9.1. Therefore, also change your cursor to node cursor and delete all the nodes circled in red. This will also delete the columns, but the top most and bottom nodes will still be intact. So connect the nodes with a beam manually as usual. The final result of what we are talking about is shown below.



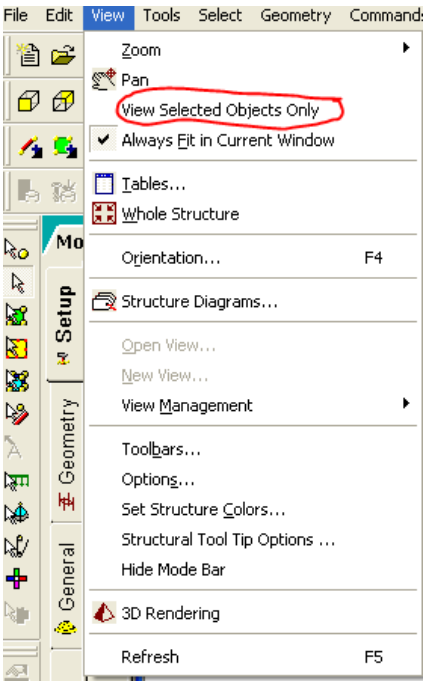
3.4.5 Creating roof beams

Change your view to +Z direction, and change to Nodes cursor. 'Rubber band' all the roof level nodes. Now connect all the beams along the Z and X axis. This creates the roof beams. Go back to 3D and see if it was successful. If there are beams that are not in existence, go back there and model them. The final arrangement of the roof beam must be as shown below.

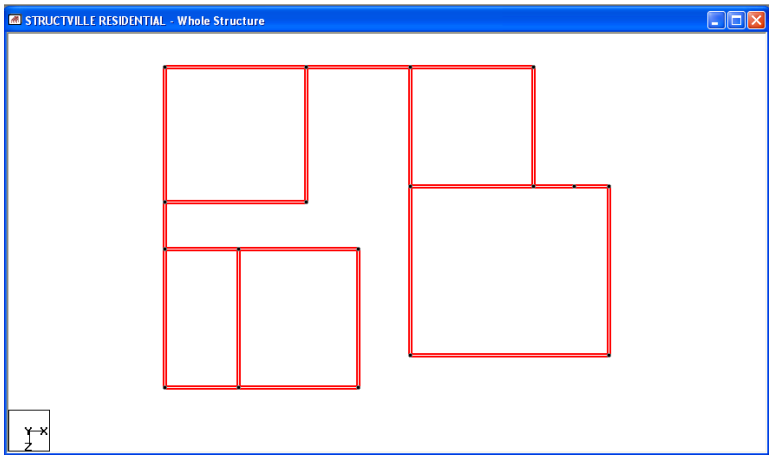


3.4.6 Completing the floor beam model

Go to +Z direction view, and change to Beam cursor, and rubber band (highlight) all the first floor level beams. Now go to **View**, and select '**View Selected Objects Only**'.

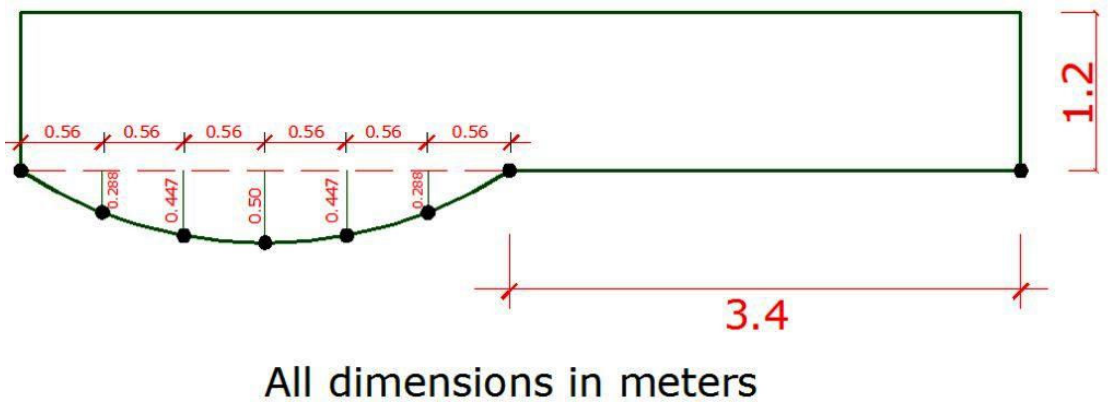


This is a very important feature in modelling in Staad Pro, as it enables someone to work freely on the level he is interested in, without complexity or interference from other elements. Now that you have only first floor level elements, go back to plan view (view from Y-direction). What we have should be the same as shown below.



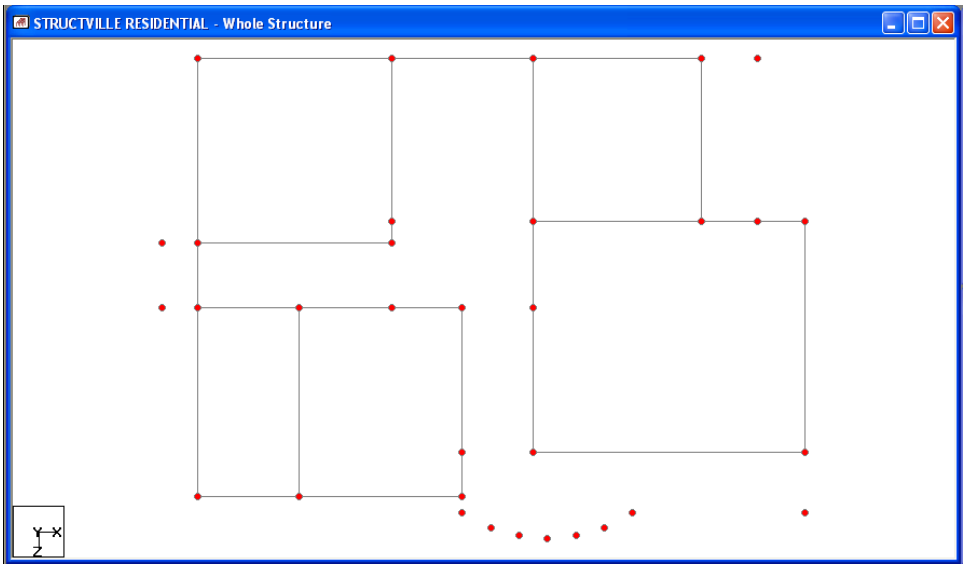
Now using the copy and paste procedure, create all the nodes where we have to insert the beams, and also create additional nodes for the cantilever sit-outs and balconies. I will give a better geometric description of the curved cantilever at the front, so that you will know how to insert your nodes to create a better curve.

I wish that we will have six segments on the curve, each separated by equal horizontal distance of 0.56m. The vertical distance of the nodes are indicated on the diagram (0.288, 0.447, and 0.50).

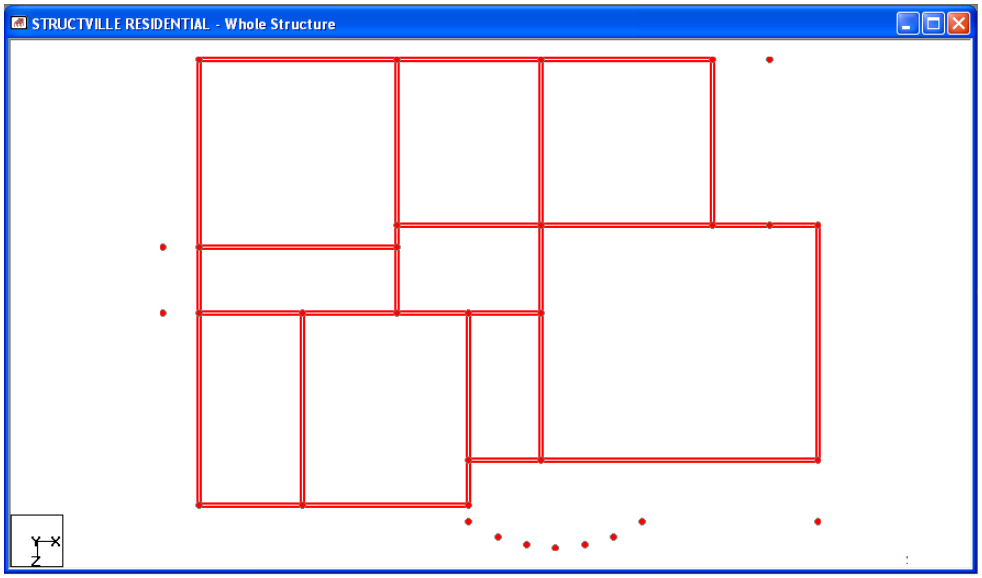


All dimensions in meters

Using any means to do the manipulation, copy and paste existing nodes to create the final diagram shown below.



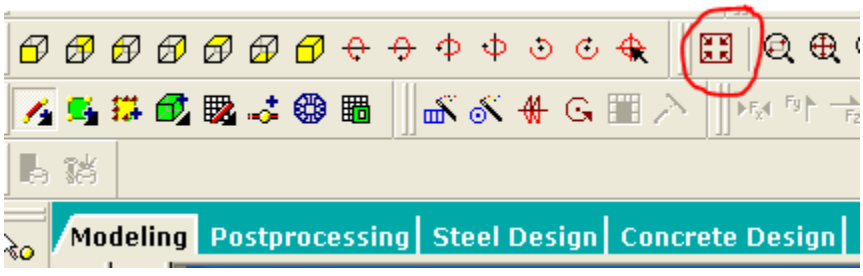
Now connect the remaining beams where they are necessary. See the final arrangement below.



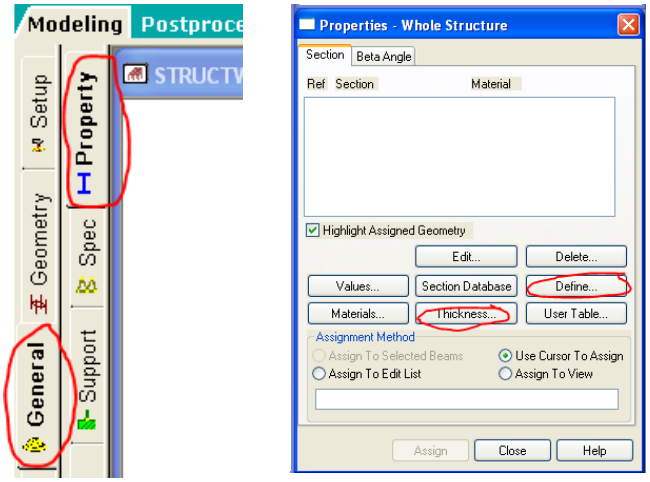
Before we go ahead and model our slabs, I will like to create the supports, assign section sizes, and load the beams first of all. I know this because once we insert the slabs, the beams will be broken into the sizes of the side of the plate they support. ***So I normally let meshing and loading of slab be the last thing I do while modelling on Staad.***

3.4.7 Creation of section properties

Go back to '**Display Whole Structure'** by clicking the icon circled in red below.



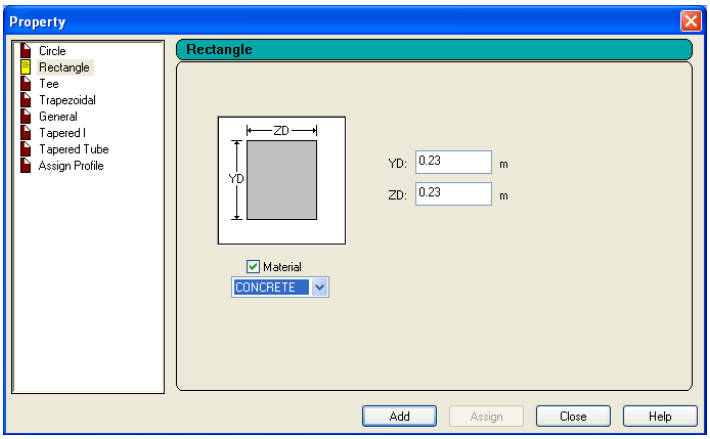
Go to '**General**', and go to the sub-page '**Property**'



From the '**Properties**' dialog box, click on '**Define**' to define the section properties. Select Rectangle, and define the dimensions of our columns as 230 x 230mm. Make sure that the material is concrete, and then you click **Add.**

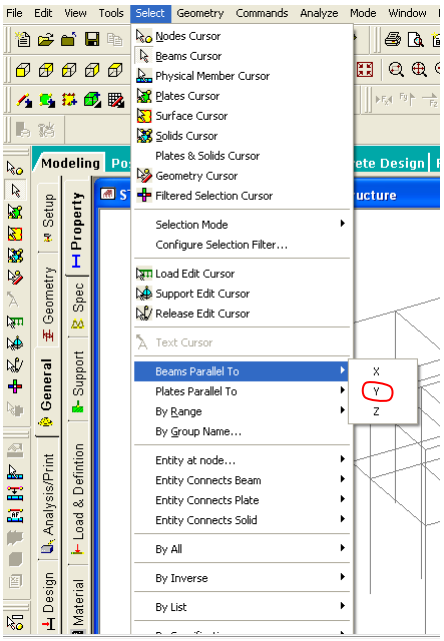
Define for our floor beams as 450 x 230mm, and define for our roof beams as 250 x 230mm. Click **Add** after defining each sections. After everything, click **Close.**

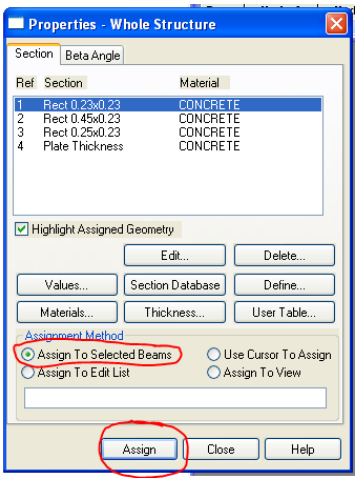
Go back to the '**Properties**' dialog box. Click on '**Thickness**', and specify the thickness of the plates as 150mm (0.15m). Click **Add**, and then close the '**Thickness**' dialog box



3.4.8 Assigning of Section Properties

3.4.8.1 Assigning Column Properties





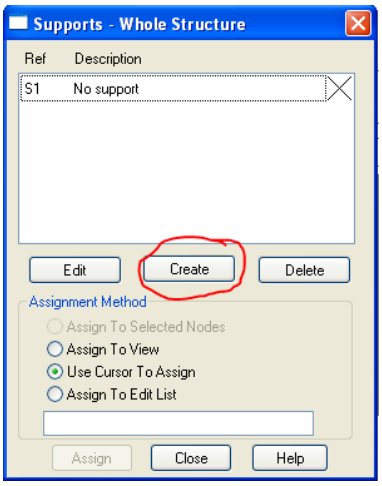
Go to '**Select**' then choose '**Beams Parallel to the Y-axis'**. This highlights all the beams in the Y axis which are our columns. Now at the Property dialog box, click on the section 0.23m x 0.23m. Now, check the '**Assign to selected beams'** button. Click on '**Assign**' button, and confirm the assignment by clicking '**Yes'**

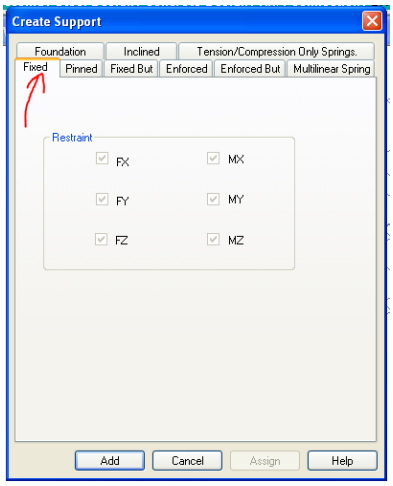
3.4.8.2 Assigning Beam Properties

Click on the floor beam section (0.45m x 0.23m). This deactivates the highlighted columns. Now, take any view of your choice (apart from floor plan view) that will enable you to rubber band all the first floor beams (you must be on beam cursor). Make sure you do not highlight the columns. You can switch to 3D view to ensure that the correct beams have been highlighted. Now assign the section as described above. Following these steps described, you can also assign the roof beam properties accurately.

3.4.9 Creation of Supports

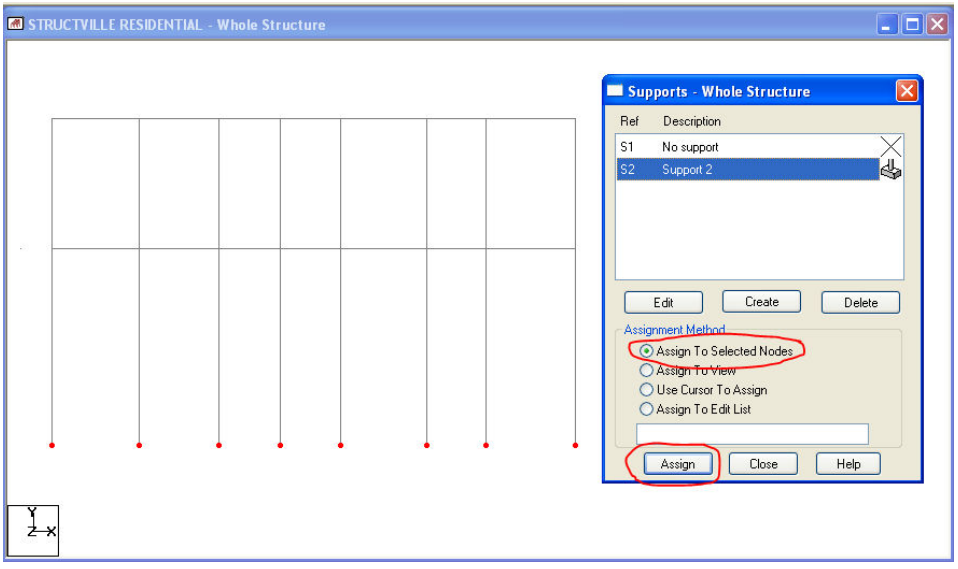
From the sub-pages of **General**, click on '**Support'**. The '**Support**' dialog box comes up. Click on '**Create**'. You can verify that the first option there is **fixed** support. Click **Add**, and then **close**. It is also at this place that you select pinned supports and other type of supports whenever you are doing any type of analysis.



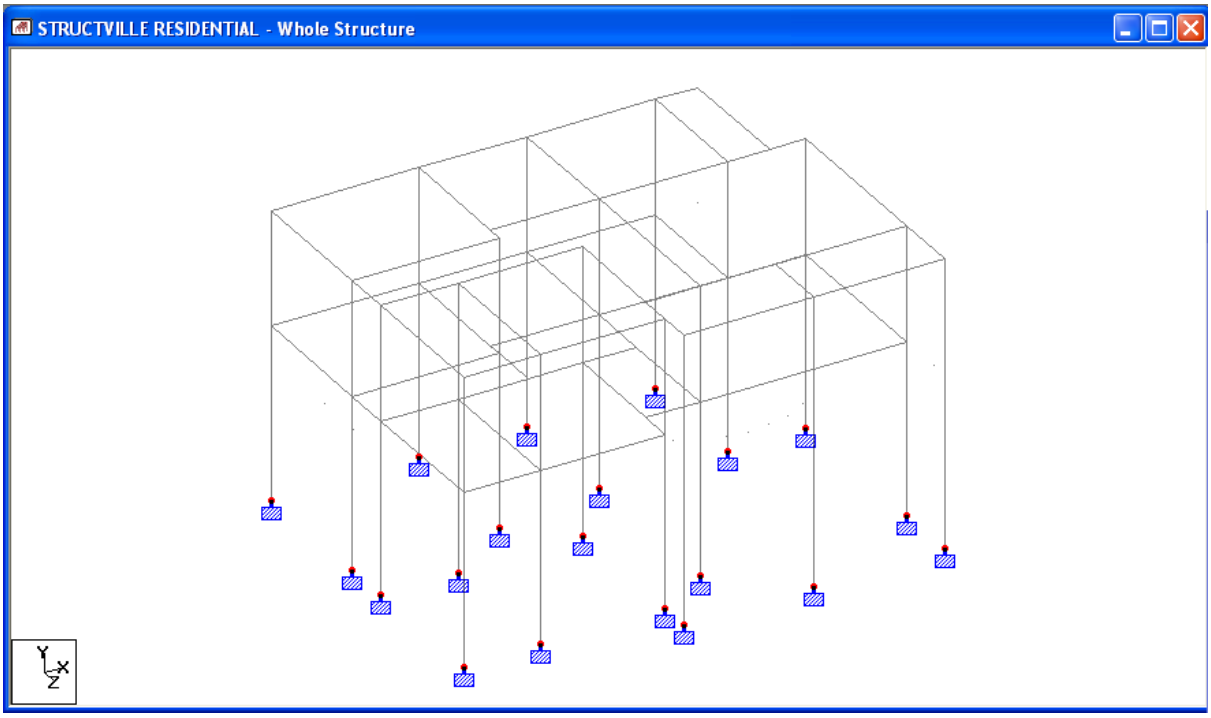


3.4.9.1 Assigning the supports

Now that you have created the supports, the next thing is to assign them to the nodes where they belong. Select any view apart from floor plan or 3D view. Change your cursor to **Node Cursor**. Highlight all the bottom nodes (foundation level). Now click on the support created (this may remove the highlights on the node; if so repeat the highlight). Now check the '**Assign to Selected Nodes**' icon, and click '**Assign**'. Confirm the assignment



See a good representation of where we are now;



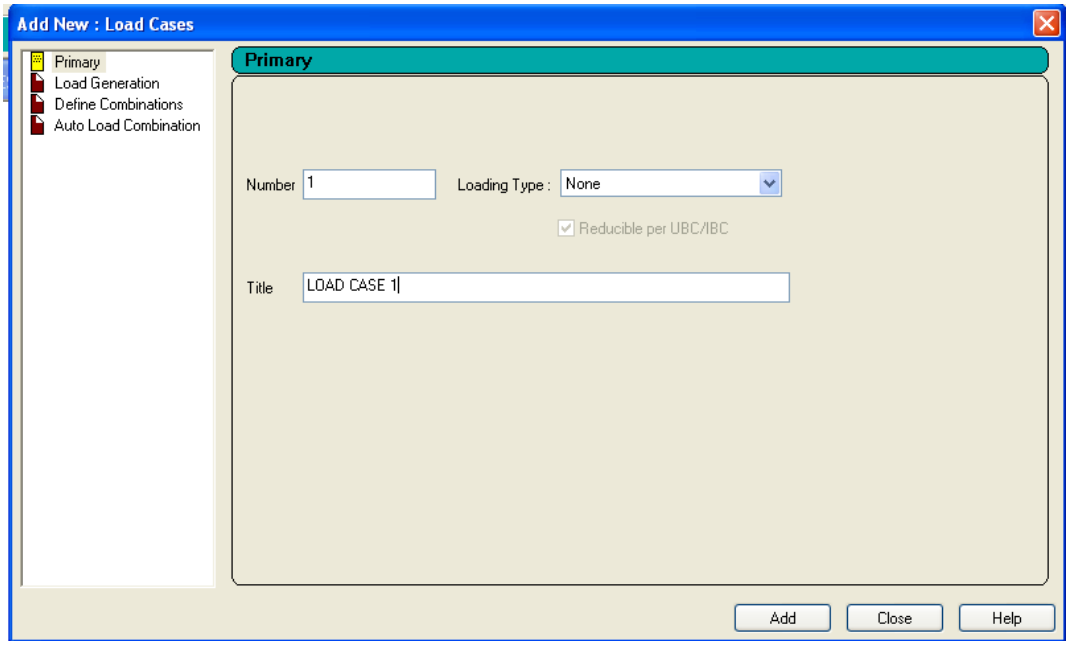
For some reasons, let me load the beams first, before modelling the slab

3.4.10 Load Case Definition

Click on '**Load Definition'** on the '**General**' Sub-page

Click on '**Load Case Details'** at the Data Window (Right Hand Side of the screen), and click **Add**

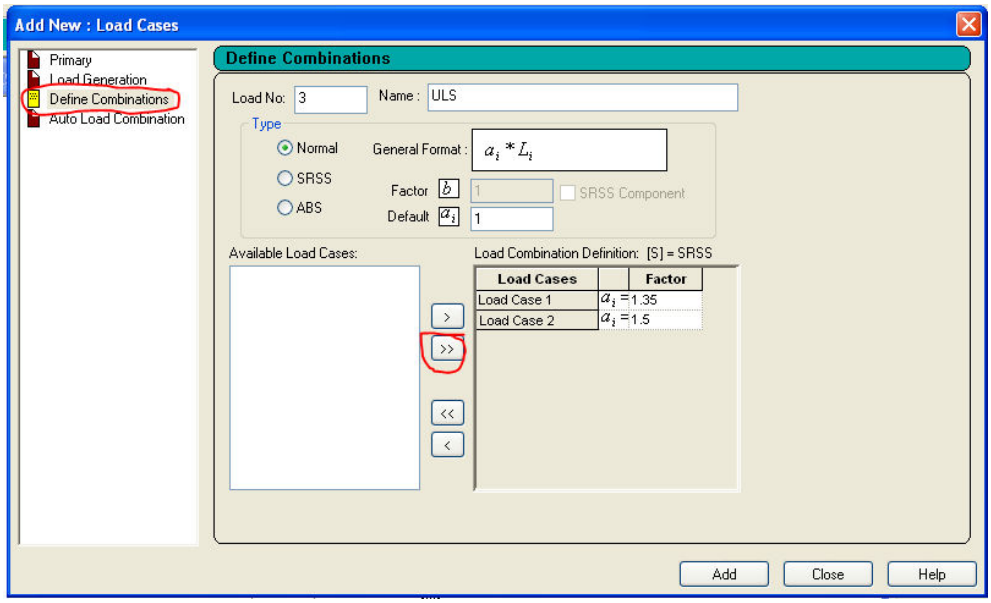
The dialog box below comes up. Rename '**LOAD CASE 1'** to '**DEAD LOAD'.** Then click **Add**



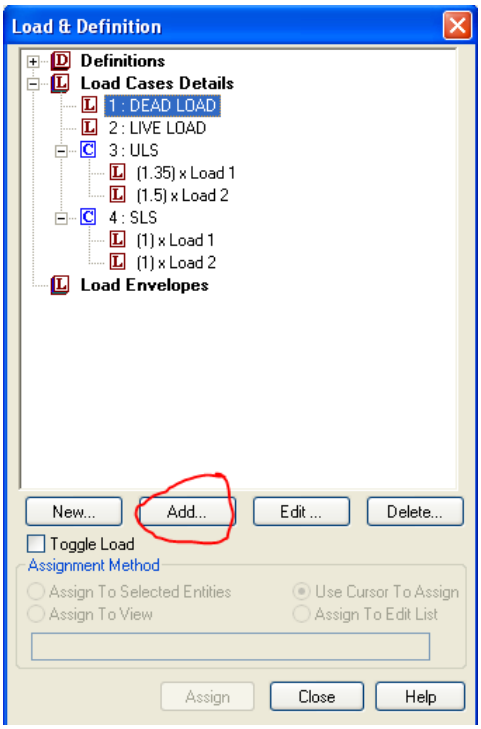
Subsequently, rename '**LOAD CASE 2'** to '**LIVE LOAD'** and also '**Add'**

Now, we have to define the critical load combination. When the **'Add New: Load Cases'** dialog box comes up again, click on '**Define Combinations'** as shown below. Rename the load case to Ultimate Limit State (ULS). Move all the available Load Cases (in this case Dead Load and Live Load) to the right for combination. Apply the Dead Load factor of 1.35 to Load Case 1, and a live load factor of 1.5 to Load Case 2. Click Add.

You can also define for Serviceability Limit State (SLS) with a Load Factor of 1.0 for both load cases. Then **Close**.

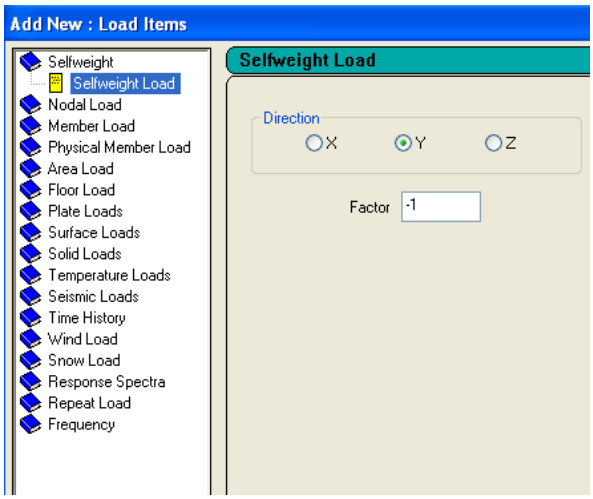


3.4.11 Load Value Definition



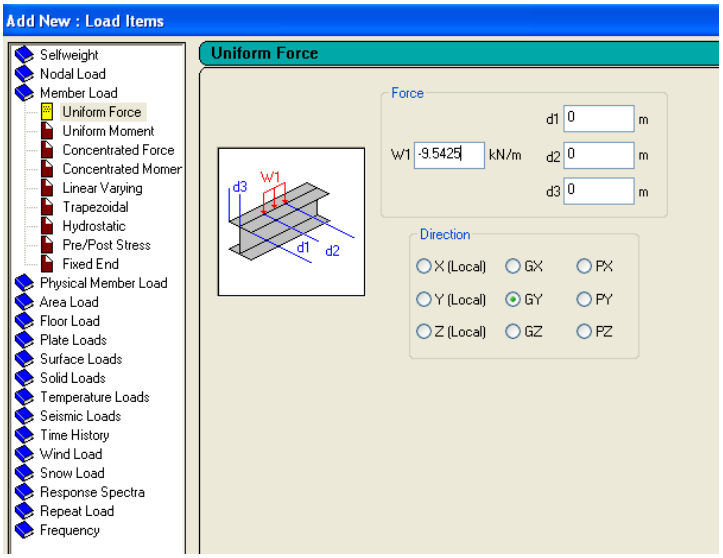
To define the dead loads, click **DEAD LOAD**, and then click **Add** as shown on the left hand side.

The dialog box below comes up. Click **Add** to '**add the self weight of elements**' as dead loads.



3.4.11.1 Beam Load

To create the beam loads, click on Member Load. Now select **Uniform Force**. You can see other types of Load Configurations for your subsequent modelling.



Beam Dead Loads

Wall load = 3.47 KN/m^2 × 2.75m = 9.5425 KN/m

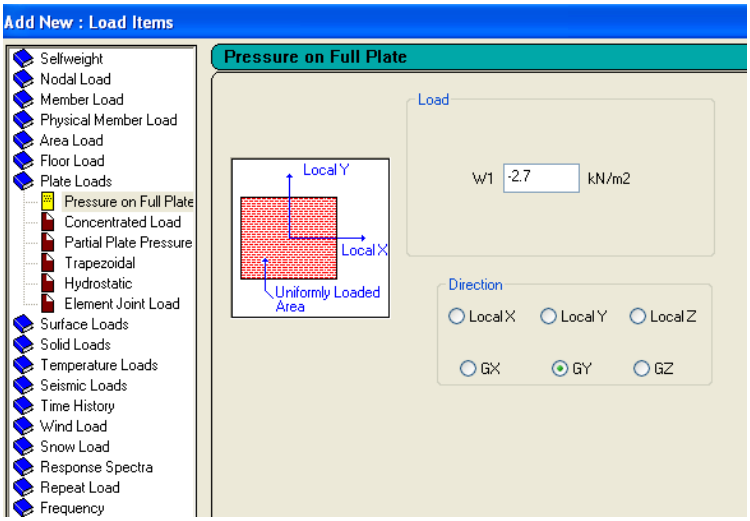
Roof dead load (external) = 4 KN/m

Roof dead load (internal) = 2 KN/m

Stair Case Dead Load (manually defined) = 10.837 KN/m

Now input the values of these UDL loads at the space provided for it. Endeavour to add minus sign, to tell Staad that it is a downward load. Hit Enter after inputting each load.

3.4.11.2 Plate Load



Click on **Plate Loads**

Select '**Pressure on Full Plate'**

The dead loads that the slabs are subjected to are; the self weight (already accounted for), weight of finishes of 1.2 KN/m^2, and a partition allowance of 1.5 KN/m^2 Therefore total additional dead load is 2.7 KN/m^2. Change the direction to Global Y (GY) and remember that it is negative.

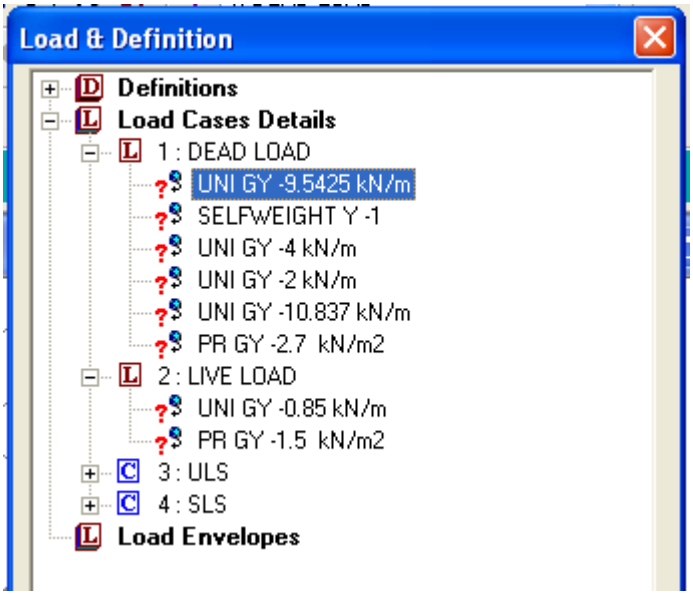
Since there are no other dead loads, click **Close**

3.4.11.3 Live Loads

Click on **LIVE LOAD,** and **click Add**
Follow the steps described above and input;
Roof live load = 0.85 KN/m
Stair Case Live Load (manually defined) = 4.9595 KN/m

Slab Live Load
Slab is subjected to a live load of −1.5 KN/m^2 **(follow the steps defined above and input these values)**

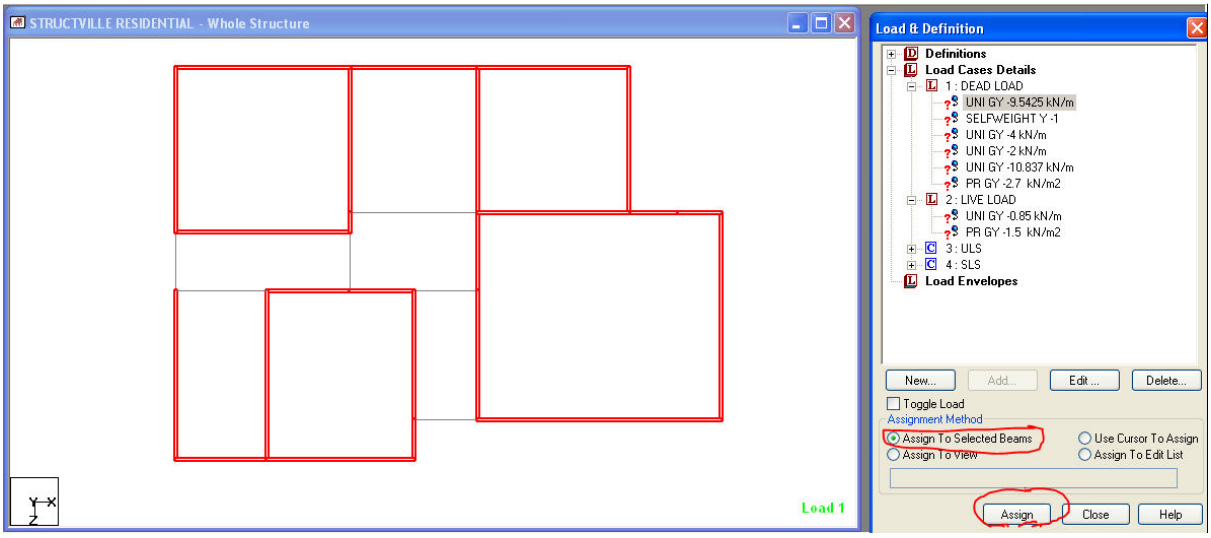
The summary of our loads is shown below.



3.4.12 Assigning Beam Loads

First of all, let us select a suitable view and rubber band (highlight) all the first floor beams. Go to '**View**' and select '**View selected Objects Only'**. After that, switch to plan view (View from +Y direction). Now click on the Wall Load (**UNI GY -9.5425 KN/m**).

Carefully select all the beams where the wall loads apply. We excluded beams running across the lobbies etc (see the selected beams as highlighted below). Now check '**Assign To Selected Beams**' option, Click '**Assign**' and confirm the '**Assignment**'.



We will apply the Staircase Load **(UNI GY -10.837 KN/m)**, to the beam it applies (Axis D: 1 – 2) by clicking on it and assigning as duly described above.

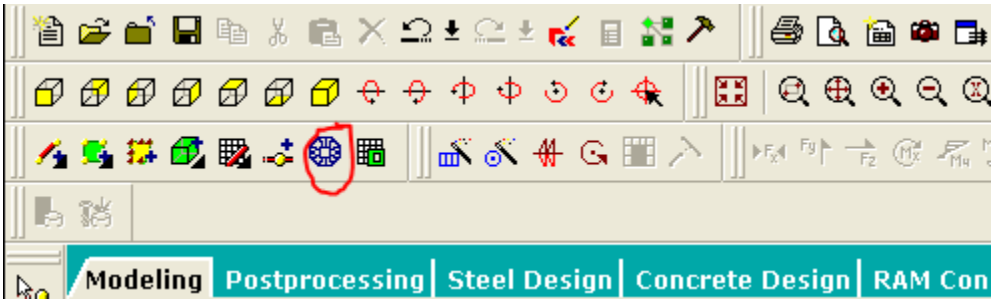
Switch to '**Display Whole Structure'**, and select only the roof beams for viewing. Now following the same procedure described above, apply the external roof beam load (-4KN/m), the internal roof beam load (-2 KN/m) and a live load 0f 0.85 KN/m to all roof beams. Click **Close**.

Note that we are yet to apply selfweight and slab loads.

3.4.13 Modelling the Floor Slab

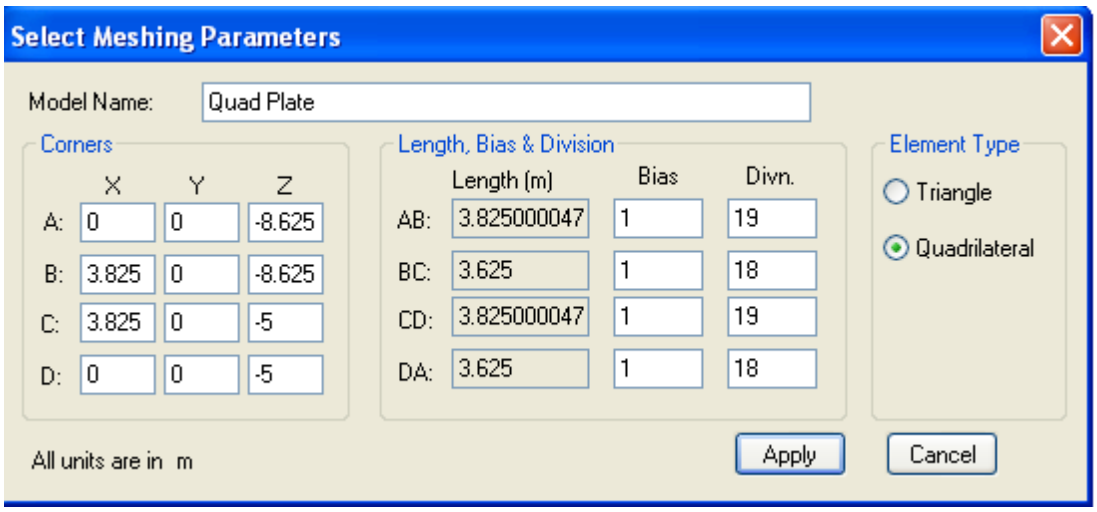
Now, select and view only the first floor beams (using the '**View selected Objects Only**' command) Switch to floor plan view.

Now, activate the plate meshing command using '**Generate surface meshing'** command. The icon is circled in red below.



Now, to mesh **Panel 1**, start at Point A1 by clicking on it. Now move to anti-clockwise direction to point A3, then to point C3, to point C1, and finally to point A1. Now an option comes up, asking you whether you wish to select '**Polygonal Meshing'** or **'Quadrilateral Meshing'**. Select **Quadrilateral Meshing** and the **'Select Meshing Parameters'** dialog box comes up. Now the length of each side is specified, and the bias is unity. Leave it at that. But we have to toggle with the number of divisions. Experience has taught me that multiplying the length with a constant factor for all the panels will give us a uniform meshing for all the panels.

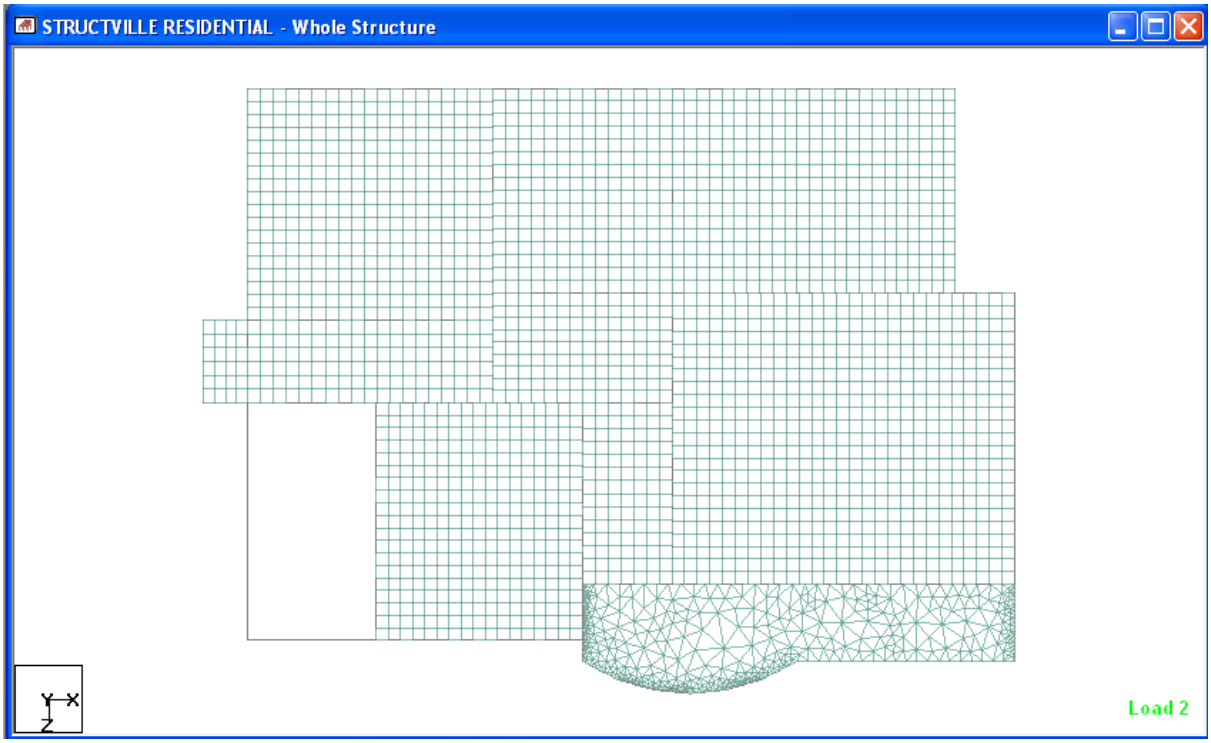
In this case, I am multiplying the length by a factor of 5. Note that the divisions cannot have decimals, and cannot be greater than 100, so you have to round up to the nearest whole number. For example in the picture below, $3.825 \times 5 = 19.125 \approx 19$.



So we have to mesh the entire panels one after the other.

However, I realised that PANEL 5 assumes a polygonal meshing by default, so let us leave it that. But you have to manipulate the number divisions to make it finer. The only thing I wish to let you know is that the finer your meshing, the more accurate your results.

The Final Meshing of the floor slab is shown below



3.4.13.1 Assigning Property to the floor slab

Go back to **Properties**
Select the plate cursor, and highlight the entire slab. Following the usual procedure, assign the thickness of 0.15m to the slab.

3.4.13.2 Assigning Load to the floor slab

Go back to **Loads and Load Definition**
Select the plate cursor, and highlight the entire slab. Following the usual procedure, assign dead load of 2.7 KN/m^2 and Live load of 1.5 KN/m^2 to the floor slab.

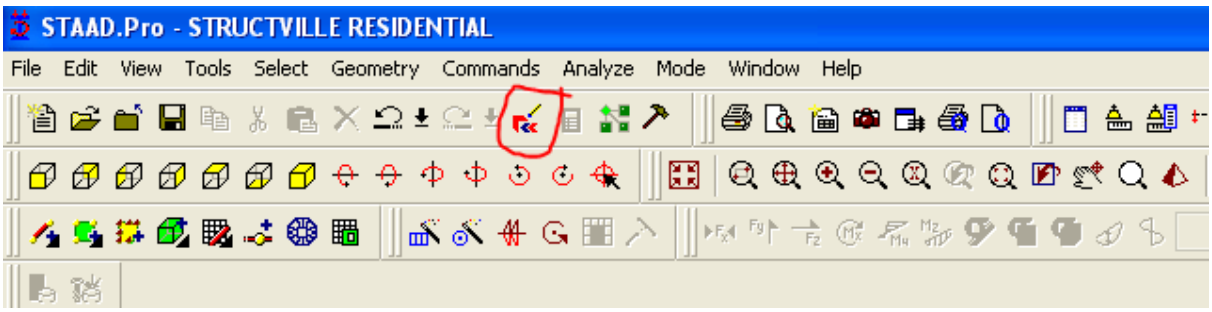
3.4.14 Assigning Self weight to the entire Structure

Select the geometry cursor, and highlight the entire structure from 3D view. Select from the dead load, the self weight of load to the entire building.

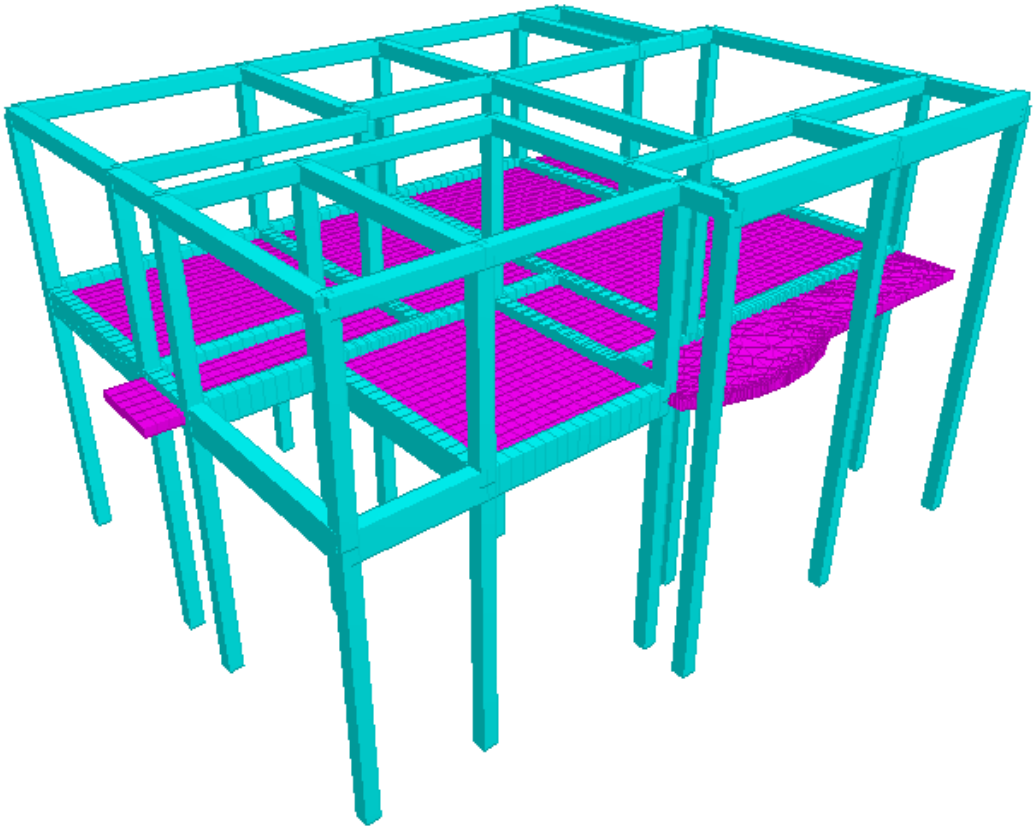
But wait, what density of concrete is being used for the calculation of self weight?

Go to **Staad Editor** (icon is circled below)

Scroll down below to **Material Properties**, and change the density of Concrete to 25 KN/m^2. Also change the Poisson ratio to 0.2. Be careful not to make any other changes to that place. It is like a program that has been written for the structure. If you violate any arrangement there, you are violating something on the original structural model.



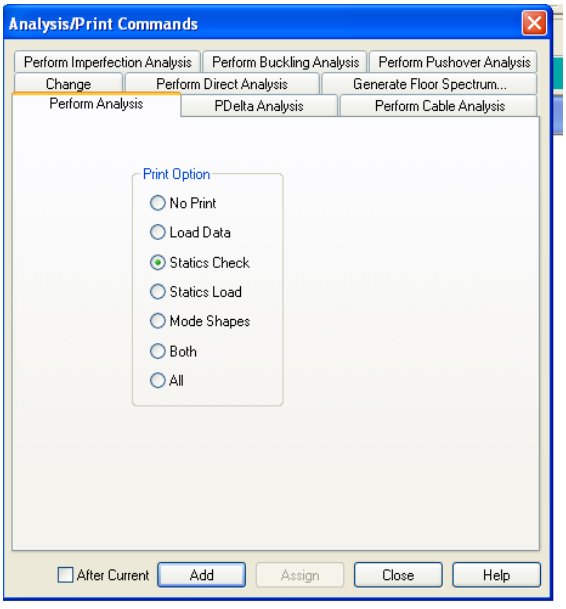
There is nothing more left to model gentlemen and ladies. The structure is ready, and fully loaded, Let me show you the 3D rendering of what we have made.



3.4.15 Analysis/Print

Go to Analysis/Print Page

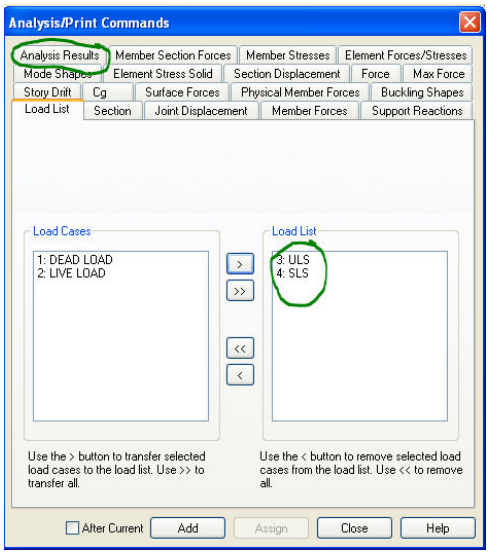
Select Statics Check as shown on the dialog box below. Click Add and Close



Go to **Post-Print** sub-page

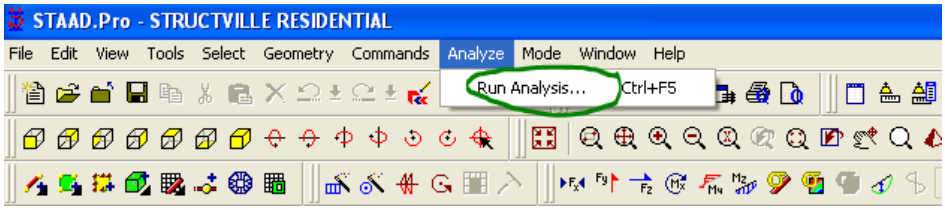
Go to the data window and click on **'Define Commands'**

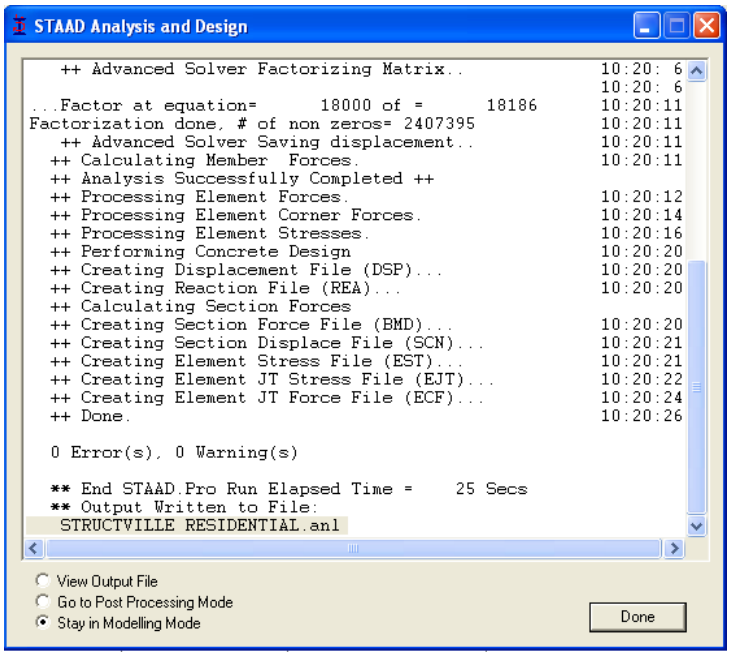
The dialog box below comes up. Move only the Limit State Loads from **Load Cases** to the **Load Lists** and click **Add**. Also Add '**Analysis Results'**. Then **Close**



3.4.16 Running the Analysis

Go to '**Analyze**', then click on '**Run Analysis...**'
Staad automatically runs the analysis.....



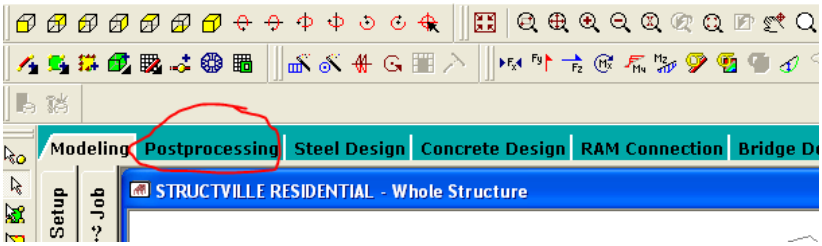


After the analysis, Staad displays the number of errors and warnings on the model. If there are warnings, then the analysis has been carried out, but there may be instability or geometric error somewhere. But if there are errors, then the analysis cannot be carried out at all. The warnings and error messages are displayed on the Output File, alongside the model data, and Analysis Results. So after viewing the warning or error message, you can then know the kind of correction to make. After making all corrections to the model, you can now reanalyse the structure, and view your results.

You can switch to the output file to view your results. However, I prefer to view my results from the post-processing mode, where I can pick any member of my choice, and view the internal stresses developed in it. Note that floor beams supporting the slabs are split into the number of plates that are supporting at the edge. So this can make viewing the bending moments a bit tedious, but it is a difficulty that can be easily overcome.

3.4.17 Post Processing

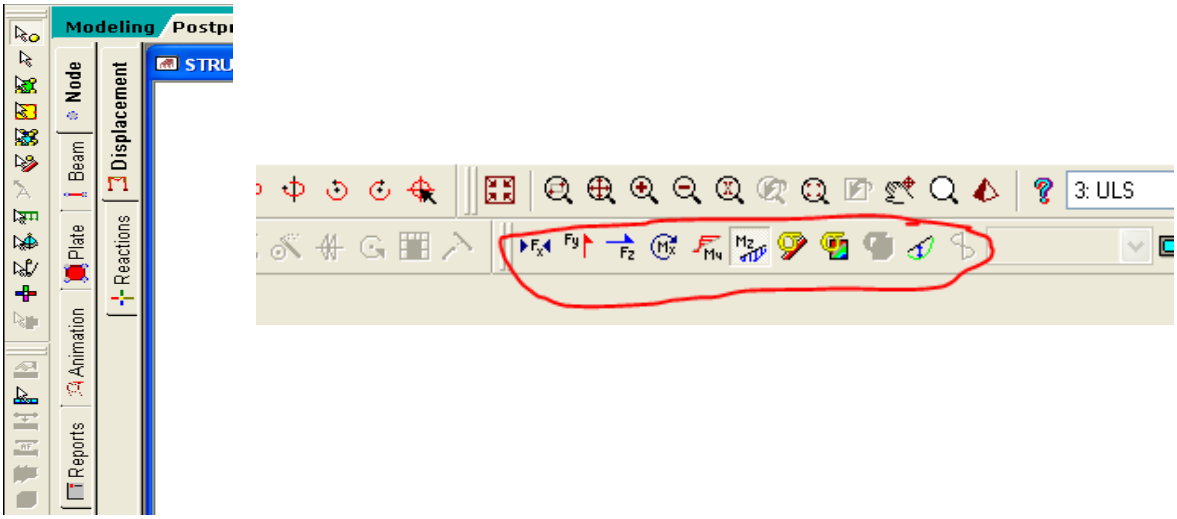
Click on '**Post Processing'** Tab



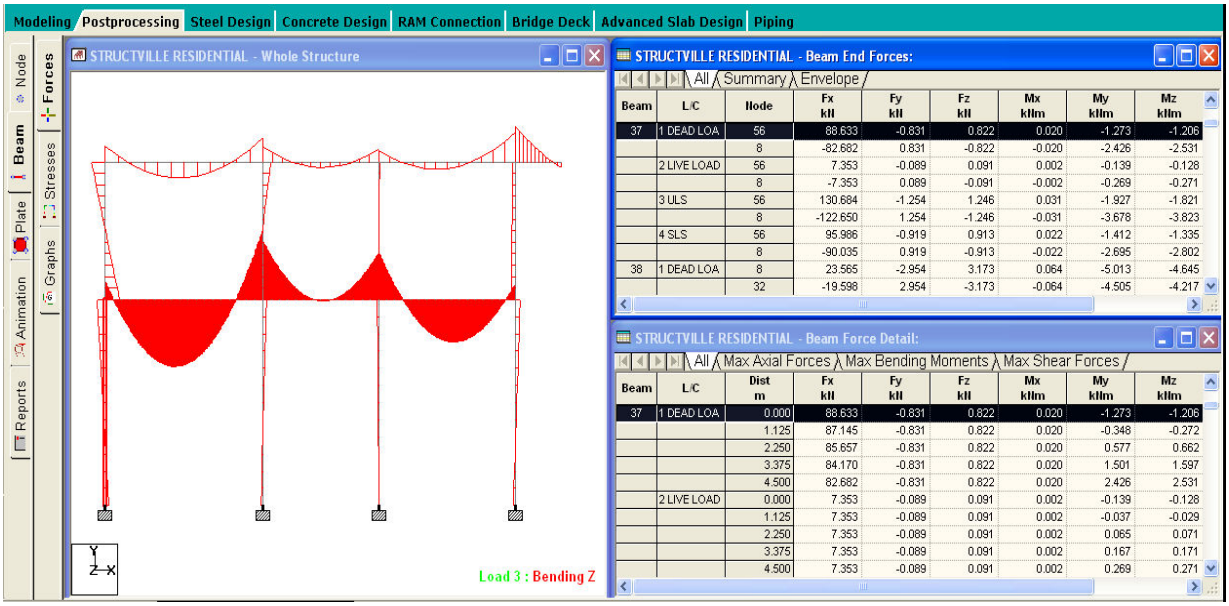
On the post processing page, we have pages for Nodes, Beams, and Plates. We view **support reactions** and **nodal displacements** on the Nodal page. To view specific displacements in a particular region, you can use the '**View Only Selected Objects'** command.

On the **beam page**, you can view the internal forces (Bending, Shear, Axial, and Torsion). For particular members you can view the stresses also in graphical form as you wish.

The icons circled in red above will give you the **Axial Force Diagram**, **Shear Force Diagram**, **Bending Moments**, **Stress Contour**, and **displacements**.

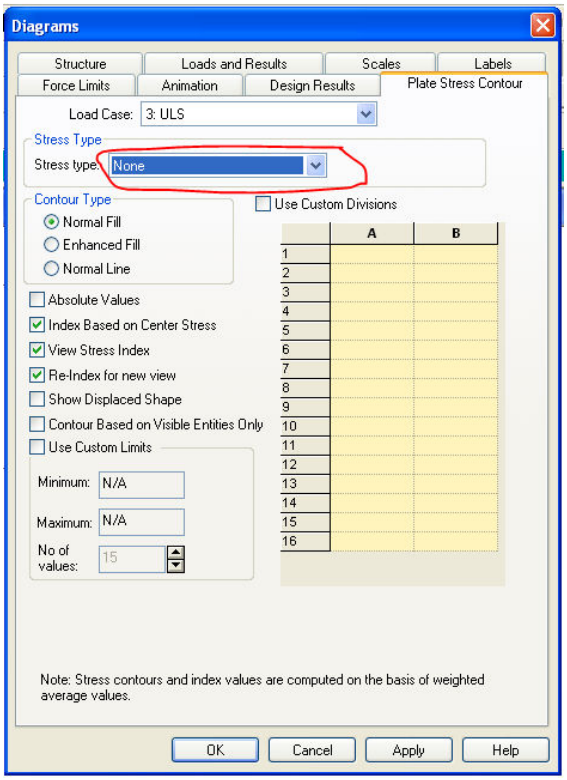


Axis A ULS bending moment diagram is as shown below (using the view selected members only) command. The highlighted column is beam No 37, and the internal forces can be viewed on the output window by the right hand side.



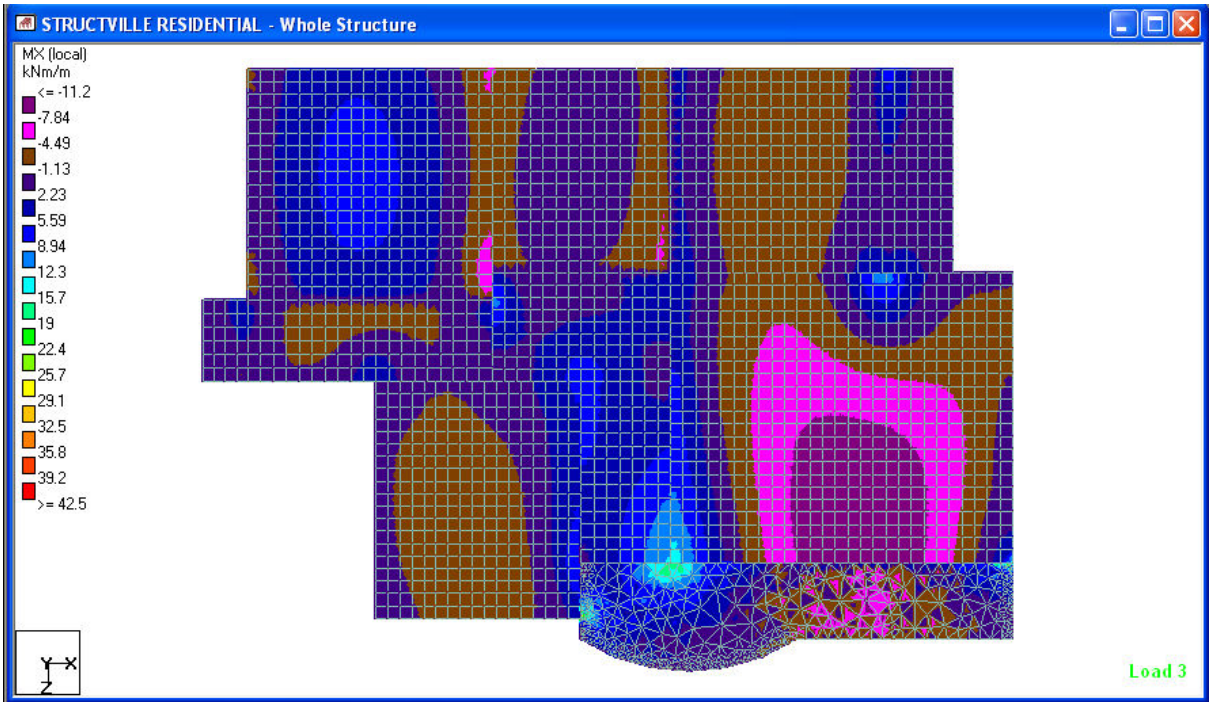
3.4.17.1 Plates Post Processing

Clicking on the **'Plate Post Processing Page'**, the dialog box below comes up.



From the '**Stress Type**' draw down list, you can select the type of stress you want to view. You can also adjust the scale as suitable to you. For instance, the contour for Mx is as shown below. The view is from the +Y direction.

The typical M*x* (local) contour plot is as shown below;



3.4.18 Design using Staad Reinforced Concrete Designer

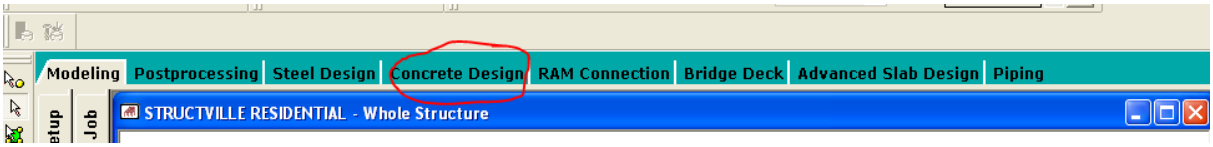
The STAAD.Pro Reinforced Concrete Designer (RC Designer) is a post processor that works from the STAAD.Pro analysis results. The data passed into RC Designer includes the geometry, section and material properties, loads and combination information and analysis results.

STAAD.Pro RC Designer is started by selecting the menu option **'Run Staad.Pro RC Design'** from STAAD.Pro. When this is done, a link file is produced that contains the basic data for creating concrete designs. This data is created each time the program is started. Additional data that is created during the use of the program is stored, such as the members, envelopes, design groups and design briefs in a persistent file. This means that if the program is closed and re-opened at a later date, the data remains available and does not need to be re-entered.

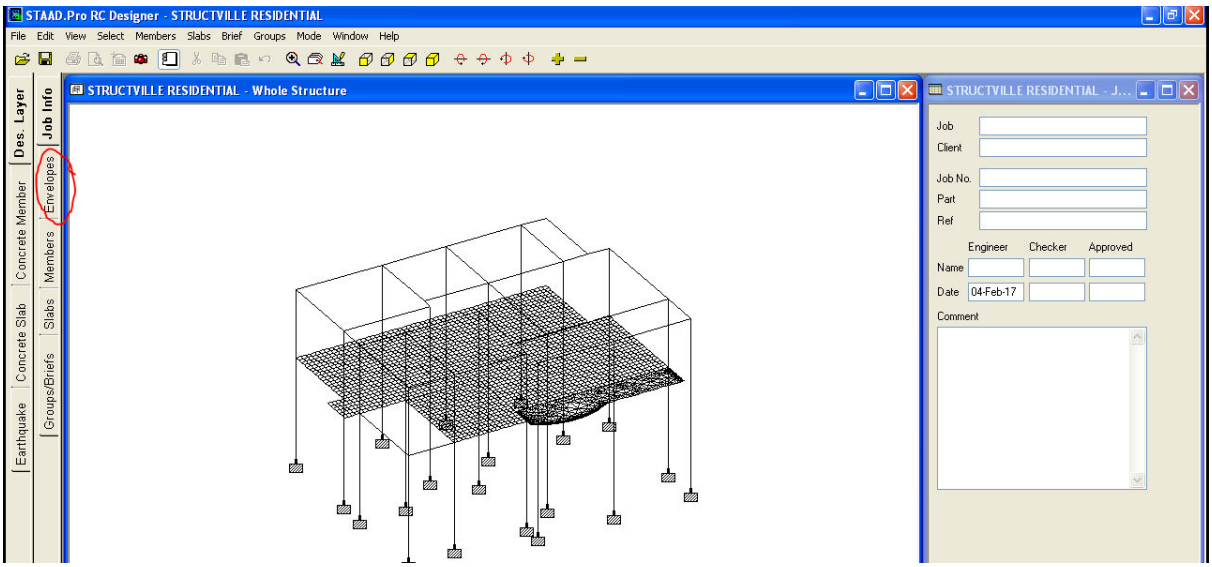
The design is performed on the selected members from the current Design Group and the parameters in the associated Design Brief. Each section is designed and a reinforcement cage is identified. The results of all the sections are then considered and a final rationalised cage is defined that requires the minimum area of reinforcement, but still matching all the design requirements including spacing and cracking checks.

If there are any issues arising from the design that results in either a partial or fully incomplete design for one or more of the design members, then these are reported in a Design Messages dialog box at the end of the design of all of the members. Note that on models where a large number of members are being designed, with a large number of load cases, then the member that is currently being designed is displayed on the lower left side of the status bar.

3.4.18.1 Launching the program

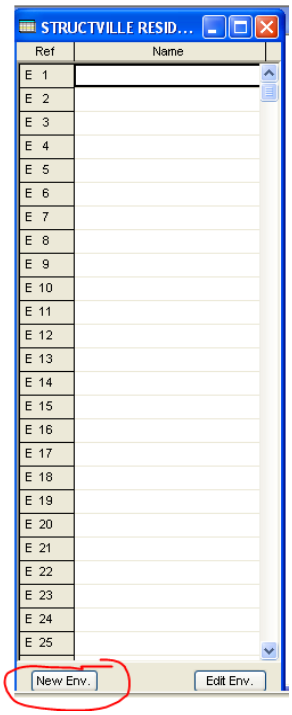


Launch the Staad RC Designer by clicking the '**Concrete Design**' tab circled in red above. The package's main window opens as shown below.

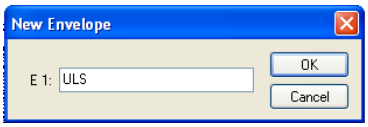


3.4.18.2 Defining the Load Envelope

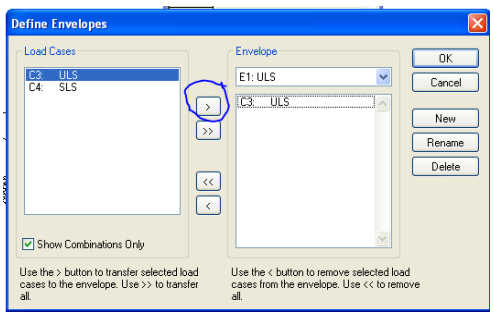
After defining the job information as specified, the next thing is to define the **'Loading Envelope'** by clicking the '**Envelopes**' Tab. The dialog box below comes up.



Click on **'New Envelope'** Tab circled by the left hand side. This is to define the load cases for the design. The dialog box below comes up. Give the new envelope name as you desire. Click **OK**



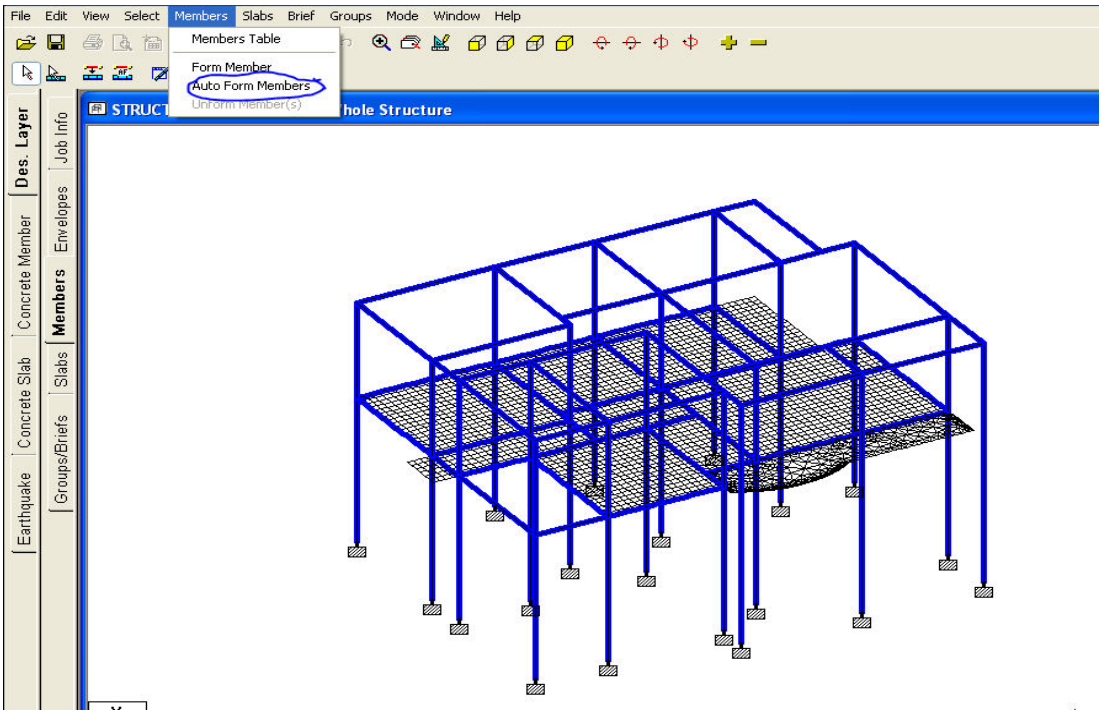
Select ULS load as the only load envelope of interest. To see other load cases, you can uncheck the 'Show Combinations Only'. Click **OK**



3.4.18.3 Forming the members

Click on the '**Members**' tab just below the '**Envelopes**' tab. The window below comes up. The cursor is on beam cursor by default. Now, rubber band (highlight) all the beams and columns in the structure. This changes the colour of the members to blue. The blue colour will show if all the members have been captured.

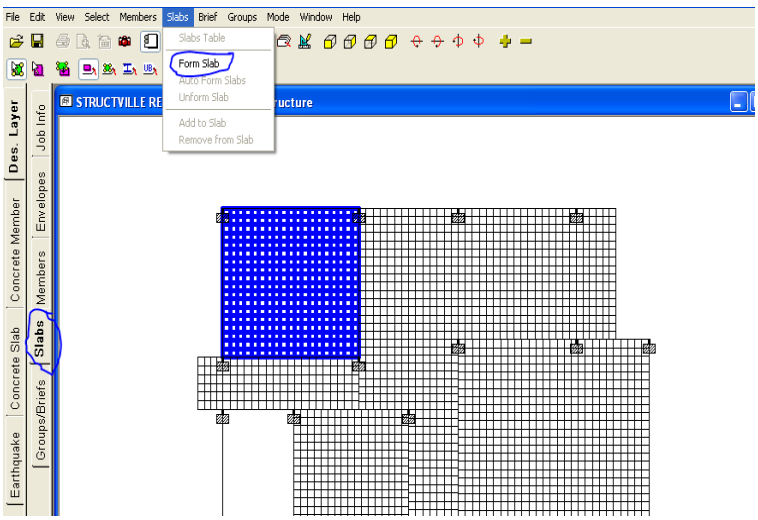
Now go to Members on the Menu Bar, Click on it, and select '**Auto Form Members'** to form the members. (Now treated as physical members)



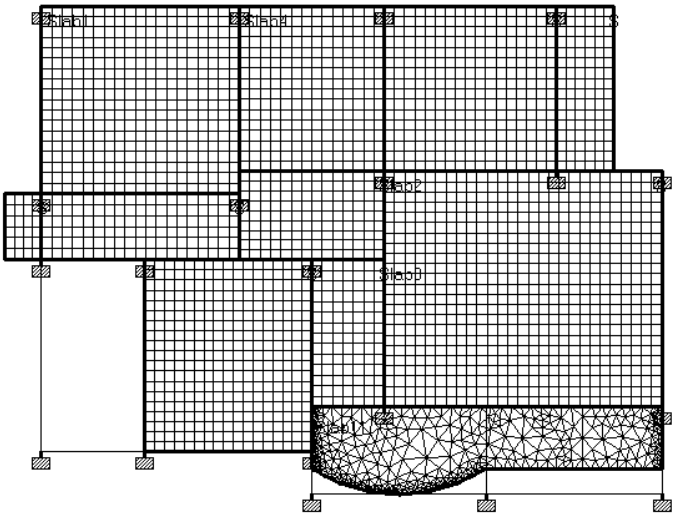
3.4.18.4 Forming the slabs

Click on **'Slabs'** just below the '**Members**' tab.

The cursor automatically changes to Plates Cursor. Now switch your view to plan view (+Y direction).



Carefully highlight Panel 1 as shown on the left hand side, making sure that all the plates in the PANEL are captured (this is to prevent us from creating Regions later). Now, go to '**Slabs**' on the Menu bar, click on it, and click on '**Form Slab'**. This automatically forms the slab, with the Title, 'Slab 1'. Now repeat this until all the panels are represented. This is shown in the second picture below.



STRUCTVILLE RESIDENTIAL - Slabs

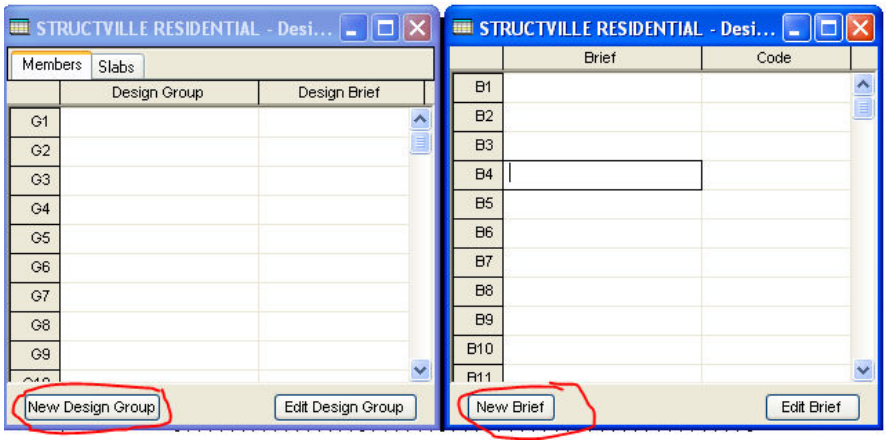
Slab	Thickness cm	Net Area m^2	Gross Area m^2	Holes	Plates
1	15.0	13.866	13.866	0	342
2	15.0	4.774	4.774	0	126
3	15.0	3.976	3.976	0	98
4	15.0	8.974	8.974	0	224
5	15.0	3.526	3.526	0	96
6	15.0	10.657	10.657	0	272
7	15.0	4.915	4.915	0	114
8	15.0	0.900	0.900	0	24
9	15.0	11.981	11.981	0	323
10	15.0	24.361	24.361	0	621
11	15.0	9.215	9.215	0	908

3.4.18.5 Design Briefs

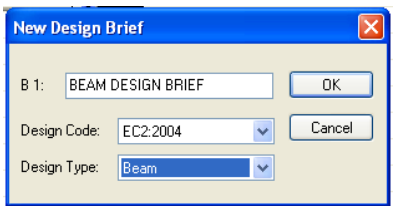
Now having formed all the members, and slabs, the time has come to define the design briefs. A design brief is where we specify all the design data including the design code, concrete cover, concrete grade etc.

Now, let us define the members design brief.

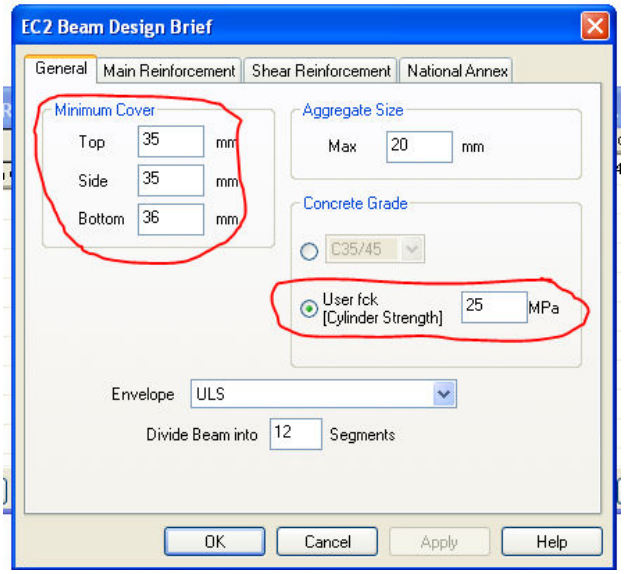
Click on '**Briefs/Groups',** and these two dialog boxes come up, one below the other.



Click on '**New Brief'**, and the dialog box below comes up. You can change the name of the brief to 'BEAM DESIGN BRIEF', and change the design code to Eurocode 2. Make sure that the '**Design Type**' is **Beam**. Click **OK**



The dialog box below comes up.



General Tab
Change the concrete cover to 35mm
Leave the maximum aggregate size to 20mm
Change the concrete grade to 25 N/mm^2

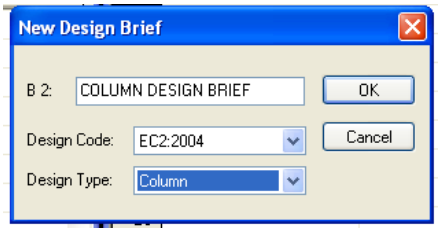
Main Reinforcement Tab
Change the minimum reinforcement size to 12mm, maximum to 32mm
Change the steel grade to 460 N/mm^2

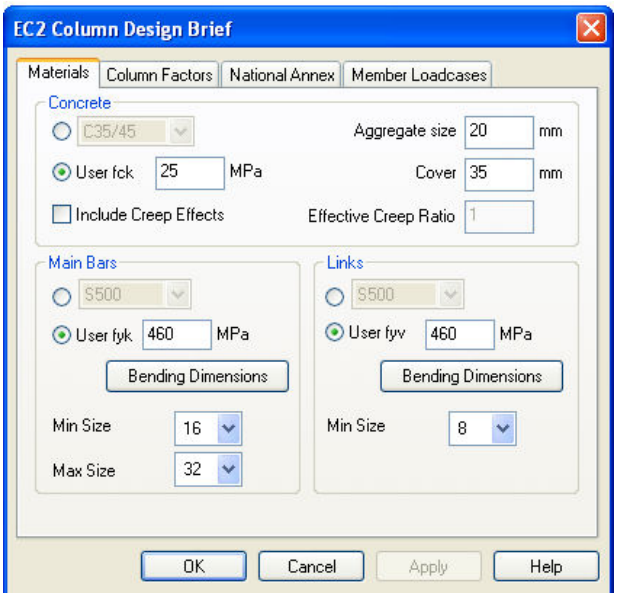
Shear Reinforcement Tab
Change the minimum reinforcement size to 8mm
Change the steel grade to 460 N/mm^2

After all changes, click **OK**

3.4.18.6 Column Design Brief

Now, click on '**New Brief'** again, and create '**Column Design Brief'.** Click **OK**





Materials Tab
Change the concrete cover to 35mm
Leave the maximum aggregate size to 20mm
Change the concrete grade to 25 N/mm^2
Change the steel grade to 460 N/mm^2 for main bars and links
Change the minimum reinforcement size to 16mm, maximum to 32mm

Leave Column Factors and National Annex the way they are

Member Load Cases
Select ULS only

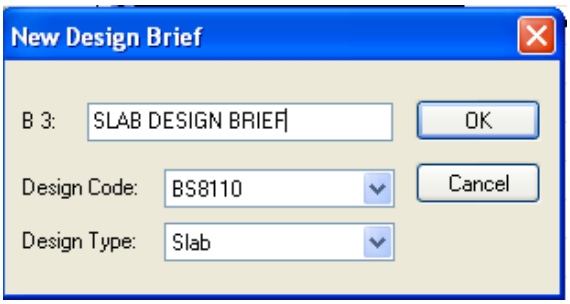
Click **OK**.

3.4.18.7 Slab Design Brief

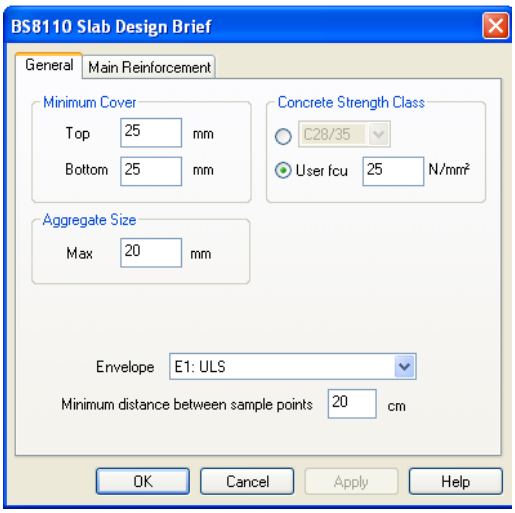
Now, click on '**New Brief**' again, and create '**Slab Design Brief**'.

You will discover that for EC2, no slab design brief has been specified for 'Design Type'. But for BS 8110, the option is available. Let us use BS 8110 for the flexural design, for the purpose of showing how it is done.

Click **OK**



The dialog box below comes up

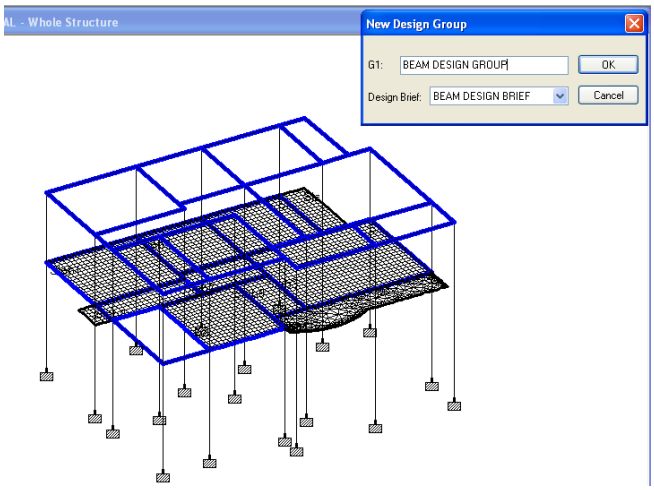


General Tab
Change the concrete cover to 25mm
Leave the maximum aggregate size to 20mm
Change the concrete grade to 25 N/mm^2

Main Reinforcement
Change the steel grade to 460 N/mm^2 for main bars and links
Use '**Fixed Bar Size**'
Check the **'Wood and Armer'** adjustment to include torsion moment on design moment
Change the bars sizes to 12mm

Click **OK**

3.4.18.8 Creating Member Design Group



Change the view to a suitable view and rubber band (highlight) all the beams (floor beams and roof beams).

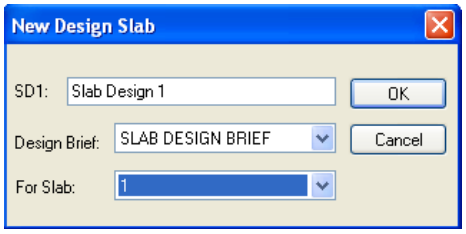
Click on **'New design Group'.** Give the design group a name, and specify the design brief as **'Beam Design Brief'**. Click **OK**.

In the same vein, highlight all the columns, and create a column design group. Assign the column design brief to it, and click **OK.**

3.4.18.9 Slab Design Group

Switch the '**Design Group'** dialog box from '**Members**' to '**Slab**'.

Click on Panel 1, and then click on **'New Design Slab',** and the dialog box below comes up**.** Click **OK** for **Slab Design 1.** Repeat this process for all the slabs.

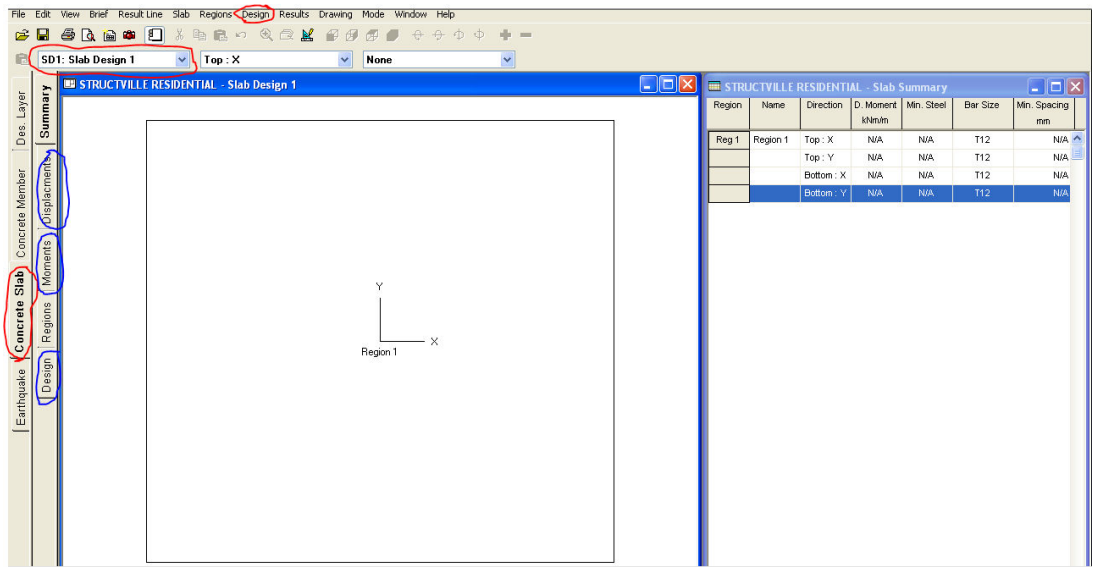


3.4.19 Slab Design

Click on '**Concrete Slab'**, and the window below comes up.

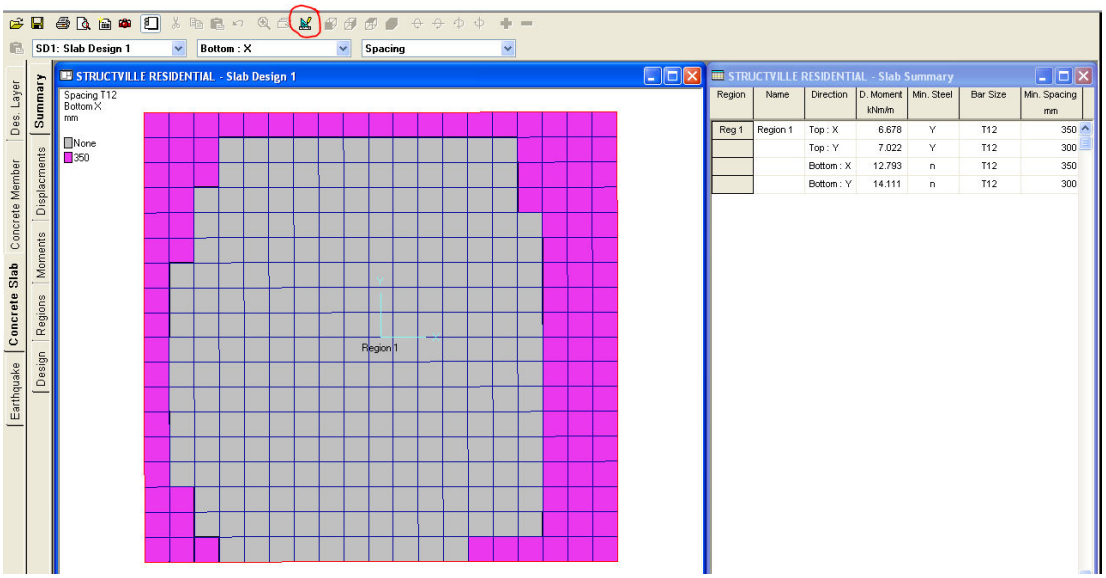
Select **'SD1: Slab Design 1'** to select PANEL 1

At the right hand side you can see that the fixed bar size has been specified (T12mm), but the spacing and minimum area of reinforcement has not been shown, because the design is yet to be carried out. The panel has been represented as a single region, which is reasonable for the purposes of design.

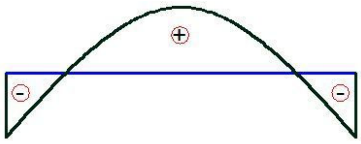


Click on '**Design**' at the Menu Bar to carry out then Design. The dialog box below confirms the design.

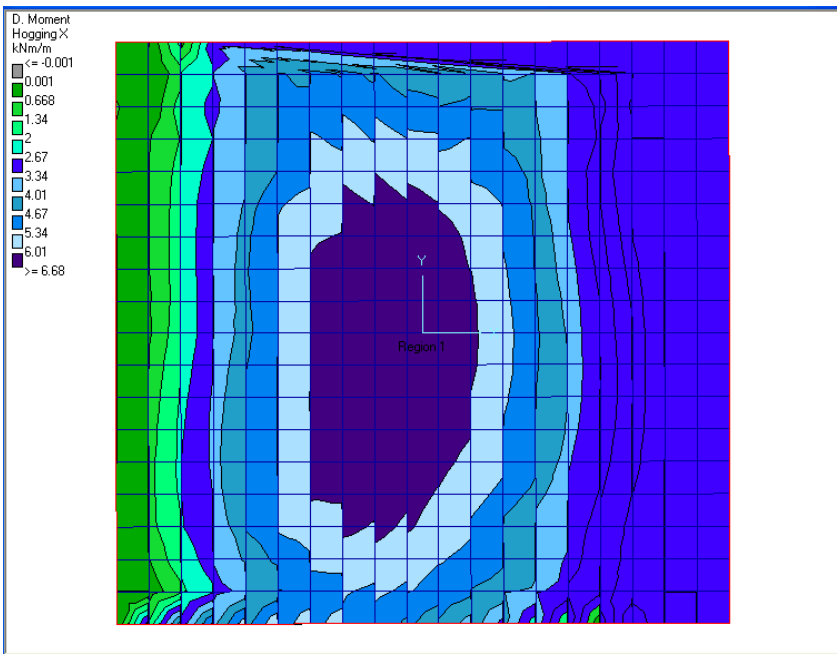
The design summary is as shown below. Initially your contour may not appear as shown below. So you have to go to the '**Scale**' icon (circled in red below) and modify the contour lines the way you want to see it. The reinforcement spacing and the design moment can be viewed explicitly by the right hand side of the screen. You can view the contour for Bottom X, Bottom Y, Top X, and Top Y by selecting from the **draw down list** at the menu bar.



Do not be confused, Staad's sign convention is like this;



So the real bottom reinforcement design moment for X – direction is shown below (note that Staad called it hogging). You can obtain it by changing from Spacing to Moment. You can view for any direction (Top or Bottom, X or Y).



You can eventually print the design detail by clicking on '**Report**'.

Design Group: - Brief Detail: SLAB DESIGN BRIEF BS8110 Slab

Group Data

Top Cover	25	Aggregate Size	20
Bottom Cover	25	Concrete strength clas	C20/25
Envelope	ULS	Wood and Armer	Yes
Sampling	0.200 m	Design Type	Fixed Bar Size

Main Reinforcement

Main bars: T(460) Material Factor γ_m: 1.15

Main Bar Criteria

	Top Bar Criteria	Bottom Bar Criteria
Outer Bar Direction	X	X
Bar Size X	12	12
Bar Size Y	12	12

Slab Information

Gross Area	13.866 m2	Number of Plates	342
Net Area	13.866 m2	Number of Holes	0
Thickness	15.0 cm		

Region Information

Region	Name	Orientation (°)	Area (m²)	Thickness (cm)	Skew (°)
Reg 1	Region 1	0.00	13.866	15.0	Orthogona

Design Summary

Region	Name	Direction	Des. Mom. (kNm/m)	Min. Steel	Bar Size	Min. Spacing (mm)
Reg 1	Region 1	Top : X	6.678	Y	T12	350
		Top : Y	7.022	Y	T12	300
		Bottom : X	12.793	n	T12	350
		Bottom : Y	14.111	n	T12	300

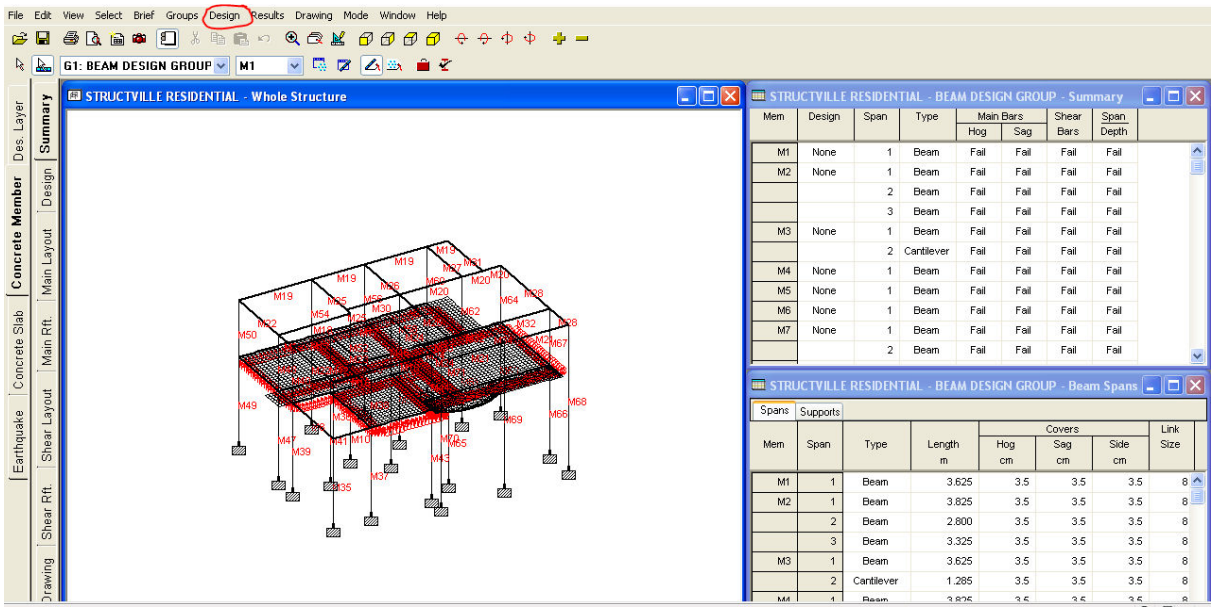
Ultimately, you can design all the panels and view all the results following the processes described above.

3.4.20 Beam Design

If for any reason your initial settings are disabled, you can follow the repeat the process of forming members, creating design brief, and creating design group.

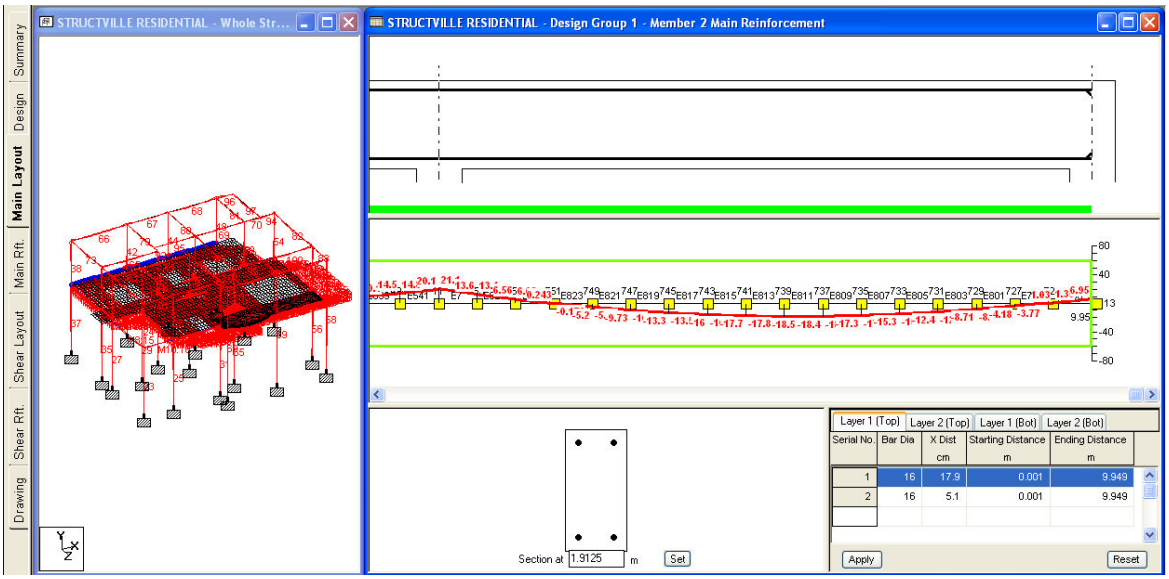
Click on **'Concrete Member'**.

The result from the data window shows that all the beam members are failing (see the picture below). This is because the design has not been done yet. Click on '**design**' on the menu bar, and then click on **'design now'**.



The design is executed at this level with all successful members being highlighted in green, and members that failed to be designed due to some different reasons are also identified. If a members fails, identify the error or failure message, and take appropriate steps to take care of it.

Beam No 1 was successful in the design, and we are going to review it at this point. Below, is the **Main Layout** of the Beam. You can view the bending moment of the beam at the Main Layout Sub-Page.



At the subsequent sub-pages, you can view the main reinforcements, shear reinforcements, and even the design drawings, which can be exported to AUTOCAD via DXF. But from the design report, you can view the detailed calculation sheet. When setting up your design report, make sure that you select only 'Design Details', to avoid having a lot of information that you may not really need.

The design report calculation sheet for beam number 1 on our GA from Staad Pro is as shown below;

Member M2 Span 1

Detailed EC2 Main Reinforcement

Hogging: at 0.000 m from the start of the member

Moment applied to section		= 8.15 kNm	
Effective depth of tension reinforcement	d	= 399 mm	
Depth to compression reinforcement	d'	= 51 mm	
Tension Bars provided		= 2T16	
Actual area of tension reinforcement		= 402 mm²	
Minimum area of tension reinforcement		= 18 mm²	9.2.1.1(1)
Maximum area of tension reinforcement		= 4140 mm²	9.2.1.1(3)
Minimum horizontal distance between top bars		= 25 mm	8.2 (2)
Smallest actual horizontal space between top bars		= 112 mm	
Minimum horizontal distance between bottom bars		= 25 mm	8.2 (2)
Smallest actual horizontal space between bottom bars		= 112 mm	
Actual neutral axis depth of section		= 54.60 mm	
Moment capacity of section		= 60.13 kNm	
		∴ **OK**	

NOTE:- No serviceablity check with Tables 7.2N and 7.3N performed.
For a hand check please use these values:-

Max. bar size		= 16 mm	
Max spacing		= 112 mm	

Sagging: at 1.611 m from the start of the member

Moment applied to section		= 29.17 kNm	
Effective depth of tension reinforcement	d	= 399 mm	
Depth to compression reinforcement	d'	= 51 mm	
Tension Bars provided		= 2T16	
Actual area of tension reinforcement		= 402 mm²	
Minimum area of tension reinforcement		= 18 mm²	9.2.1.1(1)
Maximum area of tension reinforcement		= 4140 mm²	9.2.1.1(3)

Member M2 Span 1
Detailed EC2 Main Reinforcement Cont...

Minimum horizontal distance between top bars		= 25 mm	8.2 (2)
Smallest actual horizontal space between top bars		= 112 mm	
Minimum horizontal distance between bottom bars		= 25 mm	8.2 (2)
Smallest actual horizontal space between bottom bars		= 112 mm	
Actual neutral axis depth of section		= 54.60 mm	
Moment capacity of section		= 60.13 kNm	
		∴ **OK**	

NOTE:- No serviceablity check with Tables 7.2N and 7.3N performed.
For a hand check please use these values:-

Max. bar size		= 16 mm	
Max spacing		= 112 mm	

Hogging: at 3.825 m from the start of the member

Moment applied to section		= 30.25 kNm	
Effective depth of tension reinforcement	d	= 399 mm	
Depth to compression reinforcement	d'	= 51 mm	
Tension Bars provided		= 2T16	
Actual area of tension reinforcement		= 402 mm^2	
Minimum area of tension reinforcement		= 18 mm^2	9.2.1.1(1)
Maximum area of tension reinforcement		= 4140 mm^2	9.2.1.1(3)
Minimum horizontal distance between top bars		= 25 mm	8.2 (2)
Smallest actual horizontal space between top bars		= 112 mm	
Minimum horizontal distance between bottom bars		= 25 mm	8.2 (2)
Smallest actual horizontal space between bottom bars		= 112 mm	
Actual neutral axis depth of section		= 54.60 mm	
Moment capacity of section		= 60.13 kNm	
		∴ **OK**	

Member M2 Span 1
Detailed EC2 Shear Reinforcement

Shear zone: 0.001 m to 3.824 m

$V_{Rd,c}$ $= [C_{Rd,c} k (100 \rho_l f_{ck})^{1/3} + k_1 \sigma_{cp}] b_w d$ (6.2.a)

$= [0.120 \times 1.708 \times (100 \times 0.0044 \times 25)^{1/3} + 0.15 \times -0.016] \times 230 \times 399$

$= 41.56$ kN

But $\geq (v_{min} + k_1 \sigma_{cp}) b_w d$ (6.2.b)

$= (0.39 + 0.15 \times -0.016) \times 230 \times 399$ $= 35.63$ kN

Hence, $V_{Rd,c}$ $= 41.56$ kN

V_{Ed} $= 51.30$ kN

$V_{Rd,max}$ $= \alpha_{cw} b_w z \nu_1 f_{cd} / (\cot\theta + \tan\theta)$ (6.9)

$= 1.00 \times 230 \times 359 \times 0.54 \times 16.7 / (2.50 + 0.40)$ $= 256.32$ kN 6.2.3 (3)

$V_{Ed} \leq V_{Rd,max}$

spacing provided, s	= 225 mm	
area of links provided (2T8), A_{sw}	= 100.53 mm^2	
Maximum allowable transverse spacing between shear legs	-157286400 mm	9.2.2 (8)
Largest actual transverse spacing between shear legs	mm	
	∴ **OK**	

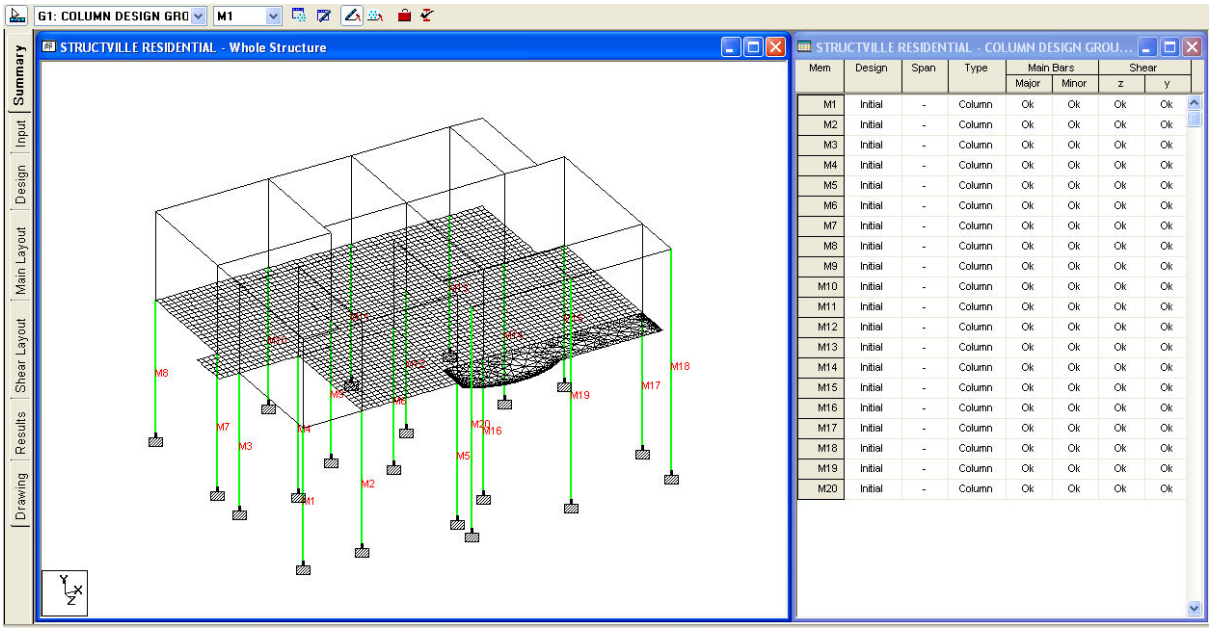
∴**OK**

3.4.21 Column Design

If for any reason your initial settings are disabled, you can follow the repeat the process of forming members, creating design brief, and creating design group.

Click on **'Concrete Member'**.

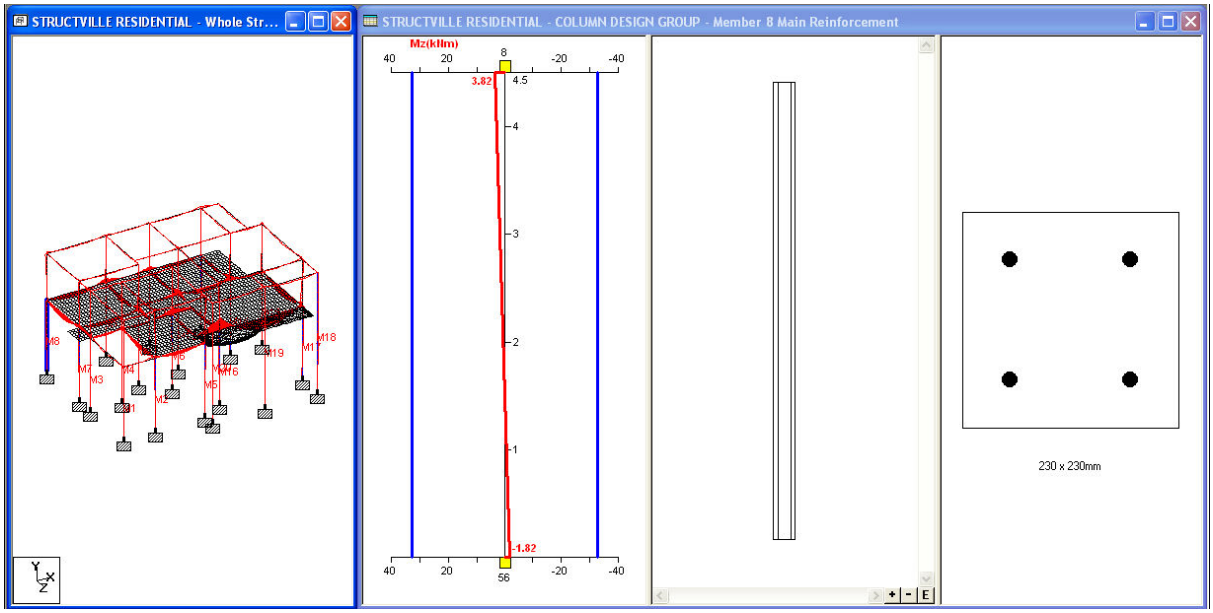
Select **'Column Design Group'**. Click on '**Design**' and then on '**Design Now'** The result from the data window shows that all the ground floor columns have passed the design. This is shown by the green colour of the all the members.



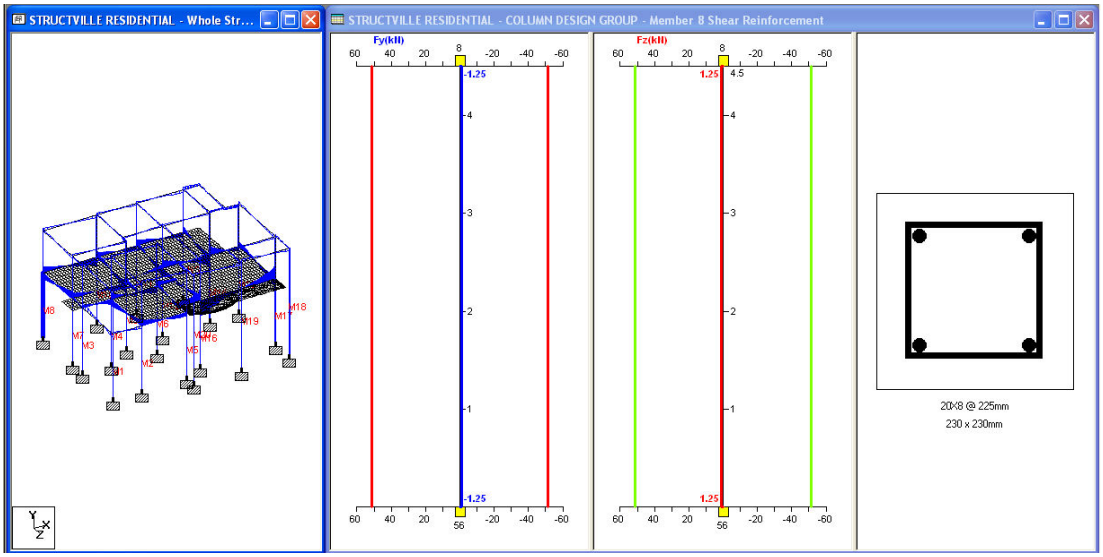
Mem	Design	Span	Type	Main Bars		Shear	
				Major	Minor	z	y
M1	Initial	-	Column	Ok	Ok	Ok	Ok
M2	Initial	-	Column	Ok	Ok	Ok	Ok
M3	Initial	-	Column	Ok	Ok	Ok	Ok
M4	Initial	-	Column	Ok	Ok	Ok	Ok
M5	Initial	-	Column	Ok	Ok	Ok	Ok
M6	Initial	-	Column	Ok	Ok	Ok	Ok
M7	Initial	-	Column	Ok	Ok	Ok	Ok
M8	Initial	-	Column	Ok	Ok	Ok	Ok
M9	Initial	-	Column	Ok	Ok	Ok	Ok
M10	Initial	-	Column	Ok	Ok	Ok	Ok
M11	Initial	-	Column	Ok	Ok	Ok	Ok
M12	Initial	-	Column	Ok	Ok	Ok	Ok
M13	Initial	-	Column	Ok	Ok	Ok	Ok
M14	Initial	-	Column	Ok	Ok	Ok	Ok
M15	Initial	-	Column	Ok	Ok	Ok	Ok
M16	Initial	-	Column	Ok	Ok	Ok	Ok
M17	Initial	-	Column	Ok	Ok	Ok	Ok
M18	Initial	-	Column	Ok	Ok	Ok	Ok
M19	Initial	-	Column	Ok	Ok	Ok	Ok
M20	Initial	-	Column	Ok	Ok	Ok	Ok

Let us now review the design results of Column A1

The Main Layout sub-page gives the output below, showing the column top and bottom design moments.



The shear Layout page of the member gave the output below, showing the shear force on the column, and he links provided



The main reinforcement design result is as shown below.

STRUCTVILLE RESIDENTIAL - COLUMN DESIGN GROUP - Main Reinforcement

Mem	L/C	Axial kN	Major kNm	Minor kNm	Design Axis	Unity Ratio	Total Bars	As Prov. mm²
M1	C3	74.201	2.427	3.218	Biaxl min	0.191	4X16	804
M2	C3	148.346	4.343	5.881	Biaxl min	0.282	4X16	804
M3	C3	100.283	3.448	2.533	Biaxl maj	0.192	4X16	804
M4	C3	183.828	4.336	5.121	Biaxl min	0.232	4X16	804
M5	C3	105.328	3.305	3.709	Biaxl min	0.223	4X16	804
M6	C3	169.082	4.831	4.588	Biaxl maj	0.243	4X16	804
M7	C3	157.085	6.380	5.033	Biaxl maj	0.308	4X16	804
M8	C3	130.684	6.436	6.292	Biaxl maj	0.374	4X16	804
M9	C3	217.672	6.301	6.448	Biaxl min	0.290	4X16	804
M10	C3	201.632	5.805	8.217	Biaxl min	0.336	4X16	804
M11	C3	163.207	3.931	6.162	Biaxl min	0.266	4X16	804

The shear reinforcement design result is as shown below.

STRUCTVILLE RESIDENTIAL - COLUMN DESIGN GROUP - Shear Reinforcement

Mem	Max Shear in Local y			Max Shear in Local z			Asw Req. mm²	Link Size	Spacing cm
	L/C	Value kN	Position m	L/C	Value kN	Position m			
M1	C3	0.349	0.000	C3	0.557	0.000	Min	8	22.5
M2	C3	0.493	0.000	C3	0.961	0.000	Min	8	22.5
M3	C3	0.492	0.000	C3	0.197	0.000	Min	8	22.5
M4	C3	0.231	0.000	C3	0.490	0.000	Min	8	22.5
M5	C3	0.364	0.000	C3	0.544	0.000	Min	8	22.5
M6	C3	0.494	0.000	C3	0.391	0.000	Min	8	22.5
M7	C3	1.082	0.000	C3	0.608	0.000	Min	8	22.5
M8	C3	1.254	0.000	C3	1.246	0.000	Min	8	22.5
M9	C3	0.644	0.000	C3	0.699	0.000	Min	8	22.5
M10	C3	0.608	0.000	C3	1.391	0.000	Min	8	22.5

The drawing can also be exported to AUTOCAD via DXF format

Column Design Report

Member 8 - Detailed EC2 Main Reinforcement

Axial N		= 130.68 kN	
Major M_y	end 1,	= -1.82 kNm	
	end 2,	= 3.82 kNm	
Minor M_z	end 1,	= -1.93 kNm	
	end 2,	= 3.68 kNm	
Slenderness			
λ_{lim}	= 20 ABC / $\sqrt{n}$		(5.13N)
n	= $N_{Ed} / (A_c f_{cd}) = 131 \times 10^3 / (52900 \times 14.2)$	= 0.174	
Major $\lambda_{lim,y}$	= $20 \times 1.00 \times 1.36 \times 2.39 / \sqrt{0.174}$	= 155.99	
Minor $\lambda_{lim,z}$	= $20 \times 1.00 \times 1.36 \times 2.42 / \sqrt{0.174}$	= 158.11	
λ	= l_0 / i		(5.14)
λ_y	= 4500 / 66.40	= 67.78	
λ_z	= 4500 / 66.40	= 67.78	
Column is	NOT SLENDER		

Axial Capacity N_{Rd}	= 1059.72 kN	
Axial capacity ratio N_{Ed} / N_{Rd}	= 0.12	
Axial Capacity ≥ Axial	∴ **OK for axial resistance**	
Major Axis Capacity M_{Rdy}	= 32.99 kNm	
Major Axis Capacity Ratio M_{Edy} / M_{Rdy}	= 0.20	
Major Axis Capacity ≥ Major Axis Moment	∴ **OK for moment resistance**	
Minor Axis Capacity M_{Rdz}	= 32.99 kNm	
Minor Axis Capacity Ratio M_{Edz} / M_{Rdz}	= 0.19	
Minor Axis Capacity ≥ Minor Axis Moment	∴ **OK for moment resistance**	
Biaxial Interaction Equation	$= \left(\frac{M_{Edz}}{M_{Rdz}}\right)^a + \left(\frac{M_{Edy}}{M_{Rdy}}\right)^a \le 1.0$	(5.39)
Where exponent a	= 1.02	
Biaxial Interaction Equation	$= \left(\frac{6.29}{32.99}\right)^{1.02} + \left(\frac{6.44}{32.99}\right)^{1.02}$	= 0.37
Biaxial Interaction Result ≤ 1.0	∴ **OK for biaxial resistance**	

Member 8 - Detailed EC2 Main Reinforcement Cont...

Max distance between bars		= 300 mm	
Largest actual space between bars		= 112 mm	
Min distance between bars	= d_g + 5 mm	= 25 mm	8.2 (2)
Smallest actual space between bars		= 112 mm	
		∴ **OK for spacing**	
Min area of reinforcement		= 106 mm²	9.5.2 (2)
Max area of reinforcement (even at laps)		= 4232 mm²	9.5.2 (3)
Actual area of main reinforcement (4T16)		= 804 mm²	
∴ area by which steel can increase at laps		= 3428 mm²	
Starter bar area		= 804 mm²	
		∴ **OK**	

Member 8 - Detailed EC2 Shear Reinforcement

Spacing provided, s		= 225 mm	
Major axis:			
Shear force, V_{Ed}		= 1.25 kN	
$V_{Rd,c}$	$= [C_{Rd,c} k (100 \rho_l f_{ck})^{1/3} + k_1 \sigma_{cp}] b_w d$		(6.2.a)
	$= [0.120 \times 2.000 \times (100 \times 0.020 \times 25)^{1/3} + 0.15 \times 2.470] \times 230 \times 179$		
		= 51.37 kN	
But	$\geq (v_{min} + k_1 \sigma_{cp}) b_w d$		(6.2.b)
	$= (0.49 + 0.15 \times 2.470) \times 230 \times 179$	= 35.63 kN	
Hence, $V_{Rd,c}$		= 51.37 kN	
$V_{Ed} \leq V_{Rd,c}$	∴only nominal links required		6.2.1(5)
Area of links provided (2T8), A_{sw}		= 100.53 mm²	
Minor axis:			
Shear force, V_{Ed}		= 1.25 kN	
$V_{Rd,c}$	$= [C_{Rd,c} k (100 \rho_l f_{ck})^{1/3} + k_1 \sigma_{cp}] b_w d$		(6.2.a)
	$= [0.120 \times 2.000 \times (100 \times 0.020 \times 25)^{1/3} + 0.15 \times 2.470] \times 230 \times 179$		
		= 51.37 kN	
But	$\geq (v_{min} + k_1 \sigma_{cp}) b_w d$		(6.2.b)
	$= (0.49 + 0.15 \times 2.470) \times 230 \times 179$	= 35.63 kN	
Hence, $V_{Rd,c}$		= 51.37 kN	
$V_{Ed} \leq V_{Rd,c}$	∴only nominal links required		6.2.1(5)
Area of links provided (2T8), A_{sw}		= 100.53 mm²	
Minimum size of link bars		= 8 mm	9.5.3 (1)
Actual size of link bars		= 8 mm	
		∴ **OK**	

You can see how detailed the calculation procedures are. Through this way, you can do the design and verification for the other members. All issues encountered and comparative analysis will be in the subsequent chapters.

CHAPTER 4

SECTION ANALYSIS AND DESIGN FORMULAE

4.1 Rectangular Section Stress Block

The rectangular section stress distribution block of EC2 is given as shown below;

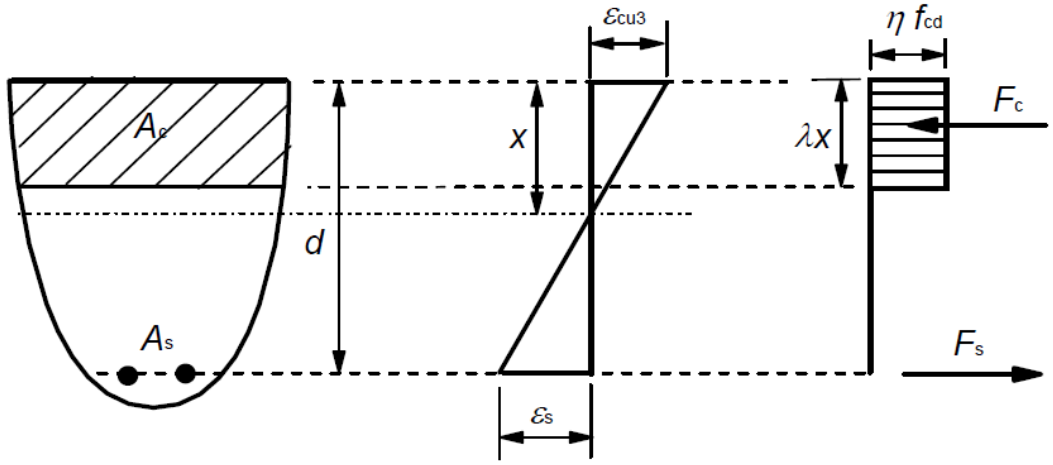


Figure 4.0: EC2 rectangular section stress block (Figure 3.5 EN 1992-1-1: 2004)

The values of λ and η are given in Table 4.0 below;

Table 4.0: Characteristic value of EC2 stress block parameters

	$F_{ck} \leq 50$ Mpa	$50 \leq F_{ck} \leq 90$ Mpa
λ	0.8	$0.8 - (f_{ck} - 50)/400$
η	1.0	$1.0 - (f_{ck} - 50)/200$

4.2 Analysis of singly reinforced section

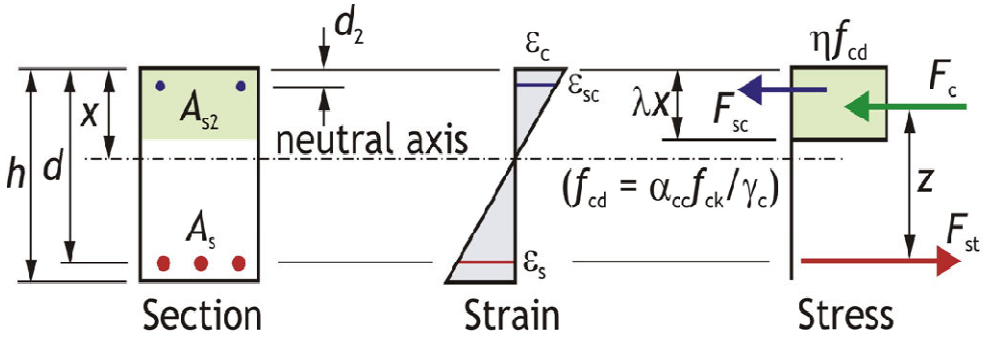


Figure 4.1: Analysis of singly reinforced section to EC2 (The Concrete Centre, 2006)

From EC2 singly reinforced concrete stress block;

$$M_{Rd} = F_C z \text{ ---------- (4.1)}$$

$f_{cd} = \frac{\alpha_{cc} f_{ck}}{\gamma_c} = \frac{0.85 \times f_{ck}}{1.5} = 0.5667 f_{ck}$

Compressive force in concrete = Design stress (fcd) x Area of compression block

$F_C = 0.5667 f_{ck} \times 0.8xb = 0.4533 f_{ck}\, xb$

From the stress block distribution;

$$z = d - 0.4x \text{ ---------- (4.2)}$$

Clause 5.6.3 of EC2 limits the depth of the neutral axis to *0.45d* for concrete class less than or equal to C50/60. Therefore for an under reinforced section (ductile);

$$x = 0.45d \text{ ---------- (4.3)}$$

Combining equation (4.1), (4.2) and (4.3), we obtain the ultimate moment of resistance (M_{Rd})

$$M_{Rd} = 0.167 f_{ck} bd^2 \text{ ---------- (4.4)}$$

Also from the reinforced concrete stress block, we can obtain the tensile design moment as;

$$M_{Ed} = F_S z \text{ ---------- (4.5)}$$

Where;

$$F_S = \frac{f_{yk}}{1.15} A_{s1} \text{ ---------- (4.6)}$$

A_{s1} = Area of tensile reinforcement required

Substituting equ (4.6) into (4.5) and making A_{s1} the subject of the formular;

$$A_{s1} = \frac{M_{Ed}}{0.87 f_{yk} z} \text{ ---------- (4.7)}$$

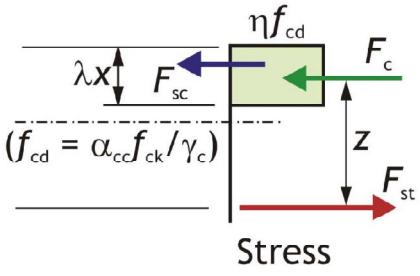
The lever arm z in EC2 is given;

From 4.2, $z = d - 0.4x$

Therefore, $x = 2.5(d - z)$

$M = 0.453 \times f_{ck} \times b \times 2.5(d - z)z$

Let $K = \frac{M}{f_{ck}\, bd^2}$



K can be considered as the normalised bending resistance

Hence;

$$\frac{M}{f_{ck}\, bd^2} = 1.1333 \left(\frac{f_{ck} bdz}{f_{ck} bd^2} - \frac{f_{ck} bz^2}{f_{ck} bd^2} \right)$$

Therefore;

$0 = 1.1333\left[\left(\frac{z}{d}\right)^2 - \left(\frac{z}{d}\right)\right] + K$

$0 = \left(\frac{z}{d}\right)^2 - \left(\frac{z}{d}\right) + 0.88235K$

Solving the quadratic equation;

$z/d = [1 + (1 - 3.529k)^{0.5}]/2$

Rearranging;

$$z = d\left[0.5 + \sqrt{\left(0.25 - \frac{k}{1.134}\right)}\right] \text{ ---------- (4.8)}$$

$z = d\left[0.5 + \sqrt{(0.25 - 0.882k)}\right]$

where ;

$$k = \frac{M_{Ed}}{f_{ck}\, bd^2} \text{ ---------- (4.9)}$$

4.3 Analysis of doubly reinforced sections

If the design ultimate moment is greater than the ultimate moment of resistance i.e. $M_{Ed} > M_{Rd}$, then compression reinforcement is required. Provided that $d_2/x \leq 0.38$ (i.e. compression steel has yielded) where d_2 is the depth of the compression steel from the compression face and $x = (d - z)/0.4$.

The area of compression reinforcement, A_{s2}, is given by:

Area of compression reinforcement $A_{S2} = \frac{M_{Ed} - M_{Rd}}{0.87 f_{yk}(d - d_2)}$ ---------- (4.10)

Area of tension reinforcement $A_{s1} = \frac{M_{Rd}}{0.87 f_{yk} z} + A_{S2}$ ---------- (4.11)

Where $z = d\left[0.5 + \sqrt{(0.25 - 0.882K')}\right]$

where

$K' = 0.167$

4.4 Shear Model of EC2

EC2 uses the variable strut inclination method for shear design. It is slightly more complex than the procedure in BS 8110 but can result in savings in the amount of shear reinforcement required. Like BS 8110, EC2 models shear behaviour on the truss analogy in which the concrete acts as the diagonal struts, the stirrups act as the vertical ties, the tension reinforcement forms the bottom chord, and the compression steel/ concrete forms the top chord. Whereas in BS 8110 the strut angle has a fixed value of 45° in EC2 it can vary between 21.8° and 45° and it is this feature which is responsible for possible reductions in the volume of shear reinforcement.

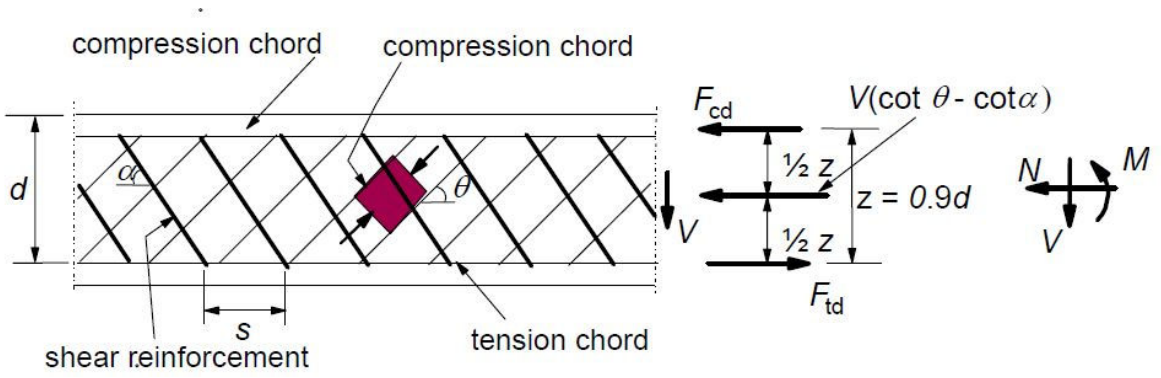


Figure 4.2: Truss model for shear reinforcement of a reinforced section to EC2 (Fig 6.5 EN 1992-1-1:2004)

where;

α = angle between shear reinforcement and the beam axis
θ = angle between the concrete compression strut and the beam axis

In EC2, the concrete resistance shear stress without shear reinforcement is given by;

$$V_{Rd,c} = [C_{Rd,c}.k.\ (100\rho_1 f_{ck})^{\frac{1}{3}} + k_1.\sigma_{cp}]bw.d \geq (V_{min} + k_1.\sigma_{cp}) \text{ ---------- (4.12)}$$

Where;

$C_{Rd,c} = 0.18/\gamma_c$;

$k = 1+\sqrt{\frac{200}{d}} < 0.02$ (d in mm);

$\rho_1 = \frac{A_{S1}}{bd} < 0.02$ (In which A_{S1} is the area of tensile reinforcement which extends $\geq (l_{bd} + d)$ beyond the section considered);

$V_{min} = 0.035k^{\frac{3}{2}}f_{ck}^{\frac{1}{2}}$

$K_1 = 0.15$; $\sigma_{cp} = N_{Ed}/Ac < 0.2fcd$

(Where N_{Ed} is the axial force at the section, Ac = cross sectional area of the concrete), f_{cd} = design compressive strength of the concrete).

The compression capacity of the compression strut ($V_{Rd,max}$) assuming θ = 21.8° (cot θ = 2.5)

$$V_{Rd,max} = \frac{b_w.z.v_1.f_{cd}}{\cot\theta + tan\theta} \text{ ---------- (4.13)}$$

$V_1 = 0.6\left(1 - \frac{f_{ck}}{250}\right)$

$f_{cd} = \frac{\alpha_{cc} f_{ck}}{\gamma_c}$

Let z = 0.9d

If $V_{Rd,c} < V_{Ed} < V_{Rd,max}$

$A_{sw}/S = \frac{V_{Ed}}{0.87\ F_{yk} z\ cot\ \theta}$

If $V_{Rd,max} > V_{Ed}$, calculate, the compression strut angle.

$$\theta = 0.5\sin^{-1}\frac{(V_{Rd,max}/b_w d)}{0.153 f_{ck}\left(1 - \frac{f_{ck}}{250}\right)} \text{ ---------- (4.14)}$$

If θ is greater than 45°, select another section.

Minimum shear reinforcement;

$A_{sw}/S = \rho_{w,min} \times b_w \times sin\alpha$ (α = 90° for vertical links)

$$\rho_{w,min} = \frac{0.08 \times \sqrt{F_{ck}}}{F_{yk}} \text{ ---------- (4.15)}$$

4.5 Deflection control

Deflection control is dealt with in Eurocode 2 in section 7.4. Section 7.4.2 deals with the control of deflection using simple 'deemed to satisfy' rules, while section 7.4.3 deals with the direct calculation of deflections.

The selection of limits to deflection which will ensure that the structure will be able to fulfil its required function is a complex process and it is not possible for a code to specify simple limits which will meet all requirements and still be economical. Limits are suggested in the code but these are for general guidance only; it remains the responsibility of the designer to check whether these are appropriate for the particular case considered or whether some other limits should be used *(Beeby and Narayanan,* 2009).

There are two basic issues which are considered to influence the choice of limits. These are appearance and function. Appearance is important because it is found that the occupants of structures find it upsetting if the floors appear to be sagging. Some research has been carried out on this highly subjective aspect of deflection control and it is generally accepted that sag will be unnoticeable provided the central deflection of a beam relative to its supports is less than about span/250.

Examples of situations where deflections may lead to impairment of function are:

1. Deflection of beams or slabs leading to cracking of partitions supported by the member considered.
2. Deflections causing doors to jam or windows to break.
3. Varying deflections leading to misalignment of apparatus or machinery mounted on the member considered.
4. Damage to brittle finishes.
5. Unacceptable vibrations or an upsetting feeling of 'liveliness' in the structure.

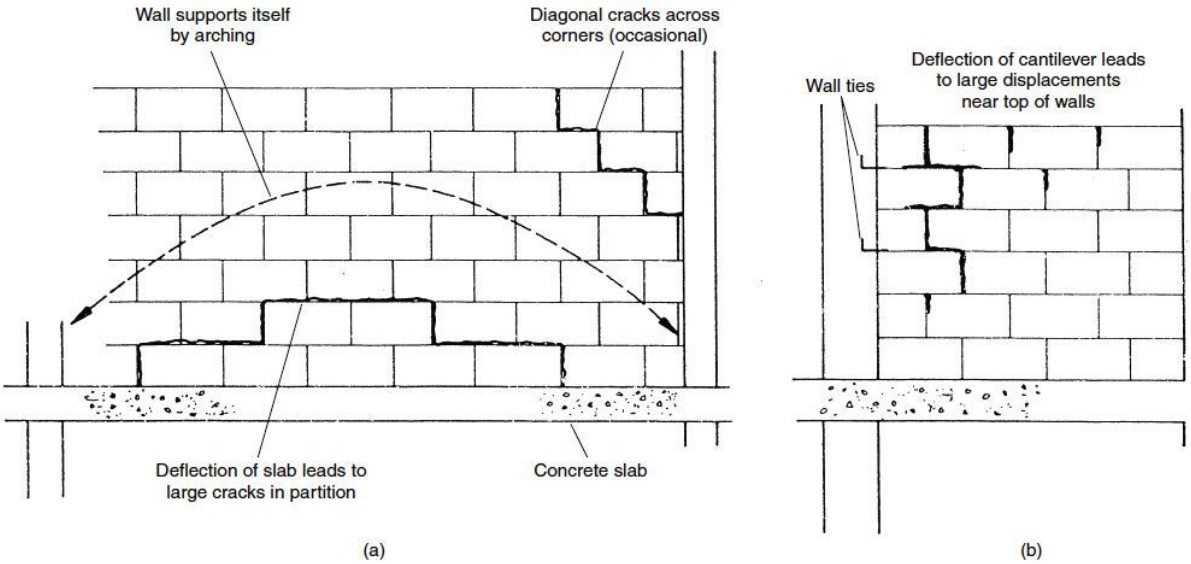


Figure 4.3: Example of damages to partition due to deflection (a) Beam (b) Cantilever (From *Beeby and Narayanan,* 2009)

According to Clause 7.4.2 of EN 1992-1-1

For deemed to satisfy basic span/effective depth (limiting to depth/250);

Actual L/d must be ≤ Limiting L/d × β_s

The limiting basic span/ effective depth ratio is given by;

$$L/d = K\left[11 + 1.5\sqrt{f_{ck}}\frac{\rho_0}{\rho} + 3.2\sqrt{f_{ck}}\left(\frac{\rho_0}{\rho} - 1\right)^{3/2}\right] \text{ if } \rho \leq \rho_0 \quad \text{---------- (4.16)}$$

$$L/d = K\left[11 + 1.5\sqrt{f_{ck}}\frac{\rho_0}{\rho - \rho'} + \frac{1}{12}\sqrt{f_{ck}}\left(\frac{\rho_0}{\rho}\right)^{1/2}\right] \text{ if } \rho > \rho_0 \quad \text{---------- (4.17)}$$

Where;

L/d is the limiting span/depth ratio
K = Factor to take into account different structural systems
ρ_0 = reference reinforcement ratio = $10^{-3}\sqrt{f_{ck}}$
ρ = Tension reinforcement ratio to resist moment due to design load
ρ' = Compression reinforcement ratio

The value of K depends on the structural configuration of the member, and relates the basic span/depth ration of reinforced concrete members. This is given in Table 4.1 below;

Table 4.1: Basic span/Effective depth ratio of different structural systems

Structural System	**k**	**Highly Stressed ρ = 1.5%**	**Lightly stressed ρ = 0.5%**
Simply supported beams and slabs	1.0	14	20
End span of interior beams and slabs	1.3	18	26
Interior span of continuous beams and slabs	1.5	20	30
Cantilever beams and slabs	0.4	6	8

$$\beta_s = \frac{500\,A_{sprov}}{F_{yk}\,A_{sreq}} \quad \text{---------- (4.18)}$$

Important hints on the calculation of deflection

1. For flanged sections with *b/bw* ≥ 3, the basic ratios for rectangular sections should be multiplied by 0.8. For values of *b/bw* < 3, the basic ratios for rectangular sections should be multiplied by (11 - *b/bw)/10*

2. The ratio should be based on the shorter span for two-way spanning slabs, and the longer span for flat slabs.

3. For beams and slabs, other than flat slabs, with spans exceeding 7 m, which support partitions liable to be damaged by excessive deflections, the basic ratio should be multiplied by 7/span. For flat slabs, where one or both spans exceeds 8.5 m, which support partitions liable to be damaged by excessive deflections, the basic ratio should be multiplied by 8.5/span.

4.6 Cracking control

According to section 7.3.1 of EN 1992-1-1:2004, cracking is normal in reinforced concrete structures subject to bending, shear, torsion or tension resulting from either direct loading or restraint or imposed deformations. However, cracking shall be limited to an extent that will not impair the proper functioning or durability of the structure or cause its appearance to be unacceptable.

In the absence of specific requirements (e.g. water-tightness), it may be assumed that limiting the calculated crack widths to the values of w_{max} given in Table 7.1N, under the quasi-permanent combination of loads, will generally be satisfactory for reinforced concrete members in buildings with respect to appearance and durability (EC2 clause 7.3.1 (5)).

Table 4.2: Recommended values of maximum crack width W_{max} (Table 7.1N EN 1992-1-1:2004)

Exposure Class	Reinforced members and prestressed members with unbonded tendons	Prestressed members with bonded tendons
	Quasi-permanent load combination	Frequent load combination
X0, XC1	0,4[1]	0,2
XC2, XC3, XC4	0,3	0,2[2]
XD1, XD2, XS1, XS2, XS3		Decompression

Note 1: For X0, XC1 exposure classes, crack width has no influence on durability and this limit is set to guarantee acceptable appearance. In the absence of appearance conditions this limit may be relaxed.
Note 2: For these exposure classes, in addition, decompression should be checked under the quasi-permanent combination of loads.

Crack widths may be calculated according to clause 7.3.4 of EC2. A simplified alternative is to limit the bar size or spacing according to clause 7.3.3. If crack control is required, a minimum amount of bonded reinforcement is required to control cracking in areas where tension is expected. The amount may be estimated from equilibrium between the tensile force in concrete just before cracking and the tensile force in reinforcement at yielding or at a lower stress if necessary to limit the crack width.

4.6.1 Control of cracking without direct calculation

For reinforced or prestressed slabs in buildings subjected to bending without significant axial tension, specific measures to control cracking are not necessary where the overall depth does not exceed 200 mm and the detailing guides are adequately followed (see Table 9.2 in chapter 9). However, the bar size and spacing for cracking control is shown in Table 4.3 and 4.4 below.

Table 4.3: Maximum bar diameter for crack control (Table 7.2N EN 1992-1-1:2004)

Steel stress[2] [MPa]	Maximum bar size [mm] w_k= 0,4 mm	w_k= 0,3 mm	w_k= 0,2 mm
160	40	32	25
200	32	25	16
240	20	16	12
280	16	12	8
320	12	10	6
360	10	8	5
400	8	6	4
450	6	5	-

Table 4.4: Maximum bar spacing for crack control (Table 7.3N EN 1992-1-1:2004)

Steel stress[2] [MPa]	Maximum bar spacing [mm]		
	w_k=0,4 mm	w_k=0,3 mm	w_k=0,2 mm
160	300	300	200
200	300	250	150
240	250	200	100
280	200	150	50
320	150	100	-
360	100	50	-

CHAPTER 5

ANALYSIS AND DESIGN OF THE FLOOR SLABS

5.1 Introduction

Floor slabs in reinforced concrete construction are normally classified as spanning in one direction or in two directions. The common criteria for judging this classification is the ratio (k) of the long span (Ly) to the short span (Lx), and if the slab is only supported on two opposite sides. When *k* is greater than 2, the slab is treated as spanning in one direction; however when *k* is less than 2, the slab is treated as spanning in two directions. Slabs are normally designed per unit width, hence in design we normally take b = 1000mm.

For one-way slabs, the bending moments are calculated in the same way as for beams. Continuity across a beam can be treated as a fixed support (see analysis of PANELS 6 and 9), if the adjacent slab is spanning in two directions, otherwise it can be analysed like a continuous beam. In detailing, if a slab is assumed to be simply supported at an end support, it is advisable to provide reinforcement for a probable negative bending moment due to the monolithic construction of beams and slabs (*Reynolds and Steedman,* 2005). In practice, this is normally attained by bending the bottom reinforcement bars to return to the top (shape code 38) and extending it into the span by 0.2*L* from the face of the support (Clause 9.3.1.2 EN 1992-1-1:2004).

Approximate analysis is normally employed in the analysis of two-way slabs. The most popular is the theory of plates utilising the elastic analysis. A slab may be assumed to be freely supported when the slab is not continuous, and the edge bears on a block work. If the slab is monolithic with the beam, then a partial fixity exists. Coefficients for analysis of factored loads on two-way slabs can be found in BS 8110-1:1997 (same with the Eurocodes) and has been extensively used in this book.

It is necessary to point out that the critical span in the analysis of solid slabs is the short span. Therefore, the main reinforcements will lie parallel to the short span, and that is where you check your deflections. For one way slabs, a transverse (secondary) reinforcement of not less than 20% of the main reinforcement should be provided as distribution bars. It is also required that the spacing of the main reinforcements should not be more than 3*h* or 400mm whichever is less. For secondary reinforcements, this should not exceed 3.5*h* or 450mm whichever is less (h is the thickness of the slab) (see clause 9.3.1.1 of EN 1992:1-1:2004). In a slab where shear reinforcement must be provided, the thickness must be greater than 200mm. More often than not, it is the control of deflection that influences the area of steel and thickness of slabs than flexural requirements. When a floor slab deflects excessively, the finishes such as tiles may crack, and of course the occupants of the building will not be happy.

All other assumptions we normally make in design of slabs will be commented on as the analysis progresses.

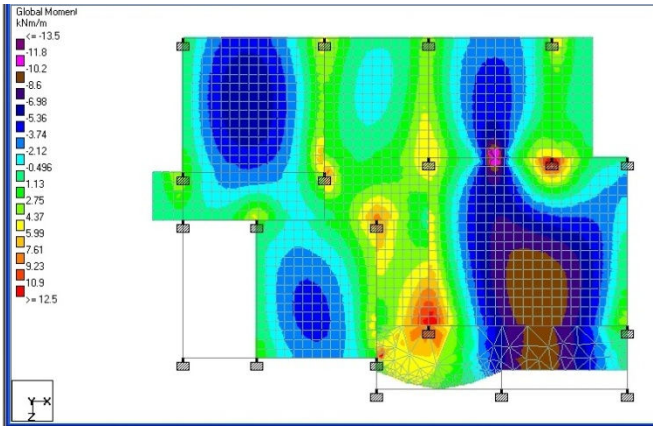


Figure 5.1: Global Moment Contour Result from Staad

5.2 Load Analysis

Permanent Loads

Self weight of slab = 25 KN/m^3 × 0.15m = 3.75 KN/m^2

Weight of finishes (assume) = 1.2 KN/m^2

Partition allowance = 1.5 KN/m^2

Total dead load (Gk) = 6.45 KN/m^2

Variable Load on slab

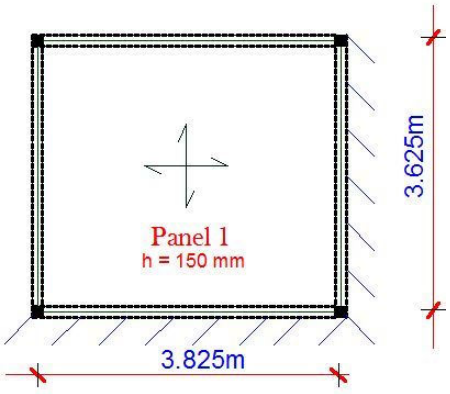
Leading variable action (Imposed load) Qk_1 = 1.5 KN/m^2

Ultimate limit state combination = $\gamma_{GK}G_k + \gamma_{QK}G_k$

Total load on slab (ULS) = 1.35Gk + 1.5Qk = 1.35(6.45) + 1.5(1.5) = 10.9575 KN/m^2

5.3 Structural Analysis and design of the slabs

PANEL 1



The floor slab (PANEL 1) is spanning in two directions, since the ratio (k) of the longer side (Ly) to the shorter side (Lx) is less than 2.

Hence, $k = \frac{Ly}{Lx} = \frac{3.825}{3.625} = 1.055$ (say 1.1)

Moment coefficient for two adjacent edges discontinuous

Short Span	Long Span
Mid-span = 0.042	Mid-span = 0.034
Continuous edge = 0.056	Continuous edge = 0.045

Design of short span

Mid span

M = $\alpha n L_x^2$ = 0.042 × 10.9575 × 3.625^2 = 6.0475 KN.m

M_{Ed} = 6.0475 KNm

d = h – Cc – ϕ/2

Assuming ϕ12mm bars will be employed for the construction

d = 150 – 25 – 6 = 119mm; b = 1000mm (designing per unit width)

$$k = \frac{M_{Ed}}{f_{ck}\, bd^2} = \frac{6.0475 \times 10^6}{25 \times 1000 \times 119^2} = 0.0171$$

Since k < 0.167 No compression reinforcement required

$$z = d\left[0.5 + \sqrt{(0.25 - 0.882K)}\right] = z = d\left[0.5 + \sqrt{(0.25 - 0.882(0.0273)}\right] = 0.95d$$

$$A_{s1} = \frac{M_{Ed}}{0.87 f_{yk} z} = \frac{6.0475 \times 10^6}{0.87 \times 460 \times 0.95 \times 119} = 133.668 \text{ mm}^2/\text{m}$$

ProvideY12mm @ 250mm c/c BOT (AS_{prov} = 452 mm^2/m)

To calculate the minimum area of steel required;
fctm = $0.3 \times f_{ck}^{2/3}$ = $0.3 \times 25^{2/3}$ = 2.5649 N/mm^2 (Table 3.1 EC2)

A_{Smin} = $0.26 \times \frac{fctm}{Fyk} \times b \times d = 0.26 \times \frac{2.5649}{460} \times 1000 \times 144$ = 208.76 mm^2/m
Check if $A_{Smin} < 0.0013 \times b \times d$ (187.2 mm^2/m)
Since, A_{Smin} = 187.2 mm^2, the provided reinforcement is adequate.

Check for deflection;

We check for deflection at the short span of slabs
K = 1.3 for slab simply supported at one end and continuous at the other end

$\rho = \frac{As}{bd} = \frac{452}{1000 \times 119}$ = 0.003798 < $10^{-3}\sqrt{25}$
Since $\rho < \rho_0$

L/d = $K\left[11 + 1.5\sqrt{f_{ck}}\frac{\rho_0}{\rho} + 3.2\sqrt{f_{ck}}\left(\frac{\rho_0}{\rho} - 1\right)^{3/2}\right]$

L/d = $1.3\left[11 + 1.5\sqrt{25} \times \frac{0.005}{0.003798} + 3.2\sqrt{25}\left(\frac{0.005}{0.003798} - 1\right)^{3/2}\right]$ = 1.3(20.873 + 2.8486) = 30.838

Modification factor $\beta_s = \frac{310}{\sigma_s}$

$\sigma_s = \frac{310 f_{yk} AS_{req}}{500\ AS_{prov}} = \frac{310 \times 460 \times 133.688}{500 \times 452}$ = 84.341 N/mm^2

$\beta_s = \frac{310}{84.342}$ = 3.67 > 2.0 (take 2.0)

Taking the distance between supports as the effective span, L = 3625mm
The allowable span/depth ratio = $\beta_s \times 30.838 = 2.0 \times 30.838$ = 61.676
Actual deflection L/d = $\frac{3625}{119}$ = 30.462
Since 61.676 < 30.462 Therefore, deflection is ok.

Shear Verification
Maximum shear force coefficient for slab(β) = 0.44
$V_{Ed} = \beta n L_x$
Ultimate shear force V_{Ed} = 0.44 × 10.9575 × 3.625 = 17.477 KN/m
$V_{Rd,c}$ = [$C_{Rd,c}$.k. $(100\rho_1 f_{ck})^{\frac{1}{3}}$ + $k_1.\sigma_{cp}$]*bw.d* ≥ (Vmin + $k_1.\sigma_{cp}$) *bw.d*

$C_{Rd,c}$ = 0.18/γ_c = 0.18/1.5 = 0.12

k = $1+\sqrt{\frac{200}{d}}$ = $1+\sqrt{\frac{200}{119}}$ = 2.296 > 2.0, therefore, k = 2
$V_{min} = 0.035 k^{\frac{3}{2}} f_{ck}^{\frac{1}{2}}$ = Vmin = $0.035 \times (2)^{\frac{3}{2}} \times (25)^{\frac{1}{2}}$ = 0.494 N/mm^2

$\rho_1 = \frac{As}{bd} = \frac{452}{1000 \times 119}$ = 0.003798 < 0.02; K_1 = 0.15

σ_{cp} = N_{Ed}/Ac < 0.2fcd (Where N_{Ed} is the axial force at the section, Ac = cross sectional area of the concrete), fcd = design compressive strength of the concrete.) Take N_{Ed} = 0
$V_{Rd,c}$ = $[0.12 \times 2 \times (100 \times 0.003798 \times 25)^{\frac{1}{3}}] \times 1000 \times 119$ = 60476.947 N = 60.476 KN
Since $V_{Rd,c}$ (60.476 KN) > V_{Ed} (17.477 KN), no shear reinforcement is required.

Note that shear reinforcements are rarely required in solid slabs supported by beams. Hence no further checks for shear will be done on slabs.

Design of short span

Continuous edge

M = $\alpha n L_x^2$ = 0.056 × 10.9575 × 3.625^2 = 8.0633 KN.m

M_{Ed} = 8.0633 KNm

K = 0.0228; La/d = 0.95; A_{S1} = 178.2236 mm^2; A_{Smin} = 172.5217 mm^2;

Provide Y12 @ 250mm c/c Top (A_{Sprov} = 452mm^2)

Design of long span

Mid span

M = $\alpha n L_x^2$ = 0.034 × 10.9575 × 3.625^2 = 4.895 KN.m

M_{Ed} = 4.895 KNm

K = 0.0138; La/d = 0.95; A_{S1} = 108.1945 mm^2; A_{Smin} = 172.5217 mm^2;

Provide Y12 @ 250mm c/c N/Bottom (A_{Sprov} = 452mm^2)

Design of long span

Continuous edge

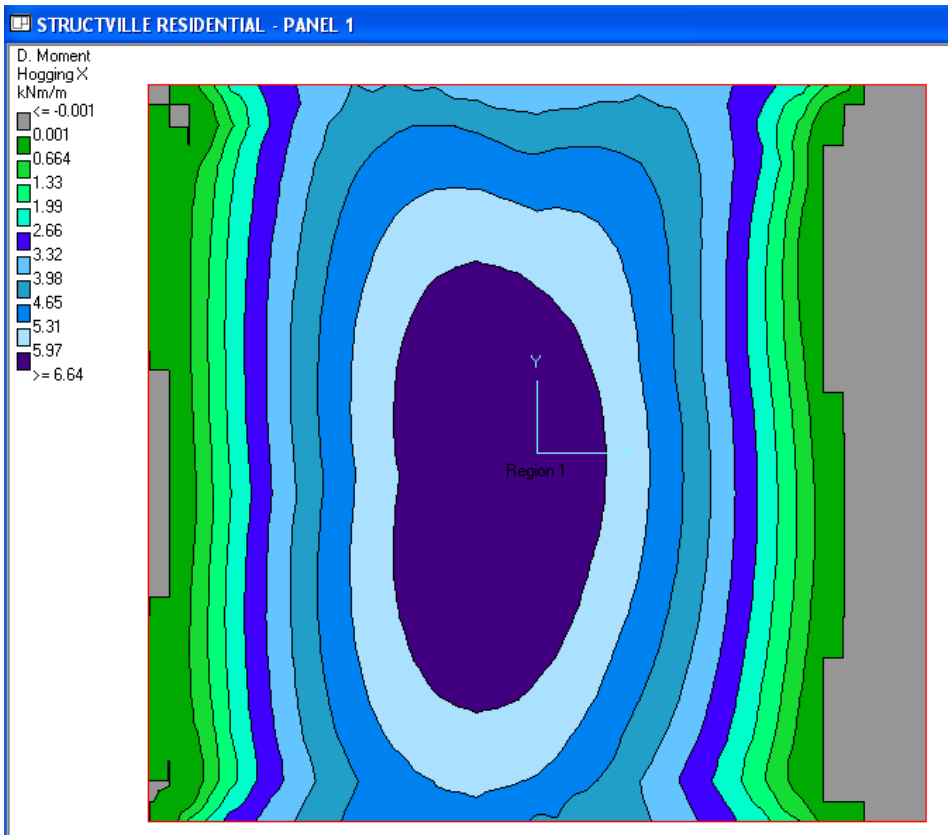
M = $\alpha n L_x^2$ = 0.045 × 10.9575 × 3.625^2 = 6.479 KN.m

M_{Ed} = 6.479 KNm

K = 0.0183; La/d = 0.95; A_{S1} = 143.2057 mm^2; A_{Smin} = 172.5217 mm^2;

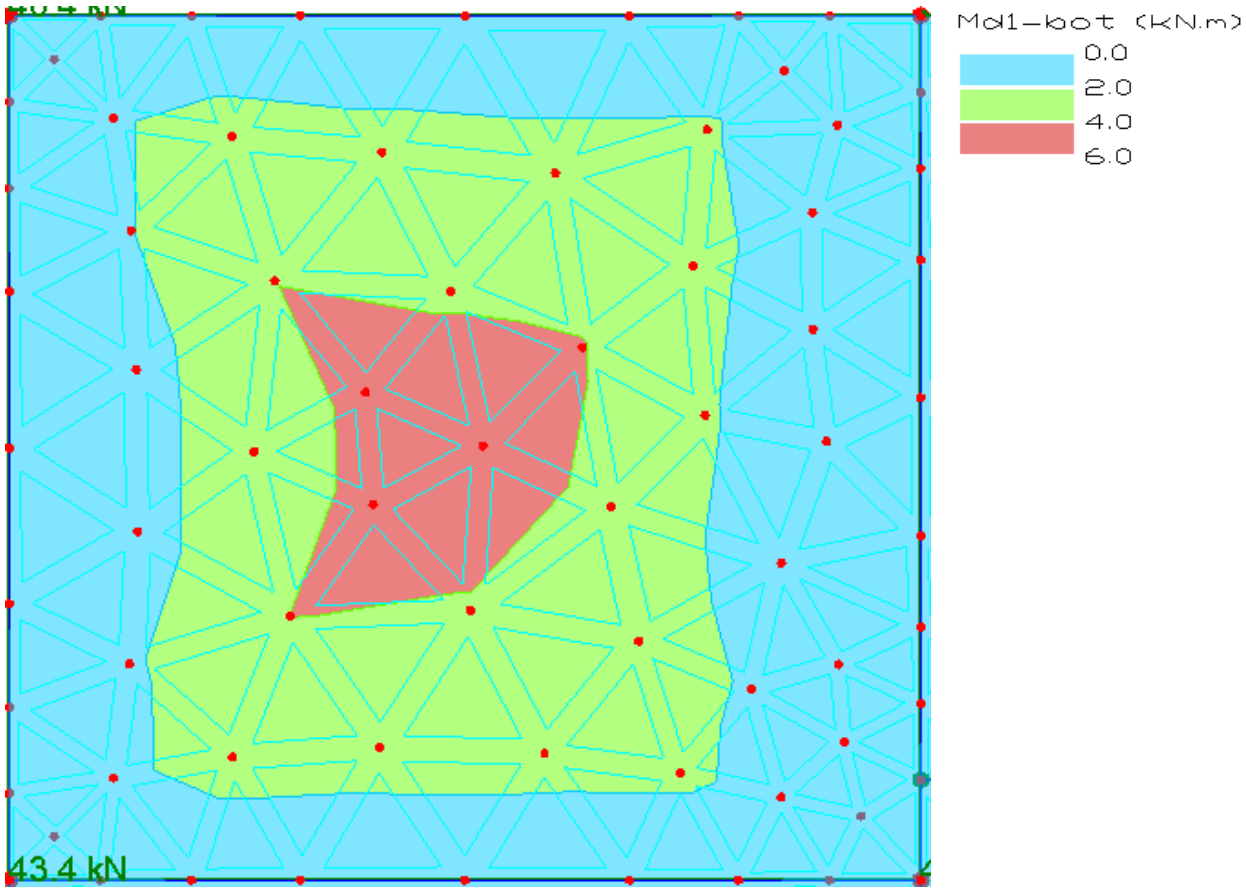
Provide Y12 @ 250mm c/c Top (A_{Sprov} = 452mm^2)

Result from Staad (PANEL 1)



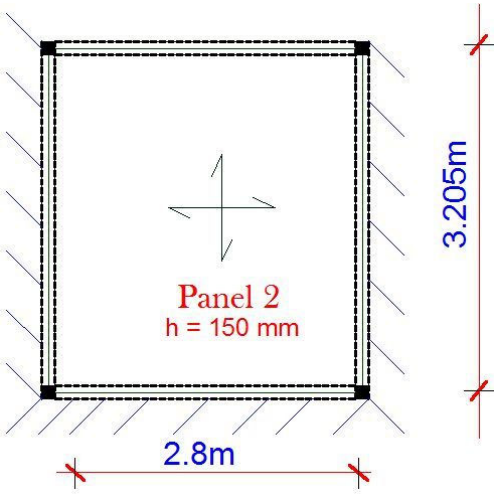
Design bottom Moment (short span) = 6.64 KNm/m

Result from Orion (PANEL 1)



As can be seen, manual and computer results are giving approximately the same answer in terms of design moment for PANEL 1.

PANEL 2



The floor slab (PANEL 2) is spanning in two directions, since the ratio (k) of the longer side (Ly) to the shorter side (Lx) is less than 2.

Hence, $k = \frac{Ly}{Lx} = \frac{3.205}{2.8} = 1.144$ (say 1.15, take average)

Moment coefficient for one short edge discontinuous

Short Span	Long Span
Mid-span = 0.0345	Mid-span = 0.028
Continuous edge = 0.046	Continuous edge = 0.037

Design of short span

Mid span

M = $\alpha n L_x^2$ = 0.0345 × 10.9575 × 2.8^2 = 2.9637 KN.m

M_{Ed} = 2.9637KNm

K = 0.0084; La/d = 0.95; A_{S1} = 65.5068 mm^2; A_{Smin} = 172.5217 mm^2;

Provide Y12 @ 250mm c/c Bottom (A_{Sprov} = 452mm^2)

Permissible L/d = 101.3240; Actual L/d = 23.5394 (Deflection is ok)

Design of short span

Continuous edge

M = $\alpha n L_x^2$ = 0.046 × 10.9575 × 2.8^2 = 3.9517 KN.m

M_{Ed} = 3.9517 KNm

K = 0.0111; La/d = 0.95; A_{S1} = 86.5489 mm^2; A_{Smin} = 172.5217 mm^2;

Provide Y12 @ 250mm c/c Top (A_{Sprov} = 452mm^2)

Design of long span

Mid span

M = $\alpha n L_x^2$ = 0.028 × 10.9575 × 2.8^2 = 2.405 KN.m

M_{Ed} = 2.405 KNm

K = 0.0068; La/d = 0.95; A_{S1} = 53.1579 mm^2; A_{Smin} = 172.5217 mm^2;

Provide Y12 @ 250mm c/c N/Bottom (A_{Sprov} = 452mm^2)

Design of long span

Continuous edge

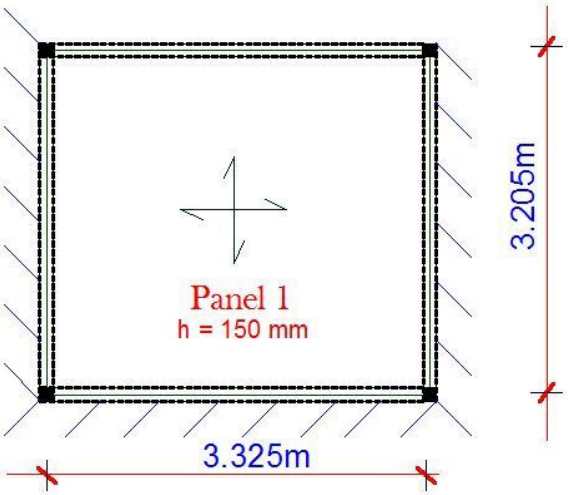
M = $\alpha n L_x^2$ = 0.037 × 10.9575 × 2.8^2 = 3.178 KN.m

M_{Ed} = 3.178 KNm

K = 0.0090; La/d = 0.95; A_{S1} = 70.2435 mm^2; A_{Smin} = 172.5217 mm^2;

Provide Y12 @ 250mm c/c Top (A_{Sprov} = 452mm^2)

PANEL 3



The floor slab (PANEL 3) is spanning in two directions, since the ratio (k) of the longer side (Ly) to the shorter side (Lx) is less than 2.

Hence, $k = \frac{Ly}{Lx} = \frac{3.325}{3.205}$ = 1.037 (say 1.0)

Moment coefficient for one long edge discontinuous

Short Span	Long Span
Mid-span = 0.030	Mid-span = 0.028
Continuous edge = 0.039	Continuous edge = 0.037

Design of short span

Mid span

M = $\alpha n L_x^2$ = 0.030 × 10.9575 × 3.205^2 = 3.376 KN.m

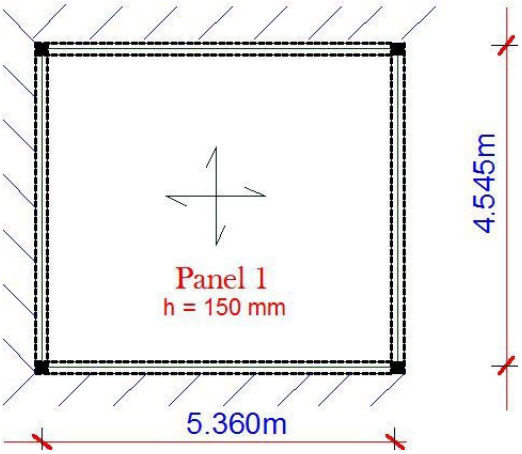
M_{Ed} = 3.376 KNm
K = 0.0095; La/d = 0.95; A_{S1} = 74.6199 mm^2; A_{Smin} = 172.5217 mm^2;
Provide Y12 @ 250mm c/c Bottom (A_{Sprov} = 452mm^2)
Permissible L/d = 87.8141; Actual L/d = 26.9328 (Deflection is ok)

Design of short span
Continuous edge
M = $\alpha n L_x^2$ = 0.039 × 10.9575 × 3.205^2 = 4.389 KN.m
M_{Ed} = 4.389 KNm
K = 0.0124; La/d = 0.95; A_{S1} = 97.0103 mm^2; A_{Smin} = 172.5217 mm^2;
Provide Y12 @ 250mm c/c Top (A_{Sprov} = 452mm^2)

Design of long span
Mid span
M = $\alpha n L_x^2$ = 0.028 × 10.9575 × 3.205^2 = 3.152 KN.m
M_{Ed} = 3.152 KNm
K = 0.0068; La/d = 0.95; A_{S1} = 53.1579 mm^2; A_{Smin} = 172.5217 mm^2;
Provide Y12 @ 250mm c/c N/Bottom (A_{Sprov} = 452mm^2)

Design of long span
Continuous edge
M = $\alpha n L_x^2$ = 0.037 × 10.9575 × 3.205^2 = 4.1645 KN.m
M_{Ed} = 4.1645KNm
K = 0.0118; La/d = 0.95; A_{S1} = 92.0482 mm^2; A_{Smin} = 172.5217 mm^2;
Provide Y12 @ 250mm c/c Top (A_{Sprov} = 452mm^2)

PANEL 4



The floor slab (PANEL 4) is spanning in two directions, since the ratio (k) of the longer side (Ly) to the shorter side (Lx) is less than 2.

Hence, $k = \frac{Ly}{Lx} = \frac{5.360}{4.545}$ = 1.179 (say 1.2)

Moment coefficient for one short edge discontinuous

Short Span	Long Span
Mid-span = 0.036	Mid-span = 0.028
Continuous edge = 0.048	Continuous edge = 0.037

Design of short span
Mid span
M = $\alpha n L_x^2$ = 0.036 × 10.9575 × 4.545^2 = 8.148 KN.m
M_{Ed} = 8.148 KNm
K = 0.0230; La/d = 0.95; A_{S1} = 180.0957 mm^2; A_{Smin} = 172.5217 mm^2;
Provide Y12 @ 250mm c/c Bottom (A_{Sprov} = 452mm^2)
Permissible L/d = 97.0628; Actual L/d = 38.1933 (Deflection is ok)

Design of short span

Continuous edge

M = $\alpha n L_x^2$ = 0.048 × 10.9575 × 4.545^2 = 10.865 KN.m

M_{Ed} = 10.865 KNm

K = 0.0307; La/d = 0.95; A_{S1} = 239.9508 mm^2; A_{Smin} = 172.5217 mm^2;

Provide Y12 @ 250mm c/c Top (A_{Sprov} = 452mm^2)

Design of long span

Mid span

M = $\alpha n L_x^2$ = 0.028 × 10.9575 × 4.545^2 = 6.337 KN.m

M_{Ed} = 6.338 KNm

K = 0.0179; La/d = 0.95; A_{S1} = 140.0671 mm^2; A_{Smin} = 172.5217 mm^2;

Provide Y12 @ 250mm c/c N/Bottom (A_{Sprov} = 452mm^2)

Design of long span

Continuous edge

M = $\alpha n L_x^2$ = 0.037 × 10.9575 × 4.545^2 = 8.3749 KN.m

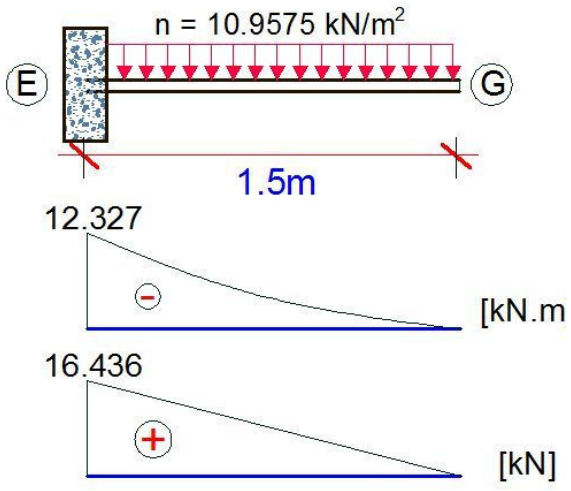
M_{Ed} = 3.178 KNm

K = 0.0237; La/d = 0.95; A_{S1} = 185.1109 mm^2; A_{Smin} = 172.5217 mm^2;

Provide Y12 @ 250mm c/c Top (A_{Sprov} = 452mm^2

PANEL 5

In this case, we are going to assume (PANEL 5) as a 1.5m long cantilever spanning along the short span. This assumption is incorrect but conservative. It is very usual to increase the live load of balconies, but in this case, let us assume that the same live load is distributed all across the entire floor slab. Let the partition allowance load of 1.5 KN/m^2 (which is not possible on balconies in this case) take care of this.



Design of short span

Support moment

M_{Ed} = 12.327 KNm

K = 0.0348; La/d = 0.95; A_{S1} = 272.4644 mm^2; A_{Smin} = 172.5217 mm^2;

Provide Y12 @ 250mm c/c Top (A_{Sprov} = 452mm^2)

Permissible L/d = 17.1086; Actual L/d = 12.6050 (Deflection is ok)

Shear Design

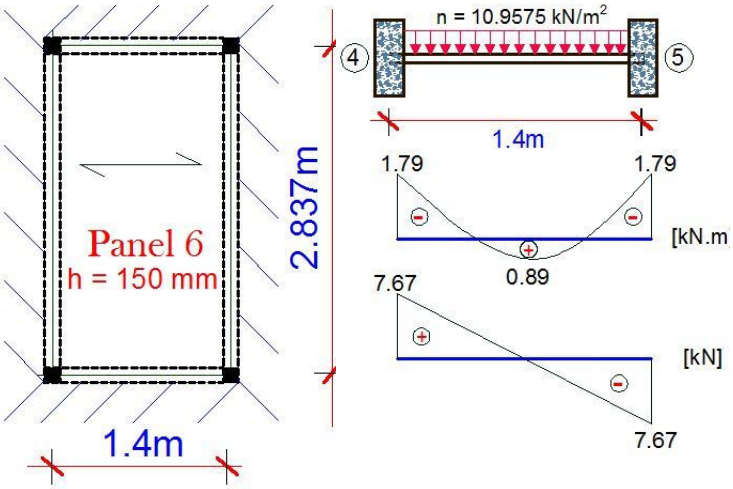
SupportE; V_{Ed} = 16.436 KN

Using the area of tension reinforcement provided (A_{Sprov} = 452 mm^2/m)

$V_{Rd,c}$ = 60.4786 KN > V_{Ed} (Therefore , no shear reinforcement is required)

This is very ok !!!!

PANEL 6

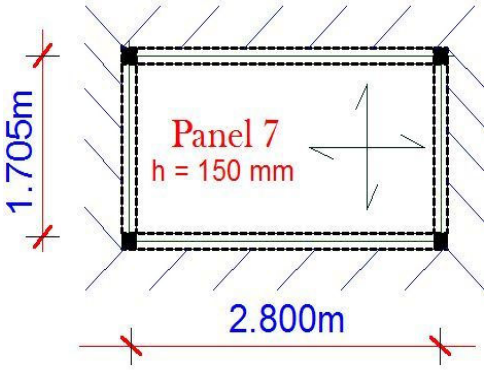


The floor slab (PANEL 6) is spanning in one direction, since the ratio (k) of the longer side (Ly) to the shorter side (Lx) is greater than 2.

Hence, $k = \frac{Ly}{Lx} = \frac{2.837}{1.4} = 2.0264$

From our previous designs without much ado, provide Y12 @ 250mm c/c Top and Bottom (A_{Sprov} = 452mm^2

PANEL 7



The floor slab (PANEL 7) is spanning in two directions, since the ratio (k) of the longer side (Ly) to the shorter side (Lx) is less than 2.

Hence, $k = \frac{Ly}{Lx} = \frac{2.8}{1.705} = 1.642$

Moment coefficient for all edges continuous

Short Span	Long Span
Mid-span = 0.042	Mid-span = 0.024
Continuous edge = 0.057	Continuous edge = 0.032

Design of short span

Mid span

M = $\alpha n L_x^2$ = 0.042 × 10.9575 × 1.705^2 = 1.338 KN.m

M_{Ed} = 1.338 KNm

K = 0.0038; La/d = 0.95; A_{S1} = 29.5739 mm^2; A_{Smin} = 172.5217 mm^2;

Provide Y12 @ 250mm c/c Bottom (A_{Sprov} = 452mm^2)

Permissible L/d = 101.3240; Actual L/d = 23.5294 (Deflection is ok)

Design of short span

Continuous edge

M = $\alpha n L_x^2$ = 0.057 × 10.9575 × 1.705^2 = 1.8156 KN.m

M_{Ed} = 1.8156 KNm

K = 0.00512; La/d = 0.95; A_{S1} = 40.1303 mm^2; A_{Smin} = 172.5217 mm^2;

Provide Y12 @ 250mm c/c Top (A_{Sprov} = 452mm^2)

Design of long span

Mid span

M = $\alpha n L_x^2$ = 0.024 × 10.9575 × 1.705^2 = 0.764 KN.m

M_{Ed} = 0.764 KNm

K = 0.00215; La/d = 0.95; A_{S1} = 16.886 mm^2; A_{Smin} = 172.5217 mm^2;

Provide Y12 @ 250mm c/c N/Bottom (A_{Sprov} = 452mm^2)

Design of long span

Continuous edge

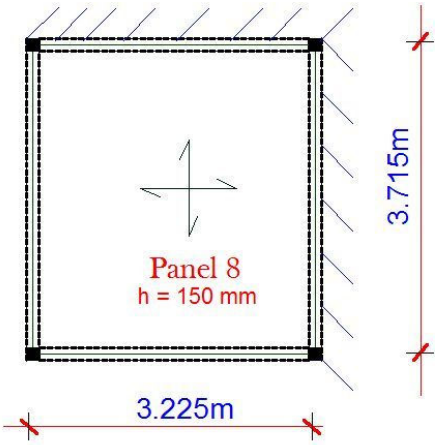
M = $\alpha n L_x^2$ = 0.032 × 10.9575 × 1.705^2 = 1.0932 KN.m

M_{Ed} = 1.01931 KNm

K = 0.00313; La/d = 0.95; A_{S1} = 24.163 mm^2; A_{Smin} = 172.5217 mm^2;

Provide Y12 @ 250mm c/c Top (A_{Sprov} = 452mm^2)

PANEL 8



The floor slab (PANEL 7) is spanning in two directions, since the ratio (k) of the longer side (Ly) to the shorter side (Lx) is less than 2.

Hence, $k = \frac{Ly}{Lx} = \frac{3.715}{3.225}$ = 1.15

Moment coefficient for all edges continuous

Short Span	Long Span
Mid-span = 0.045	Mid-span = 0.034
Continuous edge = 0.060	Continuous edge = 0.045

Design of short span

Mid span

M = $\alpha n L_x^2$ = 0.045 × 10.9575 × 3.225^2 = 5.1284 KN.m

M_{Ed} = 5.1284 KNm

K = 0.0144; La/d = 0.95; A_{S1} = 113.353 mm^2; A_{Smin} = 172.5217 mm^2;

Provide Y12 @ 250mm c/c Bottom (A_{Sprov} = 452mm^2)

Permissible L/d = 87.8141; Actual L/d = 27.101 (Deflection is ok)

Design of short span

Continuous edge

M = $\alpha n L_x^2$ = 0.060 × 10.9575 × 3.225^2 = 6.8379 KN.m

M_{Ed} = 6.8379 KNm

K = 0.0193; La/d = 0.95; A_{S1} = 151.138 mm^2; A_{Smin} = 172.5217 mm^2;

Provide Y12 @ 250mm c/c Top (A_{Sprov} = 452mm^2)

Design of long span

Mid span

M = $\alpha n L_x^2$ = 0.034 × 10.9575 × 3.225^2 = 3.8748 KN.m

M_{Ed} = 3.8748 KNm

K = 0.0109; La/d = 0.95; A_{S1} = 85.644 mm^2; A_{Smin} = 172.5217 mm^2;

Provide Y12 @ 250mm c/c N/Bottom (A_{Sprov} = 452mm^2)

Design of long span

Continuous edge

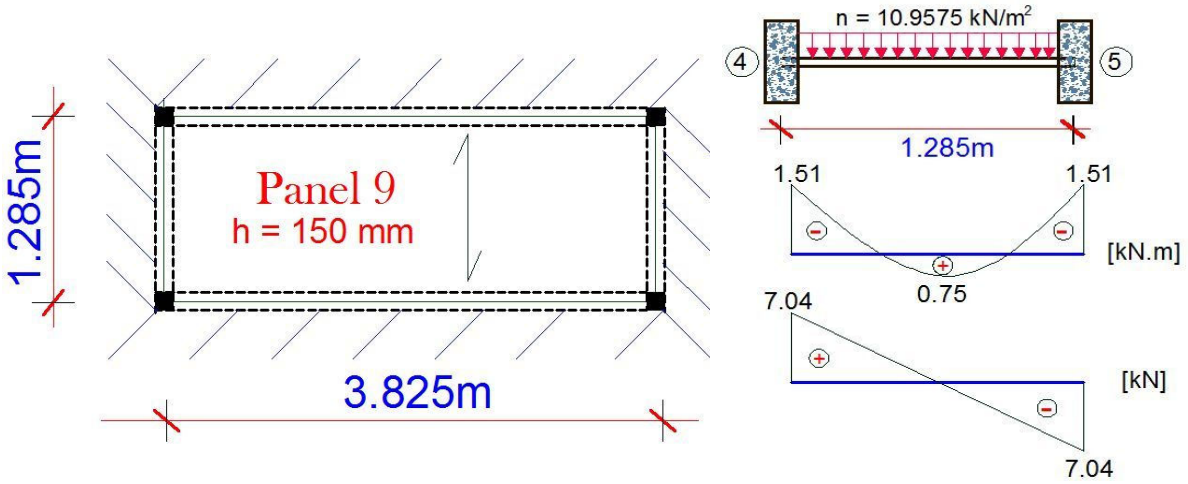
M = $\alpha n L_x^2$ = 0.045 × 10.9575 × 3.225^2 = 5.1284 KN.m

M_{Ed} = 5.1284 KNm

K = 0.0144; La/d = 0.95; A_{S1} = 113.353 mm^2; A_{Smin} = 172.5217 mm^2;

Provide Y12 @ 250mm c/c Top (A_{Sprov} = 452mm^2)

PANEL 9

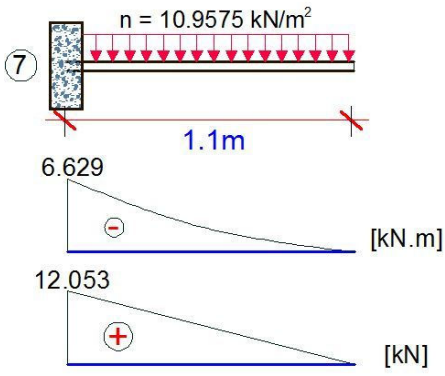


The floor slab (PANEL 6) is spanning in one direction, since the ratio (k) of the longer side (Ly) to the shorter side (Lx) is greater than 2.

Hence, $k = \frac{Ly}{Lx} = \frac{3.825}{1.285}$ = 2.9766

Judging from our previous designs, provide Y12 @ 250mm c/c Top and Bottom (A_{Sprov} = 452mm^2)

PANEL 10 (1st floor Kitchen Balcony)



In this case, we are going to assume (PANEL 10) as a 1.1m long cantilever spanning along the short span. It is very usual to increase the live load of balconies, but in this case, let us assume that the same live load is distributed all across the entire floor slab. Let the partition allowance load of 1.5 KN/m^2 (which is not likely on balconies in this case) take care of this.

Design of short span

Support moment

M_{Ed} = 6.629 KNm

K = 0.0187; La/d = 0.95; A_{S1} = 146.5212 mm^2; A_{Smin} = 172.5217 mm^2;

Provide Y12 @ 250mm c/c Top (A_{Sprov} = 452mm^2)

Permissible L/d = 27.0197; Actual L/d = 9.2437 (Deflection is ok)

Shear Design

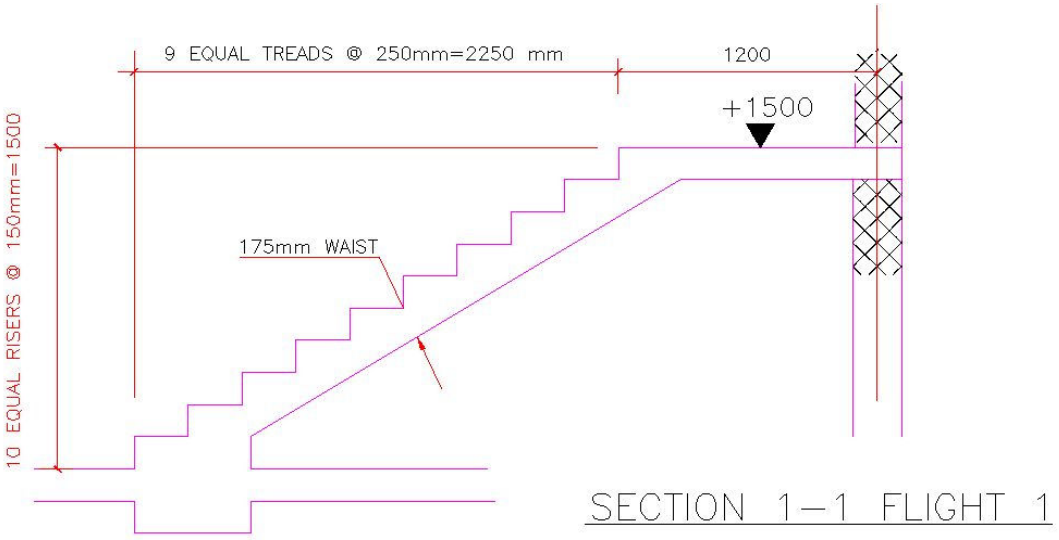
SupportE; V_{Ed} = 12.053KN

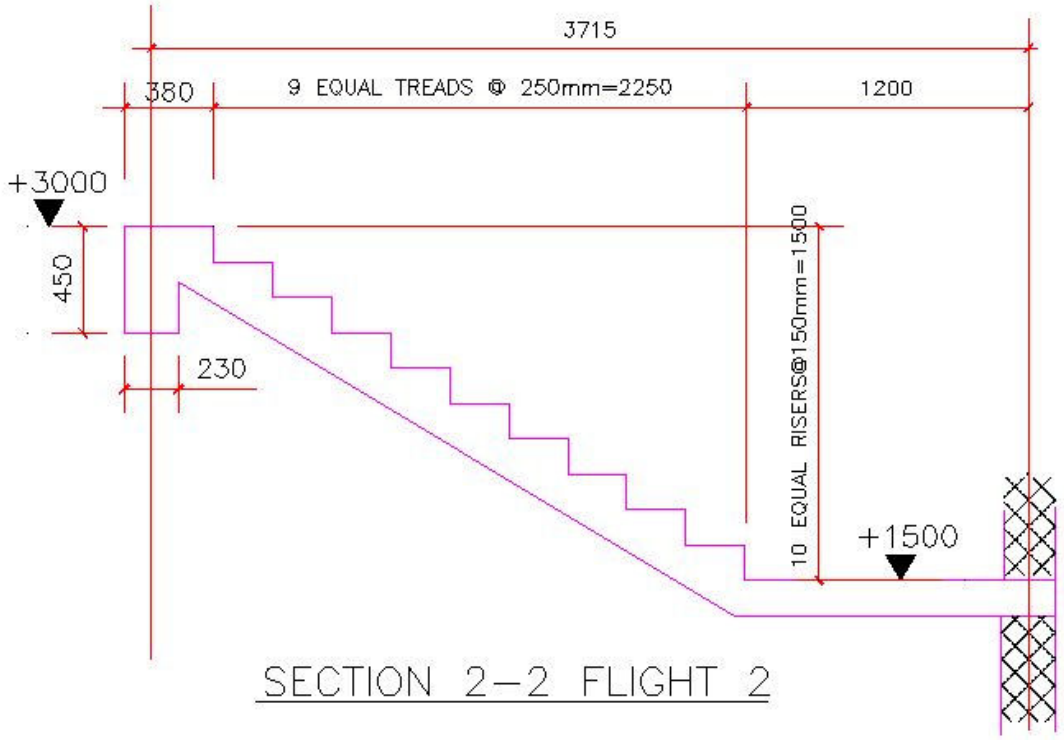
Using the area of tension reinforcement provided (A_{Sprov} = 452 mm^2/m)

$V_{Rd,c}$ = 60.4786 KN > V_{Ed} (Therefore , no shear reinforcement is required)

This is very ok !!!!

5.4 Design of the Staircase





5.4.1 Staircase Loading

Thickness of waist and landing = 175mm

Depth of riser = 150mm

Load actions on the stairs

Concrete self weight (waist area) = 0.175 × 25 = 4.375 KN/m^2 (normal to the inclination)
Stepped area = $^1/_2 \times 0.15 \times 25 = 1.875$ KN/m^2 (global vertical direction)
Finishes (say) = 1.2 KN/m^2

We intend to apply all gravity loads purely in the global y-direction, therefore we convert the load at the waist of the stair from local to global direction by considering the angle of inclination of the flight area to the horizontal;

$\alpha = \tan^{-1}\left(\frac{1.5}{2.25}\right)$ = 33.69°

Therefore UDL from waist of the stair in the global direction is given by = (4.375 × cos 33.69) = 3.640 KN/m^2

∴ Total dead load on flight area (g_k) = 3.640 + 1.875 + 1.2 = 6.715 KN/m
Variable load on staircase (q_k) = 3 KN/m^2 (Table 6.2 prEN 1991-1-2001, permits staircase variable load to range from 2 KN/m^2 to 4 KN/m^2)

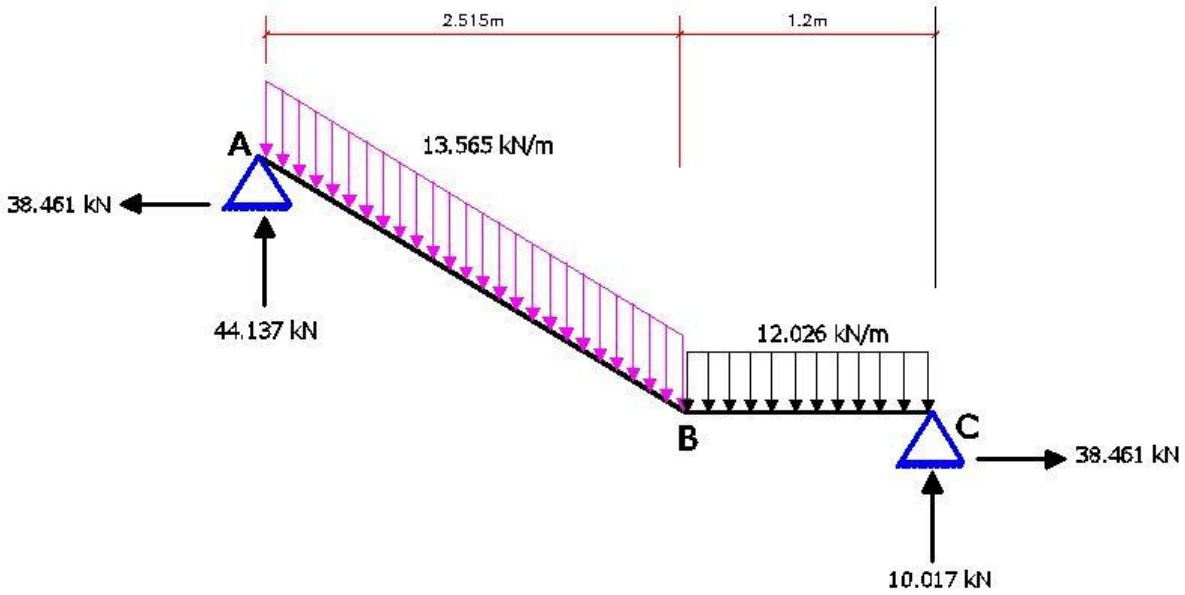
The load on the flight area at ultimate limit state = 1.35gk + 1.5qk
n = 1.35(6.715) + 1.5(3) = 13.565 KN/m^2

On the landing;

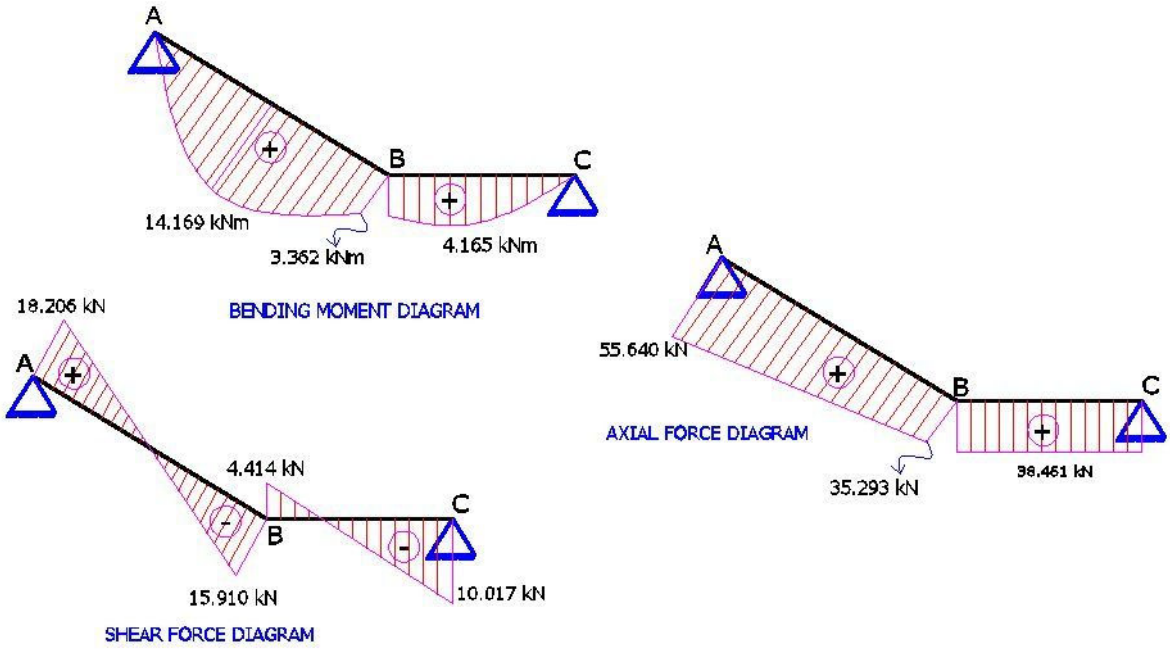
gk = 4.375 + 1.2 = 5.575 KN/m^2; qk = 3KN/m^2

The load on the landing at ultimate limit state = 1.35gk + 1.5qk

n = 1.35(5.575) + 1.5(3) = 12.026 KN/m^2



Analysis by force method gave the internal forces below



5.4.2 Structural Design

Flexural Design of the span

M_{Ed} = 14.169 KNm

d = h – Cc – ϕ/2

Assuming ϕ12mm bars will be employed for the construction

d = 175 – 25 – 6 = 144 mm; b = 1000mm (designing per unit width)

$$k = \frac{M_{Ed}}{f_{ck}\, bd^2} = \frac{14.169 \times 10^6}{25 \times 1000 \times 144^2} = 0.0273$$

Since k < 0.167 No compression reinforcement required

$$z = d\left[0.5 + \sqrt{(0.25 - 0.882K)}\right] = z = d\left[0.5 + \sqrt{(0.25 - 0.882(0.0273)}\right] = 0.95d$$

$$A_{s1} = \frac{M_{Ed}}{0.87 f_{yk} z} = \frac{14.16 \times 10^6}{0.87 \times 460 \times 0.95 \times 144} = 259\ mm^2/m$$

ProvideY12mm @ 200mm c/c BOT (AS_{prov} = 565 mm^2/m)

To calculate the minimum area of steel required;

fctm = $0.3 \times {f_{ck}}^{2/3} = 0.3 \times 25^{2/3}$ = 2.5649 N/mm^2 (Table 3.1 EC2)

$$A_{Smin} = 0.26 \times \frac{fctm}{Fyk} \times b \times d = 0.26 \times \frac{2.5649}{460} \times 1000 \times 144 = 208.76\ mm^2/m$$

Check if A_{Smin} < 0.0013 × b × d (187.2 mm^2/m)

Since, A_{Smin} = 187.2 mm^2, the provided reinforcement is adequate.

Provide Y10 @ 250mm c/c as distribution bars (AS_{prov} = 314 mm^2/m)

Check for deflection

K = 1.0 for simply supported beams and slab

$\rho = \frac{As}{bd} = \frac{565}{1000 \times 144} = 0.0039236 < 10^{-3}\sqrt{25}$

Since $\rho < \rho_0$

$L/d = K\left[11 + 1.5\sqrt{f_{ck}}\frac{\rho_0}{\rho} + 3.2\sqrt{f_{ck}}\left(\frac{\rho_0}{\rho} - 1\right)^{3/2}\right]$

$L/d = 1.0\left[11 + 1.5\sqrt{25} \times \frac{0.005}{0.0039236} + 3.2\sqrt{25}\left(\frac{0.005}{0.0039236} - 1\right)^{3/2}\right]$ = 1.0(11+ 9.5575 + 2.299) = 22.856

Modification factor $\beta_s = \frac{310}{\sigma_s}$

$\sigma_s = \frac{310 f_{yk} AS_{req}}{500\ AS_{prov}} = \frac{310 \times 460 \times 259}{500 \times 565} = 130.738$ N/mm^2

$\beta_s = \frac{310}{130.737} = 2.37$ (take 2.0)

Taking the distance between supports as the effective span, L = 3.715m
The allowable span/depth ratio = $\beta_s \times 22.856 = 2 \times 22.856 = 45.712$
Actual deflection L/d = $\frac{3715}{144} = 25.98$
Since 45.712 < 25.98, deflection is ok.

Shear Design

Ultimate shear force V_{Ed} = 18.206 KN

$V_{Rd,c} = [C_{Rd,c}.k.\ (100\rho_1 f_{ck})^{\frac{1}{3}} + k_1.\sigma_{cp}]bw.d \geq (V_{min} + k_1.\sigma_{cp})\ bw.d$

$C_{Rd,c} = 0.18/\gamma_c = 0.18/1.5 = 0.12$

$k = 1+\sqrt{\frac{200}{d}} = 1+\sqrt{\frac{200}{144}} = 2.1785 > 2.0$, therefore, k = 2

$V_{min} = 0.035k^{\frac{3}{2}}f_{ck}^{\frac{1}{2}} = V_{min} = 0.035 \times (2)^{\frac{3}{2}} \times (25)^{\frac{1}{2}} = 0.494$ N/mm^2

$\rho_1 = \frac{As}{bd} = \frac{565}{1000 \times 144} = 0.0039236 < 0.02$; $K_1 = 0.15$

$\sigma_{cp} = N_{Ed}/Ac < 0.2fcd$ (Where N_{Ed} is the axial force at the section, Ac = cross sectional area of the concrete), f_{cd} = design compressive strength of the concrete.) Take N_{Ed} = 55.640

$V_{Rd,c}$ (including the negative effect of axial force) = 82.326 KN

Since $V_{Rd,c}$ (82.326 KN) > V_{Ed} (35.358 KN), no shear reinforcement is required. Shear is ok

CHAPTER 6

ANALYSIS AND DESIGN OF THE FLOOR BEAMS

6.1 Introduction

Floor beams in a reinforced concrete framed building normally receive load from the floor slabs, carry their own self weight, weight of the blocks or bricks (walls), and the weight of finishes. In typical reinforced concrete buildings, these beams are usually T – beams, L – beams, or rectangular beams. T- beams are usually internal beams, while external beams are usually L – beams. Beams that are not carrying any slab load are more often rectangular sections.

In manual analysis of floor beams, loads are transferred from slab to beams based on the yield line assumption. This depends on if the slab is spanning in one-way or two-way. Typically, for a two-way slab, the loads are either triangular (for the beam parallel to the short span direction of the slab) or trapezoidal (for the beam parallel to the long span direction of the slab). This can be seen in Figure 4.1, and the typical load distribution on the beam is shown in Figure 4.2. Those who are experienced in manual analysis can verify that such analysis can be tedious especially for continuous beams with variable loading and unequal spans. However for beams of equal span and similar loading, coefficients for bending moment and shear can be obtained from Chapter 12 of *Reynolds and Steedman* (2005). However, for the sake of convenience, the load transferred from the slab to the beam can be treated as a uniformly distributed load, and the formulas for the transfer of such loads are given in Chapter 13 of *Reynolds and Steedman* (2005). They are presented here on Table 4.1 below;

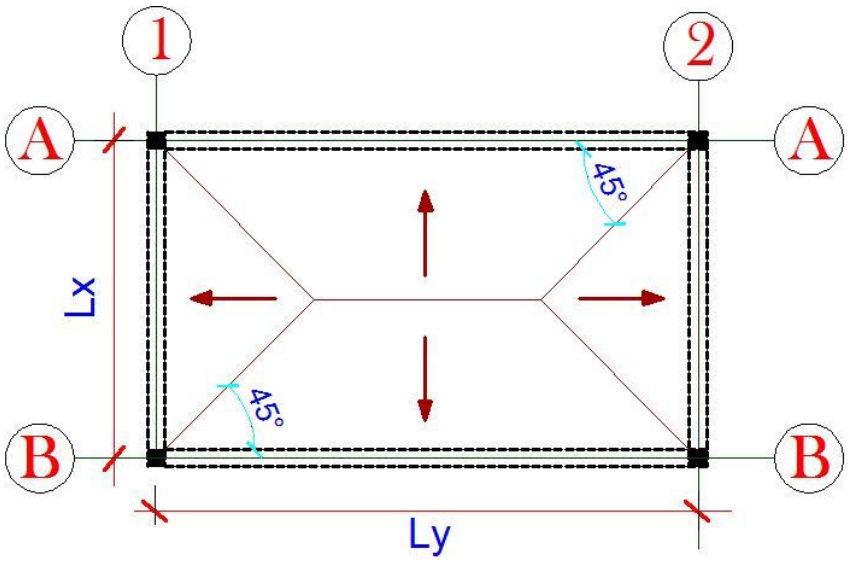


Figure 4.1: Load transfer from slab to beams based on yield line method

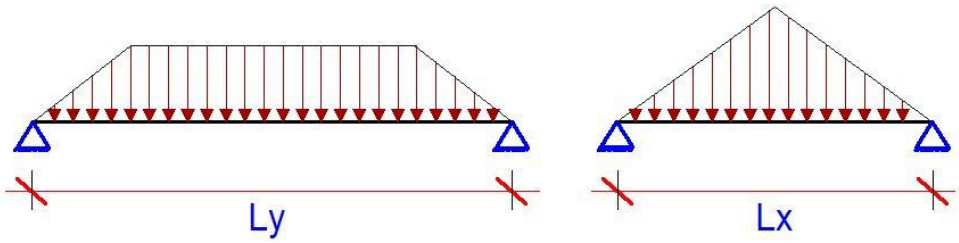


Figure 4.2: Slab loading on the Long Span (Ly) and short span (Lx)

Table 4.1: Formular for Equivalent slab load as UDL (Reynolds and Steedman, 2008)

Type of slab	Load to beam on long span (kN/m)	Load to beam on short Span (kN/m)
One-way slab	$\frac{nl_x}{2}$	$\frac{nl_x}{5}$
Two-way slab	$\frac{nl_x}{2}(1-\frac{1}{3k^2})$	$\frac{nl_x}{3}$

In CSC Orion (version 18), we have the option of transferring the slab load using the yield line method, or finite element method. To use the finite element method, load FE decomposition must be done. However in STAAD PRO v8*i*, once the floor slab has been modelled, the load from slab is transferred automatically to the beam (but the beams are broken into the respective size of the mesh sizes of the plates). However, if the floor slabs are not modelled, slab loads can be modelled as floor loads (See Figure 4.3 below). In this case also, the loads are transferred using the yield line method, and it is very beneficial, if you do not want to pass through the hassle of modelling the floor slab using finite element mesh. So you can utilise any of the medium for your designs.

6.2 Effective flange width of beams

Clause 5.3.2.1 of EN 1991-1- covers the effective flange width of beams for all limit states. For T-beams, the effective flange width, over which uniform conditions of stress can be assumed, depends on the web and flange dimensions, the type of loading, the span, the support conditions and the transverse reinforcement. The effective width of flange should be based on the distance l_o between points of zero moment, which is shown in Figure 4.3 below.

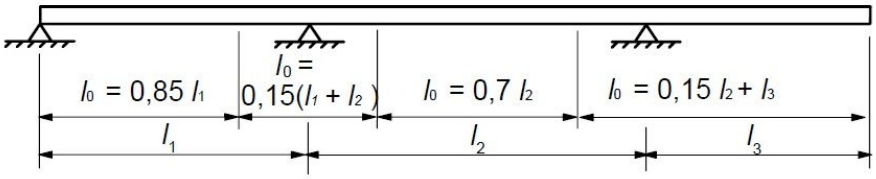


Figure 4.3: Assumed points of zero moment for beams (Figure 5.2 of EN 1992-1-1:2004 (E))

The flange width for T-beams and L-beams can be derived as shown below. The notations are shown in Figure 4.4.

$$b_{eff} = \sum b_{eff,i} + b_w \leq b \text{ ---------- (4.1)}$$

where;

$$b_{eff,i} = 0.2b_i + 0.1l_o \leq 0.2l_o \text{ ---------- (4.2)}$$

and

$b_{eff,i} \leq b_i$

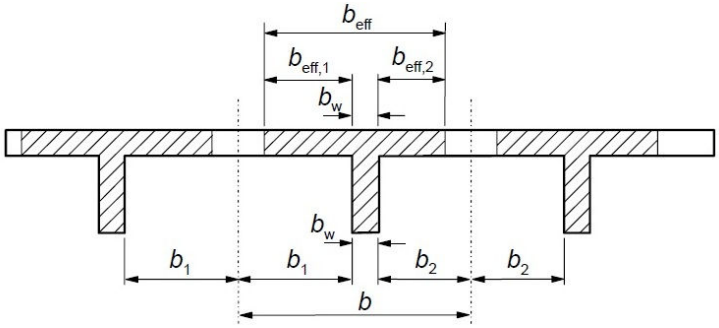


Figure 4.4: Effective flange width parameters ((Figure 5.3 of EN 1992-1-1:2004 (E))

Clause 5.3.2.1 (4) permits us to assume a constant width when great accuracy is not desired. The value applicable to the span section should be adopted.

The floor beam layout and numbering from our GA is given in Figure 4.5 below;

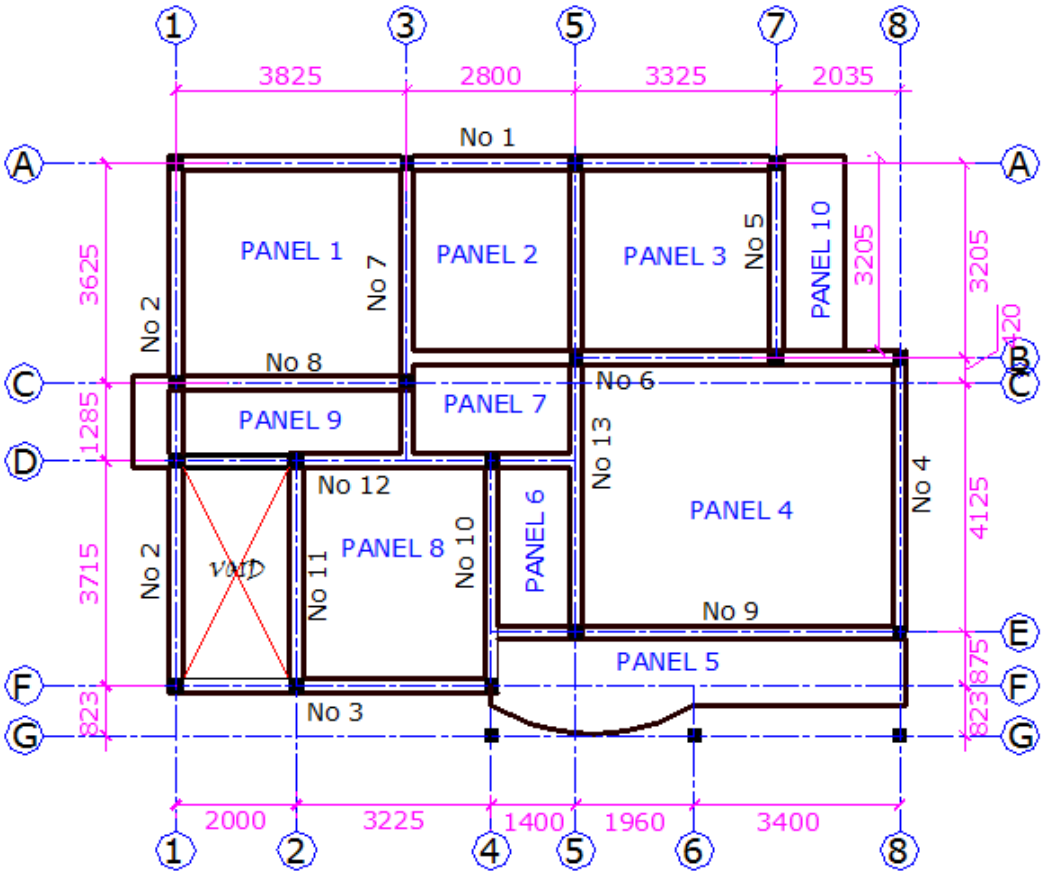


Figure 4.5: Floor beam Layout and numbering

6.3 Loading of beams

It is pertinent to point out the approach that has been adopted in the modelling of the self weight of the beam. As we already know, the self weight of the slab has already been accounted for in the transfer of the load from the slab to the beam. So the weight of the beam that we account for is the 'drop' of the beam of the beneath the slab. In this case, if the total depth (h) of the beam is 450mm, and the thickness of the slab (hf) is 150mm, then the depth (drop) to be considered on the beam is;

H_d = 450 – 150mm = 300mm

If the width of the web (b_w) of the beam is 230mm, and the density of concrete (ρ_c) is 25 KN/m^3 then the self weight of the beam is given by;

Sw = $\rho_c \times H_d \times b_w$ = 25 KN/m^3 × 0.3m × 0.23m = 1.725 KN/m

At ultimate limit state, the factor for unfavourable load combination γ_{Gk} = 1.35

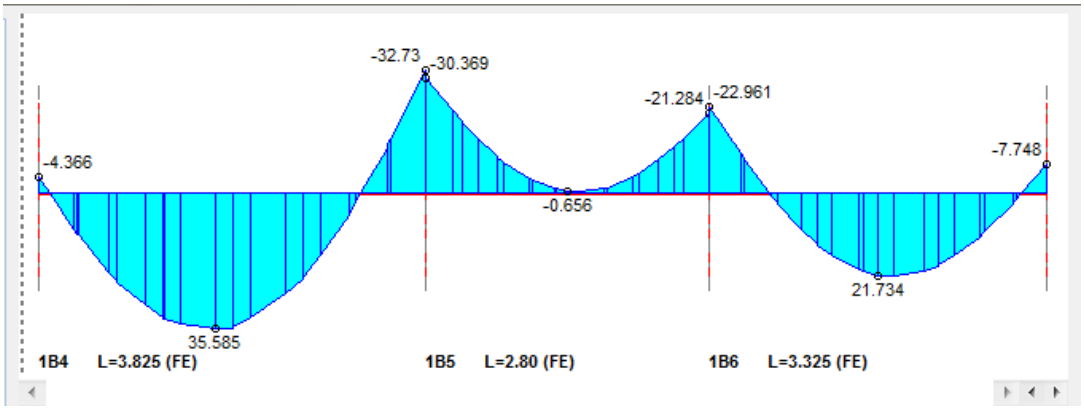
Hence Gk = 1.35 × 1.725 = 2.32875 KN/m

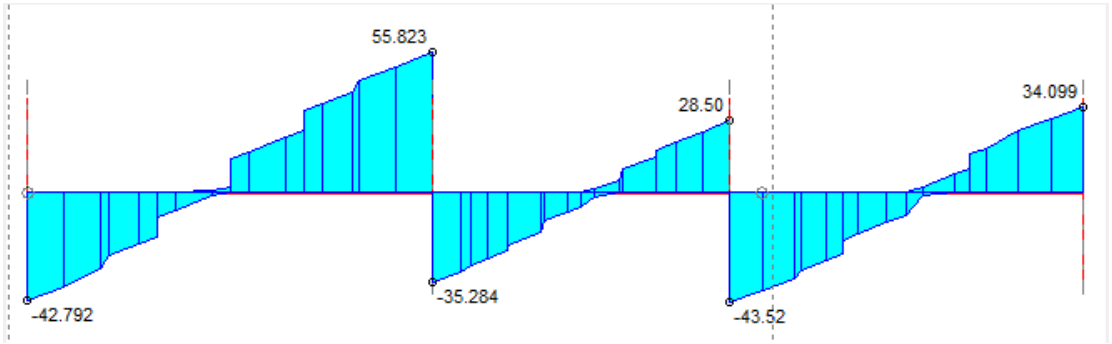
The weight of block work is also taken into account. According to *Oyenuga* (2009), the unit weight of block work in Nigeria (finishes inclusive) is 3.47 KN/m^2. When we multiply this with the height of the wall, we can rightly obtain the load of the wall. We also factor it as a dead load with γ_{Gk} = 1.35.

6.4 Analysis and Design of the floor beams

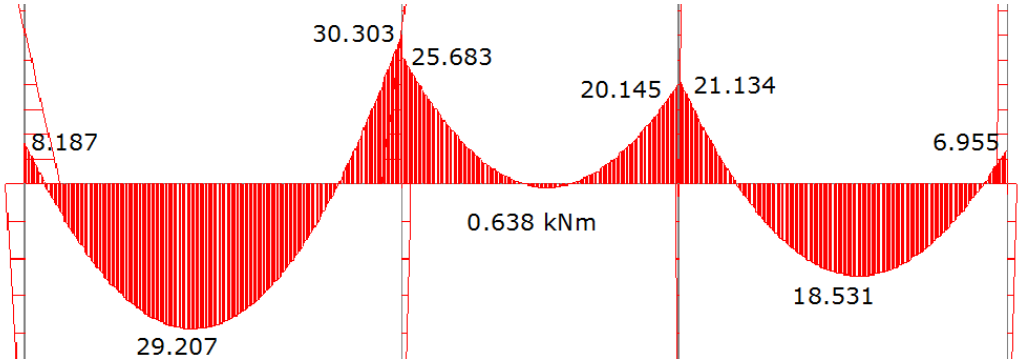
Beam NO 1: Axis 1: 1 – 7

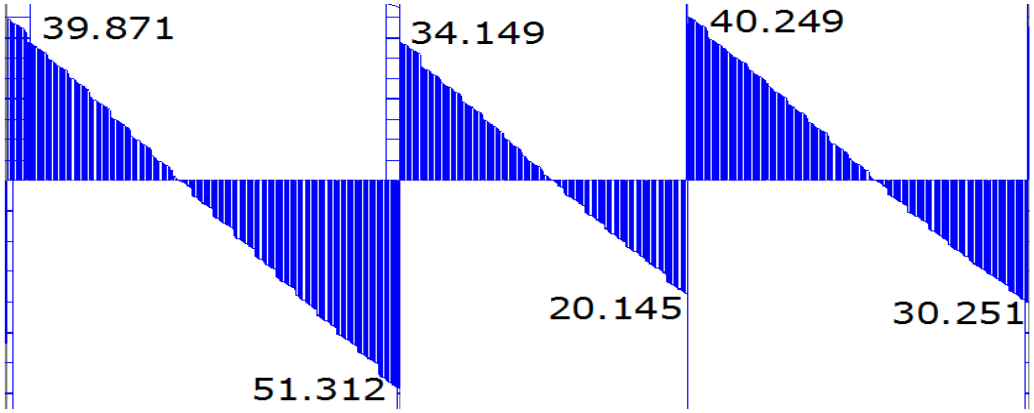
Result from Orion (ULS)





Result from Staad (ULS)





Manual Load Analysis (ULS)

Span 1- 3

The floor slabs (PANEL 1) are spanning in two directions, since the ratio (k) of the longer side (Ly) to the shorter side (Lx) is less than 2.

Hence, k = $\frac{Ly}{Lx} = \frac{3.825}{3.625}$ = 1.055

We can observe from the general arrangement of the structure that the span is parallel to the long span direction of the slab. Therefore, the equivalent load that is transferred from the slab to the beam can be represented by;

$p = \frac{nLx}{2}\left(1 - \frac{1}{3k^2}\right) = \frac{10.9575 \times 3.625}{2}\left(1 - \frac{1}{3(1.055)^2}\right)$ = 13.912 KN/m

Self weight of the beam (ULS) = 1.35 × 0.3m × 0.23m × 25 KN/m^3 = 2.32875 KN/m

Weight of block work (ULS) = 1.35 × 2.75m × 3.47 KN/m^2 = 12.882 KN/m

Therefore total load on span 1 - 3 (ULS) = 13.912 + 2.32875 + 12.882 = 29.123 KN/m

Span 3 - 5

A little consideration will show that the floor slabs (PANEL 2) are spanning in two directions, since the ratio (k) of the longer side (Ly) to the shorter side (Lx) is less than 2.

Hence, k = $\frac{Ly}{Lx} = \frac{3.625}{2.8}$ = 1.295

We can observe from the general arrangement of the structure that the span is parallel to the short span direction of the slab. Therefore, the equivalent load that is transferred from the slab to the beam can be represented by;

$p = \frac{nLx}{3} = \frac{10.9575 \times 2.8}{3}$ = 10.227 KN/m

Self weight of the beam (ULS) = 1.35 × 0.3m × 0.23m × 25 KN/m^3 = 2.32875 KN/m

Weight of block work (ULS) = 1.35 × 2.75m × 3.47 KN/m^2 = 12.882 KN/m

Therefore total load on span 3 - 5 = 10.227 + 2.32875 + 12.882 = 25.437 KN/m

Span 5-7

The floor slab (PANEL 3) is spanning in two directions, since the ratio (k) of the longer side (Ly) to the shorter side (Lx) is less than 2.

Hence, k = $\frac{Ly}{Lx} = \frac{3.625}{3.325}$ = 1.09

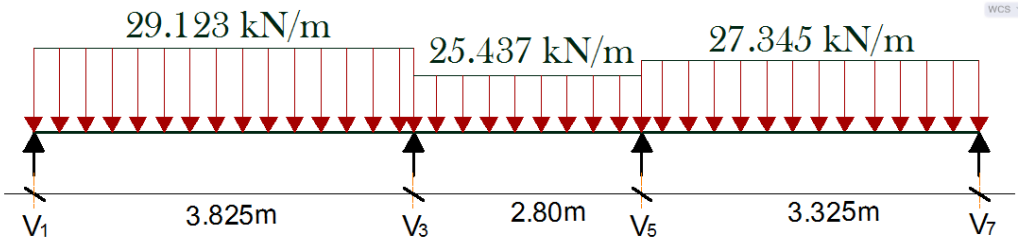
We can observe from the general arrangement of the structure that the span is parallel to the short span direction of the slab. Therefore, the equivalent load that is transferred from the slab to the beam can be represented by;

$p = \frac{nLx}{3} = \frac{10.9575 \times 3.325}{3}$ = 12.144 KN/m

Self weight of the beam (ULS) = 1.35 × 0.3m × 0.23m × 25 KN/m^3 = 2.32875 KN/m

Weight of block work = 1.35 × 2.75m × 3.47 KN/m^2 = 12.882 KN/m

Therefore total load on span 5 - 7 = 12.144 + 2.32875 + 12.882 = 27.354 KN/m



By slope deflection method;

We already know that the moment at supports 1 and 7 are zero;

$M_{3-1} = \frac{3EI}{L}(\theta_3) + F_{31}$

$M_{3-5} = \frac{EI}{L}(4\theta_3 + 2\theta_5) + F_{35}$

$M_{5-3} = \frac{EI}{L}(2\theta_3 + 4\theta_5) + F_{53}$

$M_{5-7} = \frac{3EI}{L}(\theta_5) + F_{57}$

Where F_{ij} are the respective fixed end moments;

$F_{31} = \frac{ql^2}{8} = \frac{29.123 \times 3.825^2}{8}$ = 53.26 KNm; $F_{35} = \frac{ql^2}{12} = -\frac{25.437 \times 2.8^2}{12}$ = -16.619 KNm;

$F_{53} = \frac{ql^2}{12} = \frac{25.437 \times 2.8^2}{12}$ = 16.619 KNm; $F_{57} = -\frac{ql^2}{8} = \frac{27.345 \times 3.325^2}{12}$ = −37.789 KNm

Therefore;

$M_{31} = \frac{3EI}{3.825}(\theta_3) + 53.26$ ---------- (1)

$M_{35} = \frac{EI}{2.8}(4\theta_3 + 2\theta_5) - 16.619$ ---------- (2)

$M_{53} = \frac{EI}{2.8}(2\theta_3 + 4\theta_5) + 16.619$ ---------- (3)

$M_{57} = \frac{3EI}{3.325}(\theta_5) - 37.789$ ---------- (4)

For equilibrium and compatibility;

$M_{31} + M_{35} = 0$

$M_{53} + M_{57} = 0$

$2.2129EI\theta_3 + 0.7143EI\theta_5 = -36.641$

$0.7143EI\theta_3 + 2.3308EI\theta_5 = 21.17$

Solving simultaneously;

$\theta_3 = \frac{-21.629}{EI}$ radians; $\theta_5 = \frac{15.711}{EI}$ radians

Substituting back into the original equations to obtain the support moments;

$M_{31} = \frac{3EI}{3.825}\left(\frac{-21.629}{EI}\right) + 53.26 = 36.296$ KNm

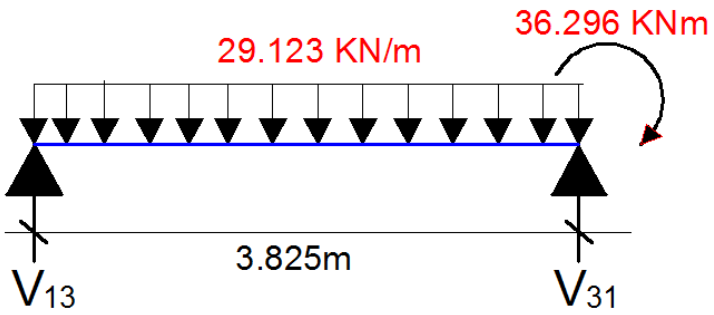
$M_{35} = \frac{EI}{2.8}\left(4 \times \frac{-21.629}{EI} + 2 \times \frac{15.711}{EI}\right) - 16.619 = -36.296$ KNm

$M_{35} = \frac{EI}{2.8}\left(2 \times \frac{-21.629}{EI} + 4 \times \frac{15.711}{EI}\right) + 16.619 = 23.614$

$M_{57} = \frac{3EI}{3.325}\left(\frac{15.711}{EI}\right) - 37.789 = -23.614$ KNm

From statics, we can then obtain the span moments and shear forces;

Span 1 - 3



Let $\sum M_3 = 0$

$3.825V_{13} - \frac{29.123 \times 3.825^2}{2} = -36.296$

Hence, $V_{13} = 46.208$ KN

Let $\sum F_Y = 0$

$V_{13} + V_{31} = (29.123 \times 3.825) = 111.395$

Hence, $V_{31} = 111.395 - 46.208 = 65.187$ KN

Maximum span Moment

Let us cut a section x-x at an arbitrary point from support 1. The bending moment at that section is given by;

$Mx = V_{13}.x - \frac{29.123 \times x^2}{2} = 46.208x - 14.5615x^2$

If we differentiate the equation for moment, we obtain the equation for shear;

$\frac{\partial M_x}{\partial x} = Qx = 46.208 - 29.123x$

Maximum moment occurs at the point of zero shear, therefore we set the equation for shear to zero, and solve for the value of x

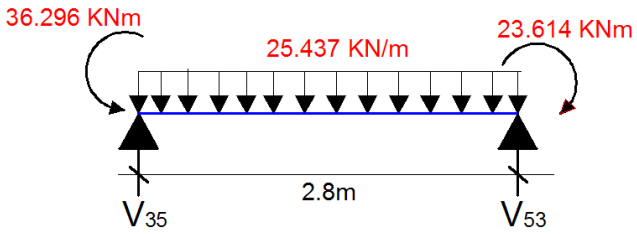
$46.208 - 29.123x = 0$

On solving, $x = 1.5866$m

We now substitute back the value of x into the equation for moment;

M_{max} = 46.208(1.5866) – 14.5615(1.5866)^2 = 36.6579 KNm

Span 3 - 5



Let $\sum M_5$ = 0

$2.8V_{35} - 36.296 - \frac{25.437 \times 2.8^2}{2} = -23.614$

Hence, V_{35} = 40.141 KN

Let $\sum F_Y$ = 0

$V_{35} + V_{53}$= (25.437 x 2.8) = 71.2236

Hence, V_{53} = 71.2236 – 40.141 = 31.085 KN

Maximum span Moment

Let us cut a section x-x at an arbitrary point from support 3. The bending moment at that section is given by;

$Mx = V_{35}.x - 36.296 - \frac{25.437 \times x^2}{2} = -36.296 + 40.141x - 12.7185x^2$

If we differentiate the equation for moment, we obtain the equation for shear;

$\frac{\partial M_x}{\partial x} = Qx = 40.141 - 25.437x$

Maximum moment occurs at the point of zero shear, therefore we set the equation for shear to zero, and solve for the value of x

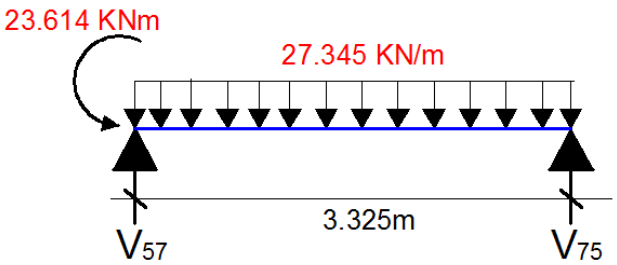
$40.141 - 25.437x = 0$

On solving, x = 1.5781m

We now substitute back the value of x into the equation for moment;

M_{max} = –36.296 + 40.141(1.5781) – 12.7185(1.5781)^2 = –4.6236KNm

Span 5 - 7



Let $\sum M_7 = 0$

$3.325V_{57} - 23.614 - \frac{27.345 \times 3.325^2}{2} = 0$

Hence, $V_{57} = 52.563$ KN

Let $\sum F_Y = 0$

$V_{57} + V_{75} = (27.345 \times 3.325) = 90.9221$

Hence, $V_{57} = 90.9221 - 52.563 = 38.3591$ KN

Maximum span Moment

Let us cut a section x-x at an arbitrary point from support 5. The bending moment at that section is given by;

$Mx = V_{57}.x - 23.614 - \frac{27.345 \times x^2}{2} = -23.614 + 52.563x - 13.6725x^2$

If we differentiate the equation for moment, we obtain the equation for shear;

$\frac{\partial M_x}{\partial x} = Qx = 52.563 - 27.345x$

Maximum moment occurs at the point of zero shear, therefore we set the equation for shear to zero, and solve for the value of x

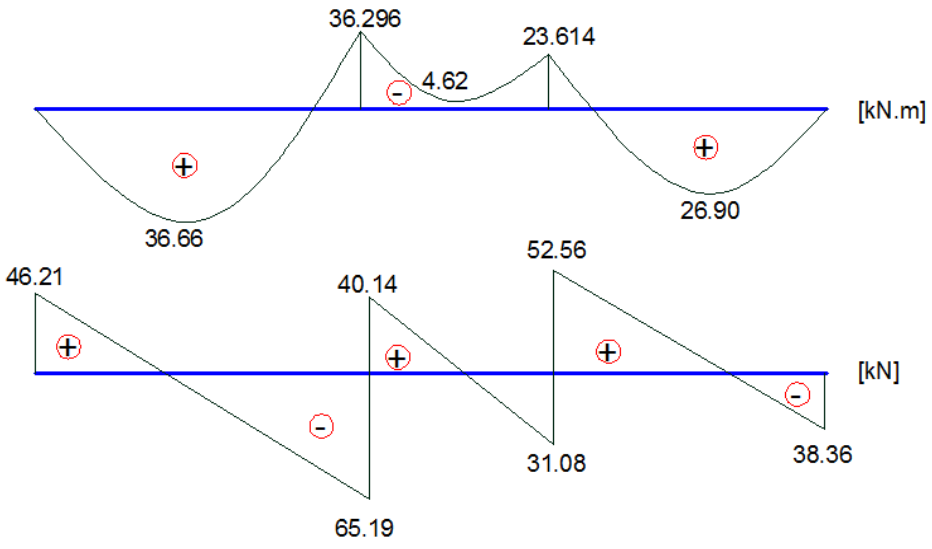
$52.563 - 27.345x = 0$

On solving, $x = 1.922$ m

We now substitute back the value of x into the equation for moment;

$M_{max} = -23.614 + 52.563(1.922) - 13.6725(1.922^2) = 26.904$ KNm

The bending moment and shear force diagrams are as shown below;



Structural Design (Using results from manual analysis)

(Span 1 - 3) L - beam

M_{Ed} = 36.66 KNm

Effective depth (d) = $450 - 35 - \frac{16}{2} - 8 = 399$mm

$B_{eff} = 0.2\left(\frac{3395}{2}\right) + 0.1(0.85 \times 3825) + 230 = 895$mm

$k = \frac{M_{Ed}}{f_{ck}\, b_{eff} d^2} = \frac{36.66 \times 10^6}{25 \times 895 \times 399^2} = 0.01029$

Since k < 0.167 No compression reinforcement required

$z = d\left[0.5 + \sqrt{(0.25 - 0.882K)}\right] = z = d\left[0.5 + \sqrt{(0.25 - 0.882(0.01029)}\right] = 0.95d$

$A_{s1} = \frac{M_{Ed}}{0.87 f_{yk} z} = \frac{36.66 \times 10^6}{0.87 \times 460 \times 0.95 \times 399} = 241.667$ mm^2

Provide 2Y16 mm BOT (AS_{prov} = 402 mm^2)

To calculate the minimum area of steel required;
fctm = $0.3 \times f_{ck}^{\,2/3} = 0.3 \times 25^{2/3} = 2.5649$ N/mm^2 (Table 3.1 EC2)

$A_{Smin} = 0.26 \times \frac{\text{fctm}}{\text{Fyk}} \times b \times d = 0.26 \times \frac{2.5649}{460} \times 230 \times 399 = 133.04$ mm^2
Check if $A_{Smin} < 0.0013 \times b \times d$ (119.301 mm^2)
Since, A_{Smin} = 168.587 mm^2, the provided reinforcement is adequate.

Check for deflection at the span;

K = 1.3 for simply supported at one end and continuous at the other end

$\rho = \frac{As}{bd} = \frac{402}{895 \times 399} = 0.0011257 < 10^{-3}\sqrt{25}$

Since $\rho < \rho_0$

$L/d = K\left[11 + 1.5\sqrt{f_{ck}}\frac{\rho_0}{\rho} + 3.2\sqrt{f_{ck}}\left(\frac{\rho_0}{\rho} - 1\right)^{3/2}\right]$

$L/d = 1.3\left[11 + 1.5\sqrt{25} \times \frac{0.005}{0.0011257} + 3.2\sqrt{25}\left(\frac{0.005}{0.0011257} - 1\right)^{3/2}\right] = 1.3(11 + 33.313 + 102.158)$
$= 190.4123$

Modification factor $\beta_s = \frac{310}{\sigma_s}$

$\sigma_s = \frac{310 f_{yk} AS_{req}}{500\, AS_{prov}} = \frac{310 \times 460 \times 241.667}{500 \times 402} = 171.451$ N/mm^2

$\beta_s = \frac{310}{171.451} = 1.808$

Since the beam is flanged, check the ratio of b/b_w = 895/230 = 3.89

Since b/bw is greater than 3, multiply the allowable L/d by 0.8

The allowable span/depth ratio = 0.8 × β_s × 190.4123 = 0.8 × 1.808 × 146.471 = 275.412

Taking the distance between supports as the effective span, L = 3825 mm

Actual deflection L/d = $\frac{3825}{399}$ = 9.5864

Since 275.412 > 9.5864, deflection is deemed to satisfy

Use 2Y16mm for the entire bottom span

Top reinforcements (Hogging moment)

Support 3

M_{Ed} = 36.296 KNm

B_{eff} = 0.2$\left(\frac{3395}{2}\right)$ + 0.1[0.15(3825 + 2800)] + 230 = 668.875 mm

But since the flange is in tension, we use the beam width to calculate the value of k (this applies to all support hogging moments)

k = $\frac{M_{Ed}}{f_{ck}\, bwd^2}$ = $\frac{36.296 \times 10^6}{25 \times 230 \times 399^2}$ = 0.0396

Since k < 0.167 No compression reinforcement required

z = *d*$\left[0.5 + \sqrt{(0.25 - 0.882K)}\right]$ = *z* = *d*$\left[0.5 + \sqrt{(0.25 - 0.882(0.0396)}\right]$ = 0.95d

A_{s1} = $\frac{M_{Ed}}{0.87 f_{yk} z}$ = $\frac{36.296 \times 10^6}{0.87 \times 460 \times 0.95 \times 399}$ = 239.268 mm^2

Provide 2Y16mm TOP (AS_{prov} = 402mm^2)

Shear Design

Using the maximum shear force for all the spans

Support A; V_{Ed} = 65.19 KN

$V_{Rd,c}$ = [$C_{Rd,c}$.k. $(100\rho_1 f_{ck})^{\frac{1}{3}}$ + $k_1.\sigma_{cp}$]*bw.d* ≥ (V_{min} + $k_1.\sigma_{cp}$) *bw.d*

$C_{Rd,c}$ = 0.18/γ_c = 0.18/1.5 = 0.12

k = 1+$\sqrt{\frac{200}{d}}$ = 1+$\sqrt{\frac{200}{399}}$ = 1.708 > 2.0, therefore, k = 1.708

V_{min} = 0.035$k^{\frac{3}{2}} f_{ck}^{\frac{1}{2}}$ = Vmin = 0.035 × $(1.708)^{\frac{3}{2}}$ × $(25)^{\frac{1}{2}}$ = 0.390 N/mm^2

ρ_1 = $\frac{As}{bd}$ = $\frac{402}{230 \times 399}$ = 0.00438 < 0.02;

σ_{cp} = N_{Ed}/Ac < 0.2fcd (Where N_{Ed} is the axial force at the section, Ac = cross sectional area of the concrete), fcd = design compressive strength of the concrete.) Take N_{Ed} = 0

$V_{Rd,c}$ = [0.12 × 1.708$(100 \times 0.00438 \times 25)^{\frac{1}{3}}$] 230 × 399 = 41767.763 N = 41.767 KN

Since $V_{Rd,c}$ (41.767) < V_{Ed} (65.19 KN), shear reinforcement is required.

The compression capacity of the compression strut ($V_{Rd,max}$) assuming θ = 21.8° (cot θ = 2.5)

$V_{Rd,max} = \frac{b_w.z.v_1.f_{cd}}{\cot\theta + tan\theta}$

$V_1 = 0.6\left(1 - \frac{f_{ck}}{250}\right) = 0.6\left(1 - \frac{25}{250}\right) = 0.54$

$f_{cd} = \frac{\alpha_{cc} f_{ck}}{\gamma_c} = \frac{0.85 \times 25}{1.5} = 14.167\ N/mm^2$

Let z = 0.9d

$V_{Rd,max} = \left[\frac{230 \times 0.9 \times 399 \times 0.54 \times 14.167}{2.5 + 0.4}\right] \times 10^{-3} = 217.879\ KN$

Since $V_{Rd,c} < V_{Ed} < V_{Rd,max}$

Hence $A_{sw}/S = \frac{V_{Ed}}{0.87 F_{yk} z \cot\theta} = \frac{65190}{0.87 \times 460 \times 0.9 \times 399 \times 2.5} = 0.18144$

Minimum shear reinforcement;

$A_{sw}/S = \rho_{w,min} \times b_w \times sin\alpha$ (α = 90° for vertical links)

$\rho_{w,min} = \frac{0.08 \times \sqrt{F_{ck}}}{F_{yk}} = \frac{0.08 \times \sqrt{25}}{460} = 0.00086956$

$A_{sw}/S_{(min)} = 0.00086956 \times 230 \times 1 = 0.2000$

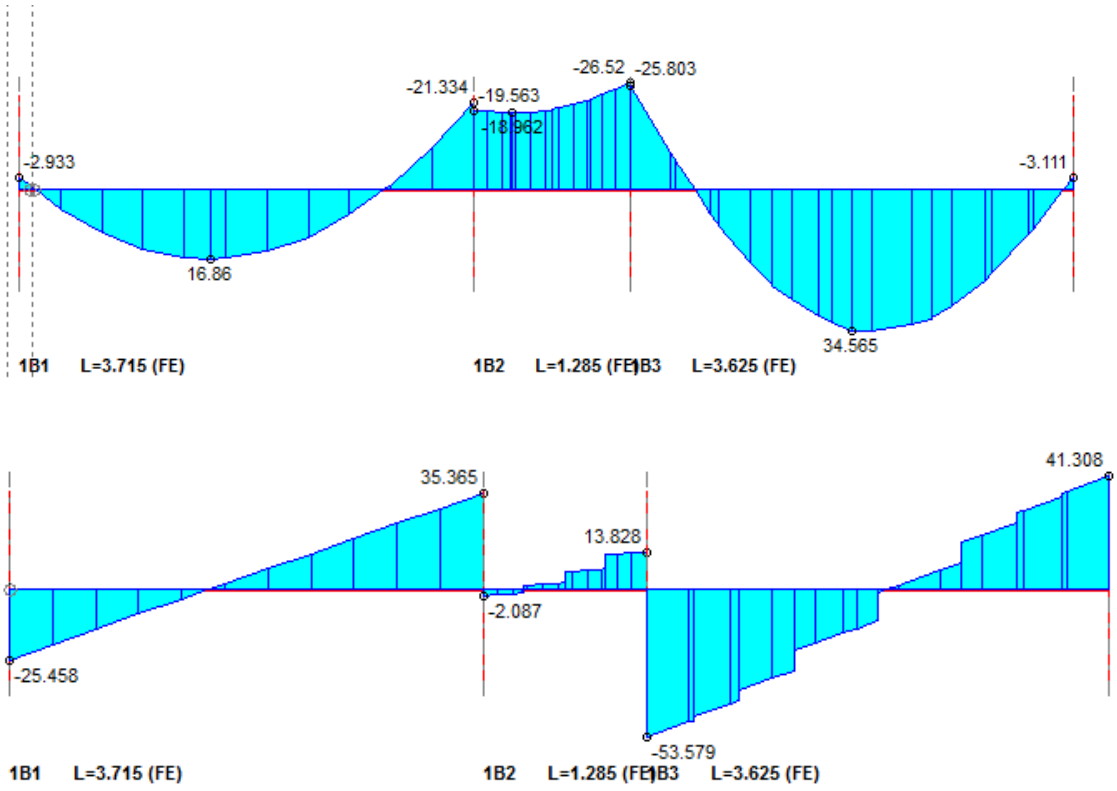
Since 0.200 > 0.18144, adopt 0.200 as the minimum shear reinforcement

Maximum spacing of shear links = 0.75d = 0.75 × 399= 299.25

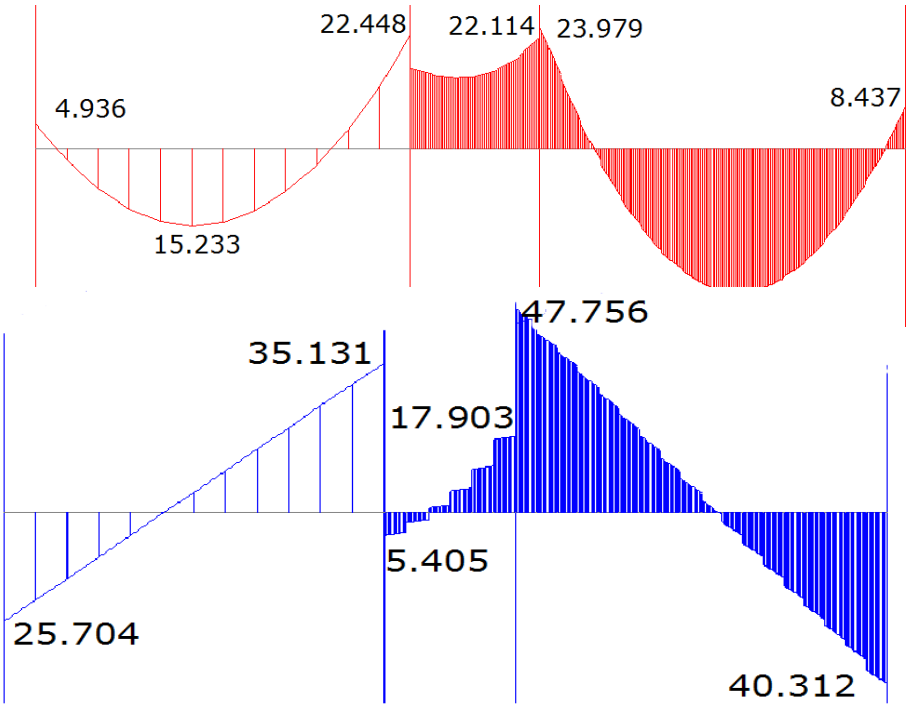
Provide Y8mm @ 250mm c/c as shear links (A_{sw}/S = 0.4021) Ok

BEAM NO 2 (AXIS 1: A - G)

RESULTS FROM ORION (ULS)



RESULTS FROM STAAD (ULS)



Manual Load Analysis (ULS)

Span A - C

You can observe that PANEL 1 is spanning in two directions, since the ratio (k) of the longer side (Ly) to the shorter side (Lx) is less than 2.

Hence, k = $\frac{Ly}{Lx}$ = $\frac{3.825}{3.625}$ = 1.055

We can observe from the general arrangement of the structure that the span is parallel to the short span direction of the slab. Therefore, the equivalent load that is transferred from the slab to the beam can be represented by;

p = $\frac{nLx}{3}$ = $\frac{10.9575 \times 3.625}{3}$ = 13.240 KN/m

Self weight of the beam (ULS) = 1.35 × 0.3m × 0.23m × 25 KN/m^3 = 2.32875 KN/m
Weight of block work (ULS) = 1.35 × 2.75m × 3.47 KN/m^2 = 12.882 KN/m
Therefore total load on **span A - C** = 13.240 + 2.32875 + 12.882 = 28.451 KN/m

Span C - D

PANEL 9 is spanning in one direction, since the ratio (k) of the longer side (Ly) to the shorter side (Lx) is greater than 2.

Hence, k = $\frac{Ly}{Lx}$ = $\frac{3.825}{1.285}$ = 2.9766

We can observe from the general arrangement of the structure that the span is parallel to the short span direction of the slab. Therefore, the equivalent load that is transferred from the slab to the beam can be represented by;

$p = \frac{nLx}{5} = \frac{10.9575 \times 1.285}{5}$ = 2.816 KN/m

Load from cantilever slab at the other side = $10.9575 \times 0.7m$ = 7.67025 KN/m

Self weight of the beam (ULS) = 1.35 × 0.3m × 0.23m × 25 KN/m^3 = 2.32875 KN/m

No block work load

Therefore total load on **span C - D** = 2.816 + 2.32875 + 7.67025 = 12.815 KN/m

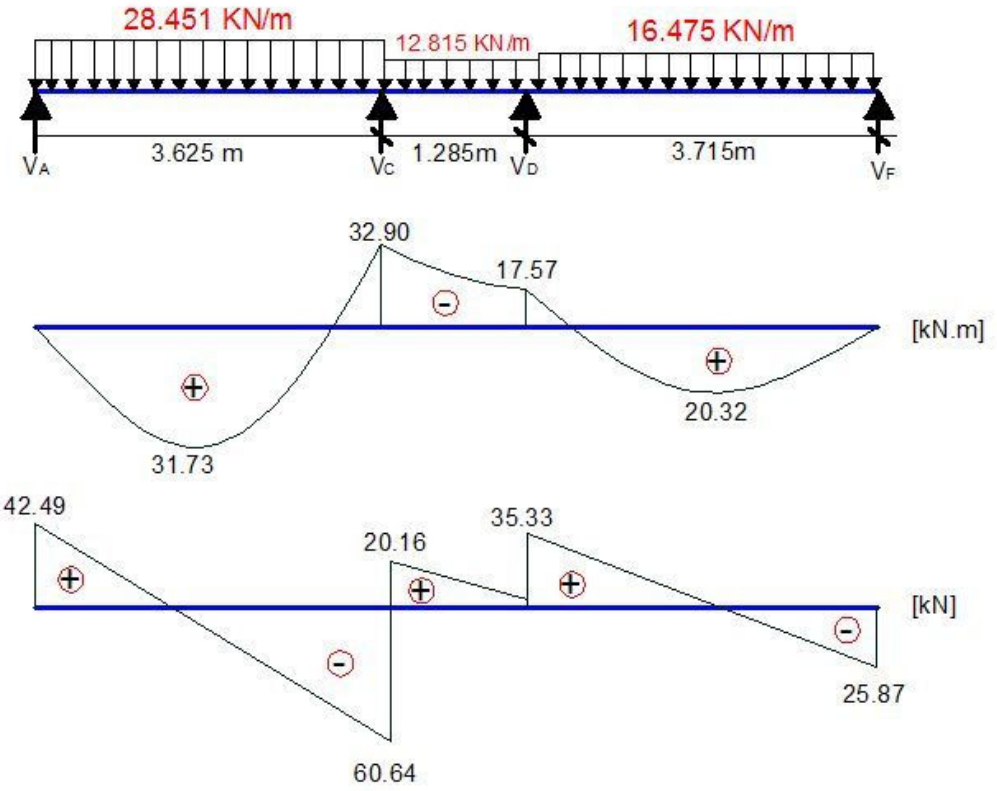
Span D - F

A little consideration will show that the area adjacent to the beam has no slab (stair case area). Therefore no load from slab will be transferred.

Self weight of the beam (ULS) = 1.35 × 0.45m × 0.23m × 25 KN/m^3 = 3.493 KN/m

Weight of block work (ULS) = 1.35 × 2.75m × 3.47 KN/m^2 = 12.882 KN/m

Therefore total load on **span D - F** = 3.493 + 12.882 = 16.475 KN/m



STRUCTURAL DESIGN

(**SPAN A-B**) (L – Beam)

M_{Ed} = 31.73 KNm

B_{eff} = 0.2(3595/2) + 0.1(0.85 × 3625) + 230 = 898mm

K = 0.0089; La/d = 0.95; A_{S1} = 209.1686 mm^2; A_{Smin} = 133.0447mm^2;

Provide 2Y16mm Bottom (A_{Sprov} = 402mm^2)

Permissible L/d = 319.896; Actual L/d = 9.0852 Deflection is OK !!!!!

(Support C - D)

M_{Ed} = 32.90 KNm (Hogging)

$B_{eff} = b_w$ = 230mm; Effective depth (d) = 450 – 35 – 8 - 8 = 399mm

K = 0.0359; La/d = 0.95; A_{S1} = 216.881 mm^2; A_{Smin} = 133.0447mm^2;

Provide 2Y16mm Top (A_{Sprov} = 402mm^2)

Also allow 2Y16 to run through the Bottom (A_{Sprov} = 402 mm^2)

(SPAN D - F) (Rectangular beam)

M_{Ed} = 20.32 KNm (Sagging)

$B_{eff} = b_w$ = 230mm; Effective depth (d) = 450 – 35 – 8 - 8 = 399mm

K = 0.0222; La/d = 0.95; A_{S1} = 133.952 mm^2; A_{Smin} = 133.0447mm^2;

Provide 2 Y16mm Top (A_{Sprov} = 402mm^2)

Permissible L/d = 86.5580; Actual L/d = 9.3108 Deflection is OK !!!!!

Shear Design

Using the maximum shear force for all the spans;

Support A; V_{Ed} = 60.64 KN

Using the area of tension reinforcement provided (A_{Sprov} = 402mm^2)

$V_{Rd,c}$ = 41.7692 > V_{Ed} (Therefore , shear reinforcement is required)

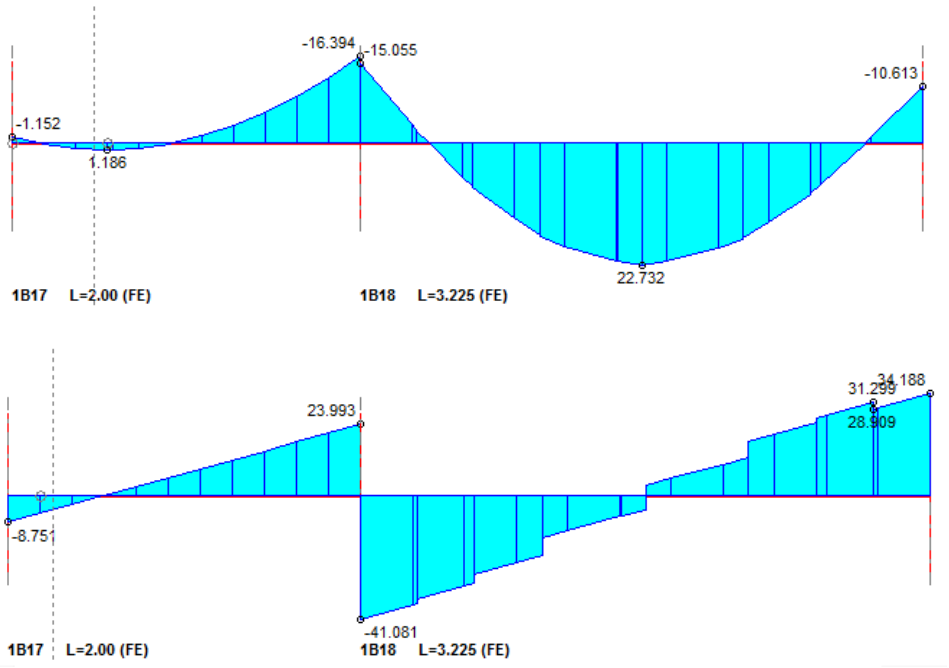
Since $V_{Rd,max}$ = 218 KN > V_{Ed} > $V_{Rd,c}$

Asw/S = 0.1688; $A_{sw,min}$/S = 0.200; Maximum spacing = 0.75d = 0.75 × 399 = 299.25

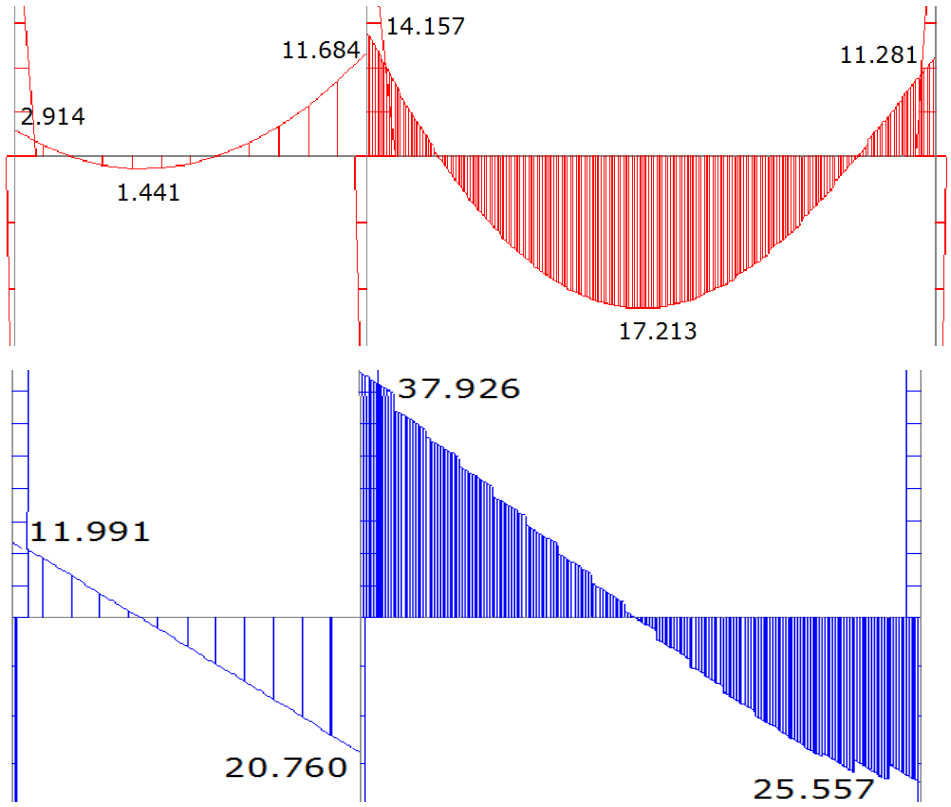
Provide 2 legs of Y8mm @ 250mm c/c (Asw/S = 0.402) This is very ok !!!!

BEAM NO 3 (AXIS F: 1 - 4)

Result From Orion (ULS)



Result from Staad (ULS)



Manual Load Analysis

Span 1 - 2

A little consideration will show that the area adjacent to this span has no slab (stair case area). Therefore no load from slab will be transferred.

Self weight of the beam (ULS) = 1.35 × 0.45m × 0.23m × 25 KN/m^3 = 3.493 KN/m

Weight of block work (ULS) = 1.35 × 2.75m × 3.47 KN/m^2 = 12.882 KN/m

Therefore total load on span 1-2 = 3.493 + 12.882 = 16.375 KN/m

Span 2 - 4

Observe that PANEL 9 is spanning in two directions, since the ratio (k) of the longer side (Ly) to the shorter side (Lx) is less than 2.

Hence, k = $\frac{Ly}{Lx} = \frac{3715}{3225}$ = 1.152

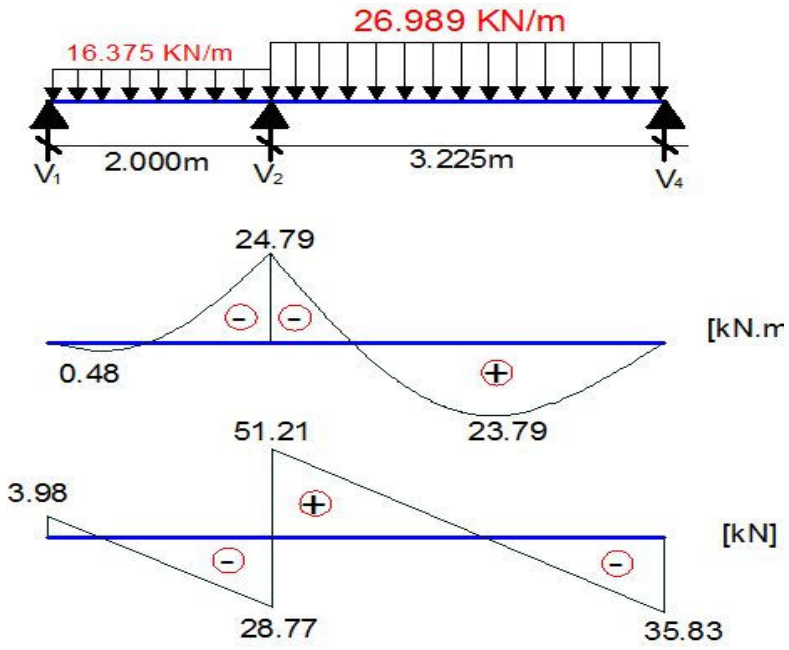
We can observe from the general arrangement of the structure that the span is parallel to the short span direction of the slab. Therefore, the equivalent load that is transferred from the slab to the beam (ULS) can be represented by;

$p = \frac{nLx}{3} = \frac{10.9575 \times 3.225}{3}$ = 11.779 KN/m

Self weight of the beam (ULS) = 1.35 × 0.3m × 0.23m × 25 KN/m^3 = 2.32875 KN/m

Weight of block work (ULS) = 1.35 × 2.75m × 3.47 KN/m^2 = 12.882 KN/m

Therefore total load on span C-D = 11.779 + 2.32875 + 12.882 = 26.989 KN/m



STRUCTURAL DESIGN

(**SPAN 1-2**) (Rectangular beam)

M_{Ed} = 0.48 KNm

Provide 2Y16mm Bottom (A_{Sprov} = 402mm^2)

(**Support 2**)

M_{Ed} = 24.79 KNm (Hogging)

B_{eff} = bw = 230mm

K = 0.0271; La/d = 0.95; A_{S1} = 163.419 mm^2; A_{Smin} = 133.0447mm^2;

Provide 2Y16mm Top (A_{Sprov} = 402mm^2)

(**SPAN 2 - 4**) (L – Beam)

M_{Ed} = 23.79 KNm

B_{eff} = 0.2(3485/2) + 0.1(0.85 × 3225) + 230 = 853 mm

K = 0.007; La/d = 0.95; A_{S1} = 156.8270 mm^2; A_{Smin} = 128.7164mm^2;

Provide 2Y16mm Bottom (A_{Sprov} = 402mm^2)

Permissible L/d = 393.4066; Actual L/d = 8.0827 Deflection is OK !!!!

Shear Design

(Using the maximum shear force for all the spans)

Support A; V_{Ed} = 51.21 KN

Using the area of tension reinforcement provided (A_{Sprov} = 402mm^2)

$V_{Rd,c}$ = 41.7692 > V_{Ed} (Therefore , shear reinforcement is required)

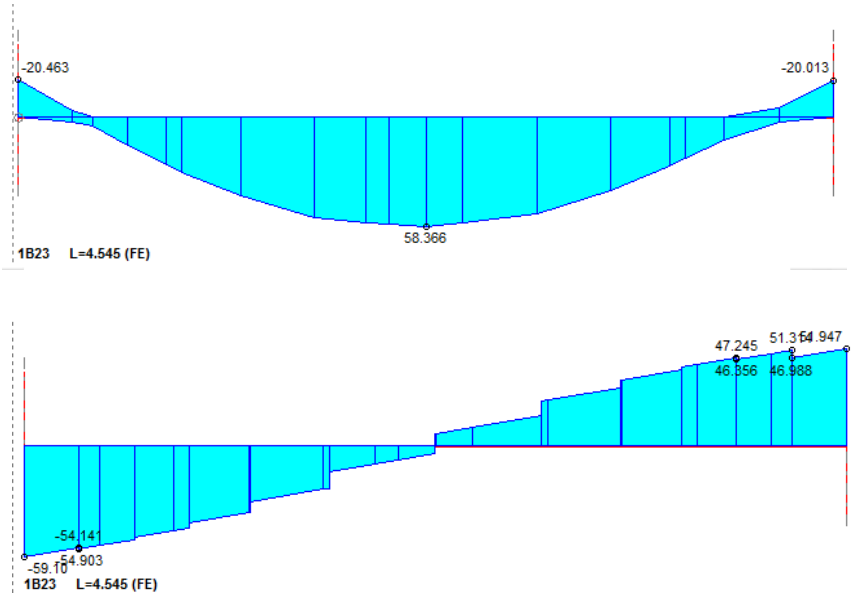
Since $V_{Rd,max}$ = 218 KN > V_{Ed} > $V_{Rd,c}$

Asw/S = 0.1492; Asw(min)/S = 0.200; Maximum spacing = 0.75d = 0.75 × 399 = 299.25

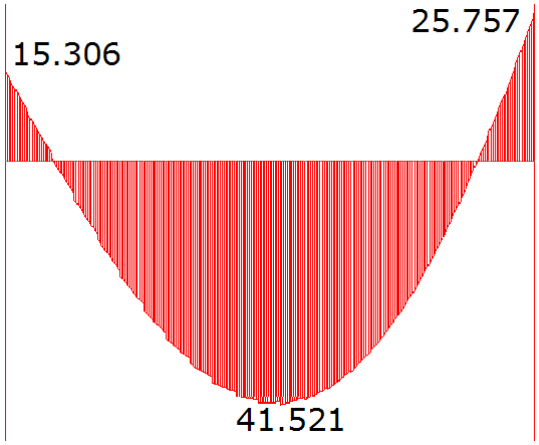
Provide 2 legs of Y8mm @ 250mm c/c (Asw/S = 0.402) all through the span. This is very ok !!!!

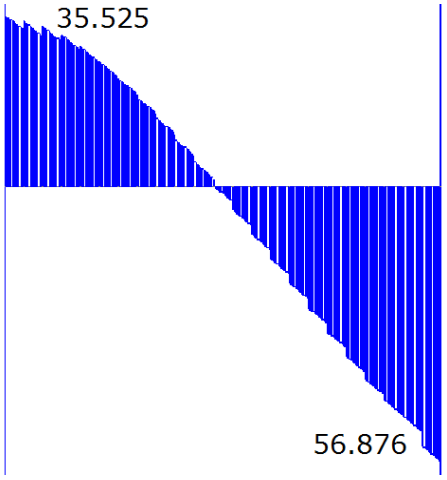
BEAM NO 4 (AXIS 8: B - E)

RESULT FROM ORION



RESULT FROM STAAD





Manual Load Analysis

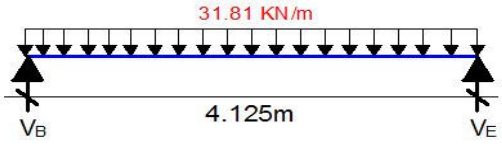
Span B - E

The floor slab (PANEL 4) is spanning in two directions, since the ratio (k) of the longer side (Ly) to the shorter side (Lx) is less than 2.

Hence, k = $\frac{Ly}{Lx}$ = $\frac{5360}{4545}$ = 1.179

The span is parallel to the short span direction of the slab. Therefore, the equivalent load that is transferred from the slab to the beam (ULS) can be represented by;

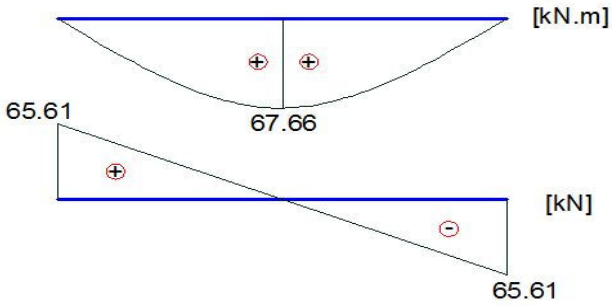
$p = \frac{nLx}{3} = \frac{10.9575 \times 4.545}{3}$ = 16.6 KN/m



Self weight of the beam (ULS) = 1.35 × 0.3m × 0.23m × 25 KN/m^3 = 2.32875 KN/m

Weight of block work (ULS) = 1.35 × 2.75m × 3.47 KN/m^2 = 12.882 KN/m

Therefore total load on **span B - E**= 16.6 + 2.32875 + 12.882 = 31.81 KN/m



STRUCTURAL DESIGN

(**SPAN B - E**) (L – Beam)

M_{Ed} = 67.66 KNm

B_{eff} = 0.2(5130/2) + 0.1(4125) + 230 = 1156mm

K = 0.0147; La/d = 0.95; A_{S1} = 446.02 mm^2; A_{Smin} = 133.0447 mm^2;

Provide 3Y16mm Bottom (A_{Sprov} = 603 mm^2)

Permissible L/d = 135.9448; Actual L/d = 10.333 Deflection is OK !!!!!

Shear Design

V_{Ed} = 65.61 KN

Using the area of tension reinforcement provided (A_{Sprov} = 603 mm^2)

$V_{Rd,c}$ = 47.8138 KN > V_{Ed} (Therefore , shear reinforcement is required)

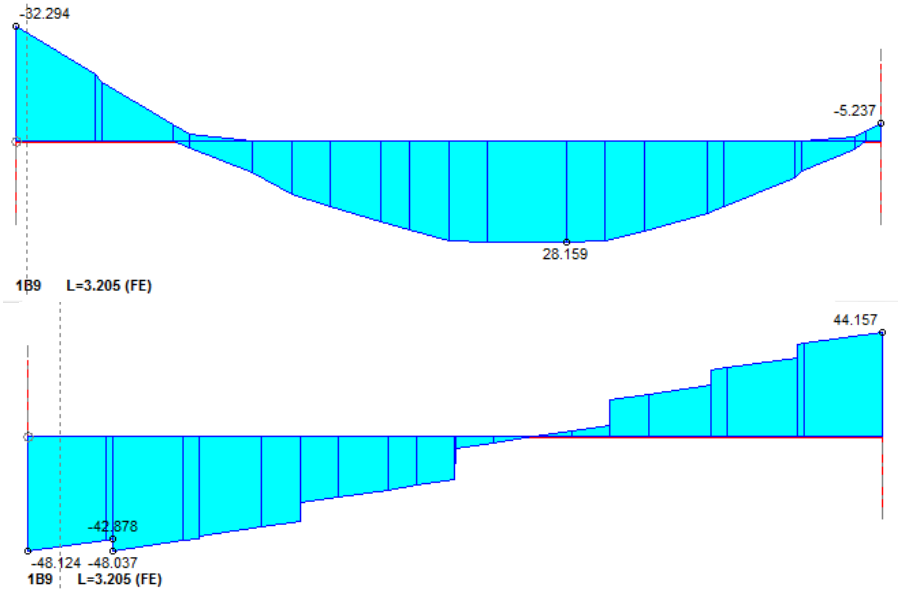
Since $V_{Rd,max}$ = 218 KN > V_{Ed} > $V_{Rd,c}$

Asw/S = 0.1826; Asw(min)/S = 0.200; Maximum spacing = 0.75d = 0.75 × 399 = 299.25

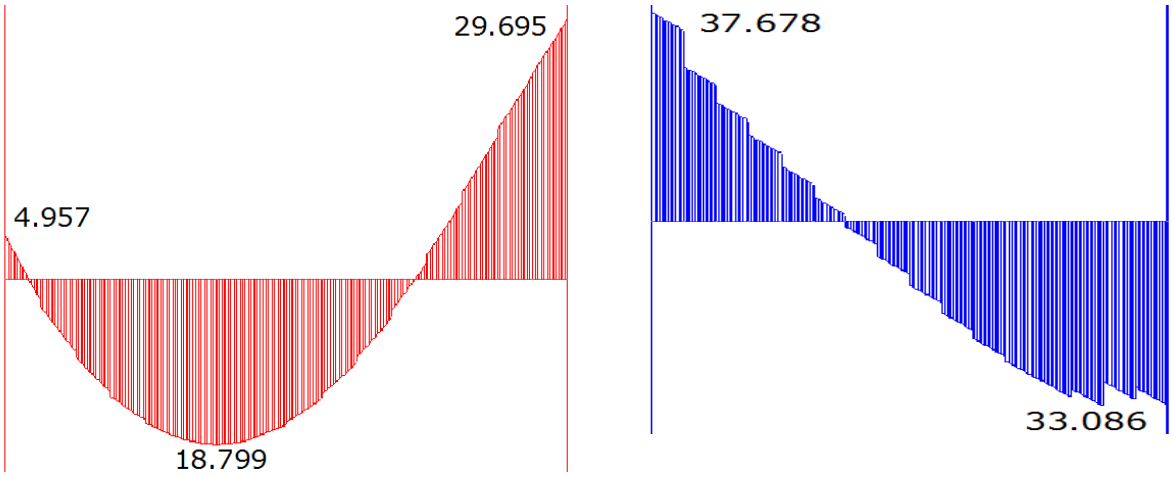
Provide 2 legs of Y8mm @ 250mm c/c (Asw/S = 0.402) all through the span. This is very ok !!!!

BEAM NO 5 (AXIS 7: A - B)

RESULT FROM ORION



RESULT FROM STAAD

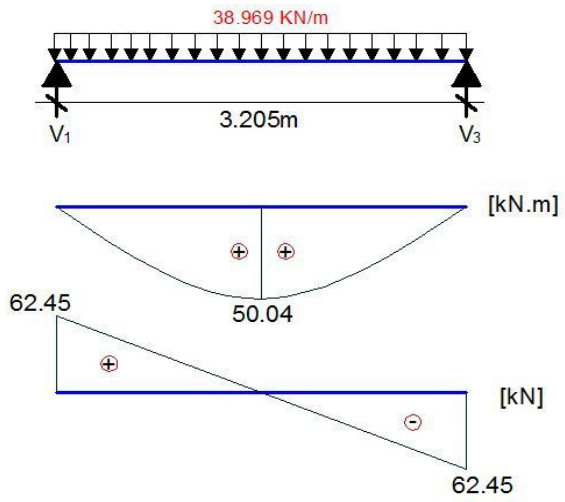


Manual Load Analysis

Span A - B

The floor slab (PANEL 3) is spanning in two directions, since the ratio (k) of the longer side (Ly) to the shorter side (Lx) is less than 2.

Hence, $k = \frac{Ly}{Lx} = \frac{3325}{3205} = 1.037$



The span is parallel to the short span direction of the slab. Therefore, the equivalent load that is transferred from the slab to the beam (ULS) can be represented by;

$p = \frac{nLx}{3} = \frac{10.9575 \times 3.205}{3} = 11.706$ KN/m

Load from adjacent cantilever slab (Assuming all is transferred to the beam) = 10.9575 × 1.1 = 12.053 KN/m

Self weight of the beam (ULS) = 1.35 × 0.3m × 0.23m × 25 KN/m^3 = 2.32875 KN/m

Weight of block work (ULS) = 1.35 × 2.75m × 3.47 KN/m^2 = 12.882 KN/m

Therefore total load on **span B – E** = 12.053 + 11.706 + 2.32875 + 12.882 = 38.969 KN/m

STRUCTURAL DESIGN

(**SPAN A - B**) (T – Beam)

M_{Ed} = 50.04 KNm

$B_{eff} = 0.2\left(\frac{985}{`2} + \frac{3095}{2}\right) + 0.1(3205) + 230 = 959$ mm

K = 0.0131; La/d = 0.95; A_{S1} = 329.8707 mm^2; A_{Smin} = 133.0447 mm^2;

Provide 2Y16mm Bottom (A_{Sprov} = 402mm^2)

Permissible L/d = 173.06; Actual L/d = 8.0326 Deflection is OK !!!!!

Shear Design

Using the maximum shear force for all the spans;

Support A; V_{Ed} = 62.45KN

Using the area of tension reinforcement provided (A_{Sprov} = 402mm^2)

$V_{Rd,c}$ = 41.7692 > V_{Ed} (Therefore , shear reinforcement is required)

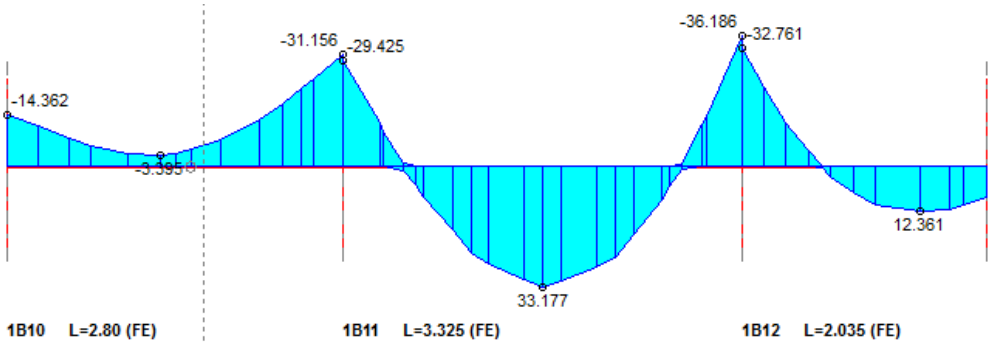
Since $V_{Rd,max}$ = 218 KN > V_{Ed} > $V_{Rd,c}$

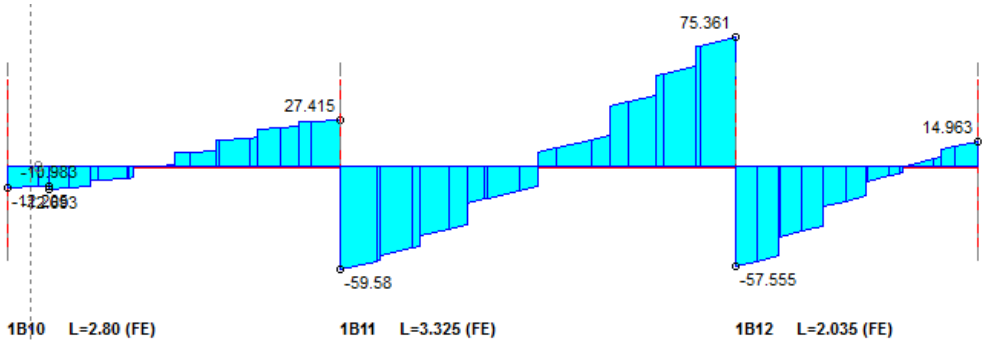
Asw/S = 0.1738; Asw(min)/S = 0.200; Maximum spacing = 0.75d = 0.75 × 399 = 299.25

Provide 2 legs of Y8mm @ 250mm c/c (Asw/S = 0.402) all through the span. This is very ok !!!!

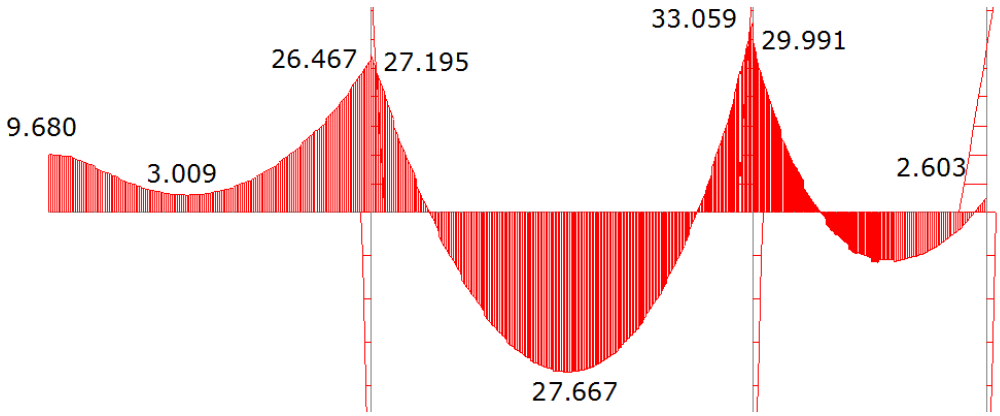
BEAM NO 6 (AXIS B: 3 - 8)

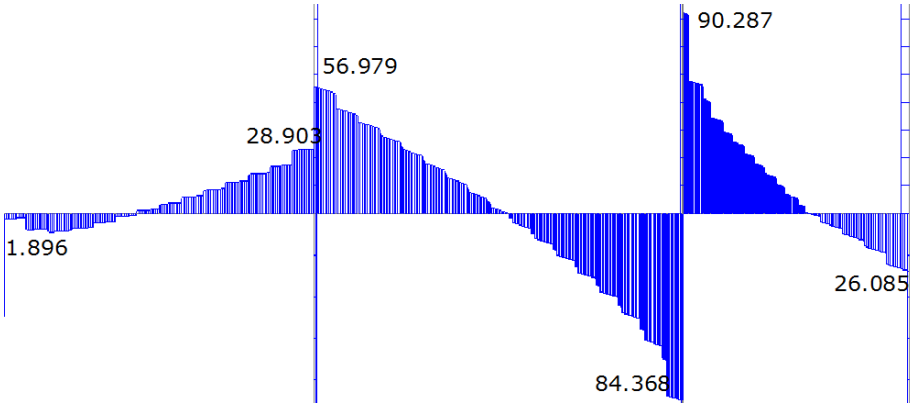
RESULTS FROM ORION





RESULTS FROM STAAD





Manual Load Analysis

Span 3 - 5

This span is receiving slab load from PANEL 2 and 7, and both are spanning in two directions, since the ratio (k) of the longer side (Ly) to the shorter side (Lx) is less than 2 in both cases.

For panel 2;

Hence, k = $\frac{Ly}{Lx} = \frac{3.625}{2.8}$ = 1.295

The span is parallel to the short span direction of the slab. Therefore, the equivalent load that is transferred from the slab to the beam can be represented by;

$p = \frac{nLx}{3} = \frac{10.9575 \times 2.8}{3}$ = 10.227 KN/m

For panel 7;

Hence, k = $\frac{Ly}{Lx} = \frac{2.8}{1.705}$ = 1.6422

The span is parallel to the long span direction of the slab. Therefore, the equivalent load that is transferred from the slab to the beam can be represented by;

$p = \frac{nLx}{2}\left(1 - \frac{1}{3k^2}\right) = \frac{10.9575 \times 1.705}{2}\left(1 - \frac{1}{3(1.6422)^2}\right)$ = 8.1867 KN/m

This span is carrying no block work load;

Self weight of the beam (ULS) = 1.35 × 0.3m × 0.23m × 25 KN/m^3 = 2.32875 KN/m

Therefore total load on the **span 3 - 5** = 10.227 + 8.1867 + 2.32875 = 20.742 KN/m

Span 5 – 8

The span is parallel to the long span direction of the slab. Therefore, the equivalent load that is transferred from the slab to the beam can be represented by;

Hence, k = $\frac{Ly}{Lx} = \frac{5360}{4545}$ = 1.179

$p = \frac{nLx}{2}\left(1 - \frac{1}{3k^2}\right) = \frac{10.9575 \times 4.545}{2}\left(1 - \frac{1}{3(1.179)^2}\right)$ = 18.929 KN/m

This span is carrying block work load;

Weight of block work = 1.35m × 2.75m × 3.47 KN/m^2 = 12.882 KN/m

Self weight of the beam (ULS) = 1.35 × 0.3m × 0.23m × 25 KN/m^3 = 2.32875 KN/m

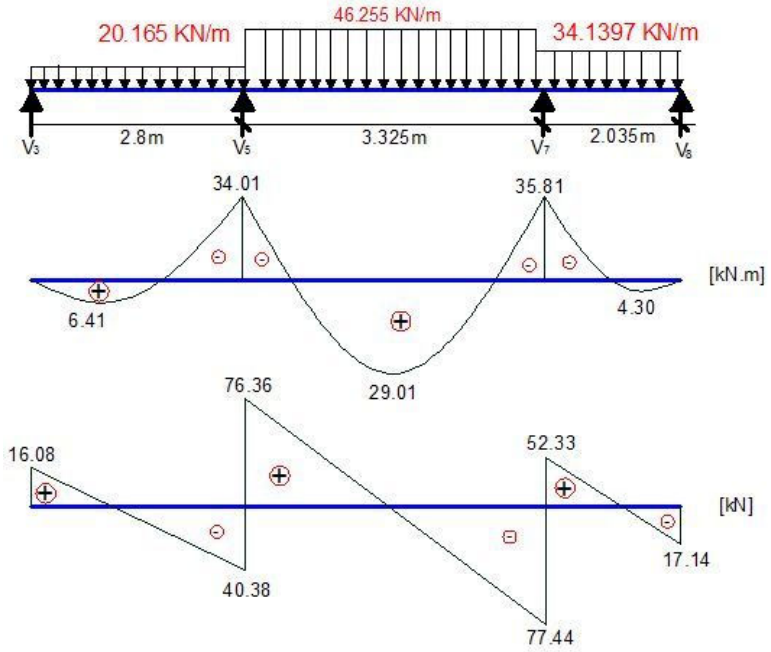
However, **span 5 – 7** is also carrying slab load from **PANEL 3** (long span direction);

Hence, k = $\frac{Ly}{Lx} = \frac{3325}{3205}$ = 1.037

$p = \frac{nLx}{2}\left(1 - \frac{1}{3k^2}\right) = \frac{10.9575 \times 3.205}{2}\left(1 - \frac{1}{3(1.037)^2}\right)$ = 12.116 KN/m

Therefore total load on the **span 5 - 7** = 18.929 + 12.882 + 2.32875 + 12.116 = 46.255 KN/m

Therefore total load on the **span 7 - 8** = 18.929 + 12.882 + 2.32875 = 34.1397 KN/m



STRUCTURAL DESIGN

(**SPAN 3 - 5**) (T – Beam)

M_{Ed} = 6.41 KNm

Provide 2Y16mm Bottom (A_{Sprov} = 402mm^2)

(**Supports 5 and 7**)

(We treat supports as rectangular sections)

M_{Ed} = 35.82 KNm (Hogging)

bw = 230mm

K = 0.0391; La/d = 0.95; A_{S1} = 236.13 mm^2; A_{Smin} = 133.0447mm^2;

Provide 2Y16mm Top (A_{Sprov} = 402mm^2)

(**SPAN 5 - 7**) (T – Beam)

M_{Ed} = 29.01 KNm (Sagging)

B_{eff} = [0.2$\left(\frac{4315}{2}\right)$+ 0.2$\left(\frac{2975}{2}\right)$ + 0.1(0.7 × 3325)] + 230 = 1192 mm

K = 0.0061; La/d = 0.95; A_{S1} = 191.238 mm^2; A_{Smin} = 159.6428 mm^2;

Provide 2Y16 Bottom (A_{Sprov} = 402 mm^2)

Permissible L/d = 785.1259; Actual L/d = 8.3333 Deflection is OK !!!!!

Shear Design

(Picking the maximum shear force for all the spans)

Support 7; V_{Ed} = 77.44 KN

Using the area of tension reinforcement provided (A_{Sprov} = 402mm^2)

$V_{Rd,c}$ = 41.7692 > V_{Ed} (Therefore , shear reinforcement is required)

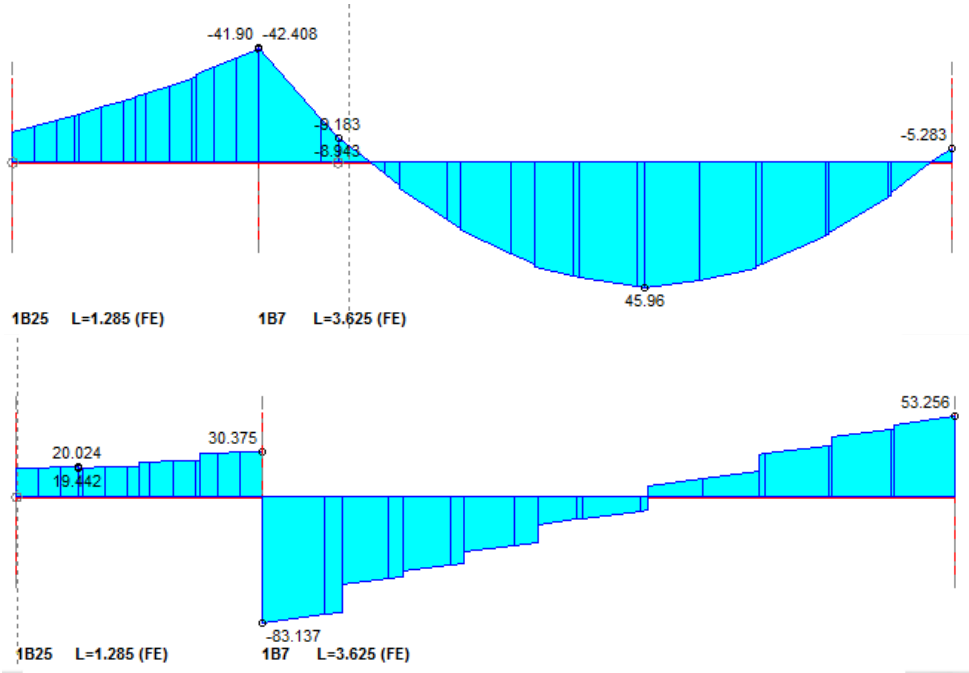
Since $V_{Rd,max}$ = 218 KN > V_{Ed} > $V_{Rd,c}$

Asw/S = 0.2155; Asw(min)/S = 0.200; Maximum spacing = 0.75d = 0.75 × 399 = 299.25

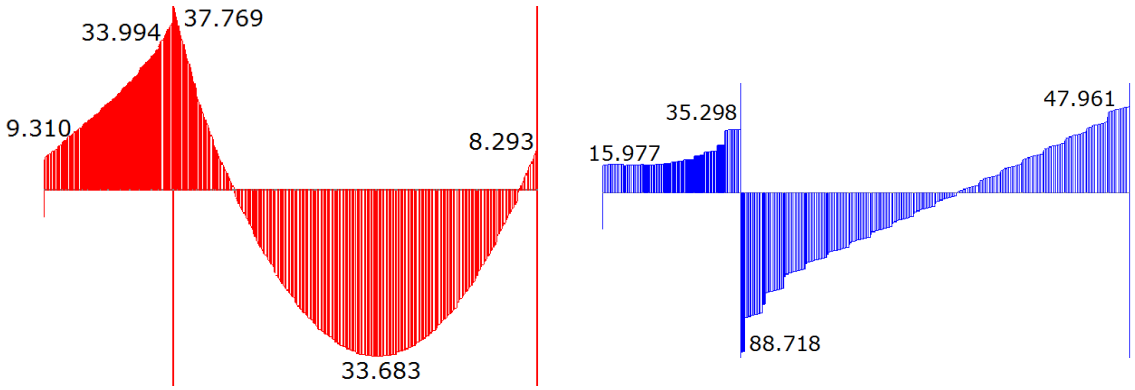
Provide 2 legs of Y8mm @ 250mm c/c (Asw/S = 0.402) This is very ok !!!!

BEAM NO 7 (AXIS 3: A - D)

RESULTS FROM ORION



RESULT FROM STAAD



Manual Load Analysis

Span A - C

This span is receiving slab load from PANEL 1 and 2, and partly from PANEL 7. There is also a concentrated load from beam NO 6. All panels are spanning in two directions, since the ratio (k) of the longer side (Ly) to the shorter side (Lx) is less than 2 in all cases.

For PANEL 1

Hence, k = $\frac{Ly}{Lx} = \frac{3.825}{3.625}$ = 1.055

We can observe from the general arrangement of the structure that the span is parallel to the short span direction of the slab. Therefore, the equivalent load that is transferred from the slab to the beam can be represented by;

$p = \frac{nLx}{3} = \frac{10.9575 \times 3.625}{3}$ = 13.240 KN/m

For PANEL 2

Hence, k = $\frac{Ly}{Lx} = \frac{3.625}{2.8}$ = 1.2946

$p = \frac{nLx}{2}\left(1 - \frac{1}{3k^2}\right) = \frac{10.9575 \times 2.8}{2}\left(1 - \frac{1}{3(1.2946)^2}\right)$ = 12.289 KN/m

Load from PANEL 7

Hence, k = $\frac{Ly}{Lx} = \frac{2.8}{1.705}$ = 1.6422

$p = \frac{nLx}{3} = \frac{10.9575 \times 1.705}{3}$ = 6.2275 KN/m

Load from PANEL 9

PANEL 9 is spanning in one direction, and parallel to the short span of the beam. Therefore the load from the slab to the beam is given by;

$p = \frac{nLx}{5} = \frac{10.9575 \times 1.705}{5}$ = 3.7365 KN/m

Concentrated load from beam NO 6;

P = 15.06 KN

Self weight of the beam (ULS) = 1.35 × 0.3m × 0.23m × 25 KN/m^3 = 2.32875 KN/m

Weight of block work (ULS) = 1.35 × 2.75m × 3.47 KN/m^2 = 12.882 KN/m

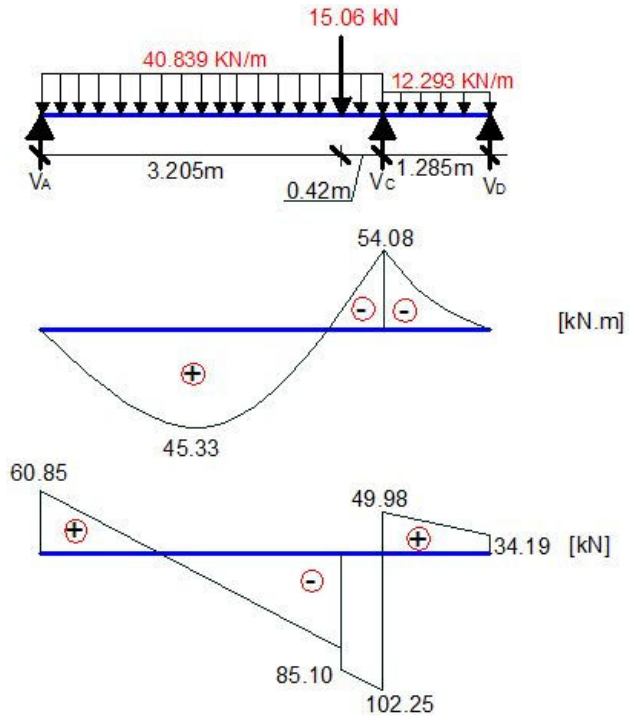
Therefore total load on section A - B = Load from PANEL 1 + Load from PANEL2 + Load from block work + self weight = 13.24 + 12.389 + 2.32875 + 12.882 = 40.839 KN/m

Concentrated load at point B = 16.38 KN

Load on section B - C = Load from PANEL 1 + Load from PANEL 7 + Load from block work + self weight = 13.24 + 6.2275 + 2.32875 + 12.882 = 34.678 KN/m

Let us conservatively assume that the 40.839 KN/m is distributed all over the span

Load on span C – D = Load from PANEL 9 + Load from PANEL 7 + Self weight = 3.7365 + 6.2275 + 2.32875 = 12.293 KN/m



I

STRUCTURAL DESIGN

(**SPAN A - C**) (T – Beam)

M_{Ed} = 45.33 KNm (Sagging)

$B_{eff} = [0.2\left(\frac{3595}{2}\right) + 0.2\left(\frac{2570}{2}\right) + 0.1(0.85 \times 3625)] + 230 = 1155mm$

K = 0.0099; La/d = 0.95; A_{S1} = 298.82 mm^2; A_{Smin} = 133.0447mm^2;

Provide 2Y16mm Bottom (A_{Sprov} = 402mm^2)

Also provide 2Y16 Bottom (A_{Sprov} = 402)

Permissible L/d = 332.5615; Actual L/d = 9.0852 Deflection is OK !!!!!

(**Support C**)

M_{Ed} = 54.08 KNm (Hogging)

bw = 230mm

K = 0.0591; La/d = 0.9467; A_{S1} = 358.44 mm^2; A_{Smin} = 133.0447mm^2;

Provide 2Y16mm Top (A_{Sprov} = 402mm^2)

Shear Design

(Picking the maximum shear force for all the spans)

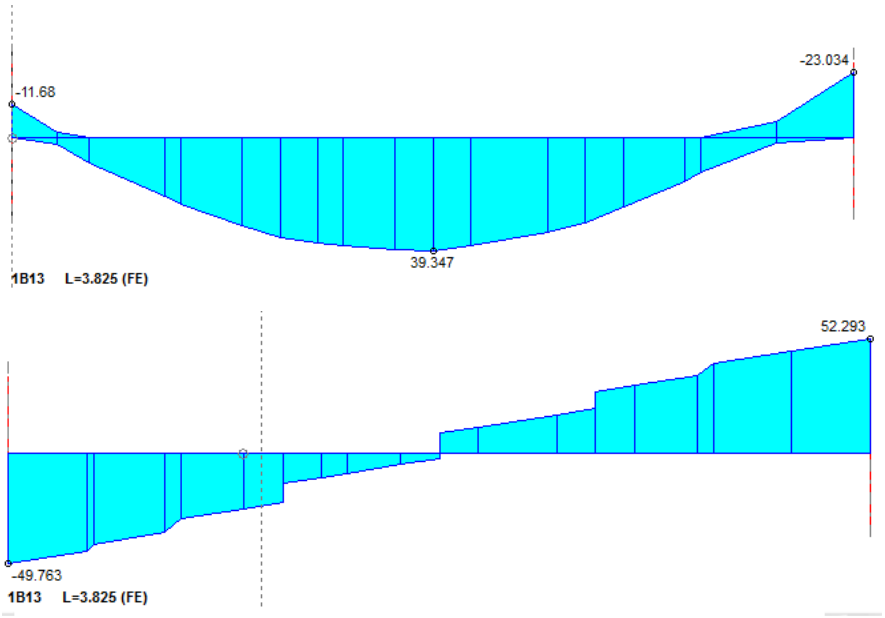
Support 7; V_{Ed} = 102.25 KN

Using the area of tension reinforcement provided (A_{Sprov} = 402mm^2)

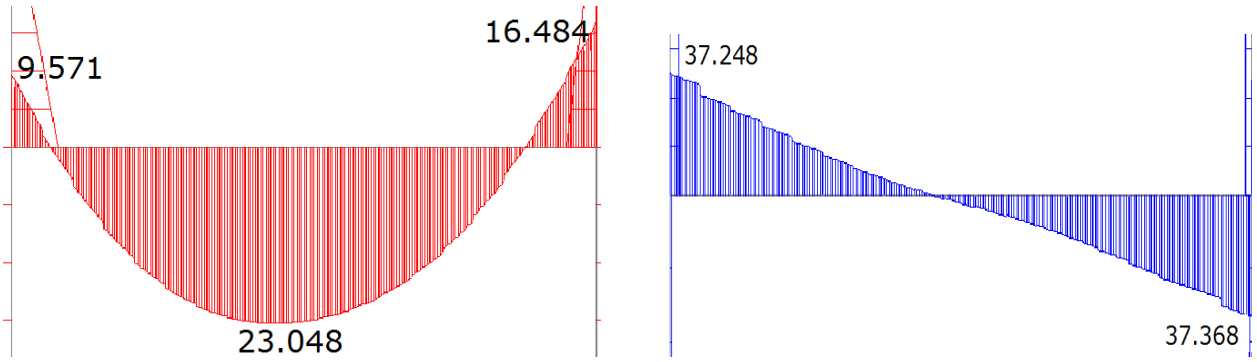
$V_{Rd,c}$ = 41.7692 > V_{Ed} (Therefore , shear reinforcement is required)
Since $V_{Rd,max}$ = 218 KN > V_{Ed} > $V_{Rd,c}$
Asw/S = 0.2846; Asw(min)/S = 0.200; Maximum spacing = 0.75d = 0.75 × 399 = 299.25
Provide 2 legs of Y8mm @ 250mm c/c (Asw/S = 0.402) This is very ok !!!!

BEAM NO 8 (AXIS C: 1 - 3)

RESULT FROM ORION



RESULT FROM STAAD



Manual Load Analysis

This span is receiving slab load from PANEL 1 and PANEL 7.

Load from PANEL 1

Hence, k = $\frac{Ly}{Lx} = \frac{3.825}{3.625}$ = 1.055

We can observe from the general arrangement of the structure that the span is parallel to the long span direction of the slab. Therefore, the equivalent load that is transferred from the slab to the beam can be represented by;

$p = \frac{nLx}{2}\left(1 - \frac{1}{3k^2}\right) = \frac{10.9575 \times 3.625}{2}\left(1 - \frac{1}{3(1.055)^2}\right)$ = 13.913 KN/m

Load from PANEL 9

PANEL 9 is spanning in one direction, and parallel to the short span of the beam. Therefore the load from the slab to the beam is given by;

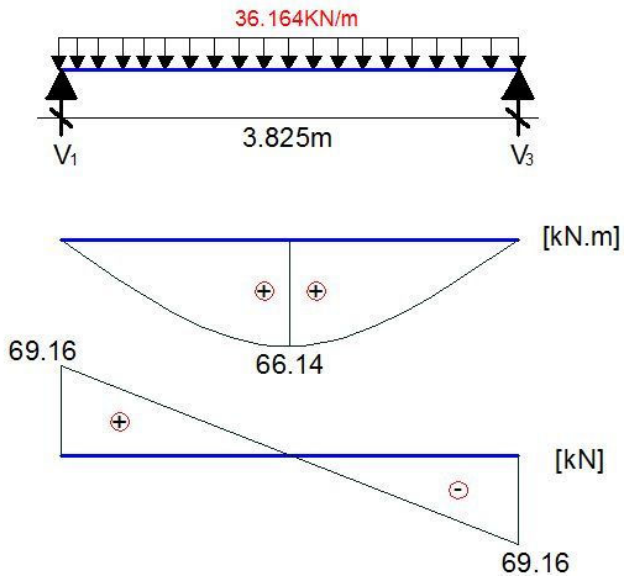
$p = \frac{nLx}{2} = \frac{10.9575 \times 1.285}{2}$ = 7.040 KN/m

Self weight of the beam (ULS) = 1.35 × 0.3m × 0.23m × 25 KN/m^3 = 2.32875 KN/m

Weight of block work (ULS) = 1.35 × 2.75m × 3.47 KN/m^2 = 12.882 KN/m

Total load on the span = 13.913+ 7.040 + 2.3287 + 12.882 = 36.164 KN/m

STRUCTURAL DESIGN



(**SPAN 1 - 3**) (T – Beam)

M_{Ed} = 66.14 KNm (Sagging)

B_{eff} = $[0.2\left(\frac{3395}{2}\right) + 0.2\left(\frac{1055}{2}\right) + 0.1(0.85 \times 3825)] + 230$ = 1000mm

K = 0.01661; La/d = 0.95; A_{S1} = 436 mm^2; A_{Smin} = 133.0447mm^2;

Provide 3 Y16mm Bottom (A_{Sprov} = 603mm^2)

Permissible L/d = 110.558; Actual L/d = 9.5865 Deflection is OK !!!!!

Shear Design

Support 7; V_{Ed} = 69.16 KN

Using the area of tension reinforcement provided (A_{Sprov} = 402mm^2)

$V_{Rd,c}$ = 47.8138 > V_{Ed} (Therefore , shear reinforcement is required)

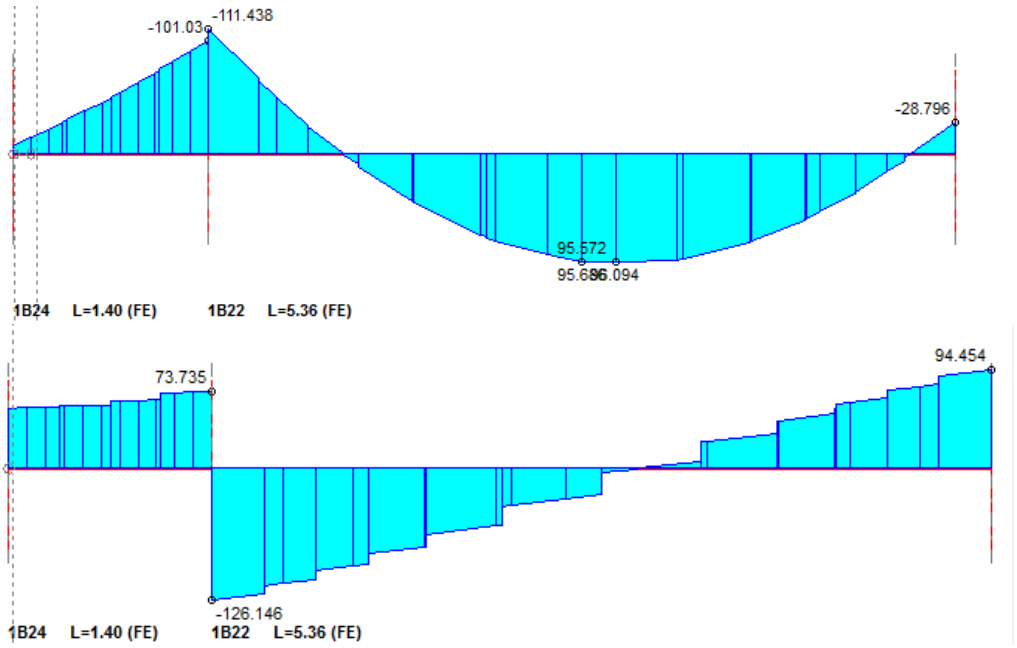
Since $V_{Rd,max}$ = 218 KN > V_{Ed} > $V_{Rd,c}$

Asw/S = 0.1925; $A_{sw(min)}$/S = 0.200; Maximum spacing = 0.75d = 0.75 × 399 = 299.25

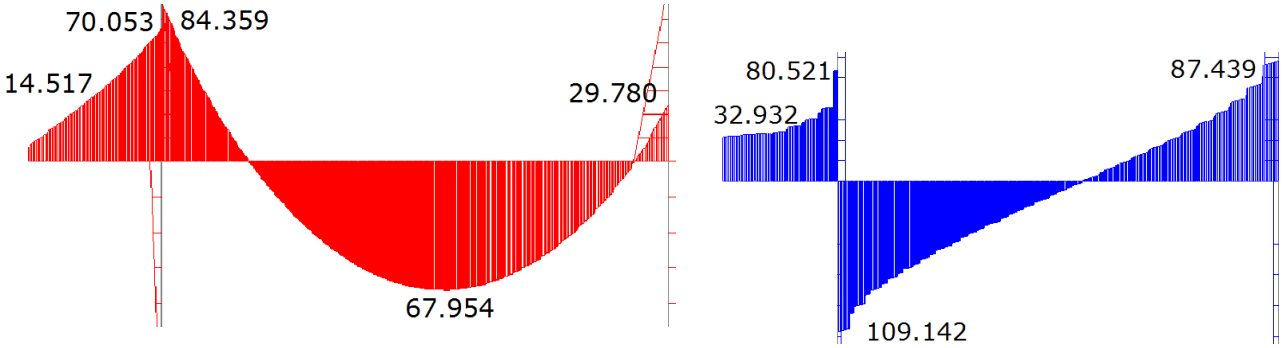
Provide 2 legs of Y8mm @ 250mm c/c (Asw/S = 0.402) This is very ok !!!!

BEAM NO 9 (AXIS E: 4 - 8)

RESULT FROM ORION



RESULT FROM STAAD



Manual Load Analysis

SPAN 4 – 5

This span is receiving load from PANELS 5 and 6

In this case, we are going to assume that the entire load on PANEL 5 (cantilever slab) are transferred completely to the beams;

Load from PANEL 5

p = 10.9575×1.21 = 13.258 KN/m

Load from PANEL 6

Hence, k = $\frac{Ly}{Lx}$ = $\frac{2.837}{1.4}$ = 2.026

Let us take it that the slab is spanning in one direction (for short span direction)

$p = \frac{nLx}{5} = \frac{10.9575 \times 1.4}{5}$ = 3.068 KN/m

Load from PANEL 4

The span is parallel to the long span direction of the slab. Therefore, the equivalent load that is transferred from the slab to the beam can be represented by;

Hence, k = $\frac{Ly}{Lx} = \frac{5360}{4545}$ = 1.179

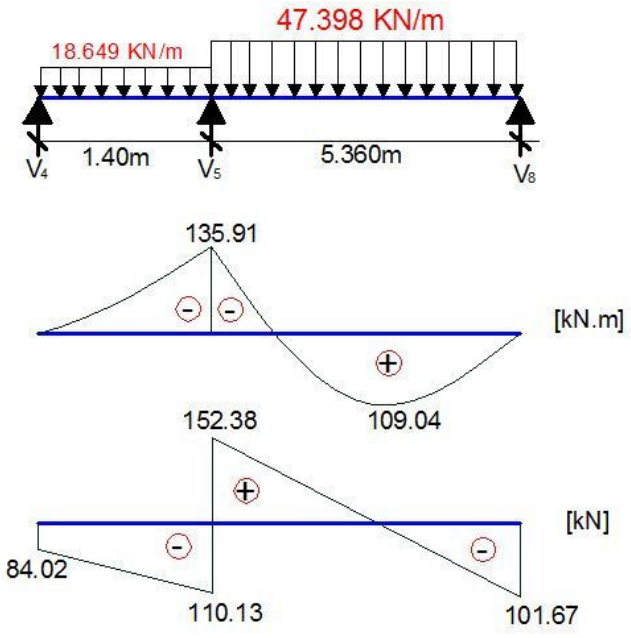
$p = \frac{nLx}{2}\left(1 - \frac{1}{3k^2}\right) = \frac{10.9575 \times 4.545}{2}\left(1 - \frac{1}{3(1.179)^2}\right)$ = 18.929 KN/m

Self weight of the beam (ULS) = 1.35 × 0.3m × 0.23m × 25 KN/m^3 = 2.32875 KN/m

Weight of block work (ULS) = 1.35 × 2.75m × 3.47 KN/m^2 = 12.882 KN/m

Total load on the **span 4 - 5** = Load from panel 5 + Load from PANEL 6 + Self weight (No block work load) = 13.258 + 3.068 + 2.3287 = 18.649 KN/m

Total load on the **span 5 - 8** = Load from panel 5 + Load from PANEL 4 + Self weight + block work load = 13.258 + 18.929 + 2.3287 + 12.882 = 47.398 KN/m



STRUCTURAL DESIGN

(**SPAN 5 - 8**) (T – Beam)

M_{Ed} = 109.04 KNm (Sagging)

B_{eff} = $[0.2\left(\frac{1200}{2}\right) + 0.2\left(\frac{4315}{2}\right) + 0.1(0.85 \times 5360)] + 230$ = 1237 mm

K = 0.0257; La/d = 0.95; A_{S1} = 718.806 mm^2; A_{Smin} = 133.0447mm^2;

Provide 3Y20mm Bottom (A_{Sprov} = 942 mm^2)

Permissible L/d = 94.266; Actual L/d = 13.4336
Deflection is OK !!!!!

Support 5

M_{Ed} = 135.91 KNm (Hogging)

b_w = 230mm

K = 0.148; La/d = 0.845; A_{S1} = 1007.22 mm^2; A_{Smin} = 133.0447mm^2;

Provide 4Y20mm Top (A_{Sprov} = 1257mm^2)

Shear Design

(Picking the maximum shear force for all the spans)

Support 7; $\mathbf{V_{Ed}}$ **= 152.38 KN**

Using the area of tension reinforcement provided (A_{Sprov} = 1020mm^2)

$V_{Rd,c}$ = 61.079 > V_{Ed} (Therefore , shear reinforcement is required)

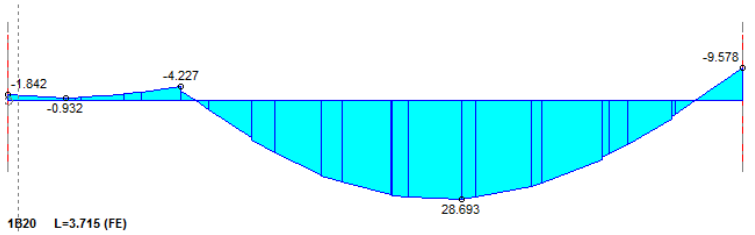
Since $V_{Rd,max}$ = 218 KN > V_{Ed} > $V_{Rd,c}$

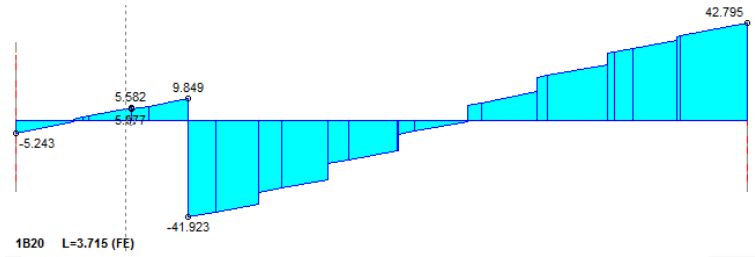
Asw/S = 0.424; Asw(min)/S = 0.200; Maximum spacing = 0.75d = 0.75 × 399 = 299.25

Provide 2 legs of Y8mm @ 200mm c/c (Asw/S = 0.502) This is very ok !!!!

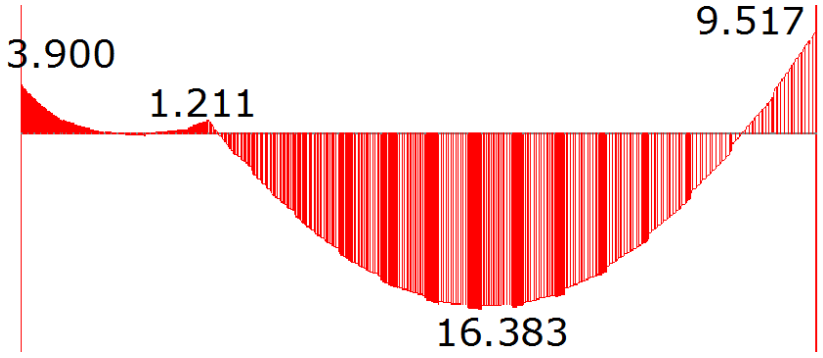
BEAM NO 10 (AXIS 4: D - G)

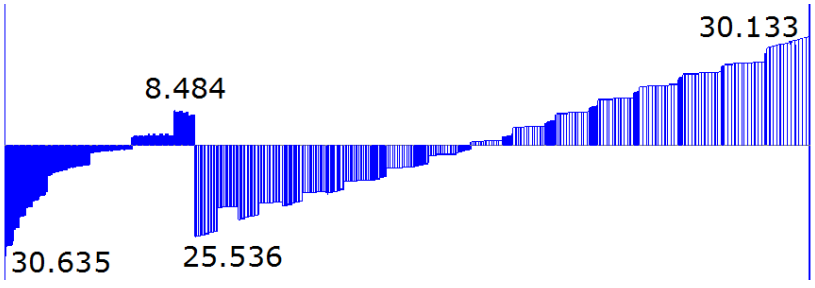
RESULTS FROM ORION





RESULT FROM STAAD





Manual Load Analysis

SPAN 4 – 5

This span is receiving load from PANELS 8 and 6, and partly from PANEL 5

In this case, we are going to assume that 30% of the load on PANEL 5 (cantilever slab) is transferred to the beams;

Load from PANEL 5

$p = 10.9575 \times 1.21 = 13.258$ KN/m

Hence, $0.3 \times 13.258 = 3.9774$ KN/m

Load from PANEL 8

The beam is parallel to the long span direction of the slab. Therefore, the equivalent load that is transferred from the slab to the beam can be represented by;

Hence, $k = \frac{Ly}{Lx} = \frac{3715}{3225} = 1.151$

$p = \frac{nLx}{2}\left(1 - \frac{1}{3k^2}\right) = \frac{10.9575 \times 3.225}{2}\left(1 - \frac{1}{3(1.151)^2}\right) = 13.223$ KN/m

Load from PANEL 6

Hence, $k = \frac{Ly}{Lx} = \frac{2.837}{1.4} = 2.026$

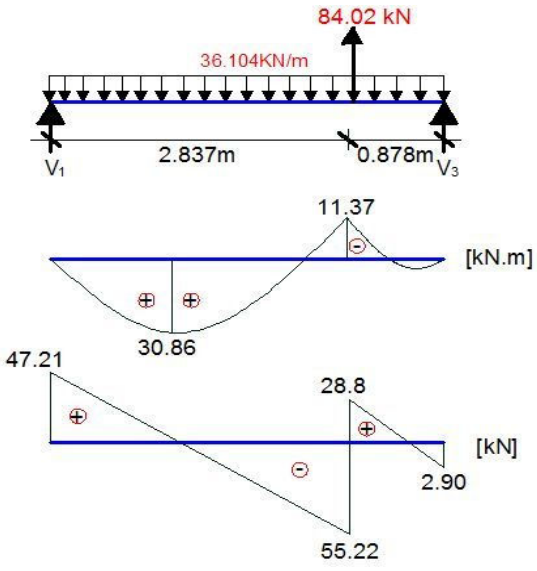
Let us take it that the slab is spanning in one direction (for long span direction)

$p = \frac{nLx}{2} = \frac{10.9575 \times 1.4}{2} = 7.670$ KN/m

Concentrated load from beam NO 9 = 77.95 KN

Self weight of the beam (ULS) = 1.35 × 0.3m × 0.23m × 25 KN/m^3 = 2.32875 KN/m

Weight of block work (ULS) = 1.35 × 2.75m × 3.47 KN/m^2 = 12.882 KN/m



Total load on **section D - E** = Load from panel 8 + Load from PANEL 6 + Self weight + block work load = 13.223 + 7.670 + 2.3287 + 12.882 = 36.104 KN/m

Total load on **section 5 - 8** = Load from panel 8+ Load from PANEL 5 + Self weight + block work load = 13.223 + 3.9744 + 2.3287 + 12.882 = 32.408 KN/m

We can conservatively assume that the load 36.104 KN/m is uniformly distributed across the span.

STRUCTURAL DESIGN

(T – Beam)

M_{Ed} = 30.86 KNm (Sagging)

B_{eff} = $[0.2\left(\frac{2995}{2}\right) + 0.2\left(\frac{1170}{2}\right) + 0.1(3715)] + 230 = 1390mm$

K = 0.00557; La/d = 0.95; A_{S1} = 203.43 mm^2; A_{Smin} = 133.0447mm^2;

Provide 2Y16mm Top and Bottom (A_{Sprov} = 402 mm^2)

Permissible L/d = 501.621; Actual L/d = 9.3108 Deflection is OK !!!!!

Shear Design

(Picking the maximum shear force for all the spans)

Support 7; V_{Ed} = 55.22 KN

Using the area of tension reinforcement provided (A_{Sprov} = 1020mm^2)

$V_{Rd,c}$ = 41.7692 > V_{Ed} (Therefore , shear reinforcement is required)

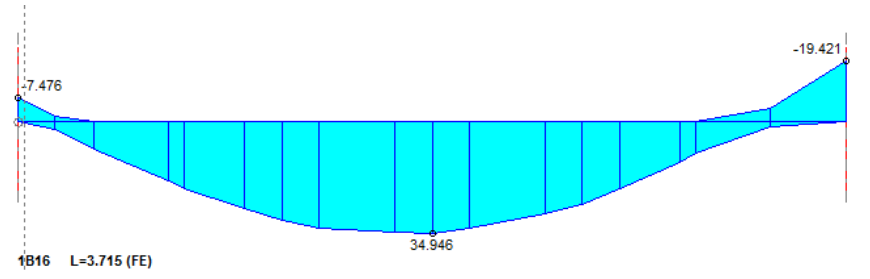
Since $V_{Rd,max}$ = 218 KN > V_{Ed} > $V_{Rd,c}$

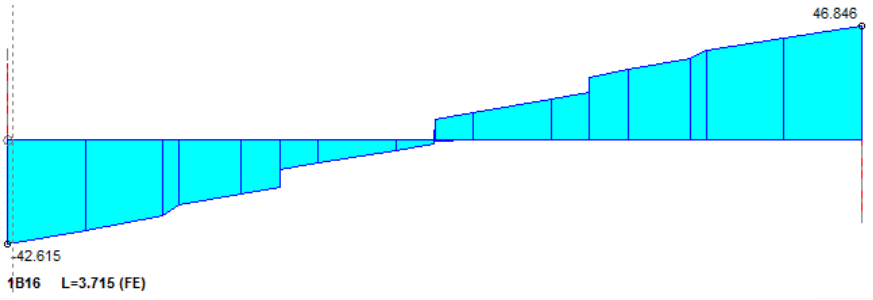
Asw/S = 0.1536; Asw(min)/S = 0.200; Maximum spacing = 0.75d = 0.75 × 399 = 299.25

Provide 2 legs of Y8mm @ 250mm c/c (Asw/S = 0.402) This is very ok !!!!

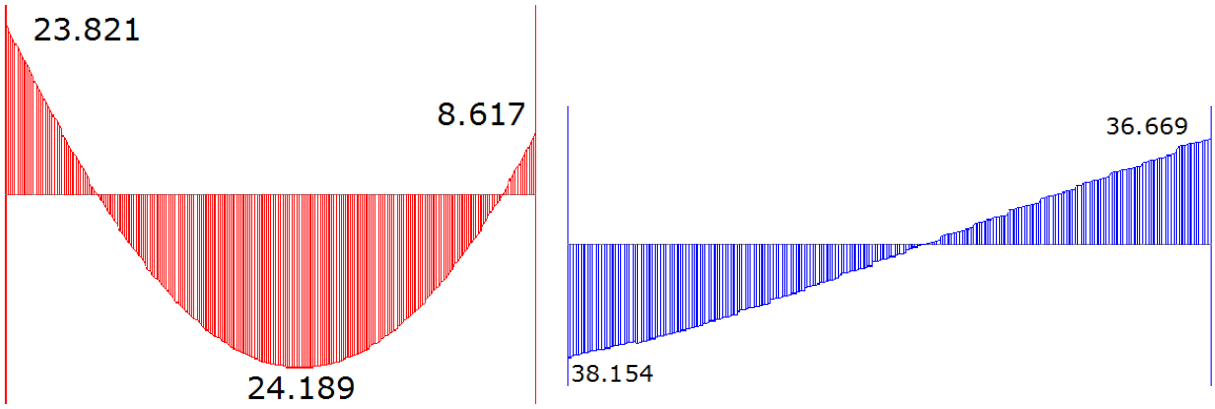
BEAM NO 11 (AXIS 2: D - F)

RESULTS FROM ORION





RESULT FROM STAAD



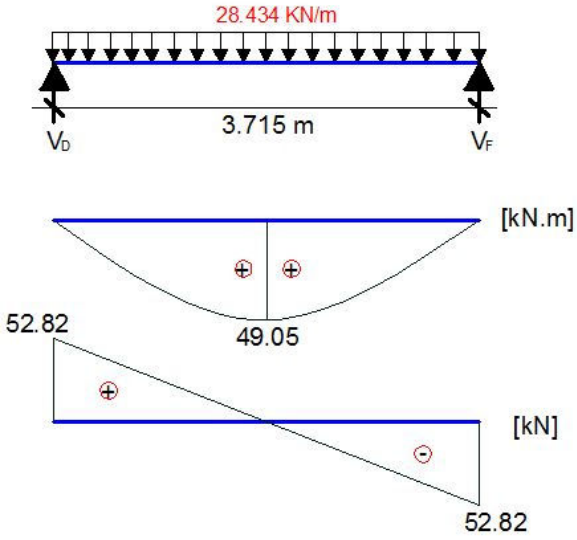
Manual Load Analysis

SPAN 4 – 5

This span is receiving load from PANEL 8 only

Load from PANEL 8

The beam is parallel to the long span direction of the slab. Therefore, the equivalent load that is transferred from the slab to the beam can be represented by;



Hence, k = $\frac{Ly}{Lx} = \frac{3715}{3225}$ = 1.151

$p = \frac{nLx}{2}\left(1 - \frac{1}{3k^2}\right) = \frac{10.9575 \times 3.225}{2}\left(1 - \frac{1}{3(1.151)^2}\right)$ = 13.223 KN/m

Self weight of the beam (ULS) = 1.35 × 0.3m × 0.23m × 25 KN/m^3 = 2.32875 KN/m

Weight of block work (ULS) = 1.35 × 2.75m × 3.47 KN/m^2 = 12.882 KN/m

Total load on **span D - F** = 13.223 + 2.3287 + 12.882 = 28.434 KN/m

STRUCTURAL DESIGN

(L – Beam)

M_{Ed} = 49.05 KNm (Sagging)

B_{eff} = [0.2$\left(\frac{2995}{2}\right)$ + 0.1(3715)] + 230 = 901 mm

K = 0.0137; La/d = 0.95; A_{S1} = 323.344 mm^2; A_{Smin} = 133.0447mm^2;

Provide 2Y16mm Bottom (A_{Sprov} = 402 mm^2)

Permissible L/d = 160.0226; Actual L/d = 9.3108 Deflection is OK !!!!!

Shear Design

V_{Ed} = 52.82 KN

Using the area of tension reinforcement provided (A_{Sprov} = 1020mm^2)

$V_{Rd,c}$ = 41.7692 > V_{Ed} (Therefore , shear reinforcement is required)

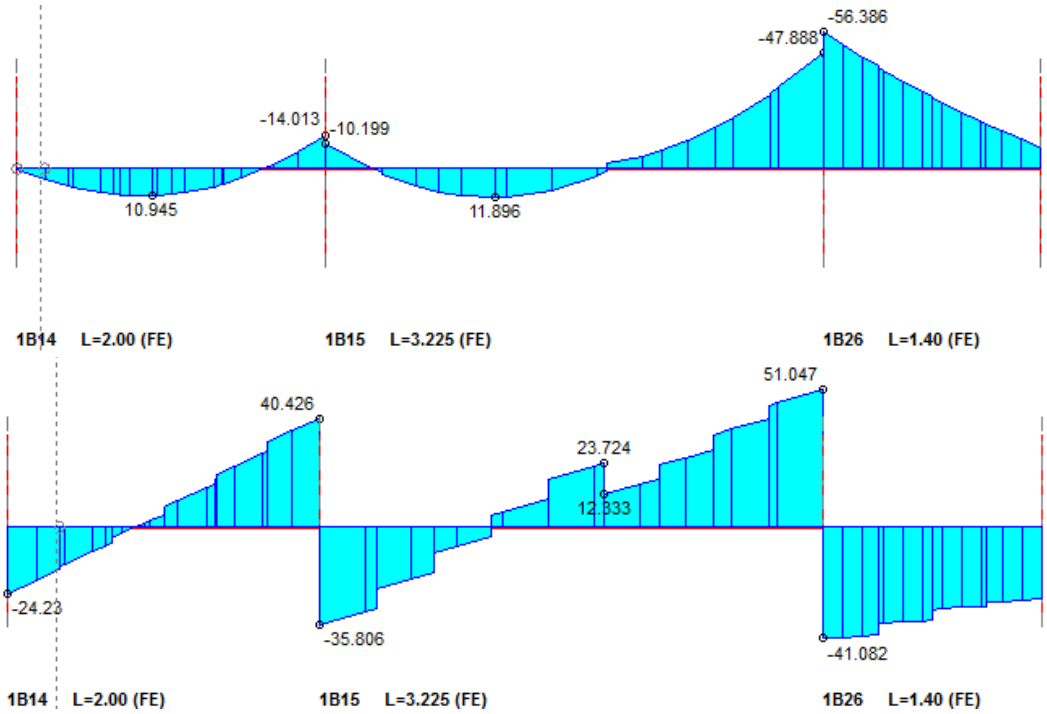
Since $V_{Rd,max}$ = 218 KN > V_{Ed} > $V_{Rd,c}$

Asw/S = 0.1470; $A_{sw,min}$/S = 0.200; Maximum spacing = 0.75d = 0.75 × 399 = 299.25

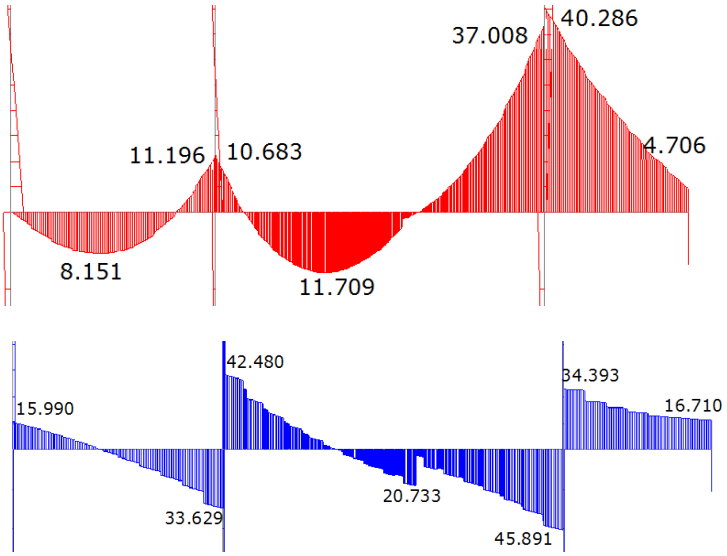
Provide 2 legs of Y8mm @ 250mm c/c (Asw/S = 0.402) This is very ok !!!!

BEAM NO 12 (AXIS D: 1 - 5)

RESULT FROM ORION



RESULT FROM STAAD



Manual Load Analysis

SPAN 1 – 2

This span is receiving load from PANEL 9 and staircase;

Reactive load from staircase = 44.147 KN

Distributing this load across the beam, *p* = 44.137/2 = 22.0685KN/m

Load from PANEL 9

PANEL 9 is spanning in one direction, and parallel to the short span of the beam. Therefore the load from the slab to the beam is given by;

$p = \frac{nLx}{2} = \frac{10.9575 \times 1.285}{2}$ = 7.040 KN/m

Load from PANEL 8

A little consideration will show that the floor slab (PANEL 9) is spanning in two directions, since the ratio (k) of the longer side (Ly) to the shorter side (Lx) is less than 2.

Hence, k = $\frac{Ly}{Lx} = \frac{3715}{3225}$ = 1.152

We can observe from the general arrangement of the structure that the span is parallel to the short span direction of the slab. Therefore, the equivalent load that is transferred from the slab to the beam (ULS) can be represented by;

$p = \frac{nLx}{3} = \frac{10.9575 \times 3.225}{3}$ = 11.779 KN/m

Load from PANEL 7

Hence, k = $\frac{Ly}{Lx} = \frac{2.8}{1.705}$ = 1.6422

The span is parallel to the long span direction of the slab. Therefore, the equivalent load that is transferred from the slab to the beam can be represented by;

$p = \frac{nLx}{2}\left(1 - \frac{1}{3k^2}\right) = \frac{10.9575 \times 1.705}{2}\left(1 - \frac{1}{3(1.6422)^2}\right)$ = 8.1866 KN/m

Load from PANEL 6

Hence, k = $\frac{Ly}{Lx} = \frac{2.837}{1.4}$ = 2.026

Let us take it that the slab is spanning in one direction (for short span direction)

$p = \frac{nLx}{5} = \frac{10.9575 \times 1.4}{5}$ = 3.068 KN/m

Concentrated Load from Beam No 7 = 34.19 KN (upwards)

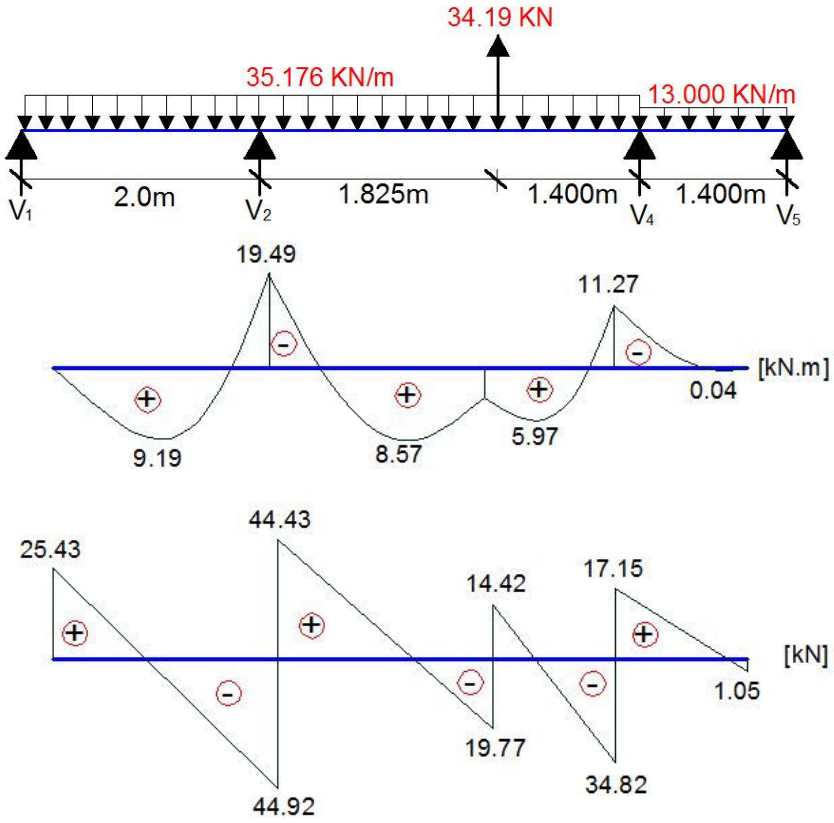
Load on span 1- 2 = Load from PANEL 9 + Staircase Load + Self weight = 22.068 + 7.040 + 2.32875 = 31.437 KN/m

Load on section 2 - 3 = Load from PANEL 9 + Load from PANEL 8 + Self weight + Block work load = 7.040 + 11.779 + 12.882 + 2.32875 = 34.030 KN/m

Load on section 3 - 4 = Load from PANEL 8 + Load from PANEL 7 + Self weight + Block work load = 11.779 + 8.1866 + 2.32875 + 12.882 = 35.176 KN/m

Load on span 4 - 5 = Load from PANEL 6 + Load from PANEL 7 + Self weight = 3.068 + 7.609 + 2.32875 = 13.00 KN/m

Let us assume that the uniformly distributed load of 35.176 KN/m runs across the span 1-4



STRUCTURAL DESIGN

Bottom Reinforcement Span (T – Beam)

M_{Ed} = 9.19 KNm (Sagging)

$B_{eff} = [0.2\left(\frac{3490}{2}\right) + 0.2\left(\frac{1060}{2}\right) + 0.1(0.7 \times 3715)] + 230 = 715mm$

K = 0.0032; La/d = 0.95; A_{S1} = 60.5818 mm^2; A_{Smin} = 133.0447mm^2;

Provide 2Y16mm Bottom (A_{Sprov} = 402 mm^2)

Permissible L/d = 404.7569; Actual L/d = 8.0827 Deflection is OK !!!!!

Top Reinforcement Span (T – Beam)

M_{Ed} = 19.49 KNm (Hogging)

B_{eff} = bw

K = 0.022; La/d = 0.95; A_{S1} = 127.5579 mm^2; A_{Smin} = 133.0447mm^2;

Provide 2Y16mm Top (A_{Sprov} = 402 mm^2)

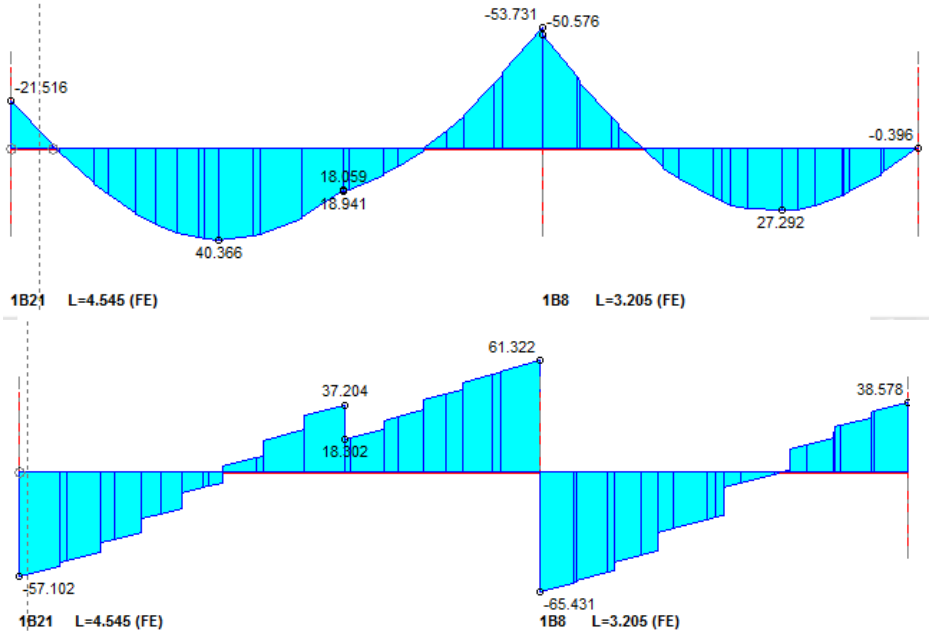
Shear Design

(Picking the maximum shear force for all the spans)

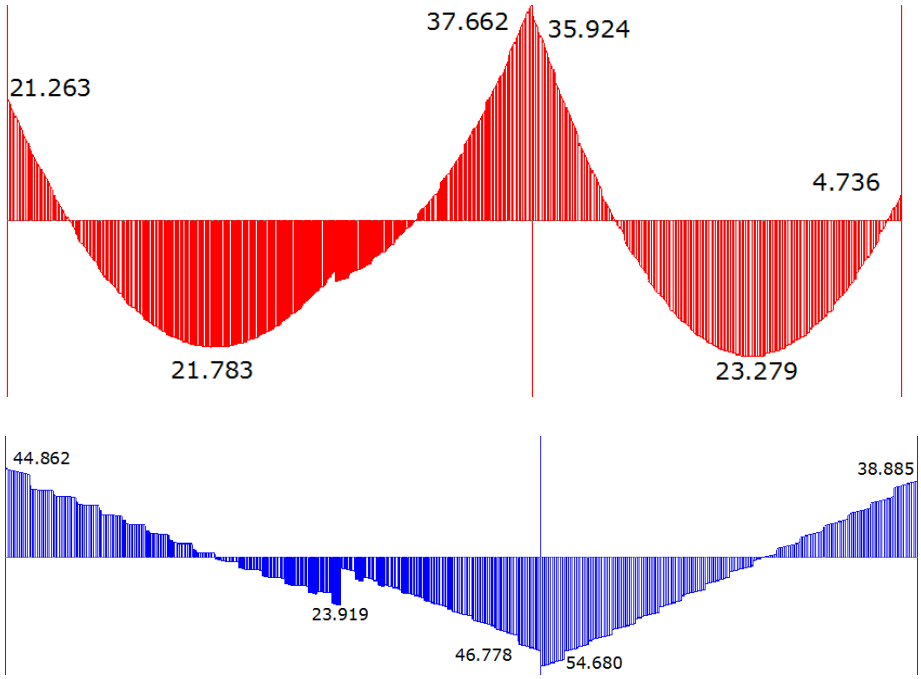
V_{Ed} = 44.92 KN
Using the area of tension reinforcement provided (A_{Sprov} = 1020mm^2)
$V_{Rd,c}$ = 41.7692 > V_{Ed} (Therefore , shear reinforcement is required)
Since $V_{Rd,max}$ = 218 KN > V_{Ed} > $V_{Rd,c}$
Asw/S = 0.1250; Asw(min)/S = 0.200; Maximum spacing = 0.75d = 0.75 × 399 = 299.25
Provide 2 legs of Y8mm @ 250mm c/c (Asw/S = 0.402) This is very ok !!!!

BEAM NO 13 (AXIS 5: A - E)

RESULT FROM ORION



RESULT FROM STAAD



Load Analysis

SPAN 1 – 2

This span is receiving load from PANEL 2 and 3

For PANEL 2

Hence, k = $\frac{Ly}{Lx} = \frac{3.625}{2.8}$ = 1.2946

$p = \frac{nLx}{2}\left(1 - \frac{1}{3k^2}\right) = \frac{10.9575 \times 2.8}{2}\left(1 - \frac{1}{3(1.2946)^2}\right)$ = 12.289 KN/m

For PANEL 3

The floor slab (PANEL 3) is spanning in two directions, since the ratio (k) of the longer side (Ly) to the shorter side (Lx) is less than 2.

Hence, k = $\frac{Ly}{Lx} = \frac{3325}{3205}$ = 1.037

The span is parallel to the short span direction of the slab. Therefore, the equivalent load that is transferred from the slab to the beam (ULS) can be represented by;

$p = \frac{nLx}{3} = \frac{10.9575 \times 3.205}{3}$ = 11.706 KN/m

Load from PANEL 7

Hence, k = $\frac{Ly}{Lx} = \frac{2.8}{1.705}$ = 1.6422

The span is parallel to the short span direction of the slab. Therefore, the equivalent load that is transferred from the slab to the beam can be represented by;

$p = \frac{nLx}{3} = \frac{10.9575 \times 1.705}{3}$ = 6.227 KN/m

Load from PANEL 6

Hence, k = $\frac{Ly}{Lx} = \frac{2.837}{1.4}$ = 2.026

Let us take it that the slab is spanning in one direction (for long span direction)

$p = \frac{nLx}{2} = \frac{10.9575 \times 1.4}{2}$ = 7.670 KN/m

Concentrated Load from beam No 12 = 1.05 KN (downwards)

Load from PANEL 4

The floor slab (PANEL 4) is spanning in two directions, since the ratio (k) of the longer side (Ly) to the shorter side (Lx) is less than 2.

Hence, k = $\frac{Ly}{Lx} = \frac{5360}{4545}$ = 1.179

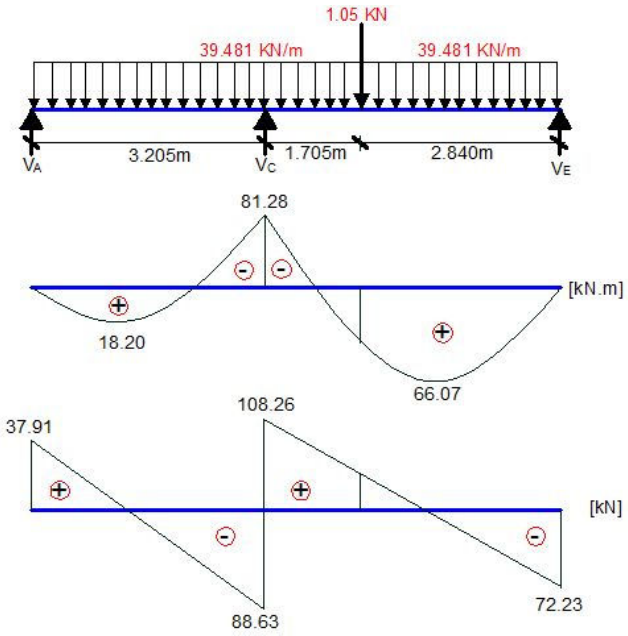
The span is parallel to the short span direction of the slab. Therefore, the equivalent load that is transferred from the slab to the beam (ULS) can be represented by;

$p = \frac{nLx}{3} = \frac{10.9575 \times 4.545}{3}$ = 16.6 KN/m

Load on span A - B = Load from PANEL 2 + Load from PANEL 3 + Self weight + Block work load = 12.289 + 11.706 + 2.32875 + 12.882 = 39.205 KN/m

Load on section C - D = Load from PANEL 7 + Load from PANEL 4 + Self weight + Block work load = 6.227 + 16.6 + 12.882 + 2.32875 = 34.030 KN/m

Load on section D - F = Load from PANEL 6 + Load from PANEL 4 + Self weight + Block work load = 7.670 + 16.6 + 2.32875 + 12.882 = 39.481 KN/m



Concentrated Load at point D = 1.05 KN

Let us conservatively allow 39.481 KN/m to run all through the span

STRUCTURAL DESIGN

Bottom Reinforcement Span (T – Beam)

M_{Ed} = 66.07 KNm (Sagging)

$B_{eff} = [0.2\left(\frac{1170}{2}\right) + 0.2\left(\frac{5130}{2}\right) + 0.1(0.85 \times 4545)] + 230 = 1246$ mm

K = 0.0133; La/d = 0.95; A_{S1} = 435.54 mm^2; A_{Smin} = 133.0447mm^2;

Provide 3Y16mm Bottom (A_{Sprov} = 603 mm^2)

Permissible L/d = 203.7604; Actual L/d = 11.3910 Deflection is OK !!!!!

Top Reinforcement Span (T – Beam)
M_{Ed} = 81.28 KNm (Sagging)
B_{eff} = bw
K = 0.0888; La/d = 0.9144; A_{S1} = 556.699 mm^2; A_{Smin} = 133.0447mm^2;
Provide 3Y16mm Top (A_{Sprov} = 603 mm^2)

Shear Design

(Picking the maximum shear force for all the spans)

Support 7; V_{Ed} = 108.26 KN
$V_{Rd,c}$ = 47.8138 > V_{Ed} (Therefore , shear reinforcement is required)
Since $V_{Rd,max}$ = 218 KN > V_{Ed} > $V_{Rd,c}$
Asw/S = 0.3013; Asw(min)/S = 0.200; Maximum spacing = 0.75d = 0.75 × 399 = 299.25
Provide 2 legs of Y8mm @ 250mm c/c (Asw/S = 0.402) This is very ok !!!!

CHAPTER 7

ANALYSIS AND DESIGN OF THE COLUMNS

7.1 Introduction

Loads from slabs and beams are transferred to the foundations through the columns. In typical cases, columns are usually rectangular or circular in shape. Normally, they are usually classified as short or slender depending on their slenderness ratio, and this in turn influences their mode of failure (see section 7.3). Columns are either subjected to axial, uniaxial, or biaxial loads depending on the location and/or loading condition. Eurocode 2 demands that we include the effects of imperfections in structural design of columns. Column design is covered in section 5.8 of EC2.

7.2 Load Analysis

To design the columns we have in our floor plan, we have to start the load analysis from the roof. The proposed roof beam layout is shown in Figure 7.1, with each roof beam numbered.

Dead load from roof trusses and concrete fascia on external roof beams = 4 KN/m (assumed)
Dead load on internal roof beam from trusses and services = 2 KN/m (assumed)
Self weight of roof beams = 25 × 0.25 × 0.23 = 1.4375 KN/m
Live load on roof = 0.85 KN/m
At ultimate limit state for external roof beams;
E_d = 1.35gk + 1.5qk = 1.35(5.4375) + 1.5(0.85) = 8.615 KN/m
At ultimate limit state for internal roof beams;
E_d = 1.35gk + 1.5qk = 1.35(3.4375) + 1.5(0.85) = 5.915 KN/m

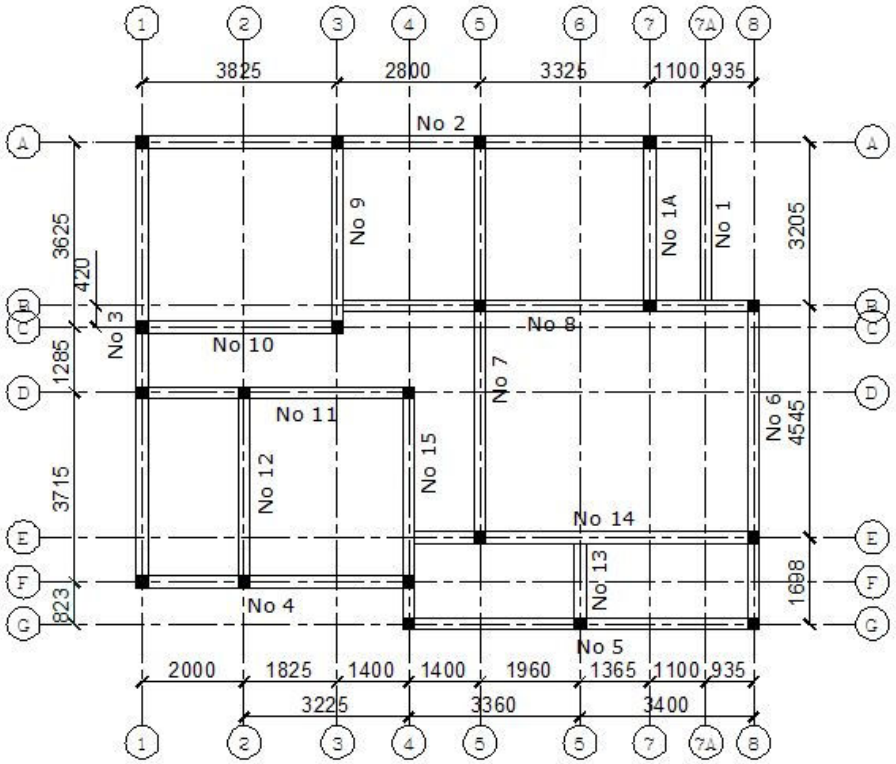


Figure 7.1: Roof beam Layout

7.2.1 Analysis of Column Axial Loads (ground floor columns)

Due to the small values of axial load that will be involved, we are going to neglect the design of first floor columns, and provide 4Y12 all through. We will transfer their loads (which will be calculated) to the ground floor columns, where we will carry out the design. The column axial loads have been obtained by summing up the reactions from all the beams supported by the columns, including the self weight of the column.

Let us use column A1 as example. At the roof level, the column is supporting beam No 2 (Support Reaction V_1 = 13.27 KN) and Beam No 3 (Support Reaction V_A = 12.99 KN). At the first floor level (see Analysis and Design of Beam No 1 and 2), the column is supporting Beam No 1 (Support Reaction V_1 = 41.38 KN), and Beam No 2 (Support Reaction V_A = 42.49 KN). Therefore the summation of all these loads gives the axial load transferred from the beams. For intermediate supports, note that the summation of the shear forces at the support gives the total support reaction (neglect the signs and use absolute value).

Another method of calculating Column Axial Load is by *Tributary Area Method*. This method has not been adopted in this work.

The summary of roof beam support reactions is given in Table 7.1.

Table 7.2: Summary of roof beam support reactions

Beam No	Axis	Support Reactions (KN)
1	7A: A - B	V_A = 13.81, V_B = 13.81
1A	7: A - B	V_A = 9.48, V_B = 9.48
2	A: 1 – 7A	V_1 = 13.27, V_3 = 35.41, V_5= 17.18, V_7 = 43.15
3	1: A - F	V_A = 12.99, V_C = 27.78, V_D = 21.89, V_F = 11.64
4	F: 1- 4	V_1 = 4.33, V_2 = 29.45, V_4 = 11.24
5	G: 4 - 8	V_4 = 12.76, V_5 = 42.96, V_8 = 13.01
6	8: B - G	V_B = 15.83, V_E = 40.7, V_G = -2.82
7	5: A - E	V_A = 5.70, V_B = 29.35, V_E = 10.78
8	B: 3 - 8	V_3 = 6.40, V_5 = 19.55, V_7 = 26.53, V_8 = 12.12
9	3: A - C	V_A = 11.46, V_C = 16.38
10	C: 1 - 3	V_1 = 11.31, V_3 = 11.31
11	D: 1 - 4	V_1 = 2.98, V_2 = 20.21, V_4 = 7.72
12	2: D - F	V_D = 10.99, V_F = 10.99
13	5: E - G	V_E = 5.00, V_G = 5.00
14	E: 4 - 8	V_4 = -10.98, V_5 = 42.23, V_8 = 13.73
15	4: D - G	V_D = 7.56, V_F = 10.88, V_G = -0.36

7.2.2 Column Self Weight

Self weight of 1st Floor Columns (ULS) = 1.35 × 25 KN/m^3 × 0.23m × 0.23m × 2.75m = 4.909 KN

Self weight of Ground Floor Columns (ULS) = 1.35 × 25 KN/m^3 × 0.23m × 0.23m × 4.050m = 7.231 KN

Total Column Self weight (1st floor and Ground floor) = 12.14 KN

COLUMN A1

Total Columns Self weight	= 12.14 KN
Load from roof beams = 13.27 + 12.99	= 26.26 KN
Load from floor beams = 46.21 + 42.49	= 88.70 KN
Total	**= 127.13 KN**

COLUMN A3

Total Columns Self weight	= 12.14 KN
Load from roof beams = 35.41 + 11.46	= 46.87 KN
Load from floor beams = 105.33 + 60.85	= 166.18 KN
Total	**= 225.19 KN**

COLUMN A5

Total Columns Self weight	= 12.14 KN
Load from roof beams = 17.19 + 5.70	= 22.89 KN
Load from floor beams = 83.64 + 37.91	= 121.55 KN
Total	**= 156.58 KN**

COLUMN A7

Total Columns Self weight	= 12.14 KN
Load from roof beams = 43.15 + 9.48	= 52.63 KN
Load from floor beams = 38.26 + 62.45	= 100.71 KN
Total	**= 165.48 KN**

COLUMN C1

Total Columns Self weight	= 12.14 KN
Load from roof beams = 27.78 + 11.31	= 39.09 KN
Load from floor beams = 80.80 + 69.16	= 149.96 KN
Total	**= 201.187 KN**

COLUMN C3

Total Columns Self weight	= 12.14 KN
Load from roof beams = 11.31 + 16.38	= 27.69 KN
Load from floor beams = 69.16 + 152.23	= 221.39 KN
Total	**= 261.217 KN**

COLUMN B5

Total Columns Self weight	= 12.14 KN
Load from roof beams = 29.35 + 19.55	= 48.90 KN
Load from floor beams = 116.74 + 196.89	= 313.63 KN
Total	**= 374.64 KN**

COLUMN B7

Total Columns Self weight	= 12.14 KN
Load from roof beams = 9.48 + 26.53	= 48.90 KN
Load from floor beams = 86.04 + 54.03	= 313.63 KN
Total	**= 188.217 KN**

COLUMN B8

Total Columns Self weight	= 12.14 KN
Load from roof beams = 15.83 + 12.12	= 27.95 KN
Load from floor beams = 15.44 + 65.61	= 81.05 KN
Total	**= 121.137 KN**

COLUMN D1

Total Columns Self weight = 12.14 KN
Load from roof beams = 21.89 + 2.98 = 24.87 KN
Load from floor beams = 35.33 + 25.43 = 60.76 KN
Total **= 141.93 KN**

COLUMN D2

Total Columns Self weight = 12.14 KN
Load from roof beams = 20.21 + 10.99 = 31.20 KN
Load from floor beams = 89.35 + 52.82 = 142.17 KN
Total **= 178.277 KN**

COLUMN D4

Total Columns Self weight = 12.14 KN
Load from roof beams = 7.72 + 7.56 = 15.28 KN
Load from floor beams = 51.97 + 47.21 = 99.18 KN
Total **= 126.597 KN**

COLUMN E5

Total Columns Self weight = 12.14 KN
Load from roof beams = 10.78 + 42.23 = 53.01 KN
Load from floor beams = 262.51 + 72.23 = 334.74 KN
Total **= 399.887 KN**

COLUMN E8

Total Columns Self weight = 12.14 KN
Load from roof beams = 40.70 + 13.73 = 54.43 KN
Load from floor beams = 101.67 + 65.61 = 167.28 KN
Total **= 233.847 KN**

COLUMN F1

Total Columns Self weight = 12.14 KN
Load from roof beams = 11.64 + 4.33 = 15.97 KN
Load from floor beams = 25.87 + 3.98 = 29.85 KN
Total **= 57.957 KN**

COLUMN F2

Total Columns Self weight = 12.14 KN
Load from roof beams = 10.99 + 29.45 = 40.44 KN
Load from floor beams = 79.98 + 52.82 = 132.80 KN
Total **= 185.377 KN**

COLUMN F4

Total Columns Self weight = 12.14 KN
Load from roof beams = 10.88 + 11.24 = 22.12 KN
Load from floor beams = 2.90 + 35.83 = 38.73 KN
Total **= 72.987 KN**

COLUMN G4

Total Columns Self weight $= 1.35 \times 25 \times 0.23 \times 0.23 \times 7.05$ = 12.586 KN
Load from roof beams = 12.76 – 0.36 = 12.40 KN
Total **= 24.986 KN**

COLUMN G6

Total Columns Self weight = 1.35 × 25 × 0.23 × 0.23 × 7.05 = 12.586 KN
Load from roof beams = 42.96 + 5.0 = 47.960 KN
Total = 60.546 KN

COLUMN G8

Total Columns Self weight = 1.35 × 25 × 0.23 × 0.23 × 7.05 = 12.586 KN
Load from roof beams = 13.01 – 2.82 = 10.19 KN
Total = 22.776 KN

7.2.3 Column Analysis for Moment (ULS)

For the manual analysis of column design moment, we are going to employ the use of sub-frames, and solve for one unknown rotation at each joint using the displacement method. The answers we are going to get here will vary from the answers from computer methods due to the nature of the loading that will be employed. We will maximise column moment by loading the frames as shown in Figure 7.2. The recommendation is that you should consider the adjacent beams fully fixed, while you reduce their stiffness by half, because we are overestimating the stiffness of the beams by considering all the ends fully fixed. (Details of this can be found in *Reynolds and Steedman, 2005, Table 1*, and *Reynolds, Steedman, and Threlfall, 2008, Table 2.57*)

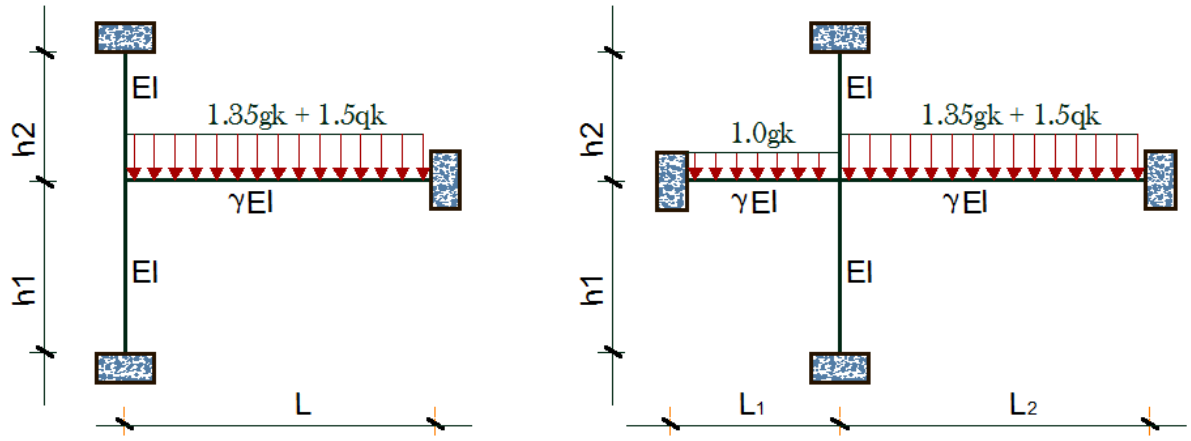
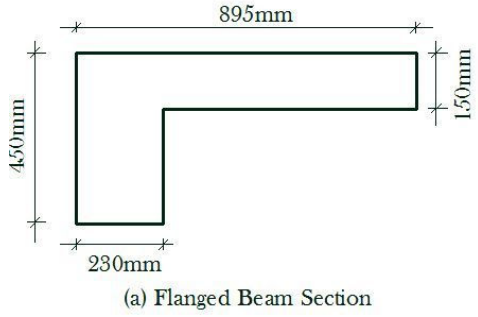
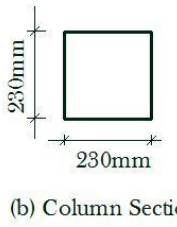


Figure 7.2: Sub-frames and loading configuration

To show example of the analysis procedure, **Axis A** ground floor columns have been analysed. We are considering the beams to be flanged. The properties of beam A: 1-3 is shown in Figure 7.3.





Moment of inertia of flanged beam I_B = 0.00307648 m^4 (See Appendix 3)

Moment of Inertia of column I_C = 0.0002332 m^4

$\frac{I_B}{I_C}$ = 13.19; Hence, I_B = 13.19I_C

Figure 7.3: Section geometrical properties

So to follow the recommendation, we are going to reduce the stiffness of the beam by half, such that $I_B = 6.595I_C$

The sub-frame loading that will maximise the column moment for ULS and the stiffness diagram (using displacement method) is as shown in Figure 7.4;

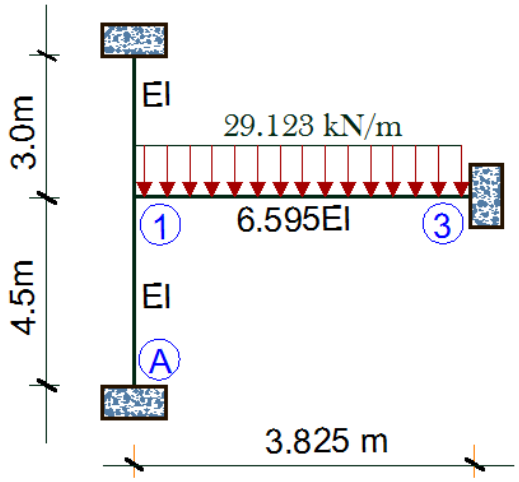


Figure 7.4(a): Loading of the sub-frame

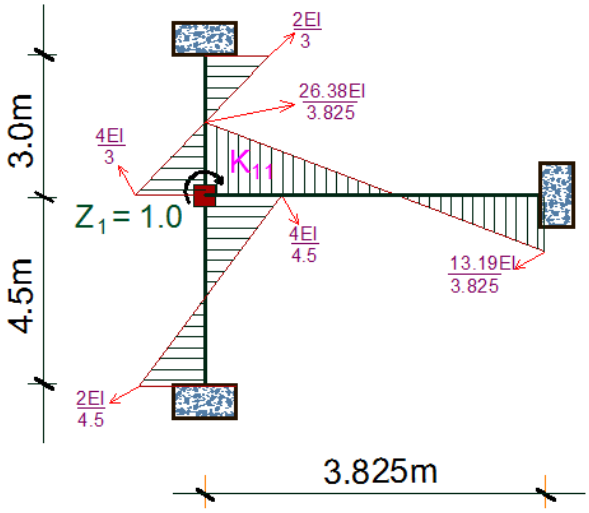


Figure 7.4(b): Stiffness diagram

Therefore the stiffness coefficient (K_{11}) – Force at point 1, due to unit rotation at point 1 is given by;

$$K_{11} = \frac{4EI}{4.5} + \frac{4EI}{3} + \frac{26.38EI}{3.825} = 9.11895EI;$$

The canonical equation is given by;

$$k_{11}Z_1 + K_{1P} = 0 \text{ ---------- (7.1)}$$

K_{1P} = Fixed end moment at point 1 due to externally applied load

$$K_{1P} = -\frac{q_1 l^2}{12} = -\frac{29.123 \times 3.825^2}{12} = -35.507 \text{ KNm}$$

Substituting the values of K_{11} and K_{1P} into equation 1, we obtain;

$$Z_1 = \frac{35.507}{9.1189} = \frac{3.8937}{EI} \text{ radians}$$

Therefore the final column moments are given by;

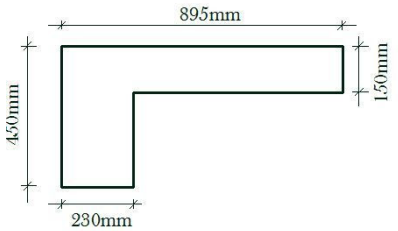
$$M_1^B = M_1Z_l = \frac{4EI}{4.5} \times \frac{3.8937}{EI} = 3.461 \text{ KNm}$$

$$M_A = M_1Z_l = \frac{2EI}{4.5} \times \frac{3.8937}{EI} = 1.7305 \text{ KNm}$$

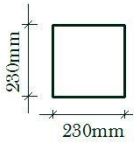
Analysis of column A3 for moment (ULS)

Geometrical properties of the adjacent floor beams are given below;

Span 1-3



(a) Flanged Beam Section



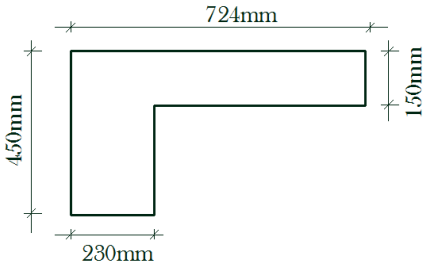
(b) Column Section

Moment of inertia of beam I_B = 0.00307648 m^4

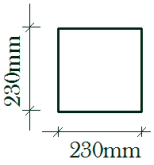
Moment of Inertia of column I_C = 0.0002332 m^4

$\frac{I_B}{I_C}$ = 13.19

Span 3 – 5



(a) Flanged Beam Section



(b) Column Section

Moment of inertia of beam I_B = 0.002857m^4

Moment of Inertia of column I_C = 0.0002332 m^4

$\frac{I_B}{I_C}$ = 12.252

We will now load the longer span with 1.35gk + 1.5qk, and load the shorter span with 1.0gk. Remember that we will have to reduce the original stiffness of the beam by half (see Figure 7.5).

Evaluation of 1.0gk

Self weight = 25 × 0.3 × 0.23 = 1.725 KN/m
Wall load = 3.47 × 2.75 = 9.5425 KN/m
Dead load from slab = Self weight of slab + partition allowance + finishes = 3.75 + 1.5 + 1.2 = 6.45 KN/m^2

Transferring it to the beam = $\frac{nLx}{3} = \frac{6.45 \times 2.8}{3}$ = 6.02 KN/m

Total load = 1.725 + 9.5425 + 6.02 = 17.2875 KN/m

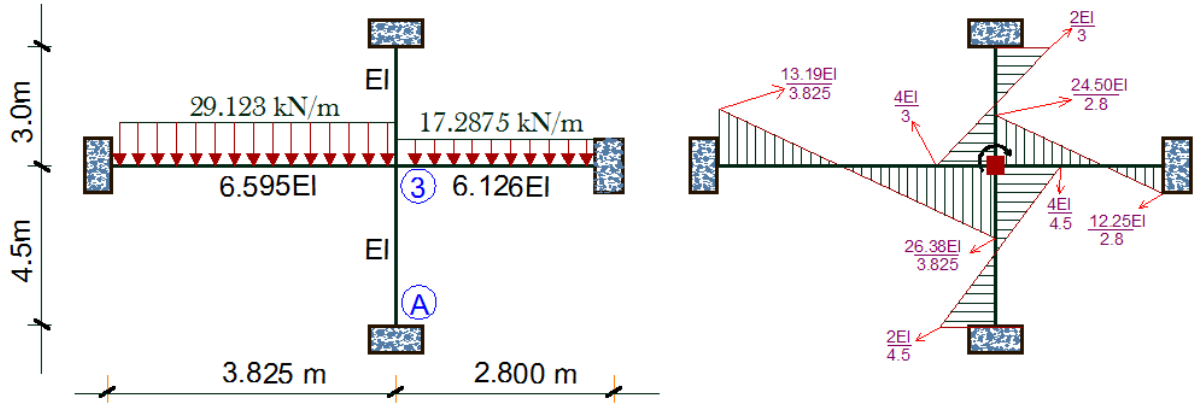


Figure 7.5: Loading and stiffness diagram of the sub-frame of column A3

$K_{33} = \frac{4EI}{4.5} + \frac{4EI}{3} + \frac{26.38EI}{3.825} + \frac{24.50EI}{2.8} = 17.869EI$

$K_{3P} = \frac{q_1 l^2}{12} - \frac{q_2 l^2}{12} = \frac{29.123 \times 3.825^2}{12} - \frac{17.2875 \times 2.8^2}{12} = 24.213$ KNm

On solving, $Z_3 = -\frac{24.213}{17.869} = \frac{-1.3550}{EI}$ radians

$M_3^B = M_3 Z_3 = \frac{4EI}{4.5} \times \frac{1.3550}{EI} = 1.204$ KNm

$M_A = M_3 Z_3 = \frac{2EI}{4.5} \times \frac{1.3550}{EI} = 0.602$ KNm

Following this arrangement, the summary of all the moments and axial forces of the columns can be obtained. The final result is shown in Table 7.2 below;

Table 7.2: Summary column design moments and axial loads

Column	Axial Load (KN)	M_X (KN.m)		M_Z (KN.m)	
		Top	Bottom	Top	Bottom
A1	127.130	3.461	-1.731	2.912	-1.456
A3	225.190	1.204	-0.602	3.926	1.963
A5	156.580	-0.662	0.331	2.744	-1.372
A7	165.480	2.260	-1.130	2.785	-1.393
B5	374.640	-1.219	0.609	-2.524	1.262
B7	188.217	-1.357	0.679	3.826	-1.913
B8	121.137	0.704	-0.352	5.661	-2.831
C1	201.187	4.123	-2.061	-0.931	0.465
C3	261.217	4.123	-2.061	3.926	-1.963
D1	141.930	0.808	-0.404	-0.613	-0.306
D2	178.277	-0.572	-0.286	3.112	-1.556
D4	126.597	3.804	-1.902	2.684	-1.342
E5	399.887	13.185	-6.592	7.138	-3.569
E8	233.847	13.185	-6.592	5.662	-2.831
F1	57.957	0.511	-0.255	2.693	-1.346
F2	185.377	-0.983	0.491	3.112	-1.556
F4	72.987	2.024	-1.012	-0.142	0.071
G4	24.986	-0.755	0.378	-0.016	0.008
G6	60.546	-0.019	0.011	-0.171	0.085
G8	22.776	-0.777	0.386	-0.249	0.124

7.3 Column Slenderness in EC2

Clause 5.8.2 of EN 1992-1-1 deals with members and structures in which the structural behaviour is significantly influenced by second order effects (e.g. columns, walls, piles, arches and shells). Global second order effects are more likely to occur in structures with a flexible bracing system.

Column design in EC2 generally involves determining the slenderness ratio (λ), of the member and checking if it lies below or above a critical value $\lambda_{\lim}$. If the column slenderness ratio lies below $(\lambda_{\lim})$, it can simply be designed to resist the axial action and moment obtained from an elastic analysis, but including the effect of geometric imperfections. These are termed first order effects. However, when

the column slenderness exceeds the critical value, additional (second order) moments caused by structural deformations can occur and must also be taken into account.

So in general, second order effects may be ignored if the slenderness λ is below a certain value $\lambda_{\lim}$.

$$\lambda_{\lim} = \frac{20.A.B.C}{\sqrt{n}} \text{ ---------- (7.2)}$$

where:

$A = \frac{1}{(1 + 0.2\phi_{ef})}$ (if ϕ_{ef} is not known, A = 0.7 may be used)

$B = 1 + 2\omega$ (if ω is not known, B = 1.1 may be used)

$C = 1.7 - r_m$ (if r_m is not known, C = 0.7 may be used)

Where;

ϕ_{ef} = effective creep ratio (0.7 may be used)

$\omega = A_s f_{yd} / (A_c f_{cd})$; mechanical reinforcement ratio;

A_s is the total area of longitudinal reinforcement

$n = N_{Ed} / (A_c f_{cd})$; relative normal force

$r_m = M_{01}/M_{02}$; moment ratio

M_{01}, M_{02} are the first order end moments, $|M_{02}| \geq |M_{01}|$

If the end moments M_{01} and M_{02} give tension on the same side, r_m should be taken positive (i.e. $C \leq 1.7$), otherwise negative (i.e. $C > 1.7$). For braced members in which the first order moments arise only from or predominantly due to imperfections or transverse loading r_m should be taken as 1.0 (i.e. $C = 0.7$):

Also, clause 5.8.3.1(2) says that for biaxial bending, the slenderness criterion may be checked separately for each direction. Depending on the outcome of this check, second order effects (a) may be ignored in both directions, (b) should be taken into account in one direction, or (c) should be taken into account in both directions.

7.3.1 Slenderness of isolated members

According to clause **5.8.3.2,** the slenderness ratio of an isolated member is defined as follows:

$$\lambda = \frac{l_o}{i} \text{ ---------- (7.3)}$$

where:

l_0 is the effective length

i is the radius of gyration of the uncracked concrete section ($i = h/\sqrt{12}$ for rectangular sections)

7.3.2 Effective length of isolated members

From Figure 5.7 of EC2, examples of effective length for isolated members with constant cross section are given as shown in the Figure below. This gives the examples of different buckling modes and corresponding effective lengths for isolated members as shown in Figure 7.5 below.

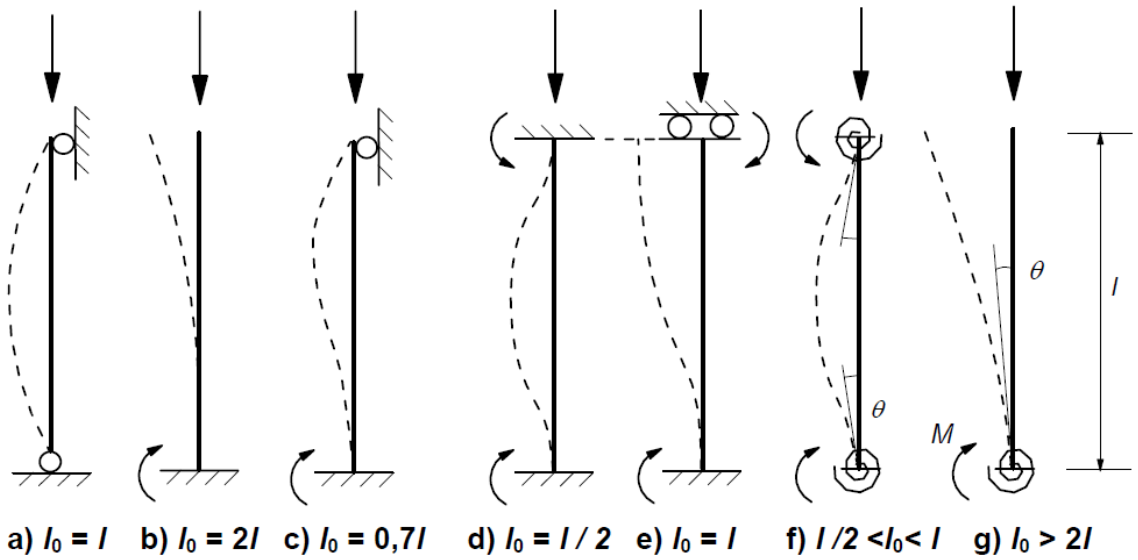


Figure 7.6: Effective length of isolated members (Fig 5.7 EC2)

However, for compression members in regular braced frames, the slenderness criterion should be checked with an effective length l_0 determined in the following way:

$$l_o = 0.5L\sqrt{\left(1+\frac{k_1}{0.45+k_1}\right)\times\left(1+\frac{k_2}{0.45+k_2}\right)} \text{ ---------- (7.4)}$$

Where;
k_1, k_2 are the relative flexibilities of rotational restraints at ends 1 and 2 respectively.

L is the clear height of the column between the end restraints

k = 0 is the theoretical limit for rigid rotational restraint, and k = ∞ represents the limit for no restraint at all. Since fully rigid restraint is rare in practise, a minimum value of 0.1 is recommended for k_1 and k_2.

According to Table 4.15 of *Reynolds*, *Steedman* and *Threlfall* (2008),

"*In the above equations, k_1 and k_2 are the relative flexibilities of rotational restraint at nodes I and 2 respectively. If the stiffness of adjacent columns does not vary significantly (say, difference not exceeding 15% of the higher value), the relative flexibility may be taken as the stiffness of the column under consideration divided by the sum of the stiffness of the beams (or, for an end column, the stiffness of the beam) attached to the column in the appropriate plane of bending. Otherwise, the effective column stiffness should be taken as the sum of the stiffness of the columns above and below the node. The stiffness of a member is 4EI/L for members fixed at the remote end, and 3EI/L for members pinned at the remote end, where I is the second moment of area of the cross-section allowing for the effect of cracking (for beams, 50% of the value for the uncracked section could be used), and L is the length of the member.*

For flat slabs, the beam stiffness should be based on the dimensions of the column strip. At nodes where the beams are considered as nominally simply-supported, and at bases not designed to resist column moments, k should be taken as 10. At bases designed to resist column moments, k may be taken as 1.0".

This procedure has been adopted in defining the value of k in this text.

7.4 Methods of analysis

According to clause 5.8.5 (1), the methods of analysis include a general method, based on non-linear second order analysis and the following two simplified methods:

(a) Method based on nominal stiffness
(b) Method based on nominal curvature

The method of nominal curvature has been used in this text, which is mainly suitable for isolated members with a constant normal force.

7.4.1 Design Bending Moment

According to clause 5.8.8.2, the design moment is:

$$M_{Ed} = M_{0Ed} + M_2 \text{ ---------- (7.5)}$$

where:
M_{0Ed} is the 1st order moment, including the effect of imperfections,
M_2 is the nominal 2nd order moment

The maximum value of M_{Ed} is given by the distributions of M_{0Ed} and M_2; the latter may be taken as parabolic or sinusoidal over the effective length.

Note: For statically indeterminate members, M_{0Ed} is determined for the actual boundary conditions, whereas M_2 will depend on boundary conditions via the effective length.

By differing first order end moments, M_{01} and M_{02} may be replaced by an equivalent first order end moment M_{0e}:

$$M_{0e} = 0.6\, M_{02} + 0.4\, M_{01} \geq 0.4\, M_{02} \text{ ---------- (7.6)}$$

M_{01} and M_{02} should have the same sign if they give tension on the same side, otherwise opposite signs. Furthermore, $|M_{02}| \geq |M_{01}|$.

The nominal second order moment M_2 in Expression (5) is;

$$M_2 = N_{Ed}\, e_2 \text{ ---------- (7.7)}$$

where:
N_{Ed} is the design value of axial force
e_2 is the deflection = $(1/r)\, l_o^2 / c$

$1/r$ is the curvature
l_o is the effective length
c is a factor depending on the curvature distribution

For constant cross section, c = 10 (≈ π2) is normally used. If the first order moment is constant, a lower value should be considered (8 is a lower limit, corresponding to constant total moment)

7.4.2 Simplified Steps for when $\lambda < \lambda_{lim}$ (Arya, 2009)

According to clause 5.8.3.1 of EC2, if the slenderness, λ, is less than $\lambda_{lim,}$ the column should be designed for the applied axial load, N_{Ed}, and the moment due to first order effects, M_{Ed}, being numerically equal to the sum of the larger elastic end moment, M_{02}, plus any moment due to geometric imperfection, $N_{Ed.ei}$, as follows:

$$M_{Ed} = M_{02} + N_{Ed.}e_i \text{ ---------- (7.8)}$$

Where;

e_1 is the geometric imperfection = $\left(\theta_i \frac{l_0}{2}\right)$

in which θ_i is the angle of inclination and can be taken as 1/200 for isolated braced columns and l_0 is the effective length (clause 5.2(7)). According to clause 6.1(4) the minimum design eccentricity, e_0, is h/30 but not less than 20 mm where h is the depth of the section.

Once N_{Ed} and M_{Ed} have been determined, the area of longitudinal steel can be calculated by strain compatibility using an iterative procedure as discussed in example 3.19 of this book. However, this approach may not be practical for everyday design and therefore The Concrete Centre has produced a series of design charts, similar to those in BS 8110:Part 3, which can be used to determine the area of longitudinal steel.

7.4.3 Curvature

According to clause 5.8.8.3, for members with constant symmetrical cross sections (including reinforcement), the following relation may be used:

$$1/r = K_r.K_\phi.1/r_0 \text{ ---------- (7.9)}$$

where:

K_r is a correction factor depending on axial load

K_ϕ is a factor for taking account of creep

$$1/r_0 = \frac{\varepsilon_{yd}}{0.45\,d}$$

$$\varepsilon_{yd} = \frac{f_{yd}}{Es}$$

d is the effective depth
E_s is the elastic modulus of steel = 200 KN/mm^2

However, if all reinforcement is not concentrated on opposite sides, but part of it is distributed parallel to the plane of bending, d is defined as

$$d = (h/2) + i_s \text{ ---------- (7.10)}$$

where i_s is the radius of gyration of the total reinforcement area

$$K_r = (n_u - n) / (n_u - n_{bal}) \leq 1 \text{ ---------- (7.11)}$$

where:
$n = N_{Ed} / (A_c\, f_{cd})$, relative axial force
N_{Ed} is the design value of axial force
$n_u = 1 + \omega$
n_{bal} is the value of n at maximum moment resistance; the value 0.4 may be used
$\omega = A_s\, f_{yd} / (A_c\, f_{cd})$

A_s is the total area of reinforcement
A_c is the area of concrete cross section

The effect of creep should be taken into account by the following factor:

$K_\phi = 1 + \beta\phi_{ef} \geq 1.0$

where:

ϕ_{ef} is the effective creep ratio
$\beta = 0.35 + f_{ck}/200 - \lambda/150$

λ is the slenderness ratio

7.4.4 Simplified Steps for when $\lambda > \lambda_{lim}$ (Arya, 2009)

When $\lambda > \lambda_{lim}$, critical conditions may occur at the top, middle or bottom of the column. The values of the design moments at these positions are, respectively (see Figure 7.7):

(i) M_{02}
(ii) $M_{0Ed} + M_2$
(iii) $M_{01} + 0.5M_2$

M_{0Ed} is the equivalent first order moment including the effect of imperfections at about mid height of the column and may be taken as M_{0e} as follows:

$$M_{0e} = (0.6M_{02} + 0.4M_{01}) \geq 0.4M_{02} \quad \text{---------- (7.12)}$$

Where;
M_{01} and M_{02} are the first order end moments including the effect of imperfections acting on the column and M_{02} is the numerically larger of the elastic end moment acting on the column i.e.

$$|M_{02}| > |M_{01}|.$$

M_2 is the nominal second order moment acting on the column and is given by

$M_2 = N_{Ed}.e_2$
Where;

N_{Ed} is the design axial load at ULS
e_2 is the deflection = $\left(\frac{1}{r}\right)\frac{l_0^2}{10}$

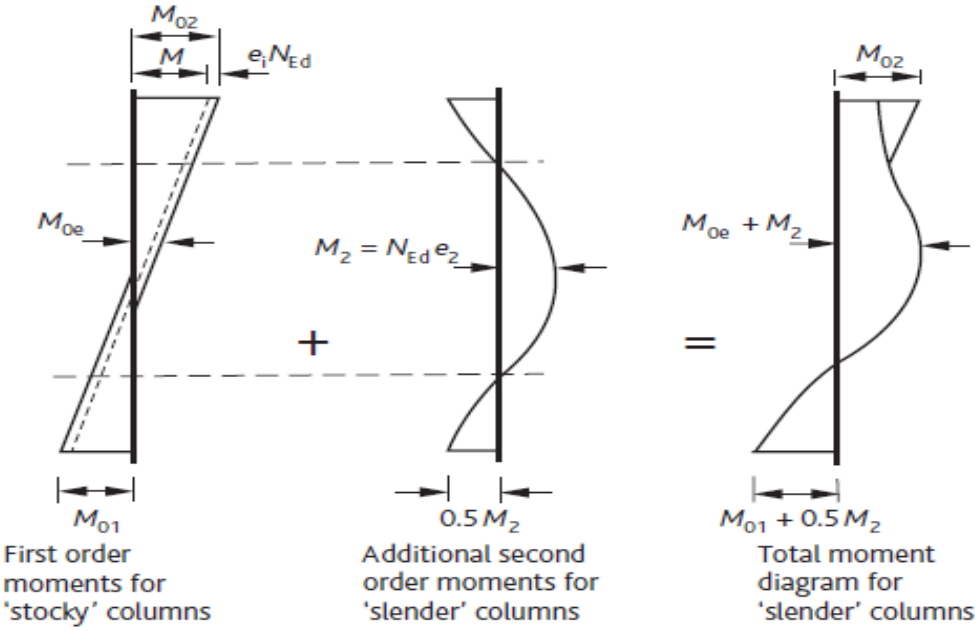


Figure 7.7: Design bending moments (Bond et al, 2006)

7.5 Biaxial bending

According to clause 5.8.9, separate design in each principal direction, disregarding biaxial bending, may be made as a first step. Imperfections need to be taken into account only in the direction where they will have the most unfavourable effect.

(3) No further check is necessary if the slenderness ratios satisfy the following two conditions

$$\frac{\lambda_y}{\lambda_z} \leq 2.0 \text{ and } \frac{\lambda_z}{\lambda_y} \leq 2.0 \quad \text{---------- (7.13)}$$

and if the relative eccentricities e_y/h and e_z/b satisfy one the following conditions:

$$\frac{e_y/h_{eq}}{e_z/b_{eq}} \leq 0.2 \text{ or } \frac{e_z/b_{eq}}{e_y/h_{eq}} \leq 0.2 \quad \text{---------- (7.14)}$$

where:

b, h are the width and depth of the section
$b_{eq} = i_y.\sqrt{12}$ and $h_{eq} = i_z.\sqrt{12}$ for an equivalent rectangular section
λ_y and λ_z are the slenderness ratios l_0/i with respect to *y*- and *z*-axis respectively
i_y, i_z are the radii of gyration with respect to *y*- and *z*-axis respectively
$e_z = M_{Edy} / N_{Ed}$; eccentricity along *z*-axis
$e_y = M_{Edz} / N_{Ed}$; eccentricity along *y*-axis
M_{Edy} is the design moment about y-axis, including second order moment
M_{Edz} is the design moment about z-axis, including second order moment
N_{Ed} is the design value of axial load in the respective load combination

If the condition of Expression (7.13 and 7.14) is not fulfilled, biaxial bending should be taken into account including the 2nd order effects in each direction (unless they may be ignored according to clauses 5.8.2 (6) or 5.8.3 of EC2). In the absence of an accurate cross section design for biaxial bending, the following simplified criterion may be used

$$\left(\frac{M_{Edz}}{M_{Rdz}}\right)^a + \left(\frac{M_{Edy}}{M_{Rdy}}\right)^a \leq 1.0 \quad \text{---------- (7.15)}$$

where:
$M_{Edz/y}$ is the design moment around the respective axis, including a 2nd order moment.
$M_{Rdz/y}$ is the moment resistance in the respective direction
a is the exponent;
for circular and elliptical cross sections: $a = 2$

For rectangular sections;

Table 7.3: Values of 'a' exponent for rectangular sections

$\frac{N_{Ed}}{N_{Rd}}$	0.1	0.7	1.0
a	1.0	1.5	2.0

with linear interpolation for intermediate values

N_{Ed} is the design value of axial force
$N_{Rd} = A_c f_{cd} + A_s f_{yd}$, design axial resistance of section.

where:
A_c is the gross area of the concrete section
A_s is the area of longitudinal reinforcement

The chart for selecting column longitudinal area of steel is as shown below. This has been adapted from Bond et al, 2006.

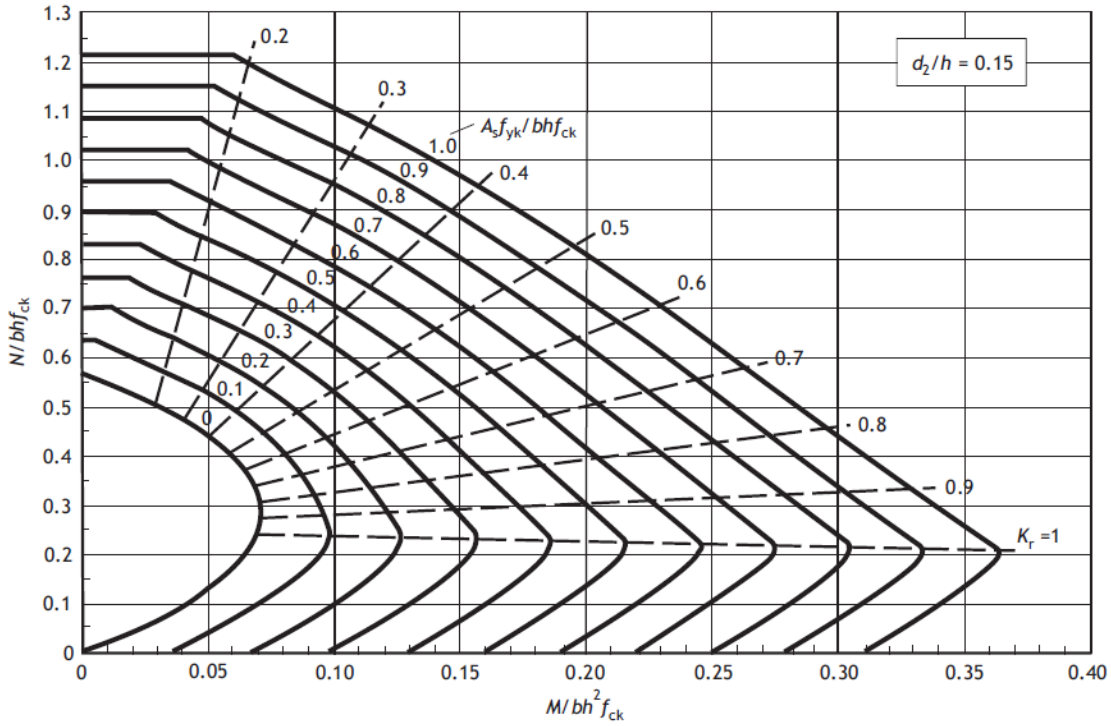


Figure 7.8: Column Design Chart (d_2/h = 0.15) (Bond et al, 2006)

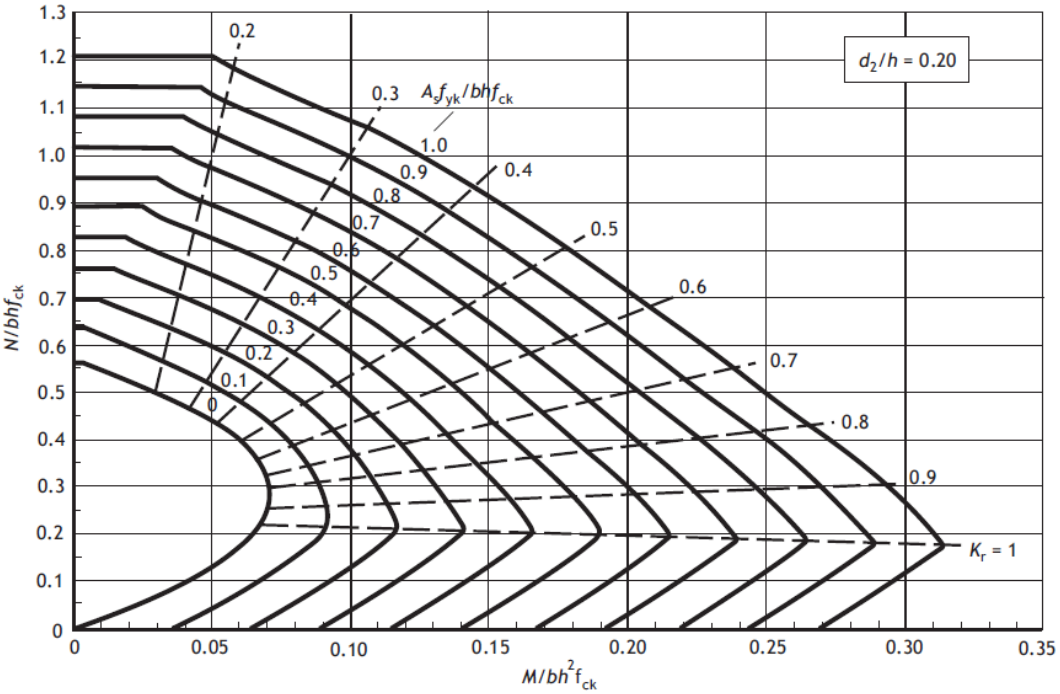


Figure 7.9: Column Design Chart (d_2/h = 0.20) (Bond et al, 2009)

For more column design charts and how to design slender columns, see Appendix 2.

7.6 Structural Design

Column E5 is the column that is subjected to the highest loading.

7.6.1 Design of Column E5

N_{Ed} = 399.887 KN

Elastic Moments

X - direction: M_{01} = 13.185 KNm; M_{02}= - 6.592 KNm

Z – direction: M_{01} = 7.138 KNm; M_{02}= - 3.569 KNm

Clear column height = 4500 – 450 = 4050 mm

To calculate the effective height of the column (l_o)

Let us now calculate the relative stiffnesses in the planes of bending

In the x-direction;

Beam 1: Axis E: 4 – 5 (Flange width) = $0.2\left(\frac{1200}{2} + \frac{2607}{2}\right)$ + (0.1 × 0.85 × 1400) + 230 = 728mm

Beam 2: Axis E: 5 – 8 (Flange width) = $0.2\left(\frac{1200}{2} + \frac{4315}{2}\right)$ + (0.1 × 0.85 × 5360) + 230 = 1237 mm

Second moment of area of beam 1 = 0.0028628 m^4

Stiffness of beam 1 (since E is constant) = $\frac{4I}{L}$ = 0.0081794

Second moment of area of beam 2 = 0.0034116 m^4

Stiffness of beam 2 (since E is constant) = $\frac{4I}{L}$ = 0.0025460

Second moment of area of column = 0.0002332 m^4
Stiffness of lower column = 0.00020728

Remember that we will have to reduce the stiffness of the beams by half to account for cracking;

k_1 = $\frac{0.00020728}{0.001273 + 0.0040897}$ = 0.0386

Since the minimum value of k_1 and k_2 is 0.1, adopt k_1 as 0.1. Let us take k_2 as 1.0 for base designed to resist moment

$$l_o = 0.5 \times 4050\sqrt{\left(1 + \frac{0.1}{0.45 + 0.1}\right) \times \left(1 + \frac{1.0}{0.45 + 1.0}\right)} = 2861.5 \text{ mm}$$

Compare with BS 8110's 0.75L = 0.75 × 4050 = 3037.5 mm

In the z-direction;

Beam 1, Axis 5: B-E

Flange width = 1246 mm

Second Moment of Area = 0.0034191 m^4

Stiffness of beam = 0.003009
Stiffness of lower column = 0.00020728

$k_1 = \frac{0.00020728}{0.0015045} = 0.1377$

$$l_o = 0.5 \times 4050\sqrt{\left(1 + \frac{0.1377}{0.45 + 0.1377}\right) \times \left(1 + \frac{1.0}{0.45 + 1.0}\right)} = 2924.34 \text{ mm}$$

Radius of gyration $i = h/\sqrt{12} = 230/\sqrt{12} = 66.395$

$\lambda_x = 2861.5/66.395 = 43.098$

$\lambda_z = 2924.34/66.395 = 44.044$

Critical Slenderness for the x-direction

$\lambda_{\lim} = \frac{20.A.B.C}{\sqrt{n}}$

$A = 0.7$
$B = 1.1$
$C = 1.7 - \frac{M_{01}}{M_{02}} = 1.7 - \left(\frac{-6.592}{13.185}\right) = 2.199$
$n = \frac{N_{Ed}}{A_c f_{cd}}$

$N_{Ed} = 399.887 \times 10^3$ N

$A_c = 230 \times 230 = 52900$ mm^2

$f_{cd} = \frac{\alpha_{cc} f_{ck}}{1.5} = \frac{0.85 \times 25}{1.5} = 14.166$ N/mm^2

$n = \frac{399.887 \times 10^3}{52900 \times 14.166} = 0.5336$

$$\lambda_{\lim} = \frac{20 \times 0.7 \times 1.1 \times 2.199}{\sqrt{0.5336}} = 46.359$$

43.098 < 46.359, second order effects need not to be considered in the x-direction

Critical Slenderness for the z-direction

$A = 0.7$
$B = 1.1$
$C = 1.7 - \frac{M_{01}}{M_{02}} = 1.7 - \left(\frac{-3.569}{7.138}\right) = 2.2$
$n = \frac{N_{Ed}}{A_c f_{cd}}$

$N_{Ed} = 399.887 \times 10^3$ N

$A_c = 230 \times 230 = 52900$ mm^2
$f_{cd} = \frac{\alpha_{cc} f_{ck}}{1.5} = \frac{0.85 \times 25}{1.5} = 14.166$ N/mm^2

$n = \frac{399.887 \times 10^3}{52900 \times 14.166} = 0.5336$

$$\lambda_{\lim} = \frac{20 \times 0.7 \times 1.1 \times 2.2}{\sqrt{0.5336}} = 46.381$$

44.044 < 46.381, second order effects need not to be considered in the z - direction

Design Moments (x-direction)

M_{01} = 6.592 KNm, M_{02} = 13.185 KNm

e_i is the geometric imperfection = $\left(\theta_i \frac{l_0}{2}\right) = \left(\frac{1}{200} \times \frac{2862}{2}\right) = 7.155$ mm

Minimum eccentricity $e_0 = \frac{h}{30} = \frac{230}{30} = 7.667$mm. Since this is less than 20mm, take minimum eccentricity = 20mm

Minimum design moment = $e_0 N_{Ed} = 20 \times 10^{-3} \times 399.87 = 7.9974$ KNm

First order end moment $M_{02} = M_{Top} + e_i N_{Ed}$

$e_i N_{Ed} = 7.155 \times 10^{-3} \times 399.87 = 2.861$ KNm

$M_{02} = M_{Top} + e_i N_{Ed} = 13.185 + 2.861 = 16.046$KNm

Longitudinal Steel Area

$d_2 = C_{nom} + \phi/2 + \phi_{links} = 35 + 8 + 8 = 51$mm

$\frac{d_2}{h} = \frac{51}{230} = 0.2217$

Reading from chart No 1; $\frac{d_2}{h} = 0.2$;

$\frac{M_{Ed}}{f_{ck} b h^2} = \frac{16.046 \times 10^6}{25 \times 230 \times 230^2} = 0.0527$

$\frac{N_{Ed}}{f_{ck} b h} = \frac{399.88 \times 10^3}{25 \times 230 \times 230} = 0.302$

From the chart, $\frac{A_s F_{yk}}{b h f_{ck}} = 0.15$

Area of longitudinal steel required (As) = $\frac{0.15 \times 25 \times 230 \times 230}{460} = 431.25$ mm^2

$A_{s,min} = 0.10\ N_{Ed}/f_{yd} = \frac{0.1 \times 399.887}{400} = 0.099$ mm^2 < $0.002 \times 230 \times 230 = 105.8$ mm^2
$A_{s,max} = 0.04bh = 2116$ mm^2

Provide 4Y16mm (Asprov = 804 mm^2) Ok

Links

Minimum size = $0.25\phi = 0.25 \times 16 = 4$mm < 6mm
We are adopting Y8mm as links

Spacing adopted = 200mm less than {b, h, 20ϕ, 400mm}

Design Moments (z-direction)

M_{01} = 3.569 KNm, M_{02} = 7.138 KNm

e_1 is the geometric imperfection = $\left(\theta_i \frac{l_0}{2}\right) = \left(\frac{1}{200} \times \frac{2865}{2}\right)$ = 7.1625 mm

Minimum eccentricity $e_0 = \frac{h}{30} = \frac{230}{30}$ = 7.667mm. Since this is less than 20mm, take minimum eccentricity = 20mm

Minimum design moment = $e_0 N_{Ed}$ = $20 \times 10^{-3} \times 399.87$ = 7.9974 KNm

First order end moment $M_{02} = M_{Top} + e_i N_{Ed}$

$e_i N_{Ed} = 7.1625 \times 10^{-3} \times 399.87 = 2.864$KNm

$M_{02} = M_{Top} + e_i N_{Ed}$ = 7.138 + 2.864 = 10.002 KNm

Longitudinal Steel Area

$d_2 = C_{nom} + \phi/2 + \phi_{links}$ = 35 + 8 + 8 = 51mm

$\frac{d_2}{h} = \frac{51}{230}$ = 0.2217

Reading from chart No 1; $\frac{d_2}{h}$ = 0.2;

$\frac{M_{Ed}}{f_{ck} b h^2} = \frac{10.002 \times 10^6}{25 \times 230 \times 230^2}$ = 0.03288

$\frac{N_{Ed}}{f_{ck} b h} = \frac{399.88 \times 10^3}{25 \times 230 \times 230}$ = 0.302

From the chart, $\frac{A_s F_{yk}}{b h f_{ck}}$ = 0.05

Area of longitudinal steel required (As) = $\frac{0.05 \times 25 \times 230 \times 230}{460}$ = 143.75 mm^2

$A_{s,min} = 0.10\ N_{Ed}/f_{yd} = \frac{0.1 \times 399.887}{400}$ = 0.099 mm^2 < $0.002 \times 230 \times 230$ = 105.8 mm^2

Provide 4Y16mm (Asprov = 804 mm^2)

Biaxial Effects

Check if $\frac{\lambda_y}{\lambda_z} \leq 2.0$ and $\frac{\lambda_z}{\lambda_y} \leq 2.0$

$\frac{43.098}{44.044}$ = 0.9785 < 2.0, and $\frac{44.044}{43.098}$ = 1.0219 < 2.0

Furthermore, let us also check;

$\frac{e_y/h_{eq}}{e_z/b_{eq}} \leq 0.2$ or $\frac{e_z/b_{eq}}{e_y/h_{eq}} \leq 0.2$

$e_x = \frac{M_{Ed,x}}{N_{Ed}} = \frac{16.046 \times 10^6}{399.887 \times 10^3}$ = 40.126 mm

$e_z = \frac{M_{Ed,z}}{N_{Ed}} = \frac{10.002 \times 10^6}{399.887 \times 10^3}$ = 25.012 mm

$h_{eq} = b_{eq} = i.\sqrt{12}$ = 66.395 × $\sqrt{12}$ = 230mm

$\frac{e_x}{h_{eq}} \div \frac{e_z}{b_{eq}} = \frac{40.126}{230} \div \frac{25.012}{230}$ = 1.6042 > 0.2

Therefore we have to check for biaxial bending;

$\frac{A_s F_{yk}}{bhf_{ck}} = \frac{804 \times 460}{25 \times 230 \times 230}$ = 0.279

Therefore from the chart; $\frac{M_{Rd}}{f_{ck} bh^2} = 0.08$

$M_{Rd} = 0.08 \times 25 \times 230 \times 230^2$ = 24.334 KNm

$N_{Rd} = A_c f_{cd} + A_s f_{yd}$

$N_{Rd} = (230^2 \times 14.1667) + (804 \times 400)$ = 1071.0 KN

$\frac{N_{Ed}}{N_{Rd}} = \frac{399.88}{1071.0}$ = 0.373

$\frac{N_{Ed}}{N_{Rd}}$	0.1	0.7	1.0
a	1.0	1.5	2.0

By linear interpolation, a = $1.0 + \frac{0.373 - 0.1}{0.7 - 0.1} \times (1.5 - 1.0)$= 1.2275

$$\left(\frac{M_{Edz}}{M_{Rdz}}\right)^a + \left(\frac{M_{Edy}}{M_{Rdy}}\right)^a \leq 1.0$$

$\left(\frac{16.046}{24.334}\right)^{1.2275} + \left(\frac{10.002}{24.334}\right)^{1.2275}$ = 0.599 + 0.3357 = 0.9347 < 1.0. This ok

A little consideration will show that this design is very representative for all the ground floor columns

Links

Minimum size = 0.25ϕ = 0.25 × 16 = 4mm < 6mm
We are adopting Y8mm as links

Spacing adopted = 200mm less than min{b, h, 20ϕ, 400mm}

7.6.2 DESIGN OF COLUMN G6

N_{Ed} = 60.546KN

Elastic Moments

X - direction: M_{01} = -0.019KNm; M_{02}= 0.011KNm

Z – direction: M_{01} = -0.171KNm; M_{02}= 0.085KNm

Clear column height = 7500– 250 = 7250 mm

Calculation of effective height;

The roof beams adjacent to the column are beams No 5 and beam No 13

Second moment of area of all beams $I_B = \frac{bh^3}{12} = \frac{0.23 \times 0.25^3}{12} = 2.995 \times 10^{-4}$ m^4

Stiffness of beam 1 (since E is constant) = $\frac{4I}{L} = \frac{4 \times 2.995 \times 10^{-4}}{3.4} = 3.52355 \times 10^{-4}$

Stiffness of beam 2 (since E is constant) = $\frac{4I}{L} = \frac{4 \times 2.995 \times 10^{-4}}{3.36} = 3.5655 \times 10^{-4}$

Stiffness of beam 3 (since E is constant) = $\frac{4I}{L} = \frac{4 \times 2.995 \times 10^{-4}}{1.689} = 7.09295 \times 10^{-4}$

Second moment of area of column = 0.0002332 m^4

Stiffness of column = $\frac{4I}{L} = \frac{4 \times 2.332 \times 10^{-4}}{7.5} = 1.2437 \times 10^{-4}$

Remember that we will have to reduce the stiffness of the beams by half to account for cracking;

In the x-direction

$k_1 = \frac{0.00012437}{0.00017618 + 0.000178275} = 0.351$

Since the minimum value of k_1 and k_2 is 0.1, adopt k_1 as 0.175. Let us take k_2 as 1.0 for base designed to resist moment

$$l_o = 0.5 \times 7250 \sqrt{\left(1 + \frac{0.351}{0.45 + 0.351}\right) \times \left(1 + \frac{1.0}{0.45 + 1.0}\right)} = 5650 \text{ mm}$$

In the z-direction

$k_1 = \frac{0.00012437}{0.00035464} = 0.350$

Since the minimum value of k_1 and k_2 is 0.1, adopt k_1 as 0.175. Let us take k_2 as 1.0 for base designed to resist moment

$$l_o = 0.5 \times 7250 \sqrt{\left(1 + \frac{0.351}{0.45 + 0.351}\right) \times \left(1 + \frac{1.0}{0.45 + 1.0}\right)} = 5650 \text{ mm}$$

Therefore, the slenderness can be taken to be the same in all directions

Radius of gyration i = h/$\sqrt{12}$ = 230/$\sqrt{12}$ = 66.395

$\lambda_x = \lambda_z$ = 5650/66.395 = 85.0967

Critical Slenderness for the x-direction

$$\lambda_{\lim} = \frac{20.A.B.C}{\sqrt{n}}$$

A = 0.7
B = 1.1
$C = 1.7 - \frac{M_{01}}{M_{02}} = 1.7 - \left(\frac{-0.011}{0.019}\right)$ = 2.2789
$n = \frac{N_{Ed}}{A_c f_{cd}}$

N_{Ed} = 60.546 × 10^3 N

A_c = 230 × 230 = 52900 mm^2

$f_{cd} = \frac{\alpha_{cc} f_{ck}}{1.5} = \frac{0.85 \times 25}{1.5}$ = 14.166 N/mm^2

$n = \frac{60.546 \times 10^3}{52900 \times 14.166}$ = 0.08079

$$\lambda_{\lim} = \frac{20 \times 0.7 \times 1.1 \times 2.2789}{\sqrt{0.08079}} = 123.471$$

123.471 < 85.0967, second order effects need not to be considered in the x-direction

Critical Slenderness for the z-direction

A = 0.7
B = 1.1
$C = 1.7 - \frac{M_{01}}{M_{02}} = 1.7 - \left(\frac{-0.085}{0.171}\right)$ = 2.197
$n = \frac{N_{Ed}}{A_c f_{cd}}$

N_{Ed} = 60.546 × 10^3 N

A_c = 230 × 230 = 52900 mm^2

$f_{cd} = \frac{\alpha_{cc} f_{ck}}{1.5} = \frac{0.85 \times 25}{1.5}$ = 14.166 N/mm^2

$n = \frac{60.546 \times 10^3}{52900 \times 14.166}$ = 0.08079

$$\lambda_{\lim} = \frac{20 \times 0.7 \times 1.1 \times 2.197}{\sqrt{0.08079}} = 119.034$$

119.034< 85.0967, second order effects need not to be considered in the z - direction

You can verify that despite the height of the front columns, they are not slender. So proceed, carry out the design and provide reinforcements as done above.

Provide 4Y16mm (Asprov = 804 mm^2) Ok

Provide Y8 @ 200mm links

7.7 Column Design Results from CSC Orion

The design internal forces, and reinforcement design for the columns from CSC Orion software are tabulated as shown in Table 7.3. You can compare the result with manual analysis result.

Table 7.3: Summary column design moments, axial loads, and reinforcement to Orion

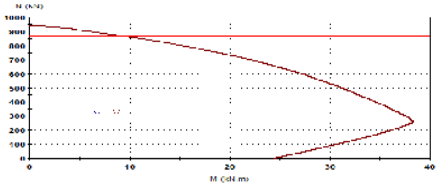
Column	N_{Ed} (KN)	M_X (KN.m)		M_Z (KN.m)		$A_{s\ (req)}$ mm^2	A_{sprov} mm^2
		Top	Bottom	Top	Bottom		
A1	126.6	5.1	-2.5	-4.7	2.3	105.8	4T16
A3	202.3	-2.1	1.0	-5.9	2.9	105.8	4T16
A5	155.9	1.0	-0.5	-3.7	1.8	105.8	4T16
A7	133.9	-2.3	1.1	-3.8	1.9	105.8	4T16
B5	283.0	1.9	-0.9	-1.1	0.6	105.8	4T16
B7	258.8	-2.0	1.0	0.3	-0.1	105.8	4T16
B8	112.7	-2.3	1.1	-8.4	4.1	105.8	4T16
C1	165.5	5.2	-2.6	2.6	-1.3	105.8	4T16
C3	223.6	-3.7	1.8	3.0	-1.5	105.8	4T16
D1	100.0	1.5	-0.7	-0.6	0.3	105.8	4T16
D2	174.3	0.4	-0.2	-3.3	1.6	105.8	4T16
D4	177.8	2.5	-1.2	-2.8	1.4	105.8	4T16
E5	332.6	7.4	-3.7	4.6	-2.3	105.8	4T16
E8	229.3	-17.7	9.3	7.6	-3.8	105.8	4T16
F1	60.0	0.8	-0.4	2.4	-1.2	105.8	4T16
F2	149.1	1.5	-0.8	4.5	-2.2	105.8	4T16
F4	100.9	-2.3	1.1	2.5	-1.2	105.8	4T16
G4	20.2	0.6	-0.4	0.1	-0.1	105.8	4T16
G6	61.7	0.5	-0.3	2.8	-1.4	105.8	4T16
G8	22.6	-1.2	0.5	0.3	-0.3	105.8	4T16

The calculation sheet for Column E5 from CSC Orion is given below;

1C16 Storey: 1 (Con.:C25/30 / Steel:Grade 460 (Type 2))

Loadings (Combination):

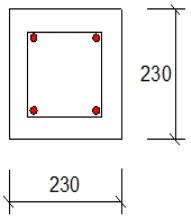
No	N (Bot)	M1 (Bot)	M2 (Bot)	N (Top)	M1 (Top)	M2 (Top)
1	332.6	-3.7	-2.3	332.6	7.4	4.6
2	332.6	-3.7	-2.3	332.6	7.4	4.6
3	332.6	-3.7	-2.3	332.6	7.4	4.6
4	332.8	-2.6	-2.3	332.8	6.4	4.6
5	332.3	-4.7	-2.3	332.3	8.4	4.6
6	332.4	-3.6	-1.2	332.4	7.3	3.7
7	243.3	-2.7	-2.7	243.3	5.5	4.3



Critical Combination:1 - (G+Q *F)

		Min	Design
N (kN)	332.6	-	332.6
M1 (kN.m)	7.4	6.7	8.7
M2 (kN.m)	4.6	6.7	6.7
N-max (kN)	0.0		

Concrete Cover = 35.0 mm



Neutral Axis:
Y = 231.4 mm
Alpha = 32.65 ^{0}

Short Column...
Lambda1/Lim1 = 22.7 < 51.2
Lambda2/Lim2 = 22.7 < 52.0
M-add(1/2)= 0.0 / 0.0 kN.m

V-Ed (1/2)= 4.4 / 2.9 kN
v-Ed = 0.11 / 0.07 N/mm2
v-Rdc = 1.20 / 1.20 N/mm2
v-RdMax = 4.86 / 4.86 N/mm2

As (Required): %0.20 (min) 105.80 mm2
As (Provided): %1.52 804.25 mm2

4T16

Links = T8-125

7.8 Column Design Results from Staad Pro

The internal forces (ULS), and reinforcement design for the columns from Staad Pro V8i software are tabulated as shown in Table 7.4. You can compare the result with manual analysis result.

Table 7.4: Summary column design moments, axial loads, and reinforcement to Staad

Column	N_{Ed} (KN)	M_X (KN.m)		M_Z (KN.m)		$A_{s\ (req)}$ mm^2	A_{sprov} mm^2
		Top	Bottom	Top	Bottom		
A1	130.684	-3.678	-1.927	-3.823	-1.821	106	4T16
A3	201.632	-4.185	-2.073	1.772	0.963	106	4T16
A5	163.207	-2.898	-1.356	-0.666	-0.250	106	4T16
A7	140.392	-2.213	0.924	2.162	1.156	106	4T16
B5	272.290	-0.053	0.059	-1.712	-0.858	106	4T16
B7	266.620	-0.552	-0.097	1.768	0.873	106	4T16
B8	118.088	-5.365	-2.437	1.802	0.891	106	4T16
C1	157.085	1.891	0.845	-3.238	-1.629	106	4T16
C3	217.672	2.095	1.051	1.947	0.950	106	4T16
D1	100.283	-0.528	-0.352	-1.443	-0.771	106	4T16
D2	183.828	-1.445	0.760	-0.659	-0.381	106	4T16
D4	169.082	-1.206	-0.552	-1.449	-0.774	106	4T16
E5	300.245	2.096	1.129	-5.744	-2.988	106	4T16
E8	221.446	5.170	2.806	11.453	5.567	106	4T16
F1	74.201	1.734	0.771	-0.943	-0.626	106	4T16
F2	148.346	2.914	1.409	-1.376	0.841	106	4T16
F4	105.328	1.602	0.846	1.199	0.441	106	4T16
G4	27.313	-1.316	-0.670	-0.898	-0.605	106	4T16
G6	64.852	-3.503	-1.827	-0.458	-0.386	106	4T16
G8	31.817	-1.886	-1.109	0.613	0.148	106	4T16

Calculation sheet for column E5 (member 16) according to Staad RC Designer

Member 16 - Detailed EC2 Main Reinforcement

Axial N		= 300.24 kN	
Major M,	end 1,	= -2.99 kNm	
	end 2,	= 5.74 kNm	
Minor M,	end 1,	= 1.13 kNm	
	end 2,	= -2.10 kNm	
Slenderness			
	λ_{lim}	= 20 ABC / $\sqrt{n}$	(5.13N)
	n	= $N_{Ed} / (A_c f_{cd}) = 300 \times 10^3 / (52900 \times 14.2)$	= 0.401
Major	$\lambda_{lim,y}$	= $20 \times 1.00 \times 1.36 \times 2.47 / \sqrt{0.401}$	= 106.20
Minor	$\lambda_{lim,z}$	= $20 \times 1.00 \times 1.36 \times 2.58 / \sqrt{0.401}$	= 111.17
	λ	= l_0 / i	(5.14)
	λ_y	= 4500 / 66.40	= 67.78
	λ_z	= 4500 / 66.40	= 67.78
Column is		NOT SLENDER	
Axial Capacity N_{Rd}			= 1059.72 kN
Axial capacity ratio N_{Ed} / N_{Rd}			= 0.28
Axial Capacity ≥ Axial			∴ **OK for axial resistance**
Major Axis Capacity M_{Rdy}			= 40.20 kNm
Major Axis Capacity Ratio M_{Edy} / M_{Rdy}			= 0.29
Major Axis Capacity ≥ Major Axis Moment			∴ **OK for moment resistance**
Minor Axis Capacity M_{Rdz}			= 40.20 kNm
Minor Axis Capacity Ratio M_{Edz} / M_{Rdz}			= 0.20
Minor Axis Capacity ≥ Minor Axis Moment			∴ **OK for moment resistance**
Biaxial Interaction Equation	$= \left(\frac{M_{Edz}}{M_{Rdz}}\right)^a + \left(\frac{M_{Edy}}{M_{Rdy}}\right)^a \le 1,0$		(5.39)
Where exponent a			= 1.15
Biaxial Interaction Equation	$= \left(\frac{8.10}{40.20}\right)^{1.15} + \left(\frac{11.75}{40.20}\right)^{1.15}$		= 0.40
Biaxial Interaction Result ≤ 1.0			∴ **OK for biaxial resistance**

Member 16 - Detailed EC2 Main Reinforcement Cont...

Max distance between bars	= 300 mm	
Largest actual space between bars	= 112 mm	
Min distance between bars $= d_g + 5$ mm	= 25 mm	8.2 (2)
Smallest actual space between bars	= 112 mm	
	∴**OK for spacing**	
Min area of reinforcement	= 106 mm²	9.5.2 (2)
Max area of reinforcement (even at laps)	= 4232 mm²	9.5.2 (3)
Actual area of main reinforcement (4T16)	= 804 mm²	
∴ area by which steel can increase at laps	= 3428 mm²	
Starter bar area	= 804 mm²	
	∴**OK**	

Member 16 - Detailed EC2 Shear Reinforcement

Spacing provided, s		= 225 mm	
Major axis:			
Shear force, V_{Ed}		= 1.94 kN	
$V_{Rd,c}$	$= [C_{Rd,c}\, k\, (100\, \rho_l\, f_{ck})^{1/3} + k_1 \sigma_{cp}]\, b_w d$		(6.2.a)
	$= [0.120 \times 2.000 \times (100 \times 0.020 \times 25)^{1/3} + 0.15 \times 2.833] \times 230 \times 179$		
		= 53.61 kN	
But	$\geq (v_{min} + k_1 \sigma_{cp})\, b_w d$		(6.2.b)
	$= (0.49 + 0.15 \times 2.833) \times 230 \times 179$	= 37.88 kN	
Hence, $V_{Rd,c}$		= 53.61 kN	
$V_{Ed} \leq V_{Rd,c}$	∴only nominal links required		6.2.1(5)
Area of links provided (2T8), A_{sw}		= 100.53 mm²	
Minor axis:			
Shear force, V_{Ed}		= 0.72 kN	
$V_{Rd,c}$	$= [C_{Rd,c}\, k\, (100\, \rho_l\, f_{ck})^{1/3} + k_1 \sigma_{cp}]\, b_w d$		(6.2.a)
	$= [0.120 \times 2.000 \times (100 \times 0.020 \times 25)^{1/3} + 0.15 \times 2.833] \times 230 \times 179$		
		= 53.61 kN	
But	$\geq (v_{min} + k_1 \sigma_{cp})\, b_w d$		(6.2.b)
	$= (0.49 + 0.15 \times 2.833) \times 230 \times 179$	= 37.88 kN	
Hence, $V_{Rd,c}$		= 53.61 kN	
$V_{Ed} \leq V_{Rd,c}$	∴only nominal links required		6.2.1(5)
Area of links provided (2T8), A_{sw}		= 100.53 mm²	
Minimum size of link bars		= 8 mm	9.5.3 (1)
Actual size of link bars		= 8 mm	
		∴ **OK**	

CHAPTER 8

ANALYSIS AND DESIGN OF THE FOUNDATIONS

8.1 Introduction

All loads from the superstructure of a building are transferred to the ground. If the foundation of a building is poorly designed, then all the efforts input in designing the superstructure is in vain. It is therefore imperative that adequate care be taken in the design of foundations. Foundation design starts from detailed field and soil investigation. It is very important to know the index and geotechnical properties of the soil, including the soil chemistry, so that the performance of the foundation can be guaranteed.

According to section 2.8(3) of Eurocode 7, the Geotechnical Design Report should normally include the following items, with cross reference to the Ground Investigation Report and to other documents, which contain more detail:

- a description of the site and surroundings;
- a description of the ground conditions;
- a description of the proposed construction, including actions (loads);
- design values of soil and rock properties, including justification, as appropriate;
- statements on the codes and standards applied;
- statements on the suitability of the site with respect to the proposed construction and the
- level of acceptable risks;
- geotechnical design calculations and drawings;
- foundation design recommendations;
- a note of items to be checked during construction or requiring maintenance or monitoring.

The design of footings is covered by Section 6 of Eurocode 7 Part 1, 'Spread foundations', whose contents are as follows:

- 6.1 General
- 6.2 Limit states
- 6.3 Actions and design situations
- 6.4 Design and construction considerations
- 6.5 Ultimate limit state design
- 6.6 Serviceability limit state design
- 6.7 Foundations on rock; additional design considerations
- 6.8 Structural design of foundations
- 6.9 Preparation of the subsoil

Section 6 of EN 1997-1 applies to pad, strip, and raft foundations and some provisions may be applied to deep foundations, such as caissons.

In section *6.2* of *EC7,* most common limit states for spread foundations are listed as shown below;

- bearing resistance failure
- failure by sliding
- combined failure in the ground and in the structure
- structural failure due to foundation movement
- excessive settlements
- excessive heave due to swelling, frost and other causes
- loss of overall stability

Actions (design loads) have to be selected in accordance to Eurocode 1 with special attention to Eurocode 0 (Basis of structural design). Special consideration is given to design situations to be

considered for the purpose of foundation design. These include all the actions, load cases and their combination, overall stability; the disposition and classification of the various soils and elements of construction; dipping bedding planes; underground structures; interbedded hard and soft strata; faults, joints and fissures; possible instability of rock blocks; solution cavities; and the environment within which the design is set that includes earthquakes, subsidence, interference with existing constructions.

Actions include: weight of soil and water; earth pressures; free water pressure, wave pressure; seepage forces; dead and imposed loads from structures; surcharges; mooring forces; removal of load and excavation of ground; and traffic loads.

8.2 Depth of foundation

According to section 6.4 of EC7, when choosing the depth of a spread foundation the following shall be considered:

- reaching an adequate bearing stratum;
- the depth above which shrinkage and swelling of clay soils, due to seasonal weather changes, or to trees and shrubs, may cause appreciable movements;
- the depth above which frost damage may occur;
- the level of the water table in the ground and the problems, which may occur if excavation for the foundation is required below this level;
- possible ground movements and reductions in the strength of the bearing stratum by seepage or climatic effects or by construction procedures;
- the effects of excavations on nearby foundations and structures;
- anticipated excavations for services close to the foundation;
- high or low temperatures transmitted from the building; the possibility of scour;
- the effects of variation of water content due to long periods of drought, and subsequent periods of rain, on the properties of volume-unstable soils in arid climatic areas;
- the presence of soluble materials, e.g. limestone, claystone, gypsum, salt rocks;

8.3 Methods of pad footing design

EC7 Section 6.4 5(P) presents three methods for design of pad footings;

(a) direct method, in which separate analyses are carried out for each limit state. When checking against an ultimate limit state, the calculation shall model as closely as possible the failure mechanism, which is envisaged. When checking against a serviceability limit state, a settlement calculation shall be used;

In direct method we have to check all limit states, including load and partial factor combinations. The formular for calculating bearing capacity is given in equation 8.1.

$$q_{ult} = c'N_cS_cD_ci_cg_cb_c + q'N_qS_qd_qi_qg_qb_q + \gamma' BN_\gamma S_\gamma d_\gamma i_\gamma g_\gamma b_\gamma /2 \text{ ---------- (8.1)}$$

Where;
C' = cohesion
q = overburden pressure
γ = Soil unit weight
N_i = bearing capacity factors
S_i = Shape factors
d_i = Depth factors
i_i = inclination factors
g_i = ground inclination factors
b_i = Base inclination factors

(b) an indirect method using comparable experience and the results of field or laboratory measurements or observations, and chosen in relation to serviceability limit state loads so as to satisfy the requirements of all relevant limit states;

(c) a prescriptive method in which a presumed bearing resistance is used (this approach has been adopted in this design)

8.4 Structural Design Process

Footings should be able to resist flexural stresses, diagonal shear, and punching shear stresses.

8.4.1 Shear at Column Face

According to Clause 6.4.5 of EC2, adjacent to the column the applied ultimate shear stress V_{Ed}, should not exceed the maximum allowable shear resistance $V_{Rd,max}$

$$v_{Ed} = \beta \frac{V_{Ed}}{u_0 d} \text{ -------- (8.2)}$$

Where;

V_{Ed} = Applied shear force

β is an eccentricity factor defined in section 6.4.3 of EC2. For footing with no eccentricity, β = 1.0. For footing with eccentricity, it can be approximated from;

$$\beta = 1 + 1.8\sqrt{\left(\frac{e_y}{b_z}\right)^2 + \left(\frac{e_z}{b_y}\right)^2} \text{ (equation 6.4.3 EC2) ---------- (8.3)}$$

e_y and e_z are the eccentricities in the y and z direction respectively
b_y and b_z are the dimensions of the control perimeter

d means the effective depth of the slab, which may be taken as (*d*y + *d*z)/2 where *d*y, *d*z effective depth of the section in the y and z direction respectively

u_0 is the column perimeter

$$V_{Rd,max} = 0.5 v f_{cd} \text{ ---------- (8.4)}$$

Where;

$v = 0.6\left[1 - \left(\frac{f_{ck}}{250}\right)\right]$

$f_{cd} = \frac{\alpha_{cc} f_{ck}}{\gamma_c} = \frac{0.85 \times 25}{1.5}$ = 14.167 N/mm^2 (for grade 25 concrete)

8.4.2 Normal shear resistance at 'd' from the column face

Normal shear resistance should also be verified on vertical sections at distance *d* from the column face extending across the full width of the base. For satisfactory performance, $v_{Ed} < V_{Rd,c}$

$$v_{Ed} = \frac{V_{Ed,red}}{ud} \text{ ---------- (8.5)}$$

Where;
u = the perimeter being considered
d = mean effective depth of the section

$V_{Ed,red} = V_{Ed} - \Delta V_{Ed}$ ---------- (8.6)

The design shear resistance, $V_{Rd,c}$

$$V_{Rd,c} = [C_{Rd,c}.k.\ (100\rho_1 f_{ck})^{\frac{1}{3}}] \times (2d/a) \geq (V_{min} + k_1.\sigma_{cp}) \text{ ---------- (8.7)}$$

$C_{Rd,c} = 0.18/\gamma_c$

$k = 1+\sqrt{\frac{200}{d}}$

$V_{min} = 0.035k^{\frac{3}{2}}f_{ck}{}^{\frac{1}{2}}$

$\rho_1 = \frac{As}{bd}$

8.4.3 Punching Shear

No punching shear reinforcement is required if the applied shear stress, v_{Ed} is less than the design punching shear resistance of concrete, $V_{Rd,c}$

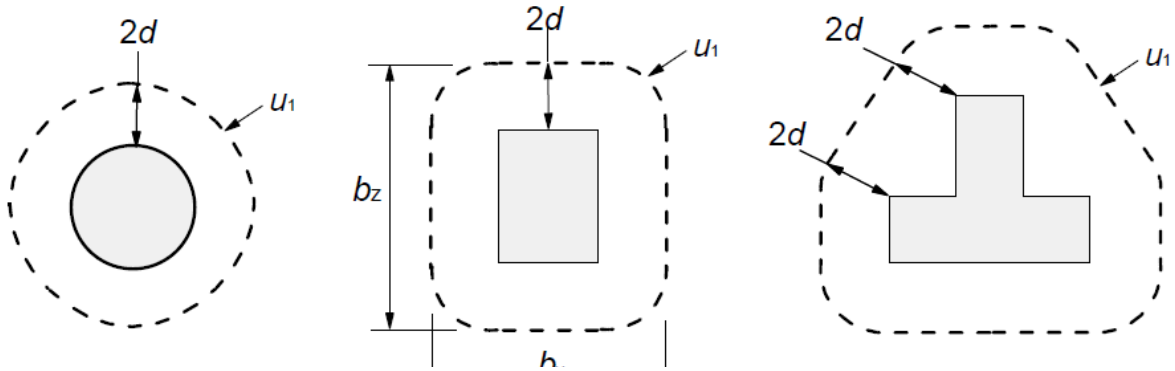


Figure 8.3: Punching shear basic control parameters around loaded areas (Fig 6.13 EC2)

$$v_{Ed} = \beta\frac{V_{Ed}}{u_1 d}$$

V_{Ed} = Applied shear force

β is an eccentricity factor defined in section 6.4.3 of EC2. See equation 8.3 in this text.

d mean effective depth of the slab, which may be taken as (*d*y + *d*z)/2 where *d*y, *d*z effective depth of the section in the y and z direction respectively

u_1 is the control perimeter. For rectangular bases, u_1 = column perimeter + $2\pi(2d)$ (see Figure 8.1)

$$V_{Rd,c} = [C_{Rd,c}.k.\ (100\rho_1 f_{ck})^{\frac{1}{3}}] \times (2d/a) \geq (V_{min} + k_1.\sigma_{cp})$$

$C_{Rd,c} = 0.18/\gamma_c$

$k = 1+\sqrt{\frac{200}{d}}$

$V_{min} = 0.035k^{\frac{3}{2}}f_{ck}^{\frac{1}{2}}$

$\rho_1 = \frac{As}{bd}$

8.5 Structural Design of foundations

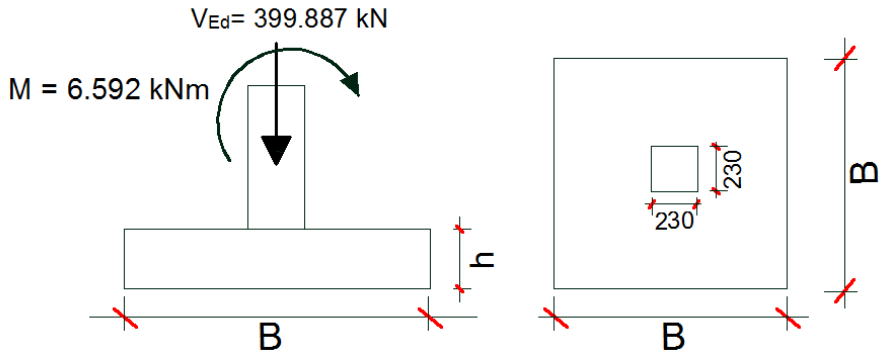
8.5.1 GROUP A DESIGN (Axial load > 300 KN)

Maximum Ultimate Axial Load from Column N_{Ed} = 399.887 KN

Maximum Elastic Column Base Moment

X - direction: M_{02} = 6.592 KNm; Z – direction M_{02} = 3.596 KNm

Bearing Capacity of soil = 150 KN/m^2
Density of soil = 18 KN/m^3
Height of surcharge (including DPC) = 1.5m



Eccentricity of loading $e_x = \frac{M_x}{N_{Ed}} = \frac{6.592}{399.887}$ = 0.0164 m = 16.48mm

Eccentricity in the z-direction $e_z = \frac{M_z}{N_{Ed}} = \frac{3.596}{399.887}$ = 0.00899 m = 8.99mm

Converting the load to serviceability limit state load

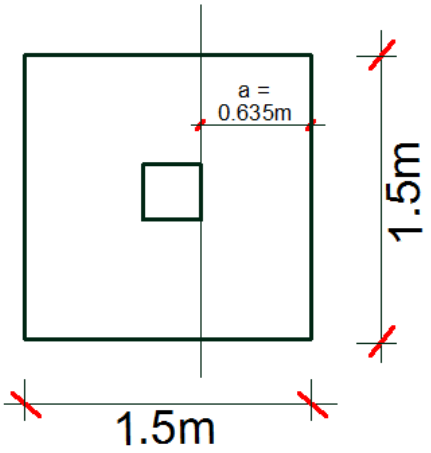
Statistical analysis has shown that a conversion factor of 1.36 is usually very adequate for residential buildings.

$V = \frac{V_{Ed}}{1.36} = \frac{399.887}{1.36}$ = 294.034 KN

Assuming 15 KN/m^2 reduction in bearing capacity to account for self weight of footing and surcharge load. Hence allowable bearing capacity = 135 KN/m^2

Area of footing required = $\frac{V + S_W + S_S}{135} = \frac{294.034}{135}$ = 2.178 m^2

For a square footing, B = L = $\sqrt{2.178}$ = 1.4758m

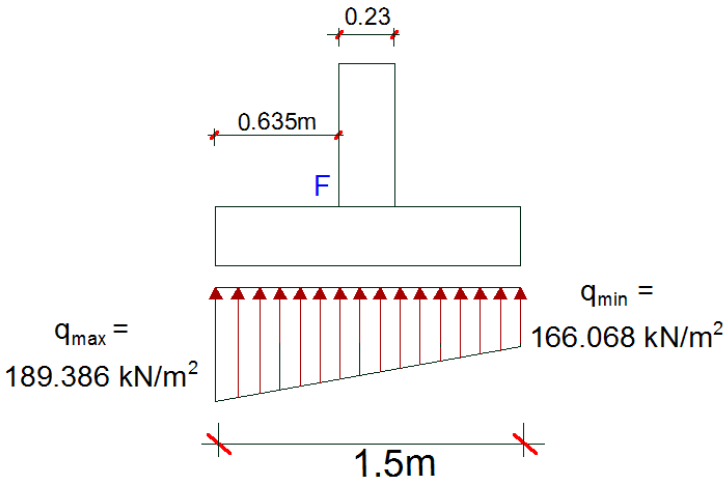


Adopt a square footing of dimensions 1500mm × 1500mm. Assume a trial depth of 400mm

Considering the maximum eccentricity (in the x-direction)

$q_{min} = \frac{P}{A}\left(1 - \frac{6e}{L}\right) = \frac{399.887}{1.5 \times 1.5}\left(1 - \frac{6(0.0164)}{1.5}\right)$ = 166.068 KN/m^2

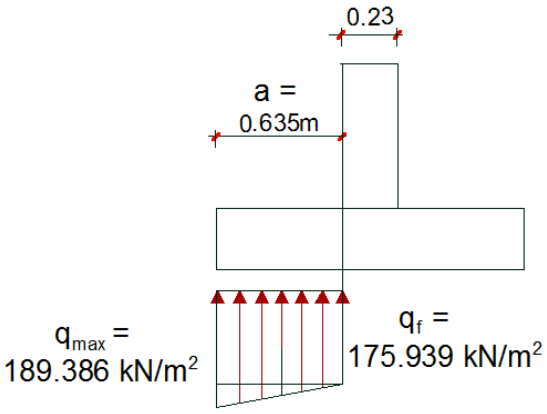
$q_{max} = \frac{P}{A}\left(1 + \frac{6e}{L}\right) = \frac{399.887}{1.5 \times 1.5}\left(1 + \frac{6(0.0164)}{1.5}\right)$ = 189.386 KN/m^2



Design Moment

Maximum moment occurs at the column face for reinforced concrete columns

From similar triangles, the earth pressure at the column face (point F) is given by;



q_f = 166.068 + $\left(\frac{(189.386 - 166.068) \times (1.5 - 0.635)}{1.5}\right)$ = 179.515 KN/m^2

$M_{Ed} = \frac{q_f a^2}{2} + \frac{a(q_{max} - q_f)}{2} \times \frac{2a}{3}$

$= \frac{179.515 \times 0.635^2}{2} + \frac{0.635(189.386 - 179.515)}{2} \times \frac{2(0.635)}{3}$

= 36.192 + 1.326 = 37.518 KN.m

Effective depth

Concrete cover = 50mm

AssumingY12mm bars,

d = 400 – 50 – 6 = 344mm

$k = \frac{M_{Ed}}{f_{ck}\, bd^2} = \frac{37.518 \times 10^6}{25 \times 1000 \times 344^2}$ = 0.01268 (designing per metre strip)

Since k < 0.167 No compression reinforcement required

$z = d[0.5 + \sqrt{(0.25 - 0.882K)}]$ = $z = d[0.5 + \sqrt{(0.25 - 0.882(0.0126)}]$ = 0.95d

$A_{s1} = \frac{M_{Ed}}{0.87 f_{yk} z} = \frac{37.518 \times 10^6}{0.87 \times 460 \times 0.95 \times 344}$ = 286.869 mm^2/m

To calculate the minimum area of steel required;
fctm = $0.3 \times f_{ck}^{2/3}$ = $0.3 \times 25^{2/3}$ = 2.5649 N/mm^2 (Table 3.1 EC2)

$A_{Smin} = 0.26 \times \frac{\text{fctm}}{\text{Fyk}} \times b \times d = 0.26 \times \frac{2.5649}{460} \times 1000 \times 344$ = 498.7 mm^2
Check if A_{Smin} < 0.0013 × b × d (447.2 mm^2)
Since, A_{Smin} = 498.7 mm^2, the provided reinforcement is adequate.

Provide Y12 @ 200mm c/c (AS_{prov} = 565 mm^2/m) each way

Shear at the column face

Ultimate Load on footing from column = 399.887 kN
Design shear stress at the column perimeter $v_{Ed} = \beta \frac{V_{Ed}}{u_0 d}$

β = 1+ $1.8\sqrt{\left(\frac{16.48}{230}\right)^2 + \left(\frac{8.99}{230}\right)^2}$ = 1.146

Where u_o is the column perimeter and d is the effective depth

$v_{Ed} = \beta \frac{V_{Ed}}{u_0 d} = \frac{1.15 \times 399.887 \times 10^3}{4(230) \times 344}$ = 1.452N/mm^2

$v_{Rd,max}$ = 0.5vf_{cd}

$v = 0.6\left[1 - \left(\frac{f_{ck}}{250}\right)\right] = 0.6\left[1 - \left(\frac{25}{250}\right)\right]$ = 0.54 N/mm^2

$f_{cd} = \frac{\alpha_{cc} f_{ck}}{\gamma_c} = \frac{0.85 \times 25}{1.5}$ = 14.167 N/mm^2

$v_{Rd,max}$ = 0.5 × 0.54 × 14.167 = 3.825 N/mm^2
v_{Ed} < $v_{Rd,max.}$ This is very ok

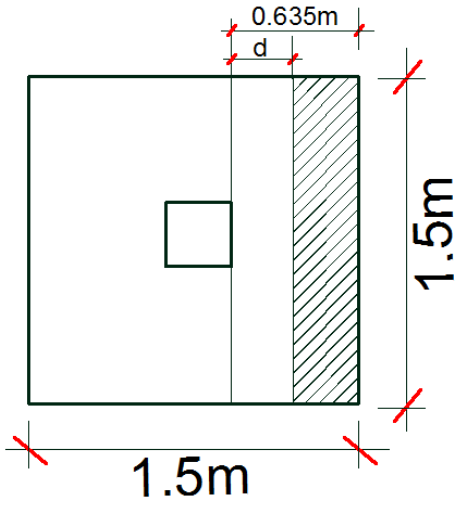
Transverse shear at 'd' from the face of column

Width of shaded area = a – d = 0.635 – 0.344 = 0.291m

Area of shaded area = (1.5m × 0.291m) = 0.4365 m^2

Therefore, $\Delta V_{Ed} = \frac{189.386 + 175.939}{2} \times 0.4365$ m^2 = 79.077 KN

$v_{Ed} = \frac{V_{Ed}}{bd} = \frac{79.077 \times 10^3}{1500 \times 344}$ = 0.15325 N/mm^2



$V_{Rd,c} = [C_{Rd,c}.k.\ (100\rho_1 f_{ck})^{\frac{1}{3}} + k_1.\sigma_{cp}] \times (2d/a) \geq (V_{min} + k_1.\sigma_{cp})\ bw.d$

$C_{Rd,c} = 0.18/\gamma_c = 0.18/1.5 = 0.12$

$k = 1+\sqrt{\frac{200}{d}} = 1+\sqrt{\frac{200}{344}} = 1.7624 > 2.0$, therefore, k = 1.7624

$V_{min} = 0.035k^{\frac{3}{2}}f_{ck}^{\frac{1}{2}}$ = Vmin = $0.035 \times (1.7624)^{\frac{3}{2}} \times (25)^{\frac{1}{2}} = 0.4094$ N/mm^2

$\rho_1 = \frac{As}{bd} = \frac{565}{1000 \times 344} = 0.001642 < 0.02$;

$V_{Rd,c} = [0.12 \times 1.7624(100 \times 0.001642 \times 25)^{\frac{1}{3}}] = 0.3386$ N/mm$^2 \times (2d/a) < V_{min}$

But in this case, d = a

Hence, $V_{Rd,c} = 2 \times 0.3386 = 0.6772$ N/mm^2

Since $V_{Rd,c}$ (0.6772) > V_{Ed} (0.1503 KN), No shear reinforcement is required.

Punching Shear at 2d

Punching shear lies outside the footing dimensions. No further check required.

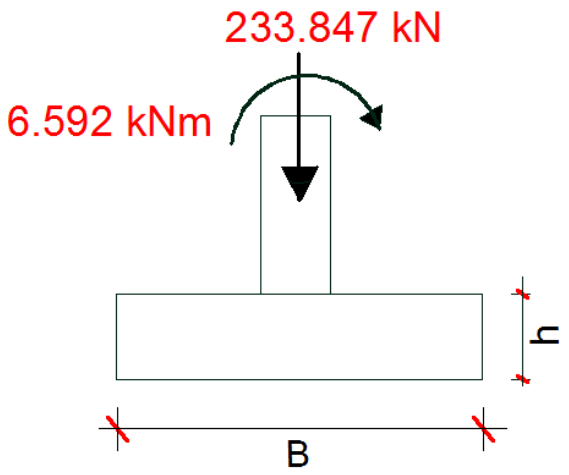
8.5.2 GROUP B DESIGN (233.847 KN ≤ Axial load ≤ 178.277 KN)

Maximum Ultimate Axial Load from Column N_{Ed} = 233.847 KN

Maximum Elastic Column Base Moment

X - direction: M_{02} = 6.592 KNm; Z – direction M_{02} = 2.831 KNm

Bearing Capacity of soil = 150 KN/m^2
Density of soil = 18 KN/m^3
Height of surcharge (including DPC) = 1.5m



Eccentricity of loading $e_x = \frac{M_x}{N_{Ed}} = \frac{6.592}{233.847} = 0.0282$ m = 28.189mm

Eccentricity in the z-direction $e_z = \frac{M_z}{N_{Ed}} = \frac{2.831}{233.847}$ = 0.0121m = 12.106 mm

Converting the load to serviceability limit state load

Statistical analysis has shown that a conversion factor of 1.36 is usually very adequate for residential buildings.

$V = \frac{V_{Ed}}{1.36} = \frac{233.847}{1.36}$ = 171.946 KN

Assuming 10% of the service load to account for the self weight of the footing.

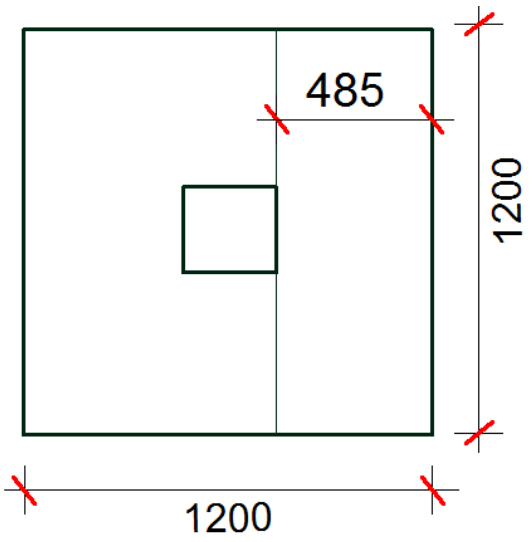
Sw = 0.1 × 171.946 = 17.1946 KN

Assuming 15 KN/m^2 reduction in bearing capacity to account for surcharge load. Hence allowable bearing capacity = 135 KN/m^2

Area of footing required = $\frac{V + S_W}{135}$ = $\frac{171.946 + 17.1946}{135}$ = 1.40104 m^2

For a square footing, B = L = $\sqrt{1.40104}$ = 1.1836

Adopt a square footing of dimensions 1200mm × 1200mm. Assume a trial depth of 300mm



ULS Earth Pressure Intensity

Considering the maximum eccentricity (in the x-direction)

$q_{min} = \frac{P}{A}\left(1 - \frac{6e}{L}\right) = \frac{233.847}{1.2 \times 1.2}\left(1 - \frac{6(0.0282)}{1.2}\right)$ = 139.496 KN/m^2

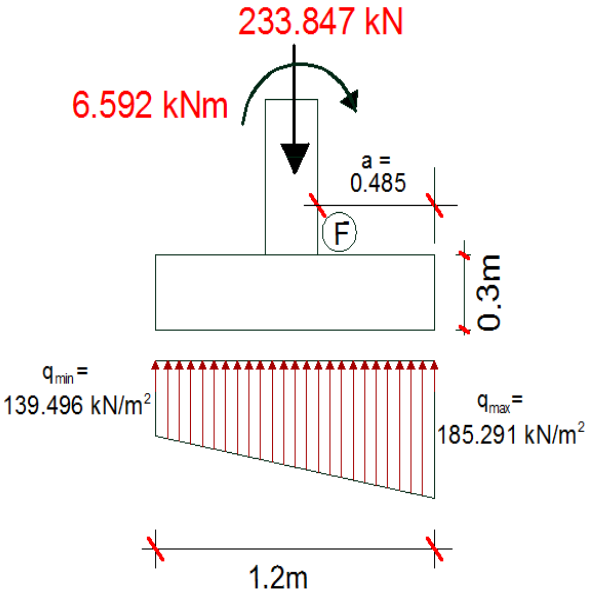
$q_{max} = \frac{P}{A}\left(1 + \frac{6e}{L}\right) = \frac{233.847}{1.2 \times 1.2}\left(1 + \frac{6(0.0282)}{1.2}\right)$ = 185.291 KN/m^2

Design Moment

Maximum moment occurs at the column face for reinforced concrete columns

From similar triangles, the earth pressure at the column face (point F) is given by;

q_f = 139.496 + $\left(\frac{(185.291 - 139.496) \times (1.2 - 0.485)}{1.2}\right)$ = 166.782 KN/m^2



$M_{Ed} = \frac{q_f a^2}{2} + \frac{a(q_{max} - q_f)}{2} \times \frac{2a}{3}$

$M_{Ed} = \frac{166.782 \times 0.485^2}{2} + \frac{0.485(166.782 - 158)}{2} \times \frac{2(0.485)}{3}$ = 19.615 + 0.688 = 20.303 KN.m

Effective depth

Concrete cover = 50mm
AssumingY12mm bars,
d = 300 – 50 – 6 = 244mm

$k = \frac{M_{Ed}}{f_{ck}\, bd^2} = \frac{20.303 \times 10^6}{25 \times 1000 \times 244^2}$ = 0.0136 (designing per metre strip)

Since k < 0.167 No compression reinforcement required

$z = d\left[0.5 + \sqrt{(0.25 - 0.882K)}\right]$ = $z = d\left[0.5 + \sqrt{(0.25 - 0.882(0.0139)}\right]$ = 0.95d

$A_{s1} = \frac{M_{Ed}}{0.87 f_{yk} z} = \frac{20.303 \times 10^6}{0.87 \times 460 \times 0.95 \times 244}$ = 218.862 mm^2/m

To calculate the minimum area of steel required;
fctm = $0.3 \times f_{ck}^{\,2/3}$ = $0.3 \times 25^{2/3}$ = 2.5649 N/mm^2 (Table 3.1 EC2)

$A_{Smin} = 0.26 \times \frac{\text{fctm}}{\text{Fyk}} \times b \times d = 0.26 \times \frac{2.5649}{460} \times 1000 \times 244$= 353.733 mm^2
Check if A_{Smin} < 0.0013 × b × d (317.2 mm^2)

Provide **Y12 @ 200mm c/c (AS_{prov} = 565 mm^2/m) each way**

Shear at the column face

Ultimate Load on footing from column = 233.847 kN
Design shear stress at the column perimeter $v_{Ed} = \beta \frac{V_{Ed}}{u_0 d}$

β = 1+ $1.8\sqrt{\left(\frac{28.189}{230}\right)^2 + \left(\frac{12.106}{230}\right)^2}$ = 1.24

Where u_o is the column perimeter and d is the effective depth

$v_{Ed} = \beta \frac{V_{Ed}}{u_0 d} = \frac{1.24 \times 233.847 \times 10^3}{4(230) \times 244}$ = 1.291 N/mm^2

$V_{Rd,max} = 0.5 v f_{cd}$

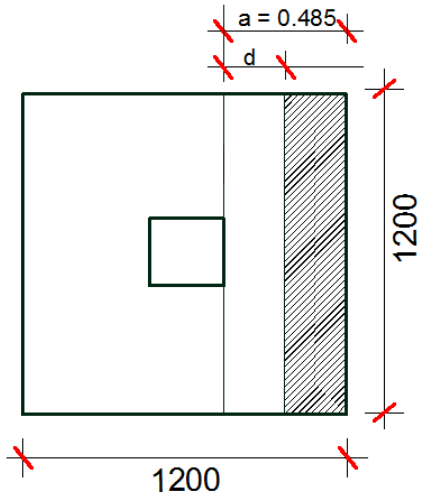
$v = 0.6\left[1 - \left(\frac{f_{ck}}{250}\right)\right] = 0.6\left[1 - \left(\frac{25}{250}\right)\right]$ = 0.54 N/mm^2

$f_{cd} = \frac{\alpha_{cc} f_{ck}}{\gamma_c} = \frac{0.85 \times 25}{1.5}$ = 14.167 N/mm^2

$v_{Rd,max}$ = 0.5 × 0.54 × 14.167 = 3.825 N/mm^2
v_{Ed} < $v_{Rd,max.}$ This is very ok

Transverse shear at 'd' from the face of column

Shaded area = a – d = 0.485 – 0.244 = 0.241m



Area of shaded area = (1.2m × 0.241m) = 0.2892 m^2

Therefore, $\Delta V_{Ed} = \frac{185.291 + 158}{2} \times 0.2892\ m^2 = 49.639$ KN

$v_{Ed} = \frac{V_{Ed}}{bd} = \frac{49.639 \times 10^3}{1200 \times 244} = 0.1695 N/mm^2$

$V_{Rd,c} = [C_{Rd,c}.k.\ (100\rho_1 f_{ck})^{\frac{1}{3}} + k_1.\sigma_{cp}] \times (2d/a) \geq (V_{min} + k_1.\sigma_{cp})\ bw.d$

$C_{Rd,c} = 0.18/\gamma_c = 0.18/1.5 = 0.12$

$k = 1+\sqrt{\frac{200}{d}} = 1+\sqrt{\frac{200}{244}} = 1.905 > 2.0$, therefore, k = 1.905

$V_{min} = 0.035k^{\frac{3}{2}}f_{ck}^{\frac{1}{2}}$ = Vmin = $0.035 \times (1.905)^{\frac{3}{2}} \times (25)^{\frac{1}{2}} = 0.460\ N/mm^2$

$\rho_1 = \frac{As}{bd} = \frac{565}{1000 \times 244} = 0.0023155 < 0.02$;

$V_{Rd,c} = [0.12 \times 1.905(100 \times 0.002316\ \times 25)^{\frac{1}{3}}] = 0.4105\ N/mm^2 \times (2d/a) > V_{min}$

Because in this case, d = a

Hence, $V_{Rd,c} = 2 \times 0.4105 = 0.8209\ N/mm^2$

Since $V_{Rd,c}$ (0.8209) > V_{Ed} (0.1695 N/mm^2), No shear reinforcement is required.

Punching Shear at 2d

Punching shear lies outside the footing dimensions. No further check required.

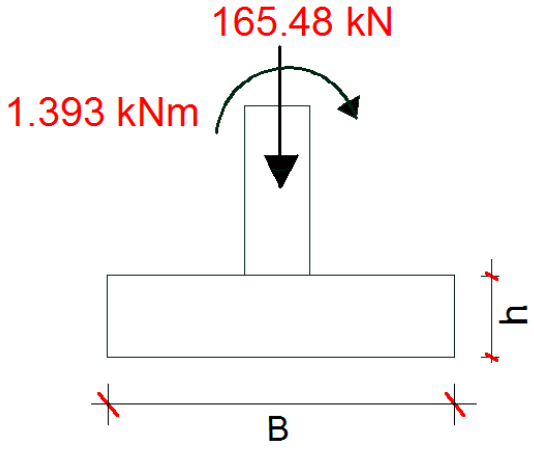
8.5.3 GROUP C DESIGN (165.48 KN ≤ Axial load < 0)

Maximum Ultimate Axial Load from Column N_{Ed} = 165.48 KN

Maximum Elastic Column Base Moment

X - direction: M_{02} = 1.130 KNm; Z – direction M_{02} = 1.393 KNm

Bearing Capacity of soil = 150 KN/m^2
Density of soil = 18 KN/m^3
Height of surcharge (including DPC) = 1.5m



Eccentricity of loading $e_x = \frac{M_x}{N_{Ed}} = \frac{1.139}{165.48}$ = 0.00688 m = 6.88 mm

Eccentricity in the z-direction $e_z = \frac{M_z}{N_{Ed}} = \frac{1.393}{165.48}$ = 0.00842m = 8.42 mm

Converting the load to serviceability limit state load

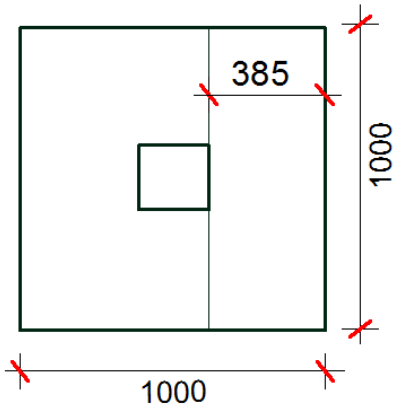
Statistical analysis has shown that a conversion factor of 1.36 is usually very adequate for residential buildings.

$V = \frac{V_{Ed}}{1.36} = \frac{165.48}{1.36}$ = 121.676 KN

Assuming 10% of the service load to account for the self weight of the footing;

Sw = 0.1 × 121.676 = 12.167 KN

Assuming 15 KN/m^2 reduction in bearing capacity to account for surcharge load. Hence allowable bearing capacity = 135 KN/m^2



Area of footing required = $\frac{V + S_w}{135} = \frac{121.676 + 12.167}{135}$ = 0.99m^2

For a square footing, B = L = $\sqrt{0.99}$ = 0.995

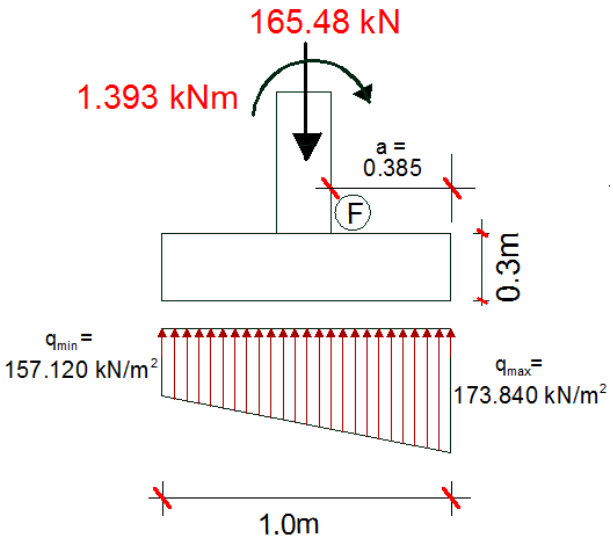
Adopt a square footing of dimensions 1000mm × 1000mm. Assume a trial depth of 300mm

ULS Earth Pressure Intensity

Considering the maximum eccentricity (in the x-direction)

$q_{min} = \frac{P}{A}\left(1 - \frac{6e}{L}\right) = \frac{165.48}{1.0 \times 1.0}\left(1 - \frac{6(0.00842)}{1.0}\right)$ = 157.120 KN/m^2

$q_{max} = \frac{P}{A}\left(1 + \frac{6e}{L}\right) = \frac{165.48}{1.0 \times 1.0}\left(1 + \frac{6(0.00842)}{1.0}\right)$ = 173.840 KN/m^2



Design Moment

Maximum moment occurs at the column face for reinforced concrete columns

From similar triangles, the earth pressure at the column face (point F) is given by;

q_f = 157.120 + $\left(\frac{(173.840 - 157.120) \times (1.0 - 0.385)}{1.0}\right)$ = 167.403 KN/m^2

$M_{Ed} = \frac{q_f a^2}{2} + \frac{a(q_{max} - q_f)}{2} \times \frac{2a}{3}$

$= \frac{167.403 \times 0.385^2}{2} + \frac{0.385(173.840 - 167.403)}{2} \times \frac{2(0.385)}{3}$ = 12.406 + 0.31804 = 12.724 KN.m

Effective depth
Concrete cover = 50mm
AssumingY12mm bars,
d = 300 – 50 – 6 = 244mm

$k = \frac{M_{Ed}}{f_{ck}\, bd^2} = \frac{12.724 \times 10^6}{25 \times 1000 \times 244^2}$ = 0.0085487(designing per metre strip)

Since k < 0.167 No compression reinforcement required

$z = d\left[0.5 + \sqrt{(0.25 - 0.882K)}\right]$ = $z = d\left[0.5 + \sqrt{(0.25 - 0.882(0.0085487)}\right]$ = 0.95d

$A_{s1} = \frac{M_{Ed}}{0.87 f_{yk} z} = \frac{12.724 \times 10^6}{0.87 \times 460 \times 0.95 \times 244}$ = 137.162 mm^2/m

To calculate the minimum area of steel required;
fctm = $0.3 \times f_{ck}^{\,2/3}$ = $0.3 \times 25^{2/3}$ = 2.5649 N/mm^2 (Table 3.1 EC2)

A_{Smin} = 0.26 × $\frac{\text{fctm}}{\text{Fyk}}$ × b × d = 0.26 × $\frac{2.5649}{460}$ × 1000 × 244= 353.733 mm^2
Check if A_{Smin} < 0.0013 × b × d (317.2 mm^2)

Provide **Y12 @ 200mm c/c (AS_{prov} = 565 mm^2/m) each way**

Shear at the column face

Ultimate Load on footing from column = 165.48 kN
Design shear stress at the column perimeter $v_{Ed} = \beta \frac{V_{Ed}}{u_0 d}$

β = 1+ $1.8\sqrt{\left(\frac{6.88}{230}\right)^2 + \left(\frac{8.42}{230}\right)^2}$ = 1.085

Where u_o is the column perimeter and d is the effective depth

$v_{Ed} = \beta \frac{V_{Ed}}{u_0 d} = \frac{1.085 \times 165.48 \times 10^3}{4(230) \times 244}$ = 0.799 N/mm^2

$V_{Rd,max} = 0.5 v f_{cd}$

$v = 0.6\left[1 - \left(\frac{f_{ck}}{250}\right)\right] = 0.6\left[1 - \left(\frac{25}{250}\right)\right] = 0.54$ N/mm^2

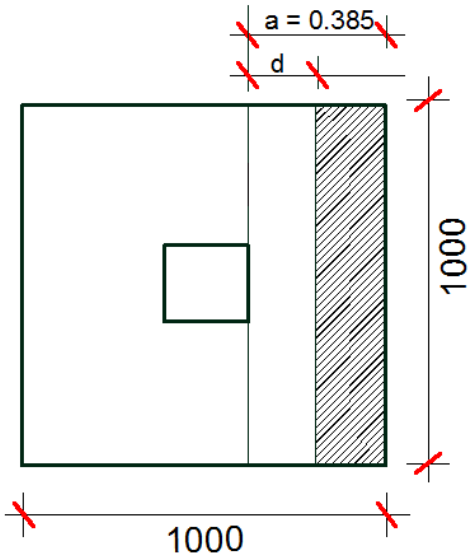
$f_{cd} = \frac{\alpha_{cc} f_{ck}}{\gamma_c} = \frac{0.85 \times 25}{1.5} = 14.167$ N/mm^2

$v_{Rd,max} = 0.5 \times 0.54 \times 14.167 = 3.825$ N/mm^2

$v_{Ed} < v_{Rd,max}$. This is very ok

Transverse shear at 'd' from the face of column

Shaded area = a – d = 0.385 – 0.244 = 0.141m



Area of shaded area = (1.0m × 0.141m) = 0.141 m^2

Therefore, $\Delta V_{Ed} = \frac{173.840 + 167.403}{2} \times 0.141$ m^2 = 24.057 KN

$v_{Ed} = \frac{V_{Ed}}{bd} = \frac{24.057 \times 10^3}{1000 \times 244} = 0.0985$ N/mm^2

$V_{Rd,c} = [C_{Rd,c}.k.\ (100\rho_1 f_{ck})^{\frac{1}{3}} + k_1.\sigma_{cp}] \times (2d/a) \geq (V_{min} + k_1.\sigma_{cp})\ bw.d$

$C_{Rd,c} = 0.18/\gamma_c = 0.18/1.5 = 0.12$

$k = 1+\sqrt{\frac{200}{d}} = 1+\sqrt{\frac{200}{244}} = 1.905 > 2.0$, therefore, k = 1.905

$V_{min} = 0.035 k^{\frac{3}{2}} f_{ck}^{\frac{1}{2}}$ = Vmin = $0.035 \times (1.905)^{\frac{3}{2}} \times (25)^{\frac{1}{2}} = 0.460$ N/mm^2

$\rho_1 = \frac{As}{bd} = \frac{565}{1000 \times 244} = 0.0023155 < 0.02$;

$V_{Rd,c} = [0.12 \times 1.905(100 \times 0.002316 \times 25)^{\frac{1}{3}}] = 0.4105$ N/mm$^2 \times (2d/a) > V_{min}$

But in this case, d = a

Hence, $V_{Rd,c} = 2 \times 0.4105 = 0.8209$ N/mm^2

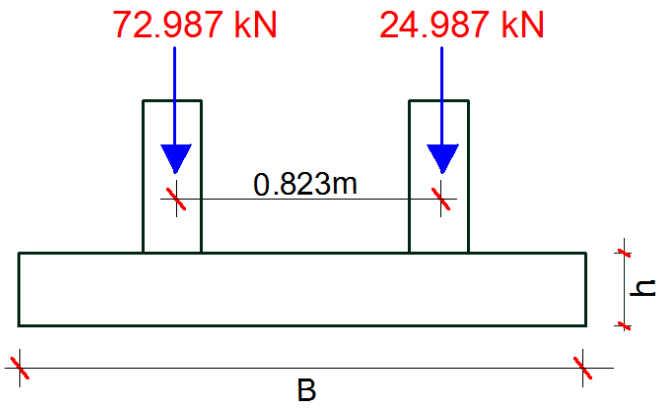
Since $V_{Rd,c}$ (0.8209) > V_{Ed} (0.0985 N/mm^2), No shear reinforcement is required.

Punching Shear at 2d

Punching shear lies outside the footing dimensions. No further check required.

8.5.4 DESIGN OF COMBINED FOOTING (FOOTING TYPE 4)

A little consideration will show that the minimum dimension of footing adopted (1000mm) will overlap at columns at column F4 and G4. Therefore, we are going to provide combined footing at those locations. You can also notice that the eccentricities we have considered have not really influenced our design if we had designed it as axially load (kindly verify). So we are going to design this axially loaded combined footing.



Total Column Axial Load (ULS) = 97.974 KN

Converting the load to serviceability limit state load

Statistical analysis has shown that a conversion factor of 1.36 is usually very adequate for residential buildings.

V = $\frac{V_{Ed}}{1.36}$ = $\frac{97.974}{1.36}$ = 72.039 KN

Assuming 10% of the service load to account for the self weight of the footing;

Sw = 0.1 × 72.039 = 7.2039 KN

Assuming 15 KN/m^2 reduction in bearing capacity to account for surcharge load. Hence allowable bearing capacity = 135 KN/m^2

Area of footing required = $\frac{V + S_w}{135}$ = $\frac{72.039 + 7.2039}{135}$ = 0.5869 m^2

For a footing length of 1.5m, provide a width of 0.391m

Provide a rectangular footing of plan 1500mm x 800mm (Area provided = 1.3 m^2)

Assume a trial depth of 300mm

Trying to locate the centroid of the footing (the resultant of the applied load)

Taking moment about column G4

$\sum M_{G4}$ = 0

Resultant of the column loads = 72.987 + 24.987 = 97.971 KN

(72.984 × 0.283) – 97.971 (0.823 – x) = 0

$20.629 - 80.630 + 97.971x = 0$

$x = \frac{60}{97.971}$ = 0.612m (from column F4)

To estimate the footing projection from the lighter column (L3), check;

$L_3 = 0.5L - \frac{F_1 L_2}{F_1 + F_2}$

Where;

L_3 is the footing projection from the lighter column

L_2 is the centre to centre distance of the two columns

L is the total length of the two columns

$L_3 = 0.5(1.5) - \frac{72.987 \times 0.823}{72.987 + 24.987}$ = 0.136m

Therefore, $L_1 = L - L_2 - L_3 = 1.5 - 0.823 - 0.136 = 0.54$m

As a verification,

Let the cantilever projection of footing at column F4 be L_1,

$L_1 + 0.612 = L/2 = 1.5/2 = 0.75$

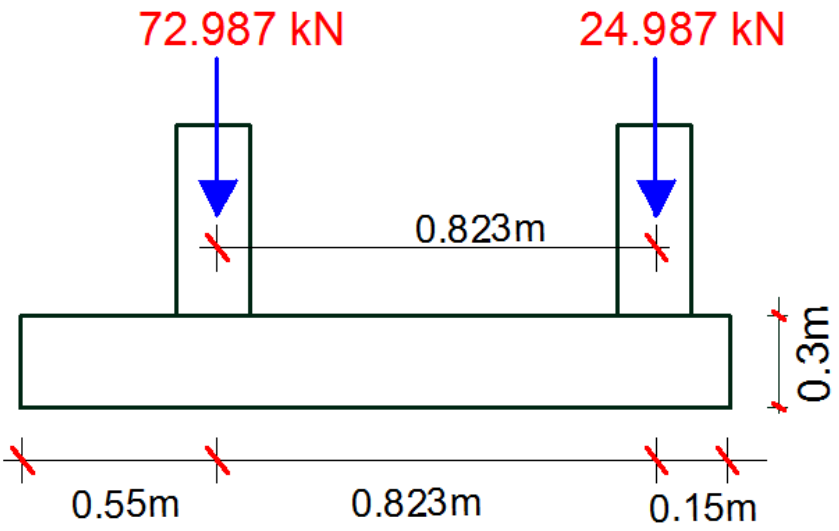
Therefore L_1 = 0.138m (say 150mm)

Let the cantilever projection of footing at column F4 be L_3,

$L_3 + (0.823 - 0.612) = L/2 = 1.5/2 = 0.75$

Therefore L_3 = 0.539m (say 550mm)

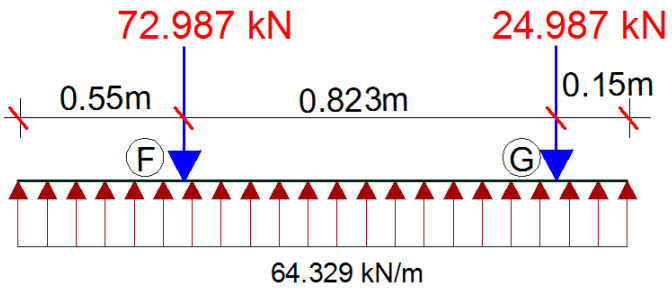
The final disposition of the column base is shown below;



Ultimate earth pressure (q) = $\frac{72.987 + 24.987}{1.523 \times 0.8}$ = 80.412 KN/m^2

Structural Analysis

For 0.8m width of the footing = 0.8 × 80.412 = 64.329 KN/m



$M_f = \frac{qL_1^2}{2} = \frac{64.329 \times 0.55^2}{2}$ = 9.729 KNm

Maximum span moment

Cutting a section x at the mid-span (x starting from the left hand side)

$Mx = \frac{64.329x^2}{2} - 72.987(x - 0.55)$

$Mx = 32.164x^2 - 72.987x + 40.142$

$\frac{\partial M_x}{\partial x}$ = Qx (shear) = $64.329x - 72.987$

Knowing that maximum moment occurs at the point of zero shear;

$\frac{\partial M_x}{\partial x}$ = Qx (shear) = $64.329x - 72.987 = 0$

On solving, x = 1.1345 m (point of maximum moment)

Plugging back x into the moment equation, we can obtain the maximum mid-span moment;

$Mmax = 32.164(1.1345)^2 - 72.987(1.1345) + 40.142 = -1.287$ KNm

$M_G = \frac{qL_3^2}{2} = \frac{64.329 \times 0.15^2}{2}$ = 0.723 KNm

Shear

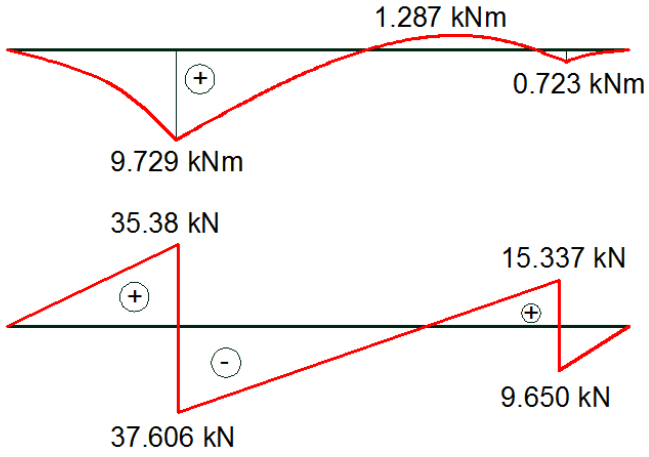
At x = 0, Q_A = 0

At x = 0.55m, Q_F^L = 64.329 × 0.55 = 35.38 KN

At x = 0.55m, Q_F^R = (64.329 × 0.55) – 72.987 = –37.606 KN

At x = 1.373m, Q_G^L = (64.329 × 1.373) – 72.987 = 15.337 KN

At x = 1.373m, Q_G^L = (64.329 × 1.373) – 72.987 – 24.987 = –9.650 KN



Structural Design

We are expected to obtain design moment at the column face (using the heavily loaded column side)

a = L_1 – (C_1/2) = 0.55 – (230/2) = 0.435m (where C_1 is the column dimension)

$M_{Ed} = \frac{qa^2}{2}$

$M_{Ed} = \frac{64.329 \times 0.435^2}{2}$ = 6.086 KN.m

Effective depth

Concrete cover = 50mm
AssumingY12mm bars,
d = 300 – 50 – 6 = 244mm

$k = \frac{M_{Ed}}{f_{ck}\, bd^2} = \frac{6.086 \times 10^6}{25 \times 800 \times 244^2}$ = 0.00511(designing per metre strip)

Since k < 0.167 No compression reinforcement required

$z = d\left[0.5 + \sqrt{(0.25 - 0.882K)}\right]$ = $z = d\left[0.5 + \sqrt{(0.25 - 0.882(0.00511)}\right]$ = 0.95d

$A_{s1} = \frac{M_{Ed}}{0.87 f_{yk} z} = \frac{6.086 \times 10^6}{0.87 \times 460 \times 0.95 \times 244}$ = 65.6 mm^2

To calculate the minimum area of steel required;
fctm = $0.3 \times f_{ck}^{\,2/3}$ = $0.3 \times 25^{2/3}$ = 2.5649 N/mm^2 (Table 3.1 EC2)

A_{Smin} = $0.26 \times \frac{\text{fctm}}{\text{Fyk}} \times b \times d = 0.26 \times \frac{2.5649}{460} \times 800 \times 244$= 282.98 mm^2
Check if A_{Smin} < 0.0013 × b × d (317.2 mm^2)

Provide **Y12 @ 200mm c/c (AS_{prov} = 565 mm^2/m) each way**

A little consideration will show that Y12@250mm will be sufficient for Top and Bottom reinforcements in the longitudinal and transverse direction.

Shear at the column face

Ultimate Load on footing from column = 72.987 kN
Design shear stress at the column perimeter $v_{Ed} = \beta \frac{V_{Ed}}{u_0 d}$
perimeter

β = 1.0 (no eccentricity considered)

$v_{Ed} = \beta \frac{V_{Ed}}{u_0 d} = \frac{1.0 \times 72.987 \times 10^3}{4(230) \times 244}$ = 0.325 N/mm^2

$V_{Rd,max} = 0.5 v f_{cd}$

$v = 0.6\left[1 - \left(\frac{f_{ck}}{250}\right)\right] = 0.6\left[1 - \left(\frac{25}{250}\right)\right]$ = 0.54 N/mm^2

$f_{cd} = \frac{\alpha_{cc} f_{ck}}{\gamma_c} = \frac{0.85 \times 25}{1.5}$ = 14.167 N/mm^2

$v_{Rd,max}$ = 0.5 × 0.54 × 14.167 = 3.825 N/mm^2
v_{Ed} < $v_{Rd,max.}$ This is very ok

Transverse shear at 'd' from the face of column

Critical Length = a – d = 0.435 – 0.244 = 0.191m

Critical area = (0.8m × 0.191m) = 0.1528 m^2

Therefore, ΔV_{Ed} = 64.329 × 0.1528 m^2 = 9.829 KN

$v_{Ed} = \frac{V_{Ed}}{bd} = \frac{9.829 \times 10^3}{1000 \times 244}$ = 0.04028 N/mm^2

$V_{Rd,c} = [C_{Rd,c}.k. (100\rho_1 f_{ck})^{\frac{1}{3}} + k_1.\sigma_{cp}] \times (2d/a) \geq (V_{min} + k_1.\sigma_{cp})\ bw.d$

$C_{Rd,c} = 0.18/\gamma_c = 0.18/1.5 = 0.12$

$k = 1+\sqrt{\frac{200}{d}} = 1+\sqrt{\frac{200}{244}}$ = 1.905 > 2.0, therefore, k = 1.905

$V_{min} = 0.035 k^{\frac{3}{2}} f_{ck}^{\frac{1}{2}}$ = Vmin = $0.035 \times (1.905)^{\frac{3}{2}} \times (25)^{\frac{1}{2}}$ = 0.460 N/mm^2

$\rho_1 = \frac{As}{bd} = \frac{565}{1000 \times 244}$ = 0.0023155 < 0.02;

$V_{Rd,c} = [0.12 \times 1.905(100 \times 0.002316 \times 25)^{\frac{1}{3}}]$ = 0.4105 N/mm^2 × *(2d/a)* > V_{min}
Because in this case, d = a
Hence, $V_{Rd,c}$ = 2 × 0.4105 = 0.8209 N/mm^2
Since $V_{Rd,c}$ (0.8209) > V_{Ed} (0.04028 N/mm^2), No shear reinforcement is required.

. **Punching Shear at 2d**

Punching shear lies outside the footing dimensions. No further check required.

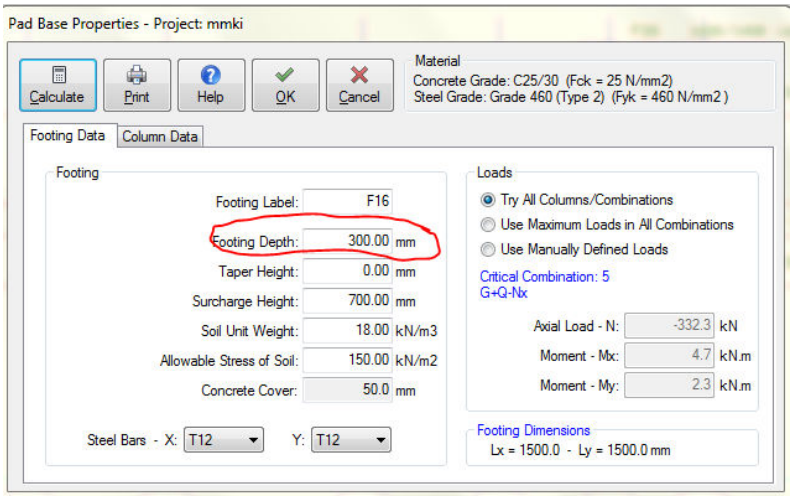
8.6 Orion Foundation Design Result

The summary of the foundation design result is as shown below;

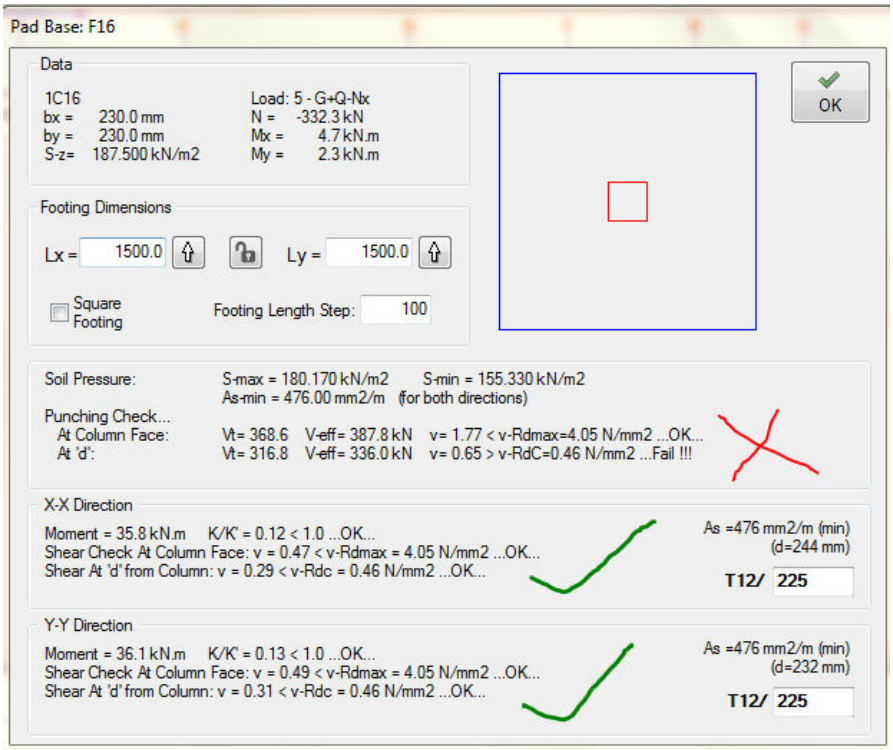
Column	N_{Ed} (KN)	Design Moment (KN.m)		Soil Pressure (KN/m^2)		Area of footing	Depth (mm)	Rebar *ϕ/Spacing*
		Mx	My	P_{max}	P_{min}			
A1	125.2	2.6	3.3	180.377	109.989	1000 x 1000	300	Y12/225
A3	202.9	1.0	3.9	177.893	143.904	1200 x 1200	300	Y12/225
A5	156.6	0.6	2.9	164.940	133.853	1100 x 1100	300	Y12/225
A7	134.7	1.1	3.0	179.196	130.210	1000 x 1000	300	Y12/225
B5	282.7	0.9	1.7	169.985	158.530	1400 x 1400	400	Y12/150
B7	258.2	2.0	0.1	178.654	166.922	1300 x 1300	300	Y12/225
B8	113.8	2.2	4.2	171.867	95.742	1000 x 1000	300	Y12/225
C1	166.2	3.6	1.2	179.158	135.541	1100 x 1100	300	Y12/225
C3	224.0	2.8	1.5	164.383	140.751	1300 x 1300	300	Y12/225
D1	101.1	1.8	0.3	133.587	108.482	1000 x 1000	300	Y12/225
D2	174.7	0.2	2.6	177.171	151.587	1100 x 1100	300	Y12/225
D4	178.3	1.2	2.4	183.734	150.892	1100 x 1100	300	Y12/225
E5	332.3	4.7	2.3	182.611	157.771	1500 x 1500	400	Y12/150
E8	229.3	9.3	3.8	168.023	110.876	1400 x 1400	400	Y12/150
F1	61.1	1.4	1.1	96.436	65.722	1000 x 1000	300	Y12/225
F2	148.7	1.9	2.2	161.060	124.628	1100 x 1000	300	Y12/225
F4 G4	121.2	2.2	32	185.734	9.775	1000 x 1500	300	Y12/225
G6	61.7	0.7	1.4	94.225	69.058	1000 x 1000	300	Y12/225
G8	22.8	0.9	0.2	49.647	35.824	1000 x 1000	300	Y12/225

8.6.1 Sample Footing Design (Column E5)

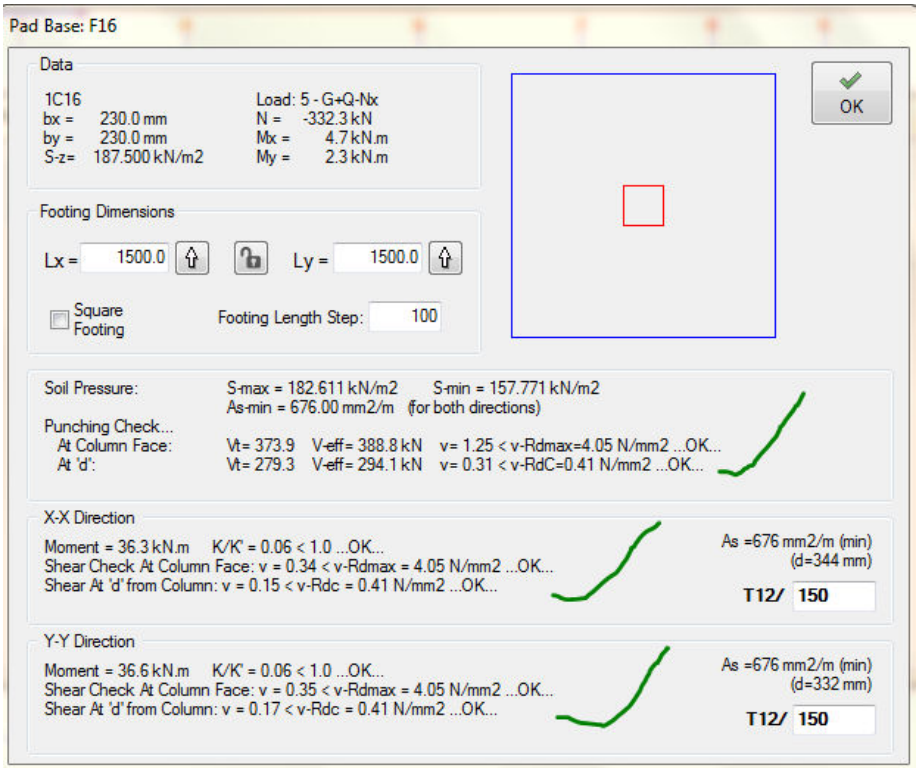
Inputting an initial trial depth of 300mm



The design result is as shown below;



Increasing the footing depth to 400mm, the result below is obtained;



CHAPTER 9

STRUCTURAL DETAILING

9.1 General Rules

9.1.1 Spacing of bars (clause 8.2 EC2)

The spacing of bars shall be such that the concrete can be placed and compacted satisfactorily for the development of adequate bond.

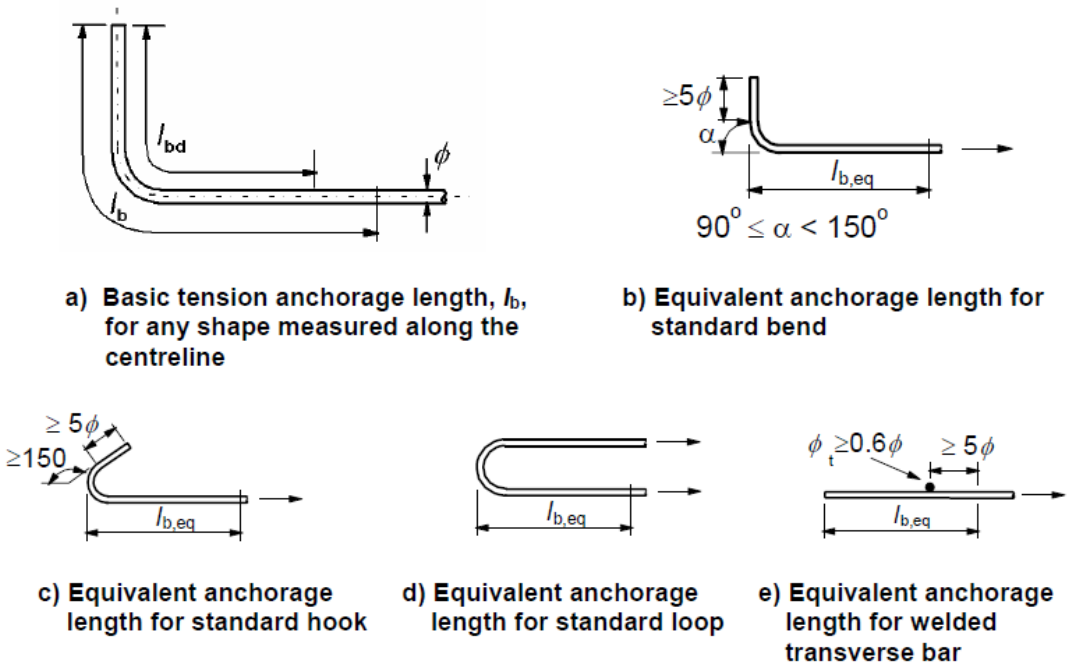
The clear distance (horizontal and vertical) between bars should be $\geq \max(\phi,\ d_g + 5\text{mm},\ 20\text{mm})$

Where;
d_g is the maximum size if aggregate
ϕ is the diameter of the bars

.
Also, where bars are positioned in separate horizontal layers, the bars in each layer should be located vertically above each other. There should be sufficient space between the resulting columns of bars to allow access for vibrators and good compaction of the concrete. However, lapped bars may be allowed to touch one another within the lap length

9.1.2 Anchorage of reinforcements

Reinforcing bars should be well anchored so that the bond forces are safely transmitted to the concrete avoiding longitudinal cracking or spalling. Transverse reinforcement shall be provided if necessary. Types of anchorage are shown in Figure 9.1 below (Figure 8.1 EC2).



a) Basic tension anchorage length, l_b, for any shape measured along the centreline

b) Equivalent anchorage length for standard bend

c) Equivalent anchorage length for standard hook

d) Equivalent anchorage length for standard loop

e) Equivalent anchorage length for welded transverse bar

Figure 9.1: Different methods of anchorage (Fig 8.1 EN 1992-1-1:2004)

For bent bars, the basic tension anchorage length is measured along the centreline of the bar from the section in question to the end of the bar, where:

$$l_{bd} = \alpha_1\alpha_2\alpha_3\alpha_4\alpha_5 l_{b,req} \geq l_{b,min} \text{ ---------- (9.1)}$$

where;

$l_{b,min}$ is the minimum anchorage length taken as follows:
In tension, the greatest of $0.3l_{b,rqd}$ or 10 ϕ or 100mm
In compression, the greatest of $0.6l_{b,rqd}$ or 10 ϕ or 100mm

$l_{b,rqd}$ is the basic anchorage length given by, $l_{b,rqd} = (\phi/4)\ \sigma_{sd}/f_{bd}$

Where;

σ_{sd} = The design strength in the bar (take $0.87f_{yk}$)

f_{bd} = The design ultimate bond stress (for ribbed bars = $2.25\eta_1\eta_2 f_{ctd}$)

f_{ctd} = Design concrete tensile strength $f_{ctd} = 0.21 f_{ck}^{\left(\frac{2}{3}\right)}$ for $f_{ck} \leq 50$ N/mm^2

η_1 is a coefficient related to the quality of the bond condition and the position of the bar during concreting

η_1 = 1.0 when 'good' conditions are obtained and
η_1 = 0.7 for all other cases and for bars in structural elements built with slip-forms, unless it can be shown that 'good' bond conditions exist

η_2 is related to the bar diameter:
η_2 = 1.0 for φ ≤ 32 mm
η_2 = (132 - φ)/100 for φ > 32 mm

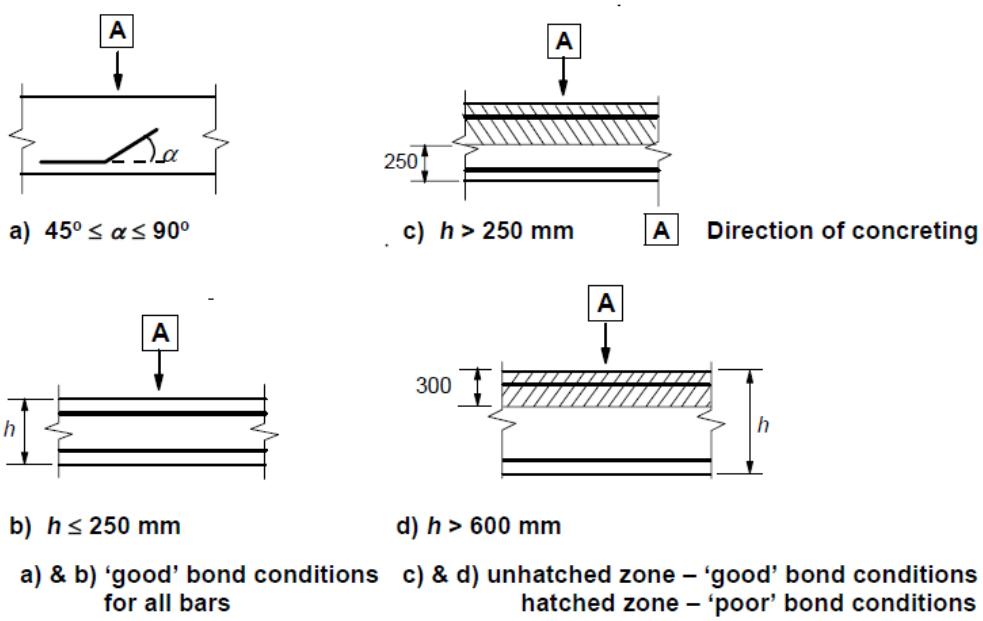


Figure 9.2: Description of bond conditions (Fig 8.2 EN 1992-1-1:2004)

α_1 is for the effect of the form of the bars assuming adequate cover (see Figure 8.1).
α_2 is for the effect of concrete minimum cover (see Figure 8.3)
α_3 is for the effect of confinement by transverse reinforcement

α_4 is for the influence of one or more welded transverse bars ($\varphi t > 0{,}6\varphi$) along the design anchorage length l_{bd}
α_5 is for the effect of the pressure transverse to the plane of splitting along the design anchorage length

Table 9.0: Values of alpha coefficients (Table 8.2 EN 1992-1-1:2004)

Influencing factor	Type of anchorage	Reinforcement bar	
		In tension	In compression
Shape of bars	Straight	$\alpha_1 = 1{,}0$	$\alpha_1 = 1{,}0$
	Other than straight (see Figure 8.1 (b), (c) and (d)	$\alpha_1 = 0{,}7$ if $c_d > 3\phi$ otherwise $\alpha_1 = 1{,}0$ (see Figure 8.3 for values of c_d)	$\alpha_1 = 1{,}0$
Concrete cover	Straight	$\alpha_2 = 1 - 0{,}15\,(c_d - \phi)/\phi$ $\geq 0{,}7$ $\leq 1{,}0$	$\alpha_2 = 1{,}0$
	Other than straight (see Figure 8.1 (b), (c) and (d))	$\alpha_2 = 1 - 0{,}15\,(c_d - 3\phi)/\phi$ $\geq 0{,}7$ $\leq 1{,}0$ (see Figure 8.3 for values of c_d)	$\alpha_2 = 1{,}0$
Confinement by transverse reinforcement not welded to main reinforcement	All types	$\alpha_3 = 1 - K\lambda$ $\geq 0{,}7$ $\leq 1{,}0$	$\alpha_3 = 1{,}0$
Confinement by welded transverse reinforcement*	All types, position and size as specified in Figure 8.1 (e)	$\alpha_4 = 0{,}7$	$\alpha_4 = 0{,}7$
Confinement by transverse pressure	All types	$\alpha_5 = 1 - 0{,}04p$ $\geq 0{,}7$ $\leq 1{,}0$	-

where:
λ = $(\Sigma A_{st} - \Sigma A_{st,min})/ A_s$
ΣA_{st} cross-sectional area of the transverse reinforcement along the design anchorage length l_{bd}
$\Sigma A_{st,min}$ cross-sectional area of the minimum transverse reinforcement = 0,25 A_s for beams and 0 for slabs
A_s area of a single anchored bar with maximum bar diameter
K values shown in Figure 8.4
p transverse pressure [MPa] at ultimate limit state along l_{bd}

* See also 8.6: For direct supports l_{bd} may be taken less than $l_{b,min}$ provided that there is at least one transverse wire welded within the support. This should be at least 15 mm from the face of the support.

Example

Calculate the design tension anchorage length of Y16mm bar (f_{yk} = 460 N/mm^2, concrete cover = 35mm, Concrete cylinder strength f_{ck} = 25 N/mm^2) for;

a. When it is a straight bar
b. When it is bent into any other shape

Assume good bond conditions

Solution

$$l_{bd} = \alpha_1\alpha_2\alpha_3\alpha_4\alpha_5 l_{b,req} \geq l_{b,min}$$

$l_{b,rqd} = (\phi/4)\,\sigma_{sd}/f_{bd}$

$f_{bd} = 2.25\eta_1\eta_2 f_{ctd}$

η_1 = 1.0 'Good' bond conditions
η_2 = 1.0 bar size ≤ 32

$$f_{ctd} = \frac{\alpha_{ct} f_{ctk\ 0.05}}{\gamma_c} \text{ ---------- (9.2)}$$

where;
$f_{ctk\ 0.05}$ = characteristic tensile strength of concrete at 28 days = 1.8 N/mm^2 (Table 3.1 EC2)
γ_c = partial (safety) factor for concrete = 1.5
α_{ct} = coefficient taking account of long-term effects on the tensile strength, this is an NDP with a recommended value of 1.

$f_{ctd} = \frac{1.0 \times 1.8}{1.5}$ = 1.2 N/mm^2

f_{bd} = 2.25 × 1.0 × 1.0 × 1.2 = 2.7 N/mm^2

$l_{b,rqd} = (\phi/4)\ \sigma_{sd}/f_{bd}$

σ_{sd} = 0.87 × 460 = 400.2 N/mm^2

$l_{b,rqd} = (\phi/4)\ \sigma_{sd}/f_{bd} = \frac{\phi \times 400.2}{4 \times 2.7}$ = 37.05ϕ

Therefore;

$$l_{bd} = \alpha_1\alpha_2\alpha_3\alpha_4\alpha_5(37.05\phi)$$

(a) For straight bar
α_1 = 1.0
α_3 = 1.0 conservative value with K= 0
α_4 = 1.0 N/A
α_5 = 1.0 conservative value
α_2 = 1.0 – 0.15 (C_d – ϕ)/ ϕ
α_2 = 1.0 – 0.15 (35 – 16)/16 = 0.8218

$$l_{bd} = 0.8218 \times (37.05\phi) = 30.4\phi$$

(a) For other shape bar
α_1 = 1.0 bC_d = 35 is ≤ 3ϕ = 3 x 16 = 48
α_3 = 1.0 conservative value with K= 0
α_4 = 1.0 N/A
α_5 = 1.0 conservative value
α_2 = 1.0 – 0.15 (C_d – 3ϕ)/ ϕ ≤ 1.0
α_2 = 1.0 – 0.15 (35 – 48)/16 = 1.121 ≤ 1.0

$$l_{bd} = 1.0 \times (37.05\phi) = 37.05\phi$$

Compression anchorage ($\alpha_1 = \alpha_2 = \alpha_3 = \alpha_4 = \alpha_5 = 1.0$)
l_{bd} = 37.05ϕ
Anchorage for 'Poor' bond conditions = 'Good'/0.7
Lap length = anchorage length x α_6

9.1.3 Anchorage of links

The anchorage of links and shear reinforcement should normally be effected by means of bends and hooks, or by welded transverse reinforcement. A bar should be provided inside a hook or bend.

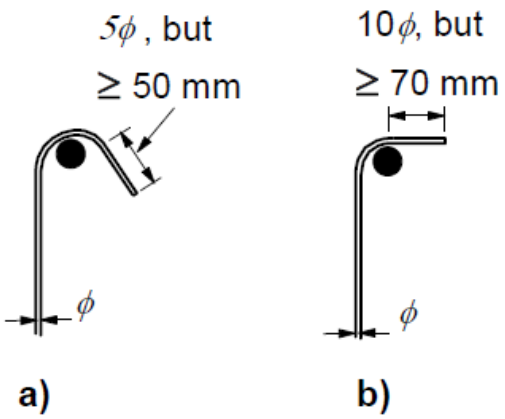


Figure 9.3: Anchorage of links (Fig 8.5 EN 1992-1-1:2004)

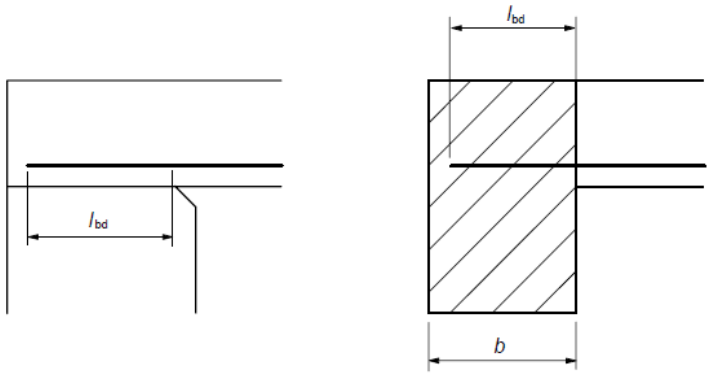
9.1.4 Anchorage of bottom reinforcement at an end support

The area of bottom reinforcement provided at supports with little or no end fixity assumed in design, should be at least 25% of the area of steel provided in the span. The tensile force to be anchored may be determined according to clause 6.2.3 (6) (members with shear reinforcement) including the contribution of the axial force if any, or according to the shift rule:

$$F_E = |V_{Ed}| \, . \, a_l \, / \, z + N_{Ed} \text{ ---------- (9.4)}$$

where N_{Ed} is the axial force, to be added to or subtracted from the tensile force

The anchorage length is l_{bd} is measured from the line of contact between beam and support. Transverse pressure may be taken into account for direct support.



a) Direct support: Beam supported by wall or column

b) Indirect support: Beam intersecting another supporting beam

Figure 9.7: Anchorage of bottom reinforcement end support (Fig 9.3 EN 1992-1-1:2004)

9.1.5 Anchorage of bottom reinforcement at intermediate supports

The anchorage length should not be less than 10ϕ (for straight bars) or not less than the diameter of the mandrel (for hooks and bends with bar diameters at least equal to 16 mm) or twice the diameter of the mandrel (in other cases).

The reinforcement required to resist possible positive moments (e.g. settlement of the support, explosion, etc.) should be specified in contract documents. This reinforcement should be continuous which may be achieved by means of lapped bars.

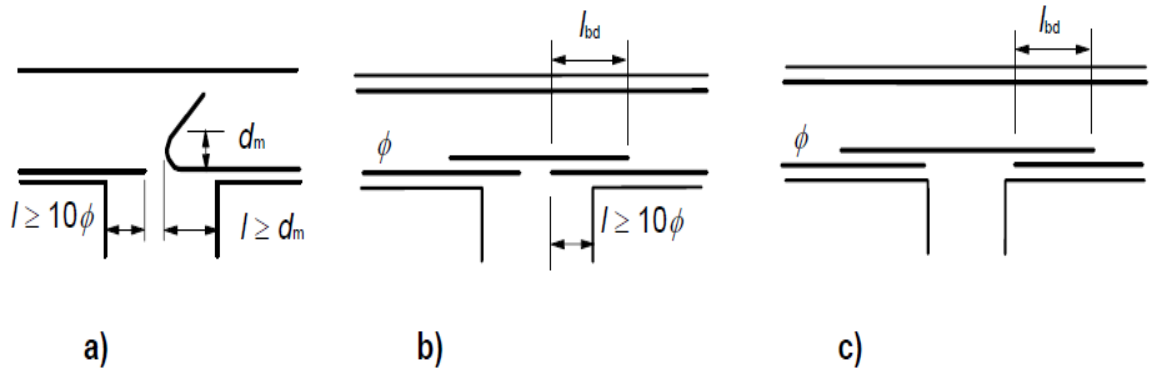


Figure 9.8: Anchorage of bottom reinforcement at intermediate support (Fig 9.4 EN 1992-1-1:2004)

9.1.6 Design Lap length

The design lap length of reinforcements is given by;

$$l_0 = \alpha_1\ \alpha_2\ \alpha_3\ \alpha_5\ \alpha_6\ l_{b,rqd} \geq l_{0,min} \text{ ---------- (9.3)}$$

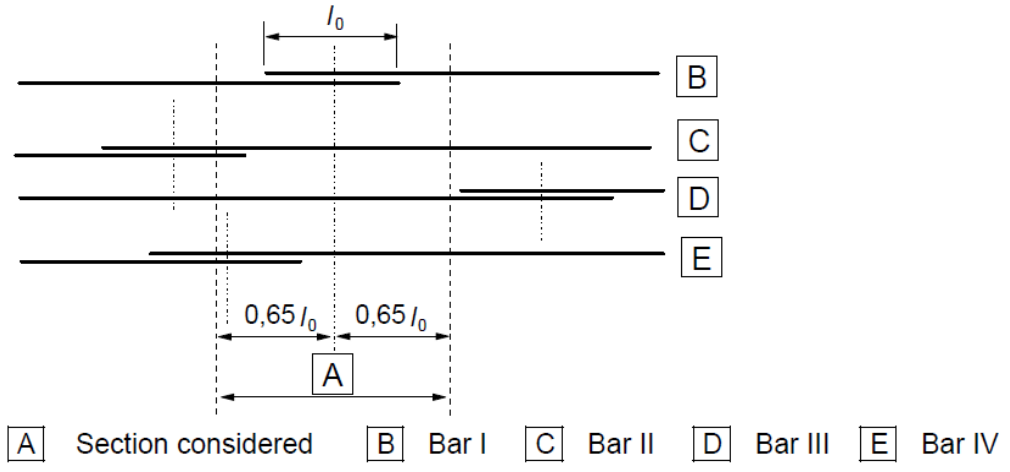
$l_{0,min} = \max\{0.3\alpha_6\ l_{b,rqd};\ 15\phi;\ 200\}$

$\alpha_6 = \sqrt{\frac{\rho_1}{25}}$ but between 1.0 and 1.5

where ρ_1 is the % of reinforcement lapped within $0.65l_0$ from the centre of the lap

Table 9.1: Values of α_6 (Table 8.3 EN 1992-1-1:2004)

Percentage of lapped bars relative to the total cross-section area	**< 25%**	**33%**	**50%**	**>50%**
α_6	1.0	1.15	1.4	1.5
Note: Intermediate values may be determined by interpolation.				



Example: Bars II and III are outside the section being considered: % = 50 and α_6 =1,4

Figure 9.4: Percentage of lapped bars in one lapped section (Fig 8.8 EN 1992-1-1:2004)

9.1.7 Arrangement of Laps

Laps between bars should normally be staggered and not located in areas of high moments /forces (e.g. plastic hinges). The arrangement of lapped bars should comply with Figure 8.7 of EC2 as shown in Figure 9.5 below:

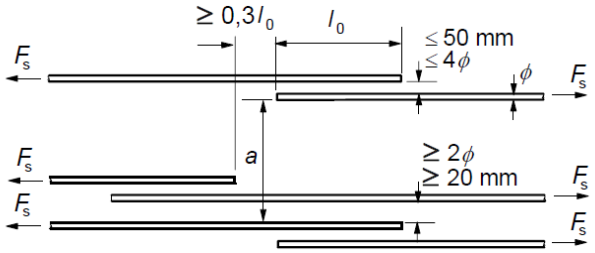


Figure 9.5: Adjacent Laps (Fig 8.7 EN 1992-1-1:2004)

- The clear distance between lapped bars should not be greater than 4φ or 50 mm, otherwise the lap length should be increased by a length equal to the clear space where it exceeds 4φ or 50 mm;
- the longitudinal distance between two adjacent laps should not be less than 0,3 times the lap length, l_0;
- In case of adjacent laps, the clear distance between adjacent bars should not be less than 2φ or 20 mm.

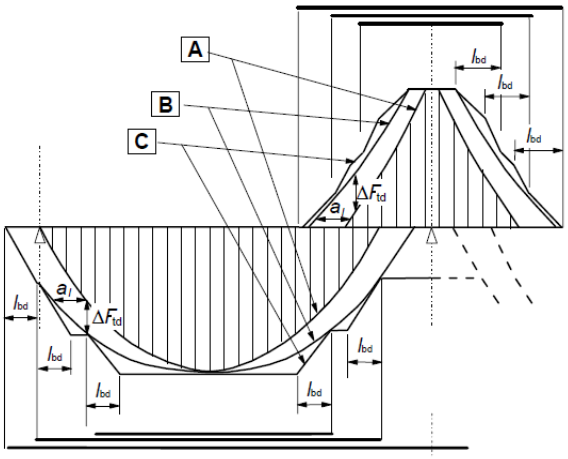
When the provisions comply with the 3 guides above, the permissible percentage of lapped bars in tension may be 100% where the bars are all in one layer. Where the bars are in several layers the percentage should be reduced to 50%. All bars in compression and secondary (distribution) reinforcement may be lapped in one section.

9.2 Curtailment

Sufficient reinforcement should be provided at all sections to resist the envelope of the acting tensile force, including the effect of inclined cracks in webs and flanges.

For members with shear reinforcement the additional tensile force, ΔF_{td}, should *be* calculated according to clause 6.2.3 (7). For members without shear reinforcement ΔF_{td} may *be* estimated by shifting the moment curve a distance $a_l = d$ according to clause 6.2.2 (5). *This "shift rule"* may also be used as an alternative for members with shear reinforcement, where:

$a_l = z\ (\cot\theta - \cot\alpha)/2 = 0.5\ z\ \cot\theta$ for vertical shear links
z= lever arm, θ = angle of compression strut
a_l = 1.125 d when cot θ = 2.5 and 0.45d when cot θ = 1



A - Envelope of $M_{Ed}/z + N_{Ed}$ B - acting tensile force F_s C - resisting tensile force F_{Rs}

Figure 9.6: Illustration of the curtailment of longitudinal reinforcement, taking into account the effect of inclined cracks and the resistance of reinforcement within anchorage lengths (Fig 8.7 EN 1992-1-1:2004)

9.3 Maximum and minimum areas of steel and spacing

The summary detailing guide for specifying minimum and maximum areas of steel, and maximum and minimum spacing of reinforcements is given in Table 9.2 below.

Table 9.2: Detailing guide

Specific Member Issue	Detailing guide
Beams	
Main bars ($A_{s,min}$)	$0.26f_{ctm}/f_{yk}bd \geq 0.0013bd$
$A_{s,max}$	$0.04bd$
Minimum spacing of main bars	d_g + 5mm, ϕ, 20mm
$A_{sw,min}$ (links)	$(0.08bSw\sqrt{f_{ck}})/f_{yk}$
Maximum spacing of links	$0.75d \leq 600mm$
Slabs	
Main bars ($A_{s,min}$)	$0.26f_{ctm}/f_{yk}bd \geq 0.0013bd$
$A_{s,max}$	$0.04bd$
Minimum area of secondary reinforcement in one way slabs	$A_{s,min} > 0.2A_{s,main}$
Minimum spacing of main bars	d_g + 5mm, ϕ, 20mm
Maximum spacing of main bars	$3h \leq 400mm$
Maximum spacing of secondary bars	$3.5h \leq 450mm$
Maximum spacing of main bars at places of maximum moment or places with concentrated load	$2h \leq 250mm$
Maximum spacing of secondary bars at places of maximum moment or places with concentrated load	$3h \leq 400mm$
Columns	
$A_{s,min}$	$\max\{0.10\ N_{Ed}/f_{yd}, 0.002bh\}$
$A_{s,max}$ outside laps	$0.04Ac$
$A_{s,max}$ at laps	$0.08Ac$
Minimum size of links	0.25ϕ or 6mm
Maximum spacing of links	$\min\{b, h, 20\phi, 400mm\}$ Reduce by a factor 0.6: i. In sections within a distance equal to the larger dimension of the column cross-section above or below a beam or slab; ii. Near lapped joints, if the maximum diameter of the longitudinal bars is greater than 14 mm. A minimum of 3 bars evenly placed in the lap length is required

9.4 Extremely simplified detailing guide for slabs

A simplified detailing guide for slabs at end support, continuous support, and cantilever ends is shown in figure 9.9 below.

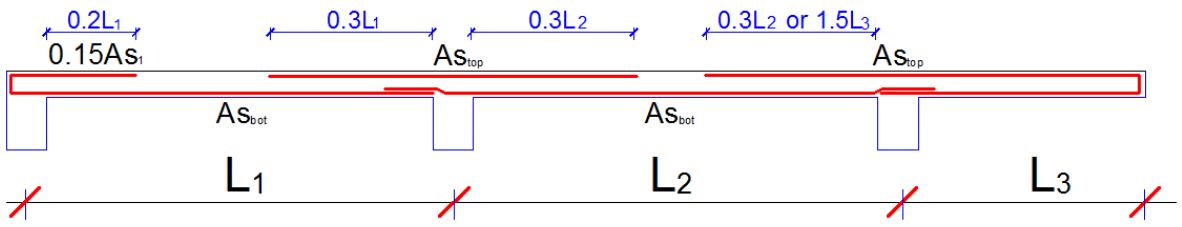


Figure 9.9: Simplified detailing guide for a slab section

9.5 Slab detailing

Detailing of panel 1

The analysis and design of the floor slab gave us the following results

Sagging areas = Y12@250mm c/c
Hogging areas = Y12@250mm c/c

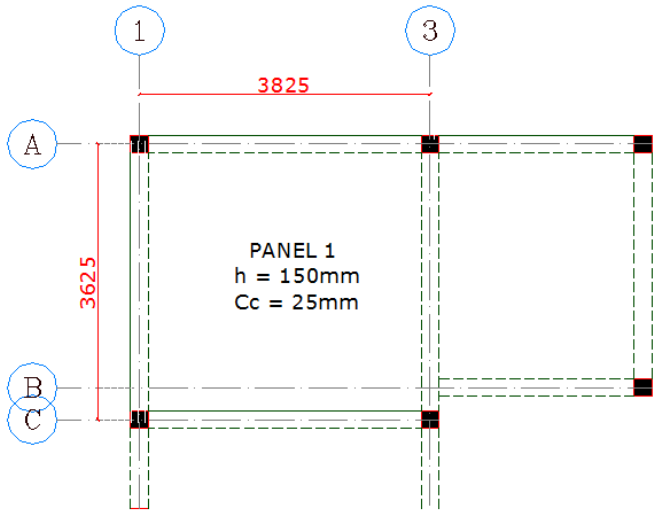


Figure 9.10: Panel 1

Note:
The bottom bars will span in short span, so let us start from there (bottom reinforcement)

Short Span
Length of span (effective length) = 3625mm

Give a spacing of 'd' from the face of the support from both sides to start laying of the reinforcements

Effective depth of slab (d) = 119mm

Hence = L_d = 3625 – 230 – 2(119) = 3157mm

Spacing recommended from design (S) = 250mm

Therefore number of bars needed = $\frac{L_d}{S} = \frac{3157}{250} + 1$ = 14 Nos

Long Span
Length of span (effective length) = 3825mm

Give a spacing of 'd' from the face of the support from both sides to start laying of the reinforcements

Effective depth of slab (d) = 119mm

Hence = L_d = 3825 – 230 – 2(119) = 3357mm

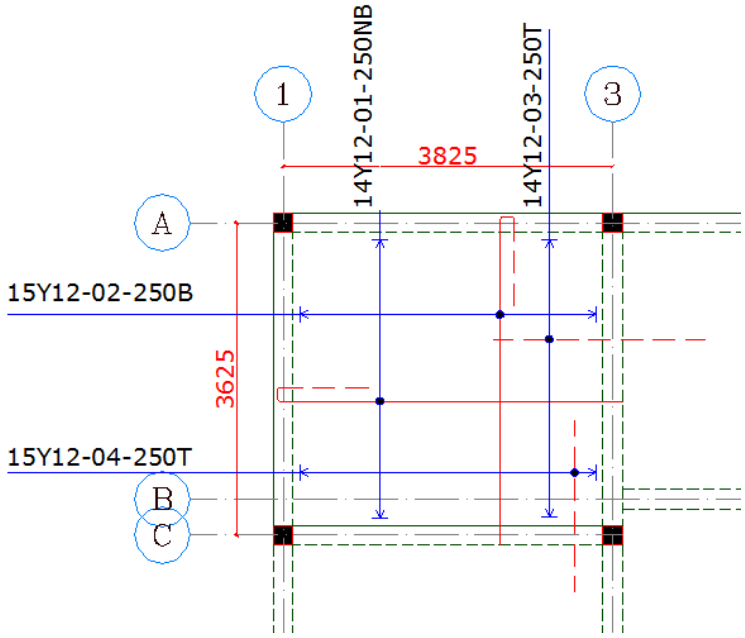
Spacing recommended from design (S) = 250mm

Therefore number of bars needed = $\frac{L_d}{S} = \frac{3357}{250} + 1$ = 15 Nos

A little consideration will show that this same procedure applies to top reinforcement.

We will be providing distribution bars of Y10@300mm (this is adequate, see table 9.2 above)

We normally use dotted lines to represent top reinforcement and continuous lines to represent bottom lines. When the drawing is too complex, top and bottom reinforcements can be separated entirely. We also allocate bar marks to the members based on the shape, size, and lengths of the bars detailed. No different bars should have the same mark, and bar marks should preferably start from 01. It is more desirable to be consistent with bar marks for each phase of construction being considered. For instance, bar marks should be consistent for foundations, and when you are done with that, you can start another bar mark afresh for columns. However, no two different bars should have the same bar mark on the same sheet, and on the same phase of construction.

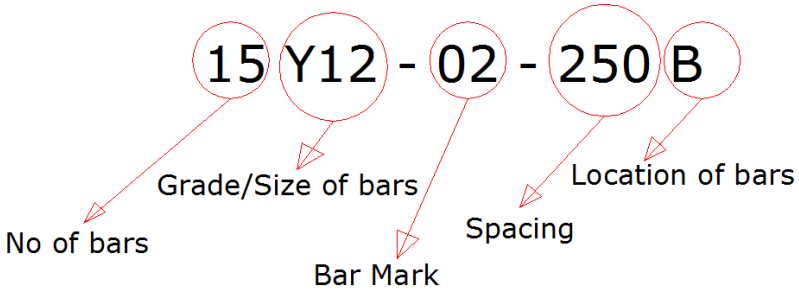


Realise how we returned the bottom bars to the top at the points where we assumed simple/discontinuous supports in our analysis (see Figure 9.11). These returned bars should extend at least 0.2L into the span. For the hogging reinforcements, they should extend at least 0.3L into the adjacent spans.

Also, at the figure by our left hand side, the top reinforcement distribution bars have not been included. We are going to show how the numbers needed will be calculated below.

Figure 9.11: Detailing of PANEL 1 showing main reinforcements only

The definition of the terms employed in detailing floor slabs is as shown below;



The location of bars should be adequately defined;

B – Outer bottom layer
NB – Inner bottom layer
T – Outer top layer
NT – Inner top Layer

Note that top reinforcements are supported by chairs or spacers (shape code 83), with a diameter of 10mm (for slabs <200mm thick) and spacing of 1000mm.

Calculation of number of distribution bars;

For bar mark 01

Returning length L_R = 0.2L = 0.2 × 3825 = 765mm

Number of bars required = $\frac{L_R - d}{spacing} = \frac{765-119}{300} + 1$ = 4 Nos of Y10@300mm spacing

For bar mark 02

Returning length L_R = 0.2L = 0.2 × 3625= 725mm

Number of bars required = $\frac{L_R - d}{spacing} = \frac{725 - 119}{300} + 1$ = 3 Nos of Y10@300mm spacing

For bar mark 03 (inside panel 1)

Returning length L_R = 0.3L = 0.3 × 3825= 1148 mm

Number of bars required = $\frac{L_R - d}{spacing} = \frac{1148 - 119}{300} + 1$ = 5 Nos of Y10@300mm spacing

For bar mark 04 (inside panel 1)

Returning length L_R = 0.3L = 0.3 × 3625= 1088 mm

Number of bars required = $\frac{L_R - d}{spacing} = \frac{1088 - 119}{300} + 1$ = 5 Nos of Y10@300mm spacing

The full final detailing of the panel is as shown below, with sections as shown in full

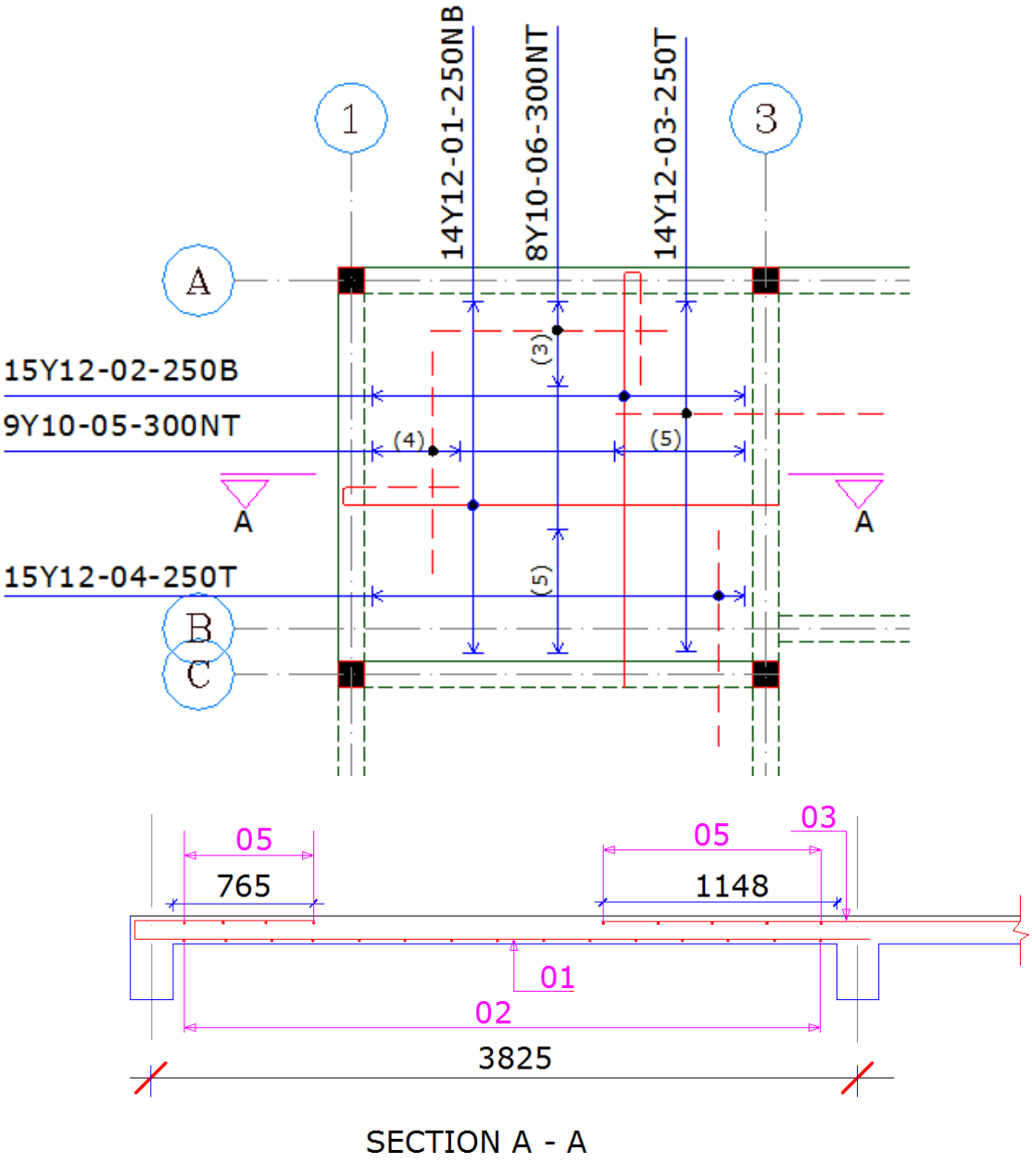


Figure 9.12: Detailing of PANEL 1 showing all reinforcements including sections

The full detailing of the entire floor slab is as shown below;

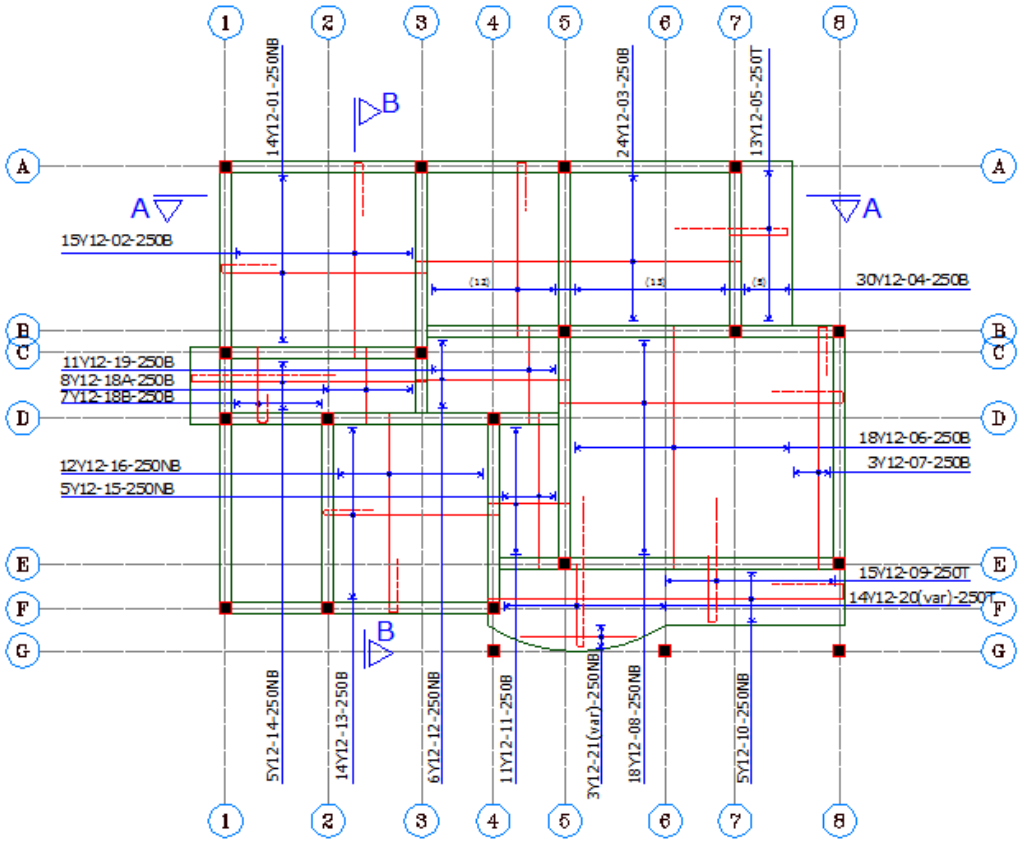


Figure 9.13: Bottom Reinforcement Detailing

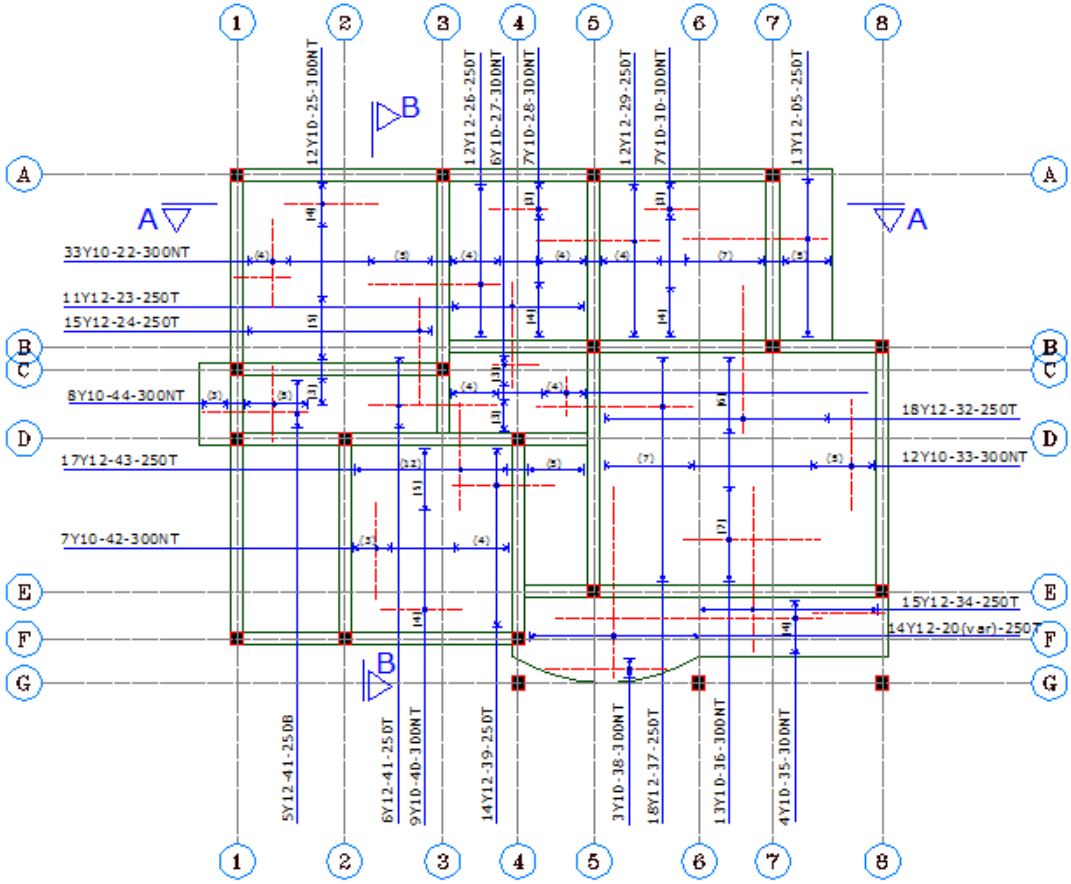


Figure 9.14: Top Reinforcement Detailing

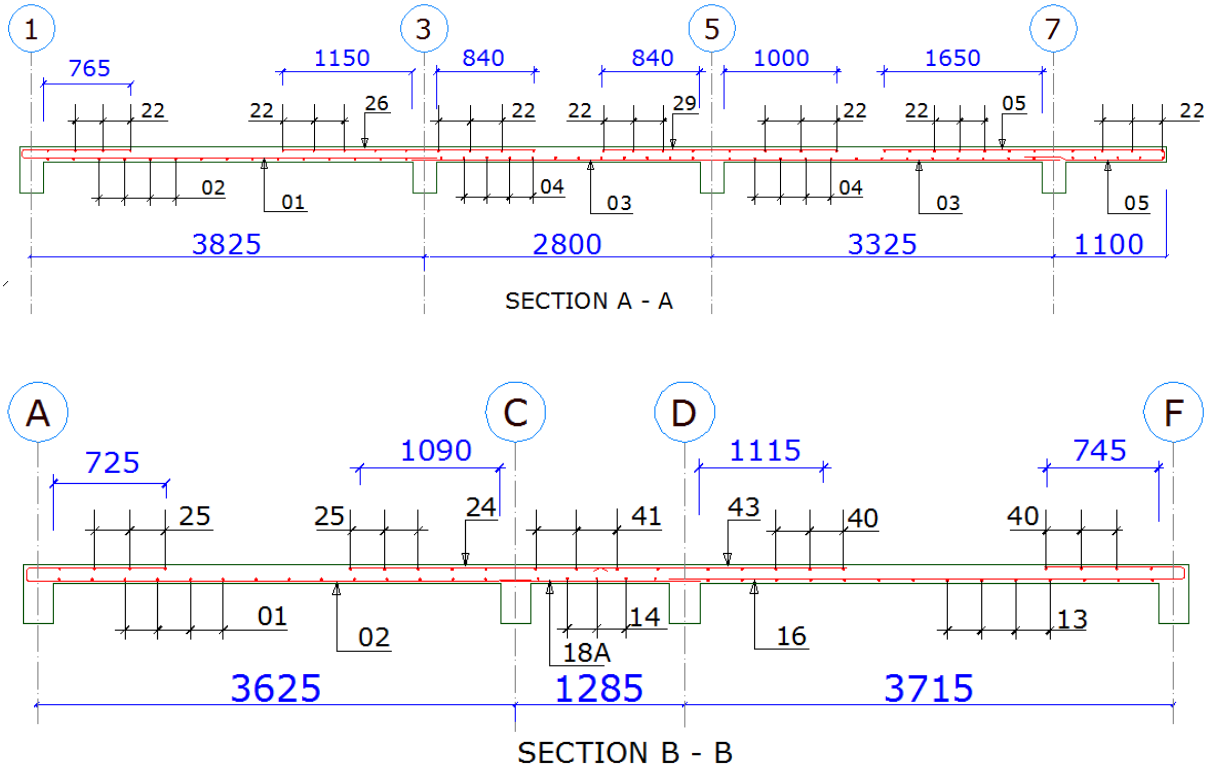


Figure 9.14: Sections of the floor slab

9.6 Beam Detailing

In the most simplified manner, the detailing guide for beams is as shown below. However, in the most practical terms for simple residential buildings with moderate spans, reinforcements are rarely curtailed (except at the supports hogging areas) for ease in construction, especially where manual iron bending is employed. Note that to satisfy anchorage requirements, take the bob length for beams as 15ϕ (15 x diameter of reinforcement).

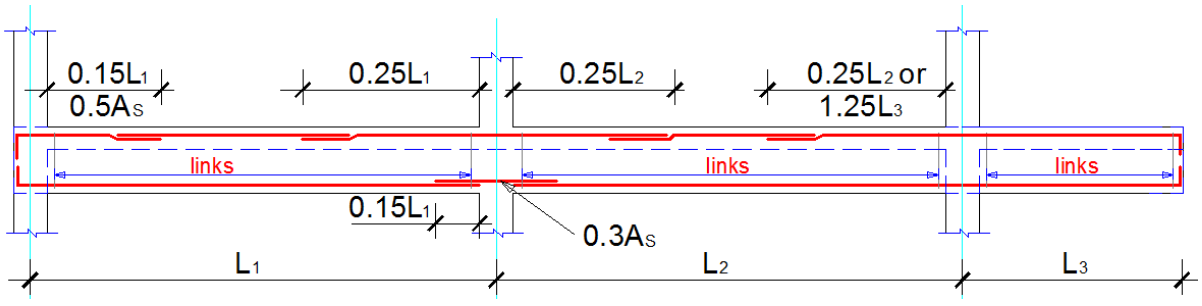


Figure 9.15: Simplified detailing guide for beams

Beam No 1 (see beam No 1 detailing)

In this member, the design recommended the provision of 2Y16 in all sagging and hogging areas. The cut length (full length) of reinforcements in Nigeria is 12m, even though sometimes they may be less than 12m (around 11.6m). The total length of the beam is 10.181m (out to out dimension). Now instead of cutting the reinforcements, why don't we provide a full length all through? However, when we want to use (say 12mm bar) as hanger bars, such curtailing may be necessary at the top. For the

links, I personally provide links at 50mm from the face of supports. However, standard detailing guide recommends providing links starting and ending at (0.5 x Spacing) from the face of support. Therefore the number of links required in a span may be given by (using 50mm start off from the face of support);

$$\text{No of links required} = \frac{Effective\ Span\ (mm) - Beam\ width(mm)\ -2(50)}{Spacing} + 1$$

The full beam detailing and detailing layout is shown below, showing the beam numbers.

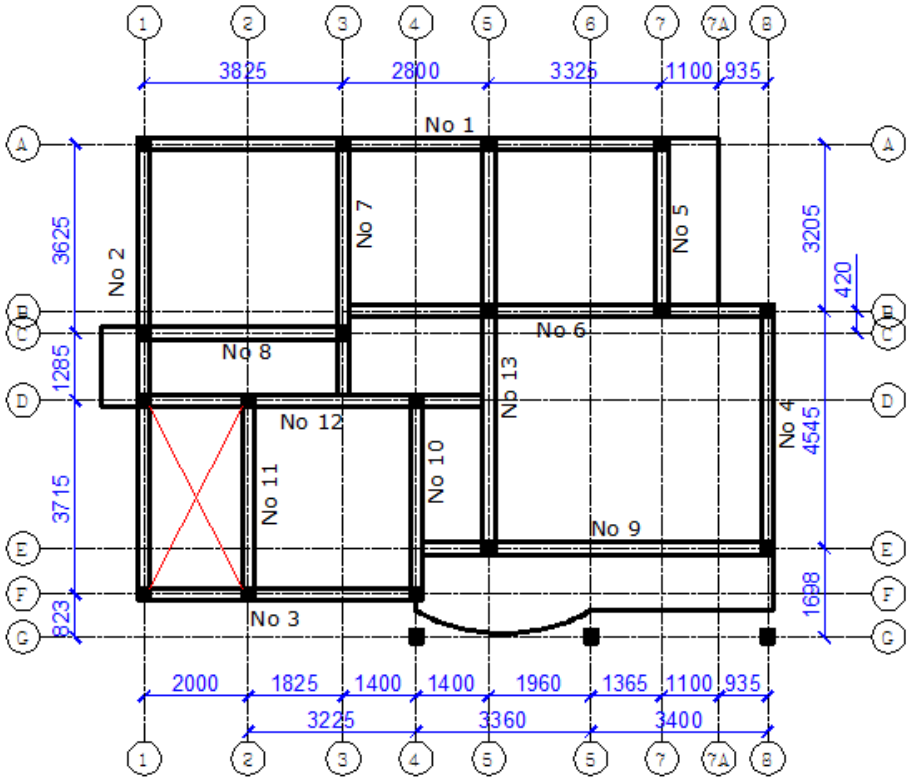
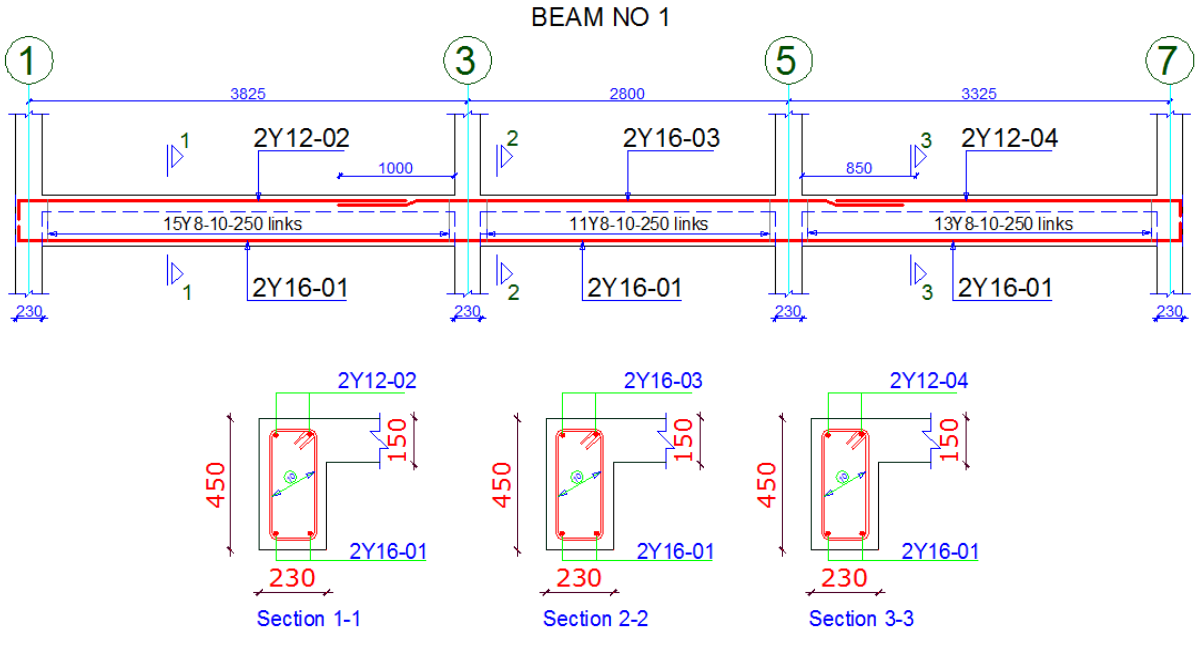
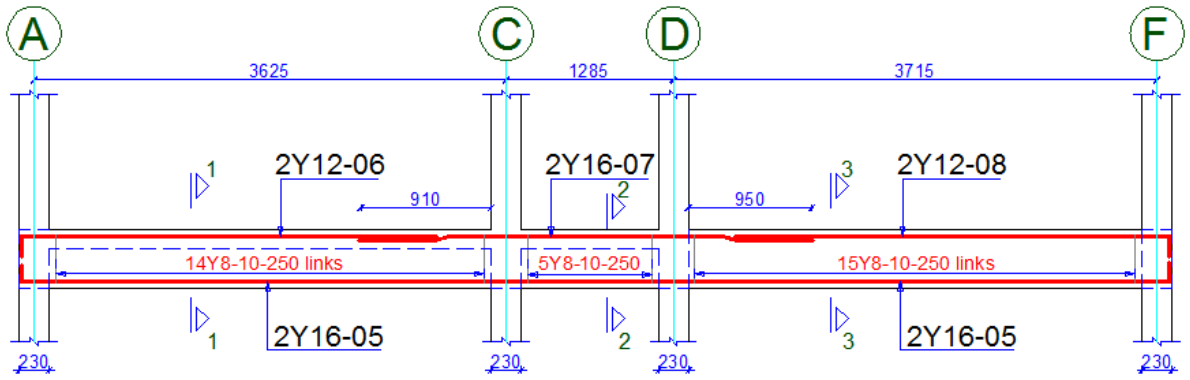
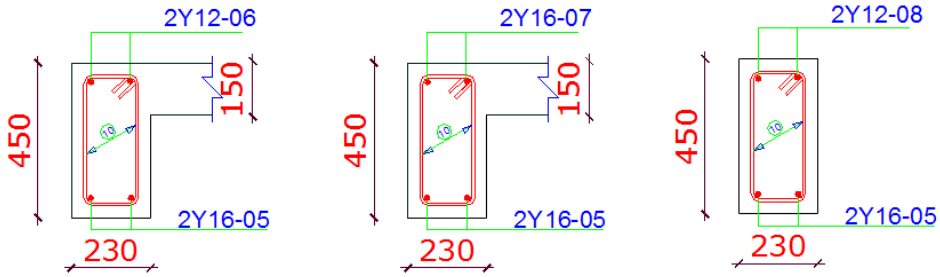


Figure 9.16: Floor beam Layout and numbering



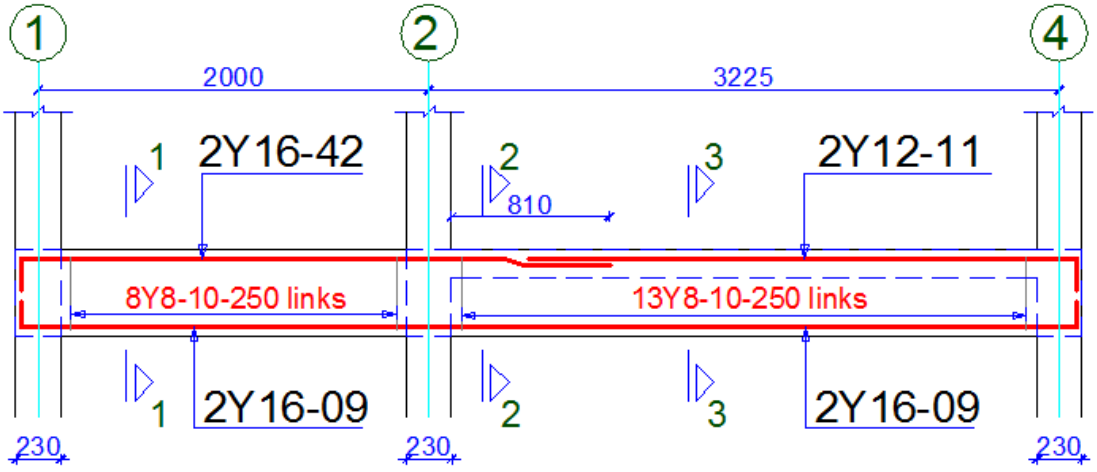
BEAM NO 2

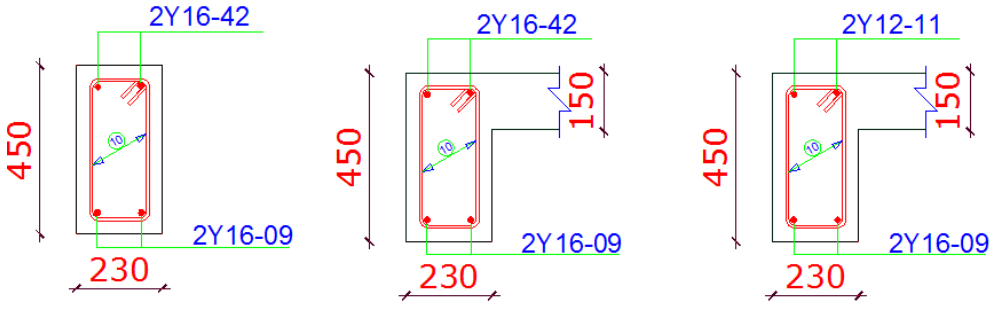




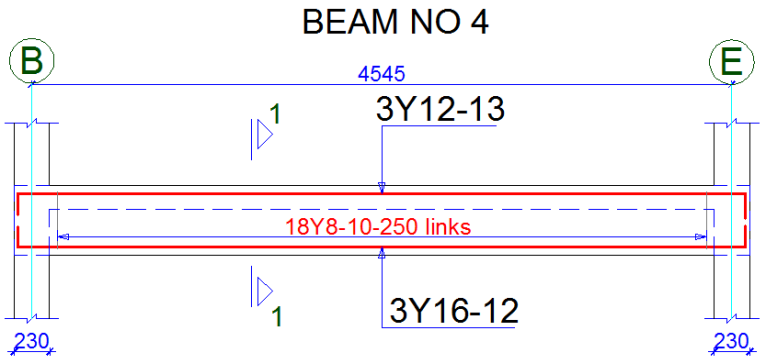
Section 1-1 Section 2-2 Section 3-3

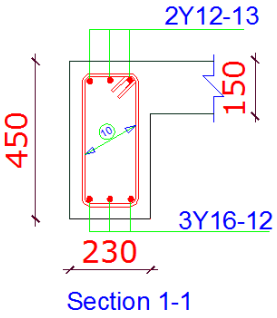
BEAM NO 3

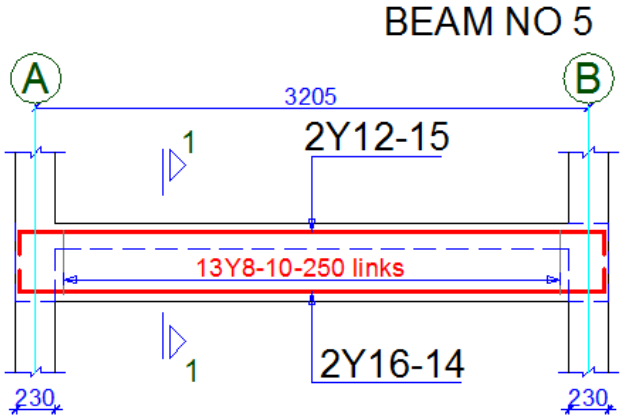


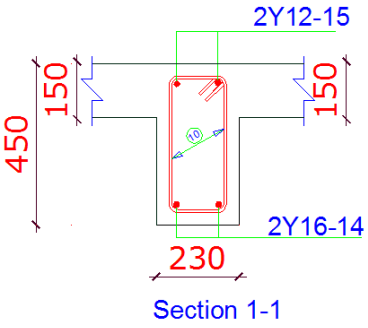


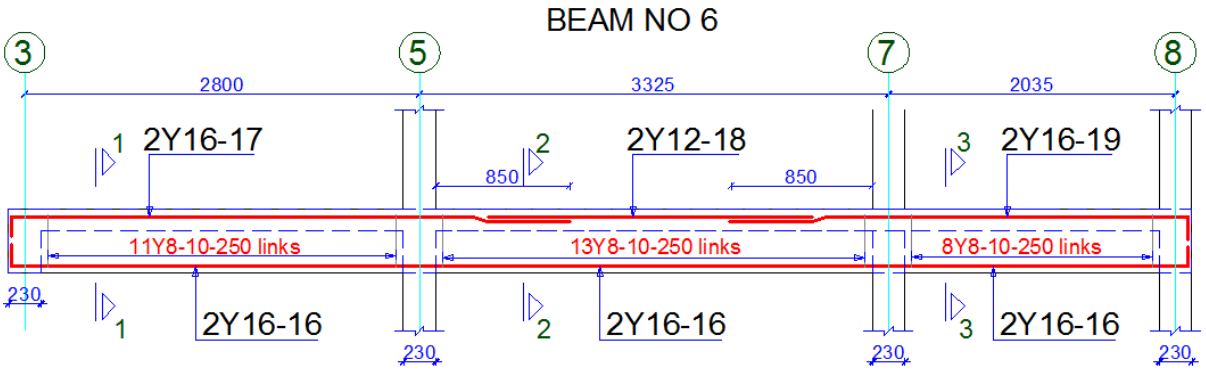
Section 1-1 Section 2-2 Section 3-3

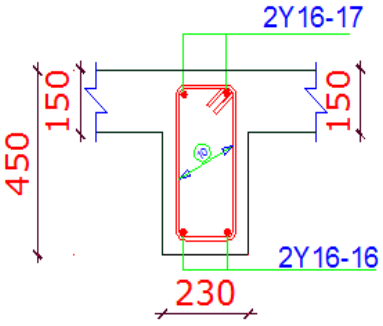

BEAM NO 4
B
E
4545
1
3Y12-13
18Y8-10-250 links
1
3Y16-12
230
230

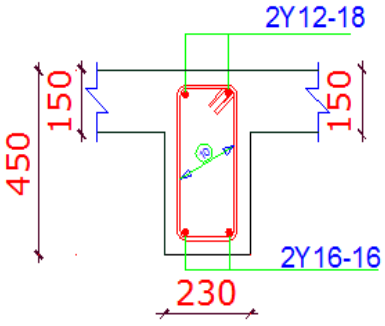

2Y12-13
150
450
3Y16-12
230
Section 1-1

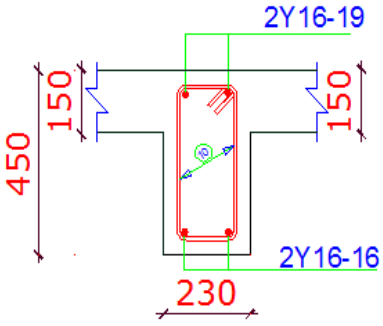

BEAM NO 5
A
B
3205
1
2Y12-15
13Y8-10-250 links
1
2Y16-14
230
230


2Y12-15
150
150
450
2Y16-14
230
Section 1-1

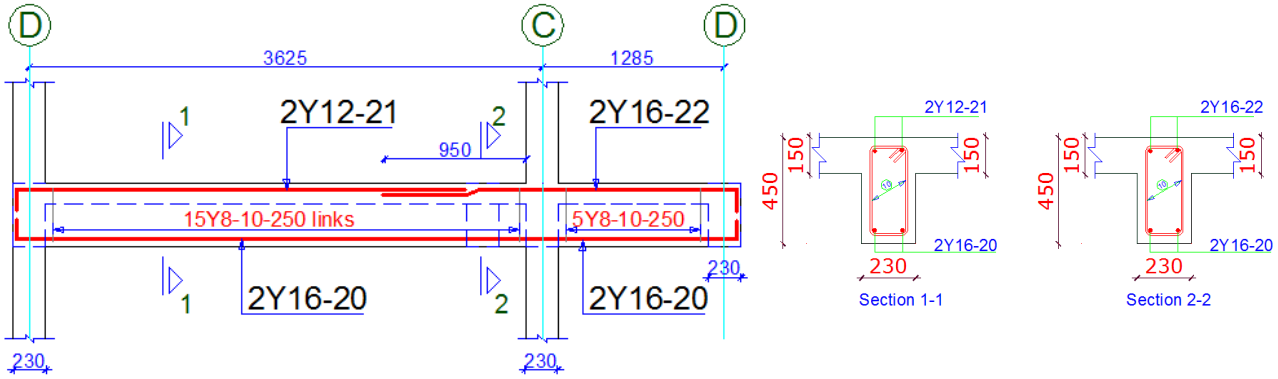

BEAM NO 6
3
5
7
8
2800
3325
2035
1
2Y16-17
2
850
2Y12-18
850
3
2Y16-19
11Y8-10-250 links
13Y8-10-250 links
8Y8-10-250 links
230
1
2Y16-16
2
2Y16-16
3
2Y16-16
230
230
230


2Y16-17
150
150
450
2Y16-16
230
Section 1-1


2Y12-18
150
150
450
2Y16-16
230
Section 2-2


2Y16-19
150
150
450
2Y16-16
230
Section 3-3

BEAM NO 7



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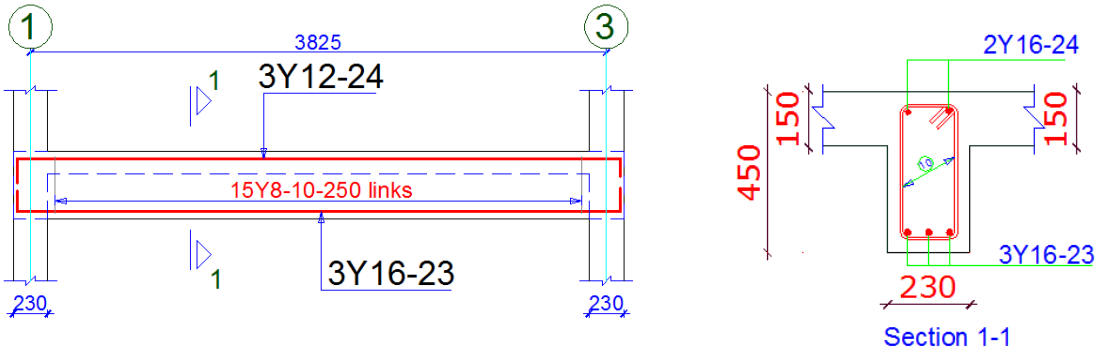
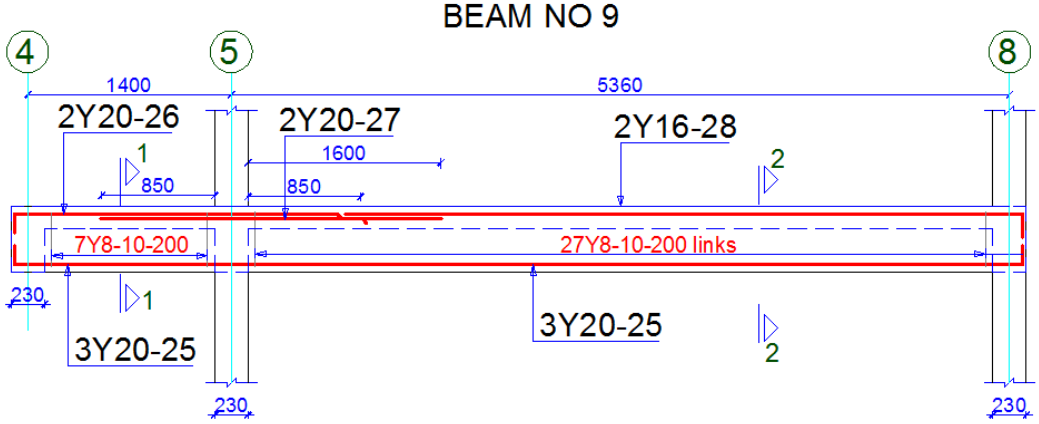
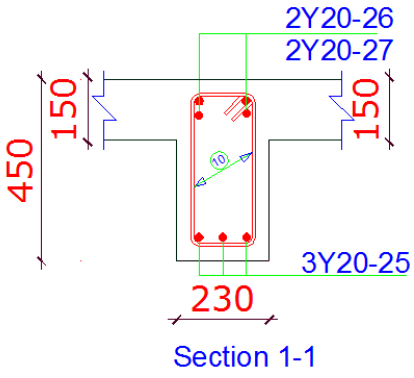
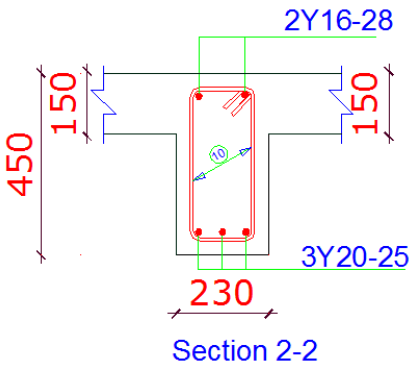


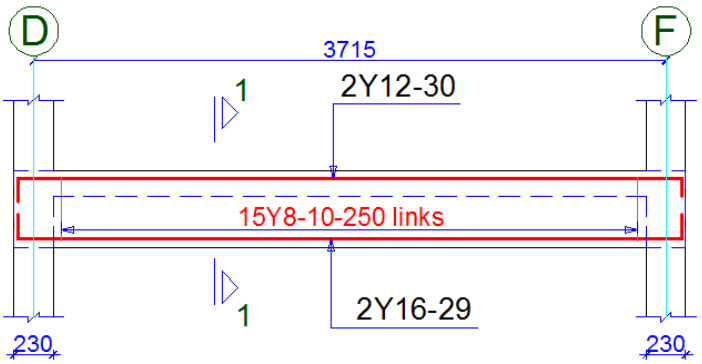
BEAM NO 9

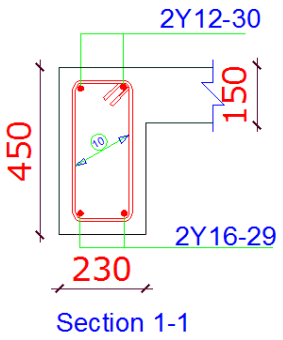






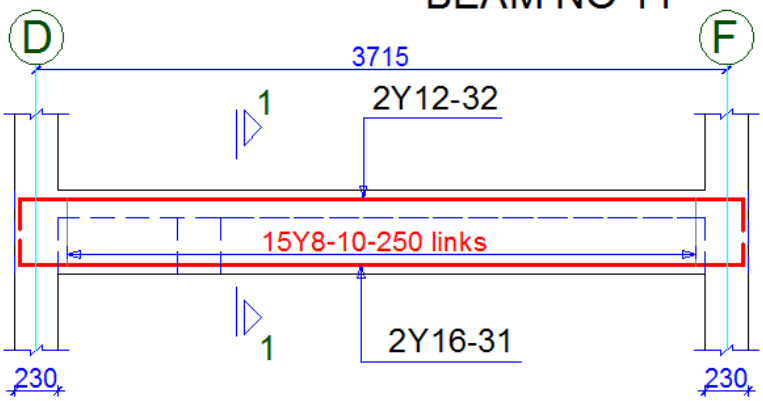
BEAM NO 10

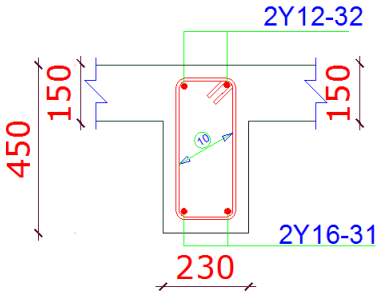




Section 1-1

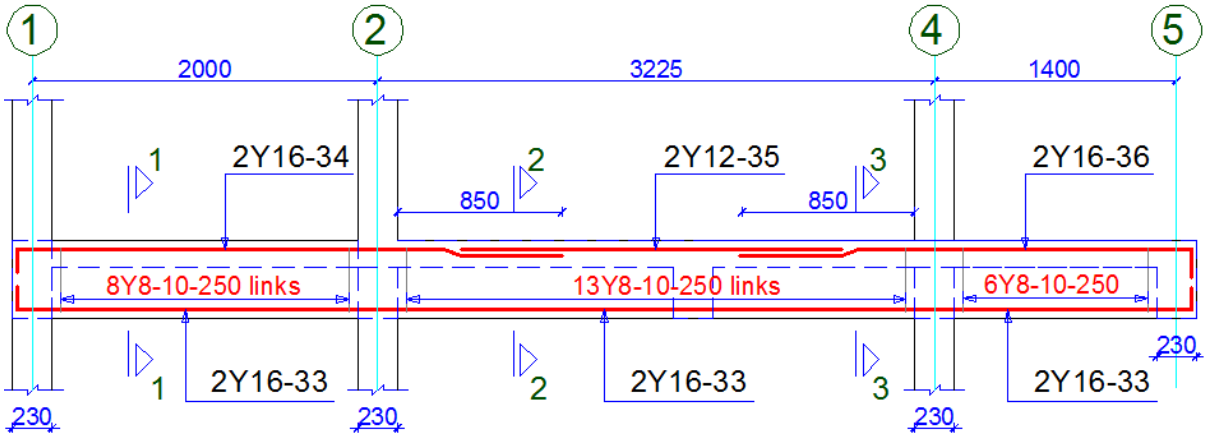
BEAM NO 11

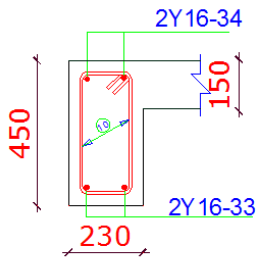




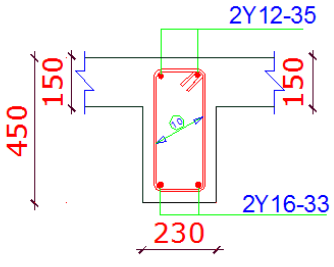
Section 1-1

BEAM NO 12

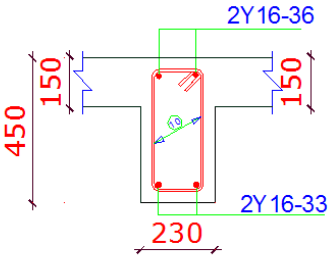




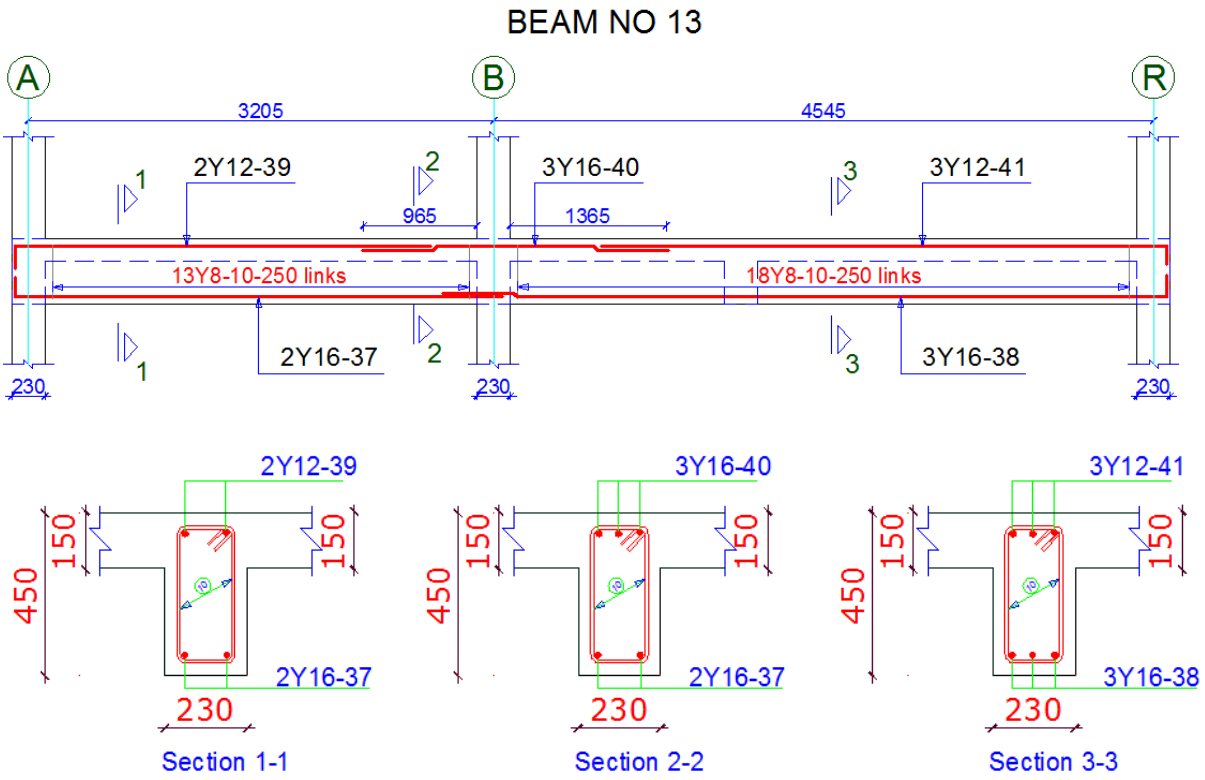
Section 1-1



Section 2-2

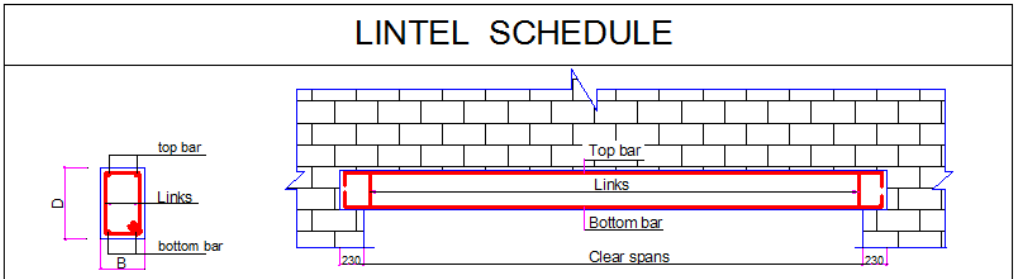


Section 3-3



9.7 Lintel Reinforcement Schedule

A simplified guide for doors and windows lintel reinforcement is as shown below;



Clear span	Depth D	Width B	Bottom bars	Top bars	Links
Up to 900	230	230	2Y12mm	2Y10mm	Y8 @ 230 ctrs
975 - 1800	230	230	2Y12mm	2Y12mm	Y8 @ 250 ctrs
1875 - 2700	250	230	2Y12mm	2Y12mm	T8 @ 250 ctrs
2775 - 2200	375	230	2Y12mm	2Y12mm	Y8 @ 250 ctrs
2275 - 5200	375	230	3Y12mm	2Y12mm	Y8 @ 250 ctrs

9.8 Column Detailing

In columns, the minimum recommended size of bars is 12mm. It is the best practice to provide even number of reinforcements in rectangular columns (with the minimum number of reinforcements being 4) while in circular columns, the minimum number of reinforcements is 6. In column detailing, the elevation and section are fully shown.

Earlier on, we verified that the anchorage length for $\phi 16$mm is 37.05ϕ

Therefore, the lap length will be; $l_0 = \alpha_6 \times 37.05\phi$

From the table, since the four bars will be lapped, $\rho > 50\%$, hence; $\alpha_6 = 1.5$

Therefore lap length $l_0 = 1.5 \times 37.05 \times 16 = 889.2$mm (take 900mm)

9.8.1 Column Type 1 Details

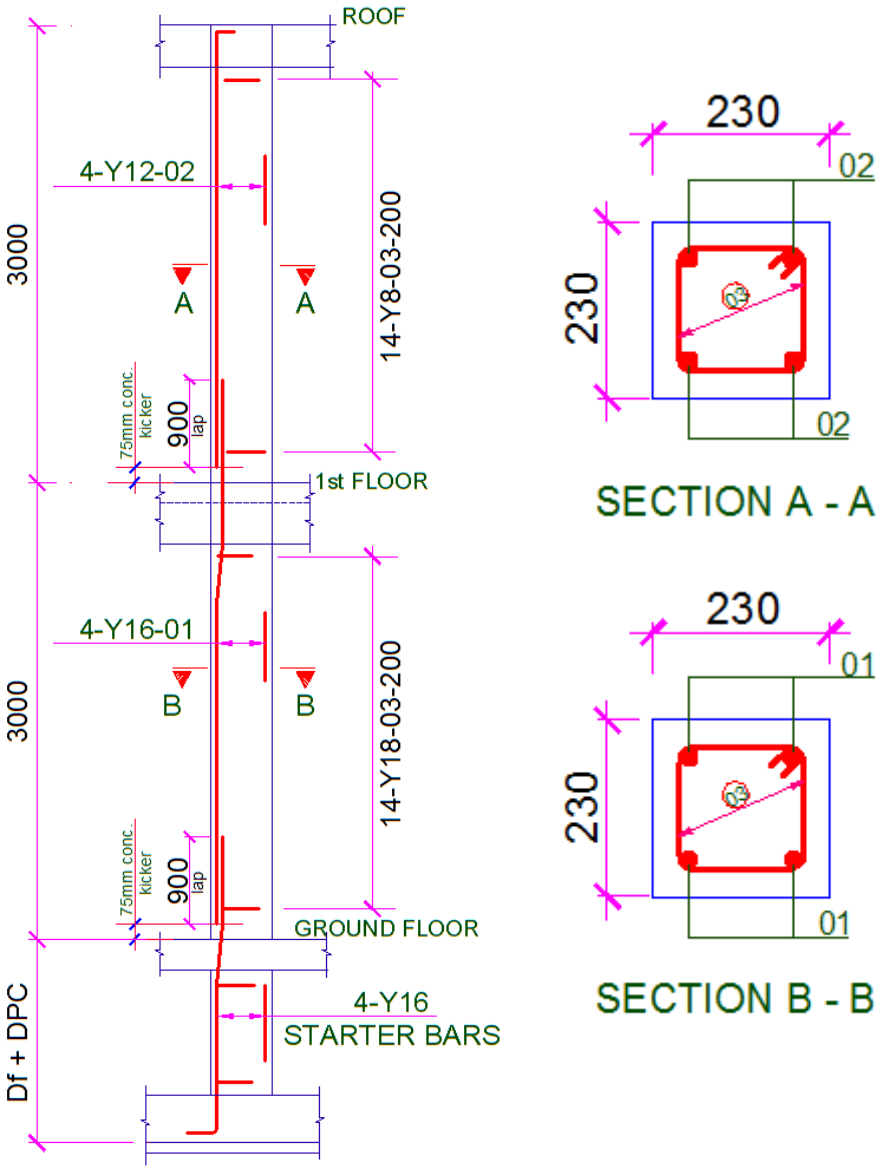


Figure 9.17: Column Type 1 detailing

9.8.2 Column Type 2 Details

For the type 2 columns (AXIS G columns), despite being a single run column running from the ground floor to the roof level, it is impractical to construct it in a single phase. This simply implies that we have to lap the bars for ease in construction. The adopted detail is shown below.

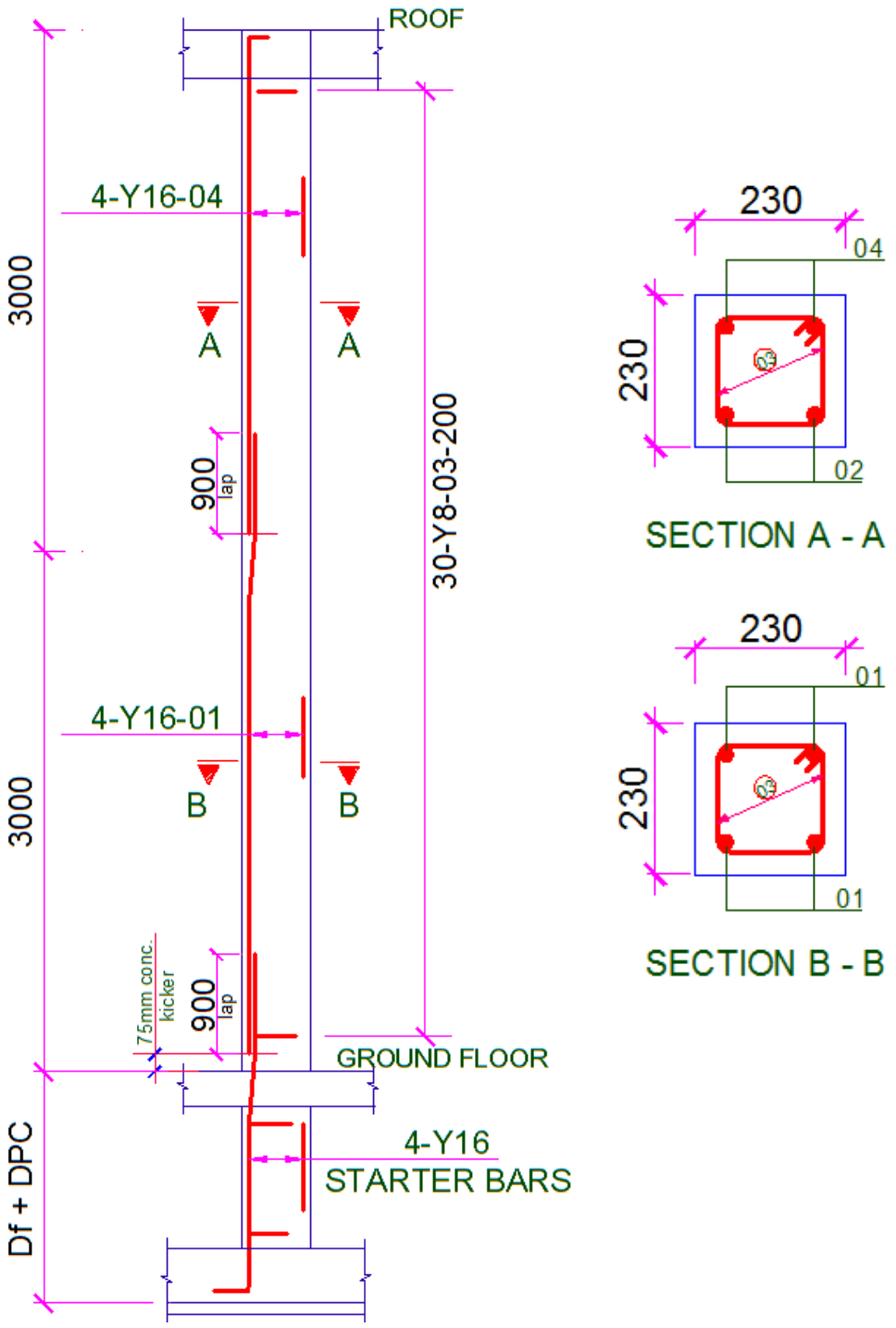


Figure 9.18: Column Type 2 Detailing

9.9 Staircase Detailing

Staircase detailing follows the simplified rules for slab detailing, including curtailment of reinforcement.

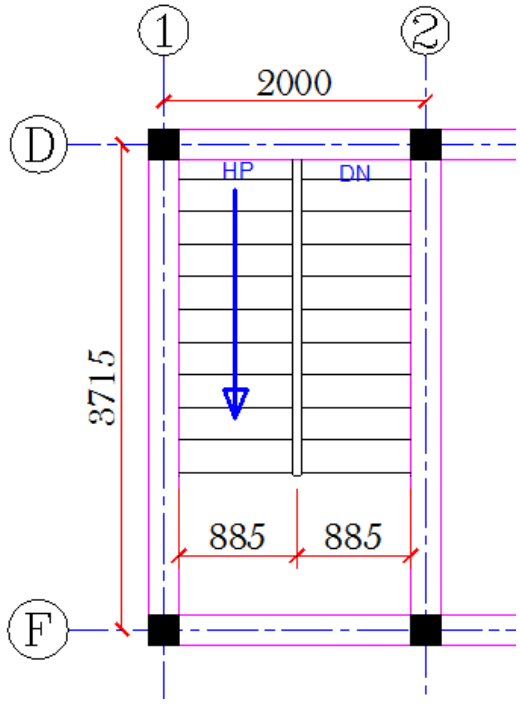


Figure 9.19: Staircase Layout

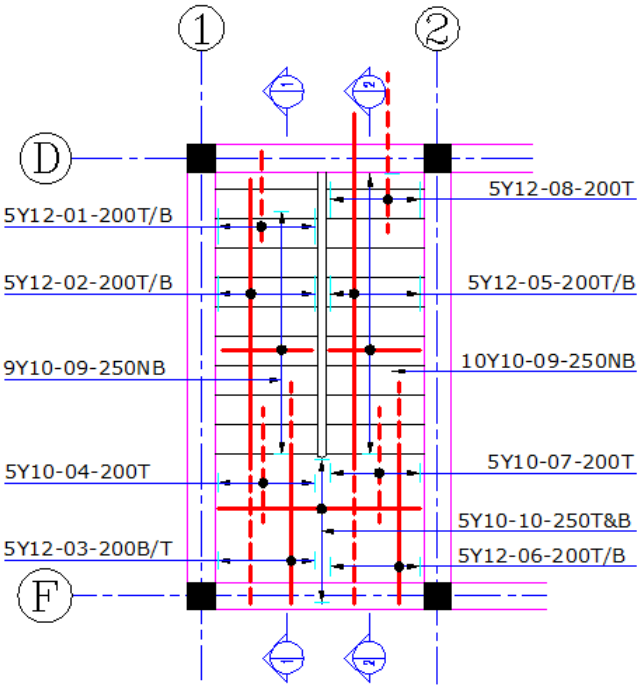
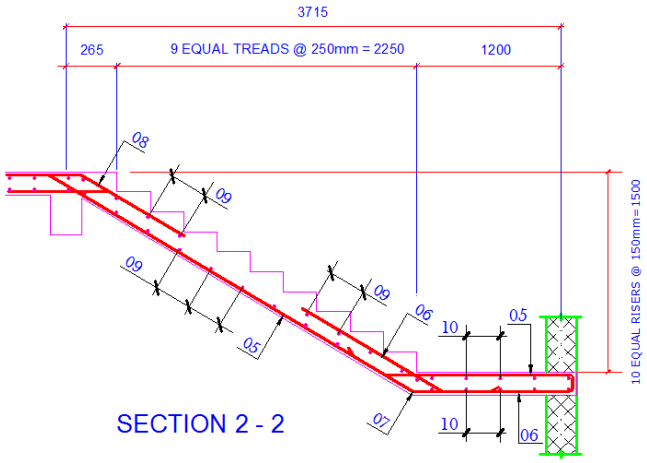
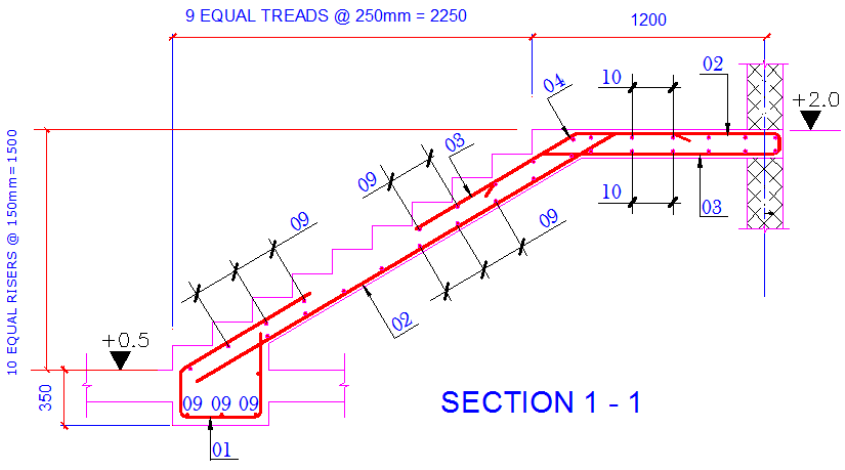


Figure 9.20: Staircase reinforcement (plan view)





9.10 Foundation Detailing

The anchorage requirement for $\phi 12$mm bars (50mm concrete cover for bars bent into any other shape is given below)

(a) For other shape bar

α_1 = 1.0 C_d = 50 is > 3ϕ = 3 x 16 = 48. Therefore α_1 = 0.7
α_3 = 1.0 conservative value with K= 0
α_4 = 1.0 N/A
α_5 = 1.0 conservative value
α_2 = 1.0 – 0.15 (C_d – 3ϕ)/ ϕ ≤ 1.0
α_2 = 1.0 – 0.15 (50 – 48)/16 = 0.981 ≤ 1.0

$$l_{bd} = 0.7 \times 0.981 \times (37.05\phi) = 25.44\phi$$

Actual Anchorage length = 25.44 x 12 = 305.28mm (say 310mm)

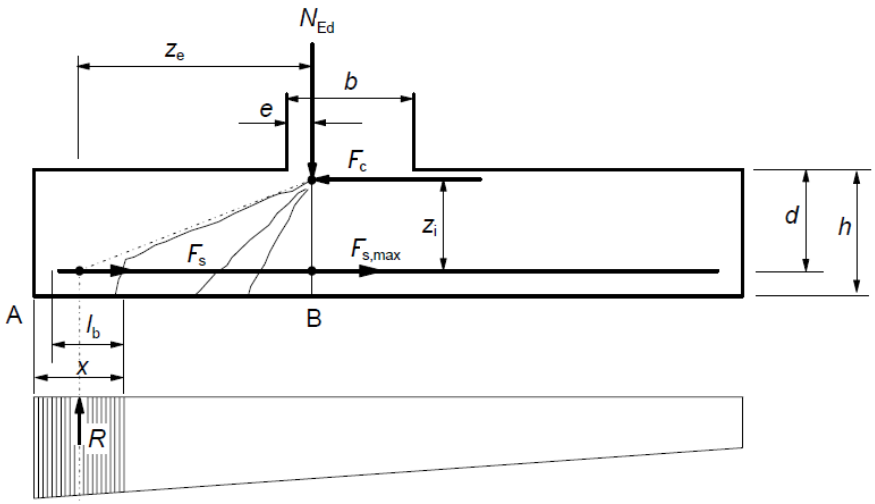


Figure 9.19: Model for anchorage requirements for footings (Fig 9.13 EN 1992-1-1:2004)

According to clause 9.8.2.2 of EC2, the tensile force to be anchored is given by:

$$Fs = R \times \frac{z_e}{z_i} \text{ ---------- (9.6)}$$

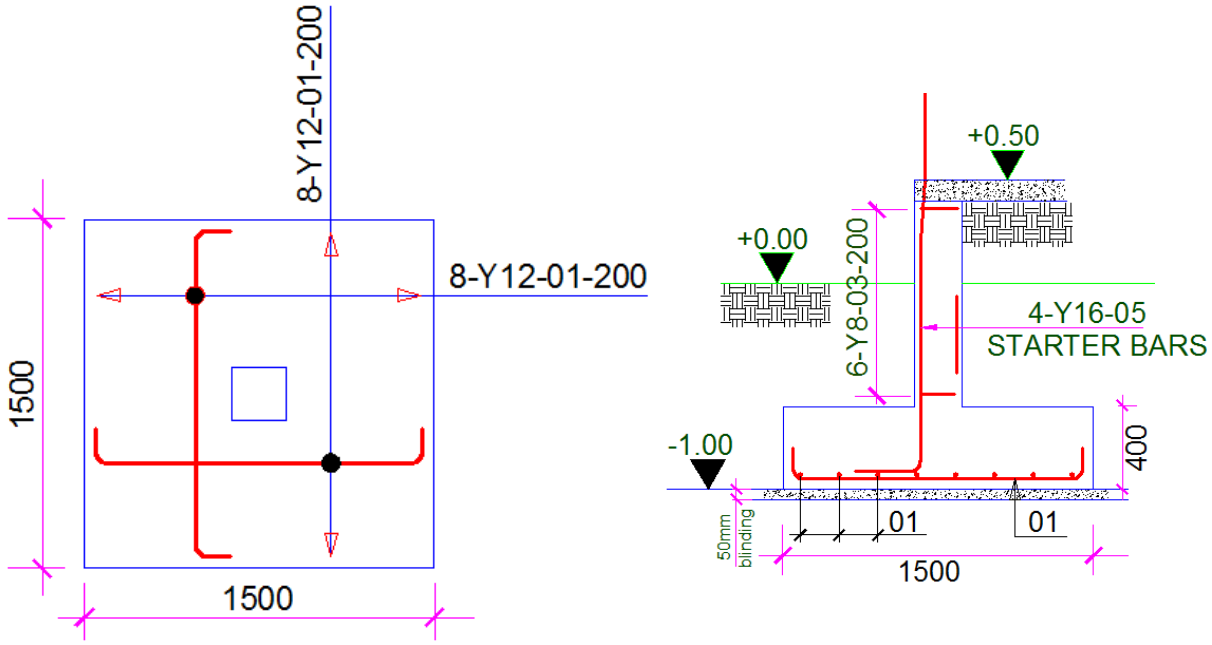
where:
R is the resultant of ground pressure within distance *x*
z_e is the external lever arm, i.e. distance between *R* and the vertical force N_{Ed}
N_{Ed} is the vertical force corresponding to total ground pressure between sections A and B
z_i is the internal lever arm, i.e. distance between the reinforcement and the horizontal force *F*c
F_c is the compressive force corresponding to maximum tensile force $F_{s,max}$

Lever arms z_e and z_i may be determined with regard to the necessary compression zones for N_{Ed} and *F*c respectively. As simplifications, z_e may be determined assuming *e* = 0.15*b*, see Figure 9.19 and z_i may be taken as 0.9*d*.

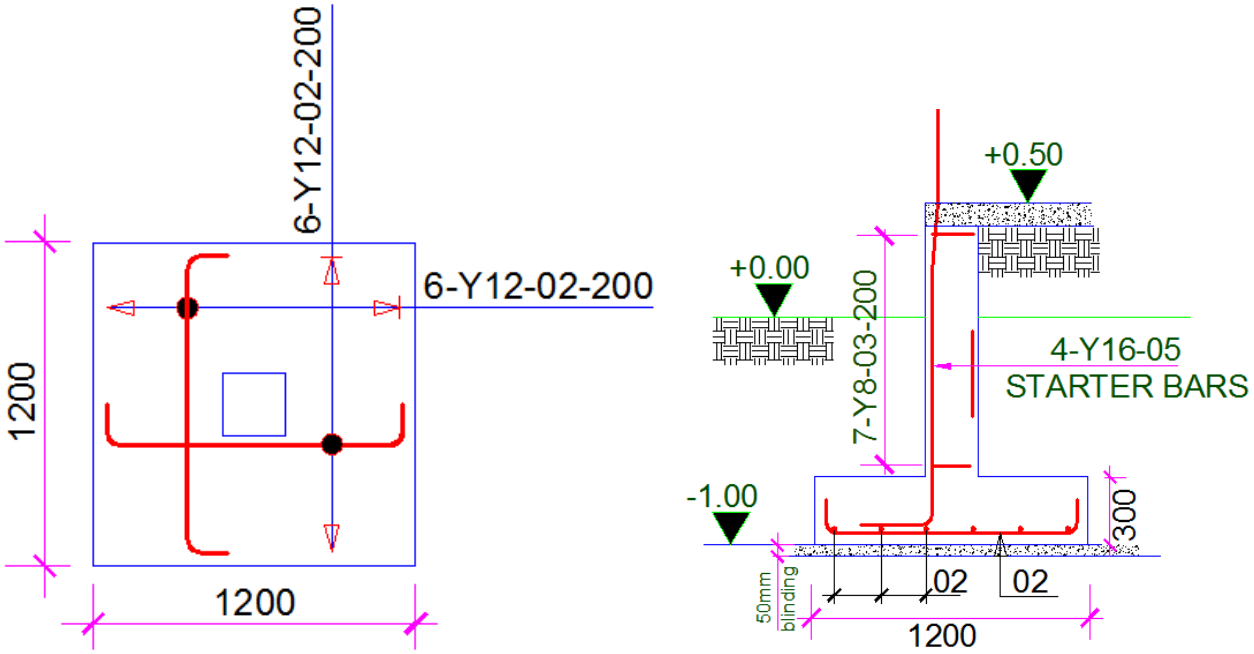
The available anchorage length for straight bars is denoted l_b in Figure 9.19. If this length is not sufficient to anchor *F*s, bars may either be bent up to increase the available length or be provided with end anchorage devices. For straight bars without end anchorage the minimum value of x is the most

critical. As a simplification $x_{min} = h/2$ may be assumed. For other types of anchorage, higher values of x may be more critical.

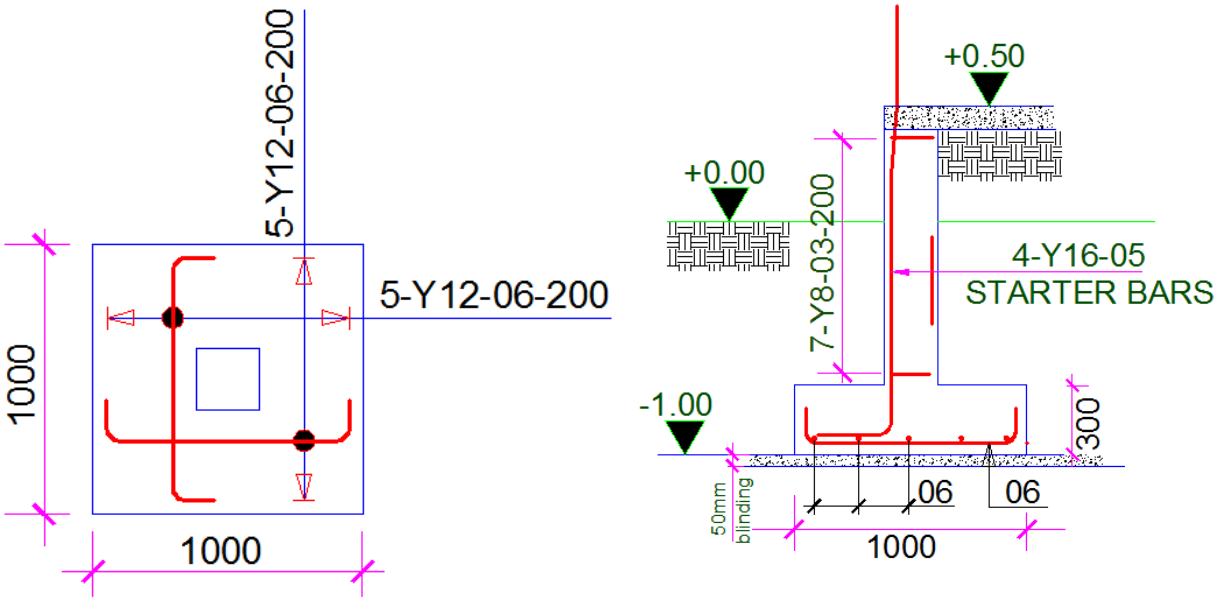
FOOTING TYPE 1



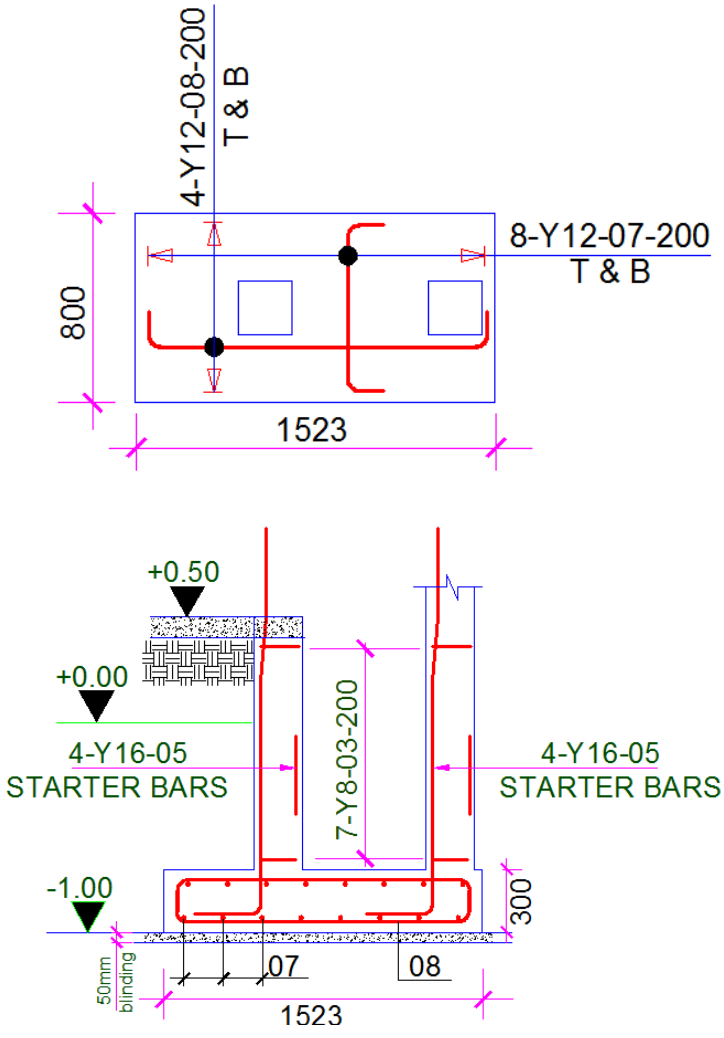
FOOTING TYPE 2



FOOTING TYPE 3



FOOTING TYPE 4



CHAPTER 10

BAR BENDING SCHEDULE

10.1 Introduction

Bar bending schedule is an important structural working document that rightly gives the disposition, bending shape, and total length of all the reinforcements that have been provided in the structural drawing, including the quantity. It is the bar mark from structural detailing drawing that is transferred to the bar bending schedule. We normally quantify reinforcements based on their unit mass.

The unit mass of the reinforcements are derived from the density of steel. The density of steel normally adopted for this purpose is 7850 kg/m^3.

For 12mm bar, the area of steel of steel is given by $\frac{\pi d^2}{4} = \frac{\pi \times 12^2}{4}$ = 113.097mm^2 = 0.0001131m^2

Considering a unit length of the bar, we can verify that the volume is 0.0001131m^3

Knowing full well that Density $= \frac{Mass}{Volume}$ = 7850 kg/m^3 = $\frac{Mass}{0.0001131}$

Therefore, the unit mass = 7850 × 0.0001131 = 0.888 kg/m

You can also rightly verify that;

Basic weight = 0.00785 kg/mm^2 per metre
Weight per metre = 0.006165 ϕ^2 kg
Weight per mm^2 at spacing s(mm) = 6.165ϕ^2/ skg

Where;
ϕ = diameter of bar in millimetres

Diameter of bar (mm)	Weight per metre (kg)	Length per tonne (m)
6	0.222	4505
8	0.395	2532
10	0.616	1623
12	0.888	1126
16	1.579	633
20	2.466	406
25	3.854	259
32	6.313	158
40	9.864	101

There are some basic standard shapes that have definite shape codes. But in modern bending schedule, shape codes are rarely included. However, the real shape of the member is drawn out for the fitters (iron benders) to follow.

To obtain the length of reinforcement bars, use the following relation;

Length of bar = Effective Length + Width of Support – Concrete cover (s) – Tolerances (say 10mm)

10.2 Bending Schedule for Foundations and Starter Bars

STRUCTVILLE ENGINEERING LIMITED

REINFORCEMENT BENDING/QUANTITY SCHEDULE

PROJECT: *Structville Residential Building*		**LOCATION:** *Owerri, Imo State*
JOB NO: *244*	**PHASE/ELEMENT:** *Foundations & Starter bars*	**DATE:** *21st, February, 2017*
PREPARED BY: *Ubani*	**CHECKED:** *Okoye*	**APPROVED:**
DESIGN CODE: *EC2*	**REBAR SPEC:** *BS EN 10080*	**DRAWING NO:** *5*	**SHEET NO:** *5*
REVISIONS:	**PAGE:** *1* of *6*	

LAST BAR MARK: *08* **OMMITTED BAR MARK:** *04*

S/No	Bar Mark	No of bars	No of members	Shape	Length (m)	Total Length (m)	**STEEL QUANTITY (kg) FOR VARIOUS SIZES OF STEEL BARS** ϕ8 0.395	ϕ10 0.616	ϕ12 0.888	ϕ16 1.579	ϕ20 2.466
1	01	16	3	150 / 1400 / 150	1.7	81.6			72.46		
2	02	12	7	150 / 1100 / 150	1.4	117.6			104.43		
3	03	7	20	160 × 160, 50	0.74	103.6	40.92				
4	05	4	20	300 / 2315	2.615	209.2				330.33	
5	06	10	8	150 / 900 / 150	1.2	96			85.25		
6	07	16	1	150 / 700 / 150	1.0	16			14.21		
7	08	8	1	150 / 1423 / 150	1.723	13.784			12.240		
						TOTAL (kg)	40.92		288.59	330.33	
						TOTAL (tonnes)	0.04092		0.2886	0.3303	
						No of pieces required (per 12m length)	**9 pcs**		**28 pcs**	**18 pcs**	

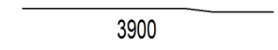
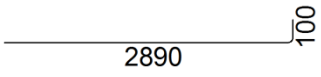
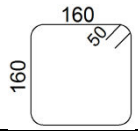
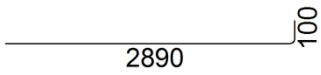
10.3 Bending Schedule for Columns

STRUCTVILLE ENGINEERING LIMITED

REINFORCEMENT BENDING/QUANTITY SCHEDULE

PROJECT: *Structville Residential Building*		**LOCATION:** *Owerri, Imo State*	
JOB NO: *244*	**PHASE/ELEMENT:** *Columns*		**DATE:** *21st, February, 2017*
PREPARED BY: *Ubani*	**CHECKED:** *Okoye*		**APPROVED:**
DESIGN CODE: *EC2*	**REBAR SPEC:** *BS EN 10080*	**DRAWING NO:** *4*	**SHEET NO:** *4*
REVISIONS:		**PAGE:** *2* of *6*	

LAST BAR MARK: *04* **OMMITTED BAR MARK:** *None*

S/ No	Bar Mark	No of bars	No of mem bers	Shape	Length (m)	Total Length (m)	STEEL QUANTITY (kg) FOR VARIOUS SIZES OF STEEL BARS ϕ8 0.395	ϕ10 0.616	ϕ12 0.888	ϕ16 1.579	ϕ20 2.466
1	01	4	20		3.9	312				492.65	
2	02	4	17		2.99	203.32			180.55		
3	03	28	20		0.74	414.40	163.69				
4	04	4	3		2.99	35.88				56.65	
						TOTAL (kg)	163.69		180.55	549.30	
						TOTAL (tonnes)	0.1639		0.1806	0.5493	
						No of pieces required (per 12m length)	**35 pcs**		**17 pcs**	**29 pcs**	

10.4 Bending Schedule for Floor beams

STRUCTVILLE ENGINEERING LIMITED

REINFORCEMENT BENDING/QUANTITY SCHEDULE

PROJECT: *Structville Residential Building*		**LOCATION:** *Owerri, Imo State*	
JOB NO: *244*	**PHASE/ELEMENT:** *Floor Beams*		**DATE:** *21st, February, 2017*
PREPARED BY: *Ubani*	**CHECKED:** *Okoye*		**APPROVED:**
DESIGN CODE: *EC2*	**REBAR SPEC:** *BS EN 10080*	**DRAWING NO:** *4*	**SHEET NO:** *4*
REVISIONS:		**PAGE:** *3* of *6*	

LAST BAR MARK: *42* **OMMITTED BAR MARK:** *None*

S/ No	Bar Mark	No of bars	No of mem bers	Shape	Length (m)	Total Length (m)	STEEL QUANTITY (kg) FOR VARIOUS SIZES OF STEEL BARS ϕ8 0.395	ϕ10 0.616	ϕ12 0.888	ϕ16 1.579	ϕ20 2.466
1	01	2	1	240 — 10110 — 240	10.49	21.18				33.44	
2	02	2	1	240 — 3365	7.21	14.42			12.805		
3	03	2	1	4880	4.880	9.76				15.411	
4	04	2	1	2825 — 240	3.065	6.13			5.443		
5	05	2	1	240 — 8765 — 240	9.245	18.49				29.196	
6	06	2	1	240 — 3270	3.51	7.02			6.234		
7	07	2	1	3375	3.375	6.750				10.658	
8	08	2	1	3320 — 240	3.560	7.120			6.322		
9	09	2	1	240 — 5365 — 240	5.845	11.69				18.458	
10	10	314	sum	150 / 380 / 50	1.16	364.24	143.87				
11	11	2	1	2970 — 240	3.21	6.42			5.701		
12	12	3	1	240 — 4455 — 240	4.935	14.805				23.377	
13	13	3	1	4455 / 240 — 240	4.935	14.805			13.146		
14	14	2	1	240 — 3115 — 240	3.595	7.190				11.353	
15	15	2	1	3115 / 240 — 240	3.595	7.190			6.385		
16	16	2	1	240 — 8300 — 240	8.780	17.56				27.727	

S/ No	Bar Mark	No of bars	No of members	Shape	Length (m)	Total Length (m)	ϕ8 0.395	ϕ10 0.616	ϕ12 0.888	ϕ16 1.579	ϕ20 2.466
17	17	2	1	240, 3835	4.075	8.15				12.869	
18	18	2	1	2595	2.595	5.190			4.608		
19	19	2	1	3070, 240	3.310	6.620				10.453	
20	20	2	1	240, 5050, 240	5.530	11.06				17.463	
21	21	2	1	240, 3230	3.470	6.94			6.163		
22	22	2	1	2420, 240	2.660	5.32				8.400	
23	23	3	1	240, 3965, 240	4.445	13.335				21.056	
24	24	3	1	240, 3965, 240	4.445	13.335			11.841		
25	25	3	1	240, 6900, 240	7.38	22.14					54.60
26	26	2	1	240, 3185	3.425	6.85					16.89
27	27	2	1	1930	1.93	3.86					9.52
28	28	2	1	4315, 240	4.555	9.110				14.384	
29	29	2	1	240, 3855, 240	4.335	8.670				13.689	
30	30	2	1	240, 3855, 240	4.335	8.670			7.699		
31	31	2	1	240, 3855, 240	4.335	8.670				13.689	
32	32	2	1	240, 3855, 240	4.335	8.670			7.699		
33	33	2	1	240, 6765, 240	7.245	14.49				22.879	
34	34	2	1	240, 3035	3.275	6.55				10.342	
35	35	2	1	2495	2.495	4.99			4.431		
36	36	2	1	2435, 240	2.675	5.35				8.448	
37	37	2	1	240, 3390	3.63	7.26				11.464	
38	38	3	1	4830, 240	5.070	15.21				24.016	
39	39	2	1	240, 2795	3.035	6.07			5.390		
40	40	3	1	2560	2.560	7.68				12.127	
41	41	3	1	3735, 240	3.975	11.925			10.589		

42	42	2	1	240 2995	3.235	6.470				10.216	
						TOTAL (kg)	143.87		114.46	381.30	81.01
						TOTAL (tonnes)	0.1438		0.1145	0.3813	0.081
						No of pieces required (per 12m length)	**31 pcs**		**11 pcs**	**21 pcs**	**3 pcs**

10.5 Bending Schedule for Slabs

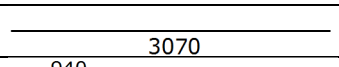
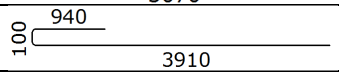
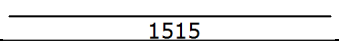
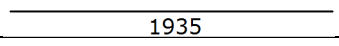
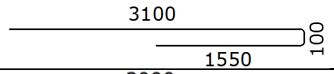
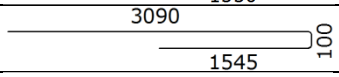
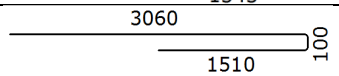
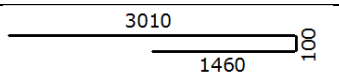
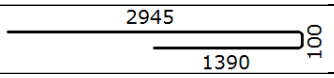
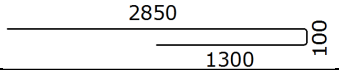
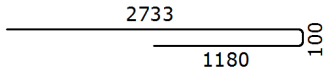
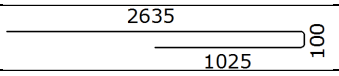
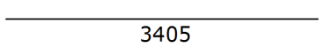
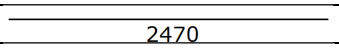
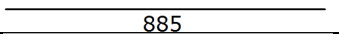
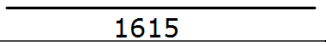
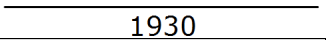
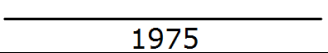
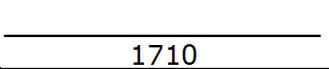
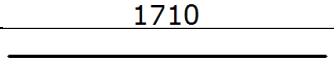
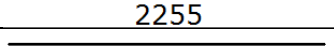
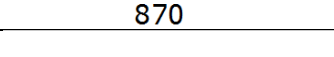
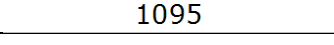
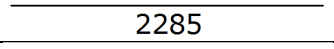
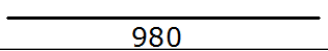
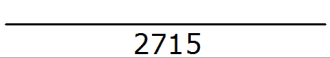
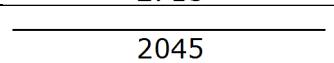
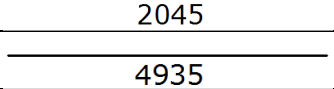
STRUCTVILLE ENGINEERING LIMITED

REINFORCEMENT BENDING/QUANTITY SCHEDULE

PROJECT: *Structville Residential Building*		**LOCATION:** *Owerri, Imo State*	
JOB NO: *244*	**PHASE/ELEMENT:** *Slabs*	**DATE:** *21st, February, 2017*	
PREPARED BY: *Ubani*	**CHECKED:** *Okoye*	**APPROVED:**	
DESIGN CODE: *EC2*	**REBAR SPEC:** *BS EN 10080*	**DRAWING NO:** *3*	**SHEET NO:** *3*
REVISIONS:	**PAGE:** *2* of *6*		

LAST BAR MARK: *45* **OMMITTED BAR MARK(S):** *17,31, 34*

S/ No	Bar Mark	No of bars	No of members	Shape	Length (m)	Total Length (m)	STEEL QUANTITY (kg) FOR VARIOUS SIZES OF STEEL BARS ϕ8 0.395	ϕ10 0.616	ϕ12 0.888	ϕ16 1.579	ϕ20 2.466
1	01	14	1	100, 950, 4020	5.07	70.98			63.030		
2	02	15	1	100, 920, 3820	4.84	72.6			64.468		
3	03	12	1	6355	6.355	76.26			67.719		
4	04	30	1	100, 836, 3400	4.336	130.08			115.51		
5	05	12	1	2845, 100, 1195	4.14	49.68			44.116		
6	06	18	1	4660	4.66	83.88			74.485		
7	07	3	1	100, 1104, 4740	5.944	17.832			15.835		
8	08	18	1	100, 1267, 5590	6.957	125.23			111.20		
9	09	15	1	2860, 100, 1395	4.355	65.325			58.009		
10	10	5	1	100, 1600, 3955	5.655	28.275			25.108		
11	11	11	1	1630	1.630	17.930			15.922		
12	12	6	1	3030	3.03	18.18			16.144		
13	13	14	1	100, 860, 3420	4.380	61.32			54.452		
14	14	5	1	100, 2045, 4605	6.750	33.750			29.97		

S/ No	Bar Mark	No of bars	No of members	Shape	Length (m)	Total Length (m)	ϕ8 0.395	ϕ10 0.616	ϕ12 0.888	ϕ16 1.579	ϕ20 2.466
15	15	5	1		3.070	15.35			13.631		
16	16	12	1		4.95	59.4			52.747		
17	18	18	1		1.515	27.27			24.213		
18	19	12	1		1.935	23.22			20.619		
19	20A	1	1		4.75	4.75			4.218		
20	20B	2	1		4.735	9.470			8.409		
21	20C	2	1		4.670	9.340			8.294		
22	20D	2	1		4.570	9.140			8.116		
23	20E	2	1		4.435	8.870			7.876		
24	20F	2	1		4.250	8.500			7.548		
25	20G	2	1		4.013	8.026			7.1271		
26	20H	2	1		3.76	7.52			6.678		
27	21A	1	1		3.405	3.405			3.024		
28	21B	1	1		2.470	2.470			2.193		
29	21C	1	1		0.885	0.885			0.786		
30	22	33	1		1.615	53.295		32.829			
31	23	11	1		1.930	21.23			18.852		
31	24	15	1		1.975	29.625			26.307		
33	25	8	1		1.710	13.68		8.626			
34	26	12	1		2.255	27.06			24.029		
35	27	6	1		0.87	5.22		3.216			
36	28	7	1		1.095	7.665		4.722			
37	29	12	1		2.285	27.42			24.348		
38	30	7	1		0.98	6.86		4.226			
39	32	18	1		2.715	48.87			43.396		
40	33	12	1		2.045	24.54		15.116			
41	35	4	1		4.935	19.74		12.159			
42	36	13	1		2.532	32.92		20.276			

S/ No	Bar Mark	No of bars	No of members	Shape	Length (m)	Total Length (m)	ϕ8 0.395	ϕ10 0.616	ϕ12 0.888	ϕ16 1.579	ϕ20 2.466
43	37	18	1	2895	2.895	52.11			46.273		
44	38A	1	1	3405	3.405	3.405		2.097			
45	38B	1	1	2470	2.470	2.470		1.522			
46	38C	1	1	885	0.885	0.885		0.5452			
47	39	14	1	1860	1.860	26.04			23.123		
48	40	9	1	1480	1.480	13.32		8.205			
49	41	6	1	2370	2.370	14.22			12.627		
50	42	9	1	1955	1.955	17.595			15.624		
51	43	7	1	1795	1.795	12.565		7.740			
52	44	17	1	1995	1.995	33.915			30.116		
53	45	8	1	1350	1.350	10.8		6.653			
						TOTAL (kg)		127.932	1196.142		
						TOTAL (tonnes)		0.1279	1.19614		
						No of pieces required (per 12m length)		**18 pcs**	**113 pcs**		

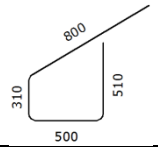
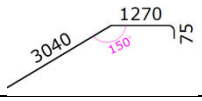
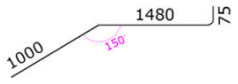
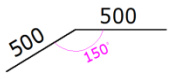
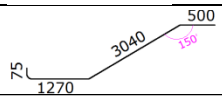
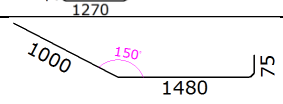
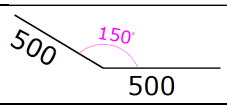
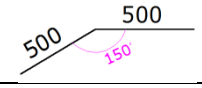
10.6 Bending Schedule for Staircase

STRUCTVILLE ENGINEERING LIMITED

REINFORCEMENT BENDING/QUANTITY SCHEDULE

PROJECT: *Structville Residential Building*		**LOCATION:** *Owerri, Imo State*
JOB NO: *244*	**PHASE/ELEMENT:** *Staircase*	**DATE:** *21st, February, 2017*
PREPARED BY: *Ubani*	**CHECKED:** *Okoye*	**APPROVED:**
DESIGN CODE: *EC2*	**REBAR SPEC:** *BS EN 10080*	**DRAWING NO:** *2* / **SHEET NO:** *2*
REVISIONS:	**PAGE:** *2* of *6*	

LAST BAR MARK: *04* **OMMITTED BAR MARK:** *None*

S/ No	Bar Mark	No of bars	No of mem bers	Shape	Length (m)	Total Length (m)	STEEL QUANTITY (kg) FOR VARIOUS SIZES OF STEEL BARS ϕ8 0.395	ϕ10 0.616	ϕ12 0.888	ϕ16 1.579	ϕ20 2.466
1	01	5	1		2.120	10.6			9.4128		
2	02	5	1		4.385	21.925			19.469		
3	03	5	1		2.555	12.775			11.344		
4	04	5	1		1.0	5		3.08			
5	05	5	1		4.885	24.425			21.689		
6	06	5	1		2.555	12.775			11.344		
7	07	5	1		1.0	5		3.08			
8	08	5	1		1.0	5		3.08			
9	09	19	1	770	0.77	14.63		9.0121			
10	10	10	1	1713	1.713	17.13		10.552			
				TOTAL (kg)				28.804	73.259		
				TOTAL (tonnes)				0.0288	0.0732		
				No of pieces required (per 12m length)				**4 pcs**	**7 pcs**		

CHAPTER 11

QUANTITY AND COST ESTIMATION

11.1 Introduction

This chapter aims to briefly outline the simple basic steps necessary for computation of the quantity of materials, and how to estimate the total cost of each phase in the construction. This is necessary because clients for a small building project will always depend on the engineer to give them an estimate on the amount of materials required. This is what this chapter will briefly highlight. You will also need this in the presentation of 'Bill of Engineering Measurement and Evaluation' (BEME) or Bill of Quantity (BOQ). If the contract does not involve 'Bill of Quantity', this estimate will guide you for the successful execution of the project.

11.2 Foundations Works

After site preparation and logistics, the setting out for the building is carried out. Setting out normally require the use of pegs, and tie members of which softwood (50mm x 50mm) is normally employed. This soft wood is normally called '2 by 2' by craftsmen – implying 2 inches by inches. You can easily calculate the quantity needed by considering the perimeter of the building. The softwood normally comes at length of 12 feet (about 3.6m). Some already made pegs are sold in bundles. This is very easy to estimate.

11.2.1 Excavation

After the setting out, the next thing is the excavation of the foundation. This normally involves the use of manual labour, and it is therefore necessary that the real amount of work to be done be estimated. The foundation layout is very helpful to estimate this correctly. We are going to use our foundation layout as an example of how this done.

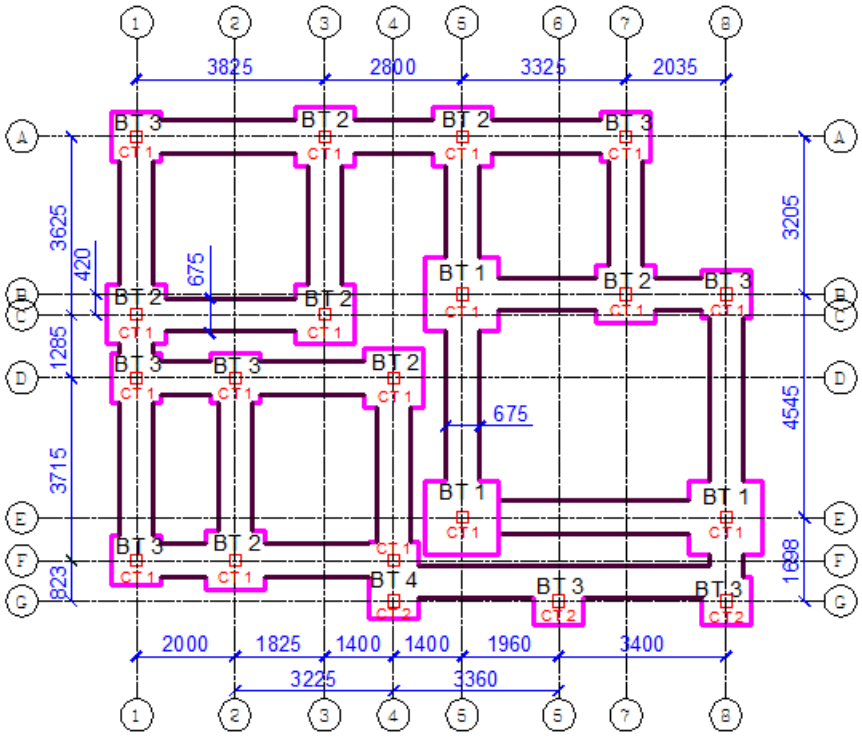


Figure 11.1: Foundation Layout

Foundation calculations simply involve the computation of volume or the computation of the total footing length. Volume calculation is important for BEME projects, while length calculation is more important when BOQ has not been employed. This process will be shown below.

Volume Computation

The volume of excavation based on the foundation has been presented in tabular form in Table 11.1 below.

Table 11.1: Volume computation for excavation

Base Type	Number	Length (m)	Width (m)	Depth (m)	Volume (m^3)
Base Type 1	3	1.5	1.5	1.0	6.75
Base Type 2	7	1.2	1.2	1.0	10.08
Base Type 3	8	1.0	1.0	1.0	8.00
Base Type 4	1	1.523	0.8	1.0	1.2184
Strip	Sum	50.830	0.675	1.0	34.310
				TOTAL	60.358 m^3

The total volume of excavation is normally taken to a round number, so take the volume as 61m^3. This total volume of excavation can then be inserted into the BOQ using the per unit volume rate.

On the other hand, the commonest method on site using direct labour is to excavate the strip footing first (covering all the perimeter of excavation) at a unit length rate, and then expanding the column base at another rate. The factors affecting cost of excavation are season of the year, terrain, type of soil, and general working environment.

For our foundation, the total perimeter of excavation (neglecting additional column base dimensions) can be calculated by summing up all the excavation length. Our layout gave us 72.214m

Let us assume that the strip footing is excavated at ₦1000 per metre, then the total cost of excavation labour is say ₦73,000

If the additional column base is excavated at an average rate of ₦500, per base, then for 19 column bases, we can be expecting ₦9500. The total cost of excavation can then be pegged at ₦73,000 + ₦9500 = ₦82,500, excluding the cost of supervision and other contingency variables.

11.2.2 Concrete works

11.2.2.1 Blinding

Blinding is normally done to receive the column base concrete and reinforcements. Blinding serves as additional protection to the reinforcement against moisture and potential chemical attack from the ground. The specified thickness of blinding in the foundation we are handling is 50mm. Blinding is normally done with a weak concrete. A ratio of 1:3:6 is usually very adequate.

Now, how do we estimate the quantity of materials needed for this operation? The first step is to calculate the volume of concrete required.

Total area of column bases = 3(1.5 x 1.5) + (7(1.2 x 1.2) + 8(1.0 x 1.0) + (1.523 x 0.8) = 26.048 m^2

Thickness of blinding = 50mm = 0.05m

Therefore the volume of concrete required = 26.048 x 0.05 = 1.3024m^3

Quantity of cement needed

For 1:3:6 mix, the total volume is (1+3+6) = $10m^3$

The ratio of cement = $\frac{1}{10}$ = $0.1m^3$

Density = $\frac{Mass}{Volume}$

Bulk Density of cement = 1440 kg/m^3

1440 = $\frac{Mass}{0.1}$; Such that the mass of cement = 144 kg

We normally make allowance of 1.54 to account for the reduction in volume when we add water to the dry concrete.

Hence mass of cement = 1.54 x 144 = 221.76 kg

But 1 bag of cement = 50 kg

Hence the number of bags required per metre cube of 1:3:6 mixture = $\frac{221.76}{50}$ = 4.4352 bags

For $1.3024m^3$ of concrete = 4.4352 x 1.3024 = 5.776 bags

Making 20% allowance for wastage = 1.2 x 5.776 = 6.93 bags (say 7 bags)

Quantity of fine aggregate needed (sharp sand)

For 1:3:6 mix, the total volume is (1 + 3 + 6) = $10m^3$

The ratio of cement = $\frac{3}{10}$ = $0.3m^3$

Making allowance of 1.5 to distinguish between the weight of fresh and hardened concrete;

Hence the volume of sand = 1.5 x 0.3 = 0.45 m^3

For $1.3024m^3$ of concrete = 0.45 x 1.3024 = 0.5861 m^3

Making 20% allowance for shrinkage and wastage = 1.2 x 0.5861 = $0.7032m^3$ of sand

Quantity of coarse aggregate needed (granite)

For 1:3:6 mix, the total volume is (1+3+6) = $10m^3$

The ratio of cement = $\frac{6}{10}$ = 0.6 m^3

Making allowance of 1.5 to distinguish between the weight of fresh and hardened concrete;

Hence volume of granite = 1.5 x 0.6 = 0.90 m^3

For $1.3024m^3$ of concrete = 0.90 x 1.3024 = 1.1722 m^3

Making 20% allowance for shrinkage and wastage = 1.2 x 1.17216 = 1.4066 m^3 of granite

11.2.2.2 Foundation Concrete Volume Calculation

Base Type 1; V_1 = 3 x (1.5 x 1.5 x 0.4) = 2.7 m^3

Base Type 2; V_2 = 7 x (1.2 x 1.2 x 0.3) = 3.024 m^3

Base Type 3; V_3 = 8 x (1.0 x 1.0 x 0.3) = 2.4 m^3

Base Type 4; V_4 = 1 x (1.523 x 0.8 x 0.3) = 0.36552 m^3

Strip footing; V_5 = (50.830 x 0.675 x 0.15) = 5.1465 m^3

Total = **13.636 m^3**

11.2.2.3 25 N/mm^2 Concrete works (mix design)

If it is specified to use a concrete grade of 25 N/mm^2 for work, I do not rely on specified mix ratios like 1:2:4 to give me my mix result. I normally like to carry out a mix design, and then specify the ratio based on the mix design results. Typically, concrete mix ratios are not usually respected on most building sites due to economy, but professionalism is professionalism, and cannot be compromised. So here, I am going to carry out a typical mix design for 25 N/mm^2 concrete.

Compressive strength of concrete is a very variable quantity. Therefore, when we are carrying out a mix design, we target a higher average strength so that every can expect every part of the structure to meet the specified strength. We normally employ the standard deviation in taking care of this, which is a measure of scatter or dispersion of strength about the mean.

Therefore, the target strength is given by;

$f_{cm} = f_{ck} + KSD$

Where f_{ck} is the compressive strength

K is the probability factor

SD is the standard deviation

The term KSD is normally referred to as the margin.

The probability of strength values in the range $f_{ck} \pm$ KSD and below f_{ck} – KSD (risk) for normal distribution is shown in Table 11.2 below;

Table 11.2: The probability of strength values in the range $f_{ck} \pm$ KSD and below f_{ck} – KSD (risk) for normal distribution (Neville and Brooks, 2010)

Probability Factor (k)	Probability of strength values in the range $f_{ck} \pm$ KSD (%)	Probability of strength below f_{ck} – KSD (%)
1.00	68.2	15.9 (1 in 6)
1.64	90.0	5.0 (1 in 20)
1.96	95.0	2.5 (1 in 40)
2.33	98.0	1.0 (1 in 100)
3.00	99.7	0.15 (1 in 700)

In the Eurocodes, the range of the risk of 1 in 20 is recommended (see section 2.2.6.2)

When the number of available test results (n) is less than 20;

- For $f_{ck} \geq 20$ N/mm^2, the standard deviation (SD) is 8 N/mm^2,
- For $f_{ck} < 20$ N/mm^2, the standard deviation is $0.4f_{ck}$

When the number of available test results is equal to or greater than 20;

- For $f_{ck} \geq 20$ N/mm^2, the standard deviation (SD) is 4 N/mm^2,
- For $f_{ck} < 20$ N/mm^2, the standard deviation is $0.2f_{ck}$

Materials Analysis

Coarse aggregate: Crushed granite of nominal maximum size of 20mm
Oven dry relative density = 2.68
Absorption = 0.4% (saturated surface dry - SSD)
Bulk density = 1650 kg/m^3

Specific gravity of cement = 3.15

Fine aggregate: Sharp sand from river
Relative density = 2.64
Absorption = 0.60 %
Bulk density = 1600 kg/m^3

Water-cement ratio

Water – Cement ratio for non-air entrained 25 N/mm^2 concrete = 0.61 (see Table 12.3)

The relationship between water-cement ratio and the 28 day compressive strength of concrete based on ACI code is as shown below in Table 12.3

Table 11.3: Relationship between water-cement ratio and 28 day compressive strength

Compressive strength at 28 days, MPa	Water-cementitious materials ratio by mass	
	Non-air-entrained concrete	Air-entrained concrete
45	0.38	0.30
40	0.42	0.34
35	0.47	0.39
30	0.54	0.45
25	0.61	0.52
20	0.69	0.60
15	0.79	0.70

For maximum size of aggregate of 19mm, and a slump of 75mm, Table 2.0 (Chapter 2) gives a water demand of 205 kg/m^3. Therefore cement content;

$\frac{205}{C} = 0.61$; Therefore the cement content C = $\frac{205}{0.61} = 336.06$ kg/m^3

Table 11.4: Relationship between bulk volume of dry-rodded coarse aggregate and fineness moduli per unit volume

Nominal maximum size of aggregate, mm (in.)	Bulk volume of dry-rodded coarse aggregate per unit volume of concrete for different fineness moduli of fine aggregate*			
	2.40	2.60	2.80	3.00
9.5 (⅜)	0.50	0.48	0.46	0.44
12.5 (½)	0.59	0.57	0.55	0.53
19 (¾)	0.66	0.64	0.62	0.60
25 (1)	0.71	0.69	0.67	0.65
37.5 (1½)	0.75	0.73	0.71	0.69
50 (2)	0.78	0.76	0.74	0.72
75 (3)	0.82	0.80	0.78	0.76
150 (6)	0.87	0.85	0.83	0.81

For 19mm aggregate with fineness modulus of 2.60, the bulk volume of dry rodded coarse aggregate per m^3 of concrete is 0.64 (Table 11.4). Therefore;

Weight per m^3 = 0.64 × 1650 = 1056 kg/m^3
Approximate air content = 2%
Mass of fine aggregate Mf = $\gamma_f \left[1000 - \left(W - \frac{C}{\gamma} + \frac{M_c}{\gamma_c} + 10A\right)\right]$
Where;
γ_f = Specific gravity of fine aggregate (saturated surface dry)

W = Mixing water requirement

C = Cement Content

γ = Specific gravity of cement (take value as 3.15 unless otherwise specified)

M_c = Coarse Aggregate content

γ_c = Specific gravity of coarse aggregate (saturated surface dry)

A = Air content (%)

Mf = 2.64 $\left[1000 - \left(205 + \frac{336.06}{3.15} + \frac{1056}{2.68} + 10(2)\right)\right]$ = 724.11 kg/m^3

Volume computations

Water = $\frac{205}{1 \times 1000} = 0.205$ m^3

Cement = $\frac{336.06}{3.15 \times 1000} = 0.1066$ m^3

Air = $\frac{2}{100} = 0.02\%$

Coarse aggregate = $\frac{1056}{2.68 \times 1000} = 0.394$ m^3

Fine aggregate = $\frac{724.11}{2.64 \times 1000} = 0.274$ m^3

Summary of trial mix

Ingredient	Weight (kg/m^3)	Volume (m^3)
Water	205	0.205
Cement	336.06	0.1066
Fine Aggregate	724.11	0.274
Coarse Aggregate	1056	0.394

We can see that the most probable mix ratio for 25 N/mm^2 concrete grade without considering any margin for test failure is 1: 2.15: 3.142 (based on weight), and 1:2.57:3.696 (based on volume).

Our quantity estimation will be based on the ratio for the volume;

Quantity of cement needed

For 1:2.57:3.696 mix, the total volume is (1+ 2.57 + 3.696) = 7.266 m^3

The ratio of cement = $\frac{1}{7.266}$ = 0.1376 m^3

Density = $\frac{Mass}{Volume}$

Bulk Density of cement = 1440 kg/m^3

1440 = $\frac{Mass}{0.1376}$; Such that the mass of cement = 198.144 kg

We normally make allowance of 1.54 to account for the reduction in volume when we add water to the dry concrete.

Hence mass of cement = 1.54 x 198.144 = 305.14 kg

But 1 bag of cement = 50 kg

Hence the number of bags required per metre cube of 1:2.57:3.696 mixture = $\frac{305.14}{50}$ = 6.1 bags

For 13.636 m^3 of concrete = 6.1 x 13.636 = 83.1796 bags

Making 20% allowance for wastage = 1.2 x 83.1796 = 100 bags of cement

Quantity of fine aggregate needed (sharp sand)

For 1:2.57:3.696 mix, the total volume is (1+ 2.57 + 3.696) = 7.266 m^3

The ratio of sand = $\frac{2.57}{7.266}$ = 0.3537 m^3

We normally make allowance of 1.54 to account for the reduction in volume when we add water to the dry concrete.

Hence volume of fine aggregate = 1.54 x 0.3537 = 0.545 m^3

For 13.636 m^3 of concrete = 0.545 x 13.636 = 7.43162 m^3

Making 20% allowance for wastage = 1.2 x 7.43162 = 8.9179 m^3

Quantity of coarse aggregate needed (granite)

For 1:2.57:3.696 mix, the total volume is (1+ 2.57 + 3.696) = 7.266 m^3

The ratio of sand = $\frac{3.696}{7.266}$ = 0.50867 m^3

We normally make allowance of 1.54 to account for the reduction in volume when we add water to the dry concrete.

Hence volume of coarse aggregate = 1.54 x 0.50867 = 0.78335 m^3

For 13.636 m^3 of concrete = 0.78335 x 13.636 = 10.682 m^3

Making 20% allowance for wastage = 1.2 x 10.682 = 12.818 m^3

Total Quantity for foundation concrete works (footing and blinding)

Cement = 100 + 7 = 107 bags

Sand = 0.7032m^3 + 8.9179 m^3 = 9.6211 m^3 (15.393 tonnes)

If the capacity of a 5 tonne tipper is 3.8m^3;

Therefore number of trips of sand = $\frac{9.6211}{3.8}$ = 2.532 trips (say 3 trips)

Granite = 1.4066 m^3 + 12.818 m^3 = 14.2246 m^3 (23.471 tonnes)

For a 5 tonne tipper, number of trips of granite = $\frac{14.2246}{3.8}$ = 3.7433 trips (say 4 trips)

11.2.3 Hint on Concrete Work Labour

At the foundation stage, labour is usually charged per bag of cement casted. So let us assume that the rate is ₦500 per bag, then the labour cost of casting the foundation will be;

107 bags x ₦500 = ₦53,500 (excluding the cost of supervision and pegging the foundation bottom to ensure adequate thickness)

11.2.4 Reinforcement

From the bending schedule already prepared;

Provide 28 pieces of Y12mm bars for the baskets (footing reinforcements) (0.2886 tonnes)

Provide 18 pieces of Y16mm bars for the starter bars (0.3303 tonnes)

Provide 9 pieces of Y8mm bars for the starter bars links (0.04092tonnes)

11.2.5 Block work

In Nigeria, 9 inches hollow block (450mm x 225mm x 225mm) is normally employed for the construction of block work of multi-storey buildings. The length of the block is 450mm, the height is 225mm, and the width is 225mm. Therefore, the plane area of each block is (0.45m x 0.225m) = 0.10125m^2. Therefore, the number of blocks in one metre square is given by;

No of blocks per metre square of wall = $\frac{1.0}{0.10125} \approx$ 10 blocks

This value is sufficiently accurate for all block work estimates.

Now take a look at the foundation layout again, we are going to raise blocks up to the DPC level (0.5m above the ground). We know that the foundation thickness will rise at least 150mm of the foundation depth (especially at the strip footing area). So let us generally assume that the height for the wall is 1500mm – 150mm = 1350mm.

The next thing is to calculate the total wall length based on the excavated perimeter.

Conservatively, let us take the total length of the strip we have calculated, which is 72.214m

Therefore, total area of wall = 1.35m x 72.214m = 97.4889 m^2

If 1 m^2 of wall needs 10 blocks, therefore 97.4889m^2 will need 975 blocks

Making 25% allowance for waste, total number of blocks needed to come up to DPC = 1.25 x 975 = 1218.75 blocks (say 1220 blocks)

11.2.5.1 Moulding Blocks

At this stage, if the client is purchasing blocks, this number of blocks can be purchased. But if the client is deciding to mould blocks by himself and asks for your advice, this is how you go about it.

The volume of hollow 9 inches block is 0.01603 m^3

It is very advisable to mould at 35 blocks per bag. This is adequate for most construction works.

Now, the volume of 35 nine inches block will be 35 x 0.01603 = 0.56105 m^3

The volume of 1 bag of cement is 33 litres = 0.033 m^3

The volume of a standard builder's wheel barrow is 0.065 m^3 (unheaped)

The volume of a standard head pan = 0.0175 m^3

Now, we can estimate the number of wheel barrow trips that the moulder should provide in order to make 35 blocks from one bag of cement.

Total volume required = 0.56105 m^3

Let the number of wheel barrow trips be x

Hence, 0.033 + x(0.065) = 0.56105 m^3

Therefore, x = 8.12 trips (but note that for accuracy, the wheel barrow must not be heaped). If you are heaping the wheel barrow, you can reduce the number of trips to 6. No account has been made for possible reduction in volume because it is negligible.

We can therefore estimate the materials for the purposes of purchase;

If 1 bag of cement is needed for 35 blocks, therefore 35 bags of cement is needed to mould 1220 blocks

The volume of sand required to make 35 blocks = 8.12 x 0.065 = 0.5278 m^3

If 0.5278 m^3 is required to make 35 blocks, therefore, 18.3976 m^3 (29.436 tonnes) is needed to make 1220 blocks

For a 5 tonne tipper of 3.8 m^3 capacity, we have to order for 5 trips.

Therefore summarily, we need 35 bags and 5 trips of sand to mould the blocks that will raise the building to DPC level (total number of blocks required is 1220 blocks)

11.2.5.2 Block work Labour

Block moulders will charge per bag of cement moulded. Let us assume that one bag of cement is moulded at ₦700, therefore for 35 bags of cement, the cost of labour for moulding will be ₦24,500.

Masons normally charge per block laid, which is better than 'day pay'. If the masons are charging ₦40 per block laid, therefore for the actual 975 blocks needed to be laid, we will pay the masons ₦39,500 for the block work. Masons normally pay the labourers serving them.

11.3 Calculation of concrete quantity for column stubs

Height of column stub from foundation to DPC = 1.35m
Dimensions of column = 0.23m x 0.23m
Number of columns = 20
Therefore the total volume = 20 x (0.23 x 0.23 x 1.35) = 1.4283m^3

Using the same mix ratio for foundation (25 N/mm^2)

For 1.4283 m^3 of concrete = 6.1 x 1.4283 = 8.712 bags

Making 20% allowance for wastage = 1.2 x 8.712 = 11 bags of cement

Quantity of fine aggregate needed (sharp sand)

For 1.4283 m^3 of concrete = 0.545 x 1.4283 = 0.7784 m^3

Making 20% allowance for wastage = 1.2 x 0.7784 = 0.934 m^3

Quantity of coarse aggregate needed (granite)

For 1.4283 m^3 of concrete = 0.78335 x 1.4283 = 1.11885 m^3

Making 20% allowance for wastage = 1.2 x 1.11885 = 1.343 m^3

These quantities should be added to the quantity for foundation.

11.4 Laterite Earth Fill

The DPC of a ground floor is normally casted using a weaker concrete mix ratio. This is normally called the 'German floor'. Hardcore stones or BRC mesh is normally provided just before the concrete casting takes place. Underneath the hardcore is laterite earth fill and the quantity required should be estimated based on the floor area and the thickness of the fill.

A little calculation will show that the ground floor area requiring fill is 92.161 m^2

Thickness of earth fill = 0.5m (ground level to DPC)

Therefore the volume of laterite required = 46.0805m^3

Making 25% allowance for bulking, losses, and wastage, we therefore provide = 1.25 x 46.0805 = 57.6 m^3

For a 5 tonne tipper, provide $\frac{57.6}{3.8}$ = 15.15 trips of laterite for filling the house to DPC level

11.5 DPC German Floor (1:3:6 concrete)

Area of German floor = 101.641 m^2

Thickness of floor = 100mm = 0.1m

Volume of concrete required = 0.1 x 101.641 = 10.1641m^2

Quantity of cement needed

For 1:3:6 mix, the total volume is (1+3+6) = 10m^3

From our previous calculations;

For 1.3024m^3 of concrete = 4.4352 x 1.3024 = 5.776 bags

Making 20% allowance for wastage = 1.2 x 5.776 = 6.93 bags (say 7 bags)

Quantity of fine aggregate needed (sharp sand)

For 1:3:6 mix, the total volume is (1 + 3 + 6) = 10m^3

The volume ratio of sand = $\frac{3}{10}$ = 0.3m^3

Making allowance of 1.54 to account for reduction in volume on adding of water

Hence volume of sand = 1.54 x 0.3 = 0.462 m^3

For 10.1641 m^3 of concrete = 0.462 x 10.1641 = 4.696 m^3

Making 20% allowance for shrinkage and wastage = 1.2 x 4.696 = 5.635m^3 of sand (9.0158 tonnes)

Quantity of coarse aggregate needed (granite)

For 1:3:6 mix, the total volume is (1+3+6) = 10m^3

The volume ratio of granite = $\frac{6}{10}$ = 0.6 m^3

Making allowance of 1.54 to account for reduction in volume on adding of water

Hence volume of granite = 1.54 x 0.6 = 0.924m^3

For 10.1641 m^3 of concrete = 0.9024 x 10.1641 = 9.3916 m^3

Making 20% allowance for shrinkage and wastage = 1.2 x 9.3916 = 11.269 m^3 of granite (18.595 tonnes)

A little consideration will show that the above steps can be repeated for the purpose of the ground floor block work and concrete works (columns, lintels, and floor slab).

11.6 Estimation of form work quantity for the construction of the floor slab

Form work for slabs that are normally employed are planks (soft wood), which are normally sold in the dimensions of 12 feet (3.6m) x 12inches (300mm), and a thickness of 1 inch (25mm). This process is popularly called boarding, and the planks are supported by bamboo or steel props. By implication, the area of each plank is (3.6m x 0.3) = 1.08m^2

Now, let us calculate the total area of the floor slab to be boarded, which gave us 107.89m^2

Therefore, the number of planks required for the boarding is given by;

Number of planks required for boarding = $\frac{107.89}{1.08}$ = Provide 100 pieces of 12' x 12'' x 1''planks

11.7 Other Estimation guides

Mortar for laying blocks

To lay 1000 blocks, you need 3 bags of cement and 0.6m^3 of sand (using the mix ratio of 1 bag of cement to 3 wheel barrows of sharp sand)

Plastering

To plaster 100 m^2 of wall (15mm thick), you need 10 bags of cement and 2 m^3 of sand (using the mix ratio of 1 bag of cement to 3 wheel barrows of plaster sand)

Binding Wire

Approximately 9 – 13kg of binding wire is required for 1 tonne of reinforcement. However, the larger the diameter of bar, the smaller the quantity of binding wire required.

CHAPTER 12

DISCUSSION AND CONCLUSION

12.1 Introduction

In this chapter, we are going to look at some of the issues that were discovered during the analysis, by going through the results obtained from the different methods of analysis and design.

12.2 Discussion on Floor Slab Analysis and design

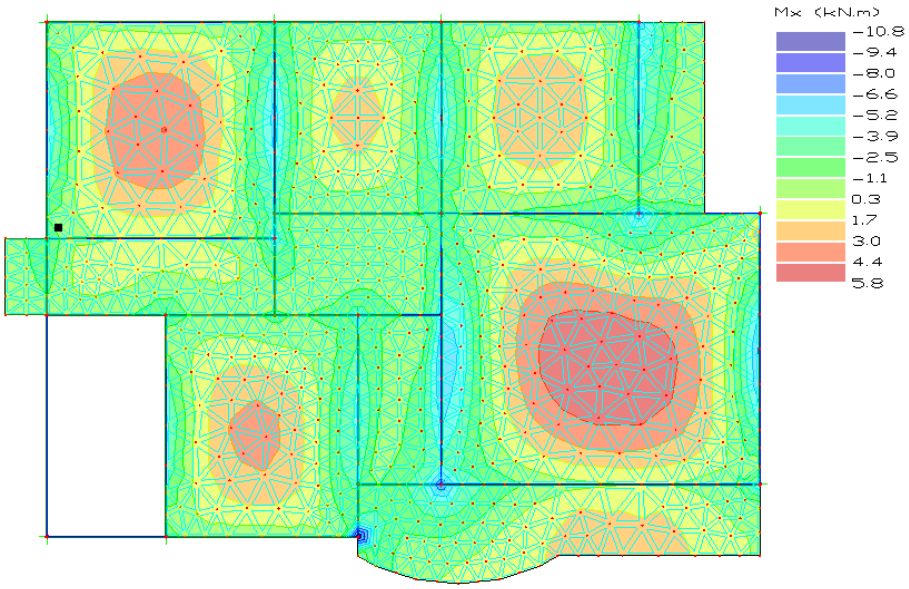


Figure 12.1: Mx Moment Contour plot from CSC Orion

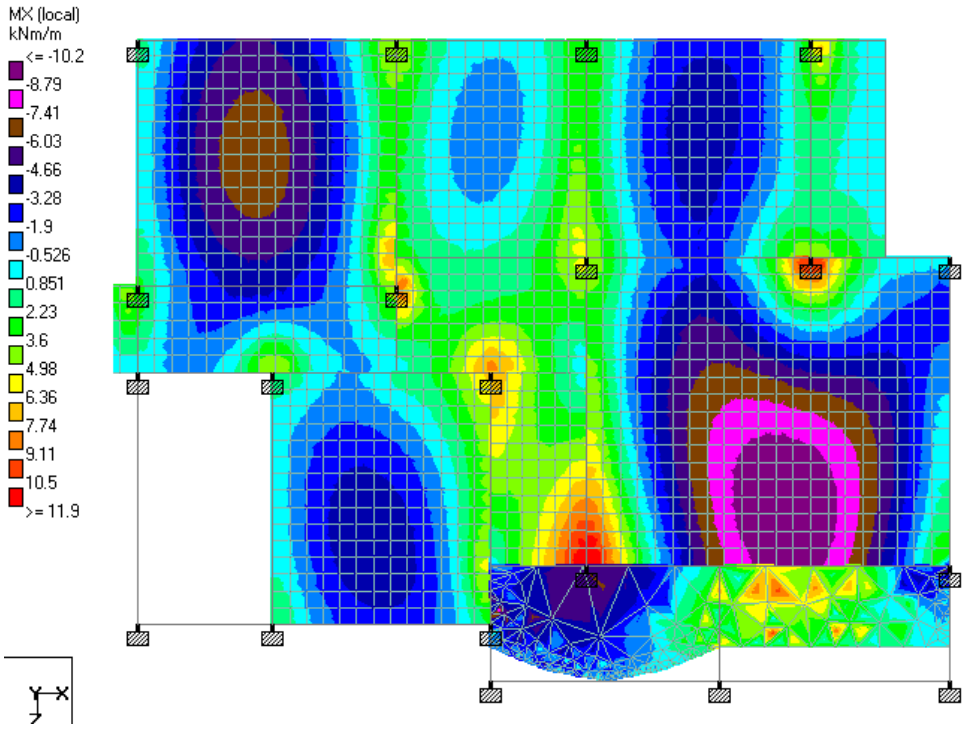


Figure 12.2: Mx Moment Contour plot from Staad

As can be seen from Chapter 5, the structural design of the floor slabs yielded approximately the same results, irrespective of approach. The finite element analysis of the results between Orion and Staad were also found to be similar irrespective of the fact that no deliberate attempt was made to make the meshing uniform. In manual analysis, we often assume the slab supports (beams) to be fully rigid, but in finite element 3D analysis such as the ones carried out in this analysis, the beams themselves are deflecting, and this in turn affects the results produced. Also, the variation in the flexural rigidities of the members can also be considered. For instance, we reduced the stiffness of the slab by 75%, and increased the beam stiffness by 50% in Orion, while no such thing was done in Staad. This can also account for the variations encountered in the analysis. However, each of the methods can be employed confidently for the purpose of analysis and design. Remember that it is very important to include the *Wood and Armer* effects (considering twisting) when using both software for slab design.

12.3 Discussion on Beam Design

Beam design using Orion can be considered to be very effective and efficient. The drawings produced are also very quality. Orion calculates the flange width automatically in reinforced concrete design, but a little consideration will show that this is not possible in Staad. For a beam to be designed as flanged in Staad, it simply implies that you will have to model the beam as flanged during your modelling, before proceeding to Staad RC design. This can be a daunting task at the onset of an analysis process. Furthermore, there were several variations in the some of the members based on the internal stresses results produced for the beams. These variations can stem from modelling assumptions or loading conditions. Otherwise, the design results from Staad are also very amazing and practical, with a very beautiful and detailed calculation sheet.

Some of the controllable variations are shown below, as we consider beam No 5;

BEAM NO 5

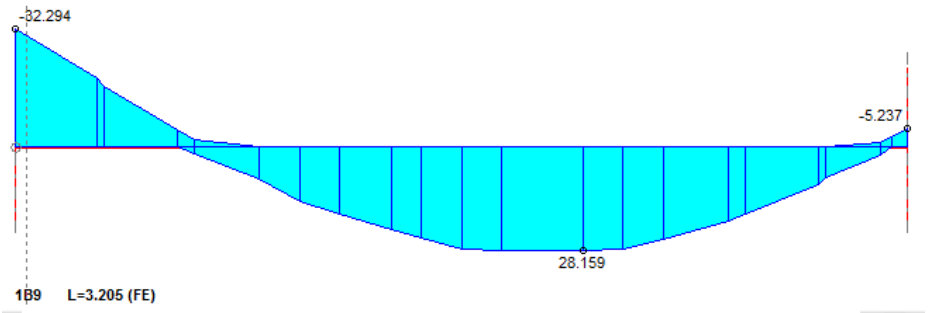


Figure 12.3: Beam No 5 bending moment diagram from Orion

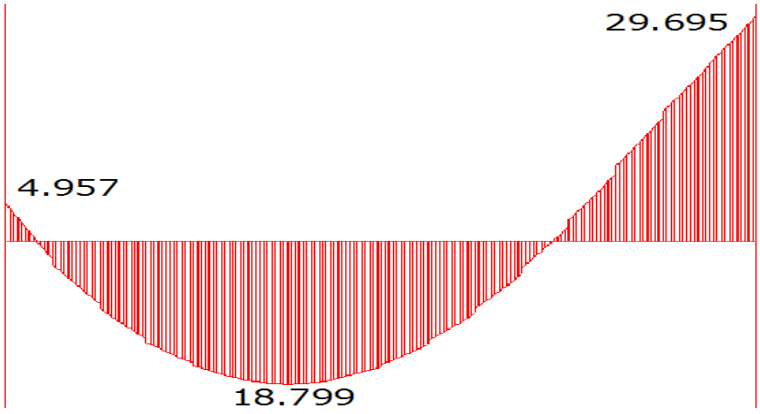


Figure 12.4: Beam No 5 bending moment diagram from Staad

Now a little consideration will show that the beam has been considered as simply supported at one end, and relatively fixed at the other end both Orion and Staad. However, in our manual analysis, we considered it as simply supported as both ends. This stems from the assumption that we treat the points where beams terminates as simply supported in our manual analysis. Now, let us carry out the analysis by considering the beam to be fixed at one end in our analysis.

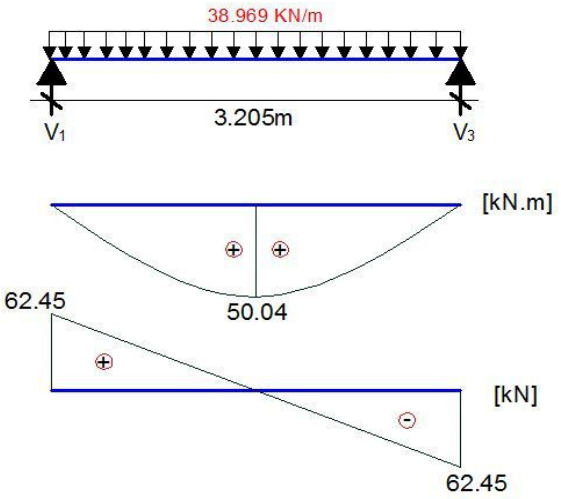


Figure 12.5: Beam No 5 internal forces diagram assuming both ends to be simply supported

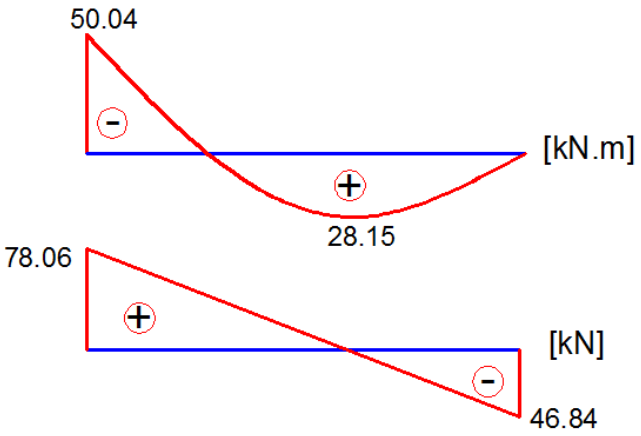


Figure 12.6: Beam No 5 internal forces diagram assuming one end to be simply supported and the other fixed

Now, if we compare the results from where we considered the beam to be fixed (Figure 12.6), we can realise that the result of the analysis is also far from the result of finite element analysis (see figures 12.3 and 12.4). The next thing therefore to look at is the loading. Remember that we have assumed that the load from the cantilever slab (PANEL 10) is totally transferred to the beam, while in reality, this is not actually the case. Let us carefully look into that;

Also, we can notice the same thing with beam No 8, the results are shown below;

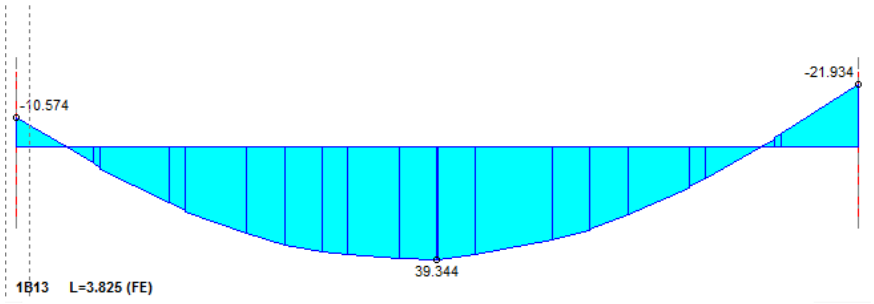


Figure 12.7: Beam No 8 bending moment diagram from Orion

Figure 12.8: Beam No 8 bending moment diagram from Staad

However, manual analysis gave the following results assuming simply supports;

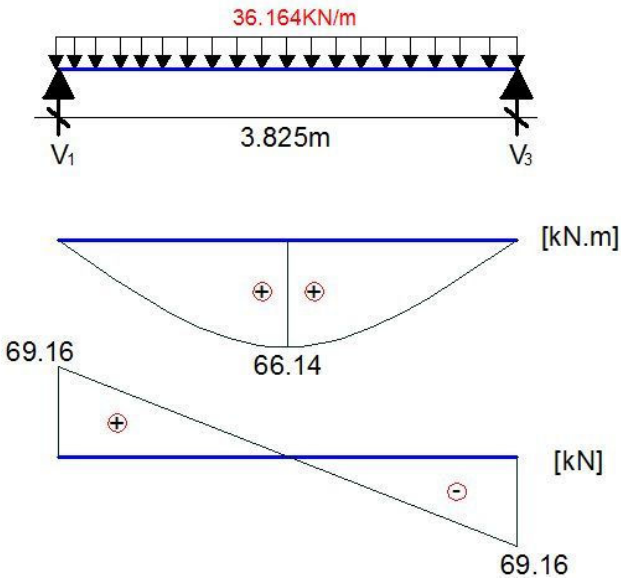


Figure 12.9: Beam No 8 internal forces diagram assuming both ends to be simply supported

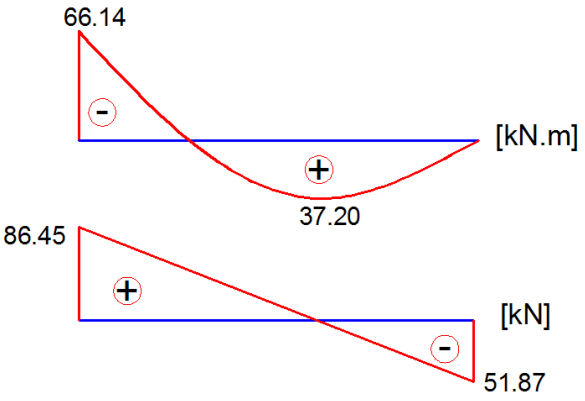


Figure 12.10: Beam No 8 internal forces diagram assuming one end to be simply supported and the other fixed

The implication of the results is that there are some very practical scenarios that cannot be really captured by manual analysis. A little consideration will show that both simply supported at both ends and simply supported and fixed at one end assumptions cannot capture the finite element results for beams in such category. This is because, the actual degree of fixity cannot be determined from manual analysis. The implication of the result is that whichever assumption you follow in manual analysis, provided that you follow the adequate detailing requirements, the structure will be good and will perform satisfactorily. However, the result that mostly captures the results is assuming the ends to be simply supported, using CSC Orion results as a guide. By assuming simply supported/fixed at the ends, the hogging bending moment is overestimated. Let me show you how simply supported assumptions with adequate detailing guides satisfies the requirements;

By following the simply supported requirements (using Figure 12.9);

Area of bottom steel required = A_{S1} = 436 mm^2; A_{Sprov} = (3Y16mm) 603 mm^2

Therefore, following the standard detailing guides, the area of steel at the supports (see section 9.6 of Chapter 9) = 0.5As_1 = 0.5 x 436 = 218 mm^2; A_{sprov} = 2Y12mm (226 mm^2)

Now considering the actual maximum hogging moments from Orion (see Figure 12.7);

M_{Ed} = 21.934 KNm (assuming the maximum from Orion); A_{s1} = 144.5920 mm^2

This will show that the provided the reinforcement for the top reinforcement is very adequate. But if we had assumed the supports as fixed at both ends, we would have overestimated the support moments.

12.4 Discussion on Column Design

The results of the column moments are very interesting. As shown below, there are negligible differences with the different methods. Manual analysis gave the most uneconomical results, because of the issue of over simplification, and the assumptions employed. If you can go through the beam results, you will encounter cases wherein the some loading cases are conservatively assumed. These

also contribute to the higher axial loads encountered in manual analysis. Another cause of error is also the nature of assumptions made in load transfer, for instance, the load transfer of cantilever panels. Also, we know that we make the unconservative assumptions when we assume beams to be continuous, other than treating them as sub-frames. The axial load comparison report is as shown below;

Table 42.1: Column Axial Load Comparison from the three methods employed

Column	Axial Load from manual analysis (KN)	Axial Load from Orion (KN)	Axial Load from Staad (KN)
A1	127.130	126.6	130.684
A3	225.190	202.3	201.632
A5	156.580	155.9	163.207
A7	165.480	133.9	140.392
B5	374.640	283.0	272.290
B7	188.217	258.8	266.620
B8	121.137	112.7	118.088
C1	201.187	165.5	157.085
C3	261.217	223.6	217.672
D1	141.930	100.0	100.283
D2	178.277	174.3	183.828
D4	126.597	177.8	169.082
E5	399.887	332.6	300.245
E8	233.847	229.3	221.446
F1	57.957	60.0	74.201
F2	185.377	149.1	148.346
F4	72.987	100.9	105.328
G4	24.986	20.2	27.313
G6	60.546	61.7	64.852
G8	22.776	22.6	31.817
Total	**3325.945**	**3090.8**	**3094.411**

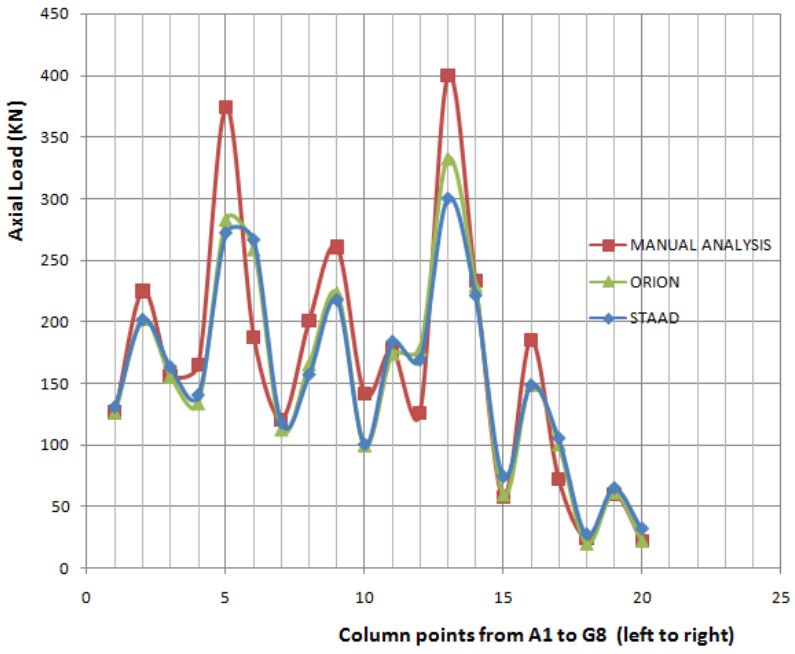


Figure 12.11: Graphical representation of Axial Load results

If you look at the Table 12.1 above, you can rightly verify that the difference in the summation of the axial load results at ultimate limit state between the computer based methods are not very much, with the results from Staad being 0.1167% higher than the result from Orion. This is very negligible.

However, the result from manual analysis is 7.07% greater than the results from Orion, and 6.961% greater than the result from Staad. The possible reasons for this variation have been highlighted above, with the maximum

variations occurring at columns B5 and E5. However, the variation in column B5 is very amazing. This calls for more investigation, and I will be glad to get your feedback on why the difference is much based on the methods employed in this book. The graphical representation of the variations from columns A1 to G8 is shown in Figure 12.11 above;

12.5 Conclusion

(1) The results from Staad for the beams are more economical than the results from Orion and manual analysis.

(2) Manual analysis can be confidently employed in the analysis and design of structures provided the detailing guides are adequately followed.

(3) Staad Pro is mainly for engineers, if you do not have adequate engineering knowledge, it will be a very big bite to chew.

(4) Staad and Orion are very reliable provided the inputs and modelling is correct.

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EN 1997-1:2004 (Eurocode 7): Geotechnical design - **Part** 1: General rules

NA to BS EN 1991-1-1:2002: UK National Annex to Eurocode 1: Actions on structures - Part 1-1: General actions - Densities, self-weight, imposed loads for buildings

Neville A.M., Brooks J.J (2010): Concrete Technology (2nd Edition). Longman Group UK Limited ISBN 978-0-273-73219-8

Oyenuga V.O. (2011): Simplified Reinforced Concrete design (A Consultant/Computer-Based Approach). *Astrols Limited (2nd Edition).*

prEN 1991-1-1: 2001 (Eurocode)1: Actions on structures - Part 1-1: General actions - Densities, self-weight, imposed loads for buildings

Reynolds C.E, Steedman J.C. (2005): "Reinforced Concrete Designer's Handbook" *Spon Press, Taylor and Francis Group (10th Edition).*

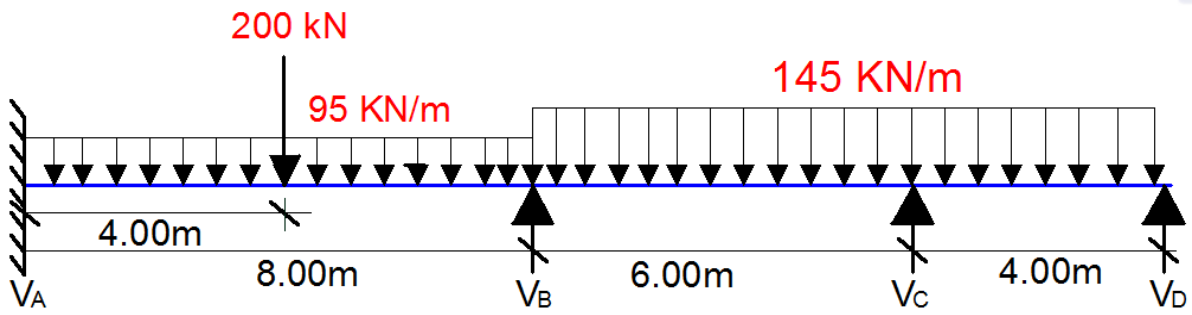
Reynolds C.E, Steedman J.C., Threlfall A.J. (2008): "Reynolds Reinforced Concrete design Handbook." *Spon Press, Taylor and Francis Group (11th Edition).*

Steven H. Kosmatka, Beatrix Kerkhoff, and William C. Panarese (2003): Design and Control of concrete mixtures (14th Edition) Portland Cement Association 2003

APPENDIX 1

ON THE DESIGN OF DOUBLY REINFORCED BEAMS

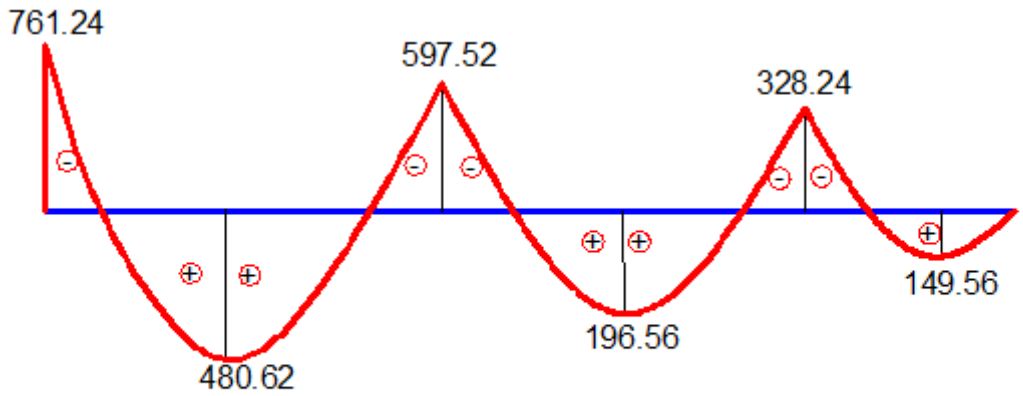
Let us consider the beam loaded with ultimate load as shown below;



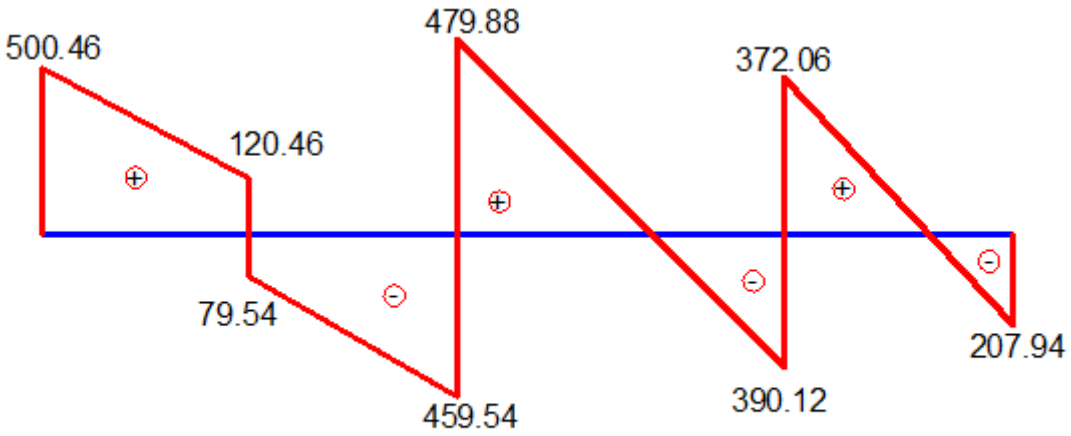
Use the following data:

Fck= 35 N/mm^2, Fyk = 460 N/mm^2, Depth of beam = 600mm, Width = 400mm, Nominal cover to reinforcement = 35mm

On analysis, the internal forces obtained from the externally applied load are as follows;



BENDING MOMENT DIAGRAM (KN.m)



SHEAR FORCE DIAGRAM (KN)

Structural Design

Span A – B (Assume Flange width = 1500mm)

M_{Ed} = 480.62 KNm

Effective depth (d) = 600 – 35 – $\frac{25}{2}$ – 10 = 543 mm (Assuming ϕ25mm bars and ϕ10mm links)

k = $\frac{M_{Ed}}{f_{ck}\, b_{eff} d^2}$ = $\frac{480.62 \times 10^6}{35 \times 1500 \times 543^2}$ = 0.0310

Since k < 0.167 No compression reinforcement required

$z = d\left[0.5 + \sqrt{(0.25 - 0.882K)}\right]$ = $z = d\left[0.5 + \sqrt{(0.25 - 0.882(0.0310)}\right]$ = 0.95d

A_{s1} = $\frac{M_{Ed}}{0.87 f_{yk} z}$ = $\frac{480.62 \times 10^6}{0.87 \times 460 \times 0.95 \times 543}$ = 2328.09 mm^2

Provide 5Y25 mm BOT (AS_{prov} = 2454 mm^2)

Minimum area of steel required;

fctm = $0.3 \times f_{ck}{}^{2/3}$ = $0.3 \times 35^{2/3}$ = 3.20996 N/mm^2 (Table 3.1 EC2)

$A_{S,min}$ = $0.26 \times \frac{\text{fctm}}{\text{Fyk}} \times b \times d = 0.26 \times \frac{3.20996}{460} \times 400 \times 543$ = 394.07 mm^2

$A_{s,max}$ = 0.04bh = 0.04 x 400 x 600 = 9600mm^2

Therefore the provided reinforcement is ok

Check for deflection at the span;

K = 1.5 for continuous spans

$\rho = \frac{As}{bd} = \frac{2454}{1500 \times 543}$ = 0.00301289 < $10^{-3}\sqrt{35}$

Since $\rho < \rho_0$

L/d = $K\left[11 + 1.5\sqrt{f_{ck}}\frac{\rho_0}{\rho} + 3.2\sqrt{f_{ck}}\left(\frac{\rho_0}{\rho} - 1\right)^{3/2}\right]$

L/d = $1.5\left[11 + 1.5\sqrt{35} \times \frac{0.00591608}{0.00301289} + 3.2\sqrt{35}\left(\frac{0.00591608}{0.00301289} - 1\right)^{3/2}\right]$ = 1.5(11 + 17.425 + 17.907) = 69.498

$\beta_s = \frac{310}{\sigma_s}$

$\sigma_s = \frac{310 f_{yk}\, AS_{req}}{500\, AS_{prov}} = \frac{310 \times 460 \times 2328.09}{500 \times 2454}$ = 270.5669 N/mm^2

$\beta_s = \frac{310}{270.5669}$ = 1.1457

Since the beam is flanged, check the ratio of b/b_w = 1500/400 = 3.75

Since b/bw is greater than 3, multiply the allowable L/d by 0.8

Since the span is greater than 7m, multiply by 7/Span

The allowable span/depth ratio = 7/8 x 0.8 × β_s × L/d = 0.875 x 0.8 × 1.1457 × 69.498 = 55.736

Taking the distance between supports as the effective span, L = 8000 mm

Actual deflection L/d = $\frac{8000}{543}$ = 14.732

Since 275.412 < 9.5864, deflection is deemed to satisfy

Top reinforcements (Hogging moment)

Support A

M_{Ed} = 761.24 KNm

But since the flange is in tension, we use the beam width to calculate the value of k (this applies to all support hogging moments)

k = $\frac{M_{Ed}}{f_{ck}\, bwd^2}$ = $\frac{761.24 \times 10^6}{35 \times 400 \times 543^2}$ = 0.184

Since k < 0.167, compression is required

Area of compression reinforcement A_{S2} = $\frac{M_{Ed} - M_{Rd}}{0.87 f_{yk}(d - d_2)}$

M_{Rd} = 0.167f_{ck}bd^2 = (0.167 x 35 x 400 x 543^2) x 10^{-6} = 689.356 KNm

d_2 = 35 + 12.5 + 10 = 57.5mm

A_{S2} = $\frac{(761.24 - 689.356) \times 10^6}{0.87 \times 460 \times (543 - 57.5)}$ = 369.969mm^2

Provide 2Y16 Bottom (A_{sprov} = 402 mm^2)

Area of tension reinforcement A_{s1} = $\frac{M_{Rd}}{0.87 f_{yk} z}$ + A_{S2}

Where *z* = *d*$\left[0.5 + \sqrt{(0.25 - 0.882K')}\right]$

where

K' = 0.167

z = *d*$\left[0.5 + \sqrt{(0.25 - 0.882(0.167)}\right]$ = 0.82d

A_{s1} = $\frac{M_{Rd}}{0.87 f_{yk} z}$ + A_{S2} = $\frac{689.356 \times 10^6}{0.87 \times 460 \times 0.82 \times 543}$ + 369.969mm = 4238.559 mm^2

Provide 6Y32mm TOP (AS_{prov} = 4825mm^2)

Shear Design

Support A; V_{Ed} = 500.46 KN

$V_{Rd,c}$ = [$C_{Rd,c}$.k. $(100\rho_1 f_{ck})^{\frac{1}{3}}$ + $k_1.\sigma_{cp}$]*bw.d* ≥ (V_{min} + $k_1.\sigma_{cp}$) *bw.d*

$C_{Rd,c}$ = 0.18/γ_c = 0.18/1.5 = 0.12

k = 1+$\sqrt{\frac{200}{d}}$ = 1+$\sqrt{\frac{200}{543}}$ = 1.606 > 2.0, therefore, k = 1.606

$V_{min} = 0.035k^{\frac{3}{2}}f_{ck}^{\frac{1}{2}}$ = $V_{min} = 0.035 \times (1.606)^{\frac{3}{2}} \times (35)^{\frac{1}{2}}$ = 0.421 N/mm^2

$\rho_1 = \frac{As}{bd} = \frac{4825}{400 \times 543}$ = 0.022 > 0.02; Therefore take 0.02

σ_{cp} = N_{Ed}/Ac < 0.2fcd (Where N_{Ed} is the axial force at the section, Ac = cross sectional area of the concrete), fcd = design compressive strength of the concrete.) Take N_{Ed} = 0

$V_{Rd,c}$ = [0.12 × 1.606 $(100 \times 0.02 \times 35)^{\frac{1}{3}}$] 400 × 543 = 172511.992 N = 172.511 KN

Since $V_{Rd,c}$ (172.511) < V_{Ed} (500.46), shear reinforcement is required.

The compression capacity of the compression strut ($V_{Rd,max}$) assuming θ = 21.8° (cot θ = 2.5)

$V_{Rd,max} = \frac{b_w.z.v_1.f_{cd}}{\cot\theta + tan\theta}$

$V_1 = 0.6\left(1 - \frac{f_{ck}}{250}\right) = 0.6\left(1 - \frac{35}{250}\right)$ = 0.516

$f_{cd} = \frac{\alpha_{cc} f_{ck}}{\gamma_c} = \frac{0.85 \times 35}{1.5}$ = 19.833 N/mm^2

Let *z* = 0.9d

$V_{Rd,max} = \left[\frac{400 \times 0.9 \times 543 \times 0.516 \times 19.833}{2.5 + 0.4}\right] \times 10^{-3}$ = 689.83 KN

Since $V_{Rd,c} < V_{Ed} < V_{Rd,max}$

Hence $A_{sw}/S = \frac{V_{Ed}}{0.87 F_{yk} z \cot\theta} = \frac{500460}{0.87 \times 460 \times 0.9 \times 543 \times 2.5}$ = 1.0235

Minimum shear reinforcement;

$A_{sw}/S = \rho_{w,min} \times b_w \times sin\alpha$ (α = 90° for vertical links)

$\rho_{w,min} = \frac{0.08 \times \sqrt{F_{ck}}}{F_{yk}} = \frac{0.08 \times \sqrt{35}}{460}$ = 0.001028

$A_{sw}/S_{(min)} = 0.001028 \times 400 \times 1$ = 0.411

Maximum spacing of shear links = 0.75d = 0.75 × 543 = 407.25

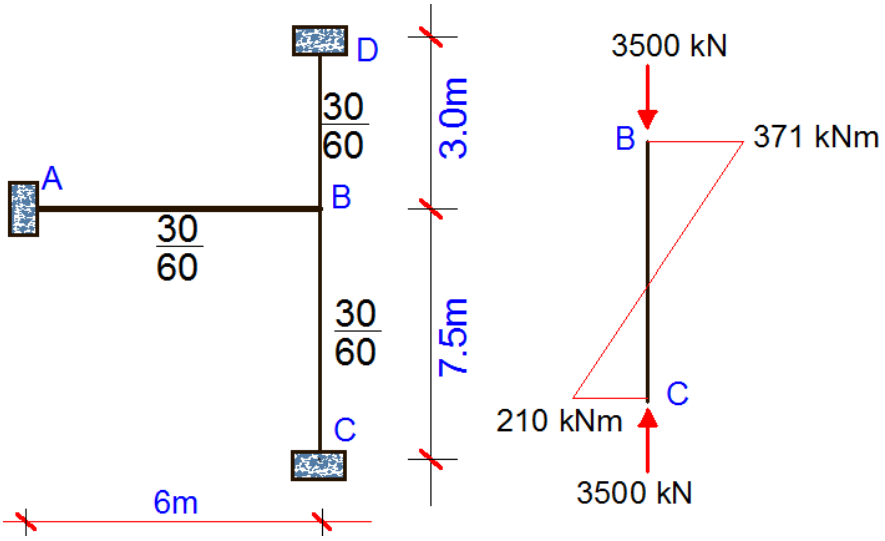
Provide Y10mm @ 150mm c/c as shear links (Asw/S = 0.10467) Ok!!!!

APPENDIX 2

ON THE DESIGN OF SLENDER COLUMNS

Let us consider the structure shown below. The effects of actions on column member BC is as shown below. It is required to design the column using the following data;

f_{ck} = 25 N/mm^2, f_{yk} = 460 N/mm^2, Concrete cover = 35mm



Second moment of area of beam AB = $\frac{0.3 \times 0.6^3}{12}$ = 0.0054 m^4

Stiffness of beam AB (since E is constant) = $\frac{4I}{L}$ = $\frac{4 \times 0.0054}{6}$ = 0.0036

Second moment of area of column BC = $\frac{0.3 \times 0.6^3}{12}$ = 0.0054 m^4

Stiffness of column BC (since E is constant) = $\frac{4I}{L}$ = $\frac{4 \times 0.0054}{7.5}$ = 0.00288

Remember that we will have to reduce the stiffness of the beams by half to account for cracking;

k_1 = $\frac{0.00288}{0.0018}$ = 1.6

Since the minimum value of k_1 and k_2 is 0.1, adopt k_1 as 1.6. Let us take k_2 as 1.0 for base designed to resist moment

Take the unrestrained clear height of column as 7000mm

$$l_o = 0.5 \times 7000\sqrt{\left(1 + \frac{1.6}{0.45 + 1.6}\right) \times \left(1 + \frac{1.0}{0.45 + 1.0}\right)} = 6071 \text{ mm}$$

Radius of gyration i = h/$\sqrt{12}$ = 600/$\sqrt{12}$ = 173.205

Slenderness ratio λ = $\frac{6071}{173.205}$ = 35.051

Critical Slenderness for the x-direction

$\lambda_{\lim} = \frac{20.A.B.C}{\sqrt{n}}$

$A = 0.7$
$B = 1.1$
$C = 1.7 - \frac{M_{01}}{M_{02}} = 1.7 - \left(\frac{-210}{371}\right) = 2.266$
$n = \frac{N_{Ed}}{A_c f_{cd}}$

$N_{Ed} = 3500 \times 10^3$ N

$A_c = 300 \times 600 = 180000$ mm^2

$f_{cd} = \frac{\alpha_{cc} f_{ck}}{1.5} = \frac{0.85 \times 25}{1.5} = 14.167$ N/mm^2

$n = \frac{3500 \times 10^3}{180000 \times 14.167} = 1.3725$

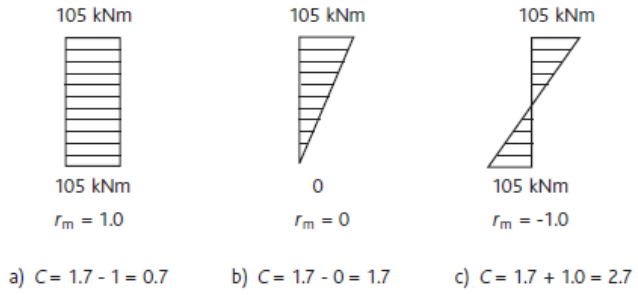


Figure A2.1: Example of how to calculate the value of C (Bond et al, 2006)

$$\lambda_{\lim} = \frac{20 \times 0.7 \times 1.1 \times 2.266}{\sqrt{1.3725}} = 29.786$$

Since 29.786 < 35.051, second order effects need to be considered in the design

Design Moments

M_{Bot} = 210 KNm, M_{Top} = 371 KNm

e_i is the geometric imperfection = $\left(\theta_i \frac{l_0}{2}\right) = \left(\frac{1}{200} \times \frac{6071}{2}\right) = 15.1775$ mm

$e_i N_{Ed} = 15.1775 \times 10^{-3} \times 3500 = 53.121$ KNm

First order end moment

$M_{01} = M_{Bot} + e_i N_{Ed} = -210 + 53.121 = -156.879$ KNm

$M_{02} = M_{Top} + e_i N_{Ed} = 371 + 53.121 = 424.121$ KNm

Equivalent first order moment M_{0Ed}

$M_{0Ed} = (0.6M_{02} + 0.4M_{01}) \geq 0.4M_{02} = 0.4 \times 424.121 = 169.648$ KNm

$= (0.6 \times 424.121 - 0.4 \times 156.879) = 191.721$ KNm

Nominal second order moment M_2

Specified concrete cover = 35mm
Diameter of longitudinal steel = 32 mm
Diameter of links = 10 mm
Thus, the effective depth (d) = h – C_{nom} – ϕ/2 – ϕ_{links}

d = 600 – 35 – 16 – 10 = 539 mm

$1/r_0 = \frac{\varepsilon_{yd}}{0.45\,d}$

$\varepsilon_{yd} = \frac{f_{yd}}{Es} = \frac{(460/1.15)}{200 \times 10^3} = 0.002$

$1/r_0 = \frac{0.002}{0.45 \times 539} = 8.2457 \times 10^{-6}$

$\beta = 0.35 + f_{ck}/200 - \lambda/150$ (λ is the slenderness ratio)

$\beta = 0.35 + 25/200 - 35.051/150 = 0.2413$

$K_\phi = 1 + \beta\phi_{ef} \geq 1.0$ (ϕ_{ef} is the effective creep ratio, assume 0.87)

$K_\phi = 1 + (0.2413 \times 0.87) = 1.2099 \geq 1.0$

Assume Kr = 0.8

$\frac{1}{r} = K\text{r}.K_\phi.\frac{1}{r_0} = 0.8 \times 1.2099 \times 8.2457 \times 10^{-6} = 7.981 \times 10^{-6}$

e_2 is the deflection $= \left(\frac{1}{r}\right)\frac{l_0^2}{10} = 7.981 \times 10^{-6} \times \frac{6071^2}{10} = 29.415$ mm

$M_2 = N_{Ed}.e_2 = 3500 \times 29.415 \times 10^{-3} = 102.954$ KNm

Design Moment M_{Ed}

M_{Ed} = maximum of $\{M_{0Ed} + M_2; M_{02}; M_{01} + 0.5M_2\}$

M_{Ed} = maximum of {191.721 + 102.954 = **294.675 kNm**; **424.121 kNm**; −156.879 + (0.5 × −102.954) = −**208.356** kNm}

Longitudinal Steel Area

$d_2 = C_{nom} + \phi/2 + \phi_{links} = 35 + 16 + 10 = 61$ mm

$\frac{d_2}{h} = \frac{61}{600} = 0.1016$

Reading from chart; $\frac{d_2}{h} = 0.10$;

$\frac{M_{Ed}}{f_{ck}bh^2} = \frac{424.121 \times 10^6}{25 \times 300 \times 600^2} = 0.1571$

$\frac{N_{Ed}}{f_{ck}bh} = \frac{3500 \times 10^3}{25 \times 300 \times 600} = 0.777$

From the chart, $\frac{A_s F_{yk}}{bhf_{ck}} = 0.55$

Area of longitudinal steel required (As) $= \frac{0.55 \times 25 \times 300 \times 600}{460} = 5380$ mm^2

Provide 6Y32 + 2Y25 ($A_{Sprov} = 5806$ mm^2)

$A_{s,min} = \frac{0.1\,N_{Ed}}{f_{yd}} = \frac{0.1 \times 3500 \times 1000}{400} = 875$mm^2, 0.002bh = 0.002 x 300 x 600 = 360 mm^2

$A_{smax} = 0.04bh = 0.04 \times 300 \times 600 = 7200\ mm^2$

Links

Minimum size = $0.25\phi = 0.25 \times 32 = 8mm < 6mm$
We are adopting Y10mm as links
Spacing adopted = 300mm less than {b, h, 20ϕ, 400mm}
Provide Y10 @ 300 mm links

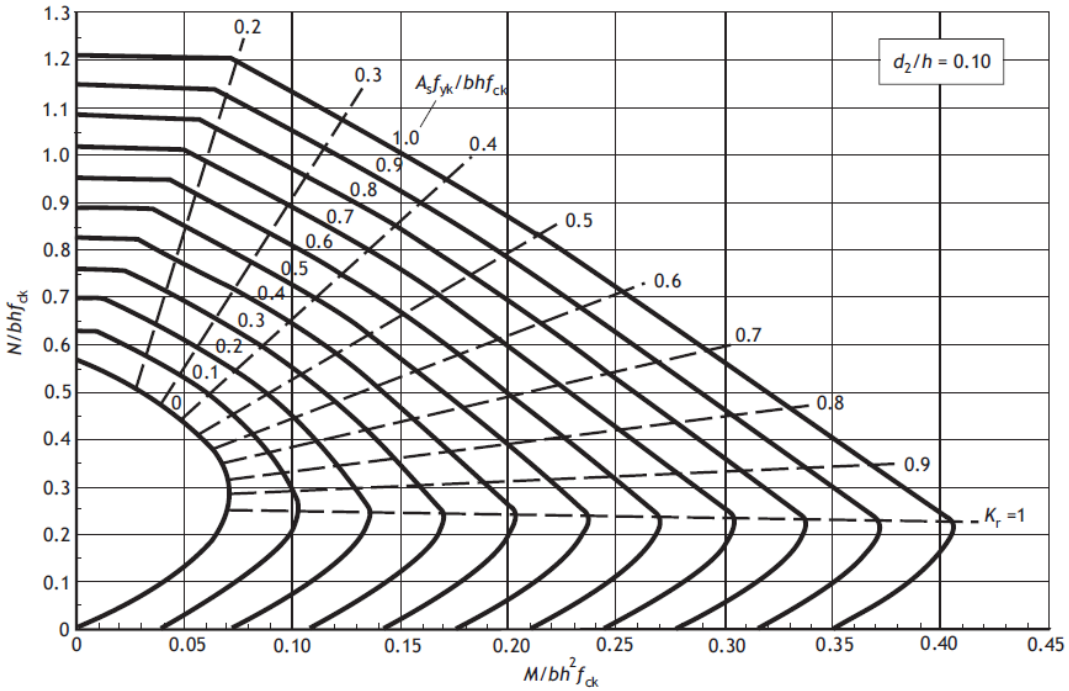


Figure A2.2: Column design chart for $d_2/h = 0.10$ (Bond et al, 2006)

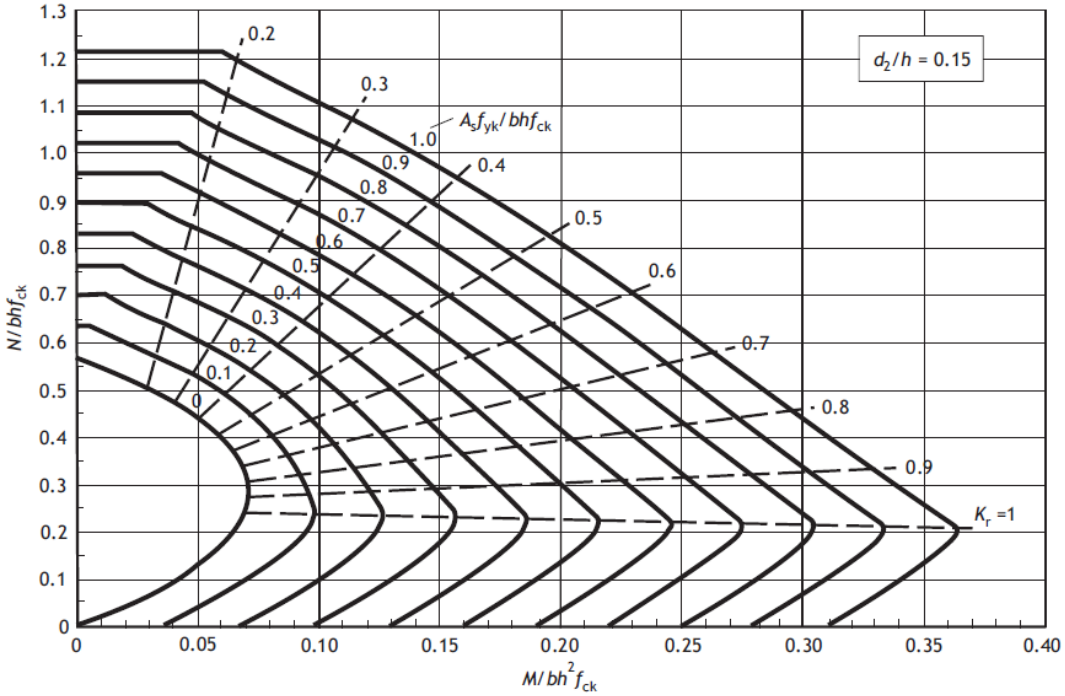


Figure A2.3: Column design chart for $d_2/h = 0.15$ (Bond et al, 2006)

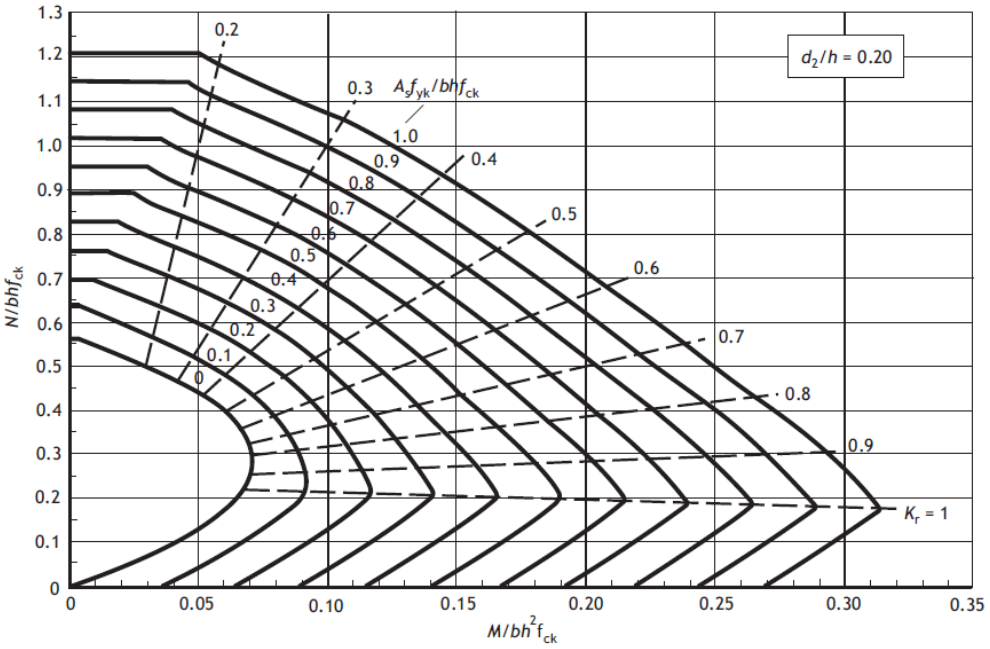


Figure A2.4: Column design chart for $d_2/h = 0.20$ (Bond et al, 2006)

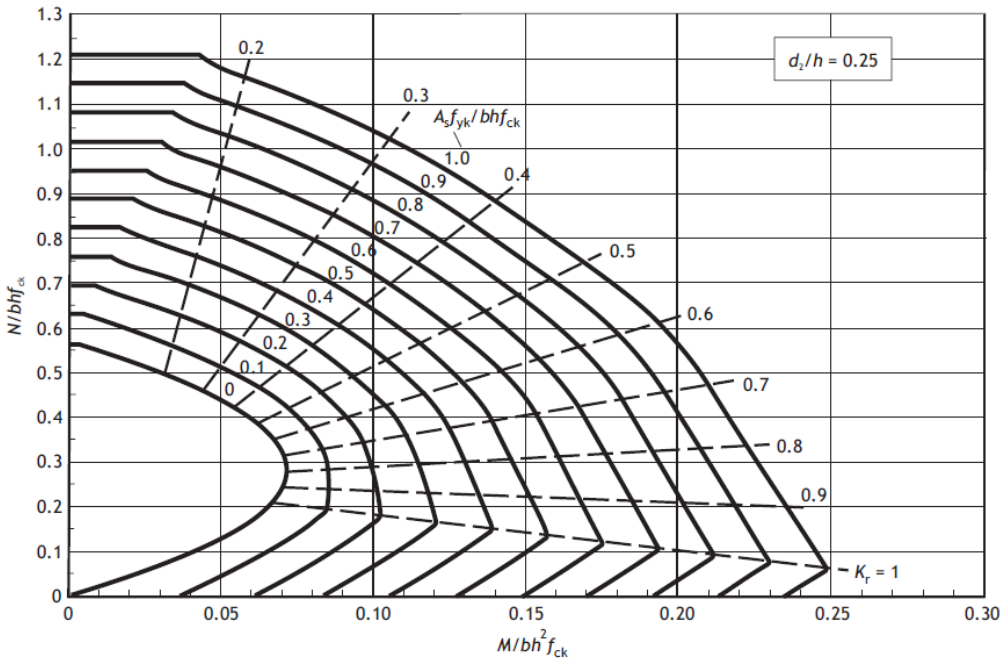


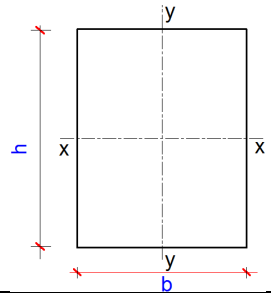
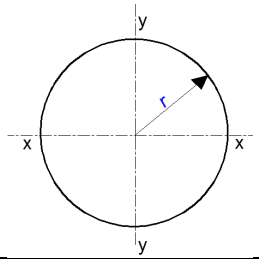
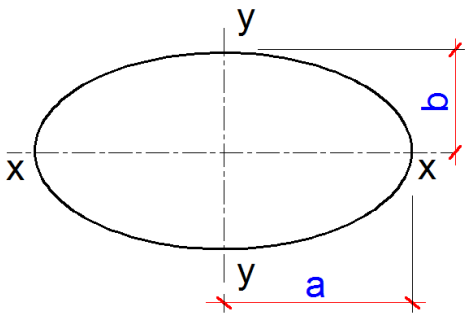
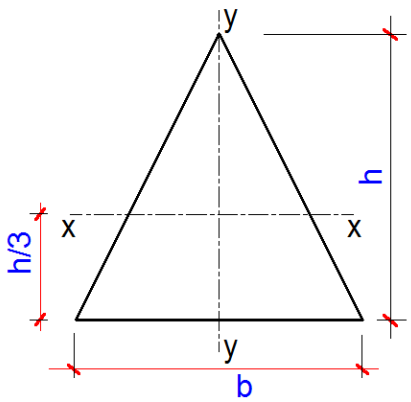
Figure A2.5: Column design chart for $d_2/h = 0.25$ (Bond et al, 2006)

APPENDIX 3

GEOMETRICAL PROPERTIES OF SECTIONS

A3.1 Moment of inertia of sections about the centroidal axes

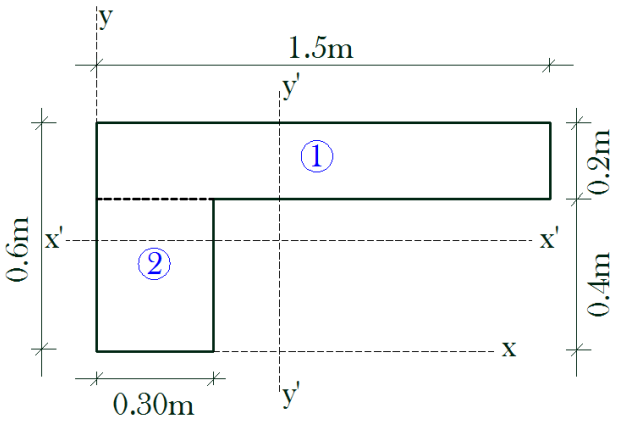
Table of moment of inertia

Shape	I_{xx}	I_{yy}
	$\frac{bh^3}{12}$	$\frac{hb^3}{12}$
	$\frac{\pi r^4}{4}$	$\frac{\pi r^4}{4}$
	$\frac{\pi ab^3}{4}$	$\frac{\pi ba^3}{4}$
	$\frac{bh^3}{36}$ Ixx (about the base) = $\frac{bh^3}{12}$	

A3.2 Moment of inertia of composite shapes

A3.2.1 L – beams

Let us calculate the moment of inertia about the x-x axis of the L-beam shown below



Member	Area A (m^2)	$\bar{y}$ (m)	$A\bar{y}$ (m^3)
1	(1.5 x 0.2) = 0.3	(0.2/2) + 0.4 = 0.5	0.15
2	(0.3 x 0.4) = 0.12	(0.4/2) = 0.2	0.024
	$\sum A$ = 0.42		**$\sum A\bar{y}$ = 0.174**

Therefore, the centroid of the shape $\bar{y} = \frac{\sum A\bar{y}}{\sum A} = \frac{0.174}{0.42}$ = 0.41428m

From parallel axis theorem,

$$I_{xx} = I_G + Ay^2$$

Therefore, the total moment of inertia;

For member 1;

$I_G = \frac{bh^3}{12} = \frac{1.5 \times 0.2^3}{12}$ = 1 x 10^{-3} m^4

y = (0.5 – 0.41428)

$I_{xx1} = I_G + Ay^2$ = 1 x 10^{-3} + 0.3(0.5 – 0.41428)^2 = 3.204 x 10^{-3} m^4

For member 2;

$I_G = \frac{bh^3}{12} = \frac{0.3 \times 0.4^3}{12}$ = 1.6 x 10^{-3} m^4

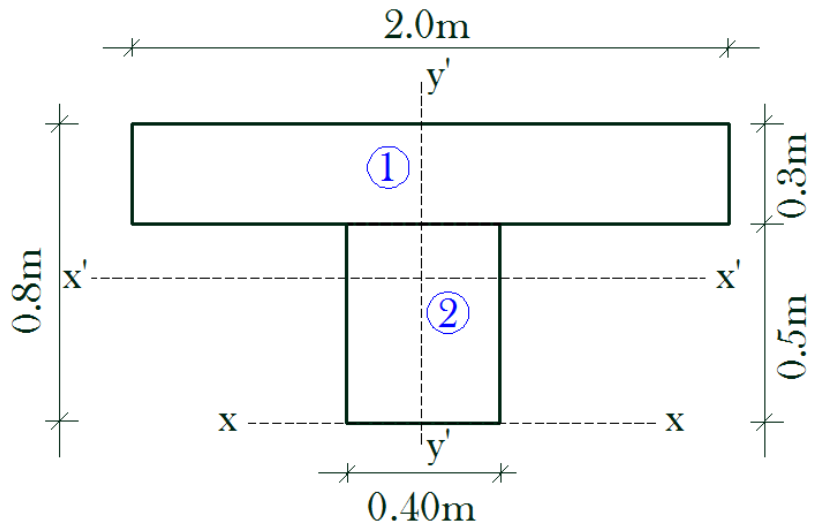
y = (0.41428 – 0.2)

$I_{xx2} = I_G + Ay^2$ = 1.6 x 10^{-3} + 0.12(0.41428 – 0.2)^2 = 7.1099 x 10^{-3} m^4

$Ixx = Ixx_1 + Ixx_2 = (3.204 \times 10^{-3}) + (7.1099 \times 10^{-3})$ = 0.0103 m^4

A3.2.1 T – beams

Let us calculate the moment of inertia about the x-x axis of the L-beam shown below



Member	Area A (m^2)	$\bar{y}$ (m)	$A\bar{y}$ (m^3)
1	(2.0 x 0.3) = 0.6	(0.3/2) + 0.5 = 0.65	0.39
2	(0.4 x 0.5) = 0.2	(0.5/2) = 0.25	0.05
	$\sum A$ = **0.80**		$\sum A\bar{y}$ = **0.44**

Therefore, the centroid of the shape $\bar{y} = \frac{\sum A\bar{y}}{\sum A} = \frac{0.44}{0.80}$ = 0.55 m

From parallel axis theorem,

$$I_{xx} = I_G + Ay^2$$

Therefore, the total moment of inertia;

For member 1;

$I_G = \frac{bh^3}{12} = \frac{2.0 \times 0.3^3}{12}$ = 4.5 x 10^{-3} m^4

y = (0.65 – 0.55)

$I_{xx1} = I_G + Ay^2$ = 4.5 x 10^{-3} + 0.6(0.65 – 0.55)^2 = 0.0105 m^4

For member 2;

$I_G = \frac{bh^3}{12} = \frac{0.4 \times 0.5^3}{12}$ = 4.167 x 10^{-3} m^4

y = (0.55 – 0.25)

$I_{xx2} = I_G + Ay^2$ = 4.167 x 10^{-3} + 0.2(0.55 – 0.25)^2 = 0.022167 m^4

$Ixx = Ixx_1 + Ixx_2$ = 0.0105 + 0.022167 = 0.032667 m^4

APPENDIX 4

MATLAB PROGRAMMING FOR DESIGN AND ANALYSIS

A4.1 MATLAB PROGRAM FOR FLEXURAL DESIGN

```
% ANALYSIS AND DESIGN OF RECTANGULAR AND FLANGED BEAMS PER EUROCODE 2
% THIS PROGRAM WILL CALCULATE THE AREA OF REINFORCEMENT REQUIRED, AND ALSO
% DO THE DEFLECTION VERIFICATION USING THE DEEMED TO SATISFY RULES OF EC2

clc
disp('THIS PROFORMA WAS WRITTEN BY O.U.R UBANI')
disp('ANALYSIS AND DESIGN OF BEAMS PER EUROCODE 2')

% INPUT MATERIAL PROPERTIES
Fck = input('Enter the grade of concrete (N/mm^2)Fck  = ');
Fyk = input('Enter the yield strength of steel (N/mm^2)Fyk = ');

% INPUT DESIGN MOMENT
MEd = input('Enter the ultimate design moment (KNm) MEd = ');

% INPUT SECTION AND DESIGN PROPERTIES
h = input('Enter the depth of beam (mm)h = ');
b = input('Enter the effective flange width of the beam (mm)b = ');
bw = input('Enter the beam width (mm)bw = ');
Cc = input('Enter concrete cover (mm) = ');
dr = input('Enter the diameter of reinforcement (mm) = ');
dl = input('Enter the diameter of links = ');

% CALCULATION OF EFFECTIVE DEPTH
disp('Effective depth d (mm)')
d = h-Cc-(dr/2)-dl % Effective depth
disp('Depth of reinforcement from the face of concrete')
do = Cc+dl+(dr/2)

% ANALYSIS FOR INTERNAL STRESSES
ko = 0.167;
k = (MEd*10^6)/(Fck*b*d^2)
if k>ko
    disp('Since k > ko, Compression reinforcement is required')
    Mcd = (Fck*b*d^2*(k-ko))*10^(-6)
    As2 = (Mcd*10^6)/(0.87*Fyk*(d-do))
    z = 0.5*d*(1+sqrt(1-3.53*ko))
    As1 = ((ko*Fck*b*d^2)/(0.87*Fyk*z))+ As2
else
    disp('Since k < ko, No Compression reinforcement is required')
    disp('Lever arm (la)')
    la = 0.5+sqrt(0.25-0.882*k)
       if la>0.95
           disp('Since la > 0.95,')
        la = 0.95
       else
        la = 0.5+sqrt(0.25-0.882*k)
       end
    As1 =(MEd*10^6)/(0.87*Fyk*la*d)
end
```

```
% MINIMUM AREA OF STEEL REQUIRED
fctm = 0.3*(Fck^(2/3)) %MEAN TENSILE STRENGTH OF CONCRETE (TABLE 3.1 EC2)
ASmin = 0.26*(fctm/Fyk)*bw*d
if ASmin < 0.0013*bw*d
    ASmin = 0.0013*bw*d
end
if As1<ASmin
    As1 = ASmin
else
    disp('Since As1 > Asmin, provide As1 which is the area of steel
required')
end

Asprov1 = input('Enter area of tension steel provided (mm^2) = ');
Asprov2 = input('Enter area of compression steel provided if any (mm^2) =
');


% CHECK FOR DEFLECTION
disp('DO YOU WANT TO CHECK FOR DEFLECTION?')
G = input('ENTER (1) FOR YES OR (0) FOR NO = ');
if G == 1;
    disp('CHECK FOR DEFLECTION')
disp('BASIC SPAN/EFFECTIVE DEPTH RATIO (K)')
disp('CANTILEVER = 0.4')
disp('SIMPLY SUPPORTED = 1.0')
disp('SIMPLY SUPPORTED AND FIXED AT ONE END = 1.3')
disp('FIXED AT BOTH ENDS = 1.5')

K = input('Enter the selected value of k = ');
L = input('Enter the deflection critical length of the member (mm) = ');
disp('Fs is the stress in tensile reinforcement under service loading')
Fs = (310*Fyk*As1)/(500*Asprov1)
disp('Bs = 310/Fs')
Bs = 310/Fs
disp('P is the reinforcement ratio in the section (Asprov1)/(b*d)')
P = (Asprov1)/(b*d)
disp('Po = sqrt(Fck)/1000')
Po = sqrt(Fck)/1000
CV = input('Is the beam is flanged? (1) for YES and (0) for NO = ')
if CV == 1
disp('S is the ratio of beff/bw')
S = b/bw
if b/bw >= 3
    disp('Since b/bw >= 3, Allowable_deflection =
0.8*Bs*K*(11+(1.5*sqrt(Fck)*(Po/P)))')
Allowable_deflection = 0.8*Bs*K*(11+(1.5*sqrt(Fck)*(Po/P)) +
3.2*sqrt(Fck)*((Po/P)-1)^1.5)
else
    bn = (11-(b/bw))/10
    Allowable_deflection = bn*Bs*K*(11+(1.5*sqrt(Fck)*(Po/P)) +
3.2*sqrt(Fck)*((Po/P)-1)^1.5)
end
else
    Allowable_deflection = Bs*K*(11+(1.5*sqrt(Fck)*(Po/P)) +
3.2*sqrt(Fck)*((Po/P)-1)^1.5)
end
if L>7000
    Allowable_deflection = Bs*K*(11+(1.5*sqrt(Fck)*(Po/P)) +
3.2*sqrt(Fck)*((Po/P)-1)^1.5)*(7000/L)
```

```
end
end
Actual_deflection = L/d
if Allowable_deflection > Actual_deflection
    disp('Deflection is satisfactory')
else
    disp('DEFLECTION IS NOT SATISFACTORY !!!!! Increase depth of beam, or
increase area of steel, or both. Then rerun proforma')
end
```

A4.2 MATLAB PROGRAM FOR SHEAR DESIGN

```
% SHEAR DESIGN IN EUROCODE 2
clc
disp('SHEAR DESIGN ACCORDING TO EC2')
disp('THIS PROFORMA WAS WRITTEN BY O.U.R. UBANI')

% MATERIALS PROPERTIES
Fck = input('Enter the grade of concrete (N/mm^2) = ');
Fyk = input('Enter the yield strength of steel (N/mm^2) = ');

VEd = input('Enter the value of shear force at ULS (KN) = ');

% SECTION PROPERTIES
h = input('Enter the depth of beam (mm) = ');
bw = input('Enter the beam width (mm) = ');
Cc = input('Enter concrete cover (mm) = ');
dr = input('Enter the diameter of reinforcement (mm) = ');
dl = input('Enter the diameter of links = ')
d = h-Cc-(dr/2)-dl % Effective depth
do = Cc+dl+(dr/2)

% CALCULATION OF THE SHEAR CAPACITY OF THE SECTION WITH NO SHEAR
REINFORCEMENT
Asprov1= input('Enter the area of steel provided in the shear zone (mm^2) =
') % The reinforcement must exceed the design anchorage length by at least
the effective depth
Crd = 0.12
k1= 1+sqrt(200/d)
if k1>2
    k1=2
end
disp('Reinforcement ratio')
P1 = (Asprov1/(bw*d))
if P1>0.02
    P1 = 0.02
End

% THIS SECTION IS TO BE CONSIDERED IF THERE IS AXIAL FORCE IN THE SECTION
disp('Axial force in the section')
N = input('Enter the value of AXIAL FORCE IF ANY(+VE FOR COMP, AND -VE FOR
TENSION)(kN) = ')
disp('Axial stress in the section (Ds)')
Ds = (N*1000)/(bw*h)
if Ds > (0.2*0.85*Fck)/1.5
    Ds = (0.2*0.85*Fck)/1.5
end
k2 = 0.15;
disp('Minimum shear stress in the section (N/mm^2)')
```

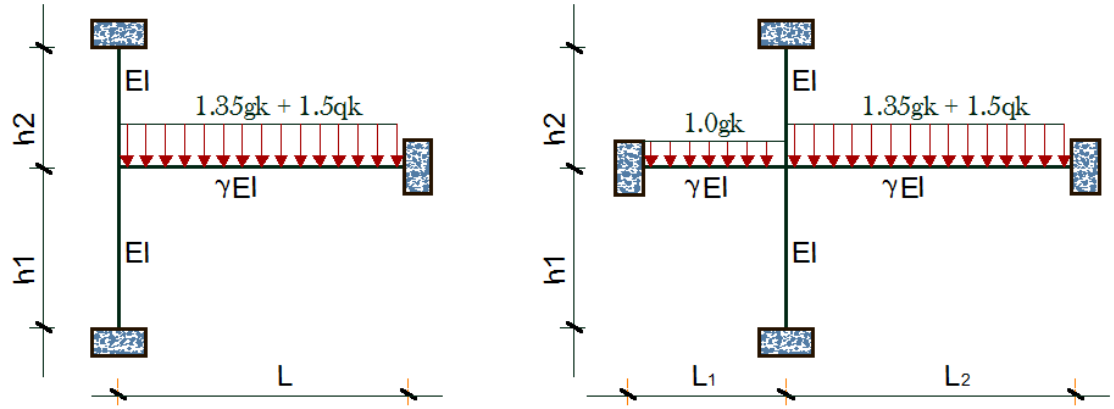
```
Vmin = 0.035*k1^(1.5)*sqrt(Fck)
disp('Concrete resistance shear stress (VRd) (N/mm^2)')
VRdQ = ((Crd*k1)*(100*P1*Fck)^(1/3)*(bw*d))/1000 % SHEAR FORCE CONTRIBUTION
VRdN = ((k2*Ds)*(bw*d))/1000 % AXIAL FORCE CONTRBUTION
VRd = VRdQ + VRdN % TOTAL SHEAR RESISTANCE
if VRd < ((Vmin+(k2*Ds))*(b*d))/1000
    VRd = ((Vmin+(k2*Ds))*(b*d))/1000
end
if VRd>VEd
    disp('Since VRd > VEd')
    disp('NO SHEAR REINFORCEMENT REQUIRED, PROVIDE NOMINAL LINKS')
else
if VRd<VEd
    disp('Since  VRd < VEd')
    disp('SHEAR REINFORCEMENT REQUIRED, CALCULATE COMPRESSION STRUT
CAPACITY')
    disp('Assume strut angle (thetha) = 21.8 deg, cot(thetha)= 2.5')
    thetha = 21.8
    V1 = 0.6*(1-(Fck/250))
    disp('Design compressive strength of concrete (fcd) (N/mm^2)')
    fcd = 0.567*Fck
    z = 0.9*d
    disp('Maximum capacity of compression strut (KN)')
    VRDmax = ((b*z*V1*fcd)/(2.9))/1000
    if VRDmax > VEd
        disp('Since VRDmax > VEd')
        disp('OK! Calculate diameter and spacing of links')
        ASMINlinks_to_spacing = ((0.08*sqrt(Fck))/Fyk)*b
        ASlinks_to_spacing = (VEd*1000)/(z*0.87*Fyk*2.5)
    else
        disp('Since VRDmax > VEd, it means we need a higher strut angle
(beta)')
        disp('By calculating the strut angle')
    disp('Shear stress at the section (N/mm^2)')
        v = (VEd*1000)/(b*d)
        beta = (0.5)*asind((v)/(0.153*Fck*(1-(Fck/250))))
    if beta>45
        disp('Since beta < 45 deg, SECTION INADEQUATE FOR SHEAR, INCREASE
DEPTH !!!!!')
    else
        disp('The ratio of Area of steel/spacing of links')
        ASlinks_to_spacing = (VEd*1000)/(z*0.87*Fyk*cotd(beta))
    end
    end
end
Area_of_legs = input('Enter the area of number of legs selected = ')
Spacing = input('Enter the spacing = ')
disp('CHECK')
T = Area_of_legs/Spacing
if T > ASlinks_to_spacing
    disp('Shear reinforcement is ok')
else
    disp('Increase area of steel, or reduce spacing')
end
end
```

A4.3 MATLAB PROGRAM FOR CALCULATING MOMENT OF INERTIA OF T-BEAMS

```
% T-BEAM SECTION PROPERTIES CALCULATOR
clc
disp('T-BEAM SECTION PROPERTIES CALCULATOR')
disp('THIS PROFORMA WAS WRITTEN BY UBANI RANKS')
B = input('Enter the width of the flange (mm) = ');
bw = input('Enter the width of the web (mm) = ');
hf = input('Enter the value of the flange thickness (mm) = ');
H = input('Enter the depth of drop of the beam (mm) = ');
A1 = B*hf
A2 = H*bw
AT = A1 + A2
Y1 = (hf/2)+H
Y2 = H/2
T = (A1*Y1) + (A2*Y2);
Ybar = (T)/(A1+A2)% CENTROID OF THE SECTION)
Ixx1 = (B*hf^3)/12
Ixx2 = (bw*H^3)/12
disp('FROM THE THEROEM OF PARALLEL AXIS')
if Ybar >= Y2
    MOI = (Ixx1 + A1*(Y1-Ybar)^2)+ (Ixx2 + A2*(Ybar - Y2)^2)
else
    MOI = (Ixx1 + A1*(Y1-Ybar)^2)+ (Ixx2 + A2*(Y2 - Ybar)^2)
end
disp('CONVERTING TO m^4')
format long
MOIB = MOI*10^(-12)

 disp('COLUMN SECTION PROPERTIES')
d = input('Enter the depth of column(mm) = ');
b = input('Enter the width of the column (mm) = ');
Ixxc = (b*d^3)/12
disp('CONVERTING TO m^4')
MOIC = Ixxc*10^(-12)
disp('RELATIVE STIFFNESS')
K = MOIB/MOIC
```

A4.4 SIMPLIFIED MATLAB PROGRAM FOR CALCULATING COLUMN MOMENT AT ULTIMATE LIMIT STATE



SUB FRAME TYPE 1 SUB FRAME TYPE 2

```
% COLUMN MOMENT CALCULATOR

clc
disp('SUB-FRAME ANALYSIS FOR MAXIMUM COLUMN MOMENT (ULS)')
TYPE = input('Enter type of sub-frame. Enter (1) for Type 1, and (2) for
Type 2 = ')
if TYPE == 1 % CALCULATION OF GEOMETRICAL PROPERTIES
    L = input ('Enter the Length of the beam (m) = ');
    H1 = input('Enter the height of the upper column (m) = ');
    H2 = input('Enter the height of the lower column (m) = ');

% CALCULATION OF THE MOMENT OF INERTIA OF THE SECTIONS
B = input('Enter the width of the flange (mm) = ');
bw = input('Enter the width of the web (mm) = ');
hf = input('Enter the thickness of the flange (mm) = ');
H = input('Enter the drop of the web (mm) = ');
A1 = B*hf;
A2 = H*bw;
AT = A1 + A2;
Y1 = (hf/2)+H;
Y2 = H/2;
T = (A1*Y1) + (A2*Y2);
Ybar = (T)/(A1+A2)% CENTROID OF THE SECTION)
Ixx1 = (B*hf^3)/12;
Ixx2 = (bw*H^3)/12;
disp('FROM THE THEROEM OF PARALLEL AXIS')
if Ybar >= Y2
    MOI = (Ixx1 + A1*(Y1-Ybar)^2)+ (Ixx2 + A2*(Ybar - Y2)^2)
else
    MOI = (Ixx1 + A1*(Y1-Ybar)^2)+ (Ixx2 + A2*(Y2 - Ybar)^2)
end
disp('CONVERTING TO m^4')
format long
MOIB = MOI*10^(-12) %Moment of inertia of the beam
MOIC =  2.332008333333333e-04 %Moment of inertia of the column
K1 = MOIB/MOIC
KD = K1/2

%Load Analysis
disp('THIS IS NOW TIME FOR LOAD ANALYSIS')
q = input('Enter the ultimate load 1.35gk + 1.5qk (KN/m) = ')
disp('Fixed end moment')
FEMF = (q*L^2)/12
J = input('Is there any other Fixed End Moment? (1) YES (0) N0 = ')
if J ==1.0
    FEMX = input('Additional FEM = ')
else
    FEMX = 0
end
FEMK = FEMF + FEMX
disp('Time to sum up the stiffness')
KT = (4/H2)+(4/H1)+(4*KD)/L
Z1 = FEMK/KT
MB = Z1*(4/H2) %TOP COLUMN MOMENT
M_support = MB/2 % BOTTOM COLUMN MOMENT
end

if TYPE ==2
    disp('Analysis of span 1 properties')
 L1 = input ('Enter the Length of the beam of span 1 (m) = ');
    H1 = input('Enter the height of the upper column (m) = ');
```

```
    H2 = input('Enter the height of the lower column (m) = ');

% CALCULATION OF THE MOMENT OF INERTIA OF THE SECTIONS
B = input('Enter the width of the flange (mm) = ');
bw = input('Enter the width of the web (mm) = ');
hf = input('Enter the thickness of the flange (mm) = ');
H = input('Enter the drop of the web (mm) = ');
A1 = B1*hf;
A2 = H*bw;
AT = A1 + A2;
Y1 = (hf/2)+H;
Y2 = H/2;
T = (A1*Y1) + (A2*Y2);
Ybar = (T)/(A1+A2)% CENTROID OF THE SECTION)
Ixx1 = (B1*hf^3)/12;
Ixx2 = (bw*H^3)/12;
disp('FROM THE THEROEM OF PARALLEL AXIS')
if Ybar >= Y2
    MOI = (Ixx1 + A1*(Y1-Ybar)^2)+ (Ixx2 + A2*(Ybar - Y2)^2)
else
    MOI = (Ixx1 + A1*(Y1-Ybar)^2)+ (Ixx2 + A2*(Y2 - Ybar)^2)
end
disp('CONVERTING TO m^4')
format long
MOIB = MOI*10^(-12)

disp('COLUMN SECTION PROPERTIES')
d = input('Enter the depth of column(mm) = ');
b = input('Enter the width of the column (mm) = ');
Ixxc = (b*d^3)/12
disp('CONVERTING TO m^4')
MOIC = Ixxc*10^(-12)
K1 = MOIB/MOIC
KD1 = K1/2

disp('Analysis of span 2 properties')
 L2 = input ('Enter the Length of the beam of span 2 (m) = ');
    H1 = input('Enter the height of the upper column (m) = ');
    H2 = input('Enter the height of the lowe column (m) = ');

% CALCULATION OF THE MOMENT OF INERTIA OF THE SECTIONS
B2 = input('Enter the width of the flange (mm) = ');
bw = input('Enter the width of the web (mm) = ');
hf = input('Enter the thickness of the flange (mm) = ');
H = input('Enter the drop of the web (mm) = ');
A1K = B2*hf;
A2K = H*bw;
ATK = A1K + A2K;
Y1K = (hf/2)+H;
Y2K = H/2;
TK = (A1K*Y1K) + (A2K*Y2K);
YbarK = (TK)/(A1K+A2K)% CENTROID OF THE SECTION)
Ixx1K = (B2*hf^3)/12;
Ixx2K = (bw*H^3)/12;
disp('FROM THE THEROEM OF PARALLEL AXIS')
if YbarK >= Y2K
    MOIK = (Ixx1K + A1K*(Y1K-YbarK)^2)+ (Ixx2K + A2K*(YbarK - Y2K)^2)
else
    MOIK = (Ixx1K + A1K*(Y1K-YbarK)^2)+ (Ixx2K + A2K*(Y2K - YbarK)^2)
end
disp('CONVERTING TO m^4')
```

```
format long
MOIBK = MOIK*10^(-12)
disp('COLUMN SECTION PROPERTIES')
d = input('Enter the depth of column(mm) = ');
b = input('Enter the width of the column (mm) = ');
Ixxc = (b*d^3)/12
disp('CONVERTING TO m^4')
MOIC = Ixxc*10^(-12)
K2 = MOIBK/MOIC
KD2 = K2/2

%Load Analysis
disp('THIS IS NOW TIME FOR LOAD ANALYSIS')
q1 = input('Enter the ultimate load 1.35gk + 1.5qk (KN/m) = ')

disp('Slab load Analysis')
Ly = input('Ly (m) = ');
Lx = input ('Lx (m) = ');
g = Ly/Lx
if g <=2.0
    disp('SLab is spanning in two direction')
G1 = input('Is beam on the Long span (1) or Short Span (0)? = ')
if G1 ==1
    Slab_load1 = ((6.45*Lx)/2)* (1-(1/3*g^2))
else
    Slab_load1 = ((6.45*Lx)/3)
end
else
    disp('SLab is spanning in one direction')
  G1 = input('Is beam on the Long span (1) or Short Span (0)? = ')
if G1 ==1
    Slab_load1 = ((6.45*Lx)/2)
else
    Slab_load1 = ((6.45*Lx)/5)
end
end


% ADJACENT SLAB LOAD
N = input('Is beam carrying an adjacent slab load? (1) YES (0) NO =    ')
if N ==1.0
Ly2 = input('Ly (m) = ');
Lx2 = input ('Lx (m) = ');
g2 = Ly2/Lx2
if g2 <=2.0
    disp('SLab is spanning in two direction')
G11 = input('Is beam on the Long span (1) or Short Span (0)? = ')
if G11 ==1
    Slab_load2 = ((6.45*Lx)/2)* (1-(1/3*g^2))
else
    Slab_load2 = ((6.45*Lx)/3)
end
else
    disp('SLab is spanning in one direction')
  G22 = input('Is beam on the Long span (1) or Short Span (0)? = ')
if G1 ==1
    Slab_load2 = ((6.45*Lx)/2)
else
    Slab_load2 = ((6.45*Lx)/5)
end
```

```
end
end
if N == 0
    Slab_load2 = 0
end
disp('Total 1.0gk = q2')
q2 = Slab_load1 + Slab_load2 + 11.2675

disp('Fixed end moment')
if L1>L2
FEM1 = -(q1*L1^2)/12
FEM2 = (q2*L2^2)/12
else
    FEM1 = -(q2*L1^2)/12
    FEM2 = (q1*L2^2)/12
end
J = input('Is there any other Fixed End Moment? (1) YES (0) N0 = ')
if J ==1.0
    FEMX = input('Additional FEM = ')
else
    FEMX = 0
end
FEMT = FEM1 + FEM2 + FEMX
if FEMT>1
    FEMT = -(FEMT)
else
    FEMT = FEMT
end

disp('Time to sum up the stiffness')
KT = (4/H2)+(4/H1)+((4*KD1)/L1)+ ((4*KD2)/L2)
Z1 = FEMT/KT
MB = Z1*(4/H2) %TOP COLUMN MOMENT
M_support = MB/2 %BOTTOM COLUMN MOMENT
end
```